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Quantum Mechanics

Non-relativistic Theory

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Contents

CHAPTER I

Fundamental Concepts of Quantum Mechanics

§ 1. The Uncertainty Principle

When classical mechanics and electrodynamics are applied in an attempt to explain atomic phenomena, they give results sharply at variance with experiment. This is already seen most clearly in the contradiction produced by applying ordinary electrodynamics to a model of the atom in which electrons move around the nucleus in classical orbits. Since any accelerated charge radiates, electrons in such orbits would be obliged to emit electromagnetic waves without interruption. Through radiation they would lose energy and eventually fall into the nucleus. Classical electrodynamics would therefore render the atom unstable—a conclusion entirely at odds with reality.

So profound a conflict between theory and experiment shows that a theory applicable to atomic phenomena—phenomena involving particles of very small mass in very small regions of space—requires a fundamental alteration of the basic classical ideas and laws.

A convenient point of departure for identifying these changes is the experimentally observed phenomenon known as electron diffraction[1]. When a uniform beam of electrons passes through a crystal, the transmitted beam displays a succession of intensity maxima and minima entirely analogous to the diffraction pattern observed for electromagnetic waves. Under certain conditions, therefore, the behavior of material particles—electrons—shows features characteristic of wave processes.

The depth of this phenomenon's conflict with ordinary ideas of motion is best shown by the following thought experiment, an idealization of electron diffraction by a crystal. Imagine a barrier impermeable to electrons, with two slits cut in it. If we observe an electron beam[2] passing through one slit while the other is closed, we obtain some intensity distribution on a solid screen behind the slit. Opening the second slit and closing the first gives another distribution. With both slits open at once, ordinary reasoning would lead us to expect simply the superposition of those two distributions: each electron, moving on its own trajectory, passes through one slit without affecting electrons passing through the other. Electron diffraction shows instead that the actual result is a diffraction pattern which, because of interference, is by no means the sum of the patterns from the separate slits. Plainly, this result cannot be reconciled in any way with the idea that electrons move along trajectories.

The mechanics governing atomic phenomena—the so-called *quantum* or *wave mechanics*—must therefore rest on concepts of motion fundamentally different from those of classical mechanics. Quantum mechanics has no concept of a particle trajectory.

[1] In fact, electron diffraction was discovered after quantum mechanics had been created. Our presentation, however, does not follow the historical sequence in which the theory developed; it is arranged instead to show as clearly as possible how the fundamental principles of quantum mechanics are connected with observed phenomena.

[2] The beam is assumed to be so dilute that interactions among its particles play no part.

This fact constitutes the content of the so-called *uncertainty principle*, a fundamental principle of quantum mechanics, due to *Heisenberg* (*W. Heisenberg*, 1927)[3]. Since it rejects the customary ideas of classical mechanics, the uncertainty principle may be said to have negative content. By itself it is naturally quite insufficient as the basis for constructing a new mechanics of particles. Such a theory must of course rest on certain positive propositions, which will be considered below (§ 2). Before those propositions can be formulated, however, the character of the problems posed to quantum mechanics must first be made clear.

For this purpose, let us first consider the special relation between quantum and classical mechanics.

Ordinarily, a more general theory can be formulated as a logically closed system independently of the less general theory that arises as its limiting case. Relativistic mechanics, for example, can be constructed from its own fundamental principles without any reference to Newtonian mechanics. The fundamental propositions of quantum mechanics, on the other hand, cannot in principle be formulated without invoking classical mechanics.

The absence of a definite trajectory for an electron[4] also deprives the electron, considered by itself, of any other dynamical characteristics[5]. It is therefore clear that no logically closed mechanics could be constructed for a system consisting solely of quantum objects. A quantitative description of electron motion requires the presence of physical objects which obey classical mechanics to sufficient accuracy. When an electron interacts with a "classical object," the state of that object is, in general, altered. Both the character and the magnitude of this alteration are governed by the electron's state and can therefore serve as its quantitative characteristic.

In this connection, the "classical object" is usually called an "apparatus," and its interaction with the electron is spoken of as a "measurement." It must be emphasized, however, that this does not mean a process of "measurement" involving a physicist as observer. In quantum mechanics, measurement means any interaction between a classical and a quantum object, occurring apart from and independently of any observer. The recognition of the profound role of the measurement concept in quantum mechanics is due to *Bohr* (*N. Bohr*).

We have defined an apparatus as a physical object which obeys classical mechanics to sufficient accuracy; a body of sufficiently great mass is one example. It should not be supposed, however, that an apparatus must be macroscopic. Under suitable conditions an object known to be microscopic can also play the part of an apparatus, since the meaning of "with sufficient accuracy" depends on the specific problem under consideration. In a Wilson chamber, for instance, one tracks the electron by its mist trail, whose thickness far exceeds atomic dimensions; at such a level of accuracy in locating the trajectory, the

[3]It is noteworthy that the complete mathematical apparatus of quantum mechanics had already been constructed by *W. Heisenberg* and *E. Schrödinger* in 1925–1926—prior to formulation of the uncertainty principle, which lays bare the physical content of that apparatus.

[4]In this and the next section, for brevity, we speak of an electron while meaning any quantum object in general: a particle or system of particles which obeys quantum but not classical mechanics.

[5]Here we mean quantities characterizing the electron's motion, not those characterizing the electron as a particle (charge and mass), which are parameters.

electron is a fully classical object.

Quantum mechanics thus occupies a very distinctive place among physical theories: it contains classical mechanics as its limiting case and, at the same time, requires that limiting case for its own foundation.

We can now formulate the problem addressed by quantum mechanics. A typical problem is to predict the outcome of a repeated measurement from the known outcomes of earlier measurements. We shall also see below that, compared with classical mechanics, quantum mechanics generally restricts the set of values that various physical quantities (energy, for example) may assume, that is, the values that may be found by measuring the quantity in question. The formalism of quantum mechanics must make it possible to determine these allowed values.

In quantum mechanics the measurement process has an essential peculiarity: it always acts upon the electron being measured, and for a given measurement accuracy this action cannot in principle be made arbitrarily weak. The more accurate the measurement, the stronger its action; only in measurements of very low accuracy can the action on the measured object be weak. This property of measurements is logically connected with the fact that the dynamical characteristics of an electron arise only through the measurement itself. It is clear that, if the action of the measurement process on the object could be made arbitrarily weak, the quantity being measured would have a definite value by itself, irrespective of the measurement.

Of the various types of measurement, the principal one is the measurement of an electron's coordinates. Within the domain of applicability of quantum mechanics, the electron's coordinates can always be measured[6] to any accuracy.

Suppose that successive measurements of the coordinates of an electron are made at specified time intervals Δt. As a rule, these results will fail to follow any smooth curve. Quite the opposite: the greater the measurement precision, the more abrupt and erratic the succession of results becomes, consistent with the absence of a path concept for the electron. A more or less smooth path is obtained only when the electron's coordinates are measured with a low degree of accuracy, for example, by the condensation of droplets of vapour in a Wilson chamber.

If instead, while keeping the measurement accuracy unchanged, the intervals Δt between measurements are reduced, adjacent measurements will, of course, give nearby coordinate values. Yet although the results of a series of successive measurements will lie in a small region of space, within that region they will be distributed quite irregularly and will by no means fit any smooth curve. In particular, as Δt tends to zero, the results of nearby measurements do not tend at all to lie on one straight line.

This last observation makes plain that a classical particle velocity does not exist in quantum mechanics: the ratio of the coordinate change between two time instants to the separation Δt tends to no well-defined limit. As we shall see later, however, quantum mechanics does admit a meaningful definition of a particle's velocity at a given instant, one that reduces to the classical velocity upon passing to classical mechanics.

In classical mechanics a particle possesses, at each instant, definite coordinates and

[6]We emphasize once again that by a "measurement made" we mean the interaction of the electron with a classical "apparatus"; this in no way presupposes the presence of an outside observer.

velocity; in quantum mechanics the situation is entirely different. If a measurement has given an electron definite coordinates, it then has no definite velocity at all. Conversely, an electron having a definite velocity cannot have a definite position in space. Indeed, the simultaneous existence of coordinates and velocity at any instant would imply the presence of a definite velocity, which the electron does not have. Thus, in quantum mechanics the electron's coordinates and velocity are quantities that cannot be measured exactly at the same time, that is, cannot simultaneously have definite values. One may say that the electron's coordinates and velocity are quantities which do not exist simultaneously. A quantitative relation governing the possibility of measuring coordinates and velocity inexactly at the same instant will be derived below. In classical mechanics, the state of a physical system is completely described by specifying all its coordinates and velocities at a given instant; from these initial data the equations of motion fully determine the system's behaviour at all future instants. In quantum mechanics such a description is impossible in principle, since the coordinates and their corresponding velocities do not exist simultaneously. Hence the state of a quantum system is described by fewer quantities than in classical mechanics, that is, its description is less detailed than a classical one.

From this follows a very important consequence regarding the character of predictions in quantum mechanics. Whereas a classical description is sufficient to predict with complete precision the future motion of a mechanical system, the less detailed quantum-mechanical description manifestly cannot be adequate for this purpose. Consequently, even when an electron is in a state described as completely as possible, its behaviour at subsequent instants is in principle not unique. Quantum mechanics therefore cannot make strictly definite predictions about the electron's future behaviour. For a specified initial state of an electron, a subsequent measurement may give different results. The task of quantum mechanics is only to determine the probability that one or another result will be obtained in that measurement. In some cases, of course, the probability of a particular measurement result may equal unity, that is, become a certainty, so that the result of the measurement is unique.

Every measurement process in quantum mechanics falls into one of two categories. The first, which includes the majority of measurements, comprises those that yield no determinate result with certainty for any state of the system. The second comprises measurements such that for every possible result there exists a state in which that result is obtained with certainty. It is precisely these latter measurements, which may be called *predictable*, that have the principal role in quantum mechanics. The quantitative characteristics of a state determined by such measurements are what quantum mechanics calls physical quantities. If in some state a measurement gives a unique result with certainty, we shall say that the corresponding physical quantity has a definite value in that state. Below we shall everywhere understand the expression "physical quantity" in precisely the sense stated here.

We shall repeatedly find below that by no means every collection of physical quantities in quantum mechanics can be measured simultaneously, that is, can have definite values simultaneously (we have already discussed one example: the velocity and coordinates of an electron).

Sets of physical quantities with the following property have an important role in quantum mechanics: the quantities can be measured simultaneously and, when they simultaneously have definite values, no other physical quantity (unless it is a function of them) can have a definite value in that state. We shall refer to such sets of physical quantities as *complete sets*.

Every description of an electron's state arises from some measurement. We now state what a complete description of a state means in quantum mechanics. Completely described states arise from the simultaneous measurement of a complete set of physical quantities. From the results of such a measurement one can, in particular, determine the probabilities of the results of any subsequent measurement, independently of everything that happened to the electron before the first measurement.

Below, everywhere except in § 14, by states of a quantum system we shall mean states described in precisely this complete manner.

§ 2. The superposition principle

The radical change in physical ideas about motion that quantum mechanics requires in comparison with classical mechanics naturally calls for an equally radical change in the theory's mathematical formalism. In this connection the first question that arises concerns the means of describing a state in quantum mechanics.

Let q stand for the totality of coordinates of a quantum system, and dq for the product of their differentials (it is called a volume element of the system's *configuration space*); for one particle, dq coincides with the volume element dV of ordinary space.

The mathematical formalism of quantum mechanics rests on the assertion that a system's state may be described by a certain (generally complex) function of the coordinates $\Psi(q)$, and that the squared modulus of this function determines the probability distribution of the coordinate values: $|\Psi|^2\,dq$ is the probability that a measurement made on the system will find coordinate values in the element dq of configuration space. The function Ψ is called the system's *wave function*[7].

Knowledge of the wave function makes it possible in principle to calculate the probabilities of the various outcomes of any measurement in general (not necessarily a coordinate measurement). Every one of these probabilities is given by an expression bilinear in Ψ and Ψ^*. In full generality, such an expression has the form

$$\iint \Psi(q)\Psi^*(q')\varphi(q,q')\,dq\,dq', \tag{2.1}$$

where the function $\varphi(q,q')$ depends on the kind and outcome of the measurement, and the integrations extend over the entire configuration space. The probability $\Psi\Psi^*$ itself for the different coordinate values is also an expression of this type[8].

As time passes, the system's state, and with it the wave function, generally changes. In this sense the wave function may also be regarded as a function of time. If the wave function is known at some initial instant, then, by the very meaning of a complete description of a state, it is thereby determined in principle at every future instant. The

[7] It was first introduced into quantum mechanics by *Schrödinger* (*E. Schrödinger*, 1926).

[8] It is obtained from (2.1) by taking $\varphi(q,q') = \delta(q-q_0)\,\delta(q'-q_0)$, where δ denotes the so-called δ-function defined below, in § 5; q_0 denotes the coordinate value whose probability is sought.

actual time dependence of the wave function is specified by equations to be derived below.

By definition, the sum of the probabilities of all possible values of the system's coordinates must equal unity. The integral of $|\Psi|^2$ over the whole configuration space must therefore equal unity:

$$\int |\Psi|^2 \, dq = 1. \tag{2.2}$$

This equality constitutes what is known as the *normalization condition* on wave functions. When the integral of $|\Psi|^2$ is convergent, by choosing a suitable constant coefficient, the function Ψ can always be normalized, as it is said. We shall see below, however, that the integral of $|\Psi|^2$ may diverge, in which case Ψ cannot be normalized by condition (2.2). Here $|\Psi|^2$ naturally does not yield the absolute coordinate probabilities; the ratio of $|\Psi|^2$ between two distinct points of configuration space, however, does determine the relative probability of the coordinate values.

Since all quantities with direct physical meaning that are calculated using the wave function have the form (2.1), in which Ψ occurs multiplied by Ψ^*, it is clear that a normalized wave function is defined only up to a constant *phase factor* of the form $e^{i\alpha}$, where α is any real number. This ambiguity is fundamental and cannot be removed; it is, however, immaterial, since it has no effect on any physical result.

The positive content of quantum mechanics is founded on a number of assertions about the properties of the wave function, as follows.

Suppose that in the state with wave function $\Psi_1(q)$ a certain measurement leads with certainty to a definite outcome, outcome 1, while in the state $\Psi_2(q)$ it leads to outcome 2. It is then postulated that every linear combination of Ψ_1 and Ψ_2, that is, every function of the form $c_1\Psi_1 + c_2\Psi_2$ (c_1, c_2 being constants), describes a state in which the same measurement yields either outcome 1 or outcome 2. Moreover, if the time dependence of the states is known, being given in one case by the function $\Psi_1(q, t)$ and in the other by $\Psi_2(q, t)$, then any linear combination of them also gives a possible time dependence of a state.

These assertions constitute the so-called *superposition principle for states*, the fundamental positive principle of quantum mechanics. It follows from this principle, in particular, that every equation satisfied by wave functions must be linear in Ψ.

Consider a system made up of two parts, and suppose its state is specified in such a way that each part is described completely[9]. One can then assert that the coordinate probabilities for q_1 in the first part are independent of those for the coordinates q_2 of the second, and therefore that the probability distribution for the whole system must equal the product of the probability distributions for its parts. It follows that the system's wave function $\Psi_{12}(q_1, q_2)$ may be written as a product of the wave functions $\Psi_1(q_1)$ and $\Psi_2(q_2)$ of its parts:

$$\Psi_{12}(q_1, q_2) = \Psi_1(q_1)\Psi_2(q_2). \tag{2.3}$$

If the two parts do not interact with each other, this relation between the wave functions

[9]This, of course, also gives a complete description of the state of the system as a whole. We emphasize, however, that the converse statement is by no means valid: in general, a complete description of the state of the system as a whole does not yet describe the states of its separate parts completely (see also § 14).

of the system and of its parts is preserved at future instants as well:

$$\Psi_{12}(q_1, q_2, t) = \Psi_1(q_1, t)\Psi_2(q_2, t). \tag{2.4}$$

§ 3. Operators

Consider some physical quantity f that characterizes the state of a quantum system. Strictly speaking, in what follows we ought to speak not of a single quantity but of an entire complete set of them at once. However, none of the reasoning is thereby essentially altered, and for the sake of brevity and simplicity we speak below of only a single physical quantity.

Those values which a given physical quantity may assume are termed its *eigenvalues* in quantum mechanics, and their aggregate is referred to as the eigenvalue *spectrum* of that quantity. In classical mechanics, quantities range, in general, over a continuous series of values. In quantum mechanics too there exist physical quantities (for example, coordinates) whose eigenvalues fill a continuous range; in such cases one speaks of a *continuous spectrum* of eigenvalues. Alongside these quantities, however, quantum mechanics also possesses quantities whose eigenvalues constitute a discrete set; one then speaks of a *discrete spectrum*.

For simplicity we shall first assume that the quantity f under consideration has a discrete spectrum; the case of a continuous spectrum is considered in § 5. We denote the eigenvalues of f by f_n, where the index n runs through the values $0, 1, 2, 3, \ldots$ We denote the wave function of the system in the state in which f has the value f_n by Ψ_n. The wave functions Ψ_n bear the name *eigenfunctions* of the physical quantity f. Every such function is taken to be normalized:

$$\int |\Psi_n|^2 \, dq = 1. \tag{3.1}$$

If the system is in some arbitrary state with wave function Ψ, then a measurement of f performed on it will yield as a result one of the eigenvalues f_n. By the superposition principle, the wave function Ψ must be a linear combination of those eigenfunctions Ψ_n that correspond to the values f_n which can be found with non-zero probability in a measurement performed on the system in the state under consideration. In the general case of an arbitrary state, Ψ may accordingly be written as the series

$$\Psi = \sum_n a_n \Psi_n, \tag{3.2}$$

where the summation is over all n, and a_n are certain constant coefficients.

Thus we arrive at the conclusion that every wave function can be, as one says, expanded in the eigenfunctions of any physical quantity. A system of functions in terms of which such an expansion can be carried out is spoken of as a *complete system of functions*.

The expansion (3.2) makes it possible to determine the probabilities of finding (by means of measurements) one or another value f_n of f for a system in the state with wave function Ψ. Indeed, according to what was said in the preceding section, these probabilities must be determined by certain expressions bilinear in Ψ and Ψ^* and must therefore be bilinear in a_n and a_n^*. Moreover, positivity of these expressions is obviously required. As for the probability associated with a given f_n: it must become unity for a system in the state $\Psi = \Psi_n$ and must vanish whenever expansion (3.2) of Ψ lacks a term containing the corresponding Ψ_n. The only essentially positive quantity satisfying this condition is the squared modulus of the coefficient a_n. Thus we arrive at the result that the squared modulus $|a_n|^2$ of every expansion coefficient in (3.2) gives the probability of the respective value f_n of f in the state with wave function Ψ. The sum of the probabilities of all possible values f_n must equal unity; in other words, the relation

$$\sum_n |a_n|^2 = 1. \tag{3.3}$$

must hold.

If Ψ were not normalized, then the relation (3.3) would also not hold. The sum $\sum_n |a_n|^2$ would then have to be determined by some expression bilinear in Ψ and Ψ^* and reducing to unity for normalized Ψ. The only such expression is the integral $\int \Psi\Psi^*\, dq$. Thus the equality

$$\sum_n a_n a_n^* = \int \Psi\Psi^*\, dq. \tag{3.4}$$

must hold.

On the other hand, multiplying the expansion $\Psi^* = \sum_n a_n^* \Psi_n^*$ of the complex conjugate function Ψ^* by Ψ and integrating, we obtain

$$\int \Psi\Psi^*\, dq = \sum_n a_n^* \int \Psi_n^* \Psi\, dq.$$

Comparing this with (3.4), we have

$$\sum_n a_n a_n^* = \sum_n a_n^* \int \Psi_n^* \Psi\, dq,$$

from which we find the following formula determining the coefficients a_n of the expansion of Ψ in the eigenfunctions Ψ_n:

$$a_n = \int \Psi\Psi_n^*\, dq. \tag{3.5}$$

Substituting (3.2) here, we obtain

$$a_n = \sum_m a_m \int \Psi_m \Psi_n^*\, dq,$$

whence it is seen that the eigenfunctions must satisfy the conditions

$$\int \Psi_m \Psi_n^*\, dq = \delta_{nm}, \tag{3.6}$$

where $\delta_{nm} = 1$ for $n = m$ and $\delta_{nm} = 0$ for $n \neq m$. The fact that the integrals of the products $\Psi_m \Psi_n^*$ with $m \neq n$ vanish is spoken of as the mutual *orthogonality* of the functions Ψ_n. The collection of eigenfunctions Ψ_n thus constitutes a complete system of normalized and mutually orthogonal (or, as one says for short, *orthonormal*) functions.

We introduce the concept of the *mean value* $\bar{f}$ of a quantity f in a given state. Following the standard definition, $\bar{f}$ is the sum of every eigenvalue f_n of the quantity in question, weighted by the respective probability $|a_n|^2$:

$$\bar{f} = \sum_n f_n |a_n|^2. \tag{3.7}$$

We write $\bar{f}$ in the form of an expression that involves not the expansion coefficients of Ψ but the function itself. Since (3.7) contains the products $a_n a_n^*$, it is clear that the desired expression must be bilinear in Ψ and Ψ^*. We introduce a certain mathematical *operator*, which we denote by $\hat{f}$[10], and define as follows. Let $(\hat{f}\Psi)$ denote the result of the action of the operator $\hat{f}$ on the function Ψ. We define $\hat{f}$ by requiring that the integral of $(\hat{f}\Psi)$ multiplied by the conjugate Ψ^* yield $\bar{f}$:

$$\bar{f} = \int \Psi^* (\hat{f}\Psi)\, dq. \tag{3.8}$$

It is easy to see that in the general case the operator $\hat{f}$ is some linear integral operator.[11] Indeed, using the expression (3.5) for a_n, the definition (3.7) of the mean value can be cast as

$$\bar{f} = \sum_n f_n a_n a_n^* = \int \Psi^* \left(\sum_n a_n f_n \Psi_n \right) dq.$$

Comparing with (3.8), we see that the result of the action of the operator $\hat{f}$ on the function Ψ has the form

$$(\hat{f}\Psi) = \sum_n a_n f_n \Psi_n. \tag{3.9}$$

Substituting here the expression (3.5) for a_n, we find that $\hat{f}$ is an integral operator of the form

$$(\hat{f}\Psi) = \int K(q, q') \Psi(q')\, dq', \tag{3.10}$$

where the function $K(q, q')$ (the so-called kernel of the operator) is

$$K(q, q') = \sum_n f_n \Psi_n^*(q') \Psi_n(q). \tag{3.11}$$

Thus, to each physical quantity in quantum mechanics there is assigned a definite linear operator.

[10]We shall everywhere denote operators by letters with a hat.

[11]A linear operator is one possessing the properties: $\hat{f}(\Psi_1 + \Psi_2) = \hat{f}\Psi_1 + \hat{f}\Psi_2$, $\hat{f}(a\Psi) = a\hat{f}\Psi$, where Ψ_1, Ψ_2 are arbitrary functions and a is an arbitrary constant.

From (3.9) it is seen that if Ψ is one of the eigenfunctions Ψ_n (so that all a_n except one are zero), then upon the action of the operator $\hat{f}$ on it, this function is simply multiplied by the eigenvalue f_n[12]:

$$\hat{f}\Psi_n = f_n\Psi_n. \tag{3.12}$$

The eigenfunctions of a physical quantity f are therefore solutions of the equation

$$\hat{f}\Psi = f\Psi,$$

in this equation f denotes a constant; the eigenvalues are those particular values of f that yield solutions obeying the necessary conditions. As will be seen later, the explicit form of operators for the various physical quantities can be established on direct physical grounds, whereupon this operator property enables one to obtain eigenfunctions and eigenvalues by solving the equations $\hat{f}\Psi = f\Psi$.

For a real physical quantity, both the eigenvalues themselves and their mean values in any state are real. This fact places a definite constraint on the properties of the associated operators. Setting the expression (3.8) equal to its own complex conjugate gives

$$\int \Psi^*(\hat{f}\Psi)\,dq = \int \Psi(\hat{f}^*\Psi^*)\,dq, \tag{3.13}$$

where $\hat{f}^*$ denotes the operator complex conjugate to $\hat{f}$[13]. For an arbitrary linear operator such a relation does not, in general, hold, so that it represents a certain restriction imposed on the possible form of the operators $\hat{f}$. For an arbitrary operator $\hat{f}$ one can define, as one says, the *transposed* operator $\tilde{\hat{f}}$, defined so that

$$\int \Phi(\hat{f}\Psi)\,dq = \int \Psi(\tilde{\hat{f}}\Phi)\,dq, \tag{3.14}$$

where Ψ, Φ are two different functions. Choosing as Φ the conjugate function Ψ^*, comparison with (3.13) reveals that

$$\tilde{\hat{f}} = \hat{f}^*. \tag{3.15}$$

An operator that satisfies this condition is called *Hermitian*[14]. Operators representing real physical quantities within the mathematical apparatus of quantum mechanics must therefore be Hermitian.

One may also formally consider complex physical quantities, i.e. quantities possessing complex eigenvalues. Suppose f is one such quantity; one may then form the complex-conjugate quantity f^*, whose eigenvalues are the complex conjugates of the eigenvalues of f. The operator corresponding to the quantity f^* will be denoted by $\hat{f}^+$. It is called the *adjoint* operator to $\hat{f}$ and must, in general, be distinguished from the complex-conjugate

[12]Below, wherever it cannot lead to misunderstanding, we shall omit the parentheses in the expression $(\hat{f}\Psi)$, the operator being understood to act on the expression written after it.

[13]By definition, if for the operator $\hat{f}$ we have $\hat{f}\psi = \varphi$, then the complex-conjugate operator $\hat{f}^*$ is the operator for which $\hat{f}^*\psi^* = \varphi^*$.

[14]For a linear integral operator of the form (3.10), Hermiticity requires that the kernel of the operator satisfy $K(q, q') = K^*(q', q)$.

operator $\hat{f}^*$. Indeed, by definition of the operator $\hat{f}^+$, the mean value of f^* in some state Ψ is

$$\overline{f^*} = \int \Psi^* \hat{f}^+ \Psi \, dq.$$

On the other hand, we have

$$(\bar{f})^* = \left[\int \Psi^* \hat{f} \Psi \, dq\right]^* = \int \Psi \hat{f}^* \Psi^* \, dq = \int \Psi^* \tilde{\hat{f}}^* \Psi \, dq.$$

Equating the two expressions, we find that

$$\hat{f}^+ = \tilde{\hat{f}}^*, \tag{3.16}$$

from which it is clear that $\hat{f}^+$ does not, in general, coincide with $\hat{f}^*$. The condition (3.15) can now be written in the form

$$\hat{f} = \hat{f}^+, \tag{3.17}$$

i.e. the operator of a real physical quantity coincides with its adjoint (Hermitian operators are also called *self-adjoint*).

Let us show how one can directly prove the mutual orthogonality of the eigenfunctions of a Hermitian operator corresponding to different eigenvalues. Let f_n, f_m be two different eigenvalues of the real quantity f, and Ψ_n, Ψ_m the corresponding eigenfunctions:

$$\hat{f}\Psi_n = f_n \Psi_n, \qquad \hat{f}\Psi_m = f_m \Psi_m.$$

Multiplying both sides of the first of these equalities by Ψ_m^*, and the complex conjugate of the second equality by Ψ_n, and subtracting these products termwise from each other, we obtain

$$\Psi_m^* \hat{f} \Psi_n - \Psi_n \hat{f}^* \Psi_m^* = (f_n - f_m) \Psi_n \Psi_m^*.$$

Integrate both sides of this equality over dq. Since $\hat{f}^* = \tilde{\hat{f}}$, by virtue of (3.14) the integral of the left-hand side of the equality vanishes, so that we obtain

$$(f_n - f_m) \int \Psi_n \Psi_m^* \, dq = 0,$$

whence, in view of $f_n \neq f_m$, the required orthogonality property of the functions Ψ_n and Ψ_m follows.

We have been speaking here all along of only a single physical quantity f, whereas we ought to speak, as was noted at the beginning of the section, of a complete system of simultaneously measurable physical quantities. Then we would find that to each of these quantities $f, g, \ldots$ there corresponds its own operator $\hat{f}, \hat{g}, \ldots$ The eigenfunctions Ψ_n correspond to states in which all the quantities under consideration have definite values, i.e. correspond to definite sets of eigenvalues $f_n, g_n, \ldots$, and are joint solutions of the system of equations

$$\hat{f}\Psi = f\Psi, \qquad \hat{g}\Psi = g\Psi, \ldots$$

§ 4. Addition and Multiplication of Operators

If $\hat{f}$ and $\hat{g}$ are the operators corresponding to two physical quantities f and g, then $\hat{f} + \hat{g}$ corresponds to their sum $f + g$. In quantum mechanics, however, the meaning of adding different physical quantities differs essentially according as the quantities are or are not measurable simultaneously. If f and g are simultaneously measurable, the operators $\hat{f}$ and $\hat{g}$ have common eigenfunctions. These are also eigenfunctions of $\hat{f} + \hat{g}$, whose eigenvalues are the sums $f_n + g_n$.

If f and g cannot have definite values simultaneously, their sum $f + g$ has a more restricted meaning. All that can be stated is that its mean value in an arbitrary state equals the sum of the separate mean values of the two terms:

$$\overline{f + g} = \bar{f} + \bar{g}. \tag{4.1}$$

The eigenvalues and eigenfunctions of $\hat{f} + \hat{g}$, on the other hand, will in general have no relation to those of f and g. Clearly, when $\hat{f}$ and $\hat{g}$ are Hermitian, so is $\hat{f} + \hat{g}$; its eigenvalues are therefore real and are the eigenvalues of the new quantity $f + g$ defined in this way.

We note the following theorem. Let f_0 and g_0 be the least eigenvalues of f and g, and let $(f + g)_0$ denote the corresponding quantity for $f + g$. Then

$$(f + g)_0 \geq f_0 + g_0. \tag{4.2}$$

The equality sign applies if f and g are simultaneously measurable. The proof follows from the evident fact that a quantity's mean value is always at least its least eigenvalue. In the state in which $(f + g)$ has the value $(f + g)_0$, we have $\overline{(f + g)} = (f + g)_0$; on the other hand, $\overline{(f + g)} = \bar{f} + \bar{g} \geq f_0 + g_0$, which gives (4.2).

Suppose again that f and g are simultaneously measurable. Besides their sum, we may introduce their product as a quantity whose eigenvalues are the products of the eigenvalues of f and g. The corresponding operator is readily seen to act by applying one operator to a function and then the other. Mathematically it is written as the product of $\hat{f}$ and $\hat{g}$. Indeed, if Ψ_n are common eigenfunctions of $\hat{f}$ and $\hat{g}$, then

$$\hat{f}\hat{g}\Psi_n = \hat{f}(\hat{g}\Psi_n) = \hat{f}g_n\Psi_n = g_n\hat{f}\Psi_n = g_n f_n\Psi_n$$

(the symbol $\hat{f}\hat{g}$ denotes the operator which acts on a function Ψ first by applying $\hat{g}$ to Ψ, and then by applying $\hat{f}$ to the function $\hat{g}\Psi$). We could equally well have used $\hat{g}\hat{f}$, whose factors occur in the opposite order. Plainly, the actions of these two operators on the functions Ψ_n give the same result. Since every wave function Ψ is expandable in the functions Ψ_n, it follows that $\hat{f}\hat{g}$ and $\hat{g}\hat{f}$ yield the same result when acting on an arbitrary function. This fact may be written symbolically as $\hat{f}\hat{g} = \hat{g}\hat{f}$, or

$$\hat{f}\hat{g} - \hat{g}\hat{f} = 0. \tag{4.3}$$

Two such operators $\hat{f}$ and $\hat{g}$ are said to *commute* with one another. We have therefore reached an important result: if two quantities f and g can have definite values simultaneously, their operators commute.

The converse theorem can also be proved (see § 11): when $\hat{f}$ and $\hat{g}$ commute, all their eigenfunctions can be chosen in common, which physically means that the corresponding physical quantities are simultaneously measurable. Thus commutation of the operators is a necessary and sufficient condition for simultaneous measurability of physical quantities.

Raising an operator to a power is a special case of multiplying operators. It follows from what has been said that the eigenvalues of $\hat{f}^p$ (where p is an integer) are the eigenvalues of $\hat{f}$ raised to the same pth power. More generally, any function of an operator, $\varphi(\hat{f})$, may be defined as the operator whose eigenvalues are the same function $\varphi(f)$ of the eigenvalues of $\hat{f}$. When $\varphi(f)$ has a Taylor-series expansion, this expansion reduces the action of $\varphi(\hat{f})$ to the actions of the various powers $\hat{f}^p$.

In particular, $\hat{f}^{-1}$ is called the operator *inverse* to $\hat{f}$. Clearly, successive application of $\hat{f}$ and $\hat{f}^{-1}$ to an arbitrary function leaves it unchanged; that is, $\hat{f}\hat{f}^{-1} = \hat{f}^{-1}\hat{f} = 1$. When f and g are not simultaneously measurable, their product has no such direct meaning. This is already apparent because $\hat{f}\hat{g}$ is then not Hermitian and consequently cannot correspond to a real physical quantity. Indeed, by the definition of the transposed operator,

$$\int \Psi\hat{f}\hat{g}\Phi\, dq = \int \Psi\hat{f}(\hat{g}\Phi)\, dq = \int (\hat{g}\Phi)(\tilde{f}\Psi)\, dq.$$

Here $\tilde{f}$ acts only on Ψ, while $\hat{g}$ acts on Φ; the integrand is therefore simply the product of the two functions $\hat{g}\Phi$ and $\tilde{f}\Psi$. Applying the definition of the transposed operator once more gives

$$\int \Psi\hat{f}\hat{g}\Phi\, dq = \int (\tilde{f}\Psi)(\hat{g}\Phi)\, dq = \int \Phi\tilde{g}\,\tilde{f}\Psi\, dq.$$

We have thus obtained an integral in which Ψ and Φ have exchanged places relative to the original one. In other words, $\tilde{g}\,\tilde{f}$ is the operator transposed to $\hat{f}\hat{g}$, so we may write

$$\widetilde{fg} = \tilde{g}\,\tilde{f}, \tag{4.4}$$

that is, the transpose of the product $\hat{f}\hat{g}$ is the product of the transposed factors written in reverse order. Taking the complex conjugate of both sides of (4.4), we find

$$(\hat{f}\hat{g})^+ = \hat{g}^+\hat{f}^+. \tag{4.5}$$

If both $\hat{f}$ and $\hat{g}$ are Hermitian, then $(\hat{f}\hat{g})^+ = \hat{g}\hat{f}$. Hence $\hat{f}\hat{g}$ is Hermitian only when the factors $\hat{f}$ and $\hat{g}$ commute.

From the products $\hat{f}\hat{g}$ and $\hat{g}\hat{f}$ of two noncommuting Hermitian operators one can nevertheless form another Hermitian operator, their *symmetrized product*

$$\frac{1}{2}(\hat{f}\hat{g} + \hat{g}\hat{f}). \tag{4.6}$$

It is also readily verified that the difference $\hat{f}\hat{g} - \hat{g}\hat{f}$ is an "anti-Hermitian" operator (that is, its transpose is equal to the negative of its complex conjugate). Multiplication by i makes it Hermitian; hence

$$i(\hat{f}\hat{g} - \hat{g}\hat{f}) \tag{4.7}$$

is likewise a Hermitian operator.

For brevity, later on we shall sometimes use the notation

$$\{\hat{f}, \hat{g}\} = \hat{f}\hat{g} - \hat{g}\hat{f} \tag{4.8}$$

for the so-called *commutator* of the operators. It is easy to verify the relation

$$\{\hat{f}\hat{g}, \hat{h}\} = \{\hat{f}, \hat{h}\}\hat{g} + \hat{f}\{\hat{g}, \hat{h}\}. \tag{4.9}$$

Notice that $\{\hat{f}, \hat{h}\} = 0$ and $\{\hat{g}, \hat{h}\} = 0$ do not in general imply that $\hat{f}$ and $\hat{g}$ commute with each other.

§ 5. Continuous spectrum

All the relations obtained in § 3, § 4 for the properties of discrete-spectrum eigenfunctions extend without difficulty to a continuous spectrum of eigenvalues.

Let f be a physical quantity possessing a continuous spectrum. We shall denote its eigenvalues simply by the same letter f without an index; the corresponding eigenfunctions will be written Ψ_f. In the same way that an arbitrary wave function Ψ admits the series expansion (3.2) in eigenfunctions of a discrete-spectrum quantity, it can also be expanded — this time as an integral — over the complete system of eigenfunctions of a continuous-spectrum quantity. Such an expansion takes the form

$$\Psi(q) = \int a_f \Psi_f(q)\, df, \tag{5.1}$$

where the integration is taken over the whole domain of values that the quantity f can assume.

More complicated than in the discrete-spectrum case is the question of normalizing the continuous-spectrum eigenfunctions. As we shall see below, the requirement that the integral of the squared modulus equal unity cannot be satisfied here. Instead, let us set ourselves the aim of normalizing the functions Ψ_f in such a way that $|a_f|^2\, df$ represents the probability for the physical quantity under consideration, in the state described by the wave function Ψ, to have a value in the given interval between f and $f + df$. Since the sum of the probabilities of all possible values of f must be equal to unity, we have

$$\int |a_f|^2\, df = 1 \tag{5.2}$$

(analogously to relation (3.3) for the discrete spectrum).

Proceeding exactly as we did in deriving formula (3.5), and using the same arguments, we write, on one hand,

$$\int \Psi\Psi^*\, dq = \int |a_f|^2\, df,$$

and, on other hand,

$$\int \Psi\Psi^*\, dq = \iint a_f^* \Psi_f^* \Psi\, df\, dq.$$

By comparing the two expressions we find the formula that determines the expansion coefficients,

$$a_f = \int \Psi(q)\Psi_f^*(q)\, dq, \tag{5.3}$$

exactly analogous to (3.5).

To derive the normalization condition we now substitute (5.1) into (5.3):

$$a_f = \int a_{f'} \left(\int \Psi_{f'} \Psi_f^* \, dq \right) df'.$$

This relation must hold for arbitrary a_f and must therefore be satisfied identically. For this to be so it is first of all necessary that the coefficient of $a_{f'}$ under the integral sign (i.e. the integral $\int \Psi_{f'} \Psi_f^* \, dq$) vanish for all $f' \neq f$. For $f' = f$ this coefficient must become infinite (otherwise the integral over df' would simply be equal to zero). Thus the integral $\int \Psi_{f'} \Psi_f^* \, dq$ is a function of the difference $f - f'$ which vanishes for non-zero values of the argument and becomes infinite when the argument is zero. We denote this function by $\delta(f' - f)$:

$$\int \Psi_{f'} \Psi_f^* \, dq = \delta(f' - f). \tag{5.4}$$

The manner in which the function $\delta(f' - f)$ becomes infinite at $f' - f = 0$ is determined by the requirement that there should be

$$\int \delta(f' - f) a_{f'} \, df' = a_f.$$

Clearly, for this to hold it is necessary that

$$\int \delta(f' - f) \, df' = 1.$$

A function defined in this way is called the *delta-function*[15]. Let us write out once more the formulas that define it. We have

$$\delta(x) = 0 \quad \text{at } x \neq 0, \qquad \delta(0) = \infty, \tag{5.5}$$

and moreover in such a way that

$$\int_{-\infty}^{+\infty} \delta(x) \, dx = 1. \tag{5.6}$$

As the limits of integration one may write any other pair between which the point $x = 0$ lies. If $f(x)$ is a function continuous at $x = 0$, then

$$\int_{-\infty}^{+\infty} \delta(x) f(x) \, dx = f(0). \tag{5.7}$$

In more general form this formula can be written as

$$\int \delta(x - a) f(x) \, dx = f(a), \tag{5.8}$$

where the region of integration includes the point $x = a$, and $f(x)$ is continuous at $x = a$. It is also obvious that the delta-function is even, i.e.

$$\delta(-x) = \delta(x). \tag{5.9}$$

[15]The delta-function was introduced into theoretical physics by *Dirac* (*P. A. M. Dirac*).

Finally, writing

$$\int_{-\infty}^{\infty} \delta(\alpha x)\, dx = \int_{-\infty}^{\infty} \delta(y) \frac{dy}{|\alpha|} = \frac{1}{|\alpha|},$$

we arrive at the conclusion that

$$\delta(\alpha x) = \frac{1}{|\alpha|}\delta(x), \tag{5.10}$$

where α is any constant. Formula (5.4) expresses the normalization rule for the eigenfunctions of the continuous spectrum; it replaces condition (3.6) of the discrete spectrum. We see that the functions Ψ_f and $\Psi_{f'}$ with $f \neq f'$ remain orthogonal to each other. The integrals of the squares $|\Psi_f|^2$ of the functions of the continuous spectrum, however, diverge.

The functions $\Psi_f(q)$ satisfy one further relation, similar to (5.4). To derive it we substitute (5.3) into (5.1), which gives

$$\Psi(q) = \int \Psi(q') \left(\int \Psi_f^*(q') \Psi_f(q)\, df \right) dq',$$

whence we immediately conclude that one must have

$$\int \Psi_f^*(q') \Psi_f(q)\, df = \delta(q - q'). \tag{5.11}$$

An analogous relation can, of course, be derived for the discrete spectrum, where it takes the form

$$\sum_n \Psi_n^*(q') \Psi_n(q) = \delta(q' - q). \tag{5.12}$$

On comparing the pair of formulas (5.1) and (5.4) with the pair (5.3) and (5.11), we see that, on the one hand, the functions $\Psi_f(q)$ effect the expansion of the function $\Psi(q)$ with the expansion coefficients a_f, while, on the other hand, formula (5.3) can be regarded as a completely analogous expansion of the function $a_f \equiv a(f)$ in the functions $\Psi_f^*(q)$, with $\Psi(q)$ playing the role of the expansion coefficients. The function $a(f)$, like $\Psi(q)$, determines the state of the system completely; it is spoken of as the wave function in the *f-representation* (and the function $\Psi(q)$ as the wave function in the *q-representation*). Just as $|\Psi(q)|^2$ determines the probability for the system to have coordinates in a given interval dq, so $|a(f)|^2$ determines the probability of values of the quantity f in a given interval df. The functions $\Psi_f(q)$, for their part, are, on the one hand, the eigenfunctions of the quantity f in the q-representation, while their complex conjugates $\Psi_f^*(q)$ serve as eigenfunctions of the coordinate q in the f-representation.

Suppose $\varphi(f)$ is a function of the quantity f, with φ and f standing in one-to-one correspondence. Every function $\Psi_f(q)$ may then equally be viewed as an eigenfunction of the quantity φ. In doing so, however, the normalization of these functions must be changed. Indeed, the eigen- functions $\Psi_\varphi(q)$ of the quantity φ must be normalized by the condition

$$\int \Psi_{\varphi(f')} \Psi_{\varphi(f)}^*\, dq = \delta[\varphi(f') - \varphi(f)],$$

whereas the functions Ψ_f are normalized by condition (5.4). The argument of the delta-function vanishes at $f' = f$. For f' close to f we have

$$\varphi(f') - \varphi(f) = \frac{d\varphi(f)}{df}(f' - f).$$

Making use of expression (5.10), one can write[16]

$$\delta[\varphi(f') - \varphi(f)] = \frac{1}{|d\varphi(f)/df|}\delta(f' - f). \tag{5.13}$$

Comparison of (5.13) with (5.4) now shows that the functions Ψ_φ and Ψ_f are related to each other by the relation

$$\Psi_{\varphi(f)} = \frac{1}{\sqrt{|d\varphi(f)/df|}}\Psi_f. \tag{5.14}$$

There exist physical quantities which, over some domain of their values, possess a discrete spectrum, and over another possess a continuous one. For the eigenfunctions of such a quantity, of course, all the same relations that were derived in this and the preceding sections continue to hold. It need only be noted that the complete system of functions is formed by the collection of the eigenfunctions of both spectra together. Therefore the expansion of an arbitrary wave function in the eigenfunctions of such a quantity has the form

$$\Psi(q) = \sum_n a_n \Psi_n(q) + \int a_f \Psi_f(q)\, df, \tag{5.15}$$

where the summation runs over the discrete spectrum, and the integration over the entire continuous spectrum.

An example of a quantity possessing a continuous spectrum is the coordinate q itself. One readily sees that the associated operator reduces to multiplication by q. Indeed, since the square $|\Psi(q)|^2$ governs the probability of different coordinate values, the mean value of the coordinate is

$$\bar{q} = \int q|\Psi|^2\, dq = \int \Psi^* q \Psi\, dq.$$

Comparing this expression with the definition of operators according to (3.8), we see that[17]

$$\hat{q} = q. \tag{5.16}$$

The eigenfunctions of this operator must be determined, according to the general rule, by the equation $q\Psi_{q_0} = q_0\Psi_{q_0}$, where q_0 is used temporarily to denote concrete values of

[16] In general, if $\varphi(x)$ is a single-valued function (its inverse, however, may be many-valued), then the formula

$$\delta[\varphi(x)] = \sum_i \frac{1}{|\varphi'(\alpha_i)|}\delta(x - \alpha_i), \tag{5.13a}$$

holds, where α_i are the roots of the equation $\varphi(x) = 0$.

[17] In what follows we shall agree, for simplicity of notation, to write operators that reduce to multiplication by some quantity everywhere simply as that quantity itself.

the coordinate, in distinction from the variable q. Since this equality can be satisfied either when $\Psi_{q_0} = 0$ or when $q = q_0$, it is clear that the normalized eigenfunctions are[18]

$$\Psi_{q_0} = \delta(q - q_0). \tag{5.17}$$

§ 6. The Limiting Transition

Classical mechanics is contained in quantum mechanics as a limiting case. We must determine how this limiting transition takes place.

In quantum mechanics a wave function describes the electron, governing the various possible values of its coordinate; thus far we know only that this function solves a certain linear partial differential equation. Classical mechanics, by contrast, treats the electron as a material particle moving along a trajectory fixed completely by the equations of motion. A relation analogous in a certain sense to that between quantum and classical mechanics occurs in electrodynamics between wave optics and geometrical optics. Wave optics describes electromagnetic waves by electric- and magnetic-field vectors satisfying a particular system of linear differential equations (Maxwell's equations). Geometrical optics, however, describes the propagation of light along definite paths—rays. This analogy leads one to conclude that the quantum-to-classical limiting transition is analogous to that from wave optics to geometrical optics.

We now recall how this latter transition is carried out mathematically (see II, § 53). Let u be any component of the field in an electromagnetic wave. It may be written $u = ae^{i\varphi}$, with real amplitude a and phase φ (the phase is called the eikonal in geometrical optics). The geometrical-optics limit corresponds to short wavelengths; mathematically, this means that φ changes greatly over short distances. In particular, its absolute magnitude may be regarded as large.

Correspondingly, assume that the classical limiting case in quantum mechanics is represented by wave functions of the form $\Psi = ae^{i\varphi}$, where a varies slowly and φ takes large values. In mechanics, as is known, a particle trajectory can be found from a variational principle according to which the so-called action S of the mechanical system must be a minimum (the principle of least action). In geometrical optics the ray path is instead fixed by Fermat's principle: the "optical path length" of the ray, that is, the difference between its phases at the end and at the beginning of the path, must be a minimum.

This analogy allows us to conclude that, in the classical limiting case, the phase φ of the wave function must be proportional to the mechanical action S of the physical system under consideration: $S = \text{const} \cdot \varphi$. The proportionality factor is called *Planck's constant* and is denoted by $\hbar$[19]. Since φ is dimensionless, it has the dimensions of action, and its value is

$$\hbar = 1{,}055 \cdot 10^{-27}\ \text{erg} \cdot \text{s}.$$

[18]The expansion coefficients of an arbitrary function Ψ in these eigenfunctions are $a_{q_0} = \int \Psi(q)\,\delta(q - q_0)\,dq = \Psi(q_0)$. The probability of values of the coordinate in a given interval dq_0 is $|a_{q_0}|^2\,dq_0 = |\Psi(q_0)|^2\,dq_0$, as it should be.

[19]It was introduced into physics by *Planck* (*M. Planck*, 1900). The constant $\hbar$ used throughout this book is, strictly speaking, Planck's constant h divided by 2π (Dirac's notation).

Thus the wave function of an “almost classical” (or, as it is called, *quasiclassical*) physical system has the form

$$\Psi = ae^{iS/\hbar}. \tag{6.1}$$

Planck’s constant has a fundamental part in every quantum phenomenon. Its relative magnitude (in comparison with other quantities having the same dimensions) determines the “degree of quantumness” of a given physical system. Passage from quantum to classical mechanics corresponds to a large phase and can formally be described by taking the limit $\hbar \to 0$ (just as the passage from wave to geometrical optics corresponds to the zero-wavelength limit, $\lambda \to 0$).

We have established the limiting form of the wave function, but must still ask how it is related to classical motion along a trajectory. In general, motion described by a wave function does not at all turn into motion along one fixed trajectory. Its connection with classical motion is this: if the wave function, and hence the probability distribution of the coordinates, is specified at an initial instant, the distribution thereafter “moves” as required by the laws of classical mechanics (see the end of § 17 for details).

To obtain motion along a definite trajectory one must begin with a wave function of a special kind, appreciably different from zero only within a very small region of space (a so-called *wave packet*); the dimensions of this region may be made to tend to zero together with $\hbar$. One can then state that, in the quasiclassical case, the wave packet moves through space along the particle’s classical trajectory.

Finally, in the limit, quantum-mechanical operators must reduce simply to multiplication by their corresponding physical quantities.

§ 7. The Wave Function and Measurements

Let us return to the measurement process, whose properties were discussed qualitatively in § 1, and show how those properties are connected with the mathematical apparatus of quantum mechanics.

Consider a system made up of two parts: a classical apparatus and an electron (treated as a quantum object). Measurement consists in bringing these two parts into interaction. The apparatus consequently passes from its initial state into some other state, and from this change we draw conclusions about the state of the electron. States of the apparatus are distinguished by the values of a physical quantity characterizing it (or quantities)—the apparatus “reading.” Denote this quantity provisionally by g, and its eigenvalues by g_n. Because the apparatus is classical, these values generally range over a continuum; solely to simplify the notation in the formulas below, however, we shall suppose that the spectrum is discrete. The states of the apparatus are described by quasiclassical wave functions, which we denote by $\Phi_n(\xi)$; the index n corresponds to the reading g_n, and ξ stands collectively for its coordinates. That the apparatus is classical manifests itself in the circumstance that at each instant one can assert with certainty that it occupies one of the known states Φ_n, with some definite value of g. Such a statement would of course be false for a quantum system.

Let $\Phi_0(\xi)$ be the wave function of the apparatus in its initial state (before measurement), and let $\Psi(q)$ be an arbitrary normalized initial wave function of the electron (q denotes its coordinates). These functions describe apparatus and electron independently, so the initial wave function of the complete system is the product

$$\Psi(q)\Phi_0(\xi). \tag{7.1}$$

The apparatus and electron then interact. In principle, the equations of quantum mechanics allow the time variation of the system's wave function to be followed. After the measurement it will plainly no longer be a product of functions of ξ and q. Expanding it in the eigenfunctions Φ_n of the apparatus (which form a complete system of functions), we obtain a sum of the form

$$\sum_n A_n(q)\Phi_n(\xi), \tag{7.2}$$

where the $A_n(q)$ are certain functions of q.

At this point the "classical nature" of the apparatus enters, together with the double role of classical mechanics as both a limiting case and a basis of quantum mechanics. As already noted, because the apparatus is classical, the quantity g (its "reading") has a definite value at every instant. It follows that after measurement the actual state of apparatus plus electron is described not by the whole sum (7.2), but by only the one term corresponding to the apparatus reading g_n:

$$A_n(q)\Phi_n(\xi). \tag{7.3}$$

Consequently, $A_n(q)$ is proportional to the electron's wave function after measurement. It is not itself that wave function, as is already evident from the fact that $A_n(q)$ is not normalized. It contains both information about the properties of the state produced and the probability, determined by the system's initial state, that the apparatus will give its nth reading.

By the linearity of the equations of quantum mechanics, the relation between $A_n(q)$ and the initial electron wave function $\Psi(q)$ is in general given by a linear integral operator,

$$A_n(q) = \int K_n(q, q')\Psi(q')\, dq', \tag{7.4}$$

whose kernel $K_n(q, q')$ characterizes the measurement process in question.

We suppose that the measurement under consideration yields a complete description of the electron's state. In other words (see § 1), in the state produced, the probabilities for all quantities must be independent of the electron's preceding state (before measurement). Mathematically, this means that the form of the functions $A_n(q)$ must be fixed by the measurement process itself and must not depend on the initial electron wave function $\Psi(q)$.

The A_n must therefore have the form

$$A_n(q) = a_n\varphi_n(q), \tag{7.5}$$

where the φ_n are certain functions that we shall take to be normalized, while only the constants a_n depend on the initial state $\Psi(q)$. In the integral relation (7.4), this corresponds to a kernel $K_n(q, q')$ which factors into functions of q and q' separately:

$$K_n(q, q') = \varphi_n(q)\Psi_n^*(q'). \tag{7.6}$$

The linear dependence of the constants a_n on $\Psi(q)$ is then expressed by formulas of the form

$$a_n = \int \Psi(q)\Psi_n^*(q)\,dq, \tag{7.7}$$

where the $\Psi_n(q)$ are certain functions fixed by the measurement process.

The functions $\varphi_n(q)$ are the normalized wave functions of the electron after measurement. We thus see how the theory's mathematical formalism represents the possibility of producing, by a measurement, an electron state described by a definite wave function. When a measurement is made on an electron having a specified wave function $\Psi(q)$, the constants a_n have a simple physical meaning: by the general rules, $|a_n|^2$ is the probability that the measurement gives the nth result. The probabilities of all results sum to unity:

$$\sum_n |a_n|^2 = 1. \tag{7.8}$$

For an arbitrary normalized function $\Psi(q)$, the validity of (7.7) and (7.8) is equivalent (compare § 3) to the statement that $\Psi(q)$ can be expanded in the functions $\Psi_n(q)$. Hence the functions $\Psi_n(q)$ form a complete normalized and mutually orthogonal set.

If the electron's initial wave function coincides with one of the functions $\Psi_n(q)$, the corresponding constant a_n is plainly unity and all the others vanish. Put differently, if the measurement is performed on an electron in the state $\Psi_n(q)$, it will with certainty give a definite (nth) result.

Taken together, these properties of the functions $\Psi_n(q)$ demonstrate that they are eigenfunctions of a certain physical quantity that characterizes the electron (call it f), and the measurement being considered may be called a measurement of that quantity.

It is essential that, in general, the functions $\Psi_n(q)$ do not coincide with the functions $\varphi_n(q)$ (indeed, the latter need not even be mutually orthogonal or form an eigenfunction system of any operator). This fact expresses first of all the non-repeatability of measurement results in quantum mechanics. If the electron was in the state $\Psi_n(q)$, a measurement of f made upon it will reveal the value f_n with certainty. After the measurement, however, the electron is in a different state $\varphi_n(q)$, in which f need no longer have any definite value. Thus, if a second measurement were made immediately after the first, the value found for f would not coincide with the result of the first measurement[20]. To predict (in the sense of calculating the probability of) the result of a repeated measurement when the first result is known, one takes from the first measurement the wave function $\varphi_n(q)$ of the state it produced, and from the second the wave function $\Psi_m(q)$ of the state whose probability is sought. More explicitly, the equations of quantum mechanics determine a wave function $\varphi_n(q,t)$ which equals $\varphi_n(q)$ at the time of the first measurement. If the second measurement is made at time t, the probability of its mth result is the squared modulus of the integral $\int \varphi_n(q,t)\Psi_m^*(q)\,dq$.

We see that measurement in quantum mechanics is "two-faced": its roles with respect to past and future do not coincide. Toward the past it "verifies" the probabilities of the

[20]There is, however, an important exception to the non-repeatability of measurements: coordinate is the one quantity whose measurement is repeatable. Two measurements of an electron's coordinate made at a sufficiently short interval must give nearby values; otherwise the electron would have infinite velocity. The mathematical origin of this exception lies in the commutativity of the coordinate with the electron–apparatus interaction energy operator, which (in the nonrelativistic theory) depends solely on coordinates.

various possible results predicted from the state created by the preceding measurement. Toward the future it creates a new state (see also § 44). A profound irreversibility is therefore inherent in the measurement process itself.

This irreversibility has important consequences of principle. As will be seen later (see the end of § 18), the fundamental equations of quantum mechanics by themselves are symmetric under reversal of the sign of time; in this respect quantum mechanics does not differ from classical mechanics. The irreversibility of measurement, however, makes the two directions of time physically inequivalent in quantum phenomena, and thus produces a distinction between future and past.

CHAPTER II

Energy and Momentum

§ 8. The Hamiltonian

In quantum mechanics the wave function Ψ fully specifies the state of a physical system. This means that prescribing Ψ at some instant not only describes all the properties of the system at that instant, but also determines its behavior at every later instant—of course, only to the degree of completeness that quantum mechanics permits at all. Mathematically, this means that at each instant the value of the time derivative $\partial\Psi/\partial t$ must be determined by the value of Ψ itself at that same instant, and, by the superposition principle, this dependence must be linear. In the most general form we may write

$$i\hbar\frac{\partial\Psi}{\partial t} = \hat{H}\Psi, \tag{8.1}$$

where $\hat{H}$ is a certain linear operator; the factor $i\hbar$ has been introduced here for a reason that will become clear below.

Since the integral $\int \Psi^*\Psi\,dq$ is constant in time, we have

$$\frac{d}{dt}\int |\Psi|^2\,dq = \int \frac{\partial\Psi^*}{\partial t}\Psi\,dq + \int \Psi^*\frac{\partial\Psi}{\partial t}\,dq = 0.$$

Substituting (8.1) here and applying the definition of the transposed operator in the first integral, we obtain (after suppressing the common factor $i/\hbar$)

$$\begin{aligned}\int \Psi\hat{H}^*\Psi^*\,dq - \int \Psi^*\hat{H}\Psi\,dq &= \int \Psi^*\tilde{\hat{H}}^*\Psi\,dq - \int \Psi^*\hat{H}\Psi\,dq \\ &= \int \Psi^*(\tilde{\hat{H}}^* - \hat{H})\Psi\,dq = 0.\end{aligned}$$

Because this equality must hold for an arbitrary function Ψ, it follows that identically $\hat{H}^+ = \hat{H}$; the operator $\hat{H}$ is therefore Hermitian. Let us determine the physical quantity to which it corresponds. We use the limiting expression (6.1) for the wave function and write

$$\frac{\partial\Psi}{\partial t} = \frac{i}{\hbar}\frac{\partial S}{\partial t}\Psi$$

(the slowly varying amplitude a need not be differentiated). Comparing this equality with definition (8.1), we see that in the limiting case $\hat{H}$ reduces to multiplication by $-\partial S/\partial t$. Thus this last quantity is the physical quantity into which the Hermitian operator $\hat{H}$ passes.

But $-\partial S/\partial t$ is precisely the Hamilton function H of the mechanical system. Hence $\hat{H}$ is the operator corresponding in quantum mechanics to the Hamilton function. This operator is termed the *Hamiltonian operator*, or simply the *Hamiltonian* of the system.

Once the form of the Hamiltonian is known, equation (8.1) fixes the wave functions of the given physical system. This fundamental equation of quantum mechanics is called the *wave equation*.

§ 9. Time Differentiation of Operators

In quantum mechanics, the concept of the time derivative of a physical quantity cannot be defined with the sense that it carries in classical mechanics. Indeed, the definition of the derivative in classical mechanics is connected with considering the values of the quantity at two close but different instants. But in quantum mechanics a quantity that has a definite value at some instant has no definite value at all at subsequent instants; this was discussed in more detail in § 1.

For this reason the concept of the time derivative has to be given a different definition in quantum mechanics. A natural course is to define the derivative $\dot{f}$ of a quantity f as that quantity whose mean value equals the time derivative of the mean value $\bar{f}$. Thus, by definition, we have

$$\bar{\dot{f}} = \dot{\bar{f}}. \tag{9.1}$$

Taking this definition as the starting point, it is not hard to derive an expression for the quantum-mechanical operator $\hat{\dot{f}}$ that corresponds to the quantity $\dot{f}$:

$$\begin{aligned} \bar{\dot{f}} = \dot{\bar{f}} = \frac{d}{dt}\int \Psi^* \hat{f}\Psi\, dq = \\ = \int \Psi^* \frac{\partial \hat{f}}{\partial t}\Psi\, dq + \int \frac{\partial \Psi^*}{\partial t}\hat{f}\Psi\, dq + \int \Psi^* \hat{f}\frac{\partial \Psi}{\partial t}\, dq. \end{aligned}$$

In this expression $\partial \hat{f}/\partial t$ stands for the operator that results from differentiating the operator $\hat{f}$ with respect to time, on which the latter may depend parametrically. Substituting for the derivatives $\partial\Psi/\partial t$, $\partial\Psi^*/\partial t$ their expressions according to (8.1), we obtain

$$\bar{\dot{f}} = \int \Psi^* \frac{\partial \hat{f}}{\partial t}\Psi\, dq + \frac{i}{\hbar}\int (\hat{H}^*\Psi^*)\hat{f}\Psi\, dq - \frac{i}{\hbar}\int \Psi^*\hat{f}(\hat{H}\Psi)\, dq.$$

Since the operator $\hat{H}$ is Hermitian,

$$\int (\hat{H}^*\Psi^*)(\hat{f}\Psi)\, dq = \int \Psi^*\hat{H}\hat{f}\Psi\, dq;$$

thus, we have

$$\bar{\dot{f}} = \int \Psi^* \left(\frac{\partial \hat{f}}{\partial t} + \frac{i}{\hbar}\hat{H}\hat{f} - \frac{i}{\hbar}\hat{f}\hat{H}\right)\Psi\, dq.$$

Since, on the other hand, by the definition of mean values we must have $\bar{\dot{f}} = \int \Psi^* \hat{\dot{f}}\Psi\, dq$, it is seen that the expression in parentheses under the integral is the required operator $\hat{\dot{f}}$[1]:

$$\hat{\dot{f}} = \frac{\partial \hat{f}}{\partial t} + \frac{i}{\hbar}(\hat{H}\hat{f} - \hat{f}\hat{H}). \tag{9.2}$$

[1] In classical mechanics, for the total time derivative of a quantity f that is a function of the generalized

If the operator $\hat{f}$ does not depend explicitly on time, then, up to a factor, $\hat{\dot{f}}$ reduces to the commutator of the operator $\hat{f}$ with the Hamiltonian.

The physical quantities whose operators have no explicit dependence on time and which, moreover, commute with the Hamiltonian—so that $\hat{\dot{f}} = 0$—form a very important category. Such quantities are called *conserved*. For these, $\bar{\dot{f}} = \dot{\bar{f}} = 0$, i.e. $\bar{f} = \text{const}$. Put differently, the quantity's mean value stays constant in time. One can assert, moreover, that if the quantity f has a definite value in a given state (i.e. if the wave function is an eigenfunction of the operator $\hat{f}$), then at later instants as well it will have a definite value—and the very same one.

§ 10. Stationary states

The Hamiltonian of a closed system (and also of a system in a constant, but not a variable, external field) cannot contain time explicitly. This follows because all instants of time are equivalent for such a physical system. Since, on the other hand, every operator commutes with itself, we conclude that the Hamilton function is conserved for systems not situated in a variable external field. A conserved Hamilton function is called the energy. The meaning of energy conservation in quantum mechanics is that if the energy has a definite value in a given state, this value remains constant in time.

States possessing definite energy values are called the *stationary states* of the system. They are described by wave functions Ψ_n that are eigenfunctions of the Hamiltonian operator, satisfying $\hat{H}\Psi_n = E_n\Psi_n$, where E_n are the energy eigenvalues. Accordingly, the wave equation (8.1) for the function Ψ_n,

$$i\hbar\frac{\partial\Psi_n}{\partial t} = \hat{H}\Psi_n = E_n\Psi_n,$$

can be integrated directly with respect to time, giving

$$\Psi_n = \exp\left(-\frac{i}{\hbar}E_n t\right)\psi_n(q), \tag{10.1}$$

---

coordinates q_i and momenta p_i of the system, we have

$$\frac{df}{dt} = \frac{\partial f}{\partial t} + \sum_i \left(\frac{\partial f}{\partial q_i}\dot{q}_i + \frac{\partial f}{\partial p_i}\dot{p}_i\right).$$

Substituting, according to Hamilton's equations, $\dot{q}_i = \partial H/\partial p_i$, $\dot{p}_i = -\partial H/\partial q_i$, we obtain

$$\frac{df}{dt} = \frac{\partial f}{\partial t} + [H, f], \qquad [H, f] \equiv \sum_i \left(\frac{\partial f}{\partial q_i}\frac{\partial H}{\partial p_i} - \frac{\partial f}{\partial p_i}\frac{\partial H}{\partial q_i}\right);$$

$[H, f]$ is the so-called *Poisson bracket* for the quantities f and H (see I, § 42). Comparing with expression (9.2), we see that in the transition to the classical limit the operator $i(\hat{H}\hat{f} - \hat{f}\hat{H})$ becomes zero to the first approximation, as it should, and in the next approximation (in $\hbar$) becomes the quantity $\hbar[H, f]$.

This result is also valid for any two quantities f and g: in the limit the operator $i(\hat{f}\hat{g} - \hat{g}\hat{f})$ becomes the quantity $\hbar[f, g]$, where $[f, g]$ is the Poisson bracket

$$[f, g] \equiv \sum_i \left(\frac{\partial g}{\partial q_i}\frac{\partial f}{\partial p_i} - \frac{\partial g}{\partial p_i}\frac{\partial f}{\partial q_i}\right).$$

This is a consequence of the fact that we can always, formally, conceive of a system whose Hamiltonian coincides with $\hat{g}$.

where ψ_n is a function of the coordinates alone. This determines the time dependence of the wave functions of stationary states.

We shall use the lower-case letter ψ for stationary-state wave functions without the time factor. These functions and the energy eigenvalues themselves are determined by

$$\hat{H}\psi = E\psi. \tag{10.2}$$

Among all stationary states, the one possessing the lowest energy is termed the *normal*, or *ground*, state of the system.

The expansion of any wave function Ψ over the stationary-state wave functions has the form

$$\Psi = \sum_n a_n \exp\left(-\frac{i}{\hbar}E_n(t)\right)\psi_n(q). \tag{10.3}$$

As usual, the squares $|a_n|^2$ give the probabilities of the various energy values of the system.

The coordinate probability distribution in a stationary state is determined by $|\Psi_n|^2 = |\psi_n|^2$ and is therefore independent of time. The same is true of the mean value $\bar{f} = \int \Psi_n^* \hat{f}\Psi_n, dq = \int \psi_n^* \hat{f}\psi_n, dq$ of any physical quantity f whose operator does not depend explicitly on time.

As stated above, the operator of every conserved quantity commutes with the Hamiltonian. Thus every conserved physical quantity can be measured simultaneously with the energy.

Among the stationary states there may be states corresponding to one and the same energy eigenvalue (or *energy level*) but differing in the values of other physical quantities. A level to which several distinct stationary states correspond is said to be *degenerate*. Physically, degeneracy is possible because the energy does not, in general, constitute by itself a complete set of physical quantities. The energy levels of a system are, in general, degenerate if there are two conserved quantities f and g whose operators do not commute. Let ψ be the wave function of a stationary state in which f, as well as the energy, has a definite value. Then $\hat{g}\psi$ cannot coincide with ψ up to a constant factor; otherwise g too would have a definite value, which is impossible because f and g cannot be measured simultaneously. On the other hand, $\hat{g}\psi$ is a Hamiltonian eigenfunction with the same energy E as ψ:

$$\hat{H}(\hat{g}\psi) = \hat{g}\hat{H}\psi = E(\hat{g}\psi).$$

Thus more than one eigenfunction corresponds to E, and the level is degenerate.

Any linear combination of wave functions belonging to the same degenerate energy level is also an eigenfunction with that energy. The choice of eigenfunctions for a degenerate eigenvalue is therefore not unique. Arbitrarily chosen eigenfunctions of a degenerate level are not in general mutually orthogonal, but suitable linear combinations can always be chosen to form a mutually orthogonal and normalized set[2].

These statements apply not only to energy eigenfunctions but to the eigenfunctions of any operator. Only functions belonging to different eigenvalues are automatically

[2]This can be done in infinitely many ways: a linear transformation of n functions involves n^2 independent coefficients, but the normalization and orthogonality conditions number only $n(n+1)/2$, which is fewer than n^2.

orthogonal; functions belonging to the same degenerate eigenvalue are not in general orthogonal.

If $\hat{H} = \hat{H}_1 + \hat{H}_2$, where one part contains only coordinates q_1 and the other only q_2, the eigenfunctions of $\hat{H}$ can be written as products of eigenfunctions of $\hat{H}_1$ and $\hat{H}_2$, and the energy eigenvalues are sums of the eigenvalues of these operators. The energy spectrum may be discrete or continuous. A stationary state in the discrete spectrum always corresponds to *finite motion*: neither the system nor any part of it recedes to infinity. Indeed, $\int |\Psi|^2, dq$ is finite over all space for a discrete-spectrum eigenfunction. Thus $|\Psi|^2$ decreases sufficiently rapidly and vanishes at infinity. Equivalently, the probability of infinite coordinate values is zero; the system performs finite motion, or is in a *bound* state.

For continuous-spectrum wave functions, $\int |\Psi|^2, dq$ diverges. $|\Psi|^2$ then determines coordinate probabilities only up to proportionality. The divergence is always associated with $|\Psi|^2$ failing to vanish at infinity, or vanishing too slowly. The integral over the region outside any arbitrarily large but finite closed surface therefore still diverges: the system, or some part of it, is at infinity. A superposition of continuous-spectrum stationary states may have a convergent integral and be localized in a finite region, but this region spreads without bound with time, and ultimately the system recedes to infinity.

Indeed, an arbitrary continuous-spectrum superposition is

$$\Psi = \int a_E \exp\left(-\frac{i}{\hbar}Et\right)\psi_E(q)\,dE,$$

and

$$|\Psi|^2 = \iint a_E a^*_{E'} \exp\left(\frac{i}{\hbar}(E' - E)t\right)\psi_E(q)\psi^*_{E'}(q)\,dE\,dE'.$$

If this expression is averaged over an interval T and then T is allowed to tend to infinity, the oscillating factors average to zero, and with them the entire integral. Thus the time-averaged probability of finding the system at any specified point of configuration space vanishes; this is possible only if the motion occupies all infinite space[3].

Thus stationary states of the continuous spectrum correspond to infinite motion of the system.

§ 11. Matrices

Assume for convenience that the system in question possesses a discrete energy spectrum (every relation derived below extends directly to the continuous-spectrum case as well). Let $\Psi = \sum_n a_n \Psi_n$ denote the expansion of an arbitrary wave function over the stationary-state wave functions Ψ_n. If this expansion is substituted into the definition (3.8) of the

[3]For a superposition of discrete-spectrum functions one would instead have

$$\overline{|\Psi|^2} = \overline{\sum_{n,m} a_n a^*_m \exp\left\{\frac{i}{\hbar}(E_m - E_n)t\right\}\psi_n\psi^*_m} = \sum_n |a_n\psi_n(q)|^2,$$

so the probability density remains finite under time averaging.

mean value of some quantity f, we obtain

$$\bar{f} = \sum_n \sum_m a_n^* a_m f_{nm}(t), \tag{11.1}$$

where $f_{nm}(t)$ denote the integrals

$$f_{nm}(t) = \int \Psi_n^* \hat{f} \Psi_m \, dq. \tag{11.2}$$

The set of quantities $f_{nm}(t)$ for all possible n, m is called the *matrix* of the quantity f, and each $f_{nm}(t)$ is spoken of as a *matrix element*, corresponding to the transition from state m to state n[4].

The time dependence of the matrix elements $f_{nm}(t)$ is determined (if the operator $\hat{f}$ does not contain t explicitly) by the time dependence of the functions Ψ_n. Substituting for them the expressions (10.1), we find that

$$f_{nm}(t) = f_{nm} e^{i\omega_{nm}t}, \tag{11.3}$$

where

$$\omega_{nm} = \frac{E_n - E_m}{\hbar} \tag{11.4}$$

is the transition frequency between states n and m, and the quantities

$$f_{nm} = \int \psi_n^* \hat{f} \psi_m \, dq \tag{11.5}$$

constitute the time-independent matrix of the quantity f, which is what one ordinarily uses[5].

One obtains the matrix elements of the derivative $\dot{f}$ by differentiating those of the quantity f with respect to time; this is an immediate consequence of the relation

$$\bar{\dot{f}} = \dot{\bar{f}} = \sum_n \sum_m a_n^* a_m \dot{f}_{nm}(t). \tag{11.6}$$

In view of (11.3) we thus have for the matrix elements of $\dot{f}$:

$$\dot{f}_{nm}(t) = i\omega_{nm} f_{nm}(t) \tag{11.7}$$

or (cancelling the time factor $\exp(i\omega_{nm}t)$ on both sides) for the time-independent matrix elements:

$$(\dot{f})_{nm} = i\omega_{nm} f_{nm} = \frac{i}{\hbar}(E_n - E_m) f_{nm}. \tag{11.8}$$

For simplicity of notation in the formulas, we derive below all relations for the time-independent matrix elements; exactly the same relations hold also for matrices that depend on time.

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[4]The matrix representation of physical quantities was introduced by *Heisenberg* (*W. Heisenberg*) in 1925, even before Schrödinger's discovery of the wave equation. "Matrix mechanics" was subsequently developed by *Born, Heisenberg and Jordan* (*M. Born, P. Jordan*).

[5]In connection with the indeterminacy of the phase factor in the normalized wave functions (see § 2), the matrix elements f_{nm} (and $f_{nm}(t)$) are likewise defined only to within factors of the form $\exp[i(\alpha_m - \alpha_n)]$. Here too this indeterminacy does not affect the physical results.

For the matrix elements of the complex conjugate quantity f^* of f, taking into account the definition of the adjoint operator, we obtain

$$(f^*)_{nm} = \int \psi_n^* \hat{f}^+ \psi_m \, dq = \int \psi_n^* \tilde{\hat{f}}^* \psi_m \, dq = \int \psi_m \hat{f}^* \psi_n^* \, dq,$$

i.e.

$$(f^*)_{nm} = (f_{mn})^*. \tag{11.9}$$

For real physical quantities, which are usually all that we consider, we therefore have

$$f_{nm} = f_{mn}^*. \tag{11.10}$$

Such matrices, like the operators corresponding to them, are called *Hermitian*.

Matrix elements with $n = m$ are called *diagonal*. These elements do not depend on time at all, and from (11.10) it is clear that they are real. The element f_{nn} is the mean value of f in the state Ψ_n.

The rule of *matrix multiplication* is readily derived. As a preliminary step, observe that the formula

$$\hat{f}\psi_n = \sum_m f_{mn}\psi_m. \tag{11.11}$$

holds. This is nothing other than the expansion of the function $\hat{f}\psi_n$ in the functions ψ_m with coefficients determined according to the general rule (3.5). Bearing this formula in mind, we write

$$\hat{f}\hat{g}\psi_n = \hat{f}(\hat{g}\psi_n) = \sum_k g_{kn}\hat{f}\psi_k = \sum_{k,m} g_{kn} f_{mk}\psi_m.$$

Since, on the other hand, we must have $\hat{f}\hat{g}\psi_n = \sum_m (fg)_{mn}\psi_m$, we obtain

$$(fg)_{mn} = \sum_k f_{mk} g_{kn}. \tag{11.12}$$

This rule matches the standard mathematical prescription for multiplying matrices: one multiplies the rows of the first factor by the columns of the second.

Specifying a matrix is equivalent to specifying the operator itself. In particular, it makes it possible in principle to determine the eigenvalues of a given physical quantity and the eigenfunctions corresponding to them.

Let us consider the values of all quantities at some definite instant of time and expand an arbitrary wave function Ψ in the eigenfunctions of the Hamiltonian, i.e. in the time-independent wave functions ψ_m of the stationary states:

$$\Psi = \sum_m c_m \psi_m, \tag{11.13}$$

where the expansion coefficients are denoted by c_m. Let us substitute this expansion into the equation $\hat{f}\Psi = f\Psi$, which determines the eigenvalues and eigenfunctions of the quantity f. We have

$$\sum_m c_m \hat{f}\psi_m = f \sum_m c_m \psi_m.$$

Multiply this equation on both sides by ψ_n^* and integrate over dq. Each of the integrals $\int \psi_n^* \hat{f} \psi_m \, dq$ on the left-hand side of the equality is the corresponding matrix element f_{nm}. On the right-hand side, all integrals $\int \psi_n^* \psi_m \, dq$ with $m \neq n$ vanish by virtue of the orthogonality of the functions ψ_m, and $\int \psi_n^* \psi_n \, dq = 1$ by virtue of their normalization[6]:

$$\sum_m f_{nm} c_m = f c_n, \tag{11.14}$$

or $\sum_m (f_{nm} - f\delta_{nm}) c_m = 0$, where $\delta_{nm} = 1$ for $n = m$ and $\delta_{nm} = 0$ for $n \neq m$.

Thus we have obtained a system of homogeneous linear algebraic equations (with unknowns c_m). As is well known, such a system has nonzero solutions only if the determinant formed from the coefficients in the equations vanishes, i.e. under the condition

$$|f_{nm} - f\delta_{nm}| = 0. \tag{11.15}$$

The roots of this equation (where f plays the role of the unknown) give the possible values of the quantity f. The set of quantities c_m satisfying equations (11.14) upon setting f equal to one of these values yields the corresponding eigenfunction. Suppose that in the definition (11.5) for the matrix elements of f we adopt, as the functions ψ_n, the eigenfunctions of this very quantity; owing to the equation $\hat{f}\psi_n = f_n \psi_n$ we obtain

$$f_{nm} = \int \psi_n^* \hat{f} \psi_m \, dq = f_m \int \psi_n^* \psi_m \, dq.$$

Since the functions ψ_m are orthogonal and normalized, it follows that $f_{nm} = 0$ for $n \neq m$ and $f_{mm} = f_m$. Thus only diagonal matrix elements turn out to be nonzero, each of them being equal to the corresponding eigenvalue of the quantity f; a matrix in which only these elements are nonzero is said to be reduced to *diagonal form*. In particular, in the ordinary representation—where the stationary-state wave functions serve as the ψ_n—the energy matrix is diagonal (along with the matrices of all other physical quantities that have definite values in the stationary states). More broadly, when the matrix of a quantity f is constructed using the eigenfunctions of an operator g, one speaks of it as the matrix of f in the representation in which g is diagonal. Everywhere where it is not otherwise specified, by the matrix of a physical quantity we shall henceforth understand the matrix in the ordinary representation, in which the energy is diagonal. All that has been said above concerning the time dependence of matrices refers, of course, only to this ordinary representation[7].

The matrix representation of operators makes it possible to prove the theorem cited in § 4: if two operators commute, they possess a common complete system of eigenfunctions.

[6] In accordance with the general rule (§ 5), the set of coefficients c_n of the expansion (11.13) can be regarded as the wave function in the "energy representation" (the variable here being the index n, which enumerates the energy eigenvalues). The matrix f_{nm} then plays the role of the operator $\hat{f}$ in this representation, whose action on the wave function is given by the expression on the left-hand side of equation (11.14). The formula $\bar{f} = \sum_n \sum_m c_n^* (f_{nm} c_m)$ then corresponds to the general formula for the mean value of a quantity expressed through its operator and the wave function of the state in question.

[7] Keeping in mind that the energy matrix is diagonal, one readily verifies that the equality (11.8) is the operator relation (9.2) written in matrix form.

Let $\hat{f}$ and $\hat{g}$ be two such operators. From $\hat{f}\hat{g} = \hat{g}\hat{f}$ and the rule of matrix multiplication (11.12) it follows that

$$\sum_k f_{mk} g_{kn} = \sum_k g_{mk} f_{kn}.$$

Taking as the system of functions ψ_n, in terms of which the matrix elements are computed, the eigenfunctions of the operator $\hat{f}$, we have $f_{mk} = 0$ for $m \neq k$, so that the written equality reduces to $f_m g_{mn} = g_{mn} f_n$, or $g_{mn}(f_m - f_n) = 0$. If all the eigenvalues f_n of the quantity f are distinct, then for all $m \neq n$ we have $f_m - f_n \neq 0$, so that we must have $g_{mn} = 0$. Thus the matrix g_{mn} likewise turns out to be diagonal, i.e. the functions ψ_n are eigenfunctions also of the physical quantity g. If, however, among the values f_n there are equal ones, then the matrix elements g_{mn} corresponding to each such group of functions ψ_n will, in general, be nonzero. However, linear combinations of the functions ψ_n corresponding to one eigenvalue of the quantity f are likewise its eigenfunctions; one can always choose these combinations in such a way as to make the corresponding nondiagonal matrix elements g_{mn} vanish, and thus in this case too we obtain a system of functions that are eigenfunctions simultaneously for the operators $\hat{f}$ and $\hat{g}$.

Let us note a useful formula

$$\left(\frac{\partial H}{\partial \lambda}\right)_{nn} = \frac{\partial E_n}{\partial \lambda}, \tag{11.16}$$

where λ — a parameter on which the Hamiltonian $\hat{H}$ depends (and with it the energy eigenvalues E_n). Indeed, differentiating the equation $(\hat{H} - E_n)\psi_n = 0$ with respect to λ and then multiplying it from the left by ψ_n^*, we obtain

$$\psi_n^*(\hat{H} - E_n)\frac{\partial \psi_n}{\partial \lambda} = \psi_n^*\left(\frac{\partial E_n}{\partial \lambda} - \frac{\partial \hat{H}}{\partial \lambda}\right)\psi_n.$$

On integrating over dq, the left-hand side of this equality vanishes, since

$$\int \psi_n^*(\hat{H} - E_n)\frac{\partial \psi_n}{\partial \lambda}\, dq = \int \frac{\partial \psi_n}{\partial \lambda}(\hat{H} - E_n)^*\psi_n^*\, dq$$

by virtue of the Hermiticity of the operator $\hat{H}$. The right-hand side gives the desired equality.

In the modern literature, a system of notation (introduced by *Dirac*) is widely used, in which matrix elements are written as[8]

$$\langle n|f|m\rangle. \tag{11.17}$$

This symbol is composed of the notation for the quantity f and the symbols $|m\rangle$ and $|n\rangle$, denoting respectively the initial and final states. If there is a complete set of state functions $|n_1\rangle, |n_2\rangle, \ldots$, then the expansion coefficients of the wave function of the state $|m\rangle$ are denoted by $\langle n_i|m\rangle$:

$$\langle n_i|m\rangle = \int \psi_{n_i}^* \psi_m\, dq. \tag{11.18}$$

[8]We will use both ways of denoting matrix elements in this book. The notation (11.17) is especially convenient when each index has to be written as a combination of several letters.

§ 12. Transformation of matrices

The matrix elements of one and the same physical quantity may be defined with respect to different sets of wave functions. These may be, for example, wave functions of stationary states described by different sets of physical quantities, or wave functions of stationary states of the same system placed in different external fields. The question therefore arises of transforming matrices from one representation to another.

Let $\psi_n(q)$ and $\psi'_n(q)$ $(n = 1, 2, \ldots)$ be two complete systems of orthonormal functions. They are related by a linear transformation

$$\psi'_n = \sum_m S_{mn}\psi_m, \tag{12.1}$$

which is simply the expansion of the functions ψ'_n in the complete system ψ_m. This transformation can be written in operator form as

$$\psi'_n = \hat{S}\psi_n. \tag{12.2}$$

The operator $\hat{S}$ must satisfy a certain condition if the functions ψ'_n are to be orthonormal whenever the functions ψ_n are. Indeed, substituting (12.2) into $\int \psi'^*_m \psi'_n \, dq = \delta_{mn}$ and using the definition (3.14) of the transposed operator, we obtain

$$\int (\hat{S}\psi_m)^* \hat{S}\psi_n \, dq = \int \psi^*_m \tilde{S}^* \hat{S}\psi_n \, dq = \delta_{mn}.$$

For this equality to hold for every m, n, we must have $\tilde{S}^*\hat{S} = 1$, or

$$\tilde{S}^* \equiv \hat{S}^+ = \hat{S}^{-1}, \tag{12.3}$$

i.e. the inverse operator coincides with the adjoint. Operators having this property are called *unitary*. By virtue of this property the transformation $\psi_n = \hat{S}^{-1}\psi'_n$, inverse to (12.1), is given by

$$\psi_n = \sum_m S^*_{nm}\psi'_m. \tag{12.4}$$

Writing $\hat{S}^+\hat{S} = 1$ or $\hat{S}\hat{S}^+ = 1$ in matrix form gives the unitarity conditions

$$\sum_l S^*_{lm} S_{ln} = \delta_{mn} \tag{12.5}$$

or

$$\sum_l S^*_{ml} S_{nl} = \delta_{mn}. \tag{12.6}$$

Now consider a physical quantity f and write its matrix elements in the "new" representation, i.e. with respect to the functions ψ'_n. They are given by

$$\int \psi'^*_m \hat{f}\psi'_n \, dq = \int (\tilde{S}^*\psi^*_m)(\hat{f}\hat{S}\psi_n) \, dq = \int \psi^*_m \hat{S}^{-1}\hat{f}\hat{S}\psi_n \, dq.$$

It follows that the matrix of $\hat{f}$ in the new representation coincides with the matrix of the operator

$$\hat{f}' = \hat{S}^{-1}\hat{f}\hat{S} \tag{12.7}$$

in the old representation[9]. The *trace* of a matrix is defined as the sum of its diagonal elements and is written Tr f[10]:

$$\operatorname{Tr} f = \sum_n f_{nn}. \tag{12.8}$$

First note that the trace of a product of two matrices does not depend on the order of the factors:

$$\operatorname{Tr}(fg) = \operatorname{Tr}(gf). \tag{12.9}$$

Indeed, by the rule for matrix multiplication,

$$\operatorname{Tr}(fg) = \sum_n \sum_k f_{nk} g_{kn} = \sum_k \sum_n g_{kn} f_{nk} = \operatorname{Tr}(gf).$$

Similarly, the trace of a product of several matrices is unchanged by a cyclic permutation of the factors; thus

$$\operatorname{Tr}(fgh) = \operatorname{Tr}(hfg) = \operatorname{Tr}(ghf). \tag{12.10}$$

The most important property of the trace is its independence of the system of functions relative to which the matrix elements are defined. Indeed,

$$\operatorname{Tr} f' = \operatorname{Tr}\left(S^{-1} f S\right) = \operatorname{Tr}\left(S S^{-1} f\right) = \operatorname{Tr} f. \tag{12.11}$$

Also, a unitary transformation leaves invariant the sum of the squared moduli of the transformed functions. From (12.6),

$$\sum_i |\psi_i'|^2 = \sum_{k,l,i} S_{ki} S_{li}^* \psi_k \psi_l^* = \sum_{k,l} \delta_{kl} \psi_k \psi_l^* = \sum_k |\psi_k|^2. \tag{12.12}$$

Every unitary operator can be represented as

$$\hat{S} = e^{i\hat{R}}, \tag{12.13}$$

where $\hat{R}$ is Hermitian; indeed, $\hat{R}^+ = \hat{R}$ gives $\hat{S}^+ = e^{-i\hat{R}^+} = e^{-i\hat{R}} = \hat{S}^{-1}$.

We also note the expansion

$$\hat{f}' = \hat{S}^{-1} \hat{f} \hat{S} = \hat{f} + \{\hat{f}, i\hat{R}\} + \frac{1}{2}\{\{\hat{f}, i\hat{R}\}, i\hat{R}\} + \ldots, \tag{12.14}$$

which is readily verified directly by expanding the factors $\exp\left(\pm i\hat{R}\right)$ in powers of $\hat{R}$. This expansion can be useful when $\hat{R}$ is proportional to a small parameter, so that (12.14) becomes an expansion in powers of that parameter.

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[9]If $\{\hat{f}, \hat{g}\} = -i\hbar\hat{c}$ is the commutation rule for two operators, then after transformation (12.7) we obtain $\{\hat{f}', \hat{g}'\} = -i\hbar\hat{c}'$, so the rule is unchanged. It was noted in the note on p. 46 that c is the quantum analogue of the classical Poisson bracket $[f, g]$. Poisson brackets in classical mechanics, however, are invariant under canonical variable transformations (see Vol. I, § 45). One may therefore say that unitary transformations in quantum mechanics serve as the counterpart of canonical transformations in classical mechanics.

[10]The source's notation comes from the German *Spur*, trace. The notation Tr, from the English *trace*, is also used. Consideration of a matrix trace of course assumes convergence of the sum over n.

§ 13. Heisenberg representation of operators

In the mathematical apparatus of quantum mechanics as here presented, the operators corresponding to the various physical quantities act on functions of the coordinates and as such ordinarily contain no explicit dependence on time. The dependence of the mean values of physical quantities on time arises only through the time dependence of the wave function of the state, in accordance with the formula

$$\bar{f}(t) = \int \Psi^*(q,t)\hat{f}\Psi(q,t)\,dq. \tag{13.1}$$

The apparatus of quantum mechanics can, however, also be formulated in a somewhat different, equivalent form, in which the dependence on time is transferred from the wave functions to the operators. Although in this book we shall not make use of such a representation of operators—the so-called Heisenberg representation, as distinct from the Schrödinger representation—we shall formulate it here, with later applications in relativistic theory in view.

Let us introduce the unitary operator (cf. (12.13))

$$\hat{S} = \exp\left(-\frac{i}{\hbar}\hat{H}t\right), \tag{13.2}$$

where $\hat{H}$ is the Hamiltonian of the system. By definition its eigenfunctions coincide with the eigenfunctions of the operator $\hat{H}$, i.e. with the wave functions $\psi_n(q)$ of the stationary states, and moreover

$$\hat{S}\psi_n(q) = \exp\left(-\frac{i}{\hbar}E_n t\right)\psi_n(q) = \Psi_n(q,t). \tag{13.3}$$

It follows that expansion (10.3) of any wave function Ψ over the stationary-state wave functions can be recast in operator form as

$$\Psi(q,t) = \hat{S}\Psi(q,0), \tag{13.4}$$

i.e. the action of the operator $\hat{S}$ carries the wave function of the system at some initial instant of time over into the wave function at an arbitrary instant of time. On introducing, in accordance with (12.7), the time-dependent operator

$$\hat{f}(t) = \hat{S}^{-1}\hat{f}\hat{S}, \tag{13.5}$$

we obtain

$$\bar{f}(t) = \int \Psi^*(q,0)\hat{f}(t)\Psi(q,0)\,dq, \tag{13.6}$$

i.e. we bring the formula for the mean value of the quantity f (which serves as the definition of the operators) into a form in which the dependence on time is wholly transferred to the operator.

It is obvious that the matrix elements of the operator (13.5), with respect to the wave functions of stationary states, coincide with the time-dependent matrix elements $f_{nm}(t)$ defined by formula (11.3).

Finally, on differentiating expression (13.5) with respect to time (assuming here that the operators $\hat{f}$ and $\hat{H}$ themselves contain no t), we obtain the equation

$$\frac{\partial \hat{f}(t)}{\partial t} = \frac{i}{\hbar}\left[\hat{H}\hat{f}(t) - \hat{f}(t)\hat{H}\right], \tag{13.7}$$

which is analogous to formula (9.2) but has a somewhat different meaning: expression (9.2) constitutes the definition of the operator $\hat{\dot{f}}$ corresponding to the physical quantity $\dot{f}$, whereas the left-hand side of equation (13.7) contains the time derivative of the operator of the quantity f itself.

§ 14. The density matrix

Describing a system by its wave function represents the fullest description that quantum mechanics allows, in the sense indicated at the end of § 1.

States not admitting such a description arise when the system considered is part of a larger closed system. Suppose the closed system as a whole is in a state described by $\Psi(q, x)$, where x denotes the coordinates of the system considered and q the remaining coordinates of the closed system. In general this function does not factor into functions of x alone and q alone, so that the system has no wave function of its own[11].

Let f be a physical quantity pertaining to our system. Its operator acts on the x coordinates only, not on q. Its mean value is

$$\bar{f} = \iint \Psi^*(q, x)\hat{f}\Psi(q, x)\, dq\, dx. \tag{14.1}$$

Introduce the function

$$\rho(x, x') = \int \Psi(q, x)\Psi^*(q, x')\, dq, \tag{14.2}$$

where integration is only over q; it is called the system's *density matrix*. From (14.2),

$$\rho^*(x, x') = \rho(x', x). \tag{14.3}$$

The "diagonal elements"

$$\rho(x, x) = \int |\Psi(q, x)|^2\, dq$$

determine the coordinate probability distribution.

In terms of the density matrix,

$$\bar{f} = \int [\hat{f}\rho(x, x')]_{x'=x}\, dx. \tag{14.4}$$

Here $\hat{f}$ acts only on the variables x in $\rho(x, x')$; after its action has been evaluated, one sets $x' = x$. Thus the density matrix determines the mean of every quantity characterizing

[11]For $\Psi(q, x)$ to factor in this way at a given instant, the measurement that produced the state must describe completely and separately the system considered and the remainder of the closed system. For the factorization to persist, these two parts must also not interact (see § 2). Neither condition is assumed here.

the system and hence also the probabilities of its possible values. A state having no wave function can therefore be described by a density matrix. The density matrix contains none of the coordinates q that do not belong to the system, although it naturally depends in substance on the state of the closed system as a whole.

The density-matrix description is the most general form of quantum-mechanical description of systems. The wave-function description is the special case $\rho(x, x') = \Psi(x)\Psi^*(x')$. There is an important distinction. For a state possessing a wave function (a *pure* state), there is always a complete system of measurements whose results can be predicted with certainty (mathematically, Ψ is an eigenfunction of some operator). For states possessing only a density matrix (*mixed* states), no complete system of measurements has uniquely predictable results.

Suppose the system is closed, or became closed at some instant. We derive an equation governing the time dependence of its density matrix, analogous to the wave equation. The desired linear differential equation for $\rho(x, x', t)$ must also hold in the special case

$$\rho(x, x', t) = \Psi(x, t)\Psi^*(x', t).$$

Differentiating and using (8.1),

$$\begin{aligned} i\hbar\frac{\partial\rho}{\partial t} &= i\hbar\Psi^*(x', t)\frac{\partial\Psi(x, t)}{\partial t} + i\hbar\Psi(x, t)\frac{\partial\Psi^*(x', t)}{\partial t} \\ &= \Psi^*(x', t)\hat{H}\Psi(x, t) - \Psi(x, t)\hat{H}'^*\Psi^*(x', t). \end{aligned}$$

Here $\hat{H}$ acts on functions of x, while $\hat{H}'$ is the same operator acting on functions of x'. Bringing the other factors under the respective operator signs gives

$$i\hbar\frac{\partial\rho(x, x', t)}{\partial t} = (\hat{H} - \hat{H}'^*)\rho(x, x', t). \tag{14.5}$$

Let $\Psi_n(x, t)$ be the stationary-state wave functions. Expanding the density matrix in these functions gives the double series

$$\rho(x, x', t) = \sum_m \sum_n a_{mn}\psi_n^*(x')\psi_m(x)\exp\left[\frac{i}{\hbar}(E_n - E_m)t\right]. \tag{14.6}$$

This plays the same role for the density matrix as (10.3) does for the wave function. In place of the single set a_n there is the double set a_{mn}, which, like the density matrix, is Hermitian:

$$a_{mn} = a_{nm}^*. \tag{14.7}$$

Substitution in (14.4) gives

$$\bar{f} = \sum_m \sum_n a_{mn}\int \Psi_n^*(x, t)\hat{f}\Psi_m(x, t)\,dx,$$

or

$$\bar{f} = \sum_m \sum_n a_{mn}f_{nm}(t) = \sum_m \sum_n a_{mn}f_{nm}\exp\left[\frac{i}{\hbar}(E_n - E_m)t\right]. \tag{14.8}$$

This is analogous to (11.1)[12].

The a_{mn} must satisfy inequalities. Since the diagonal elements $\rho(x, x)$ are probabilities, they must be positive. Hence the quadratic form

$$\sum_n \sum_m a_{mn} \xi_m \xi_n^*$$

must be positive definite for arbitrary complex ξ_n. The standard conditions for quadratic forms therefore apply; in particular,

$$a_{nn} > 0, \tag{14.9}$$

and for every a_{mm}, a_{nn}, a_{mn},

$$a_{mm} a_{nn} \geq |a_{mn}|^2. \tag{14.10}$$

The pure case, in which the density matrix is a product of functions, corresponds to

$$a_{mn} = a_m a_n^*. \tag{14.11}$$

A simple criterion distinguishes a pure from a mixed state. In the pure case,

$$(a^2)_{mn} = \sum_k a_{mk} a_{kn} = \sum_k a_m a_k^* a_k a_n^* = a_m a_n^* \sum_k |a_k|^2 = a_m a_n^*,$$

and therefore

$$(a^2)_{mn} = a_{mn}. \tag{14.12}$$

Thus the square of the density matrix coincides with the matrix itself.

§ 15. Momentum

Consider a closed system of particles in no external field. Since all positions of the system as a whole are equivalent, its Hamiltonian is unchanged by an arbitrary parallel displacement. It is enough to require this for every infinitesimal displacement; it then holds for finite displacements as well.

Under an infinitesimal displacement $\delta\mathbf{r}$, every radius vector $\mathbf{r}_a$ acquires the same increment. Thus $\mathbf{r}_a \to \mathbf{r}_a + \delta\mathbf{r}$, and

$$\psi(\mathbf{r}_1 + \delta\mathbf{r}, \mathbf{r}_2 + \delta\mathbf{r}, \ldots) = \left(1 + \delta\mathbf{r} \cdot \sum_a \boldsymbol{\nabla}_a\right) \psi(\mathbf{r}_1, \mathbf{r}_2, \ldots).$$

Here a labels the particles and $\boldsymbol{\nabla}_a$ differentiates with respect to $\mathbf{r}_a$. The expression in parentheses is the operator of an infinitesimal displacement.

If a transformation does not change the Hamiltonian, transforming $\hat{H}\psi$ gives the same result as first transforming ψ and then applying $\hat{H}$. If $\hat{O}$ produces the transformation, then $\hat{O}(\hat{H}\psi) = \hat{H}(\hat{O}\psi)$, so $\hat{O}\hat{H} - \hat{H}\hat{O} = 0$: $\hat{H}$ commutes with $\hat{O}$.

[12] The quantities a_{mn} constitute the density matrix in the energy representation. Description of states by such a matrix was introduced independently by Landau and Bloch (*F. Bloch*) in 1927.

Here $\hat{O}$ is the infinitesimal-displacement operator. The unit operator commutes with every operator, and the constant $\delta\mathbf{r}$ may be taken outside $\hat{H}$; hence

$$\left(\sum_a \nabla_a\right)\hat{H} - \hat{H}\left(\sum_a \nabla_a\right) = 0. \tag{15.1}$$

An explicitly time-independent operator commuting with the Hamiltonian corresponds to a conserved quantity. The quantity whose conservation follows from the homogeneity of space is momentum (cf. Vol. I, § 7). Thus (15.1) expresses momentum conservation; $\sum_a \nabla_a$ corresponds, up to a constant factor, to the total momentum, and each term to the momentum of one particle.

The factor relating the momentum operator to ∇ follows from the classical limit and equals $-i\hbar$:

$$\hat{\mathbf{p}} = -i\hbar\nabla, \tag{15.2}$$

or

$$\hat{p}_x = -i\hbar\frac{\partial}{\partial x}, \qquad \hat{p}_y = -i\hbar\frac{\partial}{\partial y}, \qquad \hat{p}_z = -i\hbar\frac{\partial}{\partial z}.$$

Indeed, using (6.1),

$$\hat{\mathbf{p}}\Psi = -i\hbar\nabla\Psi = \Psi\nabla S.$$

Thus classically the operator acts by multiplication by ∇S, which is the classical momentum (see Vol. I, § 43).

The operator (15.2) is Hermitian, as required. For functions $\psi(x)$ and $\varphi(x)$ vanishing at infinity, integration by parts gives

$$\int \varphi\hat{p}_x\psi\, dx = -i\hbar\int \varphi\frac{\partial\psi}{\partial x}\, dx = i\hbar\int \psi\frac{\partial\varphi}{\partial x}\, dx = \int \psi\hat{p}_x^*\varphi\, dx,$$

which is the Hermiticity condition.

Since differentiations with respect to different variables commute,

$$\hat{p}_x\hat{p}_y - \hat{p}_y\hat{p}_x = 0, \quad \hat{p}_x\hat{p}_z - \hat{p}_z\hat{p}_x = 0, \quad \hat{p}_y\hat{p}_z - \hat{p}_z\hat{p}_y = 0. \tag{15.3}$$

Thus all three momentum components may have definite values simultaneously.

Their eigenfunctions and eigenvalues satisfy

$$-i\hbar\nabla\psi = \mathbf{p}\psi, \tag{15.4}$$

with solutions

$$\psi = \text{const} \cdot e^{i\mathbf{p}\cdot\mathbf{r}/\hbar}. \tag{15.5}$$

The three components completely determine the wave function and form one possible complete set. Their eigenvalues form a continuous spectrum from $-\infty$ to ∞.

By (5.4), $\int \psi_{\mathbf{p}'}^*\psi_{\mathbf{p}}, dV$ over all space should equal $\delta(\mathbf{p}' - \mathbf{p})$[13]. For later applications it is more natural to normalize to the delta function of the momentum difference divided by $2\pi\hbar$:

$$\int \psi_{\mathbf{p}'}^*\psi_{\mathbf{p}}, dV = \delta\left(\frac{\mathbf{p}' - \mathbf{p}}{2\pi\hbar}\right),$$

[13] For a vector argument, $\delta(\mathbf{a}) = \delta(a_x)\delta(a_y)\delta(a_z)$.

or equivalently

$$\int \psi^*_{\mathbf{p}'}\psi_{\mathbf{p}}, dV = (2\pi\hbar)^3\delta(\mathbf{p}' - \mathbf{p}). \tag{15.6}$$

Each one-dimensional factor obeys, for example, $\delta[(p'_x - p_x)/(2\pi\hbar)] = 2\pi\hbar\delta(p'_x - p_x)$. Integration uses[14]

$$\frac{1}{2\pi}\int_{-\infty}^{\infty} e^{i\alpha\xi}\, d\xi = \delta(\alpha). \tag{15.7}$$

It follows that const = 1 in (15.5)[15]:

$$\psi_{\mathbf{p}} = e^{i\mathbf{p}\cdot\mathbf{r}/\hbar}. \tag{15.8}$$

Expansion of an arbitrary $\psi(\mathbf{r})$ in momentum eigenfunctions is its Fourier integral:

$$\psi(\mathbf{r}) = \int a(\mathbf{p})\psi_{\mathbf{p}}(\mathbf{r})\frac{d^3p}{(2\pi\hbar)^3} = \int a(\mathbf{p})e^{i\mathbf{p}\cdot\mathbf{r}/\hbar}\frac{d^3p}{(2\pi\hbar)^3}. \tag{15.9}$$

Here $d^3p = dp_x dp_y dp_z$, and

$$a(\mathbf{p}) = \int \psi(\mathbf{r})\psi^*_{\mathbf{p}}(\mathbf{r})\, dV = \int \psi(\mathbf{r})e^{-i\mathbf{p}\cdot\mathbf{r}/\hbar}\, dV. \tag{15.10}$$

The function $a(\mathbf{p})$ is the momentum-representation wave function; $|a(\mathbf{p})|^2 d^3p/(2\pi\hbar)^3$ is the probability that the momentum lies in d^3p. Just as $\hat{\mathbf{p}}$ represents momentum in coordinate space, introduce $\hat{\mathbf{r}}$ for the coordinates in momentum space, defined by

$$\bar{\mathbf{r}} = \int a^*(\mathbf{p})\hat{\mathbf{r}}a(\mathbf{p})\frac{d^3p}{(2\pi\hbar)^3}. \tag{15.11}$$

But also $\bar{\mathbf{r}} = \int \psi^*\mathbf{r}\psi, dV$. Substituting (15.9) and integrating by parts,

$$\mathbf{r}\psi(\mathbf{r}) = \int i\hbar e^{i\mathbf{p}\cdot\mathbf{r}/\hbar}\frac{\partial a(\mathbf{p})}{\partial\mathbf{p}}\frac{d^3p}{(2\pi\hbar)^3},$$

and hence

$$\bar{\mathbf{r}} = \int i\hbar a^*(\mathbf{p})\frac{\partial a(\mathbf{p})}{\partial\mathbf{p}}\frac{d^3p}{(2\pi\hbar)^3}.$$

Comparison with (15.11) gives

$$\hat{\mathbf{r}} = i\hbar\frac{\partial}{\partial\mathbf{p}}. \tag{15.12}$$

In this representation momentum is simply multiplication by $\mathbf{p}$.

[14]The formula means that its left-hand side has the defining property (5.8) of the delta function. Substitution in (5.8) gives the Fourier formula

$$f(a) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x)e^{i\xi(x-a)}\, dx\, \frac{d\xi}{2\pi}.$$

[15]With this normalization $|\psi|^2 = 1$: the function is normalized to "one particle per unit volume." This agreement is not accidental; cf. the note on p. 221.

Finally, express the operator of translation through an arbitrary finite distance $\mathbf{a}$ in terms of $\hat{\mathbf{p}}$. By definition, $\hat{T}_{\mathbf{a}}\psi(\mathbf{r}) = \psi(\mathbf{r}+\mathbf{a})$. Expanding $\psi(\mathbf{r}+\mathbf{a})$ in a Taylor series gives

$$\psi(\mathbf{r}+\mathbf{a}) = \psi(\mathbf{r}) + \mathbf{a}\cdot\frac{\partial\psi(\mathbf{r})}{\partial\mathbf{r}} + \dots$$

or, on introducing $\hat{\mathbf{p}} = -i\hbar\boldsymbol{\nabla}$,

$$\psi(\mathbf{r}+\mathbf{a}) = \left[1 + \frac{i}{\hbar}\mathbf{a}\cdot\hat{\mathbf{p}} + \frac{1}{2}\left(\frac{i}{\hbar}\mathbf{a}\cdot\hat{\mathbf{p}}\right)^2 + \dots\right]\psi(\mathbf{r}).$$

Thus

$$\hat{T}_{\mathbf{a}} = \exp\left(\frac{i}{\hbar}\mathbf{a}\cdot\hat{\mathbf{p}}\right). \tag{15.13}$$

This is the required *finite-displacement operator*.

§ 16. Uncertainty Relations

Let us derive the commutation rules for the momentum and coordinate operators. Successive differentiation with respect to one of the variables x, y, z and multiplication by another of them gives a result independent of the order of these operations. Hence

$$\hat{p}_x y - y\hat{p}_x = 0, \qquad \hat{p}_x z - z\hat{p}_x = 0, \tag{16.1}$$

and similarly for $\hat{p}_y$ and $\hat{p}_z$.

To derive the commutation rule for $\hat{p}_x$ and x, write

$$(\hat{p}_x x - x\hat{p}_x)\psi = -i\hbar\frac{\partial}{\partial x}(x\psi) + i\hbar x\frac{\partial\psi}{\partial x} = -i\hbar\psi.$$

We see that the action of $\hat{p}_x x - x\hat{p}_x$ reduces to multiplication of the function by $-i\hbar$; the same is of course true for the commutation of $\hat{p}_y$ with y and of $\hat{p}_z$ with z. Thus[16]

$$\hat{p}_x x - x\hat{p}_x = -i\hbar, \qquad \hat{p}_y y - y\hat{p}_y = -i\hbar, \qquad \hat{p}_z z - z\hat{p}_z = -i\hbar. \tag{16.2}$$

Relations (16.1) and (16.2) can be written together as

$$\hat{p}_i x_k - x_k\hat{p}_i = -i\hbar\delta_{ik}, \qquad i, k = x, y, z. \tag{16.3}$$

Before examining the physical meaning of these relations and their consequences, we give two formulas useful below. If $f(\mathbf{r})$ is a function of the coordinates, then

$$\hat{\mathbf{p}}f(\mathbf{r}) - f(\mathbf{r})\hat{\mathbf{p}} = -i\hbar\nabla f. \tag{16.4}$$

Indeed,

$$(\hat{\mathbf{p}}f - f\hat{\mathbf{p}})\psi = -i\hbar[\nabla(f\psi) - f\nabla\psi] = -i\hbar\psi\nabla f.$$

[16] These relations, discovered in matrix form by *Heisenberg* in 1925, served as the starting point in the creation of quantum mechanics.

An analogous relation holds for the commutator with a function of the momentum operator:

$$f(\mathbf{p})\hat{\mathbf{r}} - \hat{\mathbf{r}}f(\mathbf{p}) = -i\hbar\frac{\partial f}{\partial \mathbf{p}}. \tag{16.5}$$

It may be derived in the same way as (16.4) by working in the momentum representation and using (15.12) for the coordinate operators.

Relations (16.1) and (16.2) show that a particle coordinate along one axis can have a definite value simultaneously with the momentum components along the other two axes; a coordinate and the momentum component along that same axis do not exist simultaneously. In particular, a particle cannot occupy a definite point in space while also having a definite momentum $\mathbf{p}$.

Let the particle occupy a finite spatial region whose dimensions along the three axes are of order Δx, Δy, Δz. Let its mean momentum be $\mathbf{p}_0$. Mathematically, this means that the wave function has the form $\psi = u(\mathbf{r})e^{i\mathbf{p}_0\cdot\mathbf{r}/\hbar}$, where $u(\mathbf{r})$ is appreciably different from zero only within the specified region.

Expand ψ in eigenfunctions of the momentum operator, that is, as a Fourier integral. The coefficients $a(\mathbf{p})$ in this expansion are given by (15.10) as integrals of functions of the form $u(\mathbf{r})e^{i(\mathbf{p}_0-\mathbf{p})\cdot\mathbf{r}/\hbar}$. For such an integral to be appreciably different from zero, the periods of the oscillating factor $e^{i(\mathbf{p}_0-\mathbf{p})\cdot\mathbf{r}/\hbar}$ must not be small compared with the dimensions Δx, Δy, and Δz of the region where $u(\mathbf{r})$ differs from zero. Thus $a(\mathbf{p})$ is appreciably different from zero only for values of $\mathbf{p}$ satisfying $(p_{0x} - p_x)\Delta x/\hbar \lesssim 1, \ldots$ Since $|a(\mathbf{p})|^2$ determines the probabilities of the various momentum values, the intervals of p_x, p_y, p_z in which $a(\mathbf{p})$ is nonzero are precisely the intervals in which the particle's momentum components may lie in the state considered. Denoting these intervals by Δp_x, Δp_y, and Δp_z, we therefore have

$$\Delta p_x \Delta x \sim \hbar, \qquad \Delta p_y \Delta y \sim \hbar, \qquad \Delta p_z \Delta z \sim \hbar. \tag{16.6}$$

These *uncertainty relations* were established by *Heisenberg* in 1927.

We see that the more accurately a particle coordinate is known (the smaller Δx is), the greater is the uncertainty Δp_x in the momentum component along the same axis, and conversely. In particular, if a particle occupies a strictly definite point of space ($\Delta x = \Delta y = \Delta z = 0$), then $\Delta p_x = \Delta p_y = \Delta p_z = \infty$. All momentum values are then equally probable. Conversely, if the particle has a strictly definite momentum $\mathbf{p}$, all its positions in space are equally probable (this also follows directly from the wave function (15.8), whose squared modulus is entirely independent of the coordinates).

If the coordinate and momentum uncertainties are characterized by their root-mean-square fluctuations,

$$\delta x = \sqrt{\overline{(x-\bar{x})^2}}, \qquad \delta p_x = \sqrt{\overline{(p_x-\bar{p}_x)^2}},$$

an exact bound can be given for the least possible value of their product (*H. Weyl*).

Consider the one-dimensional case, a packet with wave function $\psi(x)$ depending on only one coordinate; for simplicity, suppose that the mean values of x and p_x in this state are zero. Begin with the evident inequality

$$\int_{-\infty}^{\infty} \left| \alpha x \psi + \frac{d\psi}{dx} \right|^2 dx \geq 0,$$

where α is an arbitrary real constant. In evaluating this integral, note that

$$\int x^2 |\psi|^2 \, dx = (\delta x)^2,$$

$$\int \left(x \frac{d\psi^*}{dx} \psi + x\psi^* \frac{d\psi}{dx} \right) dx = \int x \frac{d|\psi|^2}{dx} \, dx = -\int |\psi|^2 \, dx = -1,$$

$$\int \frac{d\psi^*}{dx} \frac{d\psi}{dx} \, dx = -\int \psi^* \frac{d^2\psi}{dx^2} \, dx = \frac{1}{\hbar^2} \int \psi^* \hat{p}_x^2 \psi \, dx = \frac{1}{\hbar^2} (\delta p_x)^2,$$

and obtain

$$\alpha^2 (\delta x)^2 - \alpha + \frac{(\delta p_x)^2}{\hbar^2} \geq 0.$$

For this quadratic trinomial in α to be nonnegative for every value of α, its discriminant must be negative. It follows that

$$\delta x \, \delta p_x \geq \hbar/2. \tag{16.7}$$

The least possible value of the product is $\hbar/2$.

This value is attained by wave packets described by functions of the form

$$\psi = \frac{1}{(2\pi)^{1/4} \sqrt{\delta x}} \exp\left(\frac{i}{\hbar} p_0 x - \frac{x^2}{4(\delta x)^2} \right), \tag{16.8}$$

where p_0 and δx are constants. The probabilities of the various coordinate values in such a state are

$$|\psi|^2 = \frac{1}{\sqrt{2\pi} \, \delta x} \exp\left(-\frac{x^2}{2(\delta x)^2} \right),$$

that is, they have a Gaussian distribution about the origin (mean value $\bar{x} = 0$), with root-mean-square fluctuation δx. The wave function in the momentum representation is

$$a(p_x) = \int_{-\infty}^{\infty} \psi(x) \exp\left(-i \frac{p_x x}{\hbar} \right) dx.$$

Evaluation of the integral gives an expression of the form

$$a(p_x) = \text{const} \cdot \exp\left(-\frac{(\delta x)^2 (p_x - p_0)^2}{\hbar^2} \right).$$

The probability distribution of the momentum values, $|a(p_x)|^2$, is also Gaussian, about the mean $\bar{p}_x = p_0$, and has root-mean-square fluctuation $\delta p_x = \hbar/(2\delta x)$. Thus the product $\delta p_x \delta x$ is exactly $\hbar/2$.

Finally, let us derive one more useful relation. Let f and g be two physical quantities whose operators obey the commutation rule

$$\hat{f}\hat{g} - \hat{g}\hat{f} = -i\hbar\hat{c}, \tag{16.9}$$

where $\hat{c}$ is the operator of a physical quantity c. The factor $\hbar$ has been introduced on the right because in the classical limit ($\hbar \to 0$) all operators of physical quantities reduce to multiplication by those quantities and commute with one another. Thus, in the "quasiclassical" case, the right-hand side of (16.9) may be taken as zero in the first approximation. In the next approximation, $\hat{c}$ may be replaced by the operator of simple multiplication by c. We then have

$$\hat{f}\hat{g} - \hat{g}\hat{f} = -i\hbar c.$$

This equation is exactly analogous to $\hat{p}_x x - x\hat{p}_x = -i\hbar$, except that the quantity $\hbar c$ occurs in place of the constant $\hbar$[17]. By analogy with $\Delta x \Delta p_x \sim \hbar$, we therefore conclude that in the quasiclassical case the quantities f and g obey the uncertainty relation

$$\Delta f \Delta g \sim \hbar c. \tag{16.10}$$

In particular, if one of the quantities is the energy ($f \equiv H$) and the operator of the other (g) has no explicit time dependence, then by (9.2) $c = \dot{g}$, and the quasiclassical uncertainty relation is

$$\Delta E \Delta g \sim \hbar \dot{g}. \tag{16.11}$$

[17]The classical quantity c is the Poisson bracket of f and g (see the note on p. 46).

CHAPTER III

The Schrödinger Equation

§ 17. The Schrödinger equation

The Hamiltonian determines the form of a physical system's wave equation and consequently acquires fundamental significance in the entire mathematical apparatus of quantum mechanics.

For a free particle, the Hamiltonian is already established by the general requirements associated with the homogeneity and isotropy of space and with the Galilean relativity principle. In classical mechanics these requirements imply a quadratic dependence of the particle energy on its momentum: $E = p^2/2m$, where the constant m is called the mass of the particle (see I, § 4). In quantum mechanics the same requirements lead to the same relation for the eigenvalues of energy and momentum, quantities which for a free particle are conserved and can be measured simultaneously.

For the relation $E = p^2/2m$ to hold for every eigenvalue of energy and momentum, however, it must also hold for their operators:

$$\hat{H} = \frac{1}{2m}\left(\hat{p}_x^2 + \hat{p}_y^2 + \hat{p}_z^2\right). \tag{17.1}$$

Substitution of (15.2) here gives the Hamiltonian of a freely moving particle in the form

$$\hat{H} = -\frac{\hbar^2}{2m}\Delta, \tag{17.2}$$

where $\Delta = \partial^2/\partial x^2 + \partial^2/\partial y^2 + \partial^2/\partial z^2$ is the Laplace operator. The Hamiltonian of a system of noninteracting particles is equal to the sum of the Hamiltonians of the individual particles:

$$\hat{H} = -\frac{\hbar^2}{2}\sum_a \frac{\Delta_a}{m_a}, \tag{17.3}$$

where the index a numbers the particles; Δ_a is the Laplace operator in which differentiation is performed with respect to the coordinates of the ath particle. In classical (nonrelativistic) mechanics, particle interaction is described by an additive term in the Hamilton function, the potential energy of interaction $U(\mathbf{r}_1, \mathbf{r}_2, \ldots)$, which is a function of the particle coordinates. Particle interaction in quantum mechanics is likewise described by adding the same function to the Hamiltonian of the system[1]:

$$\hat{H} = -\frac{\hbar^2}{2}\sum_a \frac{\Delta_a}{m_a} + U(\mathbf{r}_1, \mathbf{r}_2, \ldots); \tag{17.4}$$

[1]This assertion is not, of course, a logical consequence of the fundamental principles of quantum mechanics and must be regarded as a consequence of experimental data.

the first term may be regarded as the kinetic-energy operator and the second as the potential-energy operator. In particular, the Hamiltonian for one particle in an external field is

$$\hat{H} = \frac{\hat{\mathbf{p}}^2}{2m} + U(x, y, z) = -\frac{\hbar^2}{2m}\Delta + U(x, y, z), \tag{17.5}$$

where $U(x, y, z)$ is the potential energy of the particle in the external field.

Substitution of expressions (17.2)–(17.5) in the general equation (8.1) yields the wave equations for the corresponding systems. Here we write out the wave equation for a particle in an external field:

$$i\hbar\frac{\partial\Psi}{\partial t} = -\frac{\hbar^2}{2m}\Delta\Psi + U(x, y, z)\Psi. \tag{17.6}$$

Equation (10.2), determining the stationary states, assumes the form

$$\frac{\hbar^2}{2m}\Delta\psi + [E - U(x, y, z)]\psi = 0. \tag{17.7}$$

Equations (17.6) and (17.7) were established by *Schrödinger* in 1926 and are known as the *Schrödinger equations*.

For a free particle equation (17.7) becomes

$$\frac{\hbar^2}{2m}\Delta\psi + E\psi = 0. \tag{17.8}$$

This equation admits solutions that remain finite over all space for any positive value of the energy E. In states with definite directions of motion these solutions are the eigenfunctions of the momentum operator, with $E = p^2/2m$. The full (time-dependent) wave functions of these stationary states have the form

$$\Psi = \text{const} \cdot \exp\left[-\frac{i}{\hbar}(Et - \mathbf{p}\cdot\mathbf{r})\right]. \tag{17.9}$$

Each of these functions, a *plane wave*, describes a state in which the particle possesses definite energy E and momentum $\mathbf{p}$. The frequency of this wave is $E/\hbar$, and its wave vector is $\mathbf{k} = \mathbf{p}/\hbar$; the associated wavelength $\lambda = 2\pi\hbar/p$ is termed the *de Broglie wavelength of the particle*[2].

The energy spectrum of a freely moving particle is therefore continuous, extending from zero to $+\infty$. Each of these eigenvalues, except only $E = 0$, is degenerate, and the degeneracy is of infinite multiplicity. Indeed, every nonzero value of E corresponds to an infinite set of eigenfunctions (17.9), differing in the directions of the vector $\mathbf{p}$ while its absolute magnitude remains the same.

Let us trace how the limiting transition to classical mechanics occurs in the Schrödinger equation, considering for simplicity just one particle in an external field. Substitution in the Schrödinger equation (17.6) of the limiting expression (6.1), $\Psi = ae^{iS/\hbar}$, for the wave function gives, after the differentiations are performed,

$$a\frac{\partial S}{\partial t} - i\hbar\frac{\partial a}{\partial t} + \frac{a}{2m}(\nabla S)^2 - \frac{i\hbar}{2m}a\Delta S - \frac{i\hbar}{m}\nabla S\cdot\nabla a - \frac{\hbar^2}{2m}\Delta a + Ua = 0.$$

[2]The concept of a wave associated with a particle was first introduced by *de Broglie* (*L. de Broglie*) in 1924.

This equation contains purely real and purely imaginary terms (recall that S and a are real); equating each group separately to zero gives two equations:

$$\frac{\partial S}{\partial t} + \frac{1}{2m}(\nabla S)^2 + U - \frac{\hbar^2}{2ma}\Delta a = 0,$$

$$\frac{\partial a}{\partial t} + \frac{a}{2m}\Delta S + \frac{1}{m}\nabla S \cdot \nabla a = 0.$$

Neglecting the term containing $\hbar^2$ in the first of these equations, we obtain

$$\frac{\partial S}{\partial t} + \frac{1}{2m}(\nabla S)^2 + U = 0, \tag{17.10}$$

that is, as expected, the classical Hamilton–Jacobi equation for the action S of the particle. We note in passing that, as $\hbar \to 0$ classical mechanics is accurate up to and including quantities of first order in $\hbar$ (rather than zeroth order).

After multiplication by $2a$, the second equation obtained above can be rewritten as

$$\frac{\partial a^2}{\partial t} + \nabla\cdot\left(a^2\frac{\nabla S}{m}\right) = 0. \tag{17.11}$$

This equation has a transparent physical meaning: a^2 is the probability density for finding the particle at one or another place in space ($|\Psi|^2 = a^2$); $\nabla S/m = \mathbf{p}/m$ is the classical velocity $\mathbf{v}$ of the particle. Equation (17.11) is therefore nothing other than the continuity equation, showing that the probability density is "transported" according to the laws of classical mechanics, with the classical velocity $\mathbf{v}$ at every point.

Problem

Find the transformation law of the wave function under a Galilean transformation.

Solution. Perform the transformation on the wave function for the free motion of a particle (a plane wave). Since every function Ψ can be expanded in plane waves, this will also determine the transformation law for an arbitrary wave function.

The plane waves in the reference frames K and K' (where K' moves with velocity $\mathbf{V}$ relative to K) are

$$\Psi(\mathbf{r}, t) = \text{const}\cdot\exp[i(\mathbf{p}\cdot\mathbf{r} - Et)/\hbar],$$

$$\Psi'(\mathbf{r}', t) = \text{const}\cdot\exp[i(\mathbf{p}'\cdot\mathbf{r}' - E't)/\hbar],$$

where $\mathbf{r} = \mathbf{r}' + \mathbf{V}t$, while the particle's momenta and energies in the two frames are related by

$$\mathbf{p} = \mathbf{p}' + m\mathbf{V}, \qquad E = E' + \mathbf{V}\cdot\mathbf{p}' + \frac{mV^2}{2}$$

(see I, § 8). Substitution of these expressions in Ψ gives

$$\begin{aligned}\Psi(\mathbf{r}, t) &= \Psi'(\mathbf{r}', t)\exp\left[\frac{i}{\hbar}\left(m\mathbf{V}\cdot\mathbf{r}' + \frac{mV^2}{2}t\right)\right]\\ &= \Psi'(\mathbf{r} - \mathbf{V}t, t)\exp\left[\frac{i}{\hbar}\left(m\mathbf{V}\cdot\mathbf{r} - \frac{mV^2}{2}t\right)\right].\end{aligned} \tag{1}$$

Written this way, the formula no longer contains quantities characterizing the free motion of the particle; it therefore furnishes the sought-after general rule governing how the wave function of an arbitrary state of the particle transforms. For a system of particles, the exponent of the exponential in (1) must contain a summation over all the particles.

§ 18. Fundamental properties of the Schrödinger equation

The conditions to be satisfied by solutions of the Schrödinger equation are quite general in nature. The wave function, to begin with, must be single-valued and continuous throughout all of space. The continuity requirement remains in force even when the field $U(x, y, z)$ itself has surfaces of discontinuity. On any such surface, the wave function together with its derivatives must still be continuous. The latter are not continuous, however, if the potential energy U becomes infinite beyond some surface. A particle cannot enter at all a region of space where $U = \infty$, that is, $\psi = 0$ must hold everywhere in that region. Continuity of ψ requires it to vanish on the boundary of this region; in this case, however, the derivatives of ψ generally undergo a jump.

If the field $U(x, y, z)$ nowhere becomes infinite, the wave function must also be finite throughout space. The same condition must be observed when U becomes infinite at some point, provided it does not do so too rapidly, namely as $1/r^s$ with $s < 2$ (see also § 35).

Let $U_{\min}$ be the minimum value of the function $U(x, y, z)$. Since the particle's Hamiltonian is the sum of two terms, the kinetic- and potential-energy operators $\hat{T}$ and U, the mean energy in an arbitrary state is the sum $\bar{E} = \bar{T} + \bar{U}$. But all eigenvalues of the operator $\hat{T}$ (which coincides with the Hamiltonian of a free particle) are positive, and therefore the mean value $\bar{T} > 0$ as well. Taking account also of the evident inequality $\bar{U} > U_{\min}$, we find that $\bar{E} > U_{\min}$. Since this inequality holds for every state, it plainly holds for all energy eigenvalues as well:

$$E_n > U_{\min}. \tag{18.1}$$

Suppose a particle moves through a force field that goes to zero at large distances; following standard convention, we fix $U(x, y, z)$ so that it too tends to zero at infinity. One readily verifies that the eigenvalue spectrum for negative energies is then discrete: every state with $E < 0$ in a field that vanishes at infinity is a bound state. Indeed, in stationary states of the continuous spectrum, which correspond to infinite motion, the particle is at infinity (see § 10). At sufficiently large distances the presence of the field may be neglected and the particle's motion may be regarded as free; in free motion, however, the energy can only be positive.

Conversely, the positive eigenvalues form a continuous spectrum and correspond to infinite motion; for $E > 0$ the Schrödinger equation generally does not have (in the field under consideration) solutions for which the integral $\int |\psi|^2 \, dV$ converges[3].

We draw attention to the fact that, in quantum mechanics, a particle in finite motion may also be present in regions of space where $E < U$. Although the probability $|\psi|^2$

[3] From a purely mathematical standpoint, however, it must be qualified that, for certain forms of the function $U(x, y, z)$ (having no physical significance), a discrete set of values may drop out of the continuous spectrum.

of finding the particle tends rapidly to zero deeper inside such a region, it nevertheless differs from zero at every finite distance. This represents a fundamental departure from classical mechanics, where a particle is altogether unable to penetrate a region with $U > E$. In classical mechanics, penetration into this region is impossible because, for $E < U$, the kinetic energy would become negative, meaning that the velocity would be imaginary. In quantum mechanics, too, the kinetic-energy eigenvalues are positive; nevertheless, no contradiction arises here, because if the measurement process localizes the particle at some definite point of space, that same process disturbs the particle's state in such a way that it ceases altogether to have any definite kinetic energy.

If $U(x, y, z) > 0$ throughout space (with $U \to 0$ at infinity), inequality (18.1) gives $E_n > 0$. On the other hand, since the spectrum must be continuous for $E > 0$, it follows that in the case under consideration the discrete spectrum is absent altogether, that is, only infinite motion of the particle is possible.

Suppose that at some point, chosen as the coordinate origin, U tends to $-\infty$ according to the law

$$U \approx -\alpha/r^s, \qquad \alpha > 0. \tag{18.2}$$

Consider a wave function that is finite in a certain small region of radius r_0 around the origin and zero outside it. The uncertainty in the particle's coordinate values in such a wave packet is of order r_0; the uncertainty in the momentum value is therefore $\sim \hbar/r_0$. The mean kinetic energy in this state is of order $\hbar^2/mr_0^2$, and the mean potential energy is of order $-\alpha/r_0^s$. Suppose first that $s > 2$. Then, for sufficiently small r_0, the sum $\hbar^2/mr_0^2 - \alpha/r_0^s$ takes negative values of arbitrarily large absolute magnitude. But if the mean energy can take such values, this means in any event that there exist negative energy eigenvalues of arbitrarily large absolute magnitude. Energy levels with large $|E|$ correspond to motion of the particle in a very small region of space around the origin. The "normal" state will correspond to the particle being located at the origin itself, that is, the particle will "fall" into the point $r = 0$.

If $s < 2$, on the other hand, the energy cannot take negative values of arbitrarily large absolute magnitude. The diagonal spectrum starts from a certain finite negative value, and the particle does not fall to the center. We note that in classical mechanics a particle can in principle fall to the center in any attractive field (that is, for any positive s). The case $s = 2$ will be considered separately in § 35.

Next, let us investigate the character of the energy spectrum as a function of the behaviour of the field at large distances. Suppose that as $r \to \infty$ the potential energy, while negative, tends to zero according to the power law (18.2) (in this formula r is now large). Consider a wave packet "filling" a spherical shell of large radius r_0 and thickness $\Delta r \ll r_0$. Then again the order of magnitude of the kinetic energy is $\hbar^2/m(\Delta r)^2$, and that of the potential energy is $-\alpha/r_0^s$. Let r_0 increase while Δr is increased at the same time (so that Δr grows in proportion to r_0). If $s < 2$, then for sufficiently large r_0 the sum $\hbar^2/m(\Delta r)^2 - \alpha/r_0^s$ becomes negative. It follows that stationary states with negative energy exist in which the particle may be found with appreciable probability at large distances from the origin. But this means that negative energy levels of arbitrarily small absolute magnitude exist (it must be remembered that wave functions decay rapidly in the region of space where $U > E$). Thus, in the case under consideration, the discrete spectrum contains an infinite set of levels which crowd together towards the level $E = 0$.

If, however, the field falls off at infinity as $-1/r^s$ with $s > 2$, there are no negative levels of arbitrarily small absolute magnitude. The discrete spectrum ends at a level with nonzero absolute magnitude, so that the total number of levels is finite.

The Schrödinger equation for the stationary-state wave functions ψ, together with the conditions placed on its solutions, is real. Its solutions can therefore always be chosen real[4]. The eigenfunctions of nondegenerate energy values automatically prove to be real up to an immaterial phase factor. Indeed, ψ^* satisfies the same equation as ψ and is therefore also an eigenfunction for the same value of the energy; hence, if that value is nondegenerate, ψ and ψ^* must be essentially identical, that is, they may differ at most by a constant factor whose modulus equals unity. Wave functions that correspond to one and the same degenerate level need not be real, but a set of real functions can always be obtained by a suitable choice of their linear combinations.

The full (time-dependent) wave functions Ψ, however, are determined by an equation whose coefficients contain i. This equation nevertheless retains its form if t is replaced by $-t$ and complex conjugation is performed at the same time[5]. The functions Ψ can therefore always be chosen so that Ψ and Ψ^* differ only in the sign of time.

As is known, the equations of classical mechanics are unchanged under *time reversal*, i.e. upon reversing the sign of time. In quantum mechanics, as we see, symmetry with respect to the two directions of time manifests itself in the invariance of the wave equation under reversal of the sign of t accompanied by the replacement of Ψ with Ψ^*. It must be remembered, however, that here this symmetry applies only to the equations and not to the concept of measurement itself, which has a fundamental role in quantum mechanics (as discussed in detail in § 7).

§ 19. Flux Density

In classical mechanics a particle's velocity $\mathbf{v}$ is related to its momentum by $\mathbf{p} = m\mathbf{v}$. As expected, quantum mechanics has the same relation between the corresponding operators. This follows at once by calculating $\hat{\mathbf{v}} = \hat{\dot{\mathbf{r}}}$ from the general rule (9.2) for differentiating operators with respect to time:

$$\hat{\mathbf{v}} = \frac{i}{\hbar}\left(\hat{H}\mathbf{r} - \mathbf{r}\hat{H}\right).$$

Using (17.5) for $\hat{H}$ together with (16.5), we obtain

$$\hat{\mathbf{v}} = \hat{\mathbf{p}}/m. \tag{19.1}$$

The same relations plainly hold between the eigenvalues of velocity and momentum, and between their mean values in any state.

Like momentum, the velocity of a particle cannot have a definite value simultaneously with its coordinates. Velocity multiplied by an infinitesimal time element dt, however,

[4] This is not true for systems in a magnetic field.

[5] It is assumed that the potential energy U has no explicit dependence on time: the system is either closed or is in a constant (nonmagnetic) field.

determines the particle's displacement during dt. Thus the nonexistence of velocity simultaneously with coordinates means that if a particle occupies a definite point of space at one instant, it no longer has a definite position at the next, infinitesimally close instant.

We record a useful formula for the operator $\hat{\dot{f}}$, the time derivative of a quantity $f(\mathbf{r})$ which is a function of the particle's radius vector. Since f commutes with $U(\mathbf{r})$, we find

$$\hat{\dot{f}} = \frac{i}{\hbar}(\hat{H}f - f\hat{H}) = \frac{i}{2m\hbar}(\hat{\mathbf{p}}^2 f - f\hat{\mathbf{p}}^2),$$

while (16.4) gives

$$\hat{\mathbf{p}}^2 f = \hat{\mathbf{p}}\,(f\hat{\mathbf{p}} - i\hbar\nabla f)\,, \qquad f\hat{\mathbf{p}}^2 = (\hat{\mathbf{p}}f + i\hbar\nabla f)\,\hat{\mathbf{p}}.$$

Hence the desired expression is

$$\hat{\dot{f}} = \frac{1}{2m}\,(\hat{\mathbf{p}}\,\nabla f + \nabla f \cdot \hat{\mathbf{p}})\,. \tag{19.2}$$

Next, let us find the acceleration operator. We have

$$\hat{\dot{\mathbf{v}}} = \frac{i}{\hbar}(\hat{H}\hat{\mathbf{v}} - \hat{\mathbf{v}}\hat{H}) = \frac{i}{m\hbar}(\hat{H}\hat{\mathbf{p}} - \hat{\mathbf{p}}\hat{H}) = \frac{i}{m\hbar}(U\hat{\mathbf{p}} - \hat{\mathbf{p}}U).$$

Using (16.4), we find

$$m\hat{\dot{\mathbf{v}}} = -\nabla U. \tag{19.3}$$

In form, this operator equation coincides exactly with the equation of motion (Newton's equation) of classical mechanics.

The integral $\int_V |\Psi|^2\,dV$ over a finite volume V is the probability of finding the particle in that volume. Let us calculate its time derivative:

$$\frac{d}{dt}\int_V |\Psi|^2\,dV = \int_V \left(\Psi\frac{\partial\Psi^*}{\partial t} + \Psi^*\frac{\partial\Psi}{\partial t}\right) dV = \frac{i}{\hbar}\int_V \left(\Psi\hat{H}^*\Psi^* - \Psi^*\hat{H}\Psi\right) dV.$$

Substitution of

$$\hat{H} = \hat{H}^* = -\frac{\hbar^2}{2m}\Delta + U(x, y, z)$$

and use of the identity

$$\Psi\Delta\Psi^* - \Psi^*\Delta\Psi = \nabla\cdot(\Psi\nabla\Psi^* - \Psi^*\nabla\Psi)$$

give

$$\frac{d}{dt}\int_V |\Psi|^2\,dV = -\int_V \nabla\cdot\mathbf{j}\,dV,$$

where $\mathbf{j}$ denotes the vector[6]

$$\mathbf{j} = \frac{i\hbar}{2m}(\Psi\nabla\Psi^* - \Psi^*\nabla\Psi) = \frac{1}{2m}(\Psi\hat{\mathbf{p}}^*\Psi^* + \Psi^*\hat{\mathbf{p}}\Psi). \tag{19.4}$$

[6] If ψ is written as $|\psi|e^{i\alpha}$, then

$$\mathbf{j} = \frac{\hbar}{m}|\psi|^2\nabla\alpha. \tag{19.4a}$$

By Gauss's theorem, the integral of $\nabla \cdot \mathbf{j}$ can be transformed into an integral over the closed surface bounding V:

$$\frac{d}{dt}\int_V |\Psi|^2\, dV = -\oint \mathbf{j}\, d\mathbf{f}. \tag{19.5}$$

It follows that $\mathbf{j}$ may be called the *probability-flux-density vector*, or simply the *flux density*. Its surface integral is the probability that the particle crosses that surface per unit time. The vector $\mathbf{j}$ and the probability density $|\Psi|^2$ obey

$$\frac{\partial |\Psi|^2}{\partial t} + \nabla \cdot \mathbf{j} = 0, \tag{19.6}$$

an equation analogous to the classical continuity equation.

The free-motion wave function—the plane wave (17.9)—can be normalized so as to describe a particle flux of unit density (a flux in which, on average, one particle per unit time passes through a unit area of its cross-section). The function is

$$\Psi = \frac{1}{\sqrt{v}} \exp\left[-\frac{i}{\hbar}(Et - \mathbf{p}\cdot\mathbf{r})\right], \tag{19.7}$$

where v is the particle's speed. Indeed, substituting this function into (19.4) gives $\mathbf{j} = \mathbf{p}/mv$, the unit vector in the direction of motion. It is useful to show how the mutual orthogonality of wave functions of states with different energies follows directly from the Schrödinger equation. Let ψ_m and ψ_n be two such functions; they satisfy

$$-\frac{\hbar^2}{2m}\Delta\psi_m + U\psi_m = E_m\psi_m,$$

$$-\frac{\hbar^2}{2m}\Delta\psi_n^* + U\psi_n^* = E_n\psi_n^*.$$

Multiply the first equation by ψ_n^* and the second by ψ_m, and subtract them term by term. This gives

$$(E_m - E_n)\psi_m\psi_n^* = \frac{\hbar^2}{2m}(\psi_m\Delta\psi_n^* - \psi_n^*\Delta\psi_m) = \frac{\hbar^2}{2m}\nabla\cdot(\psi_m\nabla\psi_n^* - \psi_n^*\nabla\psi_m).$$

Integrating both sides over all space, the right-hand side vanishes after application of Gauss's theorem, and we obtain

$$(E_m - E_n)\int \psi_m\psi_n^*\, dV = 0.$$

Since $E_m \neq E_n$ by assumption, the required orthogonality relation follows:

$$\int \psi_m\psi_n^*\, dV = 0.$$

§ 20. The Variational Principle

The Schrödinger equation in its general form, $\hat{H}\psi = E\psi$, can be obtained from the variational principle

$$\delta \int \psi^*(\hat{H} - E)\psi \, dq = 0. \tag{20.1}$$

Since ψ is complex, variations with respect to ψ and ψ^* may be made independently. Variation with respect to ψ^* gives

$$\int \delta\psi^*(\hat{H} - E)\psi \, dq = 0,$$

and the arbitrariness of $\delta\psi^*$ then yields the required equation $\hat{H}\psi = E\psi$. Variation with respect to ψ gives nothing new. Indeed, varying ψ and using the Hermitian character of $\hat{H}$, we have

$$\int \psi^*(\hat{H} - E)\delta\psi \, dq = \int \delta\psi(\hat{H}^* - E)\psi^* \, dq = 0,$$

which yields the complex-conjugate equation $\hat{H}^*\psi^* = E\psi^*$.

The variational principle (20.1) requires an unconstrained extremum of the integral. It may be put in another form by treating E as a Lagrange multiplier in the problem of finding a constrained extremum of

$$\delta \int \psi^* \hat{H}\psi \, dq = 0 \tag{20.2}$$

subject to the supplementary condition

$$\int \psi\psi^* \, dq = 1. \tag{20.3}$$

The minimum value of the integral (20.2), subject to (20.3), is the first energy eigenvalue, that is, the energy E_0 of the normal state. The function ψ which realizes this minimum is correspondingly the normal-state wave function ψ_0[7]. The wave functions ψ_n ($n > 0$) of the succeeding stationary states correspond only to extrema, not to true minima of the integral.

To obtain from the minimum condition for (20.2) the wave function ψ_1 and energy E_1 of the state immediately above the normal state, the trial functions ψ must be restricted not only by the normalization condition (20.3), but also by orthogonality to the normal-state wave function ψ_0: $\int \psi\psi_0 \, dq = 0$. In general, suppose that the wave functions $\psi_0, \psi_1, \ldots, \psi_{n-1}$ of the first n states are known (the states being ordered by increasing energy). The wave function of the next state minimizes (20.2) under the supplementary conditions

$$\int \psi^2 \, dq = 1, \qquad \int \psi\psi_m \, dq = 0, \qquad m = 0, 1, 2, \ldots, n - 1. \tag{20.4}$$

[7]For the remainder of this section we take the wave functions ψ to be real, as they can always be chosen when no magnetic field is present.

We give here several general theorems which may be proved from the variational principle[8].

The normal-state wave function ψ_0 does not vanish (or, as one says, has no nodes) at any finite values of the coordinates[9]. In other words, it has the same sign throughout space. It follows that the wave functions ψ_n $(n > 0)$ of the other stationary states, being orthogonal to ψ_0, necessarily have nodal points (if ψ_n also had a constant sign, the integral $\int \psi_0\psi_n \, dq$ could not vanish).

The absence of nodes in ψ_0 also implies that the normal energy level cannot be degenerate. Suppose otherwise, and let ψ_0 and ψ_0' be two different eigenfunctions belonging to the level E_0. Every linear combination $c\psi_0 + c'\psi_0'$ would also be an eigenfunction; but by a suitable choice of c and c', this function can always be made to vanish at any prescribed point of space, producing an eigenfunction with a node.

If motion is confined to a bounded region of space, then $\psi = 0$ on its boundary (see § 18). To determine the energy levels one must use the variational principle to find the minimum of (20.2) with this boundary condition. In this case the theorem that the normal-state wave function has no nodes states that ψ_0 vanishes nowhere inside the region.

Notice that increasing the size of the region of motion lowers every energy level E_n. This follows directly because enlarging the region enlarges the class of trial functions available for minimizing the integral, so its minimum value can only decrease. For discrete-spectrum states of a system of particles, the expression

$$\int \psi \hat{H} \psi \, dq = \int \left[-\sum_a \frac{\hbar^2}{2m_a} \psi \Delta_a \psi + U\psi^2 \right] dq$$

can be transformed into a form more convenient for actual variation. In the first term of the integrand, write

$$\psi \Delta_a \psi = \nabla_a \cdot (\psi \nabla_a \psi) - (\nabla_a \psi)^2.$$

The integral of $\nabla_a \cdot (\psi \nabla_a \psi)$ over dV_a becomes an integral over a closed surface at infinity. Since discrete-spectrum wave functions vanish sufficiently rapidly at infinity, this integral disappears. Thus

$$\int \psi \hat{H} \psi \, dq = \int \left[\sum_a \frac{\hbar^2}{2m_a} (\nabla_a \psi)^2 + U\psi^2 \right] dq. \tag{20.5}$$

§ 21. General Properties of One-Dimensional Motion

When a particle's potential energy depends on just one coordinate x, the wave function may be sought as a product of a function of y, z and a function of x alone. The

[8] Proofs of the theorems about zeros of eigenfunctions (see also the next section) may be found in: *M. A. Lavrent'ev and L. A. Lyusternik*, Course of the Calculus of Variations, Moscow: Gostekhizdat, 1950, Ch. IX; *R. Courant and D. Hilbert*, Methods of Mathematical Physics, Moscow: Gostekhizdat, 1951, Vol. I, Ch. VI.

[9] This theorem (and the consequences drawn from it below) is not valid in general for wave functions of systems containing several identical particles; see the end of § 63.

first is determined by the free-motion Schrödinger equation, and the second by the one-dimensional Schrödinger equation

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}[E - U(x)]\psi = 0. \tag{21.1}$$

The problem of motion in a field with potential energy $U(x, y, z) = U_1(x)+U_2(y)+U_3(z)$, separated into a sum of functions each depending on only one coordinate, plainly reduces to the same kind of one-dimensional equations. In §§ 22–24 we shall consider several specific examples of such "one-dimensional" motion. First we establish some of its general properties.

We begin by showing that every discrete-spectrum energy level in a one-dimensional problem is nondegenerate. Suppose the contrary, and let ψ_1 and ψ_2 be two different eigenfunctions belonging to the same energy value. Since both satisfy the same equation (21.1),

$$\frac{\psi_1''}{\psi_1} = \frac{2m}{\hbar^2}(U - E) = \frac{\psi_2''}{\psi_2},$$

or $\psi_2\psi_1'' - \psi_1\psi_2'' = 0$ (a prime denotes differentiation with respect to x). Integration gives

$$\psi_1\psi_2' - \psi_2\psi_1' = \text{const}\,. \tag{21.2}$$

Since $\psi_1 = \psi_2 = 0$ at infinity, the constant must be zero, and hence

$$\psi_1\psi_2' - \psi_2\psi_1' = 0,$$

or $\psi_1'/\psi_1 = \psi_2'/\psi_2$. A second integration gives $\psi_1 = \text{const}\cdot\psi_2$; the two functions are therefore essentially identical.

For the discrete-spectrum wave functions $\psi_n(x)$, the following result, known as the *oscillation theorem*, can be stated: the function $\psi_n(x)$ corresponding to the $(n + 1)$th eigenvalue E_n in increasing order vanishes n times at finite x[10].

Suppose that $U(x)$ tends to finite limits as $x \to \pm\infty$ (it need not in any way be monotonic). Take $U(+\infty)$ as the zero of energy, so that $U(+\infty) = 0$, and denote $U(-\infty)$ by U_0, with $U_0 > 0$. The discrete spectrum lies in the range of energies for which the particle cannot escape to infinity. The energy must therefore be less than both limiting values and, in particular, negative:

$$E < 0. \tag{21.3}$$

Of course $E > U_{\min}$ must also hold in every case, so $U(x)$ must possess at least one minimum with $U_{\min} < 0$.

Now consider the positive energies below U_0:

$$0 < E < U_0. \tag{21.4}$$

Here the spectrum is continuous and the particle motion in the corresponding stationary states is infinite, with the particle escaping toward $x = +\infty$. Every energy eigenvalue in

[10]If the particle can occupy only a bounded interval of the x axis, the zeros of $\psi_n(x)$ inside that interval are to be counted.

this part of the spectrum is likewise nondegenerate. To see this, it suffices to observe that the proof given above for the discrete spectrum requires ψ_1 and ψ_2 to vanish at only one of the two infinities (here they vanish as $x \to -\infty$).

At sufficiently large positive x, $U(x)$ can be neglected in the Schrödinger equation (21.1):

$$\psi'' + \frac{2m}{\hbar^2}E\psi = 0.$$

This equation has real solutions in the form of a standing plane wave,

$$\psi = a\cos(kx + \delta), \tag{21.5}$$

where a and δ are constants, and the "wave vector" is $k = p/\hbar = \sqrt{2mE}/\hbar$. This formula gives the asymptotic form, as $x \to +\infty$, of the wave functions for the nondegenerate energy levels in the continuous-spectrum interval (21.4). At large negative x, the Schrödinger equation becomes

$$\psi'' - \frac{2m}{\hbar^2}(U_0 - E)\psi = 0.$$

The solution which does not become infinite as $x \to -\infty$ is

$$\psi = be^{\kappa x}, \qquad \kappa = \frac{1}{\hbar}\sqrt{2m(U_0 - E)}. \tag{21.6}$$

This is the asymptotic wave function as $x \to -\infty$. The wave function thus decays exponentially into the region in which $E < U_0$.

Finally, when

$$E > U_0, \tag{21.7}$$

the spectrum is continuous and the motion is infinite in both directions. Every level in this part of the spectrum is doubly degenerate. The reason is that the corresponding wave functions are determined by the second-order equation (21.1), and both independent solutions meet the required conditions at infinity (whereas, for example, in the preceding case one solution became infinite as $x \to -\infty$ and had to be rejected). As $x \to +\infty$, the asymptotic wave function is

$$\psi = a_1 e^{ikx} + a_2 e^{-ikx}, \tag{21.8}$$

and similarly as $x \to -\infty$. The term containing e^{ikx} corresponds to a particle moving to the right, and that containing e^{-ikx} to a particle moving to the left.

Suppose $U(x)$ is even, $U(-x) = U(x)$. The Schrödinger equation (21.1) is then unchanged on reversing the sign of the coordinate. Hence, if $\psi(x)$ is a solution, $\psi(-x)$ is also a solution and differs from $\psi(x)$ only by a constant factor: $\psi(-x) = c\psi(x)$. Reversing x once more gives $\psi(x) = c^2\psi(x)$, so $c = \pm 1$. Thus, for a potential energy symmetric about $x = 0$, stationary-state wave functions may be either even, $\psi(-x) = \psi(x)$, or odd, $\psi(-x) = -\psi(x)$[11]. In particular, the ground-state wave function is even: it cannot have a node, whereas an odd function necessarily vanishes at $x = 0$, since $\psi(0) = -\psi(0) = 0$.

[11] This argument assumes that the stationary state is nondegenerate, that is, that its motion is not infinite in both directions. Otherwise the two wave functions belonging to the given energy level may transform into one another when the sign of x is reversed. Even then, the stationary-state wave functions are not necessarily even or odd, but definite parity can always be achieved by forming suitable linear combinations of the two solutions.

There is a simple method for normalizing one-dimensional-motion wave functions in the continuous spectrum, which determines the normalization coefficient directly from the asymptotic wave function at large $|x|$.

Consider the wave function for motion infinite in one direction ($x \to +\infty$). The normalization integral diverges as $x \to \infty$ (at $x \to -\infty$ the function decays exponentially, so the integral converges rapidly). To determine the normalization constant, one may therefore replace ψ by its asymptotic value for large positive x and take any finite value of x, say zero, as the lower limit of integration; this is equivalent to discarding a finite contribution beside a divergent one. We now show that a wave function normalized by

$$\int \psi_p^* \psi_{p'}\, dx = \delta\left(\frac{p-p'}{2\pi\hbar}\right) = 2\pi\hbar\delta(p-p') \tag{21.9}$$

(where p is the particle momentum at infinity) must have the asymptotic form (21.5) with $a = 2$:

$$\psi_p \approx 2\cos(kx+\delta) = e^{i(kx+\delta)} + e^{-i(kx+\delta)}. \tag{21.10}$$

Since we are not seeking to verify mutual orthogonality of functions corresponding to different p, when substituting (21.10) into the normalization integral we take p and p' arbitrarily close and may put $\delta = \delta'$ (in general, δ is a function of p). We then retain in the integrand only the terms which diverge when $p = p'$; equivalently, we drop terms containing factors $e^{\pm i(k+k')x}$. We thus obtain

$$\int \psi_p^* \psi_{p'}\, dx = \int_0^\infty e^{i(k'-k)x}\, dx + \int_0^\infty e^{-i(k'-k)x}\, dx = \int_{-\infty}^\infty e^{i(k'-k)x}\, dx,$$

which, by (15.7), agrees with (21.9). According to (5.14), conversion to normalization by an energy delta function is effected by multiplying ψ_p by

$$\left(\frac{d(p/2\pi\hbar)}{dE}\right)^{1/2} = \frac{1}{\sqrt{2\pi\hbar v}},$$

where v is the particle's speed at infinity. Thus

$$\psi_E = \frac{1}{\sqrt{2\pi\hbar v}}\psi_p = \frac{1}{\sqrt{2\pi\hbar v}}\left(e^{i(kx+\delta)} + e^{-i(kx+\delta)}\right). \tag{21.11}$$

Notice that each of the two traveling waves into which the standing wave (21.11) separates has flux density $1/(2\pi\hbar)$.

We may therefore state the following rule for normalizing, to an energy delta function, the wave function of motion infinite in one direction: write its asymptotic expression as the sum of two plane waves traveling in opposite directions, and choose the normalization coefficient so that the flux density in the wave traveling toward the origin (or away from it) equals $1/(2\pi\hbar)$.

The analogous argument gives the corresponding rule for wave functions of motion infinite in both directions. Such a wave function is normalized to an energy delta function when the sum of the fluxes in the waves traveling toward the origin from the positive and negative sides of the x axis equals $1/(2\pi\hbar)$.

§ 22. The Potential Well

As a simple example of one-dimensional motion, consider motion in a rectangular *potential well*, that is, in a field with the function $U(x)$ shown in Fig. 1: $U(x) = 0$ for $0 < x < a$, and $U(x) = U_0$ for $x < 0$ and $x > a$. It is evident in advance that the spectrum is discrete when $E < U_0$, whereas for $E > U_0$ there is a continuous spectrum of doubly degenerate levels.

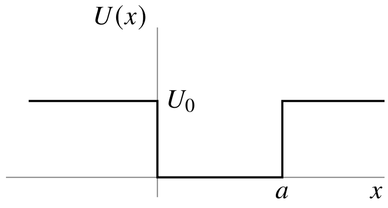

Fig. 1

In the region $0 < x < a$, the Schrödinger equation is

$$\psi'' + \frac{2m}{\hbar^2}E\psi = 0, \tag{22.1}$$

where a prime denotes differentiation with respect to x; outside the well it is

$$\psi'' + \frac{2m}{\hbar^2}(E - U_0)\psi = 0. \tag{22.2}$$

At $x = 0, a$, the solutions of these equations must pass continuously into one another, with continuous first derivative; at $x = \pm\infty$, the solution of (22.2) must remain finite (and for the discrete spectrum, $E < U_0$, it must vanish).

For $E < U_0$, the solution of (22.2) that vanishes at infinity is $\psi = \text{const} \cdot e^{\mp\kappa x}$, where

$$\kappa = \frac{1}{\hbar}\sqrt{2m(U_0 - E)} \tag{22.3}$$

(the minus and plus signs in the exponent refer respectively to the regions $x > a$ and $x < 0$). The probability $|\psi|^2$ of finding the particle decays exponentially into the region where $E < U(x)$. Instead of requiring ψ and ψ' to be continuous at the boundary of the potential well, it is convenient to require continuity of ψ and of the logarithmic derivative ψ'/ψ. With (22.3), the boundary condition becomes

$$|\psi'|/\psi = \mp\kappa. \tag{22.4}$$

Determining the energy levels for a well of arbitrary depth U_0 is left aside here (see Problem 2); we analyze in full only the limiting case of infinitely high walls ($U_0 \to \infty$).

When $U_0 = \infty$, motion occurs only on the interval bounded by $x = 0, a$ and, as stated in § 18, the boundary condition at these points is

$$\psi = 0. \tag{22.5}$$

(This condition also follows readily from the general condition (22.4). Indeed, when $U_0 \to \infty$, we also have $\kappa \to \infty$, and hence $\psi'/\psi \to \infty$; since ψ' cannot become infinite, it follows that $\psi = 0$.) Seek a solution of (22.1) inside the well in the form

$$\psi = c \sin(kx + \delta), \qquad k = \frac{\sqrt{2mE}}{\hbar}. \tag{22.6}$$

The condition $\psi = 0$ at $x = 0$ gives $\delta = 0$. The same condition at $x = a$ then gives $\sin ka = 0$, whence $ka = n\pi$ (n takes the positive integral values beginning with unity[12]), or

$$E_n = \frac{\pi^2\hbar^2}{2ma^2}n^2, \qquad n = 1, 2, 3, \ldots \tag{22.7}$$

The energy levels of a particle inside the potential well are thus determined. After normalization, the stationary-state wave functions are

$$\psi_n = \sqrt{\frac{2}{a}} \sin\left(\frac{\pi n}{a}x\right). \tag{22.8}$$

From these results one can write down directly the energy levels of a particle in a rectangular "potential box," that is, for three-dimensional motion in a field whose potential energy is $U = 0$ for $0 < x < a$, $0 < y < b$, $0 < z < c$, and $U = \infty$ outside this region. Namely, the levels are the sums

$$E_{n_1n_2n_3} = \frac{\pi^2\hbar^2}{2m}\left(\frac{n_1^2}{a^2} + \frac{n_2^2}{b^2} + \frac{n_3^2}{c^2}\right), \qquad n_1, n_2, n_3 = 1, 2, 3, \ldots, \tag{22.9}$$

and the corresponding wave functions are the products

$$\psi_{n_1n_2n_3} = \sqrt{\frac{8}{abc}} \sin\left(\frac{\pi n_1}{a}x\right) \sin\left(\frac{\pi n_2}{b}y\right) \sin\left(\frac{\pi n_3}{c}z\right). \tag{22.10}$$

Notice that, according to (22.7) or (22.9), the energy of the ground state is of order $E_0 \sim \hbar^2/ml^2$, where l denotes the linear dimensions of the region in which the particle moves. This result agrees with the uncertainty relations: for a coordinate uncertainty of order l, the momentum uncertainty, and with it the order of the momentum itself, is $\sim \hbar/l$; the corresponding energy is $\sim (\hbar/l)^2/m$.

Problem

1. Determine the probability distribution of the possible momentum values for the normal state of a particle in an infinitely deep rectangular potential well.

Solution. The coefficients $a(p)$ in the expansion of the function ψ_1 (22.8) in momentum eigenfunctions are

$$a(p) = \int \psi_p^* \psi_1 \, dx = \sqrt{\frac{2}{a}} \int_0^a \sin\left(\frac{\pi}{a}x\right) \exp\left(-i\frac{px}{\hbar}\right) dx.$$

[12] For $n = 0$, one would obtain identically $\psi = 0$.

Evaluating the integral and taking the square of its modulus gives the required probability distribution

$$|a(p)|^2 \frac{dp}{2\pi\hbar} = \frac{4\pi\hbar^3 a}{\left(p^2a^2 - \pi^2\hbar^2\right)^2} \cos^2\left(\frac{pa}{2\hbar}\right) dp.$$

PROBLEM

2. Determine the energy levels for the potential well shown in Fig. 2.

SOLUTION. The discrete part of the spectrum is the range $E < U_1$, which is the range we consider. In the region $x < 0$, the wave function is

$$\psi = c_1 e^{\kappa_1 x}, \qquad \kappa_1 = \frac{1}{\hbar}\sqrt{2m(U_1 - E)},$$

and in the region $x > a$ it is

$$\psi = c_2 e^{-\kappa_2 x}, \qquad \kappa_2 = \frac{1}{\hbar}\sqrt{2m(U_2 - E)}.$$

Inside the well ($0 < x < a$), seek ψ in the form

$$\psi = c\sin(kx + \delta), \qquad k = \sqrt{2mE}/\hbar.$$

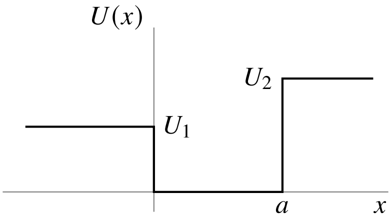

Fig. 2

Continuity of ψ'/ψ at the boundaries of the well gives

$$k\cot\delta = \kappa_1 = \sqrt{\frac{2m}{\hbar^2}U_1 - k^2},$$

$$k\cot(ak + \delta) = -\kappa_2 = -\sqrt{\frac{2m}{\hbar^2}U_2 - k^2},$$

or

$$\sin\delta = \frac{k\hbar}{\sqrt{2mU_1}}, \qquad \sin(ka + \delta) = -\frac{k\hbar}{\sqrt{2mU_2}}.$$

Eliminating δ, we obtain the transcendental equation

$$ka = n\pi - \arcsin\frac{k\hbar}{\sqrt{2mU_1}} - \arcsin\frac{k\hbar}{\sqrt{2mU_2}} \tag{1}$$

(where $n = 1, 2, 3, \ldots$, and the values of arcsin are taken between 0 and $\pi/2$), whose roots determine the energy levels $E = k^2\hbar^2/2m$. In general there is one root for each n; the values of n number the levels in increasing order.

Since the argument of arcsin cannot exceed unity, it is clear that the values of k can lie only in the interval from zero to $\sqrt{2mU_1}/\hbar$. The left-hand side of (1) is a monotonically increasing function of k, while the right-hand side is monotonically decreasing. Consequently, for a root of (1) to exist, its right-hand side at $k = \sqrt{2mU_1}/\hbar$ must be less than its left-hand side. In particular, the inequality

$$a\frac{\sqrt{2mU_1}}{\hbar} \geq \frac{\pi}{2} - \arcsin\sqrt{\frac{U_1}{U_2}}, \tag{2}$$

obtained for $n = 1$, is the condition for the well to contain at least one energy level. Thus, for given $U_1 \neq U_2$, there always exist sufficiently small values of the width a for which the well has no discrete energy level at all. When $U_1 = U_2$, condition (2) is plainly always satisfied.

When $U_1 = U_2 \equiv U_0$ (a symmetric well), equation (1) reduces to

$$\arcsin\frac{\hbar k}{\sqrt{2mU_0}} = \frac{n\pi - ka}{2}. \tag{3}$$

Introducing the variable $\xi = ka/2$, for odd n we obtain

$$\cos\xi = \pm\gamma\xi, \qquad \gamma = \frac{\hbar}{a}\sqrt{\frac{2}{mU_0}}, \tag{4}$$

where the roots for which $\tan\xi > 0$ must be selected. For even n we obtain

$$\sin\xi = \pm\gamma\xi, \tag{5}$$

where the roots for which $\tan\xi < 0$ must be selected. The energy levels are found from the roots of these two equations as $E = 2\xi^2\hbar^2/ma^2$; for $\gamma \neq 0$ only finitely many levels exist.

Consider as a special case a shallow well, in which $U_0 \ll \hbar^2/ma^2$, we have $\gamma \gg 1$, and equation (5) has no roots at all. Equation (4), however, has one root (with the upper sign on its right-hand side), equal to $\xi \approx (1/\gamma)(1 - 1/2\gamma^2)$. The well therefore contains only one energy level,

$$E_0 \approx U_0 - \frac{ma^2}{2\hbar^2}U_0^2,$$

lying close to its "top."

Problem

3. Determine the pressure exerted on the walls of a rectangular "potential box" by a particle inside it.

Solution. On a wall normal to the x axis, the force is given by the mean of the derivative $-\partial H/\partial a$ of the particle's Hamilton function with respect to the length of the box along the x axis; division of this force by the wall area bc gives the pressure. By (11.16), the required mean value is found by differentiating the energy eigenvalue (22.9). The resulting pressure is

$$p^{(x)} = \frac{\pi^2\hbar^2}{ma^3bc}n_1^2.$$

§ 23. The Linear Oscillator

Consider a particle executing small one-dimensional oscillations (the so-called *linear oscillator*). Its potential energy is $m\omega^2x^2/2$, where in classical mechanics ω is the natural frequency of the oscillations. The oscillator Hamiltonian is consequently

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{m\omega^2x^2}{2}. \tag{23.1}$$

Since the potential energy tends to infinity at $x = \pm\infty$, the particle can execute only finite motion. Accordingly, the entire energy spectrum of the oscillator is discrete.

Let us determine the oscillator's energy levels by the matrix method[13]. We begin with the equations of motion in the form (19.3), which in this case give

$$\hat{\ddot{x}} + \omega^2 x = 0. \tag{23.2}$$

In matrix form this equation is

$$(\ddot{x})_{mn} + \omega^2 x_{mn} = 0.$$

For the matrix elements of the acceleration, (11.8) gives $(\ddot{x})_{mn} = i\omega_{mn}(\dot{x})_{mn} = -\omega_{mn}^2 x_{mn}$. Hence

$$(\omega_{mn}^2 - \omega^2)x_{mn} = 0.$$

Thus all the matrix elements x_{mn} vanish except those for which $\omega_{mn} = \pm\omega$. Number all stationary states so that the frequencies $\pm\omega$ correspond to the transitions $n \to n \mp 1$, that is, $\omega_{n,n\mp1} = \pm\omega$. The only nonzero matrix elements are then $x_{n,n\mp1}$.

We shall suppose that the wave functions ψ_n have been chosen real. Since x is a real quantity, every matrix element x_{mn} is then real as well. The Hermiticity condition (11.10) now makes the matrix x_{mn} symmetric:

$$x_{mn} = x_{nm}.$$

To calculate the nonzero coordinate matrix elements, use the commutation rule

$$\hat{\dot{x}}\,\hat{x} - \hat{x}\,\hat{\dot{x}} = -i\frac{\hbar}{m},$$

written in matrix form as

$$(\dot{x}x)_{mn} - (x\dot{x})_{mn} = -\frac{i\hbar}{m}\delta_{mn}.$$

Using the matrix-multiplication rule (11.12), for $m = n$ this gives

$$i\sum_l(\omega_{nl}x_{nl}x_{ln} - x_{nl}\omega_{ln}x_{ln}) = 2i\sum_l \omega_{nl}x_{nl}^2 = -\frac{i\hbar}{m}.$$

Only the terms with $l = n \pm 1$ are nonzero in this sum, and therefore

$$(x_{n+1,n})^2 - (x_{n,n-1})^2 = \frac{\hbar}{2m\omega}. \tag{23.3}$$

[13]This was done by Heisenberg (1925), before Schrödinger's discovery of the wave equation.

This equality shows that the quantities $(x_{n+1,n})^2$ form an arithmetic progression, unbounded above but necessarily bounded below, since it can contain only positive terms. Thus far we have established only the relative ordering of the state labels n, not their absolute values; we may therefore choose arbitrarily the value of n assigned to the oscillator's first—the normal—state. Set it equal to zero. Correspondingly, $x_{0,-1}$ must be regarded as identically zero, and successive application of (23.3) for $n = 0, 1, \ldots$ gives

$$(x_{n,n-1})^2 = \frac{n\hbar}{2m\omega}.$$

The final expression for the nonzero coordinate matrix elements is therefore[14]

$$x_{n,n-1} = x_{n-1,n} = \sqrt{\frac{n\hbar}{2m\omega}}. \tag{23.4}$$

The matrix of $\hat{H}$ is diagonal, and its matrix elements H_{nn} are the required oscillator energy eigenvalues E_n. To calculate them, write

$$\begin{aligned} H_{nn} = E_n &= \frac{m}{2}\left[(\dot{x}^2)_{nn} + \omega^2 (x^2)_{nn}\right] \\ &= \frac{m}{2}\left[\sum_l i\omega_{nl}x_{nl}i\omega_{ln}x_{ln} + \omega^2 \sum_l x_{nl}x_{ln}\right] = \frac{m}{2}\sum_l (\omega^2 + \omega_{nl}^2)x_{ln}^2. \end{aligned}$$

Only the terms $l = n \pm 1$ are nonzero in the sum. Substitution of (23.4) gives

$$E_n = (n + 1/2)\hbar\omega, \qquad n = 0, 1, 2, \ldots \tag{23.5}$$

Thus the oscillator energy levels are separated by equal intervals $\hbar\omega$. The energy of the normal state ($n = 0$) is $\hbar\omega/2$; we emphasize that it is nonzero.

The result (23.5) may also be obtained by solving the Schrödinger equation. For the oscillator this equation is

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}\left(E - \frac{m\omega^2 x^2}{2}\right)\psi = 0. \tag{23.6}$$

It is convenient here to replace x by a dimensionless variable ξ, defined by

$$\xi = \sqrt{\frac{m\omega}{\hbar}}x. \tag{23.7}$$

We then obtain

$$\psi'' + \left(\frac{2E}{\hbar\omega} - \xi^2\right)\psi = 0. \tag{23.8}$$

(A prime here denotes differentiation with respect to ξ.)

For large ξ, the term $2E/\hbar\omega$ may be omitted in comparison with ξ^2; the equation $\psi'' = \xi^2\psi$ has the asymptotic integrals $\psi = e^{\pm\xi^2/2}$. (Indeed, differentiation of this

[14] We choose the undetermined phases α_n (see the note on p. 52) so that every matrix element in (23.4) has a plus sign before the root. A phase convention of this kind can always be arranged when a matrix has nonzero entries solely for transitions connecting states whose labels are adjacent.

function gives $\psi'' = \xi^2\psi$ when lower-order terms in ξ are neglected.) Since the wave function ψ must remain finite at $\xi = \pm\infty$, the minus sign must be chosen in the exponent. It is therefore natural to make the substitution

$$\psi = e^{-\xi^2/2}\chi(\xi) \tag{23.9}$$

in (23.8). For $\chi(\xi)$ we obtain the equation (introducing the notation $2E/\hbar\omega - 1 = 2n$; since we know in advance that $E > 0$, we have $n > -1/2$)

$$\chi'' - 2\xi\chi' + 2n\chi = 0, \tag{23.10}$$

where χ must be finite for every finite ξ and, at $\xi = \pm\infty$, may become infinite no faster than a finite power of ξ (so that ψ tends to zero).

Such solutions of (23.10) exist only for integer positive values of n, including zero (see § a of the mathematical supplements); this gives the energy eigenvalues (23.5), already known to us. For the different integral values of n, the corresponding solutions of (23.10) are

$$\chi = \text{const}\cdot H_n(\xi),$$

where $H_n(\xi)$ are the so-called Hermite polynomials, polynomials of degree n in ξ defined by

$$H_n(\xi) = (-1)^n e^{\xi^2}\frac{d^n e^{-\xi^2}}{d\xi^n}. \tag{23.11}$$

Choosing the constant so that ψ_n obeys the normalization condition

$$\int_{-\infty}^{+\infty} \psi_n^2(x)\,dx = 1,$$

we obtain (see (a.7))

$$\psi_n(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4}\frac{1}{\sqrt{2^n n!}}\exp\left(-\frac{m\omega}{2\hbar}x^2\right)H_n\left(x\sqrt{\frac{m\omega}{\hbar}}\right). \tag{23.12}$$

Thus, the normal-state wave function is

$$\psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4}\exp\left(-\frac{m\omega}{2\hbar}x^2\right). \tag{23.13}$$

As expected, it has no zeros at finite x.

Evaluation of the integrals $\int_{-\infty}^{+\infty}\psi_n\psi_m\xi\,d\xi$ determines the coordinate matrix elements; the calculation naturally gives the same values (23.4).

We conclude by demonstrating how the wave functions ψ_n can be computed via the matrix method. In the matrices of the operators $\hat{\dot{x}} \pm i\omega\hat{x}$, the only nonzero elements are

$$(\dot{x} - i\omega x)_{n-1,n} = -(\dot{x} + i\omega x)_{n,n-1} = -i\sqrt{\frac{2\omega\hbar n}{m}}. \tag{23.14}$$

Using the general formula (11.11) and recalling that $\psi_{-1} \equiv 0$, we obtain

$$(\hat{\dot{x}} - i\omega x)\psi_0 = 0.$$

Substitution of $\hat{\dot{x}} = -i(\hbar/m)d/dx$ then gives

$$\frac{d\psi_0}{dx} = -\frac{m\omega}{\hbar}x\psi_0,$$

whose normalized solution is (23.13). Further, since

$$(\hat{\dot{x}} + i\omega\hat{x})\psi_{n-1} = (\dot{x} + i\omega x)_{n,n-1}\psi_n = i\sqrt{\frac{2\omega\hbar n}{m}}\psi_n,$$

we obtain the recurrence formula

$$\begin{aligned}\psi_n &= \sqrt{\frac{m}{2\omega\hbar n}}\left(-\frac{\hbar}{m}\frac{d}{dx} + \omega x\right)\psi_{n-1} = \frac{1}{\sqrt{2n}}\left(-\frac{d}{d\xi} + \xi\right)\psi_{n-1} \\ &= -\frac{1}{\sqrt{2n}}\exp\frac{\xi^2}{2}\frac{d}{d\xi}\left(\exp\left(-\frac{\xi^2}{2}\right)\psi_{n-1}\right),\end{aligned}$$

whose n-fold application to (23.13) gives (23.12) for the normalized functions ψ_n.

Problem

1. Determine the probability distribution of the possible momentum values for the oscillator.

Solution. Rather than expanding a stationary-state wave function in momentum eigenfunctions, for the oscillator it is simpler to begin directly with the Schrödinger equation in the momentum representation. Substituting the coordinate operator (15.12), $\hat{x} = i\hbar d/dp$, into (23.1), we obtain the Hamiltonian in the momentum representation

$$\hat{H} = \frac{p^2}{2m} - \frac{m\omega^2\hbar^2}{2}\frac{d^2}{dp^2}.$$

The corresponding Schrödinger equation $\hat{H}a(p) = Ea(p)$ for the momentum-representation wave function $a(p)$ is

$$\frac{d^2a(p)}{dp^2} + \frac{2}{m\omega^2\hbar^2}\left(E - \frac{p^2}{2m}\right)a(p) = 0.$$

This equation has exactly the same form as (23.6), so its solutions can be written down directly by analogy with (23.12). The required probability distribution is therefore

$$|a_n(p)|^2\frac{dp}{2\pi\hbar} = \frac{1}{2^n n!\sqrt{\pi m\omega\hbar}}\exp\left(-\frac{p^2}{m\omega\hbar}\right)H_n^2\left(\frac{p}{\sqrt{m\omega\hbar}}\right)dp.$$

Problem

2. Use the uncertainty relation (16.7) to determine a lower bound on the possible energy values of the oscillator.

Solution. Noting that $\overline{x^2} = \bar{x}^2 + (\delta x)^2$ and $\overline{p^2} = \bar{p}^2 + (\delta p)^2$, and using (16.7), for the oscillator's mean energy we have

$$\bar{E} = \frac{m\omega^2}{2}\overline{x^2} + \frac{\overline{p^2}}{2m} \geq \frac{m\omega^2}{2}(\delta x)^2 + \frac{1}{2m}(\delta p)^2 \geq \frac{m\omega^2\hbar^2}{8(\delta p)^2} + \frac{(\delta p)^2}{2m}.$$

Finding the minimum of this expression as a function of δp, we obtain the lower bound for the mean values, and hence for all possible energy values: $E \geq \hbar\omega/2$.

Problem

3. Find the wave functions of linear-oscillator states that minimize the uncertainty relation, that is, states in which the root-mean-square fluctuations of coordinate and momentum in the wave packet are related by $\delta p\, \delta x = \hbar/2$ (E. Schrödinger, 1926)[15].

Solution. The required wave functions must have the form

$$\Psi(x,t) = \frac{1}{(2\pi)^{1/4}(\delta x)^{1/2}} \exp\left\{\frac{i\bar{p}x}{\hbar} - \frac{(x-\bar{x})^2}{4(\delta x)^2} - i\varphi(t)\right\}. \tag{1}$$

At every instant their coordinate dependence corresponds to (16.8), where $\bar{x} = \bar{x}(t)$ and $\bar{p} = \bar{p}(t) = m\dot{\bar{x}}(t)$ are the mean coordinate and momentum. By (19.3), for the linear oscillator ($U = m\omega^2x^2/2$) we have $\dot{p} = -m\omega^2 x$, and hence for the mean values $\dot{\bar{p}} = -m\omega^2\bar{x}$, or

$$\ddot{\bar{x}} + \omega^2\bar{x} = 0, \tag{2}$$

that is, $\bar{x}(t)$ satisfies the classical equation of motion. The constant factor in (1) is fixed by the normalization condition $\int_{-\infty}^{\infty} |\Psi|^2 dx = 1$; besides this factor, Ψ may contain a phase factor with a time-dependent phase $\varphi(t)$. To find the unknown δx and the function $\varphi(t)$, we substitute (1) into the wave equation

$$-\frac{\hbar^2}{2m}\frac{\partial^2\Psi}{\partial x^2} + \frac{m\omega^2x^2}{2}\Psi = i\hbar\frac{\partial\Psi}{\partial t}.$$

Taking account of (2), the substitution gives

$$\left(\frac{x^2}{2} - x\bar{x}\right)\left(\frac{m^2\omega^2}{\hbar^2} - \frac{1}{4(\delta x)^4}\right) + \left[\frac{m^2\dot{\bar{x}}^2}{2\hbar^2} - \frac{\bar{x}^2}{8(\delta x)^4} + \frac{1}{4(\delta x)^2} - \frac{m}{\hbar}\dot{\varphi}(t)\right] = 0.$$

It follows that $(\delta x)^2 = \hbar/(2m\omega)$ and then

$$\dot{\varphi} = \frac{m}{2\hbar}(\dot{\bar{x}}^2 - \omega^2\bar{x}^2) + \frac{\omega}{2}, \qquad \varphi = \frac{1}{2\hbar}\bar{p}\bar{x} + \frac{\omega}{2}t.$$

[15] These states are called *coherent*.

Thus, finally,

$$\Psi(x,t) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \exp\left\{\frac{i\bar{p}x}{\hbar} - \frac{m\omega(x-\bar{x})^2}{2\hbar}\right\} \exp\left\{-\frac{i\omega t}{2} - \frac{i\bar{p}\bar{x}}{2\hbar}\right\}. \tag{3}$$

For $\bar{x} = 0$, $\bar{p} = 0$, this function becomes $\psi_0(x)e^{-i\omega t/2}$, the wave function of the oscillator ground state.

The oscillator's mean energy in a coherent state is

$$\bar{E} = \frac{\overline{p^2}}{2m} + \frac{m\omega^2\overline{x^2}}{2} = \frac{\bar{p}^2}{2m} + \frac{m\omega^2\bar{x}^2}{2} + \frac{\hbar\omega}{2} = \hbar\omega\left(\bar{n} + \frac{1}{2}\right); \tag{4}$$

the quantity $\bar{n}$ introduced here is the mean "number of quanta" $\hbar\omega$ in the state. We see that a coherent state is specified completely by giving a dependence $\bar{x}(t)$ that satisfies the classical equation (2). Its general form can be written

$$\frac{m\omega\bar{x} + i\bar{p}}{\sqrt{2m\hbar\omega}} = ae^{-i\omega t}, \qquad |a|^2 = \bar{n}. \tag{5}$$

The function (3) may be expanded in the wave functions of the oscillator's stationary states:

$$\Psi = \sum_{n=0}^{\infty} a_n\Psi_n, \qquad \Psi_n(x,t) = \psi_n(x)\exp\left\{-i\left(n+\frac{1}{2}\right)\omega t\right\}.$$

The coefficients of this expansion are[16]

$$a_n = \int_{-\infty}^{\infty} \Psi_n^*\Psi\, dx. \tag{6}$$

It follows that the probability for the oscillator to be in its nth state is

$$w_n = |a_n|^2 = e^{-\bar{n}}\frac{\bar{n}^n}{n!}, \tag{7}$$

that is, it is given by the familiar Poisson distribution.

Problem

4. Determine the energy levels of a particle moving in a field with the potential energy shown in Fig. 3 (Ph. Morse, 1929).

Solution. The spectrum of positive energy eigenvalues is continuous (and the levels are nondegenerate), while the spectrum of negative values is discrete.

The Schrödinger equation is

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}(E - Ae^{-2\alpha x} + 2Ae^{-\alpha x})\psi = 0.$$

Introduce the new variable

$$\xi = \frac{2\sqrt{2mA}}{\alpha\hbar}e^{-\alpha x}$$

[16] Compare the calculation in Problem 1 of § 41.

(which ranges from zero to $+\infty$), and the notation (we consider the discrete spectrum, so $E < 0$)

$$s = \frac{\sqrt{-2mE}}{\alpha\hbar}, \qquad n = \frac{\sqrt{2mA}}{\alpha\hbar} - \left(s + \frac{1}{2}\right). \tag{1}$$

The Schrödinger equation then becomes

$$\psi'' + \frac{1}{\xi}\psi' + \left(-\frac{1}{4} + \frac{n + s + 1/2}{\xi} - \frac{s^2}{\xi^2}\right)\psi = 0.$$

As $\xi \to \infty$, ψ behaves asymptotically as $e^{\pm\xi/2}$, while as $\xi \to 0$ it is proportional to $\xi^{\pm s}$. Finiteness requires the solution behaving as $e^{-\xi/2}$ at $\xi \to \infty$ and as ξ^s at $\xi \to 0$.

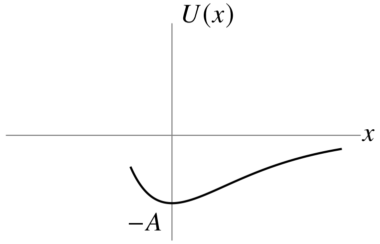

Fig. 3

Make the substitution

$$\psi = e^{-\xi/2}\xi^s w(\xi),$$

which gives for w the equation

$$\xi w'' + (2s + 1 - \xi)w' + nw = 0, \tag{2}$$

to be solved subject to the conditions that w is finite at $\xi = 0$, and that as $\xi \to \infty$ it becomes infinite no faster than a finite power of ξ. Equation (2) is the confluent hypergeometric equation (see § d of the mathematical supplements),

$$w = F(-n, 2s + 1, \xi).$$

The solution satisfying the required condition is obtained for a nonnegative integer n (and F then reduces to a polynomial). From the definitions (1), the energy levels are consequently

$$-E_n = A\left[1 - \frac{\alpha\hbar}{\sqrt{2mA}}\left(n + \frac{1}{2}\right)\right]^2,$$

where n takes integer positive values beginning at zero and ending at the greatest value for which

$$\frac{\sqrt{2mA}}{\alpha\hbar} > n + \frac{1}{2}$$

(so that the parameter s is positive, in agreement with its definition). Thus the discrete spectrum contains a finite sequence of levels. If

$$\frac{\sqrt{2mA}}{\alpha\hbar} < \frac{1}{2},$$

there is no discrete spectrum at all.

Problem

5. The same problem for $U = -U_0/\cosh^2 \alpha x$ (Fig. 4).

Solution. The positive-energy spectrum is continuous and the negative-energy spectrum is discrete; we consider the latter. The Schrödinger equation is

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}\left(E + \frac{U_0}{\cosh^2 \alpha x}\right)\psi = 0.$$

Make the change of variable $\xi = \tanh \alpha x$ and introduce the notation

$$\varepsilon = \frac{\sqrt{-2mE}}{\hbar\alpha}, \qquad \frac{2mU_0}{\alpha^2\hbar^2} = s(s+1), \qquad s = \frac{1}{2}\left(-1 + \sqrt{1 + \frac{8mU_0}{\alpha^2\hbar^2}}\right).$$

We then obtain

$$\frac{d}{d\xi}\left[(1-\xi^2)\frac{d\psi}{d\xi}\right] + \left[s(s+1) - \frac{\varepsilon^2}{1-\xi^2}\right]\psi = 0.$$

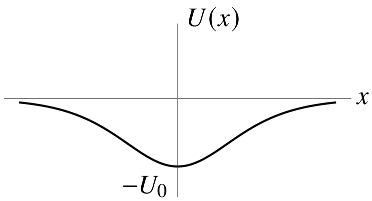

Fig. 4

This is the equation of the generalized Legendre functions. Reduce it to hypergeometric form by the substitution

$$\psi = (1-\xi^2)^{\varepsilon/2} w(\xi)$$

and the temporary change of variable $\frac{1}{2}(1-\xi) = u$:

$$u(1-u)w'' + (\varepsilon+1)(1-2u)w' - (\varepsilon - s)(\varepsilon + s + 1)w = 0.$$

The solution finite at $\xi = 1$ (that is, at $x = \infty$) is

$$\psi = (1-\xi^2)^{\varepsilon/2} F[\varepsilon - s, \varepsilon + s + 1, \varepsilon + 1, (1-\xi)/2].$$

For ψ to remain finite also at $\xi = -1$ (that is, at $x = -\infty$), we must have $\varepsilon - s = -n$, where $n = 0, 1, 2, \ldots$ (then F is a polynomial of degree n, finite at $\xi = -1$).

Thus the energy levels are fixed by $s - \varepsilon = n$, which gives

$$E_n = -\frac{\hbar^2\alpha^2}{8m}\left[-(1+2n) + \sqrt{1 + \frac{8mU_0}{\alpha^2\hbar^2}}\right]^2.$$

There is a finite number of levels, determined by the condition $\varepsilon > 0$, that is, $n < s$.

§ 24. Motion in a Uniform Field

Consider the motion of a particle in a uniform external field. Choose the direction of the field as the x axis, and let F be the force exerted by the field on the particle; in an electric field of strength E, this force is $F = eE$, where e is the particle's charge.

The potential energy of the particle in a uniform field has the form $U = -Fx + \text{const}$. Choosing the constant so that $U = 0$ at $x = 0$, we have $U = -Fx$. The Schrödinger equation for the problem is

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}(E + Fx)\psi = 0. \tag{24.1}$$

Since U tends to $+\infty$ as $x \to -\infty$ and to $-\infty$ as $x \to \infty$, it is evident in advance that the energy levels form a continuous spectrum filling the entire interval from $-\infty$ to $+\infty$. All these eigenvalues are nondegenerate and correspond to motion that is finite on the side $x = -\infty$ and infinite in the direction $x \to +\infty$.

In place of the coordinate x, introduce the dimensionless variable

$$\xi = \left(x + \frac{E}{F}\right)\left(\frac{2mF}{\hbar^2}\right)^{1/3}. \tag{24.2}$$

Equation (24.1) then becomes

$$\psi'' + \xi\psi = 0. \tag{24.3}$$

This equation contains no energy parameter at all. Consequently, once a solution satisfying the required finiteness conditions has been found, it provides an eigenfunction for every value of the energy.

The solution of (24.3) that is finite for all x is (see § b of the mathematical supplements)

$$\psi(\xi) = A\Phi(-\xi), \tag{24.4}$$

where

$$\Phi(\xi) = \frac{1}{\sqrt{\pi}}\int_0^\infty \cos\left(\frac{u^3}{3} + u\xi\right) du$$

is the so-called Airy function, and A is a normalization factor to be determined below.

As $\xi \to -\infty$, the function $\psi(\xi)$ tends exponentially to zero. The asymptotic expression determining $\psi(\xi)$ for negative ξ of large absolute magnitude is (see (b.4))

$$\psi(\xi) \approx \frac{A}{2|\xi|^{1/4}} \exp\left(-\frac{2}{3}|\xi|^{3/2}\right). \tag{24.5}$$

For large positive ξ, the asymptotic expression for $\psi(\xi)$ is instead (see (b.5)[17]):

$$\psi(\xi) = \frac{A}{\xi^{1/4}} \sin\left(\frac{2}{3}\xi^{3/2} + \frac{\pi}{4}\right). \tag{24.6}$$

[17] We note in anticipation that the asymptotic expressions (24.5) and (24.6) correspond precisely to the quasiclassical forms of the wave function in classically forbidden and allowed regions, respectively (§ 47).

Following the general rule (5.4) for normalizing eigenfunctions of a continuous spectrum, put the functions (24.4) into a form normalized to a delta function of the energy:

$$\int_{-\infty}^{+\infty} \psi(\xi)\psi(\xi')\,dx = \delta(E'-E). \tag{24.7}$$

In § 21 a simple method was given for determining the normalization coefficient from the asymptotic form of the wave functions. Following that method, write (24.6) as a sum of two traveling waves:

$$\psi(\xi) \approx \frac{A}{2\xi^{1/4}}\left\{\exp\left[i\left(\frac{2}{3}\xi^{3/2}-\frac{\pi}{4}\right)\right]+\exp\left[-i\left(\frac{2}{3}\xi^{3/2}-\frac{\pi}{4}\right)\right]\right\}.$$

The flux density calculated for each of these two terms is

$$v\left(\frac{A}{2\xi^{1/4}}\right)^2 = \sqrt{\frac{2}{m}(E+Fx)}\left(\frac{A}{2\xi^{1/4}}\right)^2 = A^2\frac{(2\hbar F)^{1/3}}{4m^{2/3}}.$$

Equating it to $1/(2\pi\hbar)$, we find

$$A = \frac{(2m)^{1/3}}{\pi^{1/2}F^{1/6}\hbar^{2/3}}. \tag{24.8}$$

Problem

Determine the momentum-representation wave functions for a particle in a uniform field.

Solution. The Hamiltonian in the momentum representation is

$$\widehat{H} = \frac{p^2}{2m} - i\hbar F\frac{d}{dp},$$

so the Schrödinger equation for the wave function $a(p)$ is

$$-i\hbar F\frac{da}{dp} + \left(\frac{p^2}{2m}-E\right)a = 0.$$

Solving this equation gives the required functions

$$a_E(p) = \frac{1}{\sqrt{2\pi\hbar F}}\exp\left\{\frac{i}{\hbar F}\left(Ep-\frac{p^3}{6m}\right)\right\}.$$

These functions are normalized by the condition

$$\int_{-\infty}^{+\infty} a_E^*(p)a_{E'}(p)\,dp = \delta(E'-E).$$

§ 25. Transmission Coefficient

We consider particle motion in a field of the type depicted in Fig. 5: $U(x)$ increases monotonically from one constant limiting value ($U = 0$ as $x \to -\infty$) to another ($U = U_0$ as $x \to +\infty$). According to classical mechanics, a particle with energy $E < U_0$ moving in such a field from left to right arrives at the *potential wall*, reflects off it, and proceeds in the reverse direction; for $E > U_0$, however, the particle keeps traveling in its original direction with a reduced velocity. Quantum mechanics introduces a new phenomenon: even when $E > U_0$, the particle may be reflected from the potential wall. In principle, the reflection probability is to be calculated as follows.

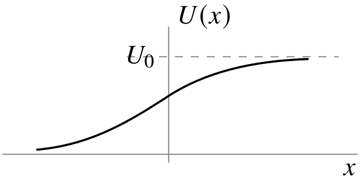

Figure 5

Let the particle move from left to right. At large positive values of x, the wave function must describe a particle that has passed "over the wall" and is moving in the positive direction of the x axis; that is, it must have the asymptotic form

$$\text{as } x \to \infty: \qquad \psi \approx Ae^{ik_2 x}, \qquad k_2 = \frac{1}{\hbar}\sqrt{2m(E-U_0)} \tag{25.1}$$

(A is a constant). Having found a solution of the Schrödinger equation that satisfies this limiting condition, we calculate its asymptotic expression as $x \to -\infty$. This is a linear combination of two solutions of the free-motion equation and therefore has the form

$$\text{as } x \to -\infty: \qquad \psi \approx e^{ik_1 x} + Be^{-ik_1 x}, \qquad k_1 = \frac{1}{\hbar}\sqrt{2mE}. \tag{25.2}$$

The first term describes the particle impinging on the wall (ψ is taken normalized so that this term has unit coefficient); the second term depicts the particle reflected from the wall. The flux density is proportional to k_1 in the incident wave, to $k_1|B|^2$ in the reflected wave, and to $k_2|A|^2$ in the transmitted wave. Define the particle's *transmission coefficient* D as the ratio of the flux density in the transmitted wave to that in the incident wave:

$$D = \frac{k_2}{k_1}|A|^2. \tag{25.3}$$

Similarly, the *reflection coefficient* R may be defined as the ratio of the reflected flux density to the incident flux density; evidently $R = 1 - D$:

$$R = |B|^2 = 1 - \frac{k_2}{k_1}|A|^2 \tag{25.4}$$

(this relation between A and B follows automatically from the constancy of the flux along the x axis).

If a particle moves from left to right with energy $E < U_0$, then k_2 is purely imaginary and the wave function decreases exponentially as $x \to +\infty$. The reflected flux equals the incident flux; that is, the particle is totally reflected from the potential wall. We stress, however, that the probability for the particle to be located in the region $E < U$ remains nonzero even in this case, though it decreases rapidly with growing x.

For the general case of an arbitrary stationary state (energy $E > U_0$), the asymptotic form of the wave function both as $x \to -\infty$ and as $x \to +\infty$ is a sum of two waves propagating in the two directions along the x axis:

$$\begin{aligned} \psi &= A_1 e^{ik_1x} + B_1 e^{-ik_1x} \quad \text{as } x \to -\infty, \\ \psi &= A_2 e^{ik_2x} + B_2 e^{-ik_2x} \quad \text{as } x \to +\infty. \end{aligned} \tag{25.5}$$

Since both expressions are asymptotic forms of one and the same solution of a linear differential equation, the coefficients A_1, B_1 and A_2, B_2 are linearly related. Let $A_2 = \alpha A_1 + \beta B_1$, where α and β are constants (in general complex) that depend on the particular form of the field $U(x)$. The analogous relation for B_2 may then be written by using the fact that the Schrödinger equation is real. By this fact, if ψ is a solution of the given Schrödinger equation, the complex-conjugate function ψ^* is also a solution of the same equation.

The asymptotic form

$$\begin{aligned} \psi^* &= A_1^* e^{-ik_1x} + B_1^* e^{ik_1x} \quad \text{as } x \to -\infty, \\ \psi^* &= A_2^* e^{-ik_2x} + B_2^* e^{ik_2x} \quad \text{as } x \to +\infty \end{aligned}$$

differs from (25.5) only in the notation of the constant coefficients; hence $B_2^* = \alpha B_1^* + \beta A_1^*$, or $B_2 = \alpha^* B_1 + \beta^* A_1$. Thus the coefficients in (25.5) are related by equations of the form

$$A_2 = \alpha A_1 + \beta B_1, \qquad B_2 = \beta^* A_1 + \alpha^* B_1. \tag{25.6}$$

The condition that the flux be constant along the x axis gives the following relation for the coefficients in (25.5):

$$k_1\big(|A_1|^2 - |B_1|^2\big) = k_2\big(|A_2|^2 - |B_2|^2\big).$$

On expressing A_2 and B_2 here in terms of A_1 and B_1 according to (25.6), we obtain

$$|\alpha|^2 - |\beta|^2 = \frac{k_1}{k_2}. \tag{25.7}$$

The relations (25.6) can be used to show that the reflection coefficients are identical (at a given energy $E > U_0$) for particles propagating in the positive or negative direction along the x axis. Indeed, the first case is obtained by setting $B_2 = 0$ in the functions (25.5); then $B_1/A_1 = -\beta^*/\alpha^*$. In the second case we put $A_1 = 0$, whereupon $A_2/B_2 = \beta/\alpha^*$. The corresponding reflection coefficients are

$$R_1 = \left|\frac{B_1}{A_1}\right|^2 = \left|\frac{\beta^*}{\alpha^*}\right|^2, \qquad R_2 = \left|\frac{A_2}{B_2}\right|^2 = \left|\frac{\beta}{\alpha^*}\right|^2,$$

and it is clear that $R_1 = R_2$.

The quantities $B_1/A_1 = -\beta^*/\alpha^*$ and $A_2/B_2 = \beta/\alpha^*$ are naturally called the *reflection amplitudes* for motion in the positive and negative directions, respectively. These amplitudes have equal moduli but may differ by a phase factor.

PROBLEM

1. For a particle with energy $E > U_0$, calculate the coefficient of reflection from a rectangular potential wall (Fig. 6).

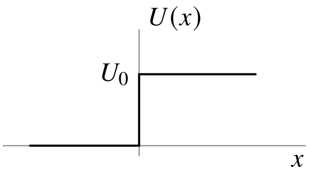

Figure 6

SOLUTION. Throughout the region $x > 0$ the wave function has the form (25.1), and in the region $x < 0$ it has the form (25.2). The constants A and B are determined by the conditions that ψ and $d\psi/dx$ be continuous at $x = 0$:

$$1 + B = A, \qquad k_1(1 - B) = k_2 A,$$

whence

$$A = \frac{2k_1}{k_1 + k_2}, \qquad B = \frac{k_1 - k_2}{k_1 + k_2}.$$

The reflection coefficient is (25.4) [18]

$$R = \left(\frac{k_1 - k_2}{k_1 + k_2}\right)^2 = \left(\frac{p_1 - p_2}{p_1 + p_2}\right)^2.$$

At $E = U_0$ ($k_2 = 0$), R becomes unity; as $E \to \infty$, it tends to zero as $R = (U_0/4E)^2$.

PROBLEM

2. Determine the transmission coefficient of a particle through a rectangular potential barrier (Fig. 7).

SOLUTION. Let $E > U_0$, and let the incident particle move from left to right. The wave function then has the following forms in the different regions:

$$\begin{aligned}
\psi &= e^{ik_1x} + Ae^{-ik_1x} && \text{for } x < 0,\\
\psi &= Be^{ik_2x} + B'e^{-ik_2x} && \text{for } 0 < x < a,\\
\psi &= Ce^{ik_1x} && \text{for } x > a.
\end{aligned}$$

(On the side $x > a$ there must be only the transmitted wave, propagating in the positive direction of the x axis.) The constants A, B, B', and C are determined by the conditions

[18] In the limiting case of classical mechanics the reflection coefficient must go to zero. However, the resulting expression does not contain a quantum constant at all. This seeming contradiction is resolved as follows. The classical limit corresponds to the situation where the particle's de Broglie wavelength $\lambda \sim \hbar/p$ is small in comparison with the characteristic dimensions of the problem, that is, with the distances over which the field $U(x)$ changes appreciably. In the schematic example under consideration this distance is zero (at $x = 0$), so that the limiting transition cannot be made.

that ψ and $d\psi/dx$ be continuous at the points $x = 0, a$. The transmission coefficient is $D = k_1|C|^2/k_1 = |C|^2$. Calculation gives

$$D = \frac{4k_1^2k_2^2}{(k_1^2 - k_2^2)^2 \sin^2 ak_2 + 4k_1^2k_2^2}.$$

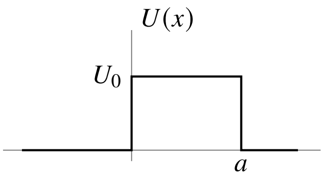

Figure 7

When $E < U_0$, k_2 is purely imaginary; the corresponding expression for D is obtained by replacing k_2 by $i\kappa_2$, where $\hbar\kappa_2 = \sqrt{2m(U_0 - E)}$:

$$D = \frac{4k_1^2\kappa_2^2}{(k_1^2 + \kappa_2^2)^2 \sinh^2(a\kappa_2) + 4k_1^2\kappa_2^2}.$$

PROBLEM

3. Determine the reflection coefficient of a particle from the potential wall defined by

$$U(x) = U_0/(1 + e^{-\alpha x})$$

(see Fig. 5); the particle energy is $E > U_0$.

SOLUTION. The Schrödinger equation is

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}\left(E - \frac{U_0}{1 + e^{-\alpha x}}\right)\psi = 0.$$

We must find the solution that, as $x \to +\infty$, has the form

$$\psi = \text{const} \cdot e^{ik_2x}.$$

Introduce the new variable

$$\xi = -e^{-\alpha x}$$

(which ranges from $-\infty$ to 0), and seek a solution in the form

$$\psi = \xi^{-ik_2/\alpha}w(\xi),$$

where $w(\xi)$ tends to a constant as $\xi \to 0$ (that is, as $x \to \infty$). For $w(\xi)$ we obtain an equation of hypergeometric type,

$$\xi(1 - \xi)w'' + \left(1 - \frac{2i}{\alpha}k_2\right)(1 - \xi)w' + \frac{1}{\alpha^2}(k_2^2 - k_1^2)w = 0,$$

whose solution is the hypergeometric function

$$w = F\left[-\frac{i}{\alpha}(k_1 - k_2), -\frac{i}{\alpha}(k_1 + k_2), -\frac{2i}{\alpha}k_2 + 1, \xi\right]$$

(a constant factor is omitted). As $\xi \to 0$, this function tends to 1 and thus satisfies the stated condition.

The asymptotic form of ψ as $\xi \to -\infty$ (that is, as $x \to -\infty$) is [19]

$$\begin{aligned}\psi &\approx \xi^{-ik_2/\alpha}\left[C_1(-\xi)^{i(k_2-k_1)/\alpha} + C_2(-\xi)^{i(k_1+k_2)/\alpha}\right]\\ &= (-1)^{ik_2/\alpha}\left[C_1 e^{ik_1x} + C_2 e^{-ik_1x}\right],\end{aligned}$$

where

$$C_1 = \frac{\Gamma(-(2i/\alpha)k_1)\Gamma(-(2i/\alpha)k_2 + 1)}{\Gamma(-(i/\alpha)(k_1 + k_2))\Gamma(-(i/\alpha)(k_1 + k_2) + 1)},$$

$$C_2 = \frac{\Gamma((2i/\alpha)k_1)\Gamma(-(2i/\alpha)k_2 + 1)}{\Gamma((i/\alpha)(k_1 - k_2))\Gamma((i/\alpha)(k_1 - k_2) + 1)}.$$

The required reflection coefficient is $R = |C_2/C_1|^2$; calculation using the familiar formula

$$\Gamma(x)\Gamma(1-x) = \frac{\pi}{\sin \pi x}$$

gives

$$R = \left\{\frac{\sinh[(\pi/\alpha)(k_1 - k_2)]}{\sinh[(\pi/\alpha)(k_1 + k_2)]}\right\}^2.$$

At $E = U_0$ ($k_2 = 0$), R becomes unity, while as $E \to \infty$ it tends to zero according to

$$R = \left(\frac{\pi U_0}{\alpha\hbar}\right)^2 \frac{2m}{E} \exp\left(-\frac{4\pi}{\alpha\hbar}\sqrt{2mE}\right).$$

In the limiting transition to classical mechanics, R vanishes, as it should.

Problem

4. Determine the transmission coefficient of a particle through the potential barrier defined by

$$U(x) = \frac{U_0}{\cosh^2(\alpha x)}$$

(Fig. 8); the particle energy is $E < U_0$.

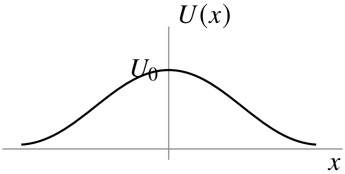

Figure 8

[19] See formula (e.6), in each of whose two terms only the first term of the expansion is to be retained; that is, the hypergeometric functions of $1/z$ are to be replaced by unity.

Solution. The Schrödinger equation for this problem is obtained from the example considered in Problem 5 of § 23 by changing the sign of U_0, with the energy E now taken to be positive.

The same method gives the solution

$$\psi = (1-\xi^2)^{-ik/2\alpha} F\left(-\frac{ik}{\alpha} - s, -\frac{ik}{\alpha} + s + 1, -\frac{ik}{\alpha} + 1, \frac{1-\xi}{2}\right), \tag{1}$$

where

$$\xi = \tanh(\alpha x), \qquad k = \frac{1}{\hbar}\sqrt{2mE}, \qquad s = \frac{1}{2}\left(-1 + \sqrt{1 - \frac{8mU_0}{\alpha^2\hbar^2}}\right).$$

This solution already satisfies the condition that as $x \to \infty$ (that is, as $\xi \to 1$, $(1-\xi) \approx 2e^{-2\alpha x}$) the wave function contain only the transmitted wave ($\sim e^{ikx}$). The asymptotic expression for the wave function as $x \to -\infty$ ($\xi \to -1$) is obtained by transforming the hypergeometric function via formula (e.7):

$$\psi \sim e^{-ikx} \frac{\Gamma(ik/\alpha)\Gamma(1-ik/\alpha)}{\Gamma(-s)\Gamma(1+s)} + e^{ikx} \frac{\Gamma(-ik/\alpha)\Gamma(1-ik/\alpha)}{\Gamma(-ik/\alpha - s)\Gamma(-ik/\alpha + s + 1)}. \tag{2}$$

Taking the squared modulus of the ratio of the coefficients in this function, we obtain the following expression for the transmission coefficient $D = 1 - R$:

$$D = \frac{\sinh^2(\pi k/\alpha)}{\sinh^2(\pi k/\alpha) + \cos^2\left(\frac{\pi}{2}\sqrt{1 - \frac{8mU_0}{\hbar^2\alpha^2}}\right)} \quad \text{when} \quad \frac{8mU_0}{\hbar^2\alpha^2} < 1,$$

$$D = \frac{\sinh^2(\pi k/\alpha)}{\sinh^2(\pi k/\alpha) + \cosh^2\left(\frac{\pi}{2}\sqrt{\frac{8mU_0}{\hbar^2\alpha^2} - 1}\right)} \quad \text{when} \quad \frac{8mU_0}{\hbar^2\alpha^2} > 1.$$

The first of these formulas also applies when $U_0 < 0$, that is, when the particle passes not over a potential barrier but over a potential well. It is noteworthy that then $D = 1$ if $1 + (8m|U_0|/\hbar^2\alpha^2) = (2n+1)^2$; thus, for certain values of the well depth $|U_0|$, particles passing over it are not reflected. This is already clear from expression (2), in which the term containing e^{-ikx} is entirely absent when s is a positive integer.

Problem

5. Determine how the transmission coefficient tends to zero as $E \to 0$, assuming that the potential energy $U(x)$ decreases rapidly at distances $|x| \gg a$, where a is the characteristic size of the interaction region.

Solution. In the range of distances $k|x| \ll 1$, the energy E may be neglected in the Schrödinger equation. If also $|x| \gg a$, the potential energy may be neglected as well, and the equation reduces to

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} = 0,$$

whose solution we write as

$$\psi = a_1 + b_1 x, \quad x < 0, \qquad \psi = a_2 + b_2 x, \quad x > 0. \tag{1}$$

Solving the equation at distances $|x| \sim a$ determines the relation among a_1, b_1 and a_2, b_2. This relation is linear and has the form

$$a_1 = \rho a_2 + \mu b_2, \qquad b_1 = \nu a_2 + \tau b_2. \tag{2}$$

The coefficients ρ, μ, ν, and τ are real and independent of the energy, since the energy no longer enters the equation [20]. Solution (1) must agree with the first two terms in the expansions of (25.1) and (25.2) in powers of x, whence

$$a_1 = 1 + B, \qquad b_1 = ik(1 - B), \qquad a_2 = A, \qquad b_2 = ikA.$$

Substituting these expressions in (2) and solving the equations for A, we find, at small k, $A \approx 2ik/\nu$, whence

$$D \approx \frac{4k^2}{\nu^2} \sim E.$$

Thus the transmission coefficient tends to zero in proportion to the particle energy. This general relation is, of course, satisfied in the examples considered in Problems 2 and 4.

[20] By constancy of the flux, they satisfy $\rho\tau - \mu\nu = 1$.

CHAPTER IV

Angular Momentum

§ 26. Angular momentum

In deriving momentum conservation in § 15, we used the homogeneity of space for a closed system of particles. Besides being homogeneous, space is also isotropic: every direction in it is equivalent. The Hamiltonian of a closed system must therefore remain unchanged when the system as a whole is rotated through an arbitrary angle about an arbitrary axis. It is enough to impose this requirement for an arbitrary infinitesimal rotation.

Let $\delta\boldsymbol{\varphi}$ be the vector of an infinitesimal rotation. Its magnitude is the angle of rotation $\delta\varphi$, and its direction is the axis about which the rotation is made. Under such a rotation, the changes $\delta\mathbf{r}_a$ of the particles' radius vectors $\mathbf{r}_a$ are

$$\delta\mathbf{r}_a = \delta\boldsymbol{\varphi} \times \mathbf{r}_a.$$

An arbitrary function $\psi(\mathbf{r}_1, \mathbf{r}_2, \ldots)$ is transformed into

$$\begin{aligned}\psi(\mathbf{r}_1 + \delta\mathbf{r}_1, \mathbf{r}_2 + \delta\mathbf{r}_2, \ldots) &= \psi(\mathbf{r}_1, \mathbf{r}_2, \ldots) + \sum_a \delta\mathbf{r}_a \boldsymbol{\nabla}_a\psi = \\ &= \psi(\mathbf{r}_1, \mathbf{r}_2, \ldots) + \sum_a (\delta\boldsymbol{\varphi} \times \mathbf{r}_a)\cdot \boldsymbol{\nabla}_a\psi = \\ &= \left(1 + \delta\boldsymbol{\varphi}\cdot \sum_a (\mathbf{r}_a \times \boldsymbol{\nabla}_a)\right)\psi(\mathbf{r}_1, \mathbf{r}_2, \ldots).\end{aligned}$$

The expression

$$1 + \delta\boldsymbol{\varphi}\cdot \sum_a (\mathbf{r}_a \times \boldsymbol{\nabla}_a)$$

is the operator of an infinitesimal rotation. The fact that such a rotation does not change the system's Hamiltonian is expressed (cf. § 15) by the commutation of the rotation operator with $\hat{H}$. Since $\delta\boldsymbol{\varphi}$ is a constant vector, this condition reduces to

$$\left(\sum_a (\mathbf{r}_a \times \boldsymbol{\nabla}_a)\right)\hat{H} - \hat{H}\left(\sum_a (\mathbf{r}_a \times \boldsymbol{\nabla}_a)\right) = 0, \qquad (26.1)$$

which expresses a conservation law.

The quantity conserved in a closed system as a consequence of the isotropy of space is the system's *angular momentum* (cf. Vol. I, § 9). Thus, apart from a constant factor, the operator $\sum_a (\mathbf{r}_a \times \boldsymbol{\nabla}_a)$ must correspond to the total angular momentum of the system's motion, while each term $(\mathbf{r}_a \times \boldsymbol{\nabla}_a)$ must correspond to the angular momentum of one particle.

The proportionality factor must be chosen as $-i\hbar$. The resulting particle angular-momentum operator $-i\hbar(\mathbf{r}\times\nabla) = \mathbf{r}\times\hat{\mathbf{p}}$ then agrees exactly with the classical expression $\mathbf{r}\times\mathbf{p}$. Henceforth we shall always measure angular momentum in units of $\hbar$. We denote the operator of the angular momentum of one particle, defined in this way, by $\hat{\mathbf{l}}$, and that of the entire system by $\hat{\mathbf{L}}$. Thus the particle angular-momentum operator is

$$\hbar\hat{\mathbf{l}} = \mathbf{r}\times\hat{\mathbf{p}} = -i\hbar(\mathbf{r}\times\nabla) \tag{26.2}$$

or, in components,

$$\hbar\hat{l}_x = y\hat{p}_z - z\hat{p}_y, \qquad \hbar\hat{l}_y = z\hat{p}_x - x\hat{p}_z, \qquad \hbar\hat{l}_z = x\hat{p}_y - y\hat{p}_x.$$

For a system in an external field, angular momentum is not generally conserved. It may nevertheless be conserved when the field has a suitable symmetry. In a centrally symmetric field, for example, all spatial directions from the center are equivalent, so angular momentum about that center is conserved. When the field has axial symmetry, the same reasoning shows that what is conserved is the angular-momentum component directed along the symmetry axis. All these conservation laws of classical mechanics remain valid in quantum mechanics.

If a system's angular momentum is not conserved, it has no definite value in a stationary state. In such cases the mean angular momentum in a given stationary state can sometimes be of interest. Its mean value is readily seen to vanish in every nondegenerate stationary state. Indeed, the energy is unchanged when the sign of time is reversed; and since only one stationary state corresponds to the given energy level, replacing t by $-t$ must leave the state of the system unchanged . The mean values of all quantities, including angular momentum, must therefore also remain unchanged. But angular momentum reverses sign under time reversal, which would give $\bar{\mathbf{L}} = -\bar{\mathbf{L}}$; hence $\bar{\mathbf{L}} = 0$. A direct calculation confirms this: the integral $\int \psi^*\hat{\mathbf{L}}\psi\, dq$, which by definition gives the mean $\bar{\mathbf{L}}$, can be evaluated using the fact that nondegenerate wave functions are real (see the end of § 18). One then finds that

$$\bar{\mathbf{L}} = -i\hbar\int \psi^*\left(\sum_a(\mathbf{r}_a\times\nabla_a)\right)\psi\, dq$$

is purely imaginary. Since $\bar{\mathbf{L}}$ must of course be real, it follows that $\bar{\mathbf{L}} = 0$.

We now determine the commutation rules between the angular-momentum operators and the coordinate and momentum operators. From (16.2) we readily obtain

$$\begin{aligned}
\{\hat{l}_x, x\} &= 0, \quad \{\hat{l}_x, y\} = iz, \quad \{\hat{l}_x, z\} = -iy,\\
\{\hat{l}_y, y\} &= 0, \quad \{\hat{l}_y, z\} = ix, \quad \{\hat{l}_y, x\} = -iz,\\
\{\hat{l}_z, z\} &= 0, \quad \{\hat{l}_z, x\} = iy, \quad \{\hat{l}_z, y\} = -ix.
\end{aligned} \tag{26.3}$$

For example,

$$\hat{l}_x y - y\hat{l}_x = \frac{1}{\hbar}(y\hat{p}_z - z\hat{p}_y)y - y(y\hat{p}_z - z\hat{p}_y)\frac{1}{\hbar} = -\frac{z}{\hbar}\{\hat{p}_y, y\} = iz.$$

All the relations (26.3) may be written in tensor form as

$$\{\hat{l}_i, x_k\} = ie_{ikl}x_l, \tag{26.4}$$

where e_{ikl} is the antisymmetric unit tensor of third rank[1] , with summation understood over each pair of repeated "dummy" indices. The angular-momentum and momentum operators satisfy analogous commutation relations:

$$\{\hat{l}_i, \hat{p}_k\} = ie_{ikl}\hat{p}_l. \tag{26.5}$$

These formulas make it easy to obtain the commutation rules among the operators for the components of angular momentum. We have

$$\begin{aligned}\hbar(\hat{l}_x\hat{l}_y - \hat{l}_y\hat{l}_x) &= \hat{l}_x(z\hat{p}_x - x\hat{p}_z) - (z\hat{p}_x - x\hat{p}_z)\hat{l}_x = \\ &= (\hat{l}_x z - z\hat{l}_x)\hat{p}_x - x(\hat{l}_x\hat{p}_z - \hat{p}_z\hat{l}_x) = -iy\hat{p}_x + ix\hat{p}_y = i\hbar\hat{l}_z.\end{aligned}$$

Thus

$$\{\hat{l}_y, \hat{l}_z\} = i\hat{l}_x, \qquad \{\hat{l}_z, \hat{l}_x\} = i\hat{l}_y, \qquad \{\hat{l}_x, \hat{l}_y\} = i\hat{l}_z \tag{26.6}$$

or

$$\{\hat{l}_i, \hat{l}_k\} = ie_{ikl}\hat{l}_l. \tag{26.7}$$

Exactly the same relations hold for the operators $\hat{L}_x$, $\hat{L}_y$, and $\hat{L}_z$ of the system's total angular momentum. Indeed, since the angular-momentum operators of different particles commute with one another, we have, for example,

$$\sum_a \hat{l}_{ay} \sum_a \hat{l}_{az} - \sum_a \hat{l}_{az} \sum_a \hat{l}_{ay} = \sum_a (\hat{l}_{ay}\hat{l}_{az} - \hat{l}_{az}\hat{l}_{ay}) = i \sum_a \hat{l}_{ax}.$$

Therefore

$$\{\hat{L}_y, \hat{L}_z\} = i\hat{L}_x, \qquad \{\hat{L}_z, \hat{L}_x\} = i\hat{L}_y, \qquad \{\hat{L}_x, \hat{L}_y\} = i\hat{L}_z. \tag{26.8}$$

Relations (26.8) show that the three components of angular momentum cannot have definite values simultaneously (except when all three components vanish at once; see below). In this respect angular momentum differs essentially from momentum, whose three components can be measured simultaneously.

From $\hat{L}_x$, $\hat{L}_y$, and $\hat{L}_z$ we form the operator for the square of the magnitude of the angular-momentum vector:

$$\hat{\mathbf{L}}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2. \tag{26.9}$$

[1]The antisymmetric unit tensor of third rank e_{ikl} (also called the unit axial tensor) is defined to be antisymmetric in all three indices, with $e_{123} = 1$. Evidently, of its 27 components, only the 6 for which i, k, l form a permutation of the numbers 1, 2, 3 are nonzero. A component is +1 if the permutation i, k, l is obtained from 1, 2, 3 by an even number of pairwise interchanges (transpositions), and is −1 for an odd number. Clearly,

$$e_{ikl}e_{ikm} = 2\delta_{lm}, \qquad e_{ikl}e_{ikl} = 6.$$

The components of the vector $\mathbf{C} = \mathbf{A} \times \mathbf{B}$, the vector product of $\mathbf{A}$ and $\mathbf{B}$, can be expressed through e_{ikl} as

$$C_i = e_{ikl}A_kB_l.$$

This operator commutes with each of $\hat{L}_x$, $\hat{L}_y$, and $\hat{L}_z$:

$$\{\hat{\mathbf{L}}^2, \hat{L}_x\} = 0, \qquad \{\hat{\mathbf{L}}^2, \hat{L}_y\} = 0, \qquad \{\hat{\mathbf{L}}^2, \hat{L}_z\} = 0. \tag{26.10}$$

Indeed, using (26.8), we find, for example,

$$\begin{aligned}
\{\hat{L}_x^2, \hat{L}_z\} &= \hat{L}_x\{\hat{L}_x, \hat{L}_z\} + \{\hat{L}_x, \hat{L}_z\}\hat{L}_x = -i(\hat{L}_x\hat{L}_y + \hat{L}_y\hat{L}_x),\\
\{\hat{L}_y^2, \hat{L}_z\} &= i(\hat{L}_x\hat{L}_y + \hat{L}_y\hat{L}_x),\\
\{\hat{L}_z^2, \hat{L}_z\} &= 0.
\end{aligned}$$

Adding these equalities gives the last relation in (26.10).

Physically, (26.10) means that the square of the angular momentum (that is, its magnitude) can have a definite value simultaneously with one of its components.

Instead of $\hat{L}_x$ and $\hat{L}_y$, it is often convenient to use their complex combinations

$$\hat{L}_+ = \hat{L}_x + i\hat{L}_y, \qquad \hat{L}_- = \hat{L}_x - i\hat{L}_y. \tag{26.11}$$

Direct calculation from (26.8) readily gives the following commutation rules for these combinations:

$$\{\hat{L}_+, \hat{L}_-\} = 2\hat{L}_z, \qquad \{\hat{L}_z, \hat{L}_+\} = \hat{L}_+, \qquad \{\hat{L}_z, \hat{L}_-\} = -\hat{L}_-. \tag{26.12}$$

It is also easy to verify that

$$\hat{\mathbf{L}}^2 = \hat{L}_+\hat{L}_- + \hat{L}_z^2 - \hat{L}_z = \hat{L}_-\hat{L}_+ + \hat{L}_z^2 + \hat{L}_z. \tag{26.13}$$

Finally, we give the frequently used expressions for the angular-momentum operator of one particle in spherical coordinates. Introducing these coordinates through the usual relations

$$x = r\sin\theta\cos\varphi, \qquad y = r\sin\theta\sin\varphi, \qquad z = r\cos\theta,$$

a straightforward calculation gives

$$\hat{l}_z = -i\frac{\partial}{\partial\varphi}, \tag{26.14}$$

$$\hat{l}_\pm = e^{\pm i\varphi}\left(\pm\frac{\partial}{\partial\theta} + i\cot\theta\frac{\partial}{\partial\varphi}\right). \tag{26.15}$$

Substitution in (26.13) gives the square of the particle's angular-momentum operator in the form

$$\hat{\mathbf{l}}^2 = -\left[\frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\varphi^2} + \frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right)\right]. \tag{26.16}$$

Notice that, apart from a factor, this is the angular part of the Laplace operator.

§ 27. Eigenvalues of angular momentum

In finding the eigenvalues of a particle's angular-momentum projection onto a chosen direction, one may orient the polar axis along that direction and employ the spherical-coordinate form of the operator. Using (26.14), the equation $\hat{l}_z\psi = l_z\psi$ takes the form

$$-i\frac{\partial\psi}{\partial\varphi} = l_z\psi. \tag{27.1}$$

Its solution is

$$\psi = f(r,\theta)e^{il_z\varphi},$$

where $f(r,\theta)$ is an arbitrary function of r and θ. For ψ to be single-valued, it must be periodic in φ, with period 2π. Hence we find[2]

$$l_z = m, \qquad m = 0,\ \pm1,\ \pm2, \dots \tag{27.2}$$

Thus the eigenvalues l_z are the positive and negative integers, including zero. We denote the factor depending on φ that is characteristic of the eigenfunctions of $\hat{l}_z$ by

$$\Phi_m(\varphi) = \frac{1}{\sqrt{2\pi}}e^{im\varphi}. \tag{27.3}$$

These functions are normalized so that

$$\int_0^{2\pi} \Phi_m^*(\varphi)\Phi_{m'}(\varphi)\,d\varphi = \delta_{mm'}. \tag{27.4}$$

The eigenvalues of the z component of the total angular momentum of a system are evidently likewise positive and negative integers:

$$L_z = M, \qquad M = 0,\ \pm1,\ \pm2, \dots \tag{27.5}$$

(this follows because $\hat{L}_z$ is the sum of the mutually commuting operators $\hat{l}_z$ of the individual particles).

Since the direction of the z axis has not been distinguished in advance, the same result clearly holds for L_x, L_y, and, in general, for the component of angular momentum along any direction: each can take only integer values. At first sight this result may seem paradoxical, particularly if it is applied to two infinitely close directions. One must bear in mind, however, that the only common eigenfunction of the operators $\hat{L}_x$, $\hat{L}_y$, and $\hat{L}_z$ corresponds to the simultaneous values

$$L_x = L_y = L_z = 0;$$

in this case the angular-momentum vector, and therefore its projection on every direction, is zero. If at least one of the eigenvalues L_x, L_y, L_z is nonzero, the corresponding operators have no common eigenfunctions. Put differently, no state exists in which definite nonzero values are possessed simultaneously by two or three angular-momentum

[2] The customary notation of the eigenvalues of the angular-momentum projection by the letter m—the same letter that also denotes the particle mass—cannot in practice lead to confusion.

components referred to distinct directions; one can therefore only discuss the integer nature of a single component at a time.

Stationary states of a system that differ only in the value of M have the same energy. This follows already from general considerations, since the direction of the z axis has not been distinguished beforehand. Thus the energy levels of a system with conserved nonzero angular momentum are in any event degenerate[3].

We now turn to finding the eigenvalues of the square of angular momentum and show how they can be obtained solely from the commutation rules (26.8). Let ψ_M denote the wave functions of stationary states having the same value of $\mathbf{L}^2$, belonging to one degenerate energy level, and differing in the value of M[4].

First, since the two directions of the z axis are physically equivalent, every possible positive value $M = |M|$ has a corresponding negative value $M = -|M|$. Let L (a nonnegative integer) denote the largest possible value of $|M|$ for a given $\mathbf{L}^2$. The very existence of such an upper bound follows from the fact that the difference $\hat{\mathbf{L}}^2 - \hat{L}_z^2 = \hat{L}_x^2 + \hat{L}_y^2$ is the operator of the necessarily nonnegative physical quantity $L_x^2 + L_y^2$, and hence its eigenvalues cannot be negative.

Applying the operator $\hat{L}_z \hat{L}_\pm$ to an eigenfunction ψ_M of $\hat{L}_z$, and using the commutation rules (26.12), we obtain

$$\hat{L}_z \hat{L}_\pm \psi_M = (M \pm 1)\hat{L}_\pm \psi_M. \tag{27.6}$$

It follows that $\hat{L}_\pm \psi_M$ is, apart from a normalization constant, an eigenfunction corresponding to the value $M \pm 1$ of L_z:

$$\psi_{M+1} = \text{const} \cdot \hat{L}_+ \psi_M, \qquad \psi_{M-1} = \text{const} \cdot \hat{L}_- \psi_M. \tag{27.7}$$

If $M = L$ is put in the first of these equalities, then we must have identically

$$\hat{L}_+ \psi_L = 0, \tag{27.8}$$

since, by definition, there are no states with $M > L$. Applying $\hat{L}_-$ to this equality and using (26.13), we obtain

$$\hat{L}_- \hat{L}_+ \psi_L = \left(\hat{\mathbf{L}}^2 - \hat{L}_z^2 - \hat{L}_z\right)\psi_L = 0.$$

But since the ψ_M are common eigenfunctions of $\hat{\mathbf{L}}^2$ and $\hat{L}_z$,

$$\hat{\mathbf{L}}^2 \psi_L = \mathbf{L}^2 \psi_L, \qquad \hat{L}_z^2 \psi_L = L^2 \psi_L, \qquad \hat{L}_z \psi_L = L \psi_L,$$

[3]This circumstance is a special case of the general theorem stated in § 10 on the degeneracy of levels in the presence of at least two conserved quantities whose operators do not commute. Here those quantities are the components of angular momentum.

[4]Here it is understood that there is no additional degeneracy by which the same energy corresponds to different values of the square of angular momentum. This is true for the discrete spectrum (apart from the so-called "accidental degeneracy" in the Coulomb field; see § 36), and is, generally speaking, not true for energy levels in the continuous spectrum. Even when there is additional degeneracy, however, the eigenfunctions can always be chosen to represent states with definite values of $\mathbf{L}^2$; from them one can then select states with the same values of E and $\mathbf{L}^2$. The mathematical basis for this is that simultaneous diagonalization of the matrices of mutually commuting operators is always achievable. In analogous cases below, for brevity, we shall speak as if there were no additional degeneracy, with the understanding that, in view of what has just been said, the results obtained do not in fact depend on this assumption.

so the resulting equation gives

$$\mathbf{L}^2 = L(L+1). \tag{27.9}$$

Formula (27.9) determines the required eigenvalues of the square of angular momentum; L takes all nonnegative integer values. For a given value of L, the angular-momentum component $L_z = M$ can take the values

$$M = L, L-1, \ldots, -L, \tag{27.10}$$

that is, $2L+1$ distinct values in all. An energy level corresponding to angular momentum L is therefore $(2L+1)$-fold degenerate; this is usually called directional degeneracy of angular momentum. A state with zero angular momentum, $L = 0$ (so that all three of its components vanish), is nondegenerate. We note that the wave function of such a state is spherically symmetric. This is already clear from the fact that its change under any infinitesimal rotation, given by $\hat{\mathbf{L}}\psi$, vanishes in this case. For brevity we shall often use the customary phrase "angular momentum L" of the system, meaning angular momentum whose square equals $L(L+1)$; the z component of angular momentum is usually called simply its "projection."

We shall denote the angular momentum of one particle by the lowercase letter l; thus for it we write (27.9) as

$$\mathbf{l}^2 = l(l+1). \tag{27.11}$$

Let us calculate the matrix elements of L_x and L_y in the representation in which, together with the energy, $\mathbf{L}^2$ and L_z are diagonal (*M. Born, W. Heisenberg, P. Jordan*, 1926). We note that, since $\hat{L}_x$ and $\hat{L}_y$ commute with the Hamiltonian, their matrices are diagonal with respect to energy; that is, every matrix element for a transition between states of different energies (and different angular momenta L) is zero. It is therefore sufficient to consider matrix elements for transitions within the group of states with different values of M corresponding to a single degenerate energy level.

From (27.7) one sees that the matrix of $\hat{L}_+$ has nonzero elements only for transitions $M-1 \to M$, and that of $\hat{L}_-$ only for $M \to M-1$. With this in mind, equating the diagonal matrix elements of the two sides of (26.13) gives[5]

$$L(L+1) = \langle M|L_+|M-1\rangle\langle M-1|L_-|M\rangle + M^2 - M.$$

Since the operators $\hat{L}_x$ and $\hat{L}_y$ are Hermitian,

$$\langle M-1|L_-|M\rangle = \langle M|L_+|M-1\rangle^*,$$

we rewrite this equality as

$$\left|\langle M|L_+|M-1\rangle\right|^2 = L(L+1) - M(M-1) = (L-M+1)(L+M),$$

whence[6]

$$\langle M|L_+|M-1\rangle = \langle M-1|L_-|M\rangle = \sqrt{(L+M)(L-M+1)}. \tag{27.12}$$

[5] In the notation for matrix elements, for brevity, we omit all indices with respect to which they are diagonal (including the index L).

[6] The choice of sign in this formula is consistent with the choice of phase factors in the angular-momentum eigenfunctions.

For the nonzero matrix elements of L_x and L_y themselves, this gives

$$\begin{aligned}\langle M|L_x|M-1\rangle &= \langle M-1|L_x|M\rangle = \frac{1}{2}\sqrt{(L+M)(L-M+1)},\\ \langle M|L_y|M-1\rangle &= -\langle M-1|L_y|M\rangle = -\frac{i}{2}\sqrt{(L+M)(L-M+1)}.\end{aligned} \tag{27.13}$$

We draw attention to the absence of diagonal elements in the matrices of L_x and L_y. Since a diagonal matrix element gives the mean value of a quantity in the corresponding state, this means that in states with definite values of L_z the mean values are $\bar{L}_x = \bar{L}_y = 0$. Thus, if the projection of angular momentum on some direction in space has a definite value, the whole vector $\bar{\mathbf{L}}$ lies along that same direction.

§ 28. Angular-Momentum Eigenfunctions

Specifying the values of l and m does not determine a particle's wave function completely. Indeed, the expressions for the operators of these quantities in spherical coordinates contain only the angles θ and φ, so their eigenfunctions may include an arbitrary factor depending on r. Here we shall consider only the angular part of the wave function characteristic of angular-momentum eigenfunctions. Denote it by $Y_{lm}(\theta, \varphi)$ and impose the normalization condition

$$\int |Y_{lm}|^2\, do = 1$$

($do = \sin\theta\, d\theta\, d\varphi$ is the element of solid angle).

The calculation below shows that the problem of finding the common eigenfunctions of $\hat{\mathbf{l}}^2$ and $\hat{l}_z$ permits separation of the variables θ and φ. These functions may therefore be sought in the form

$$Y_{lm} = \Phi_m(\varphi)\Theta_{lm}(\theta), \tag{28.1}$$

where $\Phi_m(\varphi)$ are the eigenfunctions of $\hat{l}_z$ defined by (27.3). Since the functions Φ_m have already been normalized by (27.4), the functions Θ_{lm} must satisfy

$$\int_0^\pi |\Theta_{lm}|^2 \sin\theta\, d\theta = 1. \tag{28.2}$$

Functions Y_{lm} with different values of l or m are automatically mutually orthogonal:

$$\int_0^{2\pi}\int_0^\pi Y^*_{l'm'} Y_{lm} \sin\theta\, d\theta\, d\varphi = \delta_{ll'}\delta_{mm'}, \tag{28.3}$$

because they are eigenfunctions of angular-momentum operators belonging to different eigenvalues. The functions $\Phi_m(\varphi)$ are also separately orthogonal (see (27.4)), since they are eigenfunctions of $\hat{l}_z$ belonging to its different eigenvalues m. The functions $\Theta_{lm}(\theta)$ by themselves, however, are not eigenfunctions of any angular-momentum operator; they are mutually orthogonal for different l, but not for different m.

The most direct way to calculate the required functions is to solve the eigenfunction problem for $\hat{\mathbf{l}}^2$ written in spherical coordinates (formula (26.16)). The equation $\hat{\mathbf{l}}^2\psi = l^2\psi$ reads

$$\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial\psi}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2\psi}{\partial\varphi^2} + l(l+1)\psi = 0.$$

Substitution of ψ in the form (28.1) gives the following equation for Θ_{lm}:

$$\frac{1}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta_{lm}}{d\theta}\right) - \frac{m^2}{\sin^2\theta}\Theta_{lm} + l(l+1)\Theta_{lm} = 0. \tag{28.4}$$

This equation is well known from the theory of spherical harmonics. It has solutions satisfying the finiteness and single-valuedness conditions for nonnegative integral values $l \geq |m|$, in agreement with the angular-momentum eigenvalues obtained above by the matrix method. The corresponding solutions are the associated Legendre polynomials $P_l^m(\cos\theta)$ (see § c of the mathematical supplements). Normalizing the solution by (28.2), we obtain[7]

$$\Theta_{lm} = (-1)^m i^l \sqrt{\frac{(2l+1)(l-m)!}{2(l+m)!}} P_l^m(\cos\theta). \tag{28.5}$$

Here $m \geq 0$ is assumed. For negative m, define Θ_{lm} by

$$\Theta_{l,-|m|} = (-1)^m \Theta_{l|m|}, \tag{28.6}$$

that is, for $m < 0$, Θ_{lm} is given by (28.5) with $|m|$ in place of m and with the factor $(-1)^m$ omitted. Thus, from the mathematical point of view, the angular-momentum eigenfunctions are spherical harmonics with a particular normalization. For later reference, we write their complete expression, incorporating all the definitions just given:

$$Y_{lm}(\theta,\varphi) = (-1)^{(m+|m|)/2} i^l \left[\frac{2l+1}{4\pi}\frac{(l-|m|)!}{(l+|m|)!}\right]^{1/2} P_l^{|m|}(\cos\theta)e^{im\varphi}. \tag{28.7}$$

In particular,

$$Y_{l0} = i^l \sqrt{\frac{2l+1}{4\pi}} P_l(\cos\theta). \tag{28.8}$$

Functions differing only in the sign of m are evidently related by

$$(-1)^{l-m} Y_{l,-m} = Y_{lm}^*. \tag{28.9}$$

For $l = 0$ (and consequently $m = 0$), the spherical harmonic reduces to a constant. In other words, the wave functions of particle states with zero angular momentum depend only on r and hence possess complete spherical symmetry, in accordance with the general statement made in § 27.

For a fixed m, the values of l beginning with $|m|$ enumerate, in ascending order, the successive eigenvalues of $\mathbf{l}^2$. The general theorem on zeros of eigenfunctions (§ 21) therefore implies that Θ_{lm} vanishes at $l - |m|$ different values of θ; in other words, its nodal lines are $l - |m|$ "circles of latitude" on the sphere. For the complete angular

[7]The normalization condition does not, of course, determine the choice of phase factor. The convention used in this book is the one most natural from the standpoint of the general theory of angular-momentum addition; it differs from the convention usually adopted by a factor i^l. The advantages of this choice will be apparent from the notes on pp. 278, 527, and 533.

functions, if real factors $\cos m\varphi$ or $\sin m\varphi$ are chosen instead of $e^{\pm i|m|\varphi}$[8], they also have $|m|$ "meridian circles" as nodal lines. The total number of nodal lines is therefore l.

Finally, let us show how the functions Θ_{lm} can be calculated by the matrix method. The procedure is analogous to the calculation of the oscillator wave functions in § 23. We start from (27.8), $\hat{l}_+ Y_{ll} = 0$. Using (26.15) for the operator $\hat{l}_+$ and substituting

$$Y_{ll} = \frac{1}{\sqrt{2\pi}} e^{il\varphi} \Theta_{ll}(\theta),$$

we obtain for Θ_{ll} the equation

$$\frac{d\Theta_{ll}}{d\theta} - l \cot\theta \cdot \Theta_{ll} = 0,$$

whence $\Theta_{ll} = \text{const} \cdot \sin^l \theta$. Determining the constant from the normalization condition gives

$$\Theta_{ll} = (-i)^l \sqrt{\frac{(2l+1)!}{2}} \frac{1}{2^l l!} \sin^l \theta. \tag{28.10}$$

Next, using (27.12), write

$$\hat{l}_- Y_{l,m+1} = (l_-)_{m,m+1} Y_{lm} = \sqrt{(l-m)(l+m+1)} Y_{lm}.$$

Repeated application of this formula gives

$$\sqrt{\frac{(l-m)!}{(l+m)!}} Y_{lm} = \frac{1}{\sqrt{(2l)!}} \hat{l}_-^{l-m} Y_{ll}.$$

The right-hand side is readily evaluated with the aid of (26.15) for $\hat{l}_-$, according to which

$$\hat{l}_- \left[f(\theta) e^{im\varphi} \right] = e^{i(m-1)\varphi} \sin^{1-m} \theta \frac{d}{d\cos\theta} \left(f \sin^m \theta \right).$$

Repeated application of this formula gives

$$\hat{l}_-^{l-m} e^{il\varphi} \Theta_{ll} = e^{im\varphi} \sin^{-m} \theta \frac{d^{l-m}}{(d\cos\theta)^{l-m}} \left(\sin^l \theta \cdot \Theta_{ll} \right).$$

Finally, using these relations together with (28.10) for Θ_{ll}, we obtain

$$\Theta_{lm}(\theta) = (-i)^l \sqrt{\frac{2l+1}{2} \frac{(l+m)!}{(l-m)!}} \frac{1}{2^l l! \sin^m \theta} \frac{d^{l-m}}{(d\cos\theta)^{l-m}} \sin^{2l} \theta, \tag{28.11}$$

which agrees with (28.5).

[8]Each such function represents a state in which l_z has no definite value, but may take the values $\pm m$ with equal probability.

§ 29. Matrix elements of vectors

Return once more to a closed system of particles[9]. Let f stand for some scalar physical quantity of this system and $\hat{f}$ for its operator. Any scalar is invariant under a rotation of the coordinate system. Hence the scalar operator $\hat{f}$ is unchanged by a rotation; that is, it commutes with the rotation operator. We know, however, that the operator of an infinitesimal rotation coincides, apart from a constant factor, with the angular-momentum operator. Therefore

$$\{\hat{f}, \hat{\mathbf{L}}\} = 0. \tag{29.1}$$

It follows from the commutation of $\hat{f}$ with the angular-momentum operator that, for transitions between states with definite L and M, the matrix of f is diagonal in these indices. Moreover, specifying M determines only the orientation of the system relative to the coordinate axes, while the value of a scalar quantity does not depend on this orientation at all. Consequently, the matrix elements $\langle n'LM|f|nLM\rangle$ do not depend on M (the letter n provisionally denotes the collection of all quantum numbers, other than L and M, that specify the state of the system). A formal proof follows from the commutation of $\hat{f}$ and $\hat{L}_+$:

$$\hat{f}\hat{L}_+ - \hat{L}_+\hat{f} = 0. \tag{29.2}$$

Take the matrix element of this equality for the transition $n, L, M \to n', L, M + 1$. Since the matrix of L_+ has only elements for transitions $n, L, M \to n, L, M + 1$, we find

$$\begin{aligned} \langle n', L, M+1|f|n, L, M+1\rangle\langle n, L, M+1|L_+|n, L, M\rangle &= \\ = \langle n', L, M+1|L_+|n', L, M\rangle\langle n', L, M|f|n, L, M\rangle&, \end{aligned}$$

and, since the matrix elements of L_+ do not depend on the index n,

$$\langle n', L, M+1|f|n, L, M+1\rangle = \langle n', L, M|f|n, L, M\rangle, \tag{29.3}$$

which shows that all the elements $\langle n', L, M|f|n, L, M\rangle$ with different M (and identical remaining indices) are equal.

Applied to the Hamiltonian itself, this result gives the already known independence of the stationary-state energy from M, that is, the $(2L + 1)$-fold degeneracy of the energy levels.

Next, let $\mathbf{A}$ be some vector physical quantity characterizing the closed system. Under a rotation of the coordinate system (in particular, an infinitesimal rotation, that is, under the action of the angular-momentum operator), the components of the vector transform into one another. The commutation of the operators $\hat{L}_i$ with the operators $\hat{A}_i$ must therefore again yield components of the same vector $\hat{A}_i$. Which components occur can be found by noting that, in the special case where $\mathbf{A}$ is the radius vector of a particle, the result must reduce to (26.4). We thus obtain the commutation rules

$$\{\hat{L}_i, \hat{A}_k\} = ie_{ikl}\hat{A}_l. \tag{29.4}$$

[9] Everything derived in this section applies equally to a particle in a centrally symmetric field (more broadly, to any situation in which the system's total angular momentum is conserved).

From these relations one can derive a number of results about the structure of the component matrices of the vector **A** (*M. Born, W. Heisenberg, P. Jordan*, 1926). First, they make it possible to find the *selection rules*, which determine the transitions for which matrix elements may be nonzero. We shall not, however, give the corresponding rather cumbersome calculations here, since it will be shown below (§ 107) that these rules are in fact direct consequences of the general transformation properties of vector quantities and can be obtained from them essentially without calculation. Here we merely state the rules without derivation.

Matrix elements of every component of a vector can be nonzero only for transitions in which the angular momentum L changes by no more than one:

$$L \to L, L \pm 1. \tag{29.5}$$

There is also an additional selection rule forbidding transitions between any two states with $L = 0$. This rule is an evident consequence of the complete spherical symmetry of states with zero angular momentum.

The selection rules for the angular-momentum projection M differ among the components of the vector. Specifically, matrix elements may be nonzero for transitions with the following changes of M:

$$\begin{aligned} &M \to M+1 \quad && \text{for } A_+ = A_x + iA_y, \\ &M \to M-1 \quad && \text{for } A_- = A_x - iA_y, \\ &M \to M \quad && \text{for } A_z. \end{aligned} \tag{29.6}$$

It is also possible to determine in general form the dependence of the matrix elements of a vector on M. We again give these important and frequently used formulas without derivation, since they too are special cases of more general relations, applying to arbitrary tensor quantities, which will be derived in § 107. The nonzero matrix elements of A_z are given by

$$\begin{aligned} \langle n'LM|A_z|nLM\rangle &= \frac{M}{\sqrt{L(L+1)(2L+1)}}\langle n'L\|A\|nL\rangle, \\ \langle n'LM|A_z|n, L-1, M\rangle &= \sqrt{\frac{L^2-M^2}{L(2L-1)(2L+1)}}\langle n'L\|A\|n, L-1\rangle, \\ \langle n', L-1, M|A_z|nLM\rangle &= \sqrt{\frac{L^2-M^2}{L(2L-1)(2L+1)}}\langle n', L-1\|A\|nL\rangle. \end{aligned} \tag{29.7}$$

Here the symbol

$$\langle n'L'\|A\|nL\rangle$$

denotes a so-called *reduced matrix element*, a quantity independent of the quantum number M[10]. They are related by

$$\langle n'L'\|A\|nL\rangle = \langle nL\|A\|n'L'\rangle^*, \tag{29.8}$$

[10]The appearance in (29.7) and (29.9) of denominators depending on L accords with the general notation introduced in § 107. The usefulness of these denominators is seen, in particular, in the simple form taken by (29.12) for the matrix elements of the scalar product of two vectors.

One should regard the notation for the reduced matrix element as an indivisible entity (in contrast to what was said about the matrix element notation (11.17)).

which follows directly from the Hermiticity of $\hat{A}_z$.

The matrix elements of A_- and A_+ can be expressed through the same reduced elements. The nonzero matrix elements of A_- are

$$\begin{aligned}
\langle n', L, M-1|A_-|nLM\rangle &= \sqrt{\frac{(L-M+1)(L+M)}{L(L+1)(2L+1)}}\langle n'L\|A\|nL\rangle, \\
\langle n', L, M-1|A_-|n, L-1, M\rangle &= \sqrt{\frac{(L-M+1)(L-M)}{L(2L-1)(2L+1)}}\langle n'L\|A\|n, L-1\rangle, \\
\langle n', L-1, M-1|A_-|nLM\rangle &= -\sqrt{\frac{(L+M-1)(L+M)}{L(2L-1)(2L+1)}}\langle n', L-1\|A\|nL\rangle.
\end{aligned} \tag{29.9}$$

The matrix elements of A_+ need no separate formulas, since the reality of A_x and A_y gives

$$\langle n'L'M'|A_+|nLM\rangle = \langle nLM|A_-|n'L'M'\rangle^*. \tag{29.10}$$

Let us write out the formula that relates the matrix elements of the scalar **AB** to the reduced matrix elements of the two vectors **A** and **B**. The calculation is conveniently made by writing the operator $\hat{\mathbf{A}}\hat{\mathbf{B}}$ as

$$\hat{\mathbf{A}}\hat{\mathbf{B}} = \frac{1}{2}(\hat{A}_+\hat{B}_- + \hat{A}_-\hat{B}_+) + \hat{A}_z\hat{B}_z. \tag{29.11}$$

The matrix of **AB**, like that of any scalar, is diagonal in L and M. Calculation using (29.7)–(29.9) gives

$$\langle n'LM|\mathbf{AB}|nLM\rangle = \frac{1}{2L+1}\sum_{n'',L''}\langle n'L\|A\|n''L''\rangle\langle n''L''\|B\|nL\rangle, \tag{29.12}$$

where L'' takes the values $L, L\pm 1$.

For reference, we give the reduced matrix elements of the vector **L** itself. Comparing (29.9) and (27.12), we find

$$\begin{aligned}
\langle L\|L\|L\rangle &= \sqrt{L(L+1)(2L+1)}, \\
\langle L-1\|L\|L\rangle &= \langle L\|L\|L-1\rangle = 0.
\end{aligned} \tag{29.13}$$

A quantity frequently encountered in applications is the unit vector **n** along the radius vector of a particle. Let us find its reduced matrix elements. It is sufficient to calculate, for example, the matrix elements of $n_z = \cos\theta$ at zero angular-momentum projection, $m = 0$. We have

$$\langle l-1, 0|n_z|l0\rangle = \int_0^\pi \Theta^*_{l-1,0}\cos\theta\cdot\Theta_{l0}\sin\theta\,d\theta$$

with the functions Θ_{l0} from (28.11). Evaluation of the integral gives[11]

$$\langle l-1, 0|n_z|l0\rangle = \frac{il}{\sqrt{(2l-1)(2l+1)}}.$$

[11]The calculation is performed by integrating by parts $(l-1)$ times with respect to $d\cos\theta$. For the general formula for integrals of this kind, see (107.14).

The matrix elements for transitions $l \to l$ are zero (as they are for any polar vector belonging to a single particle; see below, (30.8)). Comparison with (29.7) now gives

$$\langle l-1\|n\|l\rangle = -\langle l\|n\|l-1\rangle = i\sqrt{l}, \qquad \langle l\|n\|l\rangle = 0. \tag{29.14}$$

Problem

Average the tensor $n_i n_k - (1/3)\delta_{ik}$ (where $\mathbf{n}$ is the unit vector along the particle's radius vector) over a state with a definite value of $|\mathbf{l}|$ but no preferred orientation (that is, l_z is unspecified).

Solution. The required mean value is an operator that can be expressed only in terms of $\hat{\mathbf{l}}$. We seek it in the form

$$\overline{n_i n_k} - \frac{1}{3}\delta_{ik} = a\left[\hat{l}_i\hat{l}_k + \hat{l}_k\hat{l}_i - \frac{2}{3}\delta_{ik} l(l+1)\right];$$

this is the most general form of a symmetric second-rank tensor with zero trace constructed from the components of $\hat{\mathbf{l}}$. To determine the constant a, multiply this equality on the left by $\hat{l}_i$ and on the right by $\hat{l}_k$, with summation over i and k. Since the vector $\mathbf{n}$ is perpendicular to the vector $\hbar\hat{\mathbf{l}} = \hat{\mathbf{r}}\times\hat{\mathbf{p}}$, we have $n_i\hat{l}_i = 0$. We replace the product $\hat{l}_i\hat{l}_i\hat{l}_k\hat{l}_k = (\hat{\mathbf{l}}^2)^2$ by its eigenvalue $l^2(l+1)^2$, and transform the product $\hat{l}_i\hat{l}_k\hat{l}_i\hat{l}_k$ using the commutation relations (26.7) as follows:

$$\begin{aligned}
\hat{l}_i\hat{l}_k\hat{l}_i\hat{l}_k &= \hat{l}_i\hat{l}_i\hat{l}_k\hat{l}_k - ie_{ikl}\hat{l}_i\hat{l}_l\hat{l}_k \\
&= (\hat{\mathbf{l}}^2)^2 - \frac{i}{2}e_{ikl}\hat{l}_i(\hat{l}_l\hat{l}_k - \hat{l}_k\hat{l}_l) \\
&= (\hat{\mathbf{l}}^2)^2 + \frac{1}{2}e_{ikl}e_{lkm}\hat{l}_i\hat{l}_m = (\hat{\mathbf{l}}^2)^2 - \hat{\mathbf{l}}^2 \\
&= l^2(l+1)^2 - l(l+1)
\end{aligned}$$

(we have used $e_{ikl}e_{mkl} = 2\delta_{im}$). Straightforward simplification then gives

$$a = -\frac{1}{(2l-1)(2l+3)}.$$

§ 30. Parity of a state

Besides translations and rotations of the coordinate system (invariance under which expresses, respectively, the homogeneity and isotropy of space), there is one more operation that preserves the Hamiltonian of a closed system: the *inversion transformation*, which reverses the sign of every coordinate at once (equivalently, it flips the direction of all axes, turning a right-handed frame into a left-handed one and conversely). Invariance of the Hamiltonian under inversion is the mathematical expression of the mirror symmetry of space[12]. In classical mechanics, the invariance of the Hamilton function under

[12] Inversion likewise leaves invariant the Hamiltonian of particles moving in a centrally symmetric field (provided the coordinate origin coincides with the center of that field).

inversion gives no new conservation laws. In quantum mechanics, however, the situation is essentially different.

Introduce the inversion operator $\hat{P}$[13], whose action on a wave function $\psi(\mathbf{r})$ consists in reversing the signs of the coordinates:

$$\hat{P}\psi(\mathbf{r}) = \psi(-\mathbf{r}). \tag{30.1}$$

The eigenvalues P of this operator, defined by the equation

$$\hat{P}\psi(\mathbf{r}) = P\psi(\mathbf{r}), \tag{30.2}$$

are easily found. Two successive applications of the inversion operator give the identity: the arguments of the function are left entirely unchanged. In other words, $\hat{P}^2\psi = P^2\psi = \psi$, so that $P^2 = 1$, whence

$$P = \pm 1. \tag{30.3}$$

An eigenfunction of the inversion operator is therefore either left entirely unaltered or acquires an overall sign change. Wave functions (and the corresponding states) of the first type are called *even*, and those of the second type *odd*.

The invariance of the Hamiltonian under inversion (that is, the commutation of $\hat{H}$ and $\hat{P}$) therefore expresses the *law of conservation of parity*: if a state of a closed system has definite parity (if it is even or odd), that parity is conserved in time[14].

The angular-momentum operator is likewise invariant under inversion: inversion reverses the signs both of the coordinates and of the operators for differentiation with respect to them, and hence the operator (26.2) remains unchanged. Put differently, inversion and angular momentum commute, so that a definite parity can coexist with definite values of L and of the projection M. Moreover, all states that differ only in M have the same parity. This is already evident from the independence of the properties of a closed system from its orientation in space; formally, it can be proved from the commutation relation $\hat{L}_+\hat{P} - \hat{P}\hat{L}_+ = 0$ by the same method used to obtain (29.3) from (29.2). There are definite *parity selection rules* for the matrix elements of various physical quantities.

Consider scalar quantities first. Here one must distinguish *true scalars*, which remain entirely unchanged under inversion, from *pseudoscalars*, which change sign under inversion (the scalar product of an axial vector and a polar vector is a pseudoscalar). The operator $\hat{f}$ of a true scalar commutes with $\hat{P}$. It follows that if the matrix of P is diagonal, then f is likewise diagonal with respect to parity: matrix elements vanish except for the $g \longrightarrow g$ and $u \longrightarrow u$ transitions (here g and u label even and odd states). For the operator of a pseudoscalar quantity, on the other hand, $\hat{P}\hat{f} = -\hat{f}\hat{P}$; the operators $\hat{P}$ and $\hat{f}$ "anticommute." The matrix element of this equality for a transition $g \longrightarrow g$ is

$$P_{gg}f_{gg} = -f_{gg}P_{gg},$$

and since $P_{gg} = 1$, we have $f_{gg} = 0$; in the same way one finds $f_{uu} = 0$. Thus the matrix of a pseudoscalar quantity has nonzero elements only for transitions in which the parity

[13] From the English word *parity*.

[14] To avoid misunderstanding, we recall that the discussion concerns nonrelativistic theory. Interactions exist in nature, and are treated in relativistic theory, that violate parity conservation.

changes. The selection rules for matrix elements of scalar quantities are therefore

$$\begin{aligned} &\text{true scalars:} && g \to g, \quad u \to u, \\ &\text{pseudoscalars:} && g \to u, \quad u \to g. \end{aligned} \tag{30.4}$$

These rules can also be obtained in another way, directly from the definition of matrix elements. Consider, for example, the integral $f_{ug} = \int \psi_u^* \hat{f} \psi_g \, dq$, where ψ_g is even and ψ_u is odd. On reversing the signs of all coordinates, the integrand changes sign if f is a true scalar. On the other hand, an integral over all space cannot change when the variables of integration are merely renamed. Hence $f_{ug} = -f_{ug}$, so that $f_{ug} = 0$.

The selection rules for vector quantities are obtained in the same way. One must remember that ordinary, polar vectors reverse sign under inversion, whereas axial vectors remain unchanged (an example is the angular-momentum vector, the cross product $\mathbf{r} \times \mathbf{p}$ of two polar vectors). Taking this into account, we find

$$\begin{aligned} &\text{polar vectors:} && g \to u, \quad u \to g, \\ &\text{axial vectors:} && g \to g, \quad u \to u. \end{aligned} \tag{30.5}$$

Let us determine the parity of a one-particle state with angular momentum l. Inversion ($x \to -x$, $y \to -y$, $z \to -z$) consists, in spherical coordinates, of the transformation

$$r \to r, \qquad \theta \to \pi - \theta, \qquad \varphi \to \varphi + \pi. \tag{30.6}$$

The angular dependence of the particle's wave function is given by the spherical function Y_{lm}, which, apart from a constant immaterial here, has the form $P_l^m(\cos\theta)e^{im\varphi}$. When φ is replaced by $\varphi + \pi$, the factor $e^{im\varphi}$ is multiplied by $(-1)^m$; when θ is replaced by $\pi - \theta$, $P_l^m(\cos\theta)$ becomes $P_l^m(-\cos\theta) = (-1)^{l-m} P_l^m(\cos\theta)$. Thus the entire function is multiplied by $(-1)^l$ (which is independent of m, in agreement with the statement above); hence the parity of a state with a given l is

$$P = (-1)^l. \tag{30.7}$$

Accordingly, even values of l yield even states and odd values yield odd ones.

For a vector physical quantity associated with a single particle, nonzero matrix elements can arise only in transitions with $l \to l, l \pm 1$ (§ 29). Combining this fact with (30.7) and with the preceding statements concerning parity changes in vector matrix elements, we conclude that matrix elements of vector quantities belonging to an individual particle are nonzero only for the transitions

$$\begin{aligned} &\text{polar vectors:} && l \to l \pm 1, \\ &\text{axial vectors:} && l \to l. \end{aligned} \tag{30.8}$$

§ 31. Addition of angular momenta

Consider a system consisting of two weakly interacting parts. When the interaction is entirely neglected, the conservation of angular momentum holds for each part independently, so that the angular momenta $\mathbf{L}_1$ and $\mathbf{L}_2$ of the two parts can be summed to give the total angular momentum $\mathbf{L}$ of the system. In the next approximation, once the weak interaction is included, the conservation laws for $\mathbf{L}_1$ and $\mathbf{L}_2$ cease to hold rigorously; nonetheless, the quantum numbers L_1 and L_2, which fix the squares of those angular momenta, remain "good" and can still serve for an approximate characterization of the system's state. In a pictorial, classical treatment of the angular momenta, one may say that in this approximation $\mathbf{L}_1$ and $\mathbf{L}_2$ precess about the direction of $\mathbf{L}$ while their magnitudes remain unchanged.

The consideration of such systems raises the question of the *rule for adding angular momenta*. What values of L are possible for given L_1 and L_2? The addition rule for the angular-momentum projections is evident: from $\hat{L}_z = \hat{L}_{1z} + \hat{L}_{2z}$ it follows that

$$M = M_1 + M_2. \tag{31.1}$$

There is no equally simple relation for the operators of the squares of the angular momenta, and we derive their "addition rule" as follows.

If the quantities $\mathbf{L}_1^2$, $\mathbf{L}_2^2$, L_{1z}, and L_{2z} are chosen as a complete set of physical quantities[15], each state is specified by the numbers L_1, L_2, M_1, and M_2. For given L_1 and L_2, the numbers M_1 and M_2 take, respectively, $2L_1 + 1$ and $2L_2 + 1$ values, so there are altogether $(2L_1 + 1)(2L_2 + 1)$ different states with the same L_1 and L_2. We denote the wave functions of the states in this description by $\varphi_{L_1 L_2 M_1 M_2}$.

Instead of the four quantities just stated, one can choose the four quantities $\mathbf{L}_1^2$, $\mathbf{L}_2^2$, $\mathbf{L}^2$, and L_z as a complete set. Each state is then characterized by the numbers L_1, L_2, L, and M (we denote the corresponding wave functions by $\psi_{L_1 L_2 L M}$). For given L_1 and L_2 there must, of course, still be $(2L_1 + 1)(2L_2 + 1)$ different states; that is, for fixed L_1 and L_2, the pair L, M can take $(2L_1 + 1)(2L_2 + 1)$ pairs of values. These values can be found by the following argument.

Adding the different allowed values of M_1 and M_2 gives the corresponding values of M:

$$\begin{array}{ccl}
M_1 & M_2 & \qquad M \\
\hline
L_1 & L_2 & \quad L_1 + L_2 \\
\begin{array}{c} L_1 \\ L_1 - 1 \end{array} & \begin{array}{c} L_2 - 1 \\ L_2 \end{array} & \left.\vphantom{\begin{array}{c} a \\ b \end{array}}\right\} L_1 + L_2 - 1 \\
\begin{array}{c} L_1 - 1 \\ L_1 \\ L_1 - 2 \end{array} & \begin{array}{c} L_2 - 1 \\ L_2 - 2 \\ L_2 \end{array} & \left.\vphantom{\begin{array}{c} a \\ b \\ c \end{array}}\right\} L_1 + L_2 - 2 \\
\cdots & \cdots & \cdots
\end{array}$$

The largest possible value of M is evidently $M = L_1 + L_2$, and it corresponds to one state φ (one pair of values M_1, M_2). Consequently, the largest possible M in the states ψ, and hence the largest L, is also $L_1 + L_2$. Next, there are two states φ with $M = L_1 + L_2 - 1$. There must therefore also be two states ψ with this value of M; one is a state with

[15] Together with a number of other quantities which, along with the four stated here, make up a complete set. These remaining quantities play no part in the following argument; for conciseness we do not mention them at all and provisionally call the system of the four stated quantities complete.

$L = L_1 + L_2$ (and $M = L - 1$), and the other has $L = L_1 + L_2 - 1$ (with $M = L$). For $M = L_1 + L_2 - 2$ there are three different states φ. This means that, in addition to $L = L_1 + L_2$ and $L = L_1 + L_2 - 1$, the value $L = L_1 + L_2 - 2$ is also possible.

This argument can be continued in the same way as long as decreasing M by one increases by one the number of states with that value of M. It is easy to see that this continues until M reaches $|L_1 - L_2|$. As M is reduced further, the number of states ceases to increase and remains equal to $2L_2 + 1$ (if $L_2 \leq L_1$). Thus $|L_1 - L_2|$ is the smallest possible value of L.

We therefore obtain the result that, for given L_1 and L_2, L can take the values

$$L = L_1 + L_2, L_1 + L_2 - 1, \ldots, |L_1 - L_2|, \tag{31.2}$$

altogether $2L_2 + 1$ different values when $L_2 \leq L_1$. One readily checks that these indeed give $(2L_1 + 1)(2L_2 + 1)$ different pairs of values M, L. It is important to note that, apart from the $2L + 1$ different values of M for a given L, each possible value of L in (31.2) corresponds to just one state.

This result can be represented pictorially by the so-called *vector model*. Introduce two vectors $\mathbf{L}_1$ and $\mathbf{L}_2$ of lengths L_1 and L_2. The values of L are represented by the integer-valued lengths of vectors $\mathbf{L}$ obtained by vector addition of $\mathbf{L}_1$ and $\mathbf{L}_2$; the largest value, $L_1 + L_2$, is obtained when $\mathbf{L}_1$ and $\mathbf{L}_2$ are parallel, and the smallest, $|L_1 - L_2|$, when they are antiparallel.

In states with definite angular momenta L_1, L_2, and definite total angular momentum L, the scalar products $\mathbf{L}_1\mathbf{L}_2$, $\mathbf{L}\mathbf{L}_1$, and $\mathbf{L}\mathbf{L}_2$ also have definite values. These values are readily found. To calculate $\mathbf{L}_1\mathbf{L}_2$, write $\hat{\mathbf{L}} = \hat{\mathbf{L}}_1 + \hat{\mathbf{L}}_2$, or, on squaring and transposing terms,

$$2\hat{\mathbf{L}}_1\hat{\mathbf{L}}_2 = \hat{\mathbf{L}}^2 - \hat{\mathbf{L}}_1^2 - \hat{\mathbf{L}}_2^2.$$

Replacing the operators on the right by their eigenvalues gives the eigenvalue of the operator on the left:

$$\mathbf{L}_1\mathbf{L}_2 = \frac{1}{2}\{L(L+1) - L_1(L_1+1) - L_2(L_2+1)\}. \tag{31.3}$$

Similarly, we find

$$\mathbf{L}\mathbf{L}_1 = \frac{1}{2}\{L(L+1) + L_1(L_1+1) - L_2(L_2+1)\}. \tag{31.4}$$

Let us now establish the *rule for combining parities*. For a system that consists of two independent parts, the wave function Ψ equals the product of the individual wave functions Ψ_1 and Ψ_2. Hence, if the latter two have the same parity (both change sign, or both remain unchanged, when all coordinates change sign), the wave function of the whole system is even. If, on the other hand, Ψ_1 and Ψ_2 have opposite parities, Ψ is odd. These statements can be written as

$$P = P_1 P_2, \tag{31.5}$$

where P is the parity of the whole system, and P_1 and P_2 are the parities of its parts. This rule plainly generalizes directly to a system consisting of any number of noninteracting parts.

In particular, for a system of particles in a centrally symmetric field (with the interaction between the particles treated as weak), the parity of the state of the whole system is

$$P = (-1)^{l_1+l_2+\ldots} \tag{31.6}$$

(see (30.7)). Note that the exponent involves the algebraic sum of the particle angular momenta, a quantity that, generally speaking, does not coincide with their "vector sum," i.e., with the angular momentum L of the system.

If a closed system decays into parts under the action of forces within the system itself, its total angular momentum and parity must be conserved. This may make the decay impossible even when it is energetically possible.

For example, consider an atom in an even state with angular momentum $L = 0$, whose energy would permit it to break up into a free electron and an ion occupying an odd state, also with $L = 0$. It is easy to show that this process is in fact impossible (one calls it *forbidden*). Indeed, by conservation of angular momentum the free electron would also have to possess zero angular momentum and would therefore be in an even state ($P = (-1)^0 = +1$). But the state ion+free electron would then be odd, whereas the initial state of the atom was even.

CHAPTER V

Motion in a Centrally Symmetric Field

§ 32. Motion in a central field

As in classical mechanics, the quantum-mechanical problem of two interacting particles can be reduced to a one-particle problem. For masses m_1, m_2 interacting through $U(r)$, where r is their separation,

$$\hat{H} = -\frac{\hbar^2}{2m_1}\Delta_1 - \frac{\hbar^2}{2m_2}\Delta_2 + U(r). \tag{32.1}$$

Here Δ_1, Δ_2 are the Laplace operators in the particle coordinates. Replace $\mathbf{r}_1, \mathbf{r}_2$ by

$$\mathbf{r} = \mathbf{r}_2 - \mathbf{r}_1, \qquad \mathbf{R} = \frac{m_1\mathbf{r}_1 + m_2\mathbf{r}_2}{m_1 + m_2}. \tag{32.2}$$

Thus $\mathbf{r}$ is the relative radius vector and $\mathbf{R}$ the center-of-mass radius vector. Direct calculation gives

$$\hat{H} = -\frac{\hbar^2}{2(m_1 + m_2)}\Delta_R - \frac{\hbar^2}{2m}\Delta + U(r), \tag{32.3}$$

where $m = m_1m_2/(m_1 + m_2)$ is the reduced mass. The Hamiltonian separates into two independent parts. Accordingly, $\psi(\mathbf{r}_1, \mathbf{r}_2) = \varphi(\mathbf{R})\psi(\mathbf{r})$: the first factor describes free motion of the center of mass, of mass $m_1 + m_2$, and the second describes relative motion as that of mass m in $U(r)$.

The Schrödinger equation in a central field is

$$\Delta\psi + \frac{2m}{\hbar^2}[E - U(r)]\psi = 0. \tag{32.4}$$

In spherical coordinates this becomes

$$\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial\psi}{\partial r}\right) + \frac{1}{r^2}\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}(\sin\theta\frac{\partial\psi}{\partial\theta}) + \frac{1}{\sin^2\theta}\frac{\partial^2\psi}{\partial\varphi^2}\right] + \frac{2m}{\hbar^2}[E - U(r)]\psi = 0. \tag{32.5}$$

Introducing the squared-angular-momentum operator (26.16),[1]

$$\frac{\hbar^2}{2m}\left[-\frac{1}{r^2}\frac{\partial}{\partial r}(r^2\frac{\partial\psi}{\partial r}) + \frac{\hat{\mathbf{l}}^2}{r^2}\psi\right] + U(r)\psi = E\psi. \tag{32.6}$$

---

[1] If the radial momentum operator is defined by

$$\hat{p}_r\psi = -i\hbar\frac{1}{r}\frac{\partial}{\partial r}(r\psi) = -i\hbar(\frac{\partial}{\partial r} + \frac{1}{r})\psi,$$

then

$$\hat{H} = \frac{1}{2m}(\hat{p}_r^2 + \frac{\hbar^2\hat{\mathbf{l}}^2}{r^2}) + U(r),$$

formally the same as the classical Hamilton function in spherical coordinates.

Angular momentum is conserved in a central field. For stationary states with definite l, m, these values fix the angular dependence. Seek

$$\psi = R(r)Y_{lm}(\theta, \varphi), \tag{32.7}$$

where Y_{lm} are spherical functions. Since $\hat{\mathbf{l}}^2 Y_{lm} = l(l+1)Y_{lm}$, the radial function obeys

$$\frac{1}{r^2}\frac{d}{dr}(r^2\frac{dR}{dr}) - \frac{l(l+1)}{r^2}R + \frac{2m}{\hbar^2}[E - U(r)]R = 0. \tag{32.8}$$

This contains no $l_z = m$, consistently with the $(2l+1)$-fold directional degeneracy. Put

$$R(r) = \frac{\chi(r)}{r}. \tag{32.9}$$

Then

$$\frac{d^2\chi}{dr^2} + [\frac{2m}{\hbar^2}(E - U) - \frac{l(l+1)}{r^2}]\chi = 0. \tag{32.10}$$

If $U(r)$ is finite everywhere, ψ and hence R must remain finite at the origin, so

$$\chi(0) = 0. \tag{32.11}$$

This condition in fact also holds for fields diverging at the origin (§ 35).

Equation (32.10) is formally the one-dimensional Schrödinger equation with

$$U_l(r) = U(r) + \frac{\hbar^2}{2m}\frac{l(l+1)}{r^2}, \tag{32.12}$$

the sum of $U(r)$ and the centrifugal energy

$$\frac{\hbar^2 l(l+1)}{2mr^2} = \frac{\hbar^2\hat{\mathbf{l}}^2}{2mr^2}.$$

Thus the central-field problem reduces to one-dimensional motion on a region bounded on one side by the condition at $r = 0$. The normalization is likewise one-dimensional:

$$\int_0^\infty |R|^2 r^2 dr = \int_0^\infty |\chi|^2 dr.$$

Energy levels for one-dimensional motion in such a region are nondegenerate (§ 21), so E completely determines the radial solution of (32.10). The angular part is fixed by l, m. Hence E, l, m completely determine the wave function: energy, squared angular momentum, and its projection form a complete set of physical quantities. The one-dimensional reduction permits use of the oscillation theorem (§ 21). At fixed l, order the discrete energy eigenvalues increasingly and number them $n_r = 0, 1, \ldots$. Then n_r is the number of finite-r nodes of the radial function, excluding $r = 0$, and is called the *radial quantum number*. The numbers l and m are also called the *azimuthal* and *magnetic quantum numbers*.

States with different l are conventionally denoted by

$$\begin{array}{c|cccccccc|c} l & 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 & \ldots \\ & s & p & d & f & g & h & i & k & \ldots \end{array} \tag{32.13}$$

The normal state in a central field is always an s state: for $l \neq 0$ the angular wave function necessarily has nodes, whereas the normal-state wave function has none. The lowest possible eigenvalue at fixed l also rises with l, since angular momentum adds the positive term $\hbar^2 l(l+1)/(2mr^2)$ to the Hamiltonian.

To find the radial function near the origin, assume

$$\lim_{r\to 0} U(r)r^2 = 0. \tag{32.14}$$

Take the leading term as $R = \text{const}\cdot r^s$. From (32.8), after multiplication by r^2 and passage to $r \to 0$,

$$\frac{d}{dr}(r^2\frac{dR}{dr}) - l(l+1)R = 0,$$

so $s(s+1) = l(l+1)$ and

$$s = l \quad \text{or} \quad s = -(l+1).$$

The latter solution is inadmissible, since it diverges at $r = 0$. Therefore

$$R_l \approx \text{const}\cdot r^l. \tag{32.15}$$

The probability of finding the particle between r and $r+dr$ is determined by $r^2|R|^2$ and is proportional to $r^{2(l+1)}$; it vanishes at the origin more rapidly for larger l.

§ 33. Spherical waves

The plane wave

$$\psi_{\mathbf{p}} = \text{const}\cdot\exp(i\mathbf{pr}/\hbar)$$

describes a free particle with definite momentum $\mathbf{p}$ and energy $E = p^2/2m$. We now consider stationary free-particle states having definite angular momentum and projection as well as energy. It is convenient to use the wave vector

$$k = \frac{p}{\hbar} = \frac{\sqrt{2mE}}{\hbar}. \tag{33.1}$$

The state with angular momentum l and projection m has

$$\psi_{klm} = R_{kl}(r)Y_{lm}(\theta,\varphi), \tag{33.2}$$

where

$$R''_{kl} + \frac{2}{r}R'_{kl} + [k^2 - \frac{l(l+1)}{r^2}]R_{kl} = 0. \tag{33.3}$$

The continuum wave functions satisfy

$$\int \psi^*_{k'l'm'}\psi_{klm}dV = \delta_{ll'}\delta_{mm'}\delta((k'-k)/2\pi).$$

Angular functions ensure orthogonality in l, m; the radial functions must obey

$$\int_0^\infty r^2 R_{k'l}R_{kl}dr = \delta((k'-k)/2\pi) = 2\pi\delta(k'-k). \tag{33.4}$$

For normalization on the energy scale instead, $\int r^2 R_{E'l} R_{El} dr = \delta(E' - E)$, (5.14) gives

$$R_{El} = R_{kl} \left(\frac{1}{2\pi} \frac{dk}{dE}\right)^{1/2} = \frac{1}{\hbar} \sqrt{\frac{m}{2\pi k}} R_{kl}. \tag{33.5}$$

For $l = 0$, (33.3) reads $d^2(rR_{k0})/dr^2 + k^2 r R_{k0} = 0$. The finite solution normalized by (33.4) is

$$R_{k0} = 2\frac{\sin kr}{r}. \tag{33.6}$$

For $l \neq 0$, put

$$R_{kl} = r^l \chi_{kl}. \tag{33.7}$$

Then

$$\chi''_{kl} + \frac{2(l+1)}{r} \chi'_{kl} + k^2 \chi_{kl} = 0.$$

Differentiation gives

$$\chi'''_{kl} + \frac{2(l+1)}{r} \chi''_{kl} + [k^2 - \frac{2(l+1)}{r^2}] \chi'_{kl} = 0.$$

Putting $\chi'_{kl} = r\chi_{k,l+1}$ yields the required equation for $\chi_{k,l+1}$, so

$$\chi_{k,l+1} = \frac{1}{r} \chi'_{kl}. \tag{33.8}$$

Hence

$$\chi_{kl} = \left(\frac{1}{r} \frac{d}{dr}\right)^l \chi_{k0},$$

where $\chi_{k0} = R_{k0}$ from (33.6), up to an arbitrary constant. Thus

$$R_{kl} = (-1)^l \, 2\frac{r^l}{k^l} \left(\frac{1}{r} \frac{d}{dr}\right)^l \frac{\sin kr}{r}. \tag{33.9}$$

The factor k^{-l} normalizes the function, and $(-1)^l$ is convenient. Equivalently,

$$R_{kl} = \sqrt{\frac{2\pi k}{r}} J_{l+1/2}(kr) = 2k j_l(kr), \tag{33.10}$$

where the spherical Bessel functions are

$$j_l(x) = \sqrt{\frac{\pi}{2x}} J_{l+1/2}(x). \tag{33.11}$$

[2]

[2]The first few are

$$j_0 = \frac{\sin x}{x}, \quad j_1 = \frac{\sin x}{x^2} - \frac{\cos x}{x}, \quad j_2 = \left(\frac{3}{x^3} - \frac{1}{x}\right) \sin x - \frac{3\cos x}{x^2}.$$

Another convention in the literature differs from (33.11) by a factor x.

At large r, the slowest-decaying term in (33.9) comes from differentiating the sine l times. Every operation $-d/dr$ shifts its argument by $-\pi/2$, giving

$$R_{kl} \approx \frac{2}{r}\sin\left(kr - \frac{\pi l}{2}\right). \tag{33.12}$$

Normalization may be performed from the asymptotic form (§ 21); comparison with (33.6) confirms the coefficient in (33.9). Near the origin, expansion of $\sin kr$ and retention of the term giving the lowest power after differentiation yields[3]

$$(\frac{1}{r}\frac{d}{dr})^l \frac{\sin kr}{r} \approx (\frac{1}{r}\frac{d}{dr})^l \frac{(-1)^l (kr)^{2l+1}}{r(2l+1)!} = \frac{(-1)^l k^{2l+1}}{(2l+1)!!}.$$

Therefore

$$R_{kl} \approx \frac{2k^{l+1}}{(2l+1)!!} r^l, \tag{33.13}$$

in agreement with (32.15).

Scattering theory also uses wave functions which are not finite in the usual sense, but describe particles emitted from or incident on the center. For $l = 0$, replace the standing wave (33.6) by an outgoing or incoming wave:

$$R^{\pm}_{k0} = \frac{A}{r} e^{\pm ikr}. \tag{33.14}$$

For arbitrary l,

$$R^{\pm}_{kl} = (-1)^l A \frac{r^l}{k^l} (\frac{1}{r}\frac{d}{dr})^l \frac{e^{\pm ikr}}{r}. \tag{33.15}$$

In terms of Hankel functions,

$$R^{\pm}_{kl} = \pm iA \sqrt{\frac{k\pi}{2r}} H^{(1,2)}_{l+1/2}(kr), \tag{33.16}$$

of the first and second kind for + and − respectively. Their asymptotic forms are

$$R^{\pm}_{kl} \approx \frac{A}{r} \exp\left[\pm i(kr - \frac{\pi l}{2})\right], \tag{33.17}$$

and near the origin

$$R^{\pm}_{kl} \approx A \frac{(2l-1)!!}{k^l} r^{-l-1}. \tag{33.18}$$

Normalize these functions to emission or absorption of one particle per unit time. Far away, each small part of a spherical wave is locally plane, with current density $j = v\psi\psi^*$ and $v = k\hbar/m$. The condition $\oint j df = 1$, or $\int jr^2 do = 1$, with do the solid-angle element, and the unchanged angular normalization require

$$A = \frac{1}{\sqrt{v}} = \sqrt{\frac{m}{k\hbar}}. \tag{33.19}$$

[3] !! denotes the product of all integers of the same parity up to and including the stated one.

An asymptotic expression like (33.12) also holds for positive-energy motion in any field decreasing sufficiently rapidly with distance.[4] At large r, neglecting both the field and centrifugal energy leaves

$$\frac{1}{r}\frac{d^2(rR_{kl})}{dr^2} + k^2 R_{kl} = 0.$$

Its general solution is

$$R_{kl} \approx 2\frac{\sin(kr - l\pi/2 + \delta_l)}{r}, \tag{33.20}$$

where δ_l is the *phase shift*; the coefficient corresponds to $k/2\pi$ normalization.[5] The exact boundary condition of finiteness at the origin fixes δ_l. These phases depend on both l and k and are essential characteristics of continuum eigenfunctions.

Problem

1. Find the energy levels for $l = 0$ motion in the spherical square well $U(r) = -U_0$ for $r < a$, $U(r) = 0$ for $r > a$.

Solution. For $l = 0$ the wave functions depend only on r. Inside,

$$\frac{1}{r}\frac{d^2}{dr^2}(r\psi) + k^2\psi = 0, \qquad k = \frac{1}{\hbar}\sqrt{2m(U_0 - |E|)},$$

and the finite solution is $\psi = A\sin(kr)/r$. Outside,

$$\frac{1}{r}\frac{d^2}{dr^2}(r\psi) - \kappa^2\psi = 0, \qquad \kappa = \frac{1}{\hbar}\sqrt{2m|E|},$$

and the decaying solution is $\psi = A'e^{-\kappa r}/r$. Continuity of the logarithmic derivative of $r\psi$ at a gives

$$k\cot ka = -\kappa = -\sqrt{\frac{2mU_0}{\hbar^2} - k^2}, \tag{1}$$

or

$$\sin ka = \pm\sqrt{\frac{\hbar^2}{2ma^2U_0}}\,ka. \tag{2}$$

These equations implicitly determine the levels, using roots for which $\cot ka < 0$. The first $l = 0$ level is also the deepest level of all, the particle's normal state.

If the well is too shallow, there are no negative-energy levels. Graphically, the roots of $\pm\sin x = \alpha x$ are intersections of $y = \alpha x$ with $y = \pm\sin x$, retaining only portions with $\cot x < 0$, shown solid in Fig. 9.

[4] It will be shown in § 124 that the field must decrease faster than $1/r$.

[5] The term $-l\pi/2$ is included so that $\delta_l = 0$ in the absence of a field. Since the overall sign of a wave function is immaterial, δ_l is defined modulo π, not 2π, and can always be put between 0 and π.

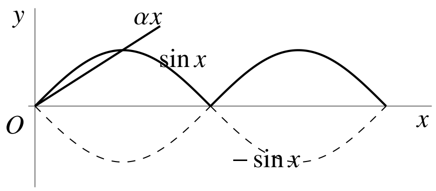

Fig. 9

For excessive α (small U_0) there are no intersections. The first appears at $\alpha = 2/\pi$, $x = \pi/2$. With $\alpha = \hbar/\sqrt{2ma^2U_0}$ and $x = ka$, the minimum depth is

$$U_{0\,\min} = \frac{\pi^2\hbar^2}{8ma^2}. \tag{3}$$

It increases as a decreases. At appearance, $ka = \pi/2$ and $E_1 = 0$. With further deepening, E_1 falls; for small $\Delta = U_0/U_{0\,\min} - 1$,

$$-E_1 = (\pi^2/16)U_{0\,\min}\Delta^2. \tag{4}$$

2. Find the order of levels with different l in a very deep spherical well, $U_0 \gg \hbar^2/ma^2$ (*W. Elsasser*, 1933).

Solution. As $U_0 \to \infty$, the boundary condition is $\psi = 0$ at the wall (§ 22). Using (33.10) inside gives

$$J_{l+1/2}(ka) = 0.$$

Its roots determine the levels above the well bottom, $U_0 - |E| = \hbar^2k^2/2m$. Starting from the ground state, their order is

$$1s,\ 1p,\ 1d,\ 2s,\ 1f,\ 2p,\ 1g,\ 2d,\ 1h,\ 3s,\ 2f, \dots$$

The preceding numeral orders levels having the same l.[6]

3. Find the order in which levels with different l appear as the well depth U_0 increases.

Solution. At first appearance a new level has $E = 0$. Outside, the solution vanishing at infinity is $R_l = \text{const}\cdot r^{-(l+1)}$. Continuity of R_l, R_l' implies

$$(r^{l+1}R_l)' = 0 \qquad \text{at } r = a.$$

This is equivalent[7] to $R_{l-1} = 0$, hence

$$J_{l-1/2}\Big(\frac{a}{\hbar}\sqrt{2mU_0}\Big) = 0;$$

for $l = 0$, replace $J_{l-1/2}$ by cos. The order of appearance is

$$1s,\ 1p,\ 1d,\ 2s,\ 1f,\ 2p,\ 1g,\ 2d,\ 3s,\ 1h,\ 2f, \dots$$

Differences from the deep-well ordering begin only for rather high levels.

[6] This notation is used for particle levels in nuclei (§ 118).

[7] From (33.7), (33.8), $(r^{-l}R_l)' \sim r^{-l}R_{l+1}$. Since (33.3) is invariant under $l \mapsto -l-1$, also $(r^{l+1}R_{-l-1})' \sim r^{l+1}R_{-l}$. Since R_{-l} and R_{l-1} obey the same equation, finally $(r^{l+1}R_l)' \sim r^{l+1}R_{l-1}$, as used here.

4. Find the energy levels of the spatial oscillator $U = \frac{1}{2}m\omega^2 r^2$, their degeneracies, and the allowed orbital angular momenta.

SOLUTION. Separation in Cartesian coordinates gives three linear oscillators, hence

$$E = \hbar\omega(n_1 + n_2 + n_3 + 3/2) \equiv \hbar\omega(n + 3/2).$$

The degeneracy is the number of ways of writing n as a sum of three nonnegative integers,[8] namely

$$\frac{(n+1)(n+2)}{2}.$$

The stationary wave functions are

$$\psi_{n_1 n_2 n_3} = \text{const}\cdot e^{-\alpha^2 r^2/2} H_{n_1}(\alpha x) H_{n_2}(\alpha y) H_{n_3}(\alpha z), \tag{5}$$

where $\alpha = \sqrt{m\omega/\hbar}$. Since H_n gains $(-1)^n$ under a coordinate sign change, (5) has parity $(-1)^n$. Linear combinations at fixed $n_1 + n_2 + n_3 = n$ yield

$$\psi_{nlm} = \text{const}\cdot r^l e^{-\alpha^2 r^2/2} Y_{lm} F(-(n-l)/2, l + 3/2, \alpha^2 r^2), \tag{6}$$

where $|m| = 0, 1, \ldots, l$. The values of l are $0, 2, \ldots, n$ for even n and $1, 3, \ldots, n$ for odd n, as also follows by matching parities $(-1)^n$ and $(-1)^l$. Thus the level sequence is

$$(1s),\ (1p),\ (1d, 2s),\ (1f, 2p),\ (1g, 2d, 3s), \ldots,$$

with mutually degenerate states in parentheses.[9]

§ 34. Expansion of a plane wave

Consider a free particle moving with definite momentum $p = k\hbar$ along the positive z axis. Its wave function is

$$\psi = \text{const}\cdot e^{ikz}.$$

Expand it in the free-motion functions ψ_{klm} with definite angular momentum. Since $E = k^2\hbar^2/2m$ is definite, only functions with the same k enter. Since e^{ikz} is axially symmetric about z, only functions independent of φ, with $m = 0$, can occur. Thus

$$e^{ikz} = \sum_{l=0}^{\infty} a_l \psi_{kl0} = \sum_{l=0}^{\infty} a_l R_{kl} Y_{l0}.$$

Substitution of (28.8) and (33.9) gives

$$e^{ikz} = \sum_{l=0}^{\infty} C_l P_l(\cos\theta) \left(\frac{r}{k}\right)^l \left(\frac{1}{r}\frac{d}{dr}\right)^l \frac{\sin kr}{kr} \qquad (z = r\cos\theta).$$

Determine C_l by comparing coefficients of $(r\cos\theta)^n$ in the power-series expansions. On the right this term occurs only in the nth summand: for $l > n$ the radial expansion

[8] Equivalently, the number of ways to distribute n identical balls among three boxes.

[9] Note the degeneracy of levels with different l; see the note on p. 164.

begins with higher powers of r, while for $n > l$, $P_l(\cos\theta)$ has lower powers of $\cos\theta$. The coefficient of $\cos^l\theta$ in $P_l(\cos\theta)$ is $(2l)!/2^l(l!)^2$ (see (c.1)). By (33.13), the relevant term is

$$(-1)^l C_l \frac{(2l)!(kr\cos\theta)^l}{2^l(l!)^2\, 1\cdot 3\ldots(2l+1)}.$$

The corresponding term in $\exp(ikr\cos\theta)$ is

$$\frac{(ikr\cos\theta)^l}{l!}.$$

Hence $C_l = (-i)^l(2l+1)$, and

$$e^{ikz} = \sum_{l=0}^{\infty}(-i)^l(2l+1)P_l(\cos\theta)\left(\frac{r}{k}\right)^l\left(\frac{1}{r}\frac{d}{dr}\right)^l\frac{\sin kr}{kr}. \tag{34.1}$$

At large distances,

$$e^{ikz} \approx \frac{1}{kr}\sum_{l=0}^{\infty} i^l(2l+1)P_l(\cos\theta)\sin\left(kr - \frac{l\pi}{2}\right). \tag{34.2}$$

In (34.1), z is along the wave vector $\mathbf{k}$. The expansion also has a coordinate-independent form. The addition theorem for spherical functions (c.11) expresses $P_l(\cos\theta)$ through spherical functions of the directions of $\mathbf{k}$ and $\mathbf{r}$:

$$e^{i\mathbf{kr}} = 4\pi\sum_{l=0}^{\infty}\sum_{m=-l}^{l} i^l j_l(kr)Y^*_{lm}\left(\frac{\mathbf{k}}{k}\right)Y_{lm}\left(\frac{\mathbf{r}}{r}\right). \tag{34.3}$$

Since $j_l(kr)$ depends only on kr, the symmetry in $\mathbf{k}$ and $\mathbf{r}$ is explicit; which spherical function is conjugated is immaterial.

Normalize e^{ikz} to unit probability-current density: one particle crosses unit area per unit time. The function is

$$\psi = \frac{1}{\sqrt{v}}e^{ikz} = \sqrt{\frac{m}{k\hbar}}e^{ikz} \tag{34.4}$$

(v is the speed; see (19.7)). Multiplying (34.1) by $\sqrt{m/k\hbar}$ and using $\psi^{\pm}_{klm} = R^{\pm}_{kl}Y_{lm}$ gives

$$\psi = \sum_{l=0}^{\infty}\sqrt{\pi(2l+1)}\,\frac{1}{ik\hbar}(\psi^{+}_{kl0} - \psi^{-}_{kl0}).$$

The squared coefficient of ψ^{-}_{kl0} (or ψ^{+}_{kl0}) gives the probability that a particle incident on the center (or leaving it) has angular momentum l. Since the incident flux density is unity, this probability has dimensions of area and can be interpreted as the target area in the xy plane for particles with angular momentum l. Denoting it by σ_l,

$$\sigma_l = \frac{\pi}{k^2}(2l+1). \tag{34.5}$$

For large l, summing over a range Δl, where $1 \ll \Delta l \ll l$, gives

$$\sum_{\Delta l} \sigma_l \approx \frac{\pi}{k^2} \cdot 2l\Delta l = 2\pi \frac{l\hbar^2}{p^2} \Delta l.$$

With the classical relation $\hbar l = \rho p$, where ρ is the impact parameter, this becomes

$$2\pi\rho\Delta\rho,$$

the classical result. This is not accidental: for large l the motion is semiclassical (§ 49).

Problem

Problem

Expand a plane wave in wave functions with definite projection m of angular momentum on the y axis and definite momentum projection p_y on that axis.

Solution. Introduce cylindrical coordinates y, ρ, φ with axis y. The functions have the form $Q_m(\rho)e^{im\varphi}e^{ip_y y/\hbar}$. Measuring φ from z,

$$e^{ikz} = e^{ik\rho\cos\varphi} = \sum_{m=-\infty}^{\infty} Q_m(\rho)e^{im\varphi}$$

(here $p_y = 0$), whence

$$Q_m(\rho) = \frac{1}{2\pi}\int_0^{2\pi} \exp[i(k\rho\cos\varphi - m\varphi)]\, d\varphi = i^m J_m(k\rho),$$

where J_m is a Bessel function. For $k\rho \gg 1$,

$$Q_m(\rho) \approx i^m \sqrt{\frac{2}{\pi k\rho}} \sin\left[k\rho - \frac{\pi}{2}\left(m - \frac{1}{2}\right)\right].$$

§ 35. Fall of a particle to the center

To clarify certain features of quantum-mechanical motion, it is useful to study a case which, although it has no immediate physical realization, involves a particle moving in a field whose potential energy becomes infinite at a point, the origin, according to $U(r) \approx -\beta/r^2$ with $\beta > 0$. The form of the field far from the origin will not matter. We saw in § 18 that this case lies precisely between those admitting ordinary stationary states and those in which the particle falls to the origin.

Near the origin the Schrödinger equation is

$$R'' + \frac{2}{r}R' + \frac{\gamma}{r^2}R = 0, \tag{35.1}$$

where $R(r)$ is the radial wave function and

$$\gamma = \frac{2m\beta}{\hbar^2} - l(l+1). \tag{35.2}$$

All terms of lower order in $1/r$ have been omitted. The energy E is assumed finite, so its term has likewise been dropped.

Put $R \sim r^s$. Then s satisfies

$$s(s+1) + \gamma = 0,$$

with roots

$$s_1 = -\frac{1}{2} + \sqrt{\frac{1}{4} - \gamma}, \qquad s_2 = -\frac{1}{2} - \sqrt{\frac{1}{4} - \gamma}. \tag{35.3}$$

For the subsequent analysis, isolate a small region of radius r_0 about the origin and replace $-\gamma/r^2$ inside it by the constant $-\gamma/r_0^2$. After determining the wave functions in this cut-off field, we shall examine the limit $r_0 \to 0$.

First suppose $\gamma < 1/4$. Then s_1 and s_2 are real negative numbers, with $s_1 > s_2$. For $r > r_0$, the general small-r solution is

$$R = Ar^{s_1} + Br^{s_2}, \tag{35.4}$$

where A and B are constants. For $r < r_0$, the solution finite at the origin of

$$R'' + \frac{2}{r}R' + \frac{\gamma}{r_0^2}R = 0$$

is

$$R = C\frac{\sin kr}{r}, \qquad k = \frac{\sqrt{\gamma}}{r_0}. \tag{35.5}$$

At $r = r_0$, both R and R' must be continuous. One condition may conveniently be written as continuity of the logarithmic derivative of rR, which gives

$$\frac{A(s_1+1)r_0^{s_1} + B(s_2+1)r_0^{s_2}}{Ar_0^{s_1+1} + Br_0^{s_2+1}} = k \cot kr_0,$$

or

$$\frac{A(s_1+1)r_0^{s_1} + B(s_2+1)r_0^{s_2}}{Ar_0^{s_1} + Br_0^{s_2}} = \sqrt{\gamma}\cot\sqrt{\gamma}.$$

Solving for B/A yields an expression of the form

$$\frac{B}{A} = \text{const} \cdot r_0^{s_1 - s_2}. \tag{35.6}$$

As $r_0 \to 0$, $B/A \to 0$ because $s_1 > s_2$. Of the two solutions of (35.1) that diverge at the origin, one must therefore choose the less rapidly divergent:

$$R = A\frac{1}{r^{|s_1|}}. \tag{35.7}$$

Now let $\gamma > 1/4$. The roots are complex:

$$s_1 = -\frac{1}{2} + i\sqrt{\gamma - \frac{1}{4}}, \qquad s_2 = s_1^*.$$

Repeating the argument again gives (35.6), which now becomes

$$\frac{B}{A} = \text{const} \cdot r_0^{i\sqrt{4\gamma-1}}. \tag{35.8}$$

This expression has no definite limit as $r_0 \to 0$, so a direct limiting procedure is impossible. Taking (35.8) into account, a real solution has the general form

$$R = \text{const} \cdot \frac{1}{\sqrt{r}} \cos\left(\sqrt{\gamma - \frac{1}{4}} \ln \frac{r}{r_0} + \text{const}\right). \tag{35.9}$$

The number of zeros of this function grows without bound as r_0 decreases. On the one hand, (35.9) applies at sufficiently small r for every finite particle energy E; on the other, the ground-state wave function must have no zeros. We therefore conclude that the ground state in this field corresponds to $E = -\infty$. In every discrete-spectrum state the particle is found mainly where $E > U$. Thus as $E \to -\infty$ it is confined to an infinitesimal region about the origin: the particle falls to the center.

The critical field U_{cr} at which fall to the center becomes possible corresponds to $\gamma = 1/4$. The smallest coefficient of $-1/r^2$ occurs for $l = 0$, and hence

$$U_{\text{cr}} = -\frac{\hbar^2}{8mr^2}. \tag{35.10}$$

Formula (35.3), with s_1, shows that the admissible Schrödinger solution near a point where $U \sim 1/r^2$ diverges no faster than $1/\sqrt{r}$ as $r \to 0$. If the field becomes infinite more slowly than $1/r^2$, then near the origin $U(r)$ may be neglected relative to the remaining terms. The free-motion solutions result, namely $\psi \sim r^l$ (§ 33). If instead the field diverges faster than $1/r^2$, as does $-1/r^s$ with $s > 2$, the wave function near the origin is proportional to $r^{s/4-1}$; see the problem in § 49. In every such case $r\psi$ vanishes at $r = 0$.

Next consider a field that at large distances decreases as $U \approx -\beta/r^2$ but is arbitrary at small distances. Suppose first that $\gamma < 1/4$. Only finitely many negative-energy levels can then exist[10]. Indeed, at $E = 0$ the large-distance Schrödinger equation is (35.1), with general solution (35.4). This function has no zeros for $r \neq 0$. All zeros of the required radial wave function therefore lie at finite distances from the origin, and their number is finite. In other words, the ordinal number of the level $E = 0$ that closes the discrete spectrum is finite.

For $\gamma > 1/4$, by contrast, the discrete spectrum contains infinitely many negative-energy levels. At $E = 0$ the large-distance wave function has the form (35.9), with infinitely many zeros, so its ordinal number is infinite. Finally, let $U = -\beta/r^2$ throughout space. If $\gamma > 1/4$, the particle falls to the center. If $\gamma < 1/4$, there are no negative-energy levels at all. The $E = 0$ wave function is then of the form (35.7) everywhere and has no zeros at finite distances; it therefore corresponds to the lowest level for the specified l.

[10] At small r the field is assumed such that the particle does not fall to the center.

§ 36. Motion in a Coulomb field (spherical coordinates)

An especially important case of motion in a central field is motion in a *Coulomb field*,

$$U = \pm\frac{\alpha}{r},$$

where $\alpha > 0$. We first consider attraction and write $U = -\alpha/r$. General considerations already show that the negative-energy spectrum is discrete, with infinitely many levels, whereas the positive-energy spectrum is continuous.

Equation (32.8) for the radial functions becomes

$$\frac{d^2R}{dr^2} + \frac{2}{r}\frac{dR}{dr} - \frac{l(l+1)}{r^2}R + \frac{2m}{\hbar^2}\left(E + \frac{\alpha}{r}\right)R = 0. \tag{36.1}$$

For the relative motion of two attracting particles, m denotes their reduced mass.

Calculations in a Coulomb field are conveniently made in special *Coulomb units*. We choose as the units of mass, length, and time

$$m, \qquad \frac{\hbar^2}{m\alpha}, \qquad \frac{\hbar^3}{m\alpha^2}.$$

All other units follow; the energy unit is $m\alpha^2/\hbar^2$. These units are used throughout this and the next section unless stated otherwise.[11] In these units (36.1) is

$$\frac{d^2R}{dr^2} + \frac{2}{r}\frac{dR}{dr} - \frac{l(l+1)}{r^2}R + 2\left(E + \frac{1}{r}\right)R = 0. \tag{36.2}$$

Discrete spectrum. Introduce

$$n = \frac{1}{\sqrt{-2E}}, \qquad \rho = \frac{2r}{n}. \tag{36.3}$$

For negative energy n is real and positive. Equation (36.2) becomes

$$R'' + \frac{2}{\rho}R' + \left[-\frac{1}{4} + \frac{n}{\rho} - \frac{l(l+1)}{\rho^2}\right]R = 0, \tag{36.4}$$

where primes mean differentiation with respect to ρ.

At small ρ the admissible solution is proportional to ρ^l (see (32.15)). At large ρ, omission of the $1/\rho$ and $1/\rho^2$ terms gives $R'' = R/4$, hence $R = e^{\pm\rho/2}$. The solution vanishing at infinity therefore behaves as $e^{-\rho/2}$. Put

$$R = \rho^l e^{-\rho/2}\omega(\rho). \tag{36.5}$$

Then

$$\rho\omega'' + (2l + 2 - \rho)\omega' + (n - l - 1)\omega = 0. \tag{36.6}$$

[11] For an electron of mass $m = 9.11 \cdot 10^{-28}$ g and $\alpha = e^2$, the Coulomb units coincide with the *atomic units*. The atomic length unit is $\hbar^2/me^2 = 0.529 \cdot 10^{-8}$ cm, the *Bohr radius*. The atomic energy unit is $me^4/\hbar^2 = 4.36 \cdot 10^{-11}$ erg = 27.21 eV; half of it is the *rydberg*, Ry. The atomic charge unit is $e = 4.80 \cdot 10^{-10}$ electrostatic units. Formally, formulas are converted to atomic units by putting $e = m = \hbar = 1$. For $\alpha = Ze^2$, Coulomb and atomic units differ.

The solution must grow no faster than a finite power at infinity and remain finite at the origin. The solution satisfying the latter condition is the confluent hypergeometric function

$$\omega = F(-n + l + 1, 2l + 2, \rho) \tag{36.7}$$

(see mathematical appendix § d).[12] The condition at infinity holds only when $-n + l + 1$ is a nonpositive integer, so that (36.7) is a polynomial of degree $n - l - 1$; otherwise it grows as e^{ρ} (see (d.14)). Thus n is a positive integer and

$$n \geq l + 1. \tag{36.8}$$

From (36.3),

$$E = -\frac{1}{2n^2}, \qquad n = 1, 2, \ldots. \tag{36.9}$$

This determines the discrete energy levels. Infinitely many lie between the normal level $E_1 = -1/2$ and zero. Their spacings decrease with n, and they accumulate at $E = 0$, where the discrete and continuous spectra meet. In ordinary units,[13]

$$E = -\frac{m\alpha^2}{2\hbar^2 n^2}. \tag{36.10}$$

The integer n is the *principal quantum number*; the radial quantum number of § 32 is $n_r = n - l - 1$. At fixed n,

$$l = 0, 1, \ldots, n - 1. \tag{36.11}$$

Since the energy depends only on n, states with different l but the same n have equal energies. Each eigenvalue is therefore degenerate not only in the magnetic quantum number m, as in every central field, but also in l. This latter, *accidental* or *Coulomb*, degeneracy is peculiar to the Coulomb field. Since each l has $2l + 1$ values of m, the degeneracy of level n is

$$\sum_{l=0}^{n-1} (2l + 1) = n^2. \tag{36.12}$$

The stationary-state wave functions follow from (36.5) and (36.7). For integer parameters the confluent hypergeometric function is, apart from a factor, a generalized Laguerre polynomial (appendix § d), so

$$R_{nl} = \text{const} \cdot \rho^l e^{-\rho/2} L_{n+l}^{2l+1}(\rho).$$

[12] The second solution of (36.6) diverges as ρ^{-2l-1} when $\rho \to 0$.

[13] Formula (36.10) was first obtained by N. Bohr in 1913, before quantum mechanics. W. Pauli derived it in quantum mechanics by the matrix method in 1926, and Schrödinger derived it from the wave equation a few months later.

With $\int_0^\infty R_{nl}^2 r^2\,dr = 1$, the final form is[14]

$$\begin{aligned} R_{nl} &= -\frac{2}{n^2}\sqrt{\frac{(n-l-1)!}{[(n+l)!]^3}}\,e^{-r/n}(2r/n)^l L_{n+l}^{2l+1}(2r/n) \\ &= \frac{2}{n^{l+2}(2l+1)!}\sqrt{\frac{(n+l)!}{(n-l-1)!}}(2r)^l e^{-r/n} F(-n+l+1, 2l+2, 2r/n). \end{aligned} \tag{36.13}$$

(For the normalization integral see appendix § f, integral (f.6).[15]) Near the origin,

$$R_{nl} \approx r^l \frac{2^{l+1}}{n^{2+l}(2l+1)!}\sqrt{\frac{(n+l)!}{(n-l-1)!}}. \tag{36.14}$$

At large distances,

$$R_{nl} \approx (-1)^{n-l-1}\frac{2^n r^{n-1} e^{-r/n}}{n^{n+1}\sqrt{(n+l)!(n-l-1)!}}. \tag{36.15}$$

The normal-state function R_{10} decays over distances $r \sim 1$, or $r \sim \hbar^2/m\alpha$ in ordinary units.

Mean powers of r are $\overline{r^k} = \int_0^\infty r^{k+2} R_{nl}^2 dr$. The general result follows from (f.7); some first values are

$$\begin{aligned} \bar{r} &= \tfrac{1}{2}[3n^2 - l(l+1)], \qquad & \overline{r^2} &= \tfrac{n^2}{2}[5n^2 + 1 - 3l(l+1)], \\ \overline{r^{-1}} &= \frac{1}{n^2}, \qquad & \overline{r^{-2}} &= \frac{1}{n^3(l+1/2)}. \end{aligned} \tag{36.16}$$

Continuous spectrum. Positive energies form a continuous spectrum from zero to infinity. Every eigenvalue is infinitely degenerate, with all integer $l \geq 0$ and all allowed m. Now n and ρ are purely imaginary:

$$n = -\frac{i}{\sqrt{2E}} = -\frac{i}{k}, \qquad \rho = 2ikr, \tag{36.17}$$

where $k = \sqrt{2E}$.[16] The continuum radial eigenfunctions are

$$R_{kl} = \frac{C_{kl}}{(2l+1)!}(2kr)^l e^{-ikr} F(i/k + l + 1, 2l+2, 2ikr), \tag{36.18}$$

[14]The first few functions are

$$R_{10} = 2e^{-r}, \quad R_{20} = \frac{1}{\sqrt{2}}e^{-r/2}(1 - r/2), \quad R_{21} = \frac{1}{2\sqrt{6}}e^{-r/2}r,$$
$$R_{30} = \frac{2}{3\sqrt{3}}e^{-r/3}(1 - 2r/3 + 2r^2/27),$$
$$R_{31} = \frac{8}{27\sqrt{6}}e^{-r/3}r(1 - r/6), \quad R_{32} = \frac{4}{81\sqrt{30}}e^{-r/3}r^2.$$

[15]It may also be evaluated by substituting (d.13) for the Laguerre polynomials and integrating by parts, as for the Legendre-polynomial integral (c.8).

[16]One could instead choose the complex conjugates $n = i/k$, $\rho = -2ikr$; the real functions R_{kl} are independent of this choice.

or, as a complex integral (appendix § d),

$$R_{kl} = C_{kl}(2kr)^l e^{-ikr}\frac{1}{2\pi i}\oint e^{\xi}\left(1-\frac{2ikr}{\xi}\right)^{-i/k-l-1}\xi^{-2l-2}d\xi, \tag{36.19}$$

over the contour in Fig. 10.[17] With $\xi = 2ikr(t+1/2)$ this becomes

$$R_{kl} = C_{kl}\frac{(-2kr)^{-l-1}}{2\pi}\oint e^{2ikrt}(t+\tfrac{1}{2})^{i/k-l-1}(t-\tfrac{1}{2})^{-i/k-l-1}dt. \tag{36.20}$$

The path encircles $t = \pm 1/2$ positively; this form directly shows that R_{kl} is real.

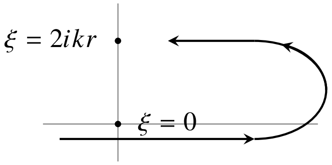

Fig. 10

Using (d.14), the asymptotic expansion is

$$R_{kl} = C_{kl}\frac{e^{-\pi/2k}}{kr}\,\mathrm{Re}\left\{\frac{\exp[-i(kr-(\pi/2)(l+1)+(1/k)\ln 2kr)]}{\Gamma(l+1-i/k)}\times G(l+1+i/k, i/k-l, -2ikr)\right\}. \tag{36.21}$$

For normalization on the $k/2\pi$ scale, condition (33.4),

$$C_{kl} = 2ke^{\pi/2k}|\Gamma(l+1-i/k)|. \tag{36.22}$$

Indeed, at large r this gives

$$R_{kl} \approx \frac{2}{r}\sin\left(kr + k^{-1}\ln 2kr - \pi l/2 + \delta_l\right), \qquad \delta_l = \arg\Gamma(l+1-i/k). \tag{36.23}$$

This agrees with the general form (33.20), apart from the logarithmic phase; since $\ln r$ grows slowly compared with r, it does not affect the divergent normalization integral.

The gamma-function modulus in (36.22) is elementary. From $\Gamma(z+1) = z\Gamma(z)$ and $\Gamma(z)\Gamma(1-z) = \pi/\sin\pi z$, one finds

$$\Gamma(l+1+i/k) = (l+i/k)\cdots(1+i/k)(i/k)\Gamma(i/k),$$
$$\Gamma(l+1-i/k) = (l-i/k)\cdots(1-i/k)\Gamma(1-i/k),$$

and

$$|\Gamma(l+1-i/k)| = \sqrt{\frac{\pi}{k}}\prod_{s=1}^{l}\sqrt{s^2+k^{-2}}(\sinh(\pi/k))^{-1/2}.$$

[17] It may be replaced by any closed loop encircling $\xi = 0$ and $\xi = 2ikr$ positively. For integral l, the function $V(\xi) = \xi^{-n-l}(\xi - 2ikr)^{n-l}$ (appendix § d) returns to its original value around such a contour.

Thus

$$C_{kl} = \left[\frac{8\pi k}{1 - e^{-2\pi/k}}\right]^{1/2} \prod_{s=1}^{l} \sqrt{s^2 + \frac{1}{k^2}}, \tag{36.24}$$

with the product replaced by 1 at $l = 0$. The $E = 0$ radial function follows by taking $k \to 0$:

$$\begin{aligned} F(i/k + l + 1, 2l + 2, 2ikr) \to F(i/k, 2l + 2, 2ikr) \\ &= 1 - \frac{2r}{(2l+1)!} + \frac{(2r)^2}{(2l+2)(2l+3)2!} - \cdots \\ &= (2l+1)!(2r)^{-l-1/2} J_{2l+1}(\sqrt{8r}), \end{aligned}$$

where J_{2l+1} is a Bessel function, while $C_{kl} \approx \sqrt{8\pi}\, k^{-l+1/2}$. Hence

$$\left.\frac{R_{kl}}{\sqrt{k}}\right|_{k\to 0} = \sqrt{\frac{4\pi}{r}} J_{2l+1}(\sqrt{8r}). \tag{36.25}$$

At large r,[18]

$$\left.\frac{R_{kl}}{\sqrt{k}}\right|_{k\to 0} = \left(\frac{8}{r^3}\right)^{1/4} \sin\left(\sqrt{8r} - l\pi - \pi/4\right). \tag{36.26}$$

The factor $\sqrt{k}$ disappears on changing to energy-scale normalization, from R_{kl} to R_{El} by (33.5); R_{El} remains finite as $E \to 0$.

For the repulsive field $U = \alpha/r$, only a continuous positive-energy spectrum exists. Formally changing r to $-r$ in the attractive-field equation gives

$$\begin{gathered} R_{kl} = \frac{C_{kl}}{(2l+1)!}(2kr)^l e^{ikr} F(i/k + l + 1, 2l + 2, -2ikr), \\ C_{kl} = 2ke^{-\pi/2k}|\Gamma(l + 1 + i/k)| = \left(\frac{8\pi k}{e^{2\pi/k} - 1}\right)^{1/2} \prod_{s=1}^{l} \sqrt{s^2 + \frac{1}{k^2}}. \end{gathered} \tag{36.27}$$

At large r,

$$R_{kl} \approx \frac{2}{r} \sin\left(kr - k^{-1} \ln 2kr - l\pi/2 + \delta_l\right), \qquad \delta_l = \arg\Gamma(l + 1 + i/k). \tag{36.28}$$

Origin of the Coulomb degeneracy. Classical motion in an attractive Coulomb field has the special conserved quantity

$$\mathbf{A} = \frac{\mathbf{r}}{r} - \mathbf{p} \times \mathbf{l} = \text{const} \tag{36.29}$$

(see I, § 15). Its quantum operator is

$$\hat{\mathbf{A}} = \frac{\mathbf{r}}{r} - \frac{1}{2}(\hat{\mathbf{p}} \times \hat{\mathbf{l}} - \hat{\mathbf{l}} \times \hat{\mathbf{p}}), \tag{36.30}$$

which commutes with $\hat{H} = \hat{\mathbf{p}}^2/2 - 1/r$.

[18] This function corresponds to the semiclassical approximation (§ 49) in $(l + 1/2)^2 \ll r \ll k^{-2}$.

Direct calculation gives

$$[\hat{l}_i, \hat{A}_k] = ie_{ikl}\hat{A}_l, \qquad [\hat{A}_i, \hat{A}_k] = -2i\hat{H}e_{ikl}\hat{l}_l. \tag{36.31}$$

The mutual noncommutativity of the $\hat{A}_i$ means that A_x, A_y, A_z cannot all be definite. Each component, say $\hat{A}_z$, commutes with $\hat{l}_z$ but not with $\hat{\mathbf{l}}^2$. A new conserved quantity which cannot be measured together with the other conserved quantities causes the additional degeneracy (§ 10): this is the accidental degeneracy peculiar to the Coulomb field.

The same origin may be stated in terms of the Coulomb problem's enlarged quantum-mechanical symmetry (V. A. Fock, 1935). At fixed negative energy, replace $\hat{H}$ on the right of (36.31) by E and define $\hat{u}_i = \hat{A}_i/\sqrt{-2E}$. Then

$$[\hat{l}_i, \hat{u}_k] = ie_{ikl}\hat{u}_l, \qquad [\hat{u}_i, \hat{u}_k] = ie_{ikl}\hat{l}_l. \tag{36.32}$$

Together with $[\hat{l}_i, \hat{l}_k] = ie_{ikl}\hat{l}_l$, these are precisely the formal commutation rules for infinitesimal rotations in four-dimensional Euclidean space.[19] This is the symmetry of the Coulomb problem.[20]

The energy levels can again be derived from (36.32).[21] Introduce

$$\hat{\mathbf{j}}_1 = \tfrac{1}{2}(\hat{\mathbf{l}} + \hat{\mathbf{u}}), \qquad \hat{\mathbf{j}}_2 = \tfrac{1}{2}(\hat{\mathbf{l}} - \hat{\mathbf{u}}). \tag{36.33}$$

Then

$$[\hat{j}_{1i}, \hat{j}_{1k}] = ie_{ikl}\hat{j}_{1l}, \quad [\hat{j}_{2i}, \hat{j}_{2k}] = ie_{ikl}\hat{j}_{2l}, \quad [\hat{j}_{1i}, \hat{j}_{2k}] = 0. \tag{36.34}$$

These are the rules for two independent three-dimensional angular momenta. Thus the eigenvalues of $\hat{\mathbf{j}}_1^2$ and $\hat{\mathbf{j}}_2^2$ are $j_1(j_1+1)$ and $j_2(j_2+1)$, with $j_1, j_2 = 0, 1/2, 1, 3/2, \ldots$.[22] Directly from $\hat{\mathbf{u}}$ and $\hat{\mathbf{l}} = \mathbf{r} \times \hat{\mathbf{p}}$,

$$\hat{\mathbf{l}}\hat{\mathbf{u}} = \hat{\mathbf{u}}\hat{\mathbf{l}} = 0, \qquad \hat{\mathbf{l}}^2 + \hat{\mathbf{u}}^2 = -1 - \frac{1}{2E} \qquad \hat{\mathbf{l}}^2 + \hat{\mathbf{u}}^2 = -1 - \frac{1}{2E}.$$

Hence

$$\hat{\mathbf{j}}_1^2 = \hat{\mathbf{j}}_2^2 = -\tfrac{1}{4}(1 + 1/2E) = j(j+1),$$

so $E = -1/[2(2j+1)^2]$. Writing

$$2j + 1 = n, \qquad n = 1, 2, 3, \ldots, \tag{36.35}$$

gives $E = -1/(2n^2)$. The degeneracy is $(2j_1+1)(2j_2+1) = n^2$. Since $\hat{\mathbf{l}} = \hat{\mathbf{j}}_1 + \hat{\mathbf{j}}_2$, for $j_1 = j_2 = (n-1)/2$ the orbital momentum ranges from $l = 0$ to $2j = n - 1$.[23]

[19] Here $\hat{l}_x, \hat{l}_y, \hat{l}_z$ generate rotations in the yz, xz, xy planes of Cartesian coordinates x, y, z, u, and $\hat{u}_x, \hat{u}_y, \hat{u}_z$ generate rotations in the xu, yu, zu planes.

[20] It appears explicitly in momentum-space wave functions; see V. A. Fock, Izv. AN SSSR, physical series, 1935, no. 2, p. 169; Zs. f. Physik 98 (1935), 145.

[21] This derivation is essentially Pauli's (1926).

[22] We anticipate § 54 in using the possible integral and half-integral values of j.

[23] Accidental degeneracy between different l also occurs in the central field $U = m\omega^2 r^2/2$ (the spatial

Problem

1. Find the probability distribution of momentum values in the ground state of the hydrogen atom.

Solution. [24] The ground-state wave function is

$$\psi = R_{10}Y_{00} = \frac{1}{\sqrt{\pi}}e^{-r}.$$

In the **p**-representation,

$$a(\mathbf{p}) = \int \psi(\mathbf{r})e^{-i\mathbf{pr}}\,dV.$$

Using spherical coordinates with polar axis along **p** gives

$$a(\mathbf{p}) = \frac{8\sqrt{\pi}}{(1+p^2)^2},$$

and the probability density in momentum space is $|a(\mathbf{p})|^2/(2\pi)^3$.

2. Find the mean potential of the field produced by the nucleus and the electron in the hydrogen ground state.

Solution. The mean potential φ_e of the "electron cloud" is most simply found as the spherical solution of Poisson's equation with charge density $\rho = -|\psi|^2$:

$$\frac{1}{r}\frac{d^2}{dr^2}(r\varphi_e) = 4e^{-2r}.$$

Choosing the integration constants so that $\varphi_e(0)$ is finite and $\varphi_e(\infty) = 0$, and adding the nuclear potential, gives

$$\varphi = 1/r + \varphi_e(r) = (1/r + 1)e^{-2r}.$$

Thus $\varphi \approx 1/r$ for $r \ll 1$, while $\varphi \approx e^{-2r}$ for $r \gg 1$.

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oscillator; Problem 4 of § 33), again because of an enlarged Hamiltonian symmetry. In $\hat{H} = \hat{\mathbf{p}}^2/2m + m\omega^2 r^2/2$, both $\hat{p}_i$ and x_i enter as sums of squares. With

$$\hat{a}_i = \frac{m\omega x_i + i\hat{p}_i}{\sqrt{2m\hbar\omega}}, \qquad \hat{a}_i^+ = \frac{m\omega x_i - i\hat{p}_i}{\sqrt{2m\hbar\omega}},$$

one obtains $\hat{H} = \hbar\omega[\hat{\mathbf{a}}^+\hat{\mathbf{a}} + 3/2]$. This is invariant under arbitrary unitary transformations of $\hat{a}_i^+$ and $\hat{a}_j$, a group larger than the three-dimensional rotation group. The quantum peculiarity of the Coulomb and oscillator fields has its classical counterpart: particle trajectories are closed in these, and only these, fields.

[24] Atomic units are used in Problems 1 and 2.

3. Find the energy levels of a particle moving in the central field $U = A/r^2 - B/r$ (Fig. 11).

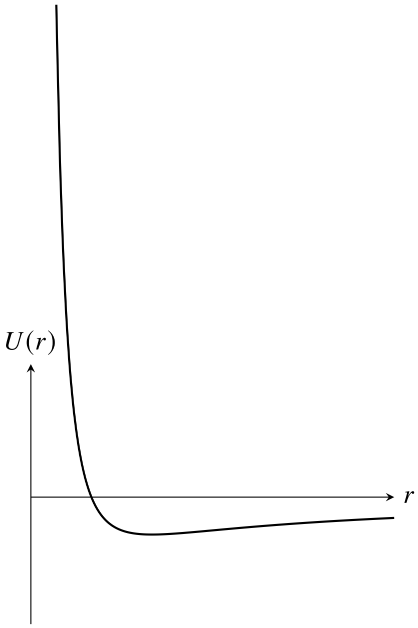


Fig. 11

SOLUTION. Positive energies form a continuous spectrum and negative energies a discrete one; consider the latter. The radial equation is

$$\frac{d^2R}{dr^2} + \frac{2}{r}\frac{dR}{dr} + \frac{2m}{\hbar^2}\left(E - \frac{\hbar^2 l(l+1)}{2mr^2} - \frac{A}{r^2} + \frac{B}{r}\right)R = 0. \tag{1}$$

Put $\rho = 2\sqrt{-mE}\,r/\hbar$ and define

$$\frac{2mA}{\hbar^2} + l(l+1) = s(s+1), \tag{2}$$

$$\frac{B}{\hbar}\sqrt{\frac{m}{-2E}} = n. \tag{3}$$

Then

$$R'' + \frac{2}{\rho}R' + (-\tfrac{1}{4} + n/\rho - s(s+1)/\rho^2)R = 0,$$

formally identical to (36.4). Hence

$$R = \rho^s e^{-\rho/2} F(-n+s+1, 2s+2, \rho),$$

where $n - s - 1 = p$ is a nonnegative integer and s is the positive root of (2). Formula (3) gives

$$-E_p = \frac{2B^2m}{\hbar^2}[2p + 1 + \sqrt{(2l+1)^2 + 8mA/\hbar^2}]^{-2}.$$

4. Repeat the calculation for $U = A/r^2 + Br^2$ (Fig. 12).

SOLUTION. There is only a discrete spectrum. The radial equation is

$$\frac{d^2R}{dr^2} + \frac{2}{r}\frac{dR}{dr} + \frac{2m}{\hbar^2}(E - \hbar^2 l(l+1)/(2mr^2) - A/r^2 - Br^2)R = 0.$$

Set

$$\xi = \frac{\sqrt{2mB}}{\hbar}r^2,$$

and define

$$l(l+1) + \frac{2mA}{\hbar^2} = 2s(2s+1), \qquad \sqrt{\frac{2m}{B}}\frac{E}{\hbar} = 4(n+s)+3.$$

Then

$$\xi R'' + \frac{3}{2}R' + [n + s + \tfrac{3}{4} - \xi/4 - s(s+1/2)/\xi]R = 0.$$

The desired solution behaves as $e^{-\xi/2}$ at infinity and as ξ^s at the origin, where

$$s = \frac{1}{4}[-1 + \sqrt{(2l+1)^2 + 8mA/\hbar^2}].$$

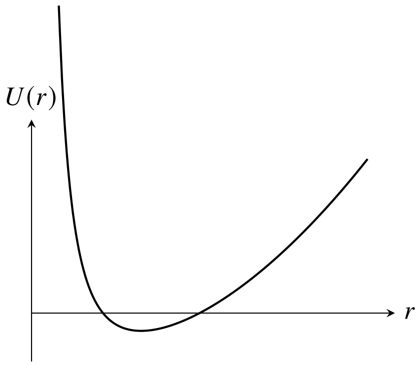


Fig. 12

Writing $R = e^{-\xi/2}\xi^s\omega$ gives

$$\xi\omega'' + (2s + 3/2 - \xi)\omega' + n\omega = 0,$$

so

$$\omega = F(-n, 2s + 3/2, \xi),$$

where n is a nonnegative integer. The equally spaced levels are

$$E_n = \hbar\sqrt{\frac{B}{2m}}[4n + 2 + \sqrt{(2l+1)^2 + 8mA/\hbar^2}], \qquad n = 0, 1, 2, \ldots.$$

§ 37. Motion in a Coulomb field (parabolic coordinates)

For motion in any centrally symmetric field, the Schrödinger equation can always be separated in spherical coordinates. In a Coulomb field, separation is also possible in the so-called parabolic coordinates. The solution of the Coulomb-field problem in these coordinates is useful in studying a number of problems in which a particular direction in space is singled out, for example by an external electric field in addition to the Coulomb field (§ 77). The parabolic coordinates ξ, η, and φ are defined by

$$\begin{gathered} x = \sqrt{\xi\eta}\cos\varphi, \qquad y = \sqrt{\xi\eta}\sin\varphi, \qquad z = \frac{1}{2}(\xi - \eta), \\ r = \sqrt{x^2 + y^2 + z^2} = \frac{1}{2}(\xi + \eta) \end{gathered} \tag{37.1}$$

or, conversely,

$$\xi = r + z, \qquad \eta = r - z, \qquad \varphi = \arctan\frac{y}{x}; \tag{37.2}$$

ξ and η range from 0 to ∞, while φ ranges from 0 to 2π. The surfaces ξ = const and η = const are paraboloids of revolution whose axis is the z axis and whose focus is at the origin. This coordinate system is orthogonal. The line element is

$$dl^2 = \frac{\xi + \eta}{4\xi}\,d\xi^2 + \frac{\xi + \eta}{4\eta}\,d\eta^2 + \xi\eta\,d\varphi^2, \tag{37.3}$$

and the volume element is

$$dV = \frac{1}{4}(\xi + \eta)\,d\xi\,d\eta\,d\varphi. \tag{37.4}$$

Equation (37.3) gives the following expression for the Laplace operator:

$$\Delta = \frac{4}{\xi + \eta}\left[\frac{\partial}{\partial\xi}\left(\xi\frac{\partial}{\partial\xi}\right) + \frac{\partial}{\partial\eta}\left(\eta\frac{\partial}{\partial\eta}\right)\right] + \frac{1}{\xi\eta}\frac{\partial^2}{\partial\varphi^2}. \tag{37.5}$$

For a particle in the attractive Coulomb field

$$U = -\frac{1}{r} = -\frac{2}{\xi + \eta},$$

the Schrödinger equation becomes

$$\frac{4}{\xi + \eta}\left[\frac{\partial}{\partial\xi}\left(\xi\frac{\partial\psi}{\partial\xi}\right) + \frac{\partial}{\partial\eta}\left(\eta\frac{\partial\psi}{\partial\eta}\right)\right] + \frac{1}{\xi\eta}\frac{\partial^2\psi}{\partial\varphi^2} + 2\left(E + \frac{2}{\xi + \eta}\right)\psi = 0. \tag{37.6}$$

We seek the eigenfunctions ψ in the form

$$\psi = f_1(\xi) f_2(\eta) e^{im\varphi}, \tag{37.7}$$

where m is the magnetic quantum number. Substitute this expression into (37.6), after multiplying that equation by $(\xi + \eta)/4$, and separate the variables ξ and η. The resulting equations for f_1 and f_2 are

$$\begin{gathered} \frac{d}{d\xi}\left(\xi\frac{df_1}{d\xi}\right) + \left[\frac{E}{2}\xi - \frac{m^2}{4\xi} + \beta_1\right] f_1 = 0, \\ \frac{d}{d\eta}\left(\eta\frac{df_2}{d\eta}\right) + \left[\frac{E}{2}\eta - \frac{m^2}{4\eta} + \beta_2\right] f_2 = 0, \end{gathered} \tag{37.8}$$

where the "separation parameters" β_1 and β_2 are related by

$$\beta_1 + \beta_2 = 1. \tag{37.9}$$

Consider the discrete energy spectrum ($E < 0$). In place of E, ξ, and η, introduce the quantities

$$n = \frac{1}{\sqrt{-2E}}, \qquad \rho_1 = \xi\sqrt{-2E} = \frac{\xi}{n}, \qquad \rho_2 = \frac{\eta}{n}, \tag{37.10}$$

after which the equation for f_1 takes the form

$$\frac{d^2 f_1}{d\rho_1^2} + \frac{1}{\rho_1}\frac{df_1}{d\rho_1} + \left[-\frac{1}{4} + \frac{1}{\rho_1}\left(\frac{|m|+1}{2} + n_1\right) - \frac{m^2}{4\rho_1^2}\right] f_1 = 0, \tag{37.11}$$

with an identical equation for f_2; here we have also introduced the notation

$$n_1 = -\frac{|m|+1}{2} + n\beta_1, \qquad n_2 = -\frac{|m|+1}{2} + n\beta_2. \tag{37.12}$$

Proceeding as for equation (36.4), we find that at large ρ_1, f_1 behaves as $e^{-\rho_1/2}$, and at small ρ_1 as $\rho_1^{|m|/2}$. We therefore seek a solution of (37.11) in the form

$$f_1(\rho_1) = e^{-\rho_1/2}\rho_1^{|m|/2} w_1(\rho_1)$$

(and analogously for f_2), obtaining for w_1 the equation

$$\rho_1 w_1'' + (|m| + 1 - \rho_1)w_1' + n_1 w_1 = 0.$$

This is again the confluent hypergeometric equation. The solution satisfying the finiteness conditions is

$$w_1 = F(-n_1, |m| + 1, \rho_1),$$

and n_1 must be a nonnegative integer.

Thus, in parabolic coordinates every stationary state of the discrete spectrum is specified by three integers: the "parabolic quantum numbers" n_1 and n_2, and the magnetic quantum number m. From (37.9) and (37.12), the number n (the "principal quantum number") is

$$n = n_1 + n_2 + |m| + 1. \tag{37.13}$$

The energy levels are, of course, the same as in (36.9).

For a given n, the number $|m|$ can take the n values from 0 through $n-1$. With n and $|m|$ fixed, n_1 takes the $n - |m|$ values from 0 through $n - |m| - 1$. For each prescribed $|m|$, functions with $m = \pm|m|$ may also be chosen. Hence the total number of distinct states for a given n is

$$2\sum_{m=1}^{n-1}(n-m) + (n-0) = n^2,$$

in agreement with the result obtained in § 36.

The discrete-spectrum wave functions $\psi_{n_1n_2m}$ must obey the normalization condition

$$\int |\psi_{n_1n_2m}|^2\, dV = \frac{1}{4}\int_0^\infty\int_0^\infty\int_0^{2\pi} |\psi_{n_1n_2m}|^2(\xi+\eta)\, d\varphi\, d\xi\, d\eta = 1. \tag{37.14}$$

The normalized functions are

$$\psi_{n_1n_2m} = \frac{\sqrt{2}}{n^2} f_{n_1m}\left(\frac{\xi}{n}\right) f_{n_2m}\left(\frac{\eta}{n}\right)\frac{e^{im\varphi}}{\sqrt{2\pi}}, \tag{37.15}$$

$$f_{pm}(\rho) = \frac{1}{|m|!}\sqrt{\frac{(p+|m|)!}{p!}}F(-p, |m|+1, \rho)e^{-\rho/2}\rho^{|m|/2}. \tag{37.16}$$

Unlike the wave functions in spherical coordinates, those in parabolic coordinates are not symmetric about the plane $z = 0$. When $n_1 > n_2$, the particle is more likely to be found on the $z > 0$ side than on the $z < 0$ side; for $n_1 < n_2$, the reverse holds.

The continuous spectrum ($E > 0$) corresponds to a continuous spectrum of real values of the parameters β_1, β_2 in equations (37.8) (still, of course, related by (37.9)); we shall not write the corresponding wave functions here. Regarded as equations for the "eigenvalues" β_1 and β_2, equations (37.8) also possess, when $E > 0$, a spectrum of complex values. The associated wave functions will be given in § 135, where they will be used to solve the problem of scattering in a Coulomb field.

The existence of the stationary states $|n_1n_2m\rangle$ is tied to the additional conservation law (36.29). In these states, besides the energy, the quantities $l_z = m$ and A_z have definite values. Evaluation of the diagonal matrix elements of the operator $\hat{A}_z$ gives

$$A_z = \frac{n_1 - n_2}{n}. \tag{37.17}$$

Here $u_z = n_1 - n_2$, while the projections of the "angular momenta" $\mathbf{j}_1$ and $\mathbf{j}_2$ are

$$j_{1z} = \frac{m + n_1 - n_2}{2} \equiv \mu_1, \qquad j_{2z} = \frac{m - n_1 + n_2}{2} \equiv \mu_2. \tag{37.18}$$

These properties of the states $|n_1n_2m\rangle$ (or, equivalently, $|n\mu_1\mu_2\rangle$) make it easy to relate their wave functions to those of the states $|nlm\rangle$. Since $\mathbf{l} = \mathbf{j}_1 + \mathbf{j}_2$, a change between the two descriptions reduces to constructing wave functions for the addition of two angular momenta (considered below in § 106). In terms of the "angular momenta" $\mathbf{j}_1$ and $\mathbf{j}_2$, the states $|nlm\rangle$ and $|n_1n_2m\rangle$ are written as $|j_1j_2lm\rangle$ and $|j_1j_2\mu_1\mu_2\rangle$, where, by (36.35) and (37.13),

$$j_1 = j_2 = \frac{n-1}{2} = \frac{n_1 + n_2 + |m|}{2}. \tag{37.19}$$

The general formulas (106.9)–(106.11) give

$$\begin{aligned}\psi_{nlm} &= \sum_{\mu_1+\mu_2=m} \langle lm|\mu_1\mu_2\rangle\psi_{n\mu_1\mu_2},\\ \psi_{n\mu_1\mu_2} &= \sum_{l=0}^{n-1}\langle l, \mu_1+\mu_2|\mu_1\mu_2\rangle\psi_{nlm}\end{aligned} \tag{37.20}$$

(*D. Park*, 1960).

CHAPTER VI

Perturbation Theory

§ 38. Time-independent perturbations

An exact solution of the Schrödinger equation can be found in only a comparatively small number of the simplest cases. Most problems in quantum mechanics lead to equations too complicated for exact solution. Frequently, however, a problem contains quantities of different orders of magnitude, including small quantities whose omission simplifies the problem enough to make an exact solution possible. The first step is then to solve this simplified problem exactly; the second is to calculate approximately the corrections due to the small terms that were omitted. The general method for calculating these corrections is called *perturbation theory*.

Suppose that the Hamiltonian of the physical system is

$$\hat{H} = \hat{H}_0 + \hat{V},$$

where $\hat{V}$ is a small correction, the *perturbation*, to the "unperturbed" operator $\hat{H}_0$. In § 38 and § 39 we consider perturbations $\hat{V}$ with no explicit time dependence; the same is assumed for $\hat{H}_0$. The conditions under which $\hat{V}$ may be regarded as small in comparison with $\hat{H}_0$ will be established below.

For the discrete spectrum, the problem may be stated as follows. The eigenfunctions $\psi^{(0)}$ and eigenvalues of $\hat{H}_0$ are assumed known; that is, the exact solutions of

$$\hat{H}_0\psi^{(0)} = E^{(0)}\psi^{(0)} \tag{38.1}$$

are known. Approximate solutions are required for

$$\hat{H}\psi = (\hat{H}_0 + \hat{V})\psi = E\psi, \tag{38.2}$$

namely approximate expressions for the eigenfunctions ψ_n and eigenvalues E_n of the perturbed operator $\hat{H}$.

Throughout this section all eigenvalues of $\hat{H}_0$ will be assumed nondegenerate. To simplify the derivation, we first suppose that the energy levels form a discrete spectrum only.

It is convenient to use matrix form from the outset. Expand the required function ψ in the functions $\psi_m^{(0)}$:

$$\psi = \sum_m c_m\psi_m^{(0)}. \tag{38.3}$$

Substitution in (38.2) gives

$$\sum_m c_m(E_m^{(0)} + \hat{V})\psi_m^{(0)} = \sum_m c_m E\psi_m^{(0)}.$$

Multiplying by $\psi_k^{(0)*}$ and integrating, we find

$$(E - E_k^{(0)})c_k = \sum_m V_{km}c_m. \tag{38.4}$$

Here V_{km} is the matrix of $\hat{V}$ in the unperturbed functions:

$$V_{km} = \int \psi_k^{(0)*}\hat{V}\psi_m^{(0)}\,dq. \tag{38.5}$$

Seek the coefficients and energy as series

$$E = E^{(0)} + E^{(1)} + E^{(2)} + \ldots, \qquad c_m = c_m^{(0)} + c_m^{(1)} + c_m^{(2)} + \ldots,$$

where $E^{(1)}, c_m^{(1)}$ have the same order of smallness as $\hat{V}$, $E^{(2)}, c_m^{(2)}$ are of second order, and so forth.

To calculate the corrections to the nth eigenvalue and eigenfunction, set $c_n^{(0)} = 1$ and $c_m^{(0)} = 0$ for $m \neq n$. For the first approximation, put $E = E_n^{(0)} + E_n^{(1)}$ and $c_k = c_k^{(0)} + c_k^{(1)}$ in (38.4), retaining first-order terms only. The equation with $k = n$ gives

$$E_n^{(1)} = V_{nn} = \int \psi_n^{(0)*}\hat{V}\psi_n^{(0)}\,dq. \tag{38.6}$$

Thus the first-order correction to $E_n^{(0)}$ is the mean value of the perturbation in the state $\psi_n^{(0)}$.

For $k \neq n$, equation (38.4) gives

$$c_k^{(1)} = \frac{V_{kn}}{E_n^{(0)} - E_k^{(0)}}, \qquad k \neq n. \tag{38.7}$$

The coefficient $c_n^{(1)}$ remains arbitrary and must be chosen so that $\psi_n = \psi_n^{(0)} + \psi_n^{(1)}$ is normalized through first order. This requires $c_n^{(1)} = 0$. Indeed,

$$\psi_n^{(1)} = \sum_m{}' \frac{V_{mn}}{E_n^{(0)} - E_m^{(0)}}\psi_m^{(0)} \tag{38.8}$$

(the prime means that $m = n$ is omitted) is orthogonal to $\psi_n^{(0)}$; hence the integral of $|\psi_n^{(0)} + \psi_n^{(1)}|^2$ differs from unity only by a second-order quantity.

Formula (38.8) gives the first-order correction to the wave functions. It also shows the condition for applicability of the method:

$$|V_{mn}| \ll |E_n^{(0)} - E_m^{(0)}|. \tag{38.9}$$

Thus perturbation matrix elements must be small compared with the corresponding separations between unperturbed energy levels.

To find the second-order correction to $E_n^{(0)}$, substitute $E = E_n^{(0)} + E_n^{(1)} + E_n^{(2)}$ and $c_k = c_k^{(0)} + c_k^{(1)} + c_k^{(2)}$ in (38.4), and retain second-order terms. The equation with $k = n$ gives

$$E_n^{(2)}c_n^0 = \sum_m{}' V_{nm}c_m^{(1)},$$

and consequently

$$E_n^{(2)} = \sum_m{}' \frac{|V_{mn}|^2}{E_n^{(0)} - E_m^{(0)}}. \tag{38.10}$$

Here (38.7) was used, together with Hermiticity, $V_{mn} = V_{nm}^*$. The second-order correction to the ground-state energy is always negative: if $E_n^{(0)}$ is the lowest eigenvalue, every term in (38.10) is negative. Higher approximations may be calculated in the same manner.

These results extend directly to an operator $\hat{H}_0$ that also has a continuous spectrum, while the perturbed state itself still belongs to the discrete spectrum. The appropriate continuous-spectrum integrals are simply added to the discrete sums. Denote continuous-spectrum states by an index ν ranging continuously; ν stands for the complete set of quantities needed to specify a state. If these states are degenerate, as they nearly always are, energy alone does not specify one.[1] For example, (38.8) becomes

$$\psi_n^{(1)} = \sum_m{}' \frac{V_{mn}}{E_n^{(0)} - E_m^{(0)}} \psi_m^{(0)} + \int \frac{V_{\nu n}}{E_n^{(0)} - E_\nu^{(0)}} \psi_\nu^{(0)}\, d\nu. \tag{38.11}$$

It is also useful to give the perturbed matrix elements of a physical quantity f, calculated through first order with $\psi_n = \psi_n^{(0)} + \psi_n^{(1)}$ and (38.8):

$$f_{nm} = f_{nm}^{(0)} + \sum_k{}' \frac{V_{nk} f_{km}^{(0)}}{E_n^{(0)} - E_k^{(0)}} + \sum_k{}' \frac{V_{km} f_{nk}^{(0)}}{E_m^{(0)} - E_k^{(0)}}. \tag{38.12}$$

In the first sum $k \neq n$, and in the second $k \neq m$.

Problem

1. Find the second-order correction to the eigenfunctions.

Solution. The coefficients $c_k^{(2)}$ for $k \neq n$ are obtained from (38.4) with $k \neq n$, written through second order; $c_n^{(2)}$ is chosen so that $\psi_n = \psi_n^{(0)} + \psi_n^{(1)} + \psi_n^{(2)}$ is normalized through second order. The result is

$$\begin{aligned}\psi_n^{(2)} = {} & \sum_m{}' \sum_k{}' \frac{V_{mk} V_{kn}}{\hbar^2 \omega_{nk} \omega_{nm}} \psi_m^{(0)} - \sum_m{}' \frac{V_{nn} V_{mn}}{\hbar^2 \omega_{nm}^2} \psi_m^{(0)} \\ & - \frac{\psi_n^{(0)}}{2} \sum_m{}' \frac{|V_{mn}|^2}{\hbar^2 \omega_{nm}^2},\end{aligned}$$

where

$$\omega_{nm} = \frac{1}{\hbar}(E_n^{(0)} - E_m^{(0)}).$$

2. Find the third-order correction to the energy eigenvalues.

[1] The wave functions $\psi_\nu^{(0)}$ must then be normalized to delta functions of the quantities ν.

Solution. Retaining the third-order terms in (38.4) with $k = n$ gives

$$E_n^{(3)} = \sum_k{}' \sum_m{}' \frac{V_{nm}V_{mk}V_{kn}}{\hbar^2\omega_{mn}\omega_{kn}} - V_{nn}\sum_m{}' \frac{|V_{nm}|^2}{\hbar^2\omega_{mn}^2}.$$

3. Find the energy levels of the anharmonic linear oscillator with Hamiltonian

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{m\omega^2 x^2}{2} + \alpha x^3 + \beta x^4.$$

Solution. Matrix elements of x^3 and x^4 follow directly from the rule for matrix multiplication and the elements of x in (23.4). The nonzero elements of x^3 are

$$(x^3)_{n-3,n} = (x^3)_{n,n-3} = \left(\frac{\hbar}{m\omega}\right)^{3/2} \sqrt{\frac{n(n-1)(n-2)}{8}},$$

$$(x^3)_{n-1,n} = (x^3)_{n,n-1} = \left(\frac{\hbar}{m\omega}\right)^{3/2} \sqrt{\frac{9n^3}{8}}.$$

There are no diagonal elements in this matrix, so the term αx^3 gives no first-order correction when regarded as a perturbation of the harmonic oscillator. Its second-order correction has the same order as the first-order correction from βx^4. The diagonal elements of x^4 are

$$(x^4)_{n,n} = \left(\frac{\hbar}{m\omega}\right)^2 \frac{3}{4}(2n^2 + 2n + 1).$$

Equations (38.6) and (38.10) then give

$$E_n = \hbar\omega\left(n + \frac{1}{2}\right) - \frac{15}{4}\frac{\alpha^2}{\hbar\omega}\left(\frac{\hbar}{m\omega}\right)^3\left(n^2 + n + \frac{11}{30}\right) + \frac{3}{2}\beta\left(\frac{\hbar}{m\omega}\right)^2\left(n^2 + n + \frac{1}{2}\right).$$

4. An infinite spherical potential well undergoes a small volume-preserving deformation into a slightly prolate or oblate ellipsoid of revolution with semiaxes $a = b$ and c. Find the splitting of its particle energy levels (*A. B. Migdal*, 1959).

Solution. The boundary equation

$$\frac{x^2 + y^2}{a^2} + \frac{z^2}{c^2} = 1$$

becomes the sphere $x^2 + y^2 + z^2 = R^2$ under $x \to ax/R$, $y \to ay/R$, $z \to cz/R$. The particle Hamiltonian $\hat{H} = \hat{\mathbf{p}}^2/2M = -\hbar^2\Delta/2M$ (M is the mass; energy is measured from the bottom of the well) becomes $\hat{H} = \hat{H}_0 + \hat{V}$, with

$$\hat{H}_0 = -\frac{\hbar^2}{2M}\Delta, \qquad \hat{V} = -\frac{\hbar^2}{2M}\left[\left(\frac{R^2}{a^2} - 1\right)\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) + \left(\frac{R^2}{c^2} - 1\right)\frac{\partial^2}{\partial z^2}\right].$$

The ellipsoidal-well problem is thereby reduced to the spherical-well problem. If the ellipsoid differs only slightly from a sphere of radius $R = (a^2c)^{1/3}$, $\hat{V}$ is a small perturbation. Introducing the "ellipticity" β, $|\beta| \ll 1$, by

$$a \approx R\left(1 - \frac{\beta}{3}\right), \qquad c \approx R\left(1 + \frac{2\beta}{3}\right),$$

we obtain

$$\hat{V} = \frac{\beta}{3M}(\hat{\mathbf{p}}^2 - 3\hat{p}_z^2).$$

In first order, the change from the spherical-well levels is

$$\Delta E_{nlm} = E_{nlm} - E_{nl}^{(0)} = \langle nlm|V|nlm\rangle.$$

Here l, m are the angular momentum and its projection on the ellipsoid axis, and n numbers the spherical-well levels for fixed l, which do not depend on m. The expression $\mathbf{p}^2 - 3p_z^2$ is the zz component of the traceless irreducible tensor $\delta_{ik}\mathbf{p}^2 - 3p_ip_k$. By (107.2) and (107.6), $\langle nlm|V|nlm\rangle$ is proportional to $(-1)^m \begin{pmatrix} l & 2 & l \\ -m & 0 & m \end{pmatrix}$, and hence

$$\langle nlm|V|nlm\rangle = \left(1 - \frac{3m^2}{l(l+1)}\right)\langle nl0|V|nl0\rangle.$$

(The table of $3j$ symbols is on p. 530.) We next write

$$\begin{aligned}\langle nl0|V|nl0\rangle &= \frac{2}{3}\beta E_{nl}^{(0)} + \beta\frac{\hbar^2}{M}\left\langle nl0\left|\frac{\partial^2}{\partial z^2}\right|nl0\right\rangle \\ &= \frac{2}{3}\beta E_{nl}^{(0)} - \frac{\beta\hbar^2}{M}\int\left|\frac{\partial\psi_{nl0}}{\partial z}\right|^2 r^2\,dr\,do.\end{aligned}$$

The Schrödinger equation $\hat{H}_0\psi_{nlm} = E_{nl}^{(0)}\psi_{nlm}$ was used in the first term, and the second was integrated by parts. Using (28.11) for Y_{l0}, the derivative of $\psi_{nl0} = R_{nl}(r)Y_{l0}(\theta,\varphi)$ is

$$\begin{aligned}\frac{\partial}{\partial z}\psi_{nl0} &= \left(\cos\theta\frac{\partial}{\partial r} - \frac{\sin\theta}{r}\frac{\partial}{\partial\theta}\right)\psi_{nl0} \\ &= -\frac{i(l+1)}{[4(l+1)^2-1]^{1/2}}\left(R'_{nl} - \frac{l}{r}R_{nl}\right)Y_{l+1,0} \\ &\quad + \frac{il}{[4l^2-1]^{1/2}}\left(R'_{nl} + \frac{l+1}{r}R_{nl}\right)Y_{l-1,0}.\end{aligned}$$

The radial integrals are evaluated from

$$\int_0^\infty R_{nl}R'_{nl}r\,dr = -\frac{1}{2}\int_0^\infty R_{nl}^2\,dr,$$

$$\int_0^\infty R'^{2}_{nl}r^2\,dr = \frac{2M}{\hbar^2}E_{nl}^{(0)} - l(l+1)\int_0^\infty R_{nl}^2\,dr,$$

by integration by parts and the radial Schrödinger equation (33.3),

$$R''_{nl} + \frac{2}{r}R'_{nl} - \frac{l(l+1)}{r^2}R_{nl} = -\frac{2M}{\hbar^2}E^{(0)}_{nl}.$$

Terms containing integrals of R^2_{nl} cancel, leaving

$$\Delta E_{nlm} = 4\beta\frac{l(l+1)}{(2l-1)(2l+3)}\left[\frac{m^2}{l(l+1)} - \frac{1}{3}\right]E^{(0)}_{nl}.$$

Finally,

$$\frac{1}{2l+1}\sum_{m=-l}^{l} E_{nlm} = E^{(0)}_{nl},$$

so the centre of gravity of the multiplet is unchanged.

§ 39. The secular equation

We now turn to the case in which the unperturbed operator $\hat{H}_0$ has degenerate eigenvalues. Let $\psi^{(0)}_n, \psi^{(0)}_{n'}, \ldots$ denote eigenfunctions belonging to the same energy eigenvalue $E^{(0)}_n$. As we know, the choice of these functions is not unique: they may be replaced by any s independent linear combinations of the same functions, where s is the degeneracy of the level $E^{(0)}_n$. This freedom disappears, however, when the wave functions are required to change only slightly under the applied small perturbation. For the moment, let $\psi^{(0)}_n$, $\psi^{(0)}_{n'}, \ldots$ denote arbitrarily chosen unperturbed eigenfunctions. The correct zeroth-order functions are linear combinations of the form

$$c^{(0)}_n\psi^{(0)}_n + c^{(0)}_{n'}\psi^{(0)}_{n'} + \ldots$$

The coefficients in these combinations and the first-order corrections to the eigenvalues are determined as follows.

Write equations (38.4) for $k = n, n', \ldots$, substituting in first order $E = E^{(0)}_n + E^{(1)}$. For the c_k it is sufficient to retain their zeroth-order values $c_n = c^{(0)}_n$, $c_{n'} = c^{(0)}_{n'}, \ldots$, while $c_m = 0$ for $m \neq n, n', \ldots$. We then obtain

$$E^{(1)}c^{(0)}_n = \sum_{n'} V_{nn'}c^{(0)}_{n'},$$

or

$$\sum_{n'}(V_{nn'} - E^{(1)}\delta_{nn'})c^{(0)}_{n'} = 0, \tag{39.1}$$

where n and n' run over all values labeling states that belong to the given unperturbed eigenvalue $E^{(0)}_n$. This homogeneous system of linear equations for the $c^{(0)}_n$ has nonzero solutions if the determinant of its coefficients vanishes. Thus

$$|V_{nn'} - E^{(1)}\delta_{nn'}| = 0. \tag{39.2}$$

This equation has degree s in $E^{(1)}$ and, in general, has s distinct real roots. These roots are the required first-order corrections to the eigenvalues. Equation (39.2) is called the *secular equation*[2]. The sum of its roots is the sum of the diagonal matrix elements V_{nn}, $V_{n'n'}, \ldots$; it is the coefficient of $E^{(1)s-1}$ in the equation.

Substituting the roots of (39.2) one by one into (39.1) and solving the resulting systems gives the coefficients $c_n^{(0)}$ and hence the correct zeroth-order eigenfunctions. As a result of the perturbation, the initially degenerate energy level is in general no longer degenerate, since the roots of (39.2) are in general different. The perturbation is said to lift the degeneracy. The lifting may be complete or partial; in the latter case the perturbed level remains degenerate, but with a lower multiplicity than before.

It may happen that, for some reason, all matrix elements for transitions within one group of mutually degenerate states $n, n', \ldots$ are especially small, or even vanish. It may then be useful, while retaining $V_{nn'}$ in first order, also to include in higher orders the matrix elements V_{nm}, with $m \neq n, n', \ldots$, for transitions to states of other energies. We shall do this by including V_{mn} through second order.

In (38.4) with $k = n$, set $E = E_n^{(0)} + E^{(1)}$ on the left-hand side; we retain the notation $E^{(1)}$ for the energy correction in the approximation now considered. Replace c_n by $c_n^{(0)}$. Since $c_m^{(0)} = 0$ for every $m \neq n, n', \ldots$, we have

$$E^{(1)} c_n^{(0)} = \sum_m V_{nm} c_m^{(1)} + \sum_{n'} V_{nn'} c_{n'}^{(0)}. \tag{39.3}$$

Equations (38.4) with $k = m \neq n, n', \ldots$, through first order, give

$$(E_n^{(0)} - E_m^{(0)}) c_m^{(1)} = \sum_{n'} V_{mn'} c_{n'}^{(0)},$$

and hence

$$c_m^{(1)} = \sum_{n'} \frac{V_{mn'}}{E_n^{(0)} - E_m^{(0)}} c_{n'}^{(0)}.$$

Substitution in (39.3) gives

$$E^{(1)} c_n^{(0)} = \sum_{n'} c_{n'}^{(0)} \left(V_{nn'} + \sum_m \frac{V_{nm} V_{mn'}}{E_n^{(0)} - E_m^{(0)}} \right).$$

This system now replaces (39.1). Its compatibility condition again gives a secular equation, differing from (39.2) by the replacement

$$V_{nn'} \longrightarrow V_{nn'} + \sum_m \frac{V_{nm} V_{mn'}}{E_n^{(0)} - E_m^{(0)}}. \tag{39.4}$$

[2] Also called the age equation (the French word *siècle* means an age); the name was borrowed from celestial mechanics.

Problem

1. Find the first-order corrections to the eigenvalue and the correct zeroth-order functions for a doubly degenerate level.

Solution. Equation (39.2) now has the form

$$\begin{vmatrix} V_{11} - E^{(1)} & V_{21} \\ V_{12} & V_{22} - E^{(1)} \end{vmatrix} = 0$$

(the indices 1 and 2 refer to two arbitrarily chosen unperturbed eigenfunctions $\psi_1^{(0)}$ and $\psi_2^{(0)}$ of the degenerate level). Solving it, we find

$$E^{(1)} = \frac{1}{2}[V_{11} + V_{22} \pm \hbar\omega^{(1)}], \tag{1}$$

where

$$\hbar\omega^{(1)} = \sqrt{(V_{11} - V_{22})^2 + 4|V_{12}|^2}$$

denotes the difference between the two values of the correction $E^{(1)}$. Solving (39.1) with these values of $E^{(1)}$, we obtain the coefficients in the normalized correct zeroth-order functions $\psi^{(0)} = c_1^{(0)}\psi_1^{(0)} + c_2^{(0)}\psi_2^{(0)}$:

$$\begin{aligned} c_1^{(0)} &= \left\{\frac{V_{12}}{2|V_{12}|}\left[1 \pm \frac{V_{11} - V_{22}}{\hbar\omega^{(1)}}\right]\right\}^{1/2}, \\ c_2^{(0)} &= \pm\left\{\frac{V_{21}}{2|V_{12}|}\left[1 \mp \frac{V_{11} - V_{22}}{\hbar\omega^{(1)}}\right]\right\}^{1/2}. \end{aligned} \tag{2}$$

Problem

2. Derive formulas for the first-order corrections to the eigenfunctions and the second-order corrections to the eigenvalues.

Solution. Let the functions $\psi_n^{(0)}$ be chosen as the correct zeroth-order functions. The matrix $V_{nn'}$ defined with these functions is evidently diagonal in the indices n, n' belonging to the same group of functions of a degenerate level. Its diagonal elements $V_{nn}, V_{n'n'}, \ldots$ equal the corresponding first-order corrections $E_n^{(1)}, E_{n'}^{(1)}, \ldots$.

Consider the perturbation of the eigenfunction $\psi_n^{(0)}$. In zeroth order, $E = E_n^{(0)}$, $c_n^{(0)} = 1$, and $c_m^{(0)} = 0$ for $m \neq n$. In first order, $E = E_n^{(0)} + V_{nn}$, $c_n = 1 + c_n^{(1)}$, and $c_m = c_m^{(1)}$. From the general system (38.4), take the equation with $k \neq n, n', \ldots$ and retain its first-order terms:

$$(E_n^{(0)} - E_k^{(0)})c_k^{(1)} = V_{kn}c_n^{(0)} = V_{kn},$$

whence

$$c_k^{(1)} = \frac{V_{kn}}{E_n^{(0)} - E_k^{(0)}} \quad \text{for} \quad k \neq n, n', \ldots \tag{1}$$

Next write the equation with $k = n'$, retaining its second-order terms:

$$E_n^{(1)} c_{n'}^{(1)} = V_{n'n'} c_{n'}^{(1)} + \sum_m{}' V_{n'm} c_m^{(1)}.$$

In the sum over m, the terms with $m = n, n', \ldots$ are omitted. Substituting $E_n^{(1)} = V_{nn}$ and (1) for $c_m^{(1)}$, we obtain, for $n' \neq n$,

$$c_{n'}^{(1)} = \frac{1}{V_{nn} - V_{n'n'}} \sum_m{}' \frac{V_{n'm} V_{mn}}{E_n^{(0)} - E_m^{(0)}}. \tag{2}$$

The coefficient $c_n^{(1)}$ is zero in this approximation. Formulas (1) and (2) determine the first-order correction $\psi_n^{(1)} = \sum_m c_m^{(1)} \psi_m^{(0)}$ to the eigenfunctions[3].

Finally, retaining the second-order terms in (38.4) with $k = n$, we obtain the second-order energy correction

$$E_n^{(2)} = \sum_m{}' \frac{V_{nm} V_{mn}}{E_n^{(0)} - E_m^{(0)}}, \tag{3}$$

which has the same formal appearance as (38.10).

Problem

3. At the initial time $t = 0$, a system is in a state $\psi_1^{(0)}$ belonging to a doubly degenerate level. Find the probability that at a later time t the system is in the other state $\psi_2^{(0)}$ of the same energy, when the transition is caused by a constant perturbation.

Solution. Construct the correct zeroth-order functions:

$$\psi = c_1\psi_1 + c_2\psi_2, \qquad \psi' = c_1'\psi_1 + c_2'\psi_2,$$

where c_1, c_2 and c_1', c_2' are the two pairs of coefficients given by (2) of Problem 1. Superscripts (0) have been omitted from all quantities for brevity. Conversely,

$$\psi_1 = \frac{c_2'\psi - c_2\psi'}{c_1c_2' - c_1'c_2}.$$

The functions ψ and ψ' belong to states with perturbed energies $E + E^{(1)}$ and $E + E^{(1)'}$, where $E^{(1)}$ and $E^{(1)'}$ are the two values of correction (1) in Problem 1. Introducing the time factors gives the time-dependent wave function

$$\Psi_1 = \frac{\exp[-(i/\hbar)Et]}{c_1c_2' - c_1'c_2} \left[c_2'\psi \exp\left(-\frac{i}{\hbar}E^{(1)}t\right) - c_2\psi' \exp\left(-\frac{i}{\hbar}E^{(1)'}t\right)\right].$$

[3] Notice that the requirement that the quantities (1) and (2) be small, and hence the condition for applicability of this perturbation method, still requires (38.9) only for transitions between states belonging to different energy levels. Transitions between states of the same degenerate level are accounted for by the secular equation in a certain sense exactly.

At $t = 0$, $\Psi_1 = \psi_1$. Expressing ψ and ψ' once more in terms of ψ_1 and ψ_2, we write Ψ_1 as a linear combination of ψ_1 and ψ_2 with time-dependent coefficients. The squared modulus of the coefficient of ψ_2 is the required transition probability w_{21}. Using (1) and (2) of Problem 1 gives

$$w_{21} = 2\frac{|V_{12}|^2}{(\hbar\omega^{(1)})^2}\left[1 - \cos\left(\omega^{(1)}t\right)\right].$$

Thus the probability oscillates periodically with frequency $\omega^{(1)}$. For times t small compared with the corresponding period, the expression in square brackets, and hence w_{21} itself, is proportional to t^2:

$$w_{21} = \frac{1}{\hbar^2}|V_{12}|^2 t^2.$$

This formula follows very simply by the method developed in the next section, using (40.4).

§ 40. Time-dependent perturbations

We now turn to perturbations with an explicit dependence on time. In this case there can be no question at all of corrections to the energy eigenvalues: when the Hamiltonian depends on time (as does the perturbed operator $\hat{H} = \hat{H}_0 + \hat{V}(t)$), energy is not conserved, and stationary states therefore do not exist. The problem is instead to calculate approximate wave functions from the wave functions of the unperturbed system's stationary states.

For this purpose we use the analogue of the familiar variation-of-constants method for solving linear differential equations (*P. A. M. Dirac*, 1926). Let $\Psi_k^{(0)}$ be the wave functions (including the time factor) of the unperturbed system's stationary states. An arbitrary solution of the unperturbed wave equation can then be written as $\Psi = \sum_k a_k \Psi_k^{(0)}$. We seek a solution of the perturbed equation

$$i\hbar\frac{\partial\Psi}{\partial t} = (\hat{H}_0 + \hat{V})\Psi \tag{40.1}$$

in the form

$$\Psi = \sum_k a_k(t)\Psi_k^{(0)}, \tag{40.2}$$

where the expansion coefficients are functions of time. Substitution of (40.2) into (40.1), with

$$i\hbar\frac{\partial\Psi_k^{(0)}}{\partial t} = \hat{H}_0\Psi_k^{(0)},$$

gives

$$i\hbar\sum_k \Psi_k^{(0)}\frac{da_k}{dt} = \sum_k a_k\hat{V}\Psi_k^{(0)}.$$

Multiplying both sides on the left by $\Psi_m^{(0)*}$ and integrating, we find

$$i\hbar\frac{da_m}{dt} = \sum_k V_{mk}(t)a_k, \tag{40.3}$$

where

$$V_{mk}(t) = \int \Psi_m^{(0)*}\hat{V}\Psi_k^{(0)}\,dq = V_{mk}e^{i\omega_{mk}t}, \qquad \omega_{mk} = \frac{E_m^{(0)} - E_k^{(0)}}{\hbar}$$

are the perturbation matrix elements including their time factor. If V itself depends explicitly on time, the quantities V_{mk} must of course also be regarded as functions of time.

Choose the wave function of the nth stationary state as the unperturbed wave function. The corresponding coefficients in (40.2) are $a_n^{(0)} = 1$ and $a_k^{(0)} = 0$ for $k \neq n$. To obtain the first approximation, write $a_k = a_k^{(0)} + a_k^{(1)}$ and put $a_k = a_k^{(0)}$ on the right-hand side of (40.3), which already contains the small quantities V_{mk}. Thus

$$i\hbar\frac{da_k^{(1)}}{dt} = V_{kn}(t). \tag{40.4}$$

To indicate the unperturbed function for which the correction is being calculated, introduce a second index on a_k and write

$$\Psi_n = \sum_k a_{kn}(t)\Psi_k^{(0)}.$$

The integrated form of (40.4) is accordingly

$$a_{kn}^{(1)} = -\frac{i}{\hbar}\int V_{kn}(t)\,dt = -\frac{i}{\hbar}\int V_{kn}e^{i\omega_{kn}t}\,dt. \tag{40.5}$$

This determines the wave functions in the first approximation.

Consider more closely the important case of a time-periodic perturbation

$$\hat{V} = \hat{F}e^{-i\omega t} + \hat{G}e^{i\omega t}, \tag{40.6}$$

where $\hat{F}$ and $\hat{G}$ do not depend on time. Hermiticity of V requires

$$\hat{F}e^{-i\omega t} + \hat{G}e^{i\omega t} = \hat{F}^+e^{i\omega t} + \hat{G}^+e^{-i\omega t},$$

whence $\hat{G} = \hat{F}^+$, or

$$G_{nm} = F_{mn}^*. \tag{40.7}$$

Using this relation, we have

$$V_{kn}(t) = V_{kn}e^{i\omega_{kn}t} = F_{kn}e^{i(\omega_{kn}-\omega)t} + F_{nk}^*e^{i(\omega_{kn}+\omega)t}. \tag{40.8}$$

Substitution in (40.5) and integration yield

$$a_{kn}^{(1)} = -\frac{F_{kn}e^{i(\omega_{kn}-\omega)t}}{\hbar(\omega_{kn}-\omega)} - \frac{F_{nk}^*e^{i(\omega_{kn}+\omega)t}}{\hbar(\omega_{kn}+\omega)}. \tag{40.9}$$

These expressions apply provided neither denominator vanishes[4]; that is, for every k (at fixed n),

$$E_k^{(0)} - E_n^{(0)} \neq \pm\hbar\omega. \tag{40.10}$$

[4]More precisely, neither may be so small that $a_{kn}^{(1)}$ ceases to be small in comparison with unity.

For a number of applications it is useful to know the matrix elements of an arbitrary quantity f, defined using the perturbed wave functions. To first order,

$$f_{nm}(t) = f^{(0)}_{nm}(t) + f^{(1)}_{nm}(t),$$

where

$$f^{(0)}_{nm}(t) = \int \Psi^{(0)*}_n \hat{f} \Psi^{(0)}_m \, dq = f^{(0)}_{nm} e^{i\omega_{nm}t},$$

$$f^{(1)}_{nm}(t) = \int \left(\Psi^{(0)*}_n \hat{f} \Psi^{(1)}_m + \Psi^{(1)*}_n \hat{f} \Psi^{(0)}_m\right) dq.$$

Inserting

$$\Psi^{(1)}_n = \sum_k a^{(1)}_{kn} \Psi^{(0)}_k$$

with the coefficients from (40.9), we obtain

$$\begin{aligned} f^{(1)}_{nm}(t) = -e^{i\omega_{nm}t} \sum_k \Bigg\{ &\left[\frac{f^{(0)}_{nk} F_{km}}{\hbar(\omega_{km} - \omega)} + \frac{f^{(0)}_{km} F_{nk}}{\hbar(\omega_{kn} + \omega)} \right] e^{-i\omega t} \\ &+ \left[\frac{f^{(0)}_{nk} F^*_{mk}}{\hbar(\omega_{km} + \omega)} + \frac{f^{(0)}_{km} F^*_{kn}}{\hbar(\omega_{kn} - \omega)} \right] e^{i\omega t} \Bigg\}. \end{aligned} \tag{40.11}$$

This formula applies if none of its terms becomes large, hence if all the frequencies ω_{kn} and ω_{km} remain sufficiently far from ω. At $\omega = 0$ it reduces to (38.12). The formulas above assume that the unperturbed energy levels form only a discrete spectrum. They extend immediately to the presence of a continuous spectrum as well (the states being perturbed still belonging to the discrete spectrum): one simply adds the corresponding integrals over the continuous spectrum to the sums over discrete levels. The denominators $\omega_{kn} \pm \omega$ in (40.9) and (40.11) must then remain nonzero as $E^{(0)}_k$ ranges through both the discrete and continuous spectra. If, as is usual, the continuous spectrum lies above all discrete levels, condition (40.10), for example, must be supplemented by

$$E^{(0)}_{\min} - E^{(0)}_n > \hbar\omega, \tag{40.12}$$

where $E^{(0)}_{\min}$ is the energy at the lower edge of the continuous spectrum.

Problem

1. Determine how the nth and mth solutions of the Schrödinger equation change under a periodic perturbation of the form (40.6), whose frequency ω satisfies $E^{(0)}_m - E^{(0)}_n = \hbar(\omega + \varepsilon)$, where ε is small.

Solution. The method developed above does not apply, since the coefficient $a^{(1)}_{mn}$ in (40.9) becomes large. Return to the exact equations (40.3), with $V_{mk}(t)$ given by (40.8). Evidently, the principal effect comes from terms on their right-hand sides whose time dependence is governed by the small frequency $\omega_{mn} - \omega$. Dropping all other terms gives

$$i\hbar \frac{da_m}{dt} = F_{mn} e^{i\varepsilon t} a_n, \qquad i\hbar \frac{da_n}{dt} = F^*_{mn} e^{-i\varepsilon t} a_m.$$

Put $a_n e^{i\varepsilon t} = b_n$. Then

$$i\hbar \dot{a}_m = F_{mn} b_n, \qquad i\hbar(\dot{b}_n - i\varepsilon b_n) = F^*_{mn} a_m,$$

and elimination of a_m gives

$$\ddot{b}_n - i\varepsilon \dot{b}_n + (1/\hbar^2)|F_{mn}|^2 b_n = 0.$$

Two independent solutions may be chosen as

$$a_n = A e^{i\alpha_1 t}, \qquad a_m = -A\frac{\hbar\alpha_1}{F_{mn}} e^{i\alpha_2 t} \tag{1}$$

and

$$a_n = B e^{-i\alpha_2 t}, \qquad a_m = B\frac{\hbar\alpha_2}{F^*_{mn}} e^{-i\alpha_1 t}, \tag{2}$$

where A and B are constants (to be fixed by normalization), and

$$\alpha_1 = -\varepsilon/2 + \Omega, \quad \alpha_2 = \varepsilon/2 + \Omega, \quad \Omega = \sqrt{\varepsilon^2/4 + |\eta|^2}, \quad \eta = F_{mn}/\hbar.$$

Thus the perturbation transforms $\Psi_n^{(0)}$ and $\Psi_m^{(0)}$ into $a_n\Psi_n^{(0)} + a_m\Psi_m^{(0)}$, with a_n, a_m taken from (1) or (2).

Suppose that at $t = 0$ the system was in the state $\Psi_m^{(0)}$. At later times its state is the linear combination of the two solutions just found which reduces to $\Psi_m^{(0)}$ at $t = 0$:

$$\Psi = e^{i\varepsilon t/2}\left(\cos\Omega t - \frac{i\varepsilon}{2\Omega}\sin\Omega t\right)\Psi_m^{(0)} - \frac{i\eta^*}{\Omega} e^{-i\varepsilon t/2}\sin\Omega t \cdot \Psi_n^{(0)}. \tag{3}$$

The squared modulus of the coefficient of $\Psi_n^{(0)}$ is

$$\frac{|\eta|^2}{2\Omega^2}\left[1 - \cos(2\Omega t)\right]. \tag{4}$$

It is the probability that the system is in $\Psi_n^{(0)}$ at time t. This probability is periodic, with frequency 2Ω, and varies from zero to $|\eta|^2/\Omega^2$.

At exact resonance, $\varepsilon = 0$, (4) becomes

$$(1/2)\left[1 - \cos(2|\eta|t)\right].$$

It varies periodically between zero and unity: the system passes periodically from $\Psi_m^{(0)}$ to $\Psi_n^{(0)}$.

§ 41. Transitions produced by a perturbation acting for a finite time

Suppose that the perturbation $V(t)$ acts only during some finite time interval (or, alternatively, that $V(t)$ decreases sufficiently rapidly as $t \to \pm\infty$). Before the perturbation begins to act (or in the limit $t \to -\infty$), let the system be in the nth stationary state of the discrete spectrum. At any later instant its state is described by

$$\Psi = \sum_k a_{kn}\Psi_k^{(0)},$$

where, to first order,

$$\begin{aligned} a_{kn} &= a_{kn}^{(1)} = -\frac{i}{\hbar}\int_{-\infty}^{t} V_{kn}e^{i\omega_{kn}t}\,dt, \qquad k \neq n, \\ a_{nn} &= 1 + a_{nn}^{(1)} = 1 - \frac{i}{\hbar}\int_{-\infty}^{t} V_{nn}\,dt. \end{aligned} \tag{41.1}$$

The integration limits in (40.5) have been selected so that every $a_{kn}^{(1)}$ vanishes as $t \to -\infty$. Once the action of the perturbation has ended (or as $t \to \infty$), the coefficients a_{kn} acquire constant values $a_{kn}(\infty)$, and the system is in the state whose wave function is

$$\Psi = \sum_k a_{kn}(\infty)\Psi_k^{(0)}.$$

This function again satisfies the unperturbed wave equation, though it differs from the original function $\Psi_n^{(0)}$. By the general rules, the squared modulus of $a_{kn}(\infty)$ gives the probability that the system has energy $E_k^{(0)}$, that is, that it is in the kth stationary state.

Thus a perturbation can carry the system from its original stationary state into any other one. The probability of a transition from the initial (ith) stationary state to the final (fth) stationary state is[5]

$$w_{fi} = \frac{1}{\hbar^2}\left|\int_{-\infty}^{+\infty} V_{fi}e^{i\omega_{fi}t}\,dt\right|^2. \tag{41.2}$$

Now consider a perturbation which, once it has arisen, continues to act indefinitely (while remaining small throughout). In other words, $V(t)$ tends to zero as $t \to -\infty$ and to a finite nonzero limit as $t \to \infty$. Formula (41.2) cannot be used directly here, since its integral diverges. From the physical standpoint, however, this divergence is immaterial and is easily removed. Integration by parts gives

$$\begin{aligned} a_{fi} &= -\frac{i}{\hbar}\int_{-\infty}^{t} V_{fi}e^{i\omega_{fi}t}\,dt \\ &= -\left.\frac{V_{fi}e^{i\omega_{fi}t}}{\hbar\omega_{fi}}\right|_{-\infty}^{t} + \int_{-\infty}^{t} \frac{\partial V_{fi}}{\partial t}\frac{e^{i\omega_{fi}t}}{\hbar\omega_{fi}}\,dt. \end{aligned}$$

The first term vanishes at the lower limit; at the upper limit it formally coincides with the expansion coefficients in (38.8). (The extra periodic factor occurs simply because a_{fi} are expansion coefficients of the full wave function Ψ, whereas the c_{fi} of § 38 are expansion coefficients of the time-independent function ψ.) It is consequently clear that the limit of this term as $t \to \infty$ describes only the change of the original wave function $\Psi_i^{(0)}$ caused by the "constant part" $V(+\infty)$ of the perturbation, and therefore has nothing to do with transitions into other states. The transition probability is instead given by the square of the second term:

$$w_{fi} = \frac{1}{\hbar^2\omega_{fi}^2}\left|\int_{-\infty}^{+\infty} \frac{\partial V_{fi}}{\partial t}e^{i\omega_{fi}t}\,dt\right|^2. \tag{41.3}$$

[5] For uniformity, whenever transition probabilities are discussed below, we shall denote the initial and final states by the indices i and f, respectively. We shall also write the indices of transition probabilities specifically in the order fi, consistently with the convention for matrix-element indices.

The formulas obtained also apply when a transition proceeds from a discrete state to a continuous-spectrum state. The only difference is that one then speaks of the probability of transition from the specified (ith) state into states for which the quantities ν_f (see the end of § 38) lie between ν_f and $\nu_f + d\nu_f$. Thus, for example, (41.2) must be written as

$$dw_{if} = \frac{1}{\hbar^2}\left|\int_{-\infty}^{\infty} V_{fi}e^{i\omega_{fi}t}\,dt\right|^2 d\nu_f. \tag{41.4}$$

If $V(t)$ varies little over time intervals of order $1/\omega_{fi}$, the integral in (41.2) or (41.3) is very small. In the limit of arbitrarily slow variation of the applied perturbation, the probability of every transition that changes the energy (that is, with nonzero ω_{fi}) tends to zero. Hence, when an applied perturbation changes sufficiently slowly (*adiabatically*), a system initially in a particular nondegenerate stationary state continues to remain in that same state (see also § 53).

In the opposite limit, when a perturbation is switched on very rapidly—that is, *suddenly*—the derivatives $\partial V_{fi}/\partial t$ become infinite at the "switching instant." In the integral of $(\partial V_{fi}/\partial t)e^{i\omega_{fi}t}$, the comparatively slow factor $e^{i\omega_{fi}t}$ may then be taken outside the integral at its value at that instant. The remaining integration is immediate and gives

$$w_{fi} = \frac{|V_{fi}|^2}{\hbar^2\omega_{fi}^2}. \tag{41.5}$$

Transition probabilities for sudden perturbations can also be found when the perturbation is not small. Let the system initially be in a state described by an eigenfunction $\psi_i^{(0)}$ of the original Hamiltonian $\hat{H}_0$. If the Hamiltonian changes suddenly (in a time short compared with the periods $1/\omega_{fi}$ of transitions from the given state i to other states), the wave function has no time to change and remains what it was before the perturbation. It is no longer, however, an eigenfunction of the system's new Hamiltonian $\hat{H}$; the state $\psi_i^{(0)}$ is therefore not stationary. By the general rules of quantum mechanics, the probabilities w_{fi} for transition into any of the new stationary states are determined by expanding $\psi_i^{(0)}$ in the eigenfunctions ψ_f of $\hat{H}$:

$$w_{fi} = \left|\int \psi_i^{(0)}\psi_f^*\,dq\right|^2. \tag{41.6}$$

We now show how this general formula reduces to (41.5) when the change $\hat{V} = \hat{H} - \hat{H}_0$ of the Hamiltonian is small. Multiply the equations

$$\hat{H}_0\psi_i^{(0)} = E_i^{(0)}\psi_i^{(0)}, \qquad \hat{H}^*\psi_f^* = E_f\psi_f^*$$

by ψ_f^* and $\psi_i^{(0)}$, respectively, integrate over dq, and subtract the two results term by term. Using also the self-adjointness of $\hat{H}$, we obtain

$$(E_f - E_i^{(0)})\int \psi_f^*\psi_i^{(0)}\,dq = \int \psi_f^*\hat{V}\psi_i^{(0)}\,dq.$$

If $\hat{V}$ is small, then to first order E_f may be replaced by the nearby unperturbed level $E_f^{(0)}$, and ψ_f on the right-hand side by the corresponding function $\psi_f^{(0)}$. It follows that

$$\int \psi_f^* \psi_i^{(0)}\, dq = \frac{1}{\hbar\omega_{fi}} \int \psi_f^{(0)*} \hat{V} \psi_i^{(0)}\, dq,$$

so that (41.6) becomes (41.5).

Problems

Problem

1. A uniform electric field is suddenly applied to a charged oscillator in its ground state. Find the probabilities for the perturbation to transfer the oscillator into its excited states.

Solution. Potential energy of an oscillator in a uniform field exerting force F upon it is

$$U(x) = \frac{m\omega^2}{2} x^2 - Fx = \frac{m\omega^2}{2} (x - x_0)^2 + \text{const},$$

where $x_0 = F/m\omega^2$. Thus it again has a purely oscillator form, but with its equilibrium position displaced. Consequently, the stationary-state wave functions of the perturbed oscillator are $\psi_k(x - x_0)$, where $\psi_k(x)$ are the oscillator functions (23.12), while the initial wave function is $\psi_0(x)$ from (23.13). Using these functions and expression (23.11) for the Hermite polynomials, we find

$$\int_{-\infty}^{+\infty} \psi_0^{(0)} \psi_k\, dx = \frac{(-1)^k}{\sqrt{2^k \pi k!}} e^{-\xi_0^2/2} \int_{-\infty}^{\infty} e^{-\xi\xi_0} \frac{d^k}{d\xi^k} e^{-\xi^2 + 2\xi\xi_0}\, d\xi,$$

where $\xi_0 = x_0\sqrt{m\omega/\hbar}$. Integrating by parts k times reduces the integral here to

$$\xi_0^k \int_{-\infty}^{\infty} \exp\Big(-\xi^2 + \xi\xi_0\Big)\, d\xi = \xi_0^k \sqrt{\pi} \exp \frac{\xi_0^2}{4}.$$

Formula (41.6) therefore gives the required transition probability as

$$w_{k0} = \frac{\bar{k}^k}{k!} e^{-\bar{k}}, \qquad \bar{k} = \frac{\xi_0^2}{2} = \frac{F^2}{2m\hbar\omega^3}.$$

As a function of the integer k, this is a Poisson distribution with mean $\bar{k}$.

The perturbative regime corresponds to small F such that $\bar{k} \ll 1$. The excitation probabilities are then small and fall rapidly as k increases; the largest is $w_{10} \approx \bar{k}$.

Conversely, when F is large ($\bar{k} \gg 1$), excitation occurs with overwhelming probability: the chance that the oscillator remains in its normal state is $w_{00} = e^{-\bar{k}}$.

Problem

2. The nucleus of an atom in its normal state receives a sudden impulse and thereby acquires velocity **v**. Assume that the duration τ of the impulse is short both in comparison with the electronic periods and with a/v, where a is an atomic dimension. Determine the probability that this "shaking" excites the atom (*A. B. Migdal*, 1939).

Solution. Pass to the reference frame K' that moves with the nucleus after the impact. Because $\tau \ll a/v$, the nucleus may be treated as having hardly moved during the impact, and the electron coordinates in K' and in the original frame K therefore coincide immediately after the perturbation. The initial wave function in K' is

$$\psi_0' = \psi_0 \exp\left(-i\mathbf{q}\sum_a \mathbf{r}_a\right), \qquad \mathbf{q} = \frac{m\mathbf{v}}{\hbar},$$

where ψ_0 is the normal-state wave function for a stationary nucleus and the sum in the exponential extends over all Z electrons of the atom. According to (41.6), the probability sought for transition into the kth excited state is now

$$w_{k0} = \left|\left\langle k \middle| \exp\left(-i\mathbf{q}\sum_a \mathbf{r}_a\right) \middle| 0\right\rangle\right|^2.$$

In particular, for $qa \ll 1$, expand the exponential under the integral. The integral of $\psi_k^*\psi_0$ vanishes by orthogonality, and hence

$$w_{k0} = \left|\left\langle k \middle| \mathbf{q}\sum_a \mathbf{r}_a \middle| 0\right\rangle\right|^2.$$

Problem

3. Find the total probability of excitation and ionization of a hydrogen atom subjected to a sudden "shaking" (see the preceding problem).

Solution. The required probability can be evaluated as the difference

$$1 - w_{00} = 1 - \left|\int \psi_0^2 e^{-i\mathbf{q}\mathbf{r}}\, dV\right|^2,$$

where w_{00} is the probability that the atom remains in the ground state; $\psi_0 = (\pi a^3)^{-1/2} e^{-r/a}$ is the hydrogen ground-state wave function, and a is the Bohr radius. Evaluation of the integral gives

$$1 - w_{00} = 1 - \frac{1}{(1 + \frac{1}{4}q^2a^2)^4}.$$

In the limiting case $qa \ll 1$, this probability tends to zero as $1 - w_{00} \approx q^2a^2$; for $qa \gg 1$, it approaches unity as $1 - w_{00} \approx 1 - (2/qa)^8$.

Problem

4. Determine the probability that an electron is ejected from the K shell of an atom with large atomic number Z during nuclear β decay. Assume the β-particle velocity to be large relative to the velocity of a K electron (*A. B. Migdal*, 1941; *E. L. Feinberg*, 1939).

Solution. [6] Under the stated conditions, the time taken by the β particle to cross the K shell is short compared with the electron's orbital period; the change in nuclear charge may therefore be regarded as instantaneous. The perturbation is the change $V = 1/r$ in the nuclear field when its charge changes by an amount small compared with Z (namely, by 1). According to (41.5), the probability for either of the two K-shell electrons, whose energy is $E_0 = -Z^2/2$,[7] to enter a continuous-spectrum state of energy $E = k^2/2$ in the interval $dE = k\,dk$ is

$$dw = 2\frac{4|V_{0k}|^2}{(k^2+Z^2)^2}\,dk.$$

The integral defining V_{0k} receives its essential contribution from distances of order $1/Z$ near the nucleus. In this region a hydrogen-like expression may also be used for the continuous-spectrum wave function. The electron's final state must have $l = 0$, the same angular momentum as the initial state. From the function R_{10} and the function R_{k0} normalized on the $k/2\pi$ scale, obtained in § 36, together with formula (f.3) of the Mathematical Appendices, we find[8]

$$\left(\frac{1}{r}\right)_{0k} = \frac{4\sqrt{2\pi k}}{1-e^{-2\pi Z/k}}\,\frac{(1+ik/Z)^{iZ/k}(1-ik/Z)^{-iZ/k}}{1+k^2/Z^2},$$

and, since

$$|(1+i\alpha)^{i/\alpha}|^2 = \exp\left(-2\frac{\arctan\alpha}{\alpha}\right),$$

we finally obtain

$$dw = \frac{2^7}{Z^4(1+k^2/Z^2)^4}\,f\left(\frac{k}{Z}\right)k\,dk,$$

where

$$f(\alpha) = \frac{1}{1-e^{-2\pi/\alpha}}\exp\left(-4\frac{\arctan\alpha}{\alpha}\right).$$

The limiting values of $f(\alpha)$ are

$$f = e^{-4} \quad \text{for } \alpha \ll 1, \qquad f = \alpha/2\pi \quad \text{for } \alpha \gg 1.$$

The total probability of ionizing the K shell follows by integrating dw over all energies of the emitted electron. Numerical evaluation yields $w = 0.65Z^{-2}$.

[6] Atomic units are used in Problems 4 and 5.

[7] Here and below the K-electron states are treated as hydrogen-like (see § 74).

[8] It is convenient during the calculation to use Coulomb units, returning to atomic units in the final result.

Problem

5. Determine the probability that an electron is ejected from the K shell of a large-Z atom during nuclear α decay. The α particle moves slowly compared with a K electron, although its time of emergence from the nucleus is short compared with the electron's orbital period (*A. B. Migdal*, 1941; *J. Levinger*, 1953).

Solution. After the α particle has emerged, the perturbation acting upon the electron is adiabatic. The effect sought is therefore determined chiefly by times close to the nonadiabatic "switching instant," when the α particle, already outside the nucleus and moving freely, is still at distances small compared with the radius of the K orbit. The ionizing perturbation V is then the departure of the combined nuclear and α-particle field from the pure Coulomb field Z/r. For particles of atomic weights 4 and $A-4$, charges 2 and $Z-2$, and separation vt (where v is the relative speed of the nucleus and α particle), the dipole moment is

$$\frac{2(A-4)-(Z-2)4}{A}vt = \frac{2(A-2Z)}{A}vt.$$

Thus the dipole term in the field of the nucleus and the α particle is [9]

$$V = \frac{2(A-2Z)}{A}vt\frac{z}{r^3},$$

where the z axis points along $\mathbf{v}$. The matrix element of this perturbation can be reduced to that of z. Taking the matrix element of the electron's equation of motion $\ddot{z} = -Zz/r^3$, we obtain

$$\left(\frac{z}{r^3}\right)_{0k} = \frac{(E-E_0)^2}{Z}z_{0k}.$$

By (41.2), the transition probability sought for either of the two K-shell electrons is

$$dw = 2\left|\int_0^\infty V_{0k}e^{i(E_0-E)t}\,dt\right|^2 dk = \frac{8(A-2Z)^2v^2}{A^2Z^2}|z_{0k}|^2\frac{dk}{2\pi}.$$

To evaluate the integral, an additional damping factor $e^{-\lambda t}$, $\lambda > 0$, is inserted in the integrand, and $\lambda \to 0$ is then taken in the result. For the matrix element of $z = r\cos\theta$, note that the initial orbital angular momentum is $l = 0$, so $\cos\theta$ has a nonzero matrix element only for transition to a state with $l = 1$. In that case

$$|(\cos\theta)_{01}|^2 = \frac{1}{3}, \qquad |z_{0k}|^2 = \frac{1}{3}|r_{0k}|^2.$$

Evaluation of r_{0k} using the radial functions R_{00} and R_{k1} then gives

$$dw = \frac{2^{11}(A-2Z)^2v^2}{3A^2Z^6(1+k^2/Z^2)^5}f\left(\frac{k}{Z}\right)k\,dk,$$

where f is the function defined in Problem 4.

[9] When the difference $A-2Z$ is small, it may also be necessary to include the next, quadrupole term.

§ 42. Transitions under a periodic perturbation

A different type of result is obtained for the probability of transition to continuous-spectrum states under a periodic perturbation. Suppose that at an initial time $t = 0$ the system is in the ith stationary state of the discrete spectrum. Let the frequency ω of the periodic perturbation satisfy

$$\hbar\omega > E_{\min} - E_i^{(0)}, \tag{42.1}$$

where $E_{\min}$ is the energy at which the continuous spectrum begins.

It follows at once from the results of § 40 that the principal role is played by continuous-spectrum states whose energies E_f lie close to the "resonant" energy $E_i^{(0)} + \hbar\omega$, that is, states for which $\omega_{fi} - \omega$ is small. For the same reason, only the first term in the perturbation matrix element (40.8), whose frequency $\omega_{fi} - \omega$ is close to zero, need be retained. Substituting this term in (40.5) and integrating gives

$$a_{fi} = -\frac{i}{\hbar}\int_0^t V_{fi}(t)\,dt = -F_{fi}\frac{\exp\left[i(\omega_{fi} - \omega)t\right] - 1}{\hbar(\omega_{fi} - \omega)}. \tag{42.2}$$

The lower integration limit has been chosen so that $a_{fi} = 0$ at $t = 0$, in accordance with the initial condition.

The squared modulus of a_{fi} is

$$|a_{fi}|^2 = |F_{fi}|^2\frac{4\sin^2\dfrac{\omega_{fi} - \omega}{2}t}{\hbar^2(\omega_{fi} - \omega)^2}. \tag{42.3}$$

It is readily seen that, at large t, the function appearing here may be represented as proportional to t.

To see this, use the formula

$$\lim_{t\to\infty}\frac{\sin^2\alpha t}{\pi t\alpha^2} = \delta(\alpha). \tag{42.4}$$

For $\alpha \neq 0$ the displayed limit is zero, while for $\alpha = 0$,

$$\frac{\sin^2\alpha t}{t\alpha^2} = t,$$

so the limit is infinite. On integrating over $d\alpha$ from $-\infty$ to $+\infty$ and making the substitution $\alpha t = \xi$, we obtain

$$\frac{1}{\pi}\int_{-\infty}^{+\infty}\frac{\sin^2\alpha t}{t\alpha^2}\,d\alpha = \frac{1}{\pi}\int_{-\infty}^{+\infty}\frac{\sin^2\xi}{\xi^2}\,d\xi = 1.$$

Thus the function on the left-hand side of (42.4) satisfies all the requirements defining the delta function.

Accordingly, for large t we may write

$$|a_{fi}|^2 = \frac{1}{\hbar^2}|F_{fi}|^2\pi t\,\delta\left(\frac{\omega_{fi} - \omega}{2}\right),$$

or, substituting $\hbar\omega_{fi} = E_f - E_i^{(0)}$ and using $\delta(ax) = \delta(x)/a$,

$$|a_{fi}|^2 = \frac{2\pi}{\hbar}|F_{fi}|^2\delta(E_f - E_i^{(0)} - \hbar\omega)t.$$

The quantity $|a_{fi}|^2 d\nu_f$ is the probability of transition from the initial state to states in a specified interval $d\nu_f$. At large t it is proportional to the interval of time elapsed since $t = 0$. The transition probability dw_{fi} per unit time is therefore[10]

$$dw_{fi} = \frac{2\pi}{\hbar}|F_{fi}|^2\delta(E_f - E_i^{(0)} - \hbar\omega)\, d\nu_f. \tag{42.5}$$

As expected, it is nonzero only for transitions to states with energy $E_f = E_i^{(0)} + \hbar\omega$. If the continuous-spectrum energy levels are nondegenerate, so that ν_f may be taken to mean the energy alone, the whole "interval" $d\nu_f$ reduces to a single state of energy $E = E_i^{(0)} + \hbar\omega$, and the probability of transition to this state is

$$w_{Ei} = \frac{2\pi}{\hbar}|F_{Ei}|^2. \tag{42.6}$$

It is also instructive to derive (42.5) in another way. Instead of switching the periodic perturbation on at the definite instant $t = 0$, let it grow slowly from $t = -\infty$ according to the exponential law $e^{\lambda t}$, where λ is positive and is ultimately allowed to tend to zero (*adiabatic switching*). The initial condition $a_{fi} = 0$ is then imposed at $t = -\infty$. The perturbation matrix element is now

$$V_{fi} = F_{fi}e^{i(\omega_{fi}-\omega)t+\lambda t},$$

and in place of (42.2) we write

$$a_{fi} = -\frac{i}{\hbar}\int_{-\infty}^{t} V_{fi}(t)\, dt = -F_{fi}\frac{\exp\left[i(\omega_{fi}-\omega)t + \lambda t\right]}{\hbar(\omega_{fi} - \omega - i\lambda)}. \tag{42.7}$$

Hence

$$|a_{fi}|^2 = \frac{1}{\hbar^2}|F_{fi}|^2\frac{e^{2\lambda t}}{(\omega_{fi}-\omega)^2 + \lambda^2}.$$

The transition probability per unit time is determined by

$$\frac{d}{dt}|a_{fi}|^2 = 2\lambda|a_{fi}|^2.$$

Now use the formula

$$\lim_{\lambda\to 0}\frac{\lambda}{\pi(\alpha^2 + \lambda^2)} = \delta(\alpha), \tag{42.8}$$

which is valid in the same sense as (42.4). Taking the limit $\lambda \to 0$ gives

$$\frac{d}{dt}|a_{fi}|^2 \longrightarrow \frac{2\pi}{\hbar^2}|F_{fi}|^2\delta(\omega_{fi} - \omega),$$

and we return once more to (42.5).

[10]It is readily verified that retaining the omitted second term in (40.8) would give additional expressions which, after division by t, tend to zero as $t \to +\infty$.

§ 43. Transitions within the continuous spectrum

One of the most important uses of perturbation theory is the calculation of transition probabilities within the continuous spectrum under a constant (time-independent) perturbation. As already noted, continuous-spectrum states are almost always degenerate. After choosing a definite set of unperturbed wave functions for a given energy level, the question can be posed as follows. The system is known to have occupied one of these states initially; we seek the probability that it passes into another state of the same energy. For transitions from an initial state i into states between ν_f and $\nu_f + d\nu_f$, (42.5), with $\omega = 0$ and a change of notation, gives directly

$$dw_{fi} = \frac{2\pi}{\hbar}|V_{fi}|^2\delta(E_f - E_i)\, d\nu_f. \tag{43.1}$$

As expected, this is nonzero only when $E_f = E_i$: a constant perturbation causes transitions solely between states of equal energy. For transitions from continuous-spectrum states, however, dw_{fi} cannot itself be treated as a transition probability. It does not even have the corresponding dimension, but has dimension $1/\text{s}$. Expression (43.1) gives the number of transitions per unit time, and its dimension depends on the chosen normalization of the continuous-spectrum wave functions.[11]

We shall calculate the perturbed wave function which, before the perturbation begins to act, coincides with the original unperturbed function $\psi_i^{(0)}$. Following the procedure stated at the end of the preceding section, take the perturbation to be switched on adiabatically as $e^{\lambda t}$, with $\lambda \to 0$. Formula (42.7), after setting $\omega = 0$ and changing notation, gives

$$a_{fi}^{(1)} = V_{fi}\frac{\exp\left\{\dfrac{i}{\hbar}(E_f - E_i)t + \lambda t\right\}}{E_i - E_f + i\lambda}. \tag{43.2}$$

The perturbed wave function is

$$\Psi_i = \Psi_i^{(0)} + \int a_{fi}^{(1)}\Psi_f^{(0)}\, d\nu_f,$$

where the integration extends over the entire continuous spectrum.[12] Substitution of (43.2) yields

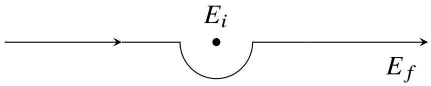
$$\Psi_i = \left[\psi_i^{(0)} + \int V_{fi}\psi_f^{(0)}\frac{d\nu_f}{E_i - E_f + i0}\right]\exp\left(-\frac{i}{\hbar}E_i t\right). \tag{43.3}$$

In the limit $\lambda \to 0$, the factor $e^{\lambda t}$ has been replaced by unity. The term $+i0$, meaning the limit of $i\lambda$ as the positive quantity λ tends to zero, specifies the integration prescription for E_f. Its differential occurs as a factor in $d\nu_f$, together with those of the other quantities characterizing continuous-spectrum states. Without $i\lambda$, the integrand in (43.3) would

[11] Phenomena covered by this theory include, for example, various collisions. In their initial and final states the system is an assemblage of free particles, while their mutual interaction serves as the perturbation. With a suitable wave- function normalization, (43.1) can then become a collision cross-section (see § 126).

[12] If a discrete spectrum is also present, the corresponding sum over its states must be added to the integral here and in the formulas below.

have a pole at $E_f = E_i$, and the integral would diverge near it. The term $i\lambda$ displaces this pole into the upper half-plane of the complex variable E_f. When $\lambda \to 0$, the pole returns to the real axis, but the integration path is now known to pass below it:



(43.4)

The time factor in (43.3) shows, as it should, that this function belongs to the same energy E_i as the original unperturbed function. Equivalently,

$$\psi_i = \psi_i^{(0)} + \int \frac{V_{fi}}{E_i - E_f + i0}\psi_f^{(0)}\, d\nu_f \tag{43.5}$$

satisfies the Schrödinger equation

$$(\hat{H}_0 + \hat{V})\psi_i = E_i\psi_i.$$

It is consequently natural that this expression agrees exactly with (38.8).[13]

The preceding calculation is the first approximation of perturbation theory. The second approximation is also readily obtained. One derives the next approximation for Ψ_i by the method of § 38, now knowing how the "divergent" integrals are to be taken. A direct calculation gives

$$\Psi_i = \left\{\psi_i^{(0)} + \int\left[V_{fi} + \int \frac{V_{f\nu}V_{\nu i}}{E_i - E_\nu + i0}\, d\nu\right]\frac{\psi_f^{(0)}\, d\nu_f}{E_i - E_f + i0}\right\}\exp\left(-\frac{i}{\hbar}E_i t\right).$$

Comparison with (43.3) lets us write the corresponding probability formula (more precisely, the number of transitions) by direct analogy with (43.1):

$$dw_{fi} = \frac{2\pi}{\hbar}\left|V_{fi} + \int \frac{V_{f\nu}V_{\nu i}}{E_i - E_\nu + i0}\, d\nu\right|^2 \delta(E_i - E_f)\, d\nu_f. \tag{43.6}$$

The matrix element V_{fi} may vanish for the transition in question. The first-order effect is then absent altogether, and (43.6) becomes

$$dw_{fi} = \frac{2\pi}{\hbar}\left|\int \frac{V_{f\nu}V_{\nu i}}{E_i - E_\nu}\, d\nu\right|^2 \delta(E_f - E_i)\, d\nu_f. \tag{43.7}$$

In applications of this formula, $E_\nu = E_i$ is usually not a pole of the integrand. The prescription for integration over dE_ν is then immaterial, and the path may be taken directly along the real axis.

States ν for which $V_{f\nu}$ and $V_{\nu i}$ are nonzero are often called *intermediate* states for the transition $i \to f$. In a pictorial description, the transition appears to take place in two stages, $i \to \nu$ and $\nu \to f$; this description must not, of course, be understood literally. A

[13] The integration prescription in (43.5) can also be fixed by requiring the large-distance asymptotic form of ψ_i to contain only an outgoing wave, and no incoming wave (see § 136).

transition $i \longrightarrow f$ may require not one but several successive intermediate states. Formula (43.7) extends directly to such cases. If two intermediate states are needed, then

$$dw_{fi} = \frac{2\pi}{\hbar}\left|\int \frac{V_{f\nu'}V_{\nu'\nu}V_{\nu i}}{(E_i - E_{\nu'})(E_i - E_\nu)}\, d\nu\, d\nu'\right|^2 \delta(E_f - E_i)\, d\nu_f. \tag{43.8}$$

Finally, the mathematical meaning of integrals taken along a path such as (43.4) is expressed by

$$\int \frac{f(x)\, dx}{x - a - i0} = \mathrm{P}\int \frac{f(x)\, dx}{x - a} + i\pi f(a), \tag{43.9}$$

where the interval of integration on the real axis contains $x = a$. Indeed, detouring around the pole $x = a$ on a semicircle of radius ρ, the full integral is the sum of the real-axis integrals from the lower limit to $a - \rho$ and from $a + \rho$ to the upper limit, plus $i\pi$ times the residue at the pole. As $\rho \to 0$, the real-axis pieces combine into the integral over the full interval understood as a principal value, as indicated by the marked integral sign, and (43.9) follows. The same formula is also written symbolically as

$$\frac{1}{x - a - i0} = P\frac{1}{x - a} + i\pi\delta(x - a); \tag{43.10}$$

the symbol P means that the principal value is to be taken when $f(x)/(x - a)$ is integrated.

§ 44. The uncertainty relation for energy

Consider a system made up of two weakly interacting parts. Suppose that at a certain instant these parts are known to have definite energies, denoted by E and ε. After an interval Δt, the energies are measured again, giving values E' and ε' which generally differ from E and ε. We can readily determine the order of magnitude of the most probable value of $E' + \varepsilon' - E - \varepsilon$ found by the measurement.

According to (42.3), with $\omega = 0$, the probability that a time-independent perturbation transfers the system during time t from energy E to energy E' is proportional to

$$\frac{\sin^2\left(\dfrac{E' - E}{2\hbar}t\right)}{(E' - E)^2}.$$

It follows that the most probable difference $E' - E$ is of order $\hbar/t$.

Applying this result to the present case, with the interaction between the two parts as the perturbation, gives

$$|E + \varepsilon - E' - \varepsilon'|\Delta t \sim \hbar. \tag{44.1}$$

Thus the shorter the interval Δt, the larger the observed energy change. Its order of magnitude, $\hbar/\Delta t$, is importantly independent of the strength of the perturbation.

The energy change specified by (44.1) will be observed however weak the interaction between the two parts may be. This is a purely quantum result with a profound physical meaning. It shows that in quantum mechanics two measurements can test energy conservation only to an accuracy of order $\hbar/\Delta t$, where Δt is the time between them.

Relation (44.1) is often called the uncertainty relation for energy. It must, however, be emphasized that its meaning is essentially different from that of the coordinate–momentum relation $\Delta p \Delta x \sim \hbar$. In the latter, Δp and Δx are the uncertainties in momentum and coordinate at one and the same instant; the relation says that these two quantities cannot have strictly definite values simultaneously.

The energies E and ε, by contrast, can be measured at each given instant with arbitrary precision. The quantity $(E + \varepsilon) - (E' + \varepsilon')$ in (44.1) is the difference between two exactly measured energy values at two different times, not an uncertainty in the energy at a particular instant.

If E is regarded as the energy of a system and ε as the energy of a "measuring apparatus," their interaction energy can be taken into account only to an accuracy $\hbar/\Delta t$. Denote the errors in the measurements of the corresponding quantities by $\Delta E, \Delta\varepsilon, \ldots$. In the favorable case where ε and ε' are known exactly ($\Delta\varepsilon = \Delta\varepsilon' = 0$),

$$\Delta(E - E') \sim \frac{\hbar}{\Delta t}. \tag{44.2}$$

This relation has important consequences for momentum measurement. Measuring the momentum of a particle (take an electron for definiteness) involves a collision with another, "measuring," particle whose momenta before and after the collision may be regarded as known exactly.[14] Applying momentum conservation to this collision gives three equations, the components of one vector equation, for six unknowns: the components of the electron momentum before and after the collision. More equations might be sought by arranging a sequence of collisions with measuring particles and applying momentum conservation to each. But the number of unknowns also grows, since it includes the electron momenta between collisions; for any number of collisions there are plainly three more unknowns than equations. Momentum measurement must therefore use energy conservation at each collision in addition to momentum conservation. As we have seen, the energy law can be applied only to an accuracy of order $\hbar/\Delta t$, with Δt the interval between the beginning and end of the process under consideration.

For simplicity, consider an idealized thought experiment in which the "measuring particle" is a perfectly reflecting plane mirror. Only the momentum component normal to the mirror then matters. Momentum and energy conservation give, for determining the particle momentum P,

$$p' + P' - p - P = 0, \tag{44.3}$$

$$|\varepsilon' + E' - \varepsilon - E| \sim \frac{\hbar}{\Delta t} \tag{44.4}$$

(P, E are the particle's momentum and energy; p, ε are those of the mirror; unprimed and primed quantities refer respectively to times before and after the collision). The quantities $p, p', \varepsilon, \varepsilon'$ of the "measuring particle" may be treated as known exactly, so their errors vanish. The equations above then give

$$\Delta P = \Delta P', \qquad |\Delta E' - \Delta E| \sim \frac{\hbar}{\Delta t}.$$

[14] For the present argument, how the energy of the "measuring" particle becomes known is immaterial.

But

$$\Delta E = \frac{\partial E}{\partial P}\Delta P = v\Delta P,$$

and similarly

$$\Delta E' = v'\Delta P' = v'\Delta P.$$

Hence

$$|(v'_x - v_x)\Delta P_x| \sim \frac{\hbar}{\Delta t}. \tag{44.5}$$

The suffix x has been attached to the velocities and momentum to stress that the relation applies separately to each component.

This is the required relation. It shows that measuring an electron's momentum to a specified accuracy ΔP is unavoidably accompanied by a change in its velocity, and therefore in the momentum itself.

The shorter the measurement process, the greater this change. The velocity change can be made arbitrarily small only as $\Delta t \to \infty$, but a momentum measurement extending over a long time can in general have meaning only for a free particle. Here the nonrepeatability of momentum measurements at short intervals, and the two-sided nature of measurement in quantum mechanics, are especially clear: one must distinguish the measured value of a quantity from the value produced by the measuring process.[15]

The argument at the beginning of this section, based on perturbation theory, can also be approached through the decay of a system under some perturbation. Let E_0 be an energy level calculated while entirely neglecting the possibility of decay. Denote by τ the *lifetime* of this state, the reciprocal of its decay probability per unit time. The same method gives

$$|E_0 - E - \varepsilon| \sim \hbar/\tau, \tag{44.6}$$

where E and ε are the energies of the two parts into which the system has decayed. Their sum $E + \varepsilon$ provides a measure of the system's energy before decay. The relation therefore shows that the energy of a decaying system in a *quasistationary* state can be defined only to an accuracy of order $\hbar/\tau$. This quantity is customarily called the level *width* Γ. Thus

$$\Gamma \sim \hbar/\tau. \tag{44.7}$$

§ 45. Potential energy treated as a perturbation

The case in which the particle's entire potential energy in an external field can be treated as the perturbation calls for separate consideration. The unperturbed Schrödinger equation is then the equation for free motion,

$$\Delta\psi^{(0)} + k^2\psi^{(0)} = 0, \qquad k = \frac{\sqrt{2mE}}{\hbar} = \frac{p}{\hbar} \tag{45.1}$$

[15] Relation (44.5), as well as the clarification of the physical meaning of the energy uncertainty relation, is due to N. Bohr (1928).

whose solutions are plane waves. Since the energy spectrum of free motion is continuous, this is a special instance of perturbation theory in a continuous spectrum. Here it is more convenient to obtain the solution directly, without using the general formulas.

The equation for the first-order correction $\psi^{(1)}$ to the wave function is

$$\Delta\psi^{(1)} + k^2\psi^{(1)} = \frac{2mU}{\hbar^2}\psi^{(0)} \tag{45.2}$$

(U is the potential energy). As is known from electrodynamics, its solution may be written in the form of "retarded potentials," namely[16]

$$\psi^{(1)}(x, y, z) = -\frac{m}{2\pi\hbar^2}\int \psi^{(0)}U(x', y', z')e^{ikr}\frac{dV'}{r}, \tag{45.3}$$

$$dV' = dx'\,dy'\,dz', \qquad r^2 = (x-x')^2 + (y-y')^2 + (z-z')^2.$$

Let us determine the conditions under which the field U may be regarded as a perturbation. Perturbation theory is applicable when $\psi^{(1)} \ll \psi^{(0)}$. Let a give the order of magnitude of the size of the region in which the field differs appreciably from zero. Suppose first that the particle's energy is so low that ak is at most of order unity. For an order-of-magnitude estimate the factor e^{ikr} in the integrand of (45.3) is then immaterial, and the whole integral is of order $\psi^{(0)}|U|a^2$. Thus $\psi^{(1)} \sim (ma^2|U|/\hbar^2)\psi^{(0)}$, and the condition becomes

$$|U| \ll \frac{\hbar^2}{ma^2} \quad \text{when} \quad ka \lesssim 1. \tag{45.4}$$

Notice that $\hbar^2/ma^2$ has a simple physical meaning: it gives the order of magnitude of the kinetic energy that a particle confined to a volume of linear dimensions a would have (for its momentum would be $\sim \hbar/a$ by the uncertainty relation).

As a particular example, consider a potential well so shallow that (45.4) is satisfied. Such a well has no negative energy levels (*R. Peierls*, 1929), as was already seen in the problem to § 33 for the special case of a spherically symmetric well. Indeed, at $E = 0$ the unperturbed wave function reduces to a constant, which may conventionally be set equal to unity: $\psi^{(0)} = 1$. Since $\psi^{(1)} \ll \psi^{(0)}$, the wave function for motion in the well, $\psi = 1 + \psi^{(1)}$, plainly vanishes nowhere. An eigenfunction without nodes belongs to the normal state, so $E = 0$ remains the lowest possible energy of the particle.

Thus, if a well is not sufficiently deep, only infinite motion of the particle is possible: the well cannot "capture" it. This result is specifically quantum-mechanical. In classical mechanics a particle can undergo finite motion in any potential well.

It must be stressed that everything just said applies only to a three-dimensional well. A one- or two-dimensional well (that is, one whose field depends on only one or two coordinates) always has negative energy levels (see the problems to this section). The reason is that in one and two dimensions the perturbation theory under consideration is wholly inapplicable at zero (or very small) energy E. [17]

[16] This is a particular integral of equation (45.2). Any solution of the equation with no right-hand side (that is, of the unperturbed equation (45.1)) may still be added to it.

[17] In two dimensions, $\psi^{(1)}$ is expressed (as follows from the theory of the two-dimensional wave equation) by

At high energies, where $ka \gg 1$, the factor e^{ikr} in the integrand is essential and greatly reduces the magnitude of the integral. In this case solution (45.3) can be put into another form. For its derivation, however, it is simpler to return directly to equation (45.2). Choose the direction of unperturbed motion as the x axis. The unperturbed wave function is then $\psi^{(0)} = e^{ikx}$ (the constant factor is conventionally set to unity). Seek a solution of

$$\Delta\psi^{(1)} + k^2\psi^{(1)} = \frac{2m}{\hbar^2}Ue^{ikx}$$

in the form $\psi^{(1)} = e^{ikx}f$. Since k is assumed large, in $\Delta\psi^{(1)}$ it suffices to retain only terms in which the factor e^{ikx} is differentiated at least once. The equation for f is then

$$2ik\frac{\partial f}{\partial x} = \frac{2mU}{\hbar^2},$$

and hence

$$\psi^{(1)} = e^{ikx}f = -\frac{im}{\hbar^2 k}e^{ikx}\int U\,dx. \tag{45.5}$$

The integral is of order $|\psi^{(1)}| \sim m|U|a/\hbar^2 k$, so in this case perturbation theory is applicable under the condition

$$|U| \ll \frac{\hbar^2}{ma^2}ka = \frac{\hbar v}{a}, \qquad ka \gg 1 \tag{45.6}$$

($v = k\hbar/m$ is the particle's velocity). This condition is weaker than (45.4). Consequently, if the field may be treated as a perturbation at low particle energies, it certainly may be so treated at high energies; the converse, generally speaking, does not hold. [18]

The applicability of the perturbation theory developed here to the Coulomb field requires separate consideration. In the field $U = \alpha/r$ there is no finite region of space outside which U is substantially smaller than inside it. The required condition follows by replacing the parameter a in (45.6) with the variable distance r; this gives

$$\frac{\alpha}{\hbar v} \ll 1. \tag{45.7}$$

Thus the Coulomb field may be treated as a perturbation at high particle energies. [19]

Finally, let us derive a formula giving an approximate wave function for a particle whose energy E is everywhere much greater than the potential energy U (no other conditions are required). To the first approximation, the coordinate dependence of the

an integral analogous to (45.3), with $i\pi H_0^{(1)}(kr)dx'\,dy'$ in place of $e^{ikr}dx'\,dy'\,dz'/r$ ($H_0^{(1)}$ is a Hankel function), and $r = \sqrt{(x'-x)^2 + (y'-y)^2}$. As $k \to 0$, the Hankel function, and with it the entire integral, tends logarithmically to infinity.

Similarly, in one dimension the integrand defining $\psi^{(1)}$ contains $2\pi i e^{ikr}dx'/k$ (where $r = |x' - x|$), and as $k \to 0$, $\psi^{(1)}$ tends to infinity as $1/k$.

[18] In one dimension the condition for applicability of perturbation theory is the inequality (45.6) for all ka. The derivation of (45.4) given above for three dimensions cannot be carried out in one dimension because of the divergence, noted in the footnote on p. 204, of the function $\psi^{(1)}$ constructed in that manner.

[19] It should be borne in mind that the integral (45.5) with $U = \alpha/r$ diverges logarithmically at large $x/\sqrt{y^2+z^2}$. The Coulomb-field wave function obtained by perturbation theory is therefore inapplicable inside a narrow cone about the x axis.

wave function is the same as in free motion, whose direction we choose as the x axis. Accordingly, write $\psi = e^{ikx}F$, where F is a function of the coordinates that varies slowly in comparison with the factor e^{ikx} (although in general it cannot be said to be close to unity). Substitution in the Schrödinger equation gives

$$2ik\frac{\partial F}{\partial x} = \frac{2m}{\hbar^2}UF, \tag{45.8}$$

whence

$$\psi = e^{ikx}F = \text{const}\cdot e^{ikx}\exp\left(-\frac{i}{\hbar v}\int U\,dx\right). \tag{45.9}$$

This is the required expression. It must be remembered, however, that it is not valid at excessively large distances. In (45.8) the term ΔF, which contains second derivatives of F, was omitted. At large distances $\partial^2 F/\partial x^2$ tends to zero together with $\partial F/\partial x$. The derivatives with respect to the transverse coordinates y, z, on the other hand, do not tend to zero, and they may be neglected only if $x \ll ka^2$.

Problems

Problem

1. Determine the energy level in a shallow one-dimensional potential well; condition (45.4) is assumed to hold.

Solution. Make the assumption, confirmed by the result, that $|E| \ll |U|$. Then E may be neglected on the right-hand side of the Schrödinger equation within the well,

$$\frac{d^2\psi}{dx^2} = \frac{2m}{\hbar^2}(U(x) - E)\psi,$$

and ψ may also be treated as a constant, which can be set equal to unity without loss of generality:

$$\frac{d^2\psi}{dx^2} = \frac{2m}{\hbar^2}U.$$

Integrate this equality with respect to x between two points $\pm x_1$ such that $a \ll x_1 \ll 1/\kappa$, where a is the width of the well and $\kappa = \sqrt{2m|E|}/\hbar$. Since the integral of $U(x)$ converges, the integration on the right may be extended over the whole interval from $-\infty$ to $+\infty$:

$$\left.\frac{d\psi}{dx}\right|_{-x_1}^{x_1} = \frac{2m}{\hbar^2}\int_{-\infty}^{+\infty} U\,dx. \tag{1}$$

Far from the well the wave function has the form $\psi = e^{\mp\kappa x}$. Substitution in (1) gives

$$-2\kappa = \frac{2m}{\hbar^2}\int_{-\infty}^{+\infty} U\,dx \qquad \text{or} \qquad |E| = \frac{m}{2\hbar^2}\left(\int_{-\infty}^{+\infty} U\,dx\right)^2.$$

In accord with the assumption made, the magnitude of the level is thus a small quantity of higher (second) order than the depth of the well.

Problem

2. Determine the energy level in a shallow two-dimensional potential well $U(r)$ (r is the polar coordinate in the plane); the integral $\int_0^\infty rU\,dr$ is assumed to converge.

Solution. Proceeding as in the preceding problem, within the well we obtain

$$\frac{1}{r}\frac{d}{dr}\left(r\frac{d\psi}{dr}\right) = \frac{2m}{\hbar^2}U.$$

Integrating with respect to r from 0 to r_1 (where $a \ll r_1 \ll 1/\kappa$), we find

$$\left.\frac{d\psi}{dr}\right|_{r=r_1} = \frac{2m}{\hbar^2 r_1}\int_0^\infty rU(r)\,dr. \tag{1}$$

Far from the well, the two-dimensional free-motion equation

$$\frac{1}{r}\frac{d}{dr}\left(r\frac{d\psi}{dr}\right) + \frac{2m}{\hbar^2}E\psi = 0$$

has the solution (vanishing at infinity) $\psi = \text{const}\cdot H_0^{(1)}(i\kappa r)$. At small values of the argument, the leading term of this function is proportional to $\ln \kappa r$. Bearing this in mind, at $r \sim a$ equate the logarithmic derivatives of ψ evaluated inside the well (the right-hand side of (1)) and outside it. This gives

$$-\frac{1}{a\ln\kappa a} \approx \frac{2m}{\hbar^2 a}\int_0^\infty U(r)r\,dr,$$

and hence

$$|E| \sim \frac{\hbar^2}{ma^2}\exp\left[-\frac{\hbar^2}{m}\left|\int_0^\infty Ur\,dr\right|^{-1}\right].$$

Thus the energy level is exponentially small in comparison with the depth of the well.

CHAPTER VII

The Semiclassical Case

§ 46. The Wave Function in the Semiclassical Case

When the de Broglie wavelengths of the particles are small compared with the characteristic dimensions L that determine the conditions of the particular problem, the system behaves nearly classically. (This is analogous to the passage from wave optics to geometrical optics as the wavelength tends to zero.)

We shall now examine the properties of semiclassical systems in greater detail. In the Schrödinger equation

$$\sum_a \frac{\hbar^2}{2m_a}\Delta_a\psi + (E - U)\psi = 0$$

make the formal substitution

$$\psi = \exp\left(\frac{i}{\hbar}\sigma\right). \tag{46.1}$$

The resulting equation for σ is

$$\sum_a \frac{1}{2m_a}(\nabla_a\sigma)^2 - \sum_a \frac{i\hbar}{2m_a}\Delta_a\sigma = E - U. \tag{46.2}$$

Since the system is assumed to have almost classical properties, we seek σ as a series

$$\sigma = \sigma_0 + \frac{\hbar}{i}\sigma_1 + \left(\frac{\hbar}{i}\right)^2\sigma_2 + \cdots, \tag{46.3}$$

in powers of $\hbar$.

We begin with the simplest situation: one-dimensional motion of a single particle. Equation (46.2) then reduces to

$$\frac{1}{2m}\sigma'^2 - \frac{i\hbar}{2m}\sigma'' = E - U(x) \tag{46.4}$$

(a prime denotes differentiation with respect to x).

In the first approximation, put $\sigma = \sigma_0$ and omit the term containing $\hbar$ from the equation:

$$\frac{1}{2m}\sigma_0'^2 = E - U(x).$$

It follows that

$$\sigma_0 = \pm\int\sqrt{2m[E - U(x)]}\,dx.$$

The integrand is precisely the classical momentum $p(x)$ of the particle, expressed as a function of position. Defining $p(x)$ by taking the positive sign of the square root, we have

$$\sigma_0 = \pm \int p\,dx, \qquad p = \sqrt{2m(E-U)}, \tag{46.5}$$

as is to be expected from the limiting expression (6.1) for the wave function[1].

The neglect made in (46.4) is justified only when the second term on the left is small compared with the first. Thus $\hbar|\sigma''/\sigma'^2| \ll 1$, or

$$\left|\frac{d}{dx}\left(\frac{\hbar}{\sigma'}\right)\right| \ll 1.$$

To the first approximation, (46.5) gives $\sigma' = p$, and hence this condition may be written

$$\left|\frac{d\lambdabar}{dx}\right| \ll 1, \tag{46.6}$$

where $\lambdabar = \lambda/2\pi$, while $\lambda(x) = 2\pi\hbar/p(x)$ is the particle's de Broglie wavelength expressed as a function of x through the classical function $p(x)$. We have therefore obtained a quantitative criterion for the semiclassical approximation: over a distance of the order of the wavelength itself, the wavelength must vary only slightly. The approximation ceases to apply in regions where this condition is not fulfilled.

Condition (46.6) can also be put into another form. Since

$$\frac{dp}{dx} = \frac{d}{dx}\sqrt{2m(E-U)} = -\frac{m}{p}\frac{dU}{dx} = \frac{mF}{p},$$

where $F = -dU/dx$ is the classical force exerted on the particle by the external field, introducing this force gives

$$\frac{m\hbar|F|}{p^3} \ll 1. \tag{46.7}$$

This shows that the semiclassical approximation fails when the particle's momentum becomes too small. In particular, it necessarily fails near turning points, that is, near points at which classical mechanics would have the particle stop and then reverse its direction of motion. Such points are determined by $p(x) = 0$, or $E = U(x)$. As $p \to 0$, the de Broglie wavelength tends to infinity and plainly cannot be regarded as small.

It should be stressed, however, that condition (46.6) or (46.7) may by itself be insufficient to justify the semiclassical approximation. It was obtained by estimating the different terms of the differential equation (46.4), whose discarded term contains the highest derivative. What must actually be small are successive terms in the expansion of the solution of this equation, and smallness of the discarded term in the equation need not ensure this. For example, suppose that the solution for $\sigma(x)$ contains a term that grows with x in an almost linear fashion. The smallness of its second derivative does not prevent that term from accumulating a large value over a sufficiently long distance.

[1] As is well known, $\int p\,dx$ is the time-independent part of the action. The complete mechanical action of the particle is $S = -Et \pm \int p\,dx$. The term Et is absent from σ because we are considering a time-independent wave function ψ.

This situation generally arises when the field extends over distances large compared with the characteristic length L on which it changes appreciably (see the remark below concerning (46.11)). The semiclassical approximation then cannot be used to follow the wave function over large distances.

We proceed to calculate the next term of expansion (46.3). Terms of first order in $\hbar$ in (46.4) give

$$\sigma_0'\sigma_1' + \frac{1}{2}\sigma_0'' = 0, \qquad \sigma_1' = -\frac{\sigma_0''}{2\sigma_0'} = -\frac{p'}{2p}.$$

Integration yields

$$\sigma_1 = -\frac{1}{2}\ln p \tag{46.8}$$

(the integration constant is omitted).

Substitution into (46.1) and (46.3) gives the wave function

$$\psi = \frac{C}{\sqrt{p}}\exp\left(\frac{i}{\hbar}\int p\,dx\right) + \frac{C'}{\sqrt{p}}\exp\left(-\frac{i}{\hbar}\int p\,dx\right). \tag{46.9}$$

The factor $1/\sqrt{p}$ has a simple interpretation. The probability of finding the particle between x and $x + dx$ is determined by $|\psi|^2$ and is

therefore, in the main, proportional to $1/p$. This is just what one expects for a "semiclassical particle," since in classical motion the time spent by the particle in an interval dx is inversely proportional to its velocity (or momentum).

In classically forbidden regions, where $E < U(x)$, the function $p(x)$ is purely imaginary and the exponential arguments are real. The general form of the wave-equation solution in these regions is

$$\psi = \frac{C}{\sqrt{|p|}}\exp\left(-\frac{1}{\hbar}\int |p|\,dx\right) + \frac{C'}{\sqrt{|p|}}\exp\left(\frac{1}{\hbar}\int |p|\,dx\right). \tag{46.10}$$

One must bear in mind, however, that the accuracy of the semiclassical approximation does not permit exponentially small terms in the wave function to be retained against a background of exponentially large ones. In this sense, keeping both terms of (46.10) at once is generally inadmissible.

Higher-order terms in the wave function are ordinarily unnecessary. We shall nevertheless find one further term of expansion (46.3), in order to point out some features concerning the accuracy of the semiclassical approximation.

The terms of order $\hbar^2$ in (46.4) give

$$\sigma_0'\sigma_2' + \frac{1}{2}\sigma_1'^2 + \frac{1}{2}\sigma_1'' = 0,$$

and therefore, on substituting (46.5) and (46.8) for σ_0 and σ_1,

$$\sigma_2' = \frac{p''}{4p^2} - \frac{3p'^2}{8p^3}.$$

Integrating (the first term being integrated by parts) and introducing the force $F = pp'/m$, we obtain

$$\sigma_2 = \frac{mF}{4p^3} + \frac{m^2}{8}\int \frac{F^2}{p^5}\,dx.$$

In this approximation the wave function is

$$\psi = \exp[(i/\hbar)\sigma] = \exp[(i/\hbar)\sigma_0 + \sigma_1](1 - i\hbar\sigma_2),$$

or

$$\psi = \frac{\text{const}}{\sqrt{p}}\left[1 - \frac{im\hbar F}{4p^3} - \frac{im^2\hbar}{8}\int \frac{F^2}{p^5}\,dx\right]\exp\left(\frac{i}{\hbar}\int p\,dx\right). \tag{46.11}$$

The appearance of imaginary correction terms in the prefactor is equivalent to a real correction to the phase of the wave function (that is, an addition to the integral $\hbar^{-1}\int p\,dx$ in its exponential). This correction is proportional to $\hbar$, and thus has order λ/L.

The second and third terms in the square brackets in (46.11) must be small compared with 1. For the former this requirement is the same as (46.7); for the latter, estimating the integral leads to (46.7) only if F^2 tends to zero sufficiently rapidly over distances $\sim L$.

§ 47. Boundary Conditions in the Semiclassical Case

Let $x = a$ be a turning point (so that $U(a) = E$), and suppose that $U > E$ for all $x > a$. The region to the right of the turning point is then classically forbidden. The wave function must decay into this region. Sufficiently far from the turning point it has the form

$$\psi = \frac{C}{2\sqrt{|p|}}\exp\left(-\frac{1}{\hbar}\int_a^x |p|\,dx\right), \qquad x > a, \tag{47.1}$$

which corresponds to the first term of (46.10). To the left of the turning point, the wave function must instead be represented by a real combination of the two semiclassical solutions (46.9) of the Schrödinger equation:

$$\psi = \frac{C_1}{\sqrt{p}}\exp\left(\frac{i}{\hbar}\int_a^x p\,dx\right) + \frac{C_2}{\sqrt{p}}\exp\left(-\frac{i}{\hbar}\int_a^x p\,dx\right), \qquad x < a. \tag{47.2}$$

To determine the coefficients in this combination, one must follow the change of the wave function from positive to negative values of $x - a$, starting in the region where (47.1) applies. This path, however, passes through the neighborhood of the stopping point, where the semiclassical approximation fails and the exact solution of the Schrödinger equation must be considered. For small $|x - a|$,

$$E - U(x) \approx F_0(x - a), \qquad F_0 = -\left.\frac{dU}{dx}\right|_{x=a} < 0; \tag{47.3}$$

in other words, the problem in this region is that of motion in a constant field. Its exact Schrödinger-equation solution was found in § 24. The relation between the coefficients in (47.1) and (47.2) can therefore be obtained by comparison with asymptotic expressions (24.5) and (24.6) for that exact solution on the two sides of the turning point. Notice here that (47.3) gives $p(x) = \sqrt{2mF_0(x - a)}$, and hence

$$\frac{1}{\hbar}\int_a^x p\,dx = \frac{2}{3\hbar}\sqrt{2mF_0}(x - a)^{3/2},$$

which coincides with the expression occurring in the argument of the exp or sin in (24.5) or (24.6). This reasoning relies essentially on a partial overlap between the domain where expansion (47.3) is valid and the semiclassical domain. If the motion is semiclassical throughout almost the whole field region, as assumed, there are values of $|x-a|$ small enough for (47.3) to apply but at the same time large enough to satisfy the semiclassical condition and permit the asymptotic forms (24.5) and (24.6)[2].

There is, however, another and methodologically more instructive procedure that avoids any need to use the exact solution. Formally regard $\psi(x)$ as a function of the complex variable x, and pass from positive to negative values of $x-a$ along a path lying entirely far from $x=a$, so that the semiclassical condition is formally satisfied all along it (A. Zwaan, 1929). We again consider values of $|x-a|$ for which expansion (47.3) is valid as well. The wave function (47.1) then takes the form

$$\psi(x) = \frac{C}{2(2m|F_0|)^{1/4}(x-a)^{1/4}} \exp\left\{-\frac{1}{\hbar}\sqrt{2m|F_0|}\int_a^x \sqrt{x-a}\,dx\right\}. \tag{47.4}$$

First follow the change of this function while passing around $x=a$ from right to left on a semicircle of radius ρ in the upper half of the complex x-plane. On this semicircle

$$x-a=\rho e^{i\varphi}, \qquad \int_a^x \sqrt{x-a}\,dx = \frac{2}{3}\rho^{3/2}\left(\cos\frac{3\varphi}{2} + i\sin\frac{3\varphi}{2}\right),$$

where the phase φ varies from 0 to π. The modulus of the exponential factor in (47.4) first increases (for $0<\varphi<2\pi/3$), and then decreases to 1. At the end of the passage the exponent is purely imaginary and equals

$$-\frac{i}{\hbar}\int_a^x \sqrt{2m|F_0|(a-x)}\,dx = -\frac{i}{\hbar}\int_a^x p(x)\,dx.$$

Meanwhile the prefactor in (47.4) changes under this continuation according to

$$(x-a)^{-1/4} \longrightarrow (a-x)^{-1/4}e^{-i\pi/4}.$$

Thus the whole function (47.4) becomes the second term in (47.2), with $C_2 = (1/2)Ce^{-i\pi/4}$.

There is a simple explanation for the fact that continuation through the upper half-plane determines only C_2 in (47.2). Follow the change of (47.2) along the same semicircle in the reverse direction, from left to right. At the start of the continuation, its first term rapidly becomes exponentially small relative to the second. But the semiclassical approximation cannot reveal exponentially small terms in ψ against the background of a large leading term. This is why the first term of (47.2) is “lost” in the continuation just performed.

To determine C_1, the continuation from right to left must be made along a semicircle in the lower half of the complex x-plane. In the same manner one finds that (47.4) then becomes the first term of (47.2), with $C_1 = (1/2)Ce^{i\pi/4}$.

[2] Indeed, expansion (47.3) is valid for $|x-a| \ll L$, where L is the characteristic distance over which the field $U(x)$ changes. The semiclassical condition (46.7), on the other hand, requires $|x-a|^{3/2} \gg \hbar/\sqrt{m|F_0|}$. These requirements are compatible because semiclassical motion far from the turning point (that is, at $|x-a|\sim L$) means that $L^{3/2} \gg \hbar/\sqrt{m|F_0|}$.

Consequently, the wave function (47.1) at $x > a$ corresponds at $x < a$ to

$$\psi = \frac{C}{\sqrt{p}} \cos\left(\frac{1}{\hbar}\int_a^x p\,dx + \frac{\pi}{4}\right).$$

The resulting connection rule may be written in a form independent of which side of the turning point contains the classically forbidden region:

$$\underset{U(x)>E}{\frac{2C}{\sqrt{|p|}} \exp\left\{-\frac{1}{\hbar}\int_a^x |p|\,dx\right\}} \longrightarrow \underset{U(x)<E}{\frac{C}{\sqrt{p}} \cos\left\{\frac{1}{\hbar}\int_a^x p\,dx - \frac{\pi}{4}\right\}} \tag{47.5}$$

(*H., A. Kramers*, 1926).

We emphasize once more a point already clear from the derivation: this rule is tied to a particular boundary condition imposed on one side of the turning point and, in this sense, is to be applied only in a specified direction. More precisely, (47.5) was obtained with the boundary condition $\psi \to 0$ deep in the classically forbidden region, and must be used to pass from that region to the classically allowed one, as the arrow in (47.5) indicates[3].

If the classically allowed region is bounded at $x = a$ by an infinitely high potential wall, the boundary condition there is $\psi = 0$ (see § 18). The semiclassical approximation is then valid all the way to the wall, and the wave function is

$$\begin{aligned} \psi &= \frac{C}{\sqrt{p}} \sin\left(\frac{1}{\hbar}\int_a^x p\,dx\right), \quad && x < a, \\ \psi &= 0, && x > a. \end{aligned} \tag{47.6}$$

§ 48. The Bohr–Sommerfeld quantization rule

States in the discrete energy spectrum are semiclassical when the quantum number n, which gives the ordinal number of the state, is large. Indeed, this number determines the number of nodes of the eigenfunction (see § 21). The distance between adjacent nodes, however, is of the order of the de Broglie wavelength. For large n this distance is small, and hence the wavelength is small compared with the dimensions of the region of motion.

Let us derive the condition that determines the quantum energy levels in the semiclassical case. Consider for this purpose the finite one-dimensional motion of a particle in a potential well; the classically accessible region $b \le x \le a$ is bounded by two turning points[4].

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[3]The reverse passage loses its meaning in the following sense: even a small change in the wave function on the right of (47.5) may produce an exponentially increasing term in the function on the left.

[4]In classical mechanics a particle in such a field would execute periodic motion with period (the time required to travel from b to a and back)

$$T = 2\int_b^a \frac{dx}{v}$$

(v is the particle velocity).

By rule (47.5), the boundary condition at $x = b$ gives, in the region to its right, the wave function

$$\psi = \frac{C}{\sqrt{p}} \cos\left(\frac{1}{\hbar}\int_b^x p\,dx - \frac{\pi}{4}\right). \tag{48.1}$$

Applying the same rule to the region to the left of $x = a$, we obtain the same wave function in the form

$$\psi = \frac{C'}{\sqrt{p}} \cos\left(\frac{1}{\hbar}\int_x^a p\,dx - \frac{\pi}{4}\right).$$

For these two expressions to agree throughout the region, the sum of their phases, which is constant, must be an integral multiple of π:

$$\frac{1}{\hbar}\int_b^a p\,dx - \frac{\pi}{2} = n\pi$$

(with $C' = (-1)^n C$). Hence

$$\frac{1}{2\pi\hbar}\oint p\,dx = n + \frac{1}{2}, \tag{48.2}$$

where the integral $\oint p\,dx = 2\int_b^a p\,dx$ is taken over a complete period of the particle's classical motion. This is the condition determining the stationary states of a particle in the semiclassical case. It corresponds to the Bohr–Sommerfeld quantization rule of the old quantum theory.

The quantity $I = (1/2\pi)\oint p\,dx$ is called the adiabatic invariant (see I, § 49), so the quantization condition (48.2) may be written

$$I(E) = \hbar(n + 1/2).$$

It was already mentioned in § 41 that, when the parameters vary sufficiently slowly, "adiabatically," the system remains in the same quantum state—in the present case, the state with a particular n. We see that in the semiclassical limit this statement becomes the classical theorem that the adiabatic invariant remains constant under a slow change of the parameters.

It is readily seen that the integer n equals the number of zeros of the wave function and is therefore the ordinal number of the stationary state. Indeed, the phase of the wave function (48.1) increases from $-\pi/4$ at $x = b$ to $(n + 1/4)\pi$ at $x = a$, so the cosine vanishes n times in this interval (outside the interval $b \ll x \ll a$, the wave function decreases monotonically and has no zeros at finite distances)[5].

As explained above, n is large in the semiclassical case. It should be stressed, nevertheless, that retaining the term $1/2$ beside n in (48.2) is legitimate: including the next correction terms in the phase of the wave functions would introduce on the right-hand side of (48.2) only terms of order λ/L, small compared with unity (see the remark at the end of § 46)[6].

[5]Strictly, the zeros must be counted using the exact form of the wave function near the turning points. Such an analysis confirms the stated result.

[6]In some cases the exact expression for the energy levels $E(n)$, as a function of the quantum number n, obtained from the exact Schrödinger equation has a form that persists as $n \to \infty$. Examples are the energy levels in a Coulomb field and those of the harmonic oscillator. It is natural that in these cases the quantization rule (48.2), applicable for large n, yields an expression for $E(n)$ that agrees with the exact one.

To normalize the wave function it is enough to integrate $|\psi|^2$ only over $b \le x \le a$, since $\psi(x)$ decays exponentially outside this interval. Because the argument of the cosine in (48.1) varies rapidly, the square of the cosine may, with sufficient accuracy, be replaced by its mean value $1/2$. We then obtain

$$\int |\psi|^2 dx \approx \frac{C^2}{2} \int_b^a \frac{dx}{p(x)} = \frac{\pi C^2}{2m\omega} = 1,$$

where $\omega = 2\pi/T$ is the frequency of the classical periodic motion. Thus the normalized semiclassical function is

$$\psi = 2\sqrt{\frac{m\omega}{\pi p}} \cos\left(\frac{1}{\hbar}\int_b^x p\,dx - \frac{\pi}{4}\right). \tag{48.3}$$

It must be borne in mind that the frequency ω is a function of the energy and is, in general, different for different levels.

Relation (48.2) also admits another interpretation. The integral $\oint p\,dx$ is the area enclosed by the closed

classical phase trajectory of the particle (that is, the curve in the p, x plane—the particle's phase space). Dividing this area into cells of area $2\pi\hbar$ gives n cells in all. But n is the number of quantum states whose energies do not exceed the specified value (the value corresponding to the phase trajectory under consideration). Thus, in the semiclassical case, we may say that every quantum state corresponds to a cell of area $2\pi\hbar$ in phase space. In other words, the number of states assigned to a phase-space volume element $\Delta p \Delta x$ is

$$\frac{\Delta p \Delta x}{2\pi\hbar}. \tag{48.4}$$

If the wave vector $k = p/\hbar$ is used instead of the momentum, this number is $\Delta k \Delta x/2\pi$. As expected, it agrees with the familiar expression for the number of eigenmodes of a wave field (see II, § 52).

The general character of the distribution of levels in the energy spectrum can be found from the quantization rule (48.2). Let ΔE be the spacing between two adjacent levels, that is, levels whose quantum numbers n differ by unity. Since ΔE is small (for large n) compared with the level energy itself, (48.2) gives

$$\Delta E \oint \frac{\partial p}{\partial E}\,dx = 2\pi\hbar.$$

But $\partial E/\partial p = v$, and therefore

$$\oint \frac{\partial p}{\partial E}\,dx = \oint \frac{dx}{v} = T.$$

Consequently,

$$\Delta E = \frac{2\pi\hbar}{T} = \hbar\omega. \tag{48.5}$$

Thus the separation of adjacent levels is $\hbar\omega$. For a sequence of neighboring levels (whose quantum numbers differ by amounts small compared with the numbers themselves), the corresponding frequencies ω may be regarded as approximately equal. We conclude that within each small segment of the semiclassical part of the spectrum the levels are equidistant, with equal intervals $\hbar\omega$. This result could, moreover, have been anticipated, since in the semiclassical case the frequencies corresponding to transitions between different energy levels must be integral multiples of the classical frequency ω.

It is instructive to examine what the matrix elements of a physical quantity f become in the classical limit. We start from the fact that in this limit the mean value of f in a quantum state must become simply the classical value of that quantity, provided the state itself tends to motion of the particle along a definite trajectory. Such a state is represented by a wave packet (see § 6) formed by superposing a number of stationary states with nearby energies. Its wave function has the form

$$\psi = \sum_n a_n \psi_n,$$

where the coefficients a_n differ appreciably from zero only in an interval Δn of values of the quantum number n, such that $1 \ll \Delta n \ll n$; the numbers n are assumed large, in accordance with the semiclassical character of the stationary states. By definition, the mean value of f is

$$\bar{f} = \int \psi^* f \psi \, dx = \sum_n \sum_m a_m^* a_n f_{mn} e^{i\omega_{mn} t},$$

or, on replacing summation over n, m by summation over n and the difference $s = m - n$,

$$\bar{f} = \sum_n \sum_s a_{n+s}^* a_n f_{n+s,n} e^{i\omega_s t},$$

where $\omega_{mn} = \omega s$ in accordance with (48.5).

Matrix elements f_{mn} calculated with semiclassical wave functions decrease rapidly in magnitude as the difference $m - n$ increases, while at the same time they vary slowly with n itself (for a fixed $m - n$). We may therefore write, approximately,

$$\bar{f} = \sum_n \sum_s a_n^* a_n f_s e^{i\omega s t} = \sum_n |a_n|^2 \sum_s f_s e^{i\omega s t},$$

where $f_s = f_{n+s,n}$ and n now denotes some mean value of the quantum number in the interval Δn. Since $\sum_n |a_n|^2 = 1$, it follows that

$$\bar{f} = \sum_s f_s e^{i\omega s t}.$$

The resulting sum has the form of an ordinary Fourier series. Since $\bar{f}$ must tend to the classical quantity $f(t)$, we find that in the limit the matrix elements f_{mn} become the components f_{m-n} in the Fourier-series expansion of the classical function $f(t)$.

Similarly, the matrix elements for transitions between continuous-spectrum states become the components of the Fourier-integral expansion of $f(t)$. In this case the wave functions of the stationary states must be normalized to a delta-function of the energy divided by $\hbar$.

All the results given above extend directly to systems with many degrees of freedom executing finite motion for which the mechanical (classical) problem admits complete separation of variables in the Hamilton–Jacobi method (the so-called conditionally periodic motion; see I, § 52). Once the variables have been separated, the problem for each degree of freedom reduces to a one-dimensional one, and the corresponding quantization conditions are

$$\oint p_i\, dq_i = 2\pi\hbar(n_i + \gamma_i), \tag{48.6}$$

where the integral is taken over a period of variation of the generalized coordinate q_i, while γ_i is a number of order unity depending on the nature of the boundary conditions for that degree of freedom[7].

For arbitrary (not conditionally periodic) multidimensional motion, a formulation of the semiclassical quantization conditions requires a deeper analysis[8]. The concept of phase-space "cells," on the other hand, always applies in the same form (within the semiclassical approximation). This is clear from its connection, noted above, with the number of eigenmodes of a wave field in a specified volume of space. In the general case of a system with s degrees of freedom, a phase-space volume element contains

$$\Delta N = \frac{\Delta q_1 \cdots \Delta q_s\, \Delta p_1 \cdots \Delta p_s}{(2\pi\hbar)^s} \tag{48.7}$$

quantum states[9].

Problem

1. Find approximately the number of discrete energy levels of a particle moving in a field $U(r)$ that satisfies the semiclassical condition.

Solution. The number of states "assigned" to the phase-space volume corresponding to momenta in the interval $0 \le p \le p_{\max}$ and particle coordinates in the volume element dV is

$$\frac{(4\pi/3)p_{\max}^3 dV}{(2\pi\hbar)^3}.$$

At a specified r, the particle may have (in its classical motion) a momentum satisfying $E = (p^2/2m) + U(r) \le 0$. Substitution of $p_{\max} = \sqrt{-2mU(r)}$ gives the total number of states in the discrete spectrum:

$$\frac{\sqrt{2}m^{3/2}}{3\pi^2\hbar^3}\int (-U)^{3/2}\, dV,$$

[7] For motion in a centrally symmetric field, for example,

$$\oint p_r\, dr = 2\pi\hbar(n_r + 1/2), \qquad \oint p_\theta\, d\theta = 2\pi\hbar(l - m + 1/2), \qquad \oint p_\varphi\, d\varphi = 2\pi\hbar m$$

(where $n_r = n - l - 1$ is the radial quantum number). The last equality follows simply because p_φ is the z-component of the angular momentum, equal to $\hbar m$.

[8] See *J., B. Keller* // Ann. Phys. 1958. V. 4. P. 180.

[9] In particular, for one particle, $d^3p/(2\pi\hbar)^3$ is the number of states assigned to the interval d^3p of momentum values per unit volume of space. This explains why the two interpretations of the normalization of the plane wave (15.8), noted in the footnote on p. 68, agree.

where the integration extends over the region of space in which $U < 0$. This integral diverges (the number of levels is infinite) if at infinity U decreases as r^{-s} with $s < 2$, in agreement with the results of § 18.

2. Find the same quantity for a semiclassical centrally symmetric field $U(r)$ (V. L. Pokrovskii).

SOLUTION. In a centrally symmetric field, the number of states does not equal the number of energy levels, because the latter are degenerate with respect to the directions of the angular momentum. The required number may be found by observing that the number of levels for a specified angular momentum M equals the number of (nondegenerate) levels of one-dimensional motion with potential energy $U_{\text{eff}} = U(r) + M^2/(2mr^2)$. At a specified r and for energies $E \leq 0$, the greatest possible value of the momentum p_r is $p_{r\,\max} = \sqrt{-2mU_{\text{eff}}}$. Hence the number of states (that is, the number of levels) is

$$\frac{1}{2\pi\hbar}\oint dr\,dp_r = \frac{\sqrt{2m}}{\pi\hbar}\int\sqrt{-U - \frac{M^2}{2mr^2}}\,dr.$$

The required total number of discrete levels is obtained by integrating this expression over $dM/\hbar$ (which replaces summation over l in the semiclassical case), and is

$$\frac{\sqrt{2m}}{4\hbar^2}\int(-U)r\,dr.$$

§ 49. Semiclassical motion in a centrally symmetric field

For motion in a centrally symmetric field, the particle's wave function separates, as we know, into angular and radial parts. We first consider the angular part.

The dependence of the angular wave function on the angle φ (specified by the quantum number m) is so simple that there is no question of seeking approximate formulas for it. Its dependence on the polar angle θ, on the other hand, is semiclassical according to the general rule if the corresponding quantum number l is large (a more precise statement of this condition is given below).

Here we shall derive a semiclassical expression for the angular function only in the case of greatest practical importance, namely states with zero magnetic quantum number ($m = 0$)[10]. Apart from a constant factor, this function is the Legendre polynomial $P_l(\cos\theta)$ (see (28.8)) and satisfies the differential equation

$$\frac{d^2P_l}{d\theta^2} + \cot\theta\frac{dP_l}{d\theta} + l(l+1)P_l = 0. \tag{49.1}$$

The substitution

$$P_l = \frac{\chi}{\sqrt{\sin\theta}} \tag{49.2}$$

[10]The opposite case, $m = l$, must in the limit correspond to motion in a classical orbit lying in the equatorial plane $\theta = \pi/2$. Indeed,

$$P_l^l(\cos\theta) = \text{const}\cdot\sin^l\theta;$$

as $l \to \infty$, this function (and with it $|\psi|^2$) tends to zero for all $\theta \neq \pi/2$.

reduces it to the equation

$$\chi'' + \left[\left(l + \frac{1}{2}\right)^2 + \frac{1}{4\sin^2\theta}\right]\chi = 0, \tag{49.3}$$

which contains no first derivative and has the form of a one-dimensional Schrödinger equation.

In (49.3), the role of the "de Broglie wavelength" is played by $\lambda \sim 1/l$.

The requirement that the derivative $d\lambda/dx$ be small (condition (46.6)) gives the inequalities

$$\theta l \gg 1, \qquad (\pi - \theta)l \gg 1 \tag{49.4}$$

(the semiclassical conditions for the angular part of the wave function). For large l these conditions hold over almost the whole interval of values of θ, except only for angles very close to zero or to π.

When condition (49.4) holds, the second term in the square brackets in (49.3) may be neglected in comparison with the first:

$$\chi'' + \left(l + \frac{1}{2}\right)^2 \chi = 0.$$

The solution of this equation is

$$\chi = \sqrt{\sin\theta}\, P_l(\cos\theta) = A\sin\left[\left(l + \frac{1}{2}\right)\theta + \alpha\right] \tag{49.5}$$

(A and α are constants).

For angles $\theta \ll 1$, we may set $\cot\theta \approx 1/\theta$ in (49.1). Also replacing $l(l+1)$ approximately by $(l+1/2)^2$, we obtain

$$\frac{d^2P_l}{d\theta^2} + \frac{1}{\theta}\frac{dP_l}{d\theta} + \left(l + \frac{1}{2}\right)^2 P_l = 0,$$

whose solution is the Bessel function of order zero,

$$P_l(\cos\theta) = J_0\left[\left(l + \frac{1}{2}\right)\theta\right], \qquad \theta \ll 1. \tag{49.6}$$

The constant factor has been set equal to unity because P_l must equal one at $\theta = 0$. The approximate expression (49.6) for P_l is valid for all angles $\theta \ll 1$. In particular, it may be used in the region $1/l \ll \theta \ll 1$, where it must agree with (49.5), which is valid whenever $\theta \gg 1/l$. When $\theta l \gg 1$, the Bessel function may be replaced by its asymptotic expression for large argument, giving

$$P_l \approx \sqrt{\frac{2}{\pi l\theta}}\sin\left[\left(l + \frac{1}{2}\right)\theta + \frac{\pi}{4}\right]$$

(in the coefficient, $1/2$ may be neglected in comparison with l). Comparison with (49.5) gives $A = \sqrt{2/\pi l}$ and $\alpha = \pi/4$. We thus arrive at the following expression for $P_l(\cos\theta)$, applicable in the semiclassical case[11]:

$$P_l(\cos\theta) = \sqrt{\frac{2}{\pi l \sin\theta}} \sin\left[\left(l + \frac{1}{2}\right)\theta + \frac{\pi}{4}\right]. \tag{49.7}$$

The normalized spherical function Y_{l0} therefore takes the form (compare (28.8))

$$Y_{l0} = \frac{1}{\pi\sqrt{\sin\theta}} \sin\left[\left(l + \frac{1}{2}\right)\theta + \frac{\pi}{4}\right]. \tag{49.8}$$

We now turn to the radial part of the wave function. It was shown in § 32 that the function $\chi(r) = rR(r)$ satisfies an equation identical with the one-dimensional Schrödinger equation having potential energy

$$U_l(r) = U(r) + \frac{\hbar^2}{2mr^2} l(l+1).$$

We may therefore apply the results of the preceding sections, with the function $U_l(r)$ understood as the potential energy.

The case $l = 0$ is the simplest. The centrifugal energy is absent, and if the field $U(r)$ satisfies the required condition (46.6), the radial wave function is semiclassical throughout space. At $r = 0$ we must have $\chi = 0$, and hence the semiclassical function $\chi(r)$ is determined in accordance with formulas (47.6).

If $l \neq 0$, the centrifugal energy must also satisfy condition (46.6). In the region of small r, where the centrifugal energy is of the order of the total energy, the wavelength is $\lambda = \hbar/p \sim r/l$, and condition (46.6) gives $l \gg 1$. Thus, when l is not large, the centrifugal energy violates the semiclassical condition in the region of small r. It is readily verified that the correct phase of the semiclassical wave function $\chi(r)$ is obtained by using the formulas for one-dimensional motion and replacing the coefficient $l(l+1)$ in the potential energy $U_l(r)$

by $(l + 1/2)^2$[12]:

$$U_l(r) = U(r) + \frac{\hbar^2}{2mr^2}\left(l + \frac{1}{2}\right)^2. \tag{49.9}$$

The applicability of the semiclassical approximation to a Coulomb field $U = \pm\alpha/r$ requires separate consideration. Of the entire region of motion, the most important part corresponds to distances r for which $|U| \sim |E|$, that is, $r \sim \alpha/|E|$. The semiclassical condition for motion in this region reduces to requiring that the wavelength $\lambda\!\!\!^{-} \sim \hbar/\sqrt{2m|E|}$ be small compared with the size $\alpha/|E|$ of the region. This gives

$$|E| \ll \frac{m\alpha^2}{\hbar^2}, \tag{49.10}$$

that is, the absolute value of the energy must be small compared with the particle's energy in the first Bohr orbit. Condition (49.10) may also be written

$$\frac{\hbar v}{\alpha} \gg 1, \tag{49.11}$$

[11] Notice that it is precisely the replacement of $l(l+1)$ by $(l+1/2)^2$ that has yielded an expression which is multiplied by $(-1)^l$ when θ is replaced by $\pi - \theta$, as the function $P_l(\cos\theta)$ must be.

[12] Thus, in the simplest case of free motion ($U = 0$), the phase of the function calculated from (48.1), with U_l from (49.9), agrees at large r, as it should, with the phase of the function (33.12).

where $v \sim \sqrt{|E|/m}$ is the particle velocity. Notice that this condition is the reverse of condition (45.7) for the applicability of perturbation theory to the Coulomb field.

As for the region of small distances ($|U(r)| \gg E$), in a repulsive Coulomb field it is of no interest at all, since for $U > E$ the semiclassical wave functions decay exponentially. In an attractive field with small l, however, the particle may penetrate into a region where $|U| \gg |E|$, and the limits of applicability of the semiclassical approximation there must be determined. We use the general condition (46.7), putting

$$F = -\frac{dU}{dr} = \frac{\alpha}{r^2}, \qquad p \approx \sqrt{2m|U|} \sim \sqrt{\frac{m\alpha}{r}}.$$

We then find that the region of applicability of the semiclassical approximation is restricted to distances

$$r \gg \frac{\hbar^2}{m\alpha}, \tag{49.12}$$

that is, distances large compared with the "radius" of the first Bohr orbit.

PROBLEM

Find the behavior of the wave function near the origin if, as $r \to 0$, the field tends to infinity as $\pm\alpha/r^s$, where $s > 2$.

SOLUTION. For sufficiently small r, the wavelength is

$$\lambda\!\!\!^{-} \sim \frac{\hbar}{\sqrt{m|U|}} \sim \frac{\hbar r^{s/2}}{\sqrt{m\alpha}},$$

and therefore

$$\frac{d\lambda\!\!\!^{-}}{dr} \sim \frac{\hbar r^{s/2-1}}{\sqrt{m\alpha}} \ll 1;$$

thus the semiclassical condition is fulfilled. In an attractive field, $U \to -\infty$ as $r \to 0$. The region near the origin is then classically accessible, and the radial wave function $\chi \sim 1/\sqrt{p}$, whence

$$\psi \sim r^{s/4-1}.$$

In a repulsive field the region of small r is classically inaccessible. The wave function then tends exponentially to zero as $r \to 0$. Omitting the factor multiplying the exponential function, we have

$$\psi \sim \exp\left[-\frac{2\sqrt{m\alpha}}{(s-2)\hbar} r^{-(s/2-1)}\right].$$

§ 50. Passage through a potential barrier

Consider the motion of a particle in a field of the type shown in Fig. 13. It contains a *potential barrier*: a region where the potential energy $U(x)$ exceeds the particle's total energy E. In classical mechanics such a barrier is impenetrable. In quantum mechanics, however, there is a nonzero probability that the particle will pass "through the barrier" (this is also called the *tunnelling effect*)[13]. If $U(x)$ meets the conditions for the semiclassical approximation, the transmission coefficient of the barrier can be found in general form. These conditions imply, in particular, that the barrier is wide; consequently, its semiclassical transmission coefficient is small.

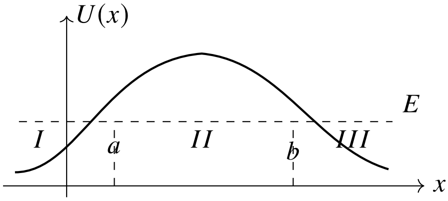


Fig. 13

Before continuing the calculation, we solve an auxiliary problem. Suppose that to the right of the turning point $x = b$, where
$U(x) < E$, the semiclassical wave function is the travelling wave

$$\psi = \frac{C}{\sqrt{p}} \exp\left(\frac{i}{\hbar}\int_b^x p\,dx + \frac{i\pi}{4}\right). \tag{50.1}$$

We must find the wave function of the same state for $x < b$. We use the same continuation around the complex x-plane as in § 47.

Putting

$$E - U(x) \approx F_0(x-b), \qquad F_0 > 0,$$

we rewrite (50.1) as

$$\psi(x) = \frac{C}{(2mF_0)^{1/4}}\frac{1}{(x-b)^{1/4}} \exp\left\{\frac{i}{\hbar}(2mF_0)^{1/2}\int_b^x \sqrt{x-b}\,dx + \frac{i\pi}{4}\right\}.$$

Continue this expression from right to left along a semicircle in the upper half-plane:

$$x - b = \rho e^{i\varphi}, \qquad i\int_b^x \sqrt{x-b}\,dx = \frac{2}{3}\rho^{3/2}\left(-\sin\frac{3\varphi}{2} + i\cos\frac{3\varphi}{2}\right),$$

with φ running from 0 to π. During this continuation $|\psi(x)|$ first decreases and then increases. At its end,

$$\psi(x) = \frac{C}{(2mF_0)^{1/4}}\frac{1}{(b-x)^{1/4}e^{i\pi/4}} \exp\left\{\frac{1}{\hbar}\int_x^b (2mF_0)^{1/2}\sqrt{b-x}\,dx + \frac{i\pi}{4}\right\}.$$

We therefore obtain the correspondence rule[14]:

$$\left.\frac{C}{\sqrt{p}} \exp\left\{\frac{i}{\hbar}\int_b^x p\,dx + \frac{i\pi}{4}\right\}\right|_{x>b} \longrightarrow \left.\frac{C}{\sqrt{|p|}} \exp\left\{\frac{1}{\hbar}\left|\int_b^x p\,dx\right|\right\}\right|_{x<b}. \tag{50.2}$$

[13]Examples of this kind were already considered in Problems 2 and 4 of § 25.

[14]If the continuation from right to left is instead made through the lower half-plane, $|\psi(x)|$ first increases and then decreases, becoming exponentially small on the negative half-axis ($\varphi \to -\pi$). It would be illegitimate to retain this small quantity against the exponentially large function (50.2). On the part of the path where $\psi(x)$ is exponentially large, the inaccuracy of the semiclassical approximation loses an exponentially small addition which, as $\varphi \to -\pi$, could have grown into an exponentially large term and is therefore likewise lost.

This rule presupposes a particular wave in the classically allowed region—a wave travelling to the right—and must be used specifically in passing from that region into the classically forbidden one.

We return to the transmission coefficient of the barrier. Let a particle be incident on it from region I, moving to the right. Beyond the barrier, in region III, there is only the transmitted right-moving wave. Write it as

$$\psi = \sqrt{\frac{D}{v}} \exp\left(\frac{i}{\hbar}\int_b^x p\,dx + \frac{i\pi}{4}\right), \tag{50.3}$$

where $v = p/m$ is the particle velocity and D is the current density in the wave. Rule (50.2) now gives, inside the barrier in region II,

$$\psi = \sqrt{\frac{D}{|v|}} \exp\left(\frac{1}{\hbar}\left|\int_x^b p\,dx\right|\right) = \sqrt{\frac{D}{|v|}} \exp\left(\frac{1}{\hbar}\left|\int_a^b p\,dx\right| - \frac{1}{\hbar}\left|\int_a^x p\,dx\right|\right). \tag{50.4}$$

Finally, application of (47.5) gives, before the barrier in region I,

$$\psi = 2\sqrt{\frac{D}{v}} \exp\left(\frac{1}{\hbar}\int_a^b |p|\,dx\right) \cos\left(\frac{1}{\hbar}\int_x^a p\,dx - \frac{\pi}{4}\right).$$

If we set

$$D = \exp\left(-\frac{2}{\hbar}\int_a^b |p|\,dx\right), \tag{50.5}$$

this becomes

$$\begin{aligned}\psi &= \frac{2}{\sqrt{v}} \cos\left(\frac{1}{\hbar}\int_a^x p\,dx + \frac{\pi}{4}\right)\\ &= \frac{1}{\sqrt{v}} \exp\left(\frac{i}{\hbar}\int_a^x p\,dx + \frac{i\pi}{4}\right) + \frac{1}{\sqrt{v}} \exp\left(-\frac{i}{\hbar}\int_a^x p\,dx - \frac{i\pi}{4}\right).\end{aligned}$$

The first term (which tends to the plane wave $\psi \sim e^{ipx/\hbar}$ as $x \longrightarrow -\infty$) describes the incident wave, and the second the reflected wave. Our normalization assigns unit current density to the incident wave. Thus D, the current density in the transmitted wave, is precisely the required barrier transmission coefficient. Formula (50.5) is applicable only when its exponent is large and D itself is consequently small[15].

So far we have assumed that $U(x)$ satisfies the semiclassical condition throughout the barrier, apart from the immediate neighbourhoods of the turning points. In practice one often meets barriers whose potential-energy curve rises so sharply on one side that the semiclassical approximation fails. The leading exponential factor in D remains the one in (50.5), but the pre-exponential factor (unity in (50.5)) changes. To calculate it, one must in principle find the exact wave function in the non-semiclassical region and match it to the semiclassical wave function within the barrier.

Problems

[15] The exponential smallness of D also explains why the incident and reflected waves in region I have equal amplitudes: their exponentially small difference is lost in the semiclassical approximation.

Problem

1. Find the transmission coefficient of the barrier in Fig. 14: $U(x) = 0$ for $x < 0$, $U(x) = U_0 - Fx$ for $x > 0$. Calculate only its exponential factor.

Solution. Direct calculation gives

$$D \sim \exp\left[-\frac{4\sqrt{2m}}{3\hbar F}(U_0 - E)^{3/2}\right].$$

Problem

2. Find the probability that a particle of zero angular momentum escapes from the spherically symmetric potential well $U(r) = -U_0$ for $r < r_0$, $U(r) = \alpha/r$ for $r > r_0$ (Fig. 15)[16].

Solution. A spherically symmetric problem reduces to a one-dimensional one, so the preceding formulas apply. Hence

$$w \sim \exp\left[-\frac{2}{\hbar}\int_{r_0}^{\alpha/E}\sqrt{2m\left(\frac{\alpha}{r} - E\right)}\,dr\right].$$

Evaluation of the integral gives

$$w \sim \exp\left\{-\frac{2\alpha}{\hbar}\sqrt{\frac{2m}{E}}\left[\arccos\sqrt{\frac{Er_0}{\alpha}} - \sqrt{\frac{Er_0}{\alpha}\left(1 - \frac{Er_0}{\alpha}\right)}\right]\right\}.$$

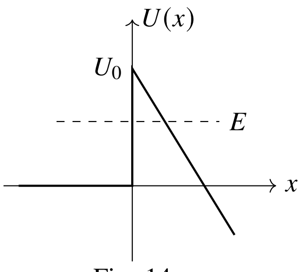


Fig. 14

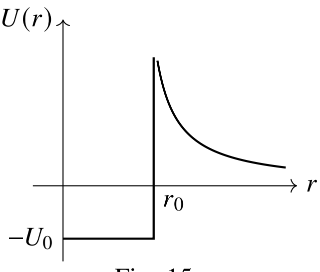


Fig. 15

As $r_0 \to 0$, this formula tends to

$$w \sim \exp\left(-\frac{\pi\alpha}{\hbar}\sqrt{\frac{2m}{E}}\right) = \exp\left(-\frac{2\pi\alpha}{\hbar v}\right).$$

These formulas require a large exponent, i.e. $\alpha/\hbar v \gg 1$. As it should, this agrees with the semiclassical condition (49.11) for motion in a Coulomb field.

[16]This problem was first considered by G. F. Gamow (1928), and by *R.W.Gurney* and *E.U.Condon* (1929), in connection with the theory of radioactive α-decay.

Problem

3. The field $U(x)$ consists of two symmetric potential wells (I and II in Fig. 16), separated by a barrier. Were the barrier impenetrable, there would be energy levels belonging to motion in one well or the other, with identical values for both wells. Passage through the barrier splits each such level into two nearby levels, corresponding to states in which the particle moves in both wells. Find the splitting, assuming that $U(x)$ is semiclassical.

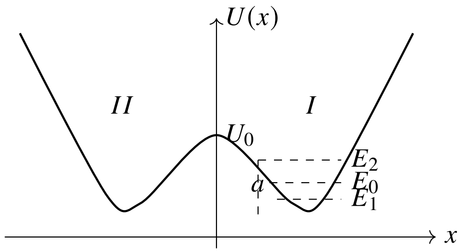


Fig. 16

Solution. Neglecting passage through the barrier, construct an approximate solution of the Schrödinger equation from a semiclassical wave function $\psi_0(x)$ describing motion, at some energy E_0, in one well (say well I). It decays exponentially on both sides of that well, and is normalized so that the integral of ψ_0^2 over well I is unity. Once the small tunnelling probability is admitted, E_0 splits into E_1 and E_2. The appropriate zeroth-order wave functions are the symmetric and antisymmetric combinations of $\psi_0(x)$ and $\psi_0(-x)$:

$$\psi_1(x) = \frac{1}{\sqrt{2}}[\psi_0(x) + \psi_0(-x)], \qquad \psi_2(x) = \frac{1}{\sqrt{2}}[\psi_0(x) - \psi_0(-x)]. \tag{1}$$

In well I, $\psi_0(-x)$ is negligibly small compared with $\psi_0(x)$; in well II the reverse is true. The product $\psi_0(x)\psi_0(-x)$ is thus negligible everywhere, and (1) are normalized so that the integrals of their squares over wells I and II are unity.

Write the Schrödinger equations

$$\psi_0'' + \frac{2m}{\hbar^2}(E_0 - U)\psi_0 = 0, \qquad \psi_1'' + \frac{2m}{\hbar^2}(E_1 - U)\psi_1 = 0.$$

Multiply the first by ψ_1, the second by ψ_0, subtract, and integrate over $0 < x < \infty$. Since at $x = 0$, $\psi_1 = \sqrt{2}\psi_0$, $\psi_1' = 0$, and

$$\int_0^\infty \psi_0\psi_1\,dx \approx \frac{1}{\sqrt{2}}\int_0^\infty \psi_0^2\,dx = \frac{1}{\sqrt{2}},$$

we obtain

$$E_1 - E_0 = -\frac{\hbar^2}{m}\psi_0(0)\psi_0'(0).$$

The same calculation gives $E_2 - E_0$ with the opposite sign. Therefore

$$E_2 - E_1 = \frac{2\hbar^2}{m}\psi_0(0)\psi_0'(0).$$

Using (47.1), with the coefficient C from (48.3), we find

$$\psi_0(0) = \sqrt{\frac{\omega}{2\pi v_0}} \exp\left(-\frac{1}{\hbar}\int_0^a |p|\,dx\right), \qquad \psi_0'(0) = \frac{mv_0}{\hbar}\psi_0(0),$$

where $v_0 = \sqrt{2(U_0 - E_0)/m}$. Hence

$$E_2 - E_1 = \frac{\omega\hbar}{\pi} \exp\left(-\frac{1}{\hbar}\int_{-a}^a |p|\,dx\right),$$

where a is the turning point for the energy E_0 (see Fig. 16).

Problem

4. Find the exact transmission coefficient D, without assuming it small, for the parabolic barrier $U(x) = -kx^2/2$ (*E.C.Kemble*, 1935)[17].

Solution. For any k and E, the motion is semiclassical at sufficiently large $|x|$, where

$$p = \sqrt{2m\left(E + \frac{1}{2}kx^2\right)} \approx x\sqrt{mk} + E\sqrt{\frac{m}{k}}\frac{1}{x},$$

and the asymptotic solution of the Schrödinger equation has the form

$$\psi = \text{const} \cdot \xi^{\pm i\varepsilon - 1/2} \exp\left(\pm i\xi^2/2\right),$$

where

$$\xi = x\left(\frac{mk}{\hbar^2}\right)^{1/4}, \qquad \varepsilon = \frac{E}{\hbar}\sqrt{\frac{m}{k}}.$$

We require the solution that, as $x \to +\infty$, contains only the wave transmitted through the barrier, travelling from left to right. Put

$$\psi = B\xi^{i\varepsilon - 1/2} \exp\left(i\xi^2/2\right) \qquad \text{for } x \to \infty, \tag{1}$$

$$\psi = (-\xi)^{-i\varepsilon - 1/2} \exp\left(-i\xi^2/2\right) + A(-\xi)^{i\varepsilon - 1/2} \exp\left(i\xi^2/2\right) \qquad \text{for } x \to -\infty. \tag{2}$$

The first term in (2) is incident and the second reflected (a wave propagates in the direction in which its phase increases). The relation between A and B follows because the asymptotic expression for ψ is valid throughout the sufficiently distant part of the complex ξ-plane. Follow (1) around a large semicircle of radius ρ in the upper half-plane:

$$\xi = \rho e^{i\varphi}, \qquad i\xi^2 = \rho^2(-\sin 2\varphi + i\cos 2\varphi),$$

where φ runs from 0 to π. After this continuation, (1) becomes the second term in (2), with

$$A = B(e^{i\pi})^{i\varepsilon - 1/2} = -iBe^{-\pi\varepsilon}. \tag{3}$$

[17] The solution also applies to passage close enough to the top of any barrier whose $U(x)$ is quadratic near its maximum.

On the portion $\pi/2 < \varphi < \pi$, where $|\exp(i\xi^2/2)|$ is exponentially large, the exponentially small quantity which should produce the first term in (2) is lost[18].

With the incident-wave normalization chosen in (2), particle-number conservation reads

$$|A|^2 + |B|^2 = 1. \tag{4}$$

Equations (3) and (4) give the desired coefficient

$$D = |B|^2 = 1/(1 + e^{-2\pi\varepsilon}).$$

This holds for every E. If E is negative and large in magnitude, $D \approx e^{-2\pi|\varepsilon|}$, in agreement with (50.5). For $E > 0$,

$$R = 1 - D = 1/(1 + e^{2\pi\varepsilon})$$

is the above-barrier reflection coefficient.

§ 51. Calculation of semiclassical matrix elements

Direct calculation of matrix elements of a physical quantity f using semiclassical wave functions presents serious difficulties. We assume that the energies of the states connected by the matrix element are not close, so the latter does not reduce to a Fourier component of f (§ 48). The difficulty arises because the exponential wave functions, with large imaginary exponents, make the integrand rapidly oscillating. Consider one-dimensional motion in $U(x)$ and, for simplicity, let the operator of f be just a function of x. Let ψ_1, ψ_2 be real wave functions for energies E_1, E_2, with $E_2 > E_1$ (Fig. 17). We must calculate

$$f_{12} = \int_{-\infty}^{+\infty} \psi_1 f \psi_2 \, dx. \tag{51.1}$$

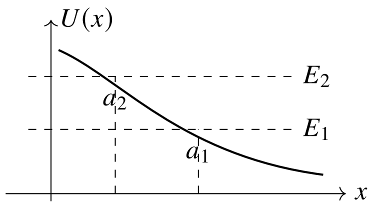


Fig. 17

By (47.5), on the two sides of $x = a_1$ and sufficiently far from it,

$$\psi_1 = \begin{cases} \dfrac{C_1}{\sqrt{|p_1|}} \exp\left(-\dfrac{1}{\hbar}\displaystyle\int_x^{a_1} |p_1|\, dx\right), & x < a_1, \\ \dfrac{2C_1}{\sqrt{p_1}} \cos\left(\dfrac{1}{\hbar}\displaystyle\int_{a_1}^{x} p_1\, dx - \dfrac{\pi}{4}\right), & x > a_1, \end{cases} \tag{51.2}$$

[18] Continuation through the lower half-plane would not determine A: on $-\pi < \varphi < -\pi/2$, adjoining the left endpoint where (2) applies, the term containing $\exp(i\xi^2/2)$ is exponentially small compared with the term containing $\exp(-i\xi^2/2)$.

and similarly for ψ_2, with 1 replaced by 2.

Substitution of these asymptotic expressions in (51.1) would give a wrong result. As shown below, the integral is exponentially small although its integrand is not. A relatively small change in the integrand may therefore change the order of the integral. This difficulty can be avoided as follows.

Write $\psi_2 = \psi_2^+ + \psi_2^-$ by resolving the cosine for $x > a_2$ into two exponentials. According to (50.2),

$$\psi_2^+ = \begin{cases} \dfrac{-iC_2}{2\sqrt{|p_2|}} \exp\left(\dfrac{1}{\hbar}\displaystyle\int_x^{a_2} |p_2|\,dx\right), & x < a_2, \\ \dfrac{C_2}{2\sqrt{p_2}} \exp\left(\dfrac{i}{\hbar}\displaystyle\int_{a_2}^{x} p_2\,dx - \dfrac{i\pi}{4}\right), & x > a_2, \end{cases} \tag{51.3}$$

and $\psi_2^- = (\psi_2^+)^*$.

Accordingly, (51.1) is the sum of two complex-conjugate integrals, $f_{12} = f_{12}^+ + f_{12}^-$. The integral

$$f_{12}^+ = \int_{-\infty}^{+\infty} \psi_1 f \psi_2^+\,dx$$

converges: although ψ_2^+ grows exponentially for $x < a_2$, ψ_1 decreases still faster for $x < a_1$, since $|p_1| > |p_2|$ throughout $x < a_2$.

Regard x as complex and displace the integration path into the upper half-plane. A positive imaginary increment of x produces a growing term in ψ_1 for $x > a_1$, but ψ_2^+ decreases faster because $p_2 > p_1$ there. The integrand therefore decreases.

The displaced path no longer crosses a_1, a_2, where the semiclassical approximation fails. Along the entire path we may use their asymptotic expressions in the upper half-plane:

$$\begin{aligned} \psi_1 &= \frac{C_1}{2[2m(U-E_1)]^{1/4}} \exp\left(\frac{1}{\hbar}\int_{a_1}^{x} \sqrt{2m(U-E_1)}\,dx\right), \\ \psi_2^+ &= \frac{-iC_2}{2[2m(U-E_2)]^{1/4}} \exp\left(-\frac{1}{\hbar}\int_{a_2}^{x} \sqrt{2m(U-E_2)}\,dx\right), \end{aligned} \tag{51.4}$$

where the roots are chosen positive on the real axis for $x < a_2$. Thus

$$\begin{aligned} f_{12}^+ = \frac{-iC_1C_2}{4\sqrt{2m}} \int \exp\Bigg[&\frac{1}{\hbar}\int_{a_1}^{x} \sqrt{2m(U-E_1)}\,dx \\ &- \frac{1}{\hbar}\int_{a_2}^{x} \sqrt{2m(U-E_2)}\,dx\Bigg] \frac{f(x)\,dx}{[(U-E_1)(U-E_2)]^{1/4}}. \end{aligned} \tag{51.5}$$

We seek to move the contour so as to reduce the exponential factor. Its exponent has an extremum only where $U(x) = \infty$; for $E_1 \neq E_2$ its derivative vanishes nowhere else. The only restriction on moving the contour upward is therefore the need to pass around the singular points of $U(x)$, which, by the general theory of linear differential equations, are also the singular points of $\psi(x)$.

The contour depends on the form of $U(x)$. If U has only one singular point x_0 in the upper half-plane, the path may be chosen as in Fig. 18. The immediate neighborhood of

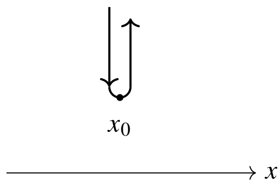


Fig. 18

the singularity gives the principal contribution, so $f_{12} = 2\,\mathrm{Re}\, f_{12}^{+}$ is mainly proportional to

$$f_{12} \sim \exp\left\{-\frac{1}{\hbar}\,\mathrm{Im}\left[\int^{x_0} \sqrt{2m(E_2-U)}\,dx - \int^{x_0} \sqrt{2m(E_1-U)}\,dx\right]\right\}. \tag{51.6}$$

(L. D. Landau, 1932).[19] The lower limits may be any points in classically accessible regions and do not affect the imaginary parts. If $U(x)$ has several upper-half-plane singularities, x_0 in (51.6) is the one giving the exponent of smallest absolute magnitude.[20]

When E_1, E_2 are close, (51.6) simplifies and the matrix element reduces, by § 48, to a time Fourier component of the classical $f[x(t)]$. Put $E_{2,1} = E \pm \hbar\omega_{21}/2$ and expand in $\hbar\omega_{21}$:

$$f_{12} \sim \exp\left(-\omega_{21}\,\mathrm{Im}\int^{x_0} \frac{dx}{v}\right) = \exp(-\omega_{21}\,\mathrm{Im}\,\tau). \tag{51.6a}$$

Here

$$\tau = \int^{x_0} \frac{dx}{v}, \qquad v(x) = \sqrt{2(E-U(x))/m},$$

may be viewed as the *complex time* needed to reach x_0 in the complex x plane, and $v(x)$ as the corresponding "complex velocity." Formula (51.6a) indeed approximates the Fourier component when $\omega_{21}\,\mathrm{Im}\,\tau \gg 1$.

For motion in a central field the calculation is similar, but $U(r)$ is now the effective potential energy, the sum of potential and centrifugal terms, and differs for different l. For later use denote the two effective potentials generally by $U_1(r), U_2(r)$. The exponent in the integrand of (51.5) then has extrema not only at infinities of U_1 or U_2, but also at points satisfying

$$U_1(r) + U_2(r) = E_1 + E_2. \tag{51.7}$$

Thus in

$$f_{12} \sim \exp\left\{-\frac{1}{\hbar}\,\mathrm{Im}\left[\int^{r_0} \sqrt{2m(E_2-U_2)}\,dr - \int^{r_0} \sqrt{2m(E_1-U_1)}\,dr\right]\right\} \tag{51.8}$$

the competing r_0 include both singular points and the roots of (51.7).

[19] Replacing the wave functions by their asymptotic expressions in deriving (51.5), (51.6) is legitimate because the order of the integral over the Fig. 18 contour is fixed by the order of its integrand; a relatively small change of the latter has no material effect.

[20] We assume that $f(x)$ itself has no singularities. Estimate (51.6) also assumes a prefactor of normal order. It may be anomalously small because of the particular problem. The simplest example is $f(x) = \text{const}$: the matrix element vanishes by orthogonality, which (51.6) does not reveal.

The central-field case also differs because the radial integral runs from 0, not $-\infty$, to $+\infty$:

$$f_{12} = \int_0^\infty \chi_1 f \chi_2 \, dr.$$

If the integrand is even, the range may formally be extended to the whole axis and nothing changes. This can occur when U_1, U_2 are even. Then χ_1, χ_2 are even or odd (see § 21),[21] and if $f(r)$ too is even or odd, their product may be even. If the integrand is not even, as necessarily happens if $U(r)$ is not even, the contour cannot be moved away from its initial point $r = 0$, and $r_0 = 0$ must also be included among the competing values in (51.8).

Problems

Problem

1. Calculate the semiclassical matrix elements, retaining only the exponential factor, in the field $U = U_0 e^{-\alpha x}$.

Solution. The only infinity of $U(x)$ is at $x \to -\infty$, so put $x_0 = -\infty$ in (51.6), extending the integrals to $+\infty$. Each separately diverges at $-\infty$; first integrate from $-x$ to $+\infty$ and then let $x \to \infty$. The result is

$$f_{12} \sim \exp\left[-\frac{\pi m}{\alpha \hbar}(v_2 - v_1)\right],$$

where $v_1 = \sqrt{2E_1/m}$ and $v_2 = \sqrt{2E_2/m}$ are the asymptotic free velocities as $x \to \infty$.

Problem

2. Do the same in the Coulomb field $U = \alpha/r$ for transitions between states with $l = 0$.

Solution. The only singular point of $U(r)$ is $r = 0$. The corresponding integral was evaluated in Problem 2 of § 50. Formula (51.8) gives

$$f_{12} \sim \exp\left[\frac{\pi\alpha}{\hbar}\left(\frac{1}{v_2} - \frac{1}{v_1}\right)\right].$$

Problem

3. Do the same for an anharmonic oscillator with

$$U(x) = \frac{m\omega^2}{2}x^2 + \beta x^4$$

under the condition

$$\hbar\omega \ll E_1, E_2 \ll \frac{m^2\omega^4}{\beta}. \tag{1}$$

[21] For even $U(r)$, the radial wave function $R(r)$ is even (odd) for even (odd) l, as follows from $R \sim r^l$ at small r.

SOLUTION. Extension of the argument to finite motion shows that (51.6) still applies. Choose $x_0 \to \pm\infty$; both points contribute at the same order. Thus

$$f_{12} \sim \exp\left\{-\frac{1}{\hbar}\left[\int_{a_2}^{-\infty} \sqrt{2m(U-E_2)}\,dx - \int_{a_1}^{-\infty} \sqrt{2m(U-E_1)}\,dx\right]\right\}.$$

Under condition (1), the main contribution comes from

$$\sqrt{\frac{E_1}{m\omega^2}}, \sqrt{\frac{E_2}{m\omega^2}} \ll |x| \ll \sqrt{\frac{m\omega^2}{\beta}}, \tag{2}$$

where

$$\frac{m\omega^2}{2}x^2 \gg E_1, E_2, \beta x^4.$$

Expanding the exponent in $E_{1,2}/U$, canceling the zeroth-order terms, and neglecting βx^4, we obtain

$$f_{12} \sim \exp\left(-\frac{E_2}{\hbar\omega}\int \frac{d|x|}{|x|} + \frac{E_1}{\hbar\omega}\int \frac{d|x|}{|x|}\right).$$

Cut off the logarithmic integrals at the boundaries of (2): above at $x \sim \sqrt{m\omega^2/\beta}$ and below at $x \sim a_{2,1} \sim \sqrt{E_{2,1}/(m\omega^2)}$. Then

$$f_{12} \sim \exp\left(-\frac{E_2}{2\hbar\omega}\ln\frac{m^2\omega^4}{\beta E_2} + \frac{E_1}{2\hbar\omega}\ln\frac{m^2\omega^4}{\beta E_1}\right).$$

Introducing the state numbers

$$n_1 \approx E_1/\hbar\omega, \qquad n_2 \approx E_2/\hbar\omega,$$

write the result as

$$f_{12} \sim \frac{n_2^{n_2/2}}{n_1^{n_1/2}}\left(\frac{\beta\hbar}{m\omega^3}\right)^{(n_2-n_1)/2}.$$

Since large x are essential to the solution, this result applies when $f(x)$ does not grow too rapidly at infinity. If $f(x)$ is a polynomial, its degree must be small compared with $n_2 - n_1$.

§ 52. Transition probability in the quasiclassical case

Passage through a potential barrier is an example of a process that is wholly impossible in classical mechanics. In the quasiclassical case the probability of such processes is exponentially small. The corresponding exponential factor can be found as follows.

When considering the transition of some system from one state to another, we solve the corresponding classical equations of motion and find the transition "trajectory," which is complex, however, in keeping with the impossibility of the process in classical mechanics. In particular, the "transition point" q_0, at which the formal change of the system from one state to the other occurs, is generally complex; its position is fixed by the classical conservation laws. We then calculate the action $S_1(q_1, q_0) + S_2(q_0, q_2)$ for motion of the system in the first state from some initial position q_1 to the "transition point" q_0, and subsequently in the second state from q_0 to the final position q_2. The probability of the process in question is then given by

$$w \sim \exp\left\{-\frac{2}{\hbar}\,\mathrm{Im}[S_1(q_1, q_0) + S_2(q_0, q_2)]\right\}. \tag{52.1}$$

If the position of the "transition point" is not unique, the one for which the exponent in (52.1) has the smallest absolute value must be selected (while this value must, of course, still be sufficiently large for formula (52.1) to be applicable at all)[22].

Formula (52.1) agrees with the rule obtained in the preceding section for calculating quasiclassical matrix elements. It must be stressed, however, that it would be incorrect to calculate the pre-exponential factor in the probability of transitions of this kind from the square of the corresponding matrix element.

The method of *complex classical trajectories* based on (52.1) is quite general and applies to transitions in systems with any number of degrees of freedom (L. D. Landau, 1932). If the transition point is real but lies in a classically inaccessible region, then, in the simplest case of one-dimensional motion, formula (52.1) coincides with expression (50.5) for the probability of passage through a potential barrier.

Over-barrier reflection. Let us apply (52.1) to the one-dimensional problem of over-barrier reflection—the reflection of a particle whose energy exceeds the height of the barrier. Here q_0 is to be understood as the complex coordinate x_0 of the "stopping point," at which the particle reverses the direction of its motion, that is, a complex root of $U(x) = E$. We shall show how in this case the reflection coefficient can also be calculated more accurately, including the pre-exponential factor.

Once again (as in § 50), we must establish the correspondence between the wave functions far to the right of the barrier (the transmitted wave) and far to its left (incident + reflected waves). This is readily done by a method similar to that already used in § 47 and 50, treating ψ as a function of the complex variable x.

Write the transmitted wave as

$$\psi_+ = \frac{1}{\sqrt{p}} \exp\left(\frac{i}{\hbar}\int_{x_1}^{x} p\,dx\right)$$

(where x_1 is any point on the real axis), and follow its change during continuation in the upper half-plane along a path C which goes round the turning point x_0 (at a sufficiently large distance), Fig. 19. The final part of this path must lie entirely in a region sufficiently far to the left that, along it, the error in the approximate (quasiclassical) wave function of the incident wave is smaller than the small quantity ψ_- being sought. Going round x_0 changes the sign of the root $\sqrt{E - U(x)}$; consequently, on returning to the real axis, ψ_+ becomes a wave ψ_- propagating to the left, that is, the reflected wave[23]. Since the amplitudes of the incident and transmitted waves may be regarded as equal, the required reflection coefficient R is simply

[22]If the potential energy of the system itself has singular points, these too must be included among the competing values of q_0.

[23]Continuation along a path passing below x_0 (for example, simply along the real axis itself) instead turns ψ_+ into the incident wave.

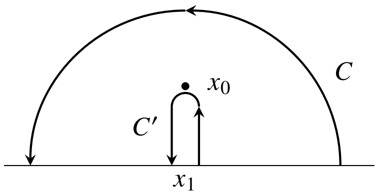


Fig. 19

the ratio of the squared moduli of ψ_- and ψ_+:

$$R = \left|\frac{\psi_-}{\psi_+}\right|^2 = \exp\left(-\frac{2}{\hbar}\,\mathrm{Im}\int_C p\,dx\right). \tag{52.2}$$

Once this formula has been obtained, the integration path in the exponent may be deformed arbitrarily. If it is changed into the path C' shown in Fig. 19, the integral reduces to twice the integral along the path from x_1 to x_0, and we obtain

$$R = e^{-4\sigma(x_1,x_0)/\hbar}, \qquad \sigma(x_1,x_0) = \mathrm{Im}\int_{x_1}^{x_0} p(x)\,dx; \tag{52.3}$$

since $p(x)$ is real over the entire real axis, the choice of x_1 is immaterial[24]. Notice that the pre-exponential factor in (52.3) is equal to unity (V. L. Pokrovsky, S. K. Savvinykh, F. R. Ulinich, 1958)[25].

As already stated, among all possible values of x_0 one must choose that for which the exponent in (52.3) has the smallest absolute value, while this value must still be sufficiently large compared with unity. (Only points x_0 for which $\sigma > 0$, that is, points lying in the upper half-plane, are of course to be considered.) It is also assumed that if the potential energy $U(x)$ itself has singular points in the upper half-plane, the integrals $\sigma(x_1, x_0)$ for them are large[26]. Otherwise such a point itself determines the exponent, but the pre-exponential coefficient is then no longer the one in (52.3). The latter condition necessarily fails as the energy E is increased if $U(x)$ becomes infinite somewhere

in the upper half-plane: a stage is reached at which the point x_0, where $U = E$, approaches the point x_∞, where $U = \infty$, so closely that the two give comparable contributions to the reflection coefficient ($\sigma(x_\infty, x_0) \sim 1$), and formula (52.3) ceases to apply. In the limiting case, when E is so large that this integral is small compared with

[24] In some cases not only the amplitude relation but also the phase relation between the incident and reflected waves is of interest. These relations are characterized by the reflection amplitude, expressed in terms of the coefficients α and β introduced in § 25. The reasoning given above readily shows, in particular, that the reflection amplitude for a wave incident from the left is

$$\left(-\frac{\beta^*}{\alpha^*}\right) = -i\exp\left[\frac{2i}{\hbar}\left(\int_{x_1}^{x_0} p\,dx + p_1x_1\right)\right], \qquad x_1 \to -\infty.$$

(The factor $(-i)$ is associated with the change in phase of the pre-exponential factor on going round the branch point; compare § 47.)

[25] The derivation of this result presented here is due to L. D. Landau (1961).

[26] The contour C in Fig. 19 must pass below the singular points of $U(x)$.

unity, perturbation theory becomes applicable (see Problem 2)[27].

Problems

Problem

1. In the quasiclassical approximation, to exponential accuracy, determine the probability of deuteron breakup in a collision with a heavy nucleus treated as a fixed centre of a Coulomb field (E. M. Lifshitz, 1939).

Solution. Solution. Collisions with zero orbital angular momentum make the largest contribution to the reaction probability. In the quasiclassical approximation these are "head-on" collisions, in which the motion of the particles reduces to one-dimensional motion.

Let E be the deuteron energy measured in units of ε, its proton–neutron binding energy, and let E_n and E_p be the energies of the released neutron and proton (in the same units). Introduce also the dimensionless coordinate $q = r/(Ze^2/\varepsilon)$ (where Ze is the nuclear charge), and denote its generally complex value at the "transition point," that is, at the "instant of breakup" of the deuteron, by q_0. Write E_n, E_p, and E in the form

$$E_n = v_n^2/2, \qquad E_p = v_p^2/2 + 1/q_0, \qquad E = v_d^2 + 1/q_0; \tag{1}$$

v_n, v_p, and v_d are the particle "velocities" at the instant of breakup, measured in units of $\sqrt{\varepsilon/m}$ (where m is the nucleon mass). Here v_n is real and equals the velocity of the released neutron, whereas v_p and v_d are complex. Conservation of energy and momentum at the transition point gives

$$E_p + E_n = E - 1, \qquad v_p + v_n = 2v_d, \tag{2}$$

whence

$$v_p = 2i + v_n, \qquad v_d = i + v_n, \qquad 1/q_0 = E + 1 - v_n^2 + 2iv_n.$$

Before the transition, the action of the system corresponds to motion of the deuteron in the nuclear field up to the breakup point; its imaginary part is

$$\begin{aligned} \operatorname{Im} S_1 &= Ze^2\sqrt{\frac{m}{\varepsilon}}\operatorname{Im}\int_\infty^{q_0}\sqrt{4\left(E-\frac{1}{q}\right)}\,dq = \\ &= Ze^2\sqrt{\frac{m}{\varepsilon}}\operatorname{Im}\left\{2q_0v_d - \frac{2}{\sqrt{E}}\cosh^{-1}\sqrt{q_0E}\right\}. \end{aligned} \tag{3}$$

After the transition, the action corresponds to motion of the neutron and proton away from the breakup point:

[27] The intermediate case was considered by V. L. Pokrovsky and I. M. Khalatnikov (ZhETF, 1961, vol. 40, p. 1713).

$$\begin{aligned}\operatorname{Im} S_2 &= Ze^2\sqrt{\frac{m}{\varepsilon}}\operatorname{Im}\left\{\int_{q_0}^{\infty} v_n\,dq + \int_{q_0}^{\infty}\sqrt{2\left(E_p - \frac{1}{q}\right)}\,dq\right\} = \\ &= Ze^2\sqrt{\frac{m}{\varepsilon}}\operatorname{Im}\left\{-v_n q_0 - v_p q_0 + \sqrt{\frac{2}{E_p}}\cosh^{-1}\sqrt{q_0 E_p}\right\}.\end{aligned} \tag{4}$$

According to (52.1), the probability of the process is

$$w \sim \exp\left\{-\frac{2Ze^2}{\hbar}\sqrt{\frac{m}{\varepsilon}}\operatorname{Im}\left[\sqrt{\frac{2}{E_p}}\cosh^{-1}\sqrt{q_0 E_p} - \frac{2}{\sqrt{E}}\cosh^{-1}\sqrt{q_0 E}\right]\right\}. \tag{5}$$

In accordance with the origin of the first and second $\cosh^{-1}$ terms in the square brackets from expressions (4) and (3), the signs of their imaginary parts must agree with the signs of $\operatorname{Im} v_p$ and $\operatorname{Im} v_d$, respectively (the signs of the latter quantities in solving equations (2) were chosen so that $\operatorname{Im}(S_1 + S_2) > 0$ results).

Because w depends exponentially on E_n, the total breakup probability (including all values of E_n and $E_p = E - 1 - E_n$) is determined by the smallest absolute value of the exponent as a function of E_n. Analysis shows that this value is attained as $E_n \to 0$. Then $q_0 = 1/(E+1)$, and (5) gives

$$w \sim \exp\left\{-\frac{2Ze^2}{\hbar}\sqrt{\frac{m}{\varepsilon}}\left[\sqrt{\frac{2}{E-1}}\arccos\sqrt{\frac{E-1}{E+1}} - \frac{2}{\sqrt{E}}\arccos\sqrt{\frac{E}{E+1}}\right]\right\}.$$

The condition for this formula to apply is that the exponent be large compared with unity.

By calculating the imaginary part of the action $S = S_1 + S_2$ at nonzero values of E_n, one can find the energy distribution of the released particles. Near $E_n = 0$ we have[28]

$$\operatorname{Im} S(E_n) - \operatorname{Im} S(0) \approx E_n\left(\frac{d\operatorname{Im} S}{dE_n}\right)_{E_n=0}.$$

Evaluation of the derivative gives

$$\frac{dw}{dE_n} \sim \exp\left\{-\frac{2Ze^2}{\hbar}\sqrt{\frac{m}{\varepsilon}}E_n\left[\frac{3-E}{(E-1)(E+1)^2} + \frac{1}{\sqrt{2(E-1)^3}}\arccos\sqrt{\frac{E-1}{E+1}}\right]\right\}.$$

Problem

2. Determine the coefficient of over-barrier reflection at particle energies for which perturbation theory is applicable.

Solution. Solution. The result follows from formula (43.1), in which the initial and final wave functions are plane waves propagating in opposite directions and normalized, respectively, to unit flux density and to a delta function of momentum divided by $2\pi\hbar$.

[28] At $E_n = 0$ the function $\operatorname{Im} S(E_n)$ has a cusp, from which it increases in both directions—for positive and negative E_n (values $E_n < 0$ would correspond to capture of the neutron by the nucleus).

Here $d\nu = dp'/2\pi\hbar$, where p' is the momentum after reflection. Carrying out the integration over dp' in (43.1), with the delta function taken into account, gives

$$R = \frac{m^2}{\hbar^2 p^2}\left|\int_{-\infty}^{\infty} U(x)e^{2ipx/\hbar}\,dx\right|^2. \tag{1}$$

This formula is valid when the condition for applicability of perturbation theory is satisfied: $Ua/\hbar v \ll 1$, where a is the width of the barrier (see the note on p. 205), and at the same time $pa/\hbar \lesssim 1$. The latter condition ensures that the dependence $R(p)$ is not exponential; otherwise, the applicability of formula (1) requires further investigation.

PROBLEM

3. Determine the coefficient of over-barrier reflection from a quasiclassical barrier when the function $U(x)$ has a kink at $x = x_0$.

SOLUTION. Solution. If $U(x)$ has a singular feature at a real value of x, the reflection coefficient is determined mainly by the field near that point. To calculate it, perturbation theory may formally be applied without requiring its applicability condition to hold for every x; it is enough that the quasiclassical condition be satisfied. We thus arrive at formula (1) of Problem 2, with the sole difference that the momentum of the incident particle in it must be replaced by the value of $p(x)$ at $x = x_0$.

Choosing the kink as the point $x = 0$, we have near it

$$U = -F_1 x \quad \text{for } x > 0, \qquad U = -F_2 x \quad \text{for } x < 0,$$

with different values of F_1 and F_2. The integration over dx is performed by introducing the damping factor $e^{\pm\lambda x}$ into the integrand (after which $\lambda \to 0$ is taken). The result is

$$R = \frac{m^2\hbar^2}{16p_0^6}(F_2 - F_1)^2,$$

where $p_0 = p(0)$.

§ 53. Transitions Caused by Adiabatic Perturbations

We noted already in § 41 that, in the limit of an arbitrarily slowly time-varying perturbation, the probability that a system passes from one state to another tends to zero. We shall now examine this question quantitatively by calculating the transition probability under a slowly varying (adiabatic) perturbation (L. D. Landau, 1961).

Let the Hamiltonian of the system be a slowly varying function of time, tending to definite limits as $t \to \pm\infty$. Further, let $\psi_n(q,t)$ and $E_n(t)$ be the energy eigenfunctions and eigenvalues (with time entering as a parameter) obtained by solving the Schrödinger equation $\widehat{H}(t)\psi_n = E_n\psi_n$. Since the time dependence of $\widehat{H}$ is adiabatic, the time dependence of E_n and ψ_n is likewise slow.

Our task is to determine the probability w_{21} that the system is found in a certain state ψ_2 as $t \to +\infty$, given that it was in the state ψ_1 as $t \to -\infty$.

The slowness of the perturbation makes the "transition process" last a long time, so the change of action during it (given by the integral $-\int E(t)dt$) is large. In this sense the problem is quasiclassical. The transition probability is governed essentially by those values $t = t_0$ for which

$$E_1(t_0) = E_2(t_0) \tag{53.1}$$

and which may be regarded as corresponding to the "instant of transition" in classical mechanics (compare § 52). In reality, of course, such a transition is classically impossible; this is expressed by the complex character of the roots of (53.1). We must therefore investigate the solutions of the Schrödinger equation for complex values of the parameter t, near a point $t = t_0$ at which two energy eigenvalues become equal.

As will be seen, near this point the eigenfunctions ψ_1, ψ_2 depend strongly on t. To determine that dependence, first introduce linear combinations of them, denoted by φ_1, φ_2, satisfying

$$\int \varphi_1^2 dq = \int \varphi_2^2 dq = 0, \qquad \int \varphi_1\varphi_2 dq = 1. \tag{53.2}$$

This can always be achieved by a suitable choice of complex coefficients (functions of t). The functions φ_1, φ_2 are no longer singular at $t = t_0$.

We now seek the eigenfunctions as linear combinations

$$\psi = a_1\varphi_1 + a_2\varphi_2. \tag{53.3}$$

It must be borne in mind that, for complex values of the "time" t, the operator $\widehat{H}(t)$ depending on it (of the form (17.4)) still equals its transpose ($\widehat{H} = \widetilde{H}$), but is no longer Hermitian ($\widehat{H} \neq \widetilde{H}^*$), since the potential energy $U(t) \neq U(t)^*$.

Substitute (53.3) in the Schrödinger equation, multiply it on the left first by φ_1 and then by φ_2, and integrate over dq. With the notation

$$H_{ik}(t) = \int \varphi_i \widehat{H}\varphi_k, dq \tag{53.4}$$

and using $H_{12} = H_{21}$, which follows from the stated property of the Hamiltonian, we obtain the system

$$\begin{aligned} H_{11}a_1 + H_{12}a_2 &= Ea_2, \\ H_{12}a_1 + H_{22}a_2 &= Ea_1. \end{aligned} \tag{53.5}$$

The solvability condition for this system gives $(H_{12} - E)^2 = H_{11}H_{22}$. Its roots determine the energy eigenvalues:

$$E = H_{12} \pm \sqrt{H_{11}H_{22}}. \tag{53.6}$$

Equation (53.5) then gives

$$a_2/a_1 = \pm\sqrt{H_{11}/H_{22}}. \tag{53.7}$$

It follows from (53.6) that, if the two eigenvalues are to coincide at $t = t_0$, either H_{11} or H_{22} must vanish there; suppose it is H_{11}. At a regular point a function generally vanishes in proportion to $t - t_0$. Hence

$$E(t) - E(t_0) = \pm\,\mathrm{const}\cdot\sqrt{t - t_0}, \tag{53.8}$$

so that $E(t)$ has a branch point at $t = t_0$. At the same time $a_2 \sim \sqrt{t - t_0}$, and consequently there is only one eigenfunction at $t = t_0$, namely φ_1.

We can now see that the present problem is formally identical to the problem of above-barrier reflection considered in § 52. Here the wave function $\Psi(t)$ is "quasiclassical in time" (instead of being quasiclassical in the coordinate, as in § 52). We have to determine the term $c_2\psi_2 \exp(-iE_2t/\hbar)$ in the wave function as $t \to +\infty$, given that as $t \to -\infty$ the wave function is $\Psi(t) = \psi_1 \exp(-iE_1t/\hbar)$. This is analogous to determining the reflected wave at $x \to -\infty$ from the transmitted wave at $x \to +\infty$; the desired transition probability is $w_{21} = |c_2|^2$. The action $S = -\int E(t)dt$ is an integral over time of a function with complex branch points, just as $p(x)$ had complex branch points in the integral $\int p, dx$. The problem is therefore solved by making a detour in the complex t plane, from large negative to large positive values, in complete analogy with the detour made in § 52 in the complex x plane. We shall not repeat the argument here.

Suppose that $E_2 > E_1$ on the real axis. The detour must then lie in the upper half-plane of the complex t plane, since on displacement into this half-plane the ratio $e^{-iE_2t/\hbar}/e^{-iE_1t/\hbar}$ increases. The result is the formula (analogous to (52.2))

$$w_{21} = \exp\left(\frac{2}{\hbar}\operatorname{Im}\int_{C'} E(t)\,dt\right), \tag{53.9}$$

where integration is along the contour shown in Fig. 19, but from left to right.

On the left-hand branch of this contour $E = E_1$, and on the right-hand branch $E = E_2$. Thus (53.9) can be written

$$w_{21} = \exp\left(-2\operatorname{Im}\int_t^{t_0} \omega_{21}(t)\,dt\right), \tag{53.10}$$

where $\omega_{21} = (E_2 - E_1)/\hbar$; t is any point on the real t axis, while t_0 is to be the root of (53.1) in the upper half-plane for which the exponent in (53.10) has the smallest absolute value[29]. In addition, "transition paths" through various intermediate states may compete with the direct transition from state 1 to state 2; their probabilities are expressed by analogous formulas. Thus, for a transition along the "path" $1 \to 3 \to 2$, the integral in (53.10) is replaced by the sum

$$\int^{t_0^{(31)}} \omega_{31}(t)\,dt + \int^{t_0^{(23)}} \omega_{23}(t)\,dt,$$

whose upper limits are the "crossing points" of the terms $E_1(t)$, $E_3(t)$ and $E_2(t)$, $E_3(t)$, respectively. This result follows by taking a contour that encloses both complex points[30].

[29]The competing values of t_0 must also include points at which $E(t)$ becomes infinite (though for such points the pre-exponential factor in (53.10) would be different).

[30]Cases in which the intermediate states belong to the continuous spectrum require separate consideration.

Problem

Determine the change of the adiabatic invariant of a classical oscillator obeying

$$\frac{d^2x}{dt^2} + \omega^2(t)x = 0 \tag{1}$$

when the frequency $\omega(t)$ varies slowly from its value ω_1 as $t \to -\infty$ to ω_2 as $t \to \infty$ (A. M. Dykhne, 1960).

Solution. Equation (1) is obtained from the Schrödinger equation by making the replacements

$$\psi \to x, \qquad x \to t; \qquad p(x)/\hbar = k(x) \to \omega(t),$$

after which the problem is formally equivalent to the potential-barrier reflection problem treated in § 25. The calculation of the change in the adiabatic invariant can therefore be reduced to calculating the reflection amplitude.

Write the solution of (1) as $t \to \mp\infty$ in the form

$$x = A_1 e^{i\omega_1 t} + A_1^* e^{-i\omega_1 t}, \quad t \to -\infty,$$
$$x = A_2 e^{i\omega_2 t} + A_2^* e^{-i\omega_2 t}, \quad t \to \infty.$$

According to (25.6),

$$A_2 = \alpha A_1 + \beta A_1^*. \tag{2}$$

The adiabatic invariant of the oscillator is E/ω, so that

$$I_1 = m\omega_1 \overline{x^2} = 2m\omega_1|A_1|^2, \qquad I_2 = 2m\omega_2|A_2|^2,$$

or, on substituting (2),

$$I_2 = 2m\omega_2[(|\alpha|^2 + |\beta|^2)|A_1|^2 + 2\,\mathrm{Re}(\alpha\beta^* A_1^2)].$$

Using relation (25.7), which in our notation reads $|\alpha|^2 = |\beta|^2 + \omega_1/\omega_2$, we find

$$I_2 - I_1 = 4m\omega_2[|\beta|^2|A_1|^2 + \mathrm{Re}(\alpha\beta^* A_1^2)]. \tag{3}$$

The present case of a slowly varying $\omega(t)$ corresponds, in the barrier-reflection problem, to the quasiclassical situation of the preceding section. In that situation β is exponentially small, while $|\alpha|^2 \approx \omega_1/\omega_2$. (It is assumed that $\omega^2(t)$ has neither singularities nor zeros on the real t axis.) The method for calculating the reflection amplitude described in the preceding section gives the estimate

$$\Delta I = I_2 - I_1 \sim |\beta| \sim \exp\left(-2\,\mathrm{Im}\int_{t_1}^{t_0} \omega(t)\,dt\right),$$

where t_0 is the singular point in the upper half of the t plane that makes the largest contribution to ΔI. This formula agrees with the results of § 51 (see Vol. I) for the present case of a harmonic oscillator. If $\omega^2(t)$ has a simple zero in the upper half-plane, the formulas of the preceding section also give the pre-exponential factor. (See the note on p. 241.)

Notice that the second—and principal—term in (3) depends on the initial phase of the oscillation. When averaged over this phase it vanishes, and hence

$$\overline{\Delta I} \approx 2RI_1,$$

where $R \approx (\omega_2/\omega_1)|\beta|^2$ is the "reflection coefficient."

CHAPTER VIII

Spin

§ 54. Spin

In classical and quantum mechanics alike, conservation of angular momentum follows from the isotropy of space for a closed system. This already displays the connection between angular momentum and rotational symmetry. In quantum mechanics, however, that connection becomes especially profound and is in essence the central content of the concept of angular momentum. Moreover, the classical definition of a particle's angular momentum as the product $\mathbf{r}\times\mathbf{p}$ loses its direct meaning because position and momentum cannot be measured simultaneously.

We saw in § 28 that specifying l and m fixes the angular dependence of a particle's wave function, and thereby all its symmetry properties under rotations. In their most general formulation, these properties amount to the transformation law of wave functions when the coordinate system is rotated.

The wave function ψ_{LM} of a system of particles, with specified angular momentum L and projection M, remains unchanged[1] only when the coordinate system is rotated about the z axis. Any rotation that changes the direction of that axis makes the angular-momentum projection on it indefinite. In the new coordinate axes the wave function therefore becomes, in general, a superposition (linear combination) of the $2L+1$ functions belonging to the possible values of M at fixed L. Thus, under rotations, the $2L + 1$ functions ψ_{LM} transform among themselves[2]. The law of this transformation, that is, the superposition coefficients as functions of the angles through which the coordinate axes are rotated, is completely fixed by L. Angular momentum thus acquires the meaning of a quantum number that classifies the states of a system by their transformation properties under coordinate rotations. This aspect is particularly important in quantum mechanics because it is not tied directly to an explicit angular dependence of the wave functions: their mutual transformation law can be stated without reference to such a dependence.

Consider a composite particle, say an atomic nucleus, at rest as a whole and in a definite internal state. Besides a definite internal energy it has an angular momentum of definite magnitude L, associated with the motion of its constituents; this angular momentum may have $2L + 1$ different orientations in space. Hence, in describing the motion of the composite particle as a whole, we must assign to it, in addition to its coordinates, a discrete variable: the projection of its internal angular momentum on a chosen direction.

With the meaning of angular momentum just described, its origin is immaterial. We are led naturally to the idea of an "intrinsic" angular momentum that must be assigned to

[1] Apart from an irrelevant phase factor.

[2] In mathematical terminology, these functions realize the so-called *irreducible representations* of the rotation group. The number of functions transforming among themselves is called the *dimension of the representation*; it is understood that no choice of other linear combinations can reduce this number.

a particle regardless of whether the particle is "composite" or "elementary."

Quantum mechanics therefore attributes to an elementary particle an "intrinsic" angular momentum unrelated to its motion through space. This is a specifically quantum property of elementary particles, disappearing in the limit $\hbar \to 0$, and in principle permits no classical interpretation [3].

The intrinsic angular momentum of a particle is called its *spin*, in contrast to the angular momentum associated with motion through space, which is called *orbital angular momentum*[4]. The particle may be elementary, or it may be composite but behave as elementary in the class of phenomena under consideration (an atomic nucleus, for example). We denote its spin, measured like orbital angular momentum in units of $\hbar$, by s.

For a particle with spin, the wave-function description of a state must give not only the probabilities of its various positions in space but also the probabilities of the possible orientations of its spin. Thus the wave function depends not only on the three continuous particle coordinates, but also on a discrete *spin variable*. This variable specifies the spin projection on a chosen direction (the z axis) and takes a finite set of discrete values, which will be denoted by σ.

Let $\psi(x, y, z; \sigma)$ be such a wave function. It is essentially a set of different coordinate functions, one for each value of σ; we shall call them the *spin components* of the wave function. The integral

$$\int |\psi(x, y, z; \sigma)|^2 \, dV$$

is the probability that the particle has the specified value σ. The probability that it lies in the volume element dV with any value of σ is

$$dV \sum_{\sigma} |\psi(x, y, z; \sigma)|^2.$$

When applied to a wave function, the quantum-mechanical spin operator acts on the spin variable σ. In other words, it transforms the wave-function components among themselves in a certain way. Its form will be obtained below. Even from general considerations, however, it is easy to see that $\hat{s}_x$, $\hat{s}_y$, and $\hat{s}_z$ obey the same commutation relations as the orbital-angular-momentum operators.

The angular-momentum operator is essentially the operator of an infinitesimal rotation. In deriving the orbital-angular-momentum operator in § 26, we considered how a rotation acts on a function of the coordinates. That derivation is inapplicable to spin, since the spin operator acts on the spin variable rather than on the coordinates. We must instead consider an infinitesimal rotation in its general form, as a rotation of the coordinate system. Carrying out infinitesimal rotations successively about the x and y axes and then about the same axes in the reverse order, direct calculation shows that the difference between the two results is equivalent to an infinitesimal rotation about the z axis, through an angle equal to the product of the rotation angles about x and y.

[3] In particular, it would be wholly meaningless to imagine the "intrinsic" angular momentum of an elementary particle as arising from its rotation "about its own axis."

[4] The physical idea that the electron possesses intrinsic angular momentum was proposed by G. Uhlenbeck and S. Goudsmit (1925). Spin was introduced into quantum mechanics by W. Pauli (1927).

We omit this simple calculation. It again gives the usual commutation relations for the components of angular momentum, and these must therefore hold for spin:

$$\{\hat{s}_x, \hat{s}_y\} = i\hat{s}_z \tag{54.1}$$

with all the physical consequences that follow from them.

The commutation relations (54.1) determine the possible magnitudes and components of spin. The whole derivation in § 27, equations (27.7)–(27.9), used only the commutation relations and applies here without change, with s in place of L. Equations (27.7) show that the eigenvalues of a spin projection form a sequence differing by unity. Here, however, we cannot infer that the values themselves must be integral, as for the orbital projection L_z: the argument at the beginning of § 27 used the specifically orbital expression (26.14) for $\hat{l}_z$ and does not apply.

The sequence of eigenvalues of s_z is bounded above and below by values of equal magnitude and opposite sign, denoted by $\pm s$. The difference $2s$ between the greatest and least values must be an integer or zero. Hence s may be 0, 1/2, 1, 3/2, ...

The eigenvalues of the square of the spin are consequently

$$s^2 = s(s+1), \tag{54.2}$$

where s may be integral, including zero, or half-integral. For given s, the spin component s_z can take the $2s+1$ values $s, s-1, \ldots, -s$. Accordingly, the wave function of a spin-s particle has $2s+1$ components[5].

Experiment shows that most elementary particles—electrons, positrons, protons, neutrons, μ mesons, and all hyperons (Λ, Σ, Ξ)—have spin 1/2. There are also elementary particles, the π and K mesons, with spin zero.

The total angular momentum of a particle is the sum of its orbital angular momentum **l** and spin **s**. Since their operators act on functions of wholly different variables, they of course commute.

The eigenvalues of the total angular momentum

$$\mathbf{j} = \mathbf{l} + \mathbf{s} \tag{54.3}$$

are determined by the same "vector-model" rule as the sum of the orbital angular momenta of two different particles (§ 31). For fixed l and s, the total angular momentum can take the values $l+s, l+s-1, \ldots, |l-s|$. Thus an electron of spin 1/2 with nonzero orbital angular momentum l may have $j = l \pm 1/2$; when $l = 0$, naturally, there is only $j = 1/2$.

The total-angular-momentum operator **J** of a system of particles is the sum of their individual operators **j**, so its values are again given by the vector-model rules. It may be written

$$\mathbf{J} = \sum_a \mathbf{j}_a = \mathbf{L} + \mathbf{S}, \qquad \mathbf{L} = \sum_a \mathbf{l}_a, \quad \mathbf{S} = \sum_a \mathbf{s}_a, \tag{54.4}$$

where **S** may be called the total spin and **L** the total orbital angular momentum of the system.

[5] For each kind of particle, s is fixed. Thus in the classical limit $\hbar \to 0$, the spin angular momentum $\hbar s$ tends to zero. This reasoning is inapplicable to orbital angular momentum because l can take arbitrary values. The classical limit involves simultaneously $\hbar \to 0$ and $l \to \infty$, with $\hbar l$ remaining finite.

If the total spin of the system is half-integral (or integral), the total angular momentum is likewise half-integral (or integral), since orbital angular momentum is always integral. In particular, a system containing an even number of identical particles necessarily has integral total spin and hence integral total angular momentum.

The total-angular-momentum operators **j** of a particle (or **J** of a system) obey the same commutation rules as orbital angular momentum or spin, since these are the general commutation rules for every angular momentum. Equations (27.13) for its matrix elements likewise hold for every angular momentum when the elements are defined with respect to its own eigenfunctions. Equations (29.7)–(29.10) for matrix elements of arbitrary vector quantities also remain valid, with the corresponding changes of notation.

Problem

A spin-$1/2$ particle is in a state with the definite value $s_z = 1/2$. Find the probabilities of the possible values of its spin projection on an axis z' inclined at an angle θ to the z axis.

Solution.

The mean spin vector $\bar{\mathbf{s}}$ plainly points along z and has magnitude $1/2$. Its projection on z' gives the mean spin in that direction, $\bar{s}_{z'} = (1/2)\cos\theta$. On the other hand, $\bar{s}_{z'} = (1/2)(w_+ - w_-)$, where $w_\pm$ are the probabilities of $s_{z'} = \pm 1/2$. Together with $w_+ + w_- = 1$, this gives

$$w_+ = \cos^2(\theta/2), \qquad w_- = \sin^2(\theta/2).$$

§ 55. The spin operator

In the remainder of this chapter we shall not be concerned with the coordinate dependence of wave functions. Thus, when discussing how $\psi(x, y, z; \sigma)$ behaves under a rotation of the coordinate system, we may suppose that the particle is at the origin. Its coordinates are then unchanged by the rotation, and the result describes specifically the dependence of ψ on the spin variable σ.

The variable σ differs from ordinary variables (coordinates) in being discrete. The most general linear operator acting on functions of a discrete variable may be written

$$(\hat{f}\psi)(\sigma) = \sum_{\sigma'} f_{\sigma\sigma'}\psi(\sigma'), \tag{55.1}$$

where $f_{\sigma\sigma'}$ are constants. The parentheses around $\hat{f}\psi$ emphasize that the spin argument following them belongs to the function produced by $\hat{f}$, not to the original function ψ. The quantities $f_{\sigma\sigma'}$ are readily seen to be the matrix elements of the operator as ordinarily defined by (11.5)[6]. Coordinate integration in (11.5) is now replaced by summation over the discrete variable, so that

$$f_{\sigma_2\sigma_1} = \sum_{\sigma} \psi^*_{\sigma_2}(\sigma)[\hat{f}\psi_{\sigma_1}(\sigma)]. \tag{55.2}$$

[6]Notice that the indices on the matrix elements on the right of (55.1) occur in an order which is, in a certain sense, the reverse of the usual order in (11.11).

Here $\psi_{\sigma_1}(\sigma)$ and $\psi_{\sigma_2}(\sigma)$ are eigenfunctions of $\hat{s}_z$ belonging to $s_z = \sigma_1$ and $s_z = \sigma_2$. Each represents a state with a definite s_z, so only one wave-function component is nonzero[7]:

$$\psi_{\sigma_1}(\sigma) = \delta_{\sigma\sigma_1}, \qquad \psi_{\sigma_2}(\sigma) = \delta_{\sigma\sigma_2}. \tag{55.3}$$

By (55.1),

$$(\hat{f}\psi_{\sigma_1})(\sigma) = \sum_{\sigma'} f_{\sigma\sigma'}\psi_{\sigma_1}(\sigma') = \sum_{\sigma'} f_{\sigma\sigma'}\delta_{\sigma'\sigma_1} = f_{\sigma\sigma_1}.$$

Substitution of this result and $\psi_{\sigma_2}(\sigma)$ in (55.2) makes that equation an identity, proving the assertion.

Operators acting on functions of σ can therefore be represented by matrices of order $2s + 1$. In particular, this applies to the spin operator itself, whose action is, by (55.1),

$$(\hat{\mathbf{s}}\psi)(\sigma) = \sum_{\sigma'} \mathbf{s}_{\sigma\sigma'}\psi(\sigma'). \tag{55.4}$$

As stated at the end of § 54, the matrices $\hat{s}_x$, $\hat{s}_y$, and $\hat{s}_z$ are the matrices L_x, L_y, and L_z obtained in § 27, with L, M replaced by s, σ:

$$\begin{aligned} (s_x)_{\sigma,\sigma-1} &= (s_x)_{\sigma-1,\sigma} = \frac{1}{2}\sqrt{(s+\sigma)(s-\sigma+1)}, \\ (s_y)_{\sigma,\sigma-1} &= -(s_y)_{\sigma-1,\sigma} = -\frac{i}{2}\sqrt{(s+\sigma)(s-\sigma+1)}, \\ (s_z)_{\sigma\sigma} &= \sigma. \end{aligned} \tag{55.5}$$

This defines the spin operator.

In the particularly important case $s = 1/2$, $\sigma = \pm 1/2$, these are two-rowed matrices. They are written

$$\hat{\mathbf{s}} = \frac{1}{2}\hat{\boldsymbol{\sigma}}, \tag{55.6}$$

where[8]

$$\hat{\sigma}_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \hat{\sigma}_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \hat{\sigma}_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{55.7}$$

The matrices (55.7) are called the *Pauli matrices*. As it should be for a matrix defined using eigenfunctions of s_z itself, $\hat{s}_z = \frac{1}{2}\hat{\sigma}_z$ is diagonal[9].

[7] More precisely one should write $\psi_{\sigma_1}(\sigma) = \psi(x, y, z)\,\delta_{\sigma_1\sigma}$; the coordinate factors irrelevant here have been omitted in (55.3).

We emphasize once again the need to distinguish the specified eigenvalue of s_z (σ_1 or σ_2) from the independent variable σ. This is precisely the reason for the difference between (11.11) and (55.1).

[8] When matrices are written as in (55.7), rows and columns are numbered by the values of σ: the row number corresponds to the first index of a matrix element and the column number to the second. Here the numbers are $+1/2, -1/2$. According to (55.4), the operator acts by multiplying the σth row of the matrix by the wave-function components arranged in the column $\psi = \begin{pmatrix} \psi(1/2) \\ \psi(-1/2) \end{pmatrix}$.

[9] Using the same letter for the spin projection and for the Pauli matrices cannot cause confusion: the Pauli matrices carry a hat.

We record several characteristic properties of the Pauli matrices. Direct multiplication of (55.7) gives

$$\hat{\sigma}_y\hat{\sigma}_z = i\hat{\sigma}_x, \qquad \hat{\sigma}_z\hat{\sigma}_x = i\hat{\sigma}_y, \qquad \hat{\sigma}_x\hat{\sigma}_y = i\hat{\sigma}_z. \tag{55.8}$$

Combining these with the general commutation rules (54.1), we find

$$\hat{\sigma}_i\hat{\sigma}_k + \hat{\sigma}_k\hat{\sigma}_i = 0 \quad (i \neq k), \tag{55.9}$$

so the Pauli matrices anticommute. These relations immediately give the useful formulas

$$\hat{\boldsymbol{\sigma}}^2 = 3, \qquad (\hat{\boldsymbol{\sigma}}\mathbf{a})(\hat{\boldsymbol{\sigma}}\mathbf{b}) = \mathbf{ab} + i\hat{\boldsymbol{\sigma}}(\mathbf{a}\times\mathbf{b}), \tag{55.10}$$

where $\mathbf{a}$ and $\mathbf{b}$ are arbitrary vectors[10]. By these relations, any scalar polynomial in the matrices $\hat{\sigma}_i$ reduces to terms independent of $\hat{\boldsymbol{\sigma}}$ and terms linear in it. Hence every scalar function of $\hat{\boldsymbol{\sigma}}$ reduces to a linear function (see problem 1). Finally, the traces (sums of diagonal components) of the Pauli matrices and their products are

$$\operatorname{Tr}\hat{\sigma}_i = 0, \qquad \operatorname{Tr}(\hat{\sigma}_i\hat{\sigma}_k) = 2\delta_{ik}. \tag{55.11}$$

The following sections examine spin wave functions in detail, including their behavior under arbitrary coordinate rotations. We note here at once an important property: their behavior under rotations about the z axis.

Make an infinitesimal rotation $\delta\varphi$ about z. In terms of angular momentum, here spin, its operator is $1 + i\delta\varphi\hat{s}_z$. The functions $\psi(\sigma)$ therefore become $\psi(\sigma) + \delta\psi(\sigma)$, with

$$\delta\psi(\sigma) = i\delta\varphi\hat{s}_z\psi(\sigma) = i\sigma\psi(\sigma)\delta\varphi.$$

Writing this as $d\psi/d\varphi = i\sigma\psi(\sigma)$ and integrating, we find that a finite rotation through φ gives

$$\psi'(\sigma) = e^{i\sigma\varphi}\psi(\sigma). \tag{55.12}$$

For $\varphi = 2\pi$, the multiplier is $e^{2\pi i\sigma}$, the same for every σ and equal to $(-1)^{2s}$, since 2σ always has the same parity as $2s$. Thus a complete rotation of the coordinate system about z restores the wave functions of an integral-spin particle, but reverses the sign of those of a half-integral-spin particle.

Problem

1. Reduce an arbitrary function of the scalar $a + \mathbf{b}\hat{\boldsymbol{\sigma}}$, which is linear in the Pauli matrices, to a linear function.

Solution.

To find the coefficients in

$$f(a + \mathbf{b}\hat{\boldsymbol{\sigma}}) = A + B\mathbf{b}\hat{\boldsymbol{\sigma}},$$

[10] Terms on the right-hand sides of (55.8)–(55.10) that do not depend on $\hat{\boldsymbol{\sigma}}$ are, of course, to be understood as constants times the two-rowed unit matrix.

choose z along $\mathbf{b}$. The eigenvalues of $a + \mathbf{b}\hat{\boldsymbol{\sigma}}$ are $a \pm b$, and those of the corresponding operator $f(a + \mathbf{b}\hat{\boldsymbol{\sigma}})$ are $f(a \pm b)$. Hence

$$A = \frac{1}{2}[f(a+b) + f(a-b)], \qquad B = \frac{1}{2b}[f(a+b) - f(a-b)].$$

2. Find the values of the scalar product $\mathbf{s}_1\mathbf{s}_2$ of the spin-$1/2$ angular momenta of two particles in states for which the total spin $\mathbf{S} = \mathbf{s}_1 + \mathbf{s}_2$ has a definite value, 0 or 1.

Solution.

The general addition formula (31.3) for any two angular momenta gives

$$\mathbf{s}_1\mathbf{s}_2 = 1/4 \quad \text{for } S = 1, \qquad \mathbf{s}_1\mathbf{s}_2 = -3/4 \quad \text{for } S = 0.$$

3. Which powers of the operator s_z for arbitrary spin s are independent?

Solution.

The operator

$$(\hat{s}_z - s)(\hat{s}_z - s + 1)\cdots(\hat{s}_z + s),$$

formed from the differences between $\hat{s}_z$ and every possible eigenvalue of s_z, gives zero on every wave function and is therefore itself zero. Consequently $(\hat{s}_z)^{2s+1}$ is expressible through lower powers of $\hat{s}_z$; only the powers from 1 through $2s$ are independent.

§ 56. Spinors

For zero spin the wave function has a single component, $\psi(0)$. The spin operator annihilates it: $\hat{\mathbf{s}}\psi = 0$. Since $\mathbf{s}$ is connected with the operator of infinitesimal rotations, this means that the wave function of a spin-zero particle is unchanged by rotations of the coordinate system; it is a scalar.

The wave function of a spin-$1/2$ particle has two components, $\psi(1/2)$ and $\psi(-1/2)$. For convenience in later generalizations, label these by upper indices 1 and 2. The two-component quantity

$$\psi = \begin{pmatrix} \psi^1 \\ \psi^2 \end{pmatrix} = \begin{pmatrix} \psi(1/2) \\ \psi(-1/2) \end{pmatrix} \tag{56.1}$$

is called a *spinor*.

Under an arbitrary rotation of the coordinate system, its components undergo the linear transformation

$$\psi^{1'} = a\psi^1 + b\psi^2, \qquad \psi^{2'} = c\psi^1 + d\psi^2. \tag{56.2}$$

This can be written

$$\psi^{\lambda'} = (\hat{U}\psi)^\lambda, \qquad \hat{U} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \tag{56.3}$$

where $\hat{U}$ is the transformation matrix[11]. Its elements are generally complex functions of the rotation angles of the coordinate axes. Relations among them follow directly from the physical requirements on a spinor as a particle wave function.

[11]The notation $\hat{U}\psi$ means multiplication of the rows of $\hat{U}$ by the column ψ.

Consider the bilinear form

$$\psi^1\varphi^2 - \psi^2\varphi^1, \tag{56.4}$$

where ψ and φ are two spinors. Direct calculation gives

$$\psi^{1'}\varphi^{2'} - \psi^{2'}\varphi^{1'} = (ad - bc)(\psi^1\varphi^2 - \psi^2\varphi^1),$$

so (56.4) transforms into itself under a coordinate rotation. A single function that transforms into itself can be regarded as belonging to spin zero; it must consequently be a scalar and remain altogether unchanged by a rotation. Hence

$$ad - bc = 1; \tag{56.5}$$

the determinant of the transformation matrix is unity[12].

Further relations follow from requiring the expression

$$\psi^1\psi^{1*} + \psi^2\psi^{2*}, \tag{56.6}$$

which gives the probability of finding the particle at a given point, to be a scalar. A transformation preserving the sum of squared moduli is unitary, so $\hat{U}^+ = \hat{U}^{-1}$ (see § 12). Subject to (56.5),

$$\hat{U}^{-1} = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Equating this to the conjugate matrix

$$\hat{U}^+ = \begin{pmatrix} a^* & c^* \\ b^* & d^* \end{pmatrix}$$

gives

$$a = d^*, \qquad b = -c^*. \tag{56.7}$$

By (56.5) and (56.7), the four complex quantities a, b, c, d contain only three independent real parameters, corresponding to the three angles that specify a rotation of a three-dimensional coordinate system.

Comparing the scalar expressions (56.4) and (56.6), we see that ψ^{1*}, ψ^{2*} must transform as $\psi^2, -\psi^1$. Relations (56.5) and (56.7) make this easy to verify[13].

Spinor algebra can be put into a form analogous to tensor algebra. Alongside the contravariant components ψ^1, ψ^2, with upper indices, introduce *covariant* components, with lower indices, by

$$\psi_1 = \psi^2, \qquad \psi_2 = -\psi^1. \tag{56.8}$$

The invariant combination (56.4) of two spinors then becomes the scalar product

$$\psi_\lambda\varphi^\lambda = \psi_1\varphi^1 + \psi_2\varphi^2. \tag{56.9}$$

[12] Such a transformation of two quantities is called *binary*.

[13] This property is closely connected with time-reversal symmetry. Time reversal corresponds (see § 18) to replacing a wave function by its complex conjugate, while also reversing angular-momentum projections. Thus the functions complex conjugate to $\psi^1 = \psi(1/2)$ and $\psi^2 = \psi(-1/2)$ must have the same properties as the components belonging respectively to spin projections $-1/2$ and $1/2$.

Here and below, repeated dummy indices are summed as in tensor algebra. One rule specific to spinor algebra must be kept in mind. Since $\psi^\lambda\varphi_\lambda = \psi^1\varphi_1 + \psi^2\varphi_2 = -\psi^2\varphi^1 - \psi^1\varphi^2$, we have

$$\psi^\lambda\varphi_\lambda = -\psi_\lambda\varphi^\lambda. \tag{56.10}$$

It follows at once that the scalar product of any spinor with itself vanishes:

$$\psi_\lambda\psi^\lambda = 0. \tag{56.11}$$

As explained above, ψ_1, ψ_2 transform as ψ^{1*}, ψ^{2*}, that is,

$$\psi'_\lambda = (\hat{U}^*\psi)_\lambda. \tag{56.12}$$

The product $\hat{U}^*\psi$ may equally be written $\psi\hat{U}^*$ using the transpose of $\hat{U}^*$. Unitarity gives $\hat{U}^* = \widetilde{\hat{U}}^{-1}$, so that $\widetilde{\psi}' = (\widetilde{\psi}\hat{U}^{-1})$, or[14]

$$\psi'_\lambda = \psi_\mu(U^{-1})^\mu{}_\lambda. \tag{56.13}$$

Just as vectors lead to tensors in ordinary tensor algebra, one can introduce *spinors of higher rank*. A second-rank spinor is a four-component quantity $\psi^{\lambda\mu}$ whose components transform like the products $\psi^\lambda\varphi^\mu$ of components of two first-rank spinors. Alongside the contravariant components $\psi^{\lambda\mu}$ one may consider covariant $\psi_{\lambda\mu}$ and mixed $\psi_\lambda{}^\mu$ components, transforming respectively like $\psi_\lambda\varphi_\mu$ and $\psi_\lambda\varphi^\mu$. Spinors of any rank are defined analogously.

The passage from contravariant to covariant components and back can be written

$$\psi_\lambda = g_{\lambda\mu}\psi^\mu, \qquad \psi^\lambda = g^{\mu\lambda}\psi_\mu, \tag{56.14}$$

where

$$g_{11} = g_{22} = 0, \qquad g_{12} = -g_{21} = 1 \tag{56.15}$$

is the *metric spinor* in a two-dimensional vector space[15]. Similarly, for example,

$$\psi_\lambda{}^\mu = g_{\lambda\nu}\psi^{\nu\mu}, \qquad \psi_{\lambda\mu} = g_{\lambda\nu}g_{\mu\rho}\psi^{\nu\rho},$$

so that $\psi_1{}^2 = -\psi^{11}$, $\psi_2{}^1 = -\psi^{21}$, $\psi_{11} = \psi^{22}$, and so forth.

The components $g_{\lambda\mu}$ themselves form the antisymmetric unit spinor of rank two. Their values are unchanged under coordinate transformations, and one readily verifies

$$g_{\lambda\nu}g^{\mu\nu} = \delta^\mu_\lambda, \tag{56.16}$$

where $\delta^1_1 = \delta^2_2 = 1$ and $\delta^2_1 = \delta^1_2 = 0$.

As in ordinary tensor algebra, spinor algebra has two basic operations: multiplication and contraction over a pair of indices. Multiplication of two spinors gives one of higher rank; for example, second- and third-rank spinors $\psi_{\lambda\mu}$ and $\varphi^{\nu\rho\sigma}$ give the fifth-rank

[14] Notation of the form $\psi\hat{U}$, with ψ to the left, means multiplying the row of components (ψ_1, ψ_2) by the columns of $\hat{U}$.

[15] The matrix (56.15) coincides with $i\hat{\sigma}_y$.

spinor $\psi_{\lambda\mu}\varphi^{\nu\rho\sigma}$. Contraction over a pair of indices, meaning summation over equal values of one covariant and one contravariant index, lowers the rank by two. Thus contraction of $\psi_{\lambda\mu}{}^{\nu\rho\sigma}$ over μ and ν gives the third-rank spinor $\psi_{\lambda\mu}{}^{\mu\rho\sigma}$; contraction of $\varphi_\lambda{}^\mu$ gives the scalar $\varphi_\lambda{}^\lambda$. A rule analogous to (56.10) holds: interchanging the upper and lower positions of the contracted indices changes the sign, so $\varphi_\lambda{}^\lambda = -\varphi^\lambda{}_\lambda$. In particular, contracting two indices over which a spinor is symmetric gives zero. Thus for a symmetric second-rank spinor $\varphi_\lambda{}^\mu$, one has $\varphi_\lambda{}^\lambda = 0$.

A *symmetric spinor* of rank n is symmetric in all its indices. A nonsymmetric spinor can be symmetrized by summing the components obtained from all possible permutations of its indices. From the preceding result, no spinor of lower rank can be formed by contracting components of a symmetric spinor.

A spinor antisymmetric in all its indices can only have rank two. Indeed, each index has only two possible values; with three or more indices at least two must have equal values, and every component then vanishes identically. Every antisymmetric second-rank spinor is a scalar multiple of the unit spinor $g_{\lambda\mu}$. One consequence is

$$g_{\lambda\mu}\psi_\nu + g_{\mu\nu}\psi_\lambda + g_{\nu\lambda}\psi_\mu = 0, \tag{56.17}$$

where ψ_λ is arbitrary. The left-hand side is simply, as can be checked directly, an antisymmetric third-rank spinor and must therefore vanish.

The spinor formed by multiplying $\psi_{\lambda\mu}$ by itself and contracting one pair of indices is antisymmetric in the other pair, since

$$\psi_{\lambda\nu}\psi_\mu{}^\nu = -\psi_\lambda{}^\nu\psi_{\mu\nu}.$$

It must consequently reduce to $g_{\lambda\mu}$ times a scalar. Determining that scalar by requiring contraction over the second pair to give the proper result, we obtain

$$\psi_{\lambda\nu}\psi_\mu{}^\nu = -\frac{1}{2}\psi_{\rho\sigma}\psi^{\rho\sigma}g_{\lambda\mu}. \tag{56.18}$$

The components of the spinor $\varphi^{\lambda\mu\cdots}$ complex conjugate to $\psi_{\lambda\mu\dots}$ transform as those of a contravariant spinor, and conversely. The sum of the squared moduli of the components of any spinor is therefore invariant.

§ 57. Wave Functions of Particles with Arbitrary Spin

Having developed the formal algebra of spinors of arbitrary rank, we can turn to our immediate task: studying the properties of wave functions for particles of arbitrary spin.

A convenient approach is to consider a collection of n particles of spin $1/2$. The greatest possible value of the z component of the system's total spin is $n/2$; it is attained when every particle has $s_z = 1/2$, so that all spins point in the same direction, along the z axis. In this case the total spin S of the system must itself equal $n/2$.

All components of the system's wave function $\psi(\sigma_1, \sigma_2, \dots, \sigma_n)$ then vanish except the single component $\psi(1/2, 1/2, \dots, 1/2)$. Write the wave function as a product of n spinors $\psi^\lambda\varphi^\mu\dots$, one for each particle. In each spinor only the component with $\lambda, \mu, \dots = 1$ is nonzero, so only the product $\psi^1\varphi^1\dots$ survives. The collection of all these products forms an nth-rank spinor symmetric in all its indices. After a transformation of the coordinate system, so that the spins no longer point along z, this becomes a general nth-rank spinor, but it remains symmetric.

The spin properties of wave functions are, in essence, their behavior under rotations of the coordinate system. They are identical for a particle of spin s and for a system of $n = 2s$ spin-$1/2$ particles oriented so that the system's total spin is s. It follows that the wave function of a particle of spin s is a symmetric spinor of rank $n = 2s$.

As required, a symmetric spinor of rank $2s$ has $2s + 1$ independent components. Indeed, the distinct components are only those whose indices contain $2s$ ones and no twos, $2s - 1$ ones and one two, and so on, down to no ones and $2s$ twos.

Mathematically, symmetric spinors classify the possible transformation laws of quantities under rotations of the coordinate system. If $2s + 1$ distinct quantities transform linearly among themselves, and no choice of linear combinations can reduce their number, their transformation law is equivalent to that of the components of a symmetric rank-$2s$ spinor. Any collection of any number of functions which transform linearly among themselves under coordinate rotations can, by a suitable linear transformation, be reduced to one or more symmetric spinors.[16]

Thus an arbitrary nth-rank spinor $\psi^{\lambda\mu\nu\cdots}$ can be reduced to symmetric spinors of ranks $n, n-2, n-4, \ldots$. The reduction can be carried out as follows. Symmetrizing $\psi^{\lambda\mu\nu\cdots}$ over all its indices gives a symmetric spinor of the same rank n. Next, contracting

the original spinor $\psi^{\lambda\mu\nu\cdots}$ over different pairs of indices gives spinors of rank $n-2$, such as $\psi^{\lambda}{}_{\lambda}{}^{\nu\cdots}$; symmetrizing these in turn produces symmetric spinors of rank $n-2$. Symmetrizing the spinors obtained after contracting $\psi^{\lambda\mu\cdots}{}_{\lambda\mu}$ over two pairs of indices gives symmetric spinors of rank $n-4$, and so forth.

It remains to connect the components of a symmetric rank-$2s$ spinor with the $2s + 1$ functions $\psi(\sigma)$, where $\sigma = s, s-1, \ldots, -s$. The component

$$\psi^{\overbrace{11\ldots1}^{s+\sigma}\overbrace{22\ldots2}^{s-\sigma}},$$

in which 1 occurs $s + \sigma$ times and 2 occurs $s - \sigma$ times, corresponds to spin projection σ on the z axis. To see this, replace the one spin-s particle once more by a system of $n = 2s$ spin-$1/2$ particles. The component displayed above corresponds to the product

$$\underbrace{\psi^1\varphi^1\ldots}_{s+\sigma}\underbrace{\chi^2\rho^2\ldots}_{s-\sigma}.$$

This product describes a state in which $s + \sigma$ particles have spin projection $+1/2$, and $s - \sigma$ have projection $-1/2$; the total projection is consequently $\frac{1}{2}(s+\sigma) - \frac{1}{2}(s-\sigma) = \sigma$. Finally, choose the factor of proportionality between this spinor component and $\psi(\sigma)$ so that

$$\sum_{\sigma=-s}^{+s} |\psi(\sigma)|^2 = \sum_{\lambda,\mu,\ldots=1}^{2} |\psi^{\lambda\mu\cdots}|^2 \tag{57.1}$$

[16] In other words, symmetric spinors realize the irreducible representations of the rotation group (see § 98).

holds. This sum is a scalar, as it must be, since it determines the probability of finding the particle at the specified point in space. In the sum on the right, components having $s+\sigma$ indices equal to 1 occur

$$\frac{(2s)!}{(s+\sigma)!(s-\sigma)!}$$

times. The correspondence between the functions $\psi(\sigma)$ and the spinor components is therefore

$$\psi(\sigma) = \left[\frac{(2s)!}{(s+\sigma)!(s-\sigma)!}\right]^{1/2} \psi^{\overbrace{11\ldots1}^{s+\sigma}\overbrace{22\ldots2}^{s-\sigma}}. \tag{57.2}$$

Relation (57.2) ensures not only (57.1), but, as is readily verified, also the more general relation

$$\psi^{\lambda\mu\cdots}\varphi_{\lambda\mu\ldots} = \sum_{\sigma}(-1)^{s-\sigma}\psi(\sigma)\varphi(-\sigma), \tag{57.3}$$

where $\psi^{\lambda\mu\cdots}$ and $\varphi_{\lambda\mu\ldots}$ are two different spinors of equal rank, and $\psi(\sigma)$, $\varphi(\sigma)$ are the functions associated with them by (57.2). The factor $(-1)^{s-\sigma}$ occurs because raising all the indices of a spinor component changes its sign once for every index equal to 2.

Equations (55.5) define the action of the spin operator on the wave functions $\psi(\sigma)$. It is straightforward to determine how these operators act when the wave function is written as a rank-$2s$ spinor. For spin $1/2$, the functions $\psi(+1/2)$ and $\psi(-1/2)$ coincide with the spinor components ψ^1 and ψ^2. According to (55.6) and (55.7), the action of the spin operators on them is

$$\begin{aligned}
&(\hat{s}_x\psi)^1 = \tfrac{1}{2}\psi^2, \quad (\hat{s}_y\psi)^1 = -\tfrac{i}{2}\psi^2, \quad (\hat{s}_z\psi)^1 = \tfrac{1}{2}\psi^1,\\
&(\hat{s}_x\psi)^2 = \tfrac{1}{2}\psi^1, \quad (\hat{s}_y\psi)^2 = \tfrac{i}{2}\psi^1, \quad (\hat{s}_z\psi)^2 = -\tfrac{1}{2}\psi^2.
\end{aligned} \tag{57.4}$$

To pass to arbitrary spin, consider again a system of $2s$ spin-$1/2$ particles and write its wave function as a product of $2s$ spinors. The spin operator of the system is the sum of the spin operators of the particles; each acts only on its corresponding spinor, with its action given by (57.4). Returning then to arbitrary symmetric spinors, or equivalently to wave functions of a spin-s particle, gives

$$\begin{aligned}
(\hat{s}_x\psi)^{\overbrace{11\ldots}^{s+\sigma}\overbrace{22\ldots}^{s-\sigma}} &= \frac{s+\sigma}{2}\psi^{\overbrace{11\ldots}^{s+\sigma-1}\overbrace{22\ldots}^{s-\sigma+1}} + \frac{s-\sigma}{2}\psi^{\overbrace{11\ldots}^{s+\sigma+1}\overbrace{22\ldots}^{s-\sigma-1}},\\
(\hat{s}_y\psi)^{\overbrace{11\ldots}^{s+\sigma}\overbrace{22\ldots}^{s-\sigma}} &= -i\frac{s+\sigma}{2}\psi^{\overbrace{11\ldots}^{s+\sigma-1}\overbrace{22\ldots}^{s-\sigma+1}} + i\frac{s-\sigma}{2}\psi^{\overbrace{11\ldots}^{s+\sigma+1}\overbrace{22\ldots}^{s-\sigma-1}},\\
(\hat{s}_z\psi)^{\overbrace{11\ldots}^{s+\sigma}\overbrace{22\ldots}^{s-\sigma}} &= \sigma\psi^{\overbrace{11\ldots}^{s+\sigma}\overbrace{22\ldots}^{s-\sigma}}.
\end{aligned} \tag{57.5}$$

So far, spinors have been discussed as wave functions of the intrinsic angular momentum of elementary particles. Formally, however, there is no distinction between the spin of one particle and the total angular momentum of any system viewed as a whole, with its internal structure disregarded. The transformation properties of spinors therefore apply equally to the behavior under spatial rotations of the wave functions ψ_{jm} of any particle, or system of particles, having total angular momentum j, regardless of whether its origin is orbital or spin. There must accordingly be a definite correspondence between the transformation laws of the eigenfunctions ψ_{jm} under rotations of the coordinate system and those of the components of a symmetric spinor of rank $2j$.

In establishing this correspondence, however, two aspects of a wave function's dependence on the angular-momentum projection m, for fixed j, must be distinguished clearly. A wave function may be viewed as a probability amplitude for the various values of m, or as an eigenfunction belonging to a specified value of m.

These two aspects appeared already at the beginning of § 55, where the eigenfunction $\delta_{\sigma\sigma_0}$ of s_z, belonging to $s_z = \sigma_0$, was considered. Their mathematical distinction is especially transparent for a particle with $s = 1/2$. As a function of the variable σ, the spin function is a contravariant spinor of first rank; in spinor notation it must therefore be written $\delta^{\lambda}{}_{\sigma_0}$. As a function of σ_0, it is consequently a covariant spinor.

This fact is plainly general: the eigenfunctions ψ_{jm} can be associated with the components of a covariant symmetric rank-$2j$ spinor by formulas analogous to (57.2).[17]

$$\psi_{jm} = \left[\frac{(2j)!}{(j+m)!(j-m)!}\right]^{1/2} \psi_{\underbrace{11\ldots}_{j+m}\underbrace{22\ldots}_{j-m}}. \tag{57.6}$$

For integral angular momentum j, the eigenfunctions are the spherical functions Y_{jm}. The case $j = 1$ is especially important.

The three spherical functions Y_{1m} are

$$Y_{10} = i\sqrt{\frac{3}{4\pi}}\cos\theta = i\sqrt{\frac{3}{4\pi}}n_z, \qquad Y_{1,\pm1} = \mp i\sqrt{\frac{3}{8\pi}}\sin\theta e^{\pm i\varphi} = \mp i\sqrt{\frac{3}{8\pi}}(n_x \pm i n_y),$$

where $\mathbf{n}$ is a unit vector in the direction of the radius vector. It is evident that, by their transformation properties, these three functions are equivalent to the components of a vector $\mathbf{a}$. We write the correspondence as

$$\psi_{10} = i a_z, \qquad \psi_{11} = -\frac{i}{\sqrt{2}}(a_x + i a_y), \qquad \psi_{1,-1} = \frac{i}{\sqrt{2}}(a_x - i a_y). \tag{57.7}$$

Comparison with (57.6) shows that the components of a symmetric second-rank spinor may be associated with those of a vector by

$$\psi_{11} = -\frac{i}{\sqrt{2}}(a_x + i a_y), \qquad \psi_{12} = \psi_{21} = \frac{i}{\sqrt{2}}a_z, \qquad \psi_{22} = \frac{i}{\sqrt{2}}(a_x - i a_y), \tag{57.8}$$

[17]The same result may be approached in another way. Expand the wave function ψ of a particle in a state of angular momentum j in the eigenfunctions ψ_{jm}: $\psi = \sum_m a_m \psi_{jm}$. The coefficients a_m are probability amplitudes for the different values of m. In this sense they correspond to the "components" $\psi(m)$ of the spin wave function, which fixes their transformation law. On the other hand, the value of ψ at a given point in space cannot depend on the choice of coordinate system; hence the sum $\sum a_m \psi_{jm}$ must be a scalar. Comparison with the scalar (57.3) shows that a_m must transform as $(-1)^{j-m}\psi_{j,-m}$.

or

$$\psi^{11} = \frac{i}{\sqrt{2}}(a_x - ia_y), \qquad \psi^{12} = \psi^{21} = -\frac{i}{\sqrt{2}}a_z, \qquad \psi^{22} = -\frac{i}{\sqrt{2}}(a_x + ia_y). \quad (57.9)$$

Conversely,

$$a_z = i\sqrt{2}\psi^{12}, \qquad a_x = \frac{i}{\sqrt{2}}(\psi^{22} - \psi^{11}), \qquad a_y = \frac{1}{\sqrt{2}}(\psi^{11} + \psi^{22}). \qquad (57.10)$$

With this definition one readily verifies that

$$\psi^{\lambda\mu}\varphi_{\lambda\mu} = \mathbf{ab}, \qquad (57.11)$$

where **a** and **b** are the vectors corresponding to the symmetric spinors $\psi^{\lambda\mu}$ and $\varphi_{\lambda\mu}$. The following spinor and vector are likewise found to correspond:[18]

$$\psi^{\lambda}{}_{\rho}\varphi^{\rho\mu} - \varphi^{\lambda}{}_{\rho}\psi^{\rho\mu} \longleftrightarrow i\sqrt{2}(\mathbf{a} \times \mathbf{b}). \qquad (57.12)$$

Equations (57.10) can be written compactly with the Pauli matrices:

$$a_i = \frac{i}{\sqrt{2}}(\hat{\sigma}_i)^{\lambda}{}_{\mu}\psi^{\mu}{}_{\lambda}. \qquad (57.13)$$

The matrix indices on σ are placed above and below to match the positions of the spinor indices on $\psi^{\mu}{}_{\lambda}$. The origin of this formula is easily understood from the special case in which the second-rank spinor $\psi^{\mu}{}_{\lambda}$ reduces to the product of a first-rank spinor ψ^{μ} and its complex conjugate ψ^{*}_{λ}. Then

$$\frac{1}{2}\psi^{*}_{\lambda}\hat{\boldsymbol{\sigma}}^{\lambda}{}_{\mu}\psi^{\mu}$$

is the mean spin (for a particle whose wave function is ψ^{μ}), so its vector character is evident.

The correspondence (57.8) or (57.9) is a special case of a general rule: any symmetric spinor of even rank $2j$, where j is an integer, can be associated with a symmetric tensor of half that rank, j, which vanishes upon contraction over any pair of indices. We shall call such a tensor irreducible. This follows already because spinor and tensor have the same number of independent components, namely $2j + 1$, as a direct count readily shows.[19] The correspondence between the spinor and tensor components can be found from (57.8)–(57.10) by treating the spinor of the given rank as a product of several second-rank spinors, and the tensor as a product of vectors.

[18] The mixed components of a symmetric spinor may be written $\psi^{\lambda}{}_{\mu}$, without distinguishing between $\psi^{\lambda\mu}$ and $\psi^{\mu\lambda}$.

[19] In other words, the $2j+1$ components of an irreducible rank-j tensor (j integral), the set of $2j+1$ spherical functions Y_{jm}, and the $2j+1$ components of a symmetric rank-$2j$ spinor all realize the same irreducible representation of the rotation group.

PROBLEM

1. Rewrite definition (57.4) of the spin-1/2 operator using the spinor components of the vector **s**.

SOLUTION.

Using (57.9), which relates the vector **s** to the spinor $s^{\lambda\mu}$, definition (57.4) becomes

$$s^{\lambda\mu}\psi^{\nu} = \frac{i}{2\sqrt{2}}(\psi^{\lambda}g^{\mu\nu} + \psi^{\mu}g^{\lambda\nu}).$$

2. Write formulas defining the action of the spin operator on the vector wave function of a spin-1 particle.

SOLUTION.

Equations (57.9) relate the components of the vector function $\boldsymbol{\psi}$ to those of the spinor $\psi^{\lambda\mu}$, while the last of equations (57.5) gives

$$\hat{s}_z\psi_+ = -\psi_+, \qquad \hat{s}_z\psi_- = \psi_-, \qquad \hat{s}_z\psi_z = 0,$$

where $\psi_\pm = \psi_x \pm i\psi_y$, or

$$\hat{s}_z\psi_x = -i\psi_y, \qquad \hat{s}_z\psi_y = i\psi_x, \qquad \hat{s}_z\psi_z = 0.$$

The remaining formulas follow by cyclic permutation of x, y, z. Together they can be written

$$\hat{s}_i\psi_k = -ie_{ik\ell}\psi_\ell.$$

The complex vector $\boldsymbol{\psi}$ can be represented as $\boldsymbol{\psi} = e^{i\alpha}(\mathbf{u} + i\mathbf{v})$, where **u** and **v** are real vectors. By a suitable choice of the common phase α, they can be made mutually perpendicular. The two vectors **u** and **v** determine a plane with the following property: the spin projection on the direction perpendicular to it can take only the values ± 1.

§ 58. The finite-rotation operator

We return to spinor transformations and show how their coefficients can be expressed in terms of the angles through which the coordinate axes are rotated.

By the definition of the angular-momentum operator (here, spin), $1 + i\delta\varphi\,\mathbf{n}\hat{\mathbf{s}}$ is the operator for a rotation through $\delta\varphi$ about the direction of the unit vector **n**. For a spin-1/2 wave function, a spinor of rank 1, one sets $\hat{\mathbf{s}} = \hat{\boldsymbol{\sigma}}/2$. A finite rotation through φ about the same direction is therefore represented by

$$\hat{U}_{\mathbf{n}} = \exp(i\varphi\,\mathbf{n}\hat{\boldsymbol{\sigma}}/2). \tag{58.1}$$

(Compare (15.13).) Like every function of the Pauli matrices (see Problem 1, § 55), this reduces to an expression linear in them:

$$\hat{U}_{\mathbf{n}} = \cos(\varphi/2) + i\mathbf{n}\hat{\boldsymbol{\sigma}}\sin(\varphi/2). \tag{58.2}$$

For a rotation about z,

$$\hat{U}_z(\varphi) = \cos\frac{\varphi}{2} + i\hat{\sigma}_z\sin\frac{\varphi}{2} = \begin{pmatrix} e^{i\varphi/2} & 0 \\ 0 & e^{-i\varphi/2} \end{pmatrix}. \tag{58.3}$$

Thus the spinor components transform as

$$\psi^{1'} = \psi^1 e^{i\varphi/2}, \qquad \psi^{2'} = \psi^2 e^{-i\varphi/2}.$$

In particular, a rotation through 2π changes the sign of the spinor components; every spinor of odd rank consequently has the same property (compare the end of § 55).

Similarly, rotations through φ about x or y have the matrices

$$\hat{U}_x(\varphi) = \begin{pmatrix} \cos(\varphi/2) & i\sin(\varphi/2) \\ i\sin(\varphi/2) & \cos(\varphi/2) \end{pmatrix}, \qquad \hat{U}_y(\varphi) = \begin{pmatrix} \cos(\varphi/2) & \sin(\varphi/2) \\ -\sin(\varphi/2) & \cos(\varphi/2) \end{pmatrix}. \tag{58.4}$$

For the special case of a rotation through π about y,

$$\psi^{1'} = \psi^2, \qquad \psi^{2'} = -\psi^1, \qquad \text{i.e.} \qquad \psi^{\lambda'} = \psi_\lambda. \tag{58.5}$$

We can now write the transformation matrix for an arbitrary rotation in terms of its Euler angles.

A rotation of the axes specified by Euler angles α, β, γ is performed in three stages: (1) rotation through α ($0 \leqslant \alpha \leqslant 2\pi$) about z; (2) rotation through β ($0 \leqslant \beta \leqslant \pi$) about the new position of y (ON in Fig. 20, the *line of nodes*); and (3) rotation through γ ($0 \leqslant \gamma \leqslant 2\pi$) about the final position z' of the z axis[20].

The angles α, β evidently coincide with the spherical angles φ, θ of the new z' axis relative to xyz: $\alpha = \varphi$, $\beta = \theta$.

For this sequence of rotations, the complete transformation matrix is

$$\hat{U}(\alpha, \beta, \gamma) = \hat{U}_z(\gamma)\hat{U}_y(\beta)\hat{U}_z(\alpha).$$

Direct multiplication gives

$$\hat{U}(\alpha, \beta, \gamma) = \begin{pmatrix} \cos(\beta/2)e^{i(\alpha+\gamma)/2} & \sin(\beta/2)e^{-i(\alpha-\gamma)/2} \\ -\sin(\beta/2)e^{i(\alpha-\gamma)/2} & \cos(\beta/2)e^{-i(\alpha+\gamma)/2} \end{pmatrix}. \tag{58.6}$$

By definition, higher-rank spinors transform as products of rank-1 spinor components. In physical applications, however, the transformation laws of the corresponding wave functions ψ_{jm} are of greater interest than those of the spinors themselves.

Let ψ_{jm}, $m = j, j-1, \dots, -j$, describe in the system xyz a state with specified angular momentum j, and let $\psi_{jm'}$ describe the same state relative to $x'y'z'$. Thus m is the value of J_z and m' that of $J_{z'}$. The two sets are related linearly; write

$$\psi_{jm} = \sum_{m'} D^{(j)}_{m'm} \psi_{jm'}. \tag{58.7}$$

The coefficients $D^{(j)}_{m'm}$ form, in the indices m', m, a matrix of order $2j+1$: the *finite-rotation matrix* $\hat{D}^{(j)}$. Its elements depend on the rotation angles α, β, γ of $x'y'z'$ relative to xyz.

[20]The systems xyz and $x'y'z'$ are, as always, right-handed; positive angles follow the direction of a right-handed screw advancing along the positive rotation axis.

This definition of the Euler angles, customary in quantum-mechanical applications, differs from that in § 35 (Volume I) because the second rotation is about y, not x. The angles α, β, γ are related to the angles φ, θ, ψ of Volume I (not to be confused with the spherical angles φ, θ) by $\psi = \gamma - \pi/2$.

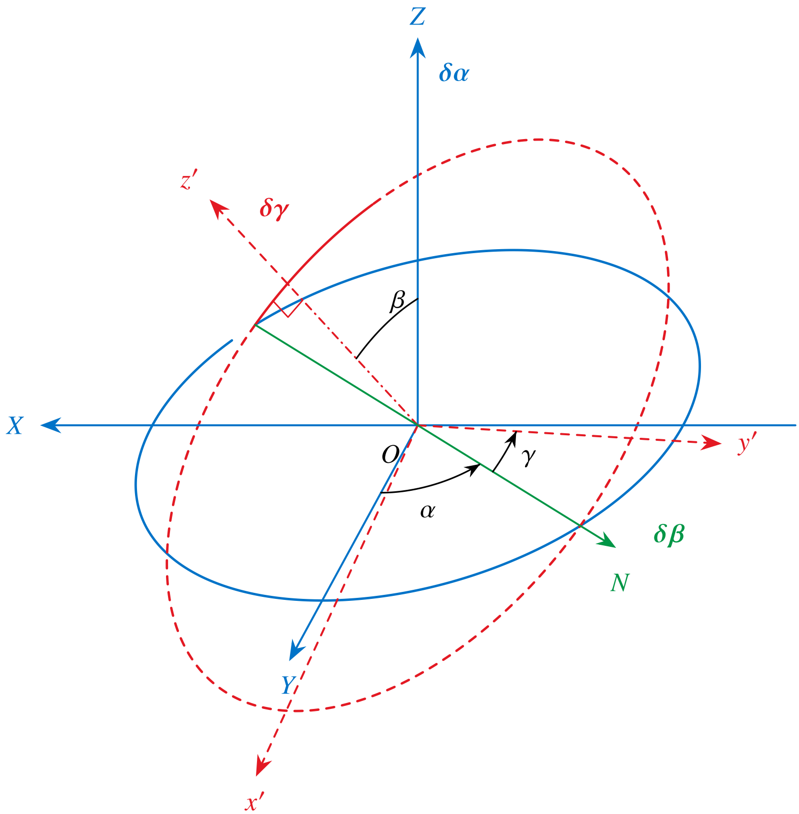


Fig. 20. Definition of the Euler angles.

The finite-rotation matrix can be constructed using the spinor representation of ψ_{jm}.

For $j = 1/2$, the two functions $\psi_{1/2,m}$ form a covariant spinor of rank 1. By (56.13), its transformation from $x'y'z'$ to xyz is effected by $\hat{U}$ in (58.6), so $\hat{D}^{(1/2)} = \hat{U}$[21]. Write its elements as

$$D^{(1/2)}_{m'm}(\alpha, \beta, \gamma) = e^{im'\gamma} d^{(1/2)}_{m'm}(\beta) e^{im\alpha},$$

where

$$\begin{array}{c|cc} d^{(1/2)}_{m'm}(\beta) & \frac{1}{2} & -\frac{1}{2} \\ \hline \frac{1}{2} & \cos(\beta/2) & \sin(\beta/2) \\ -\frac{1}{2} & -\sin(\beta/2) & \cos(\beta/2) \end{array} . \tag{58.8}$$

For arbitrary j, (57.6) relates ψ_{jm} to a symmetric covariant spinor of rank $2j$. Its transformation matrix is a product of $2j$ matrices $\hat{D}^{(1/2)}$, one acting on each spinor index. Multiplying and returning to ψ_{jm} gives

$$D^{(j)}_{m'm}(\alpha, \beta, \gamma) = e^{im'\gamma} d^{(j)}_{m'm}(\beta) e^{im\alpha}, \tag{58.9}$$

[21]The matrix indices in (58.7) are ordered precisely as required for multiplying the columns of $\hat{D}^{(j)}$ by the functions $\psi_{jm'}$ arranged in a row. Symbolically, (58.7) would read $\psi_{jm} = (\boldsymbol{\psi}_j \hat{D}^{(j)})_m$, consistently with (56.13).

where[22]

$$d^{(j)}_{m'm}(\beta) = \left[\frac{(j+m')!(j-m')!}{(j+m)!(j-m)!}\right]^{1/2} \left(\cos\frac{\beta}{2}\right)^{m'+m} \times \left(\sin\frac{\beta}{2}\right)^{m'-m} P^{(m'-m,m'+m)}_{j-m'}(\cos\beta), \tag{58.10}$$

and

$$P^{(a,b)}_n(\cos\beta) = \frac{(-1)^n}{2^n n!}(1-\cos\beta)^{-a}(1+\cos\beta)^{-b} \times \left(\frac{d}{d\cos\beta}\right)^n [(1-\cos\beta)^{a+n}(1+\cos\beta)^{b+n}] \tag{58.11}$$

are the Jacobi polynomials[23]. Notice that

$$d^{(j)}_{m'm}(\beta) = 0 \quad \text{if } |m| > j \text{ or } |m'| > j. \tag{58.12}$$

The functions $d^{(j)}_{m'm}$ have several symmetry properties. Although these can be read from (58.11), (58.12), it is simpler to derive them directly from their definition as rotation coefficients.

The rotation matrix $\hat{D}^{(j)}$ is unitary. The inverse of the rotation (α, β, γ) is $(-\gamma, -\beta, -\alpha)$; since $\hat{d}$ is real, this gives

$$d^{(j)}_{m'm}(-\beta) = d^{(j)}_{mm'}(\beta). \tag{58.13}$$

We also have

$$d^{(j)}_{m'm}(\beta) = d^{(j)}_{-m,-m'}(\beta), \tag{58.14}$$

and

$$d^{(j)}_{m'm}(\pi) = (-1)^{j+m}\delta_{m',-m}, \qquad d^{(j)}_{m'm}(-\pi) = (-1)^{j-m}\delta_{m',-m}, \qquad d^{(j)}_{m'm}(0) = \delta_{m'm}. \tag{58.15}$$

For $j = 1/2$ these follow directly from (58.8), and for arbitrary j from the construction of the transformation matrix given above.

Perform a rotation through $\pi - \beta$ as successive rotations through π and $-\beta$:

$$d^{(j)}_{m'm}(\pi - \beta) = \sum_{m''} d^{(j)}_{m'm''}(\pi) d^{(j)}_{m''m}(-\beta) = (-1)^{j-m'} d^{(j)}_{-m',m}(-\beta),$$

or, using (58.13),

$$d^{(j)}_{m'm}(\pi - \beta) = (-1)^{j-m'} d^{(j)}_{m,-m'}(\beta). \tag{58.16}$$

Two rotations about the same axis give the same result in either order. Reversing the order of the rotations $-\beta$ and π and comparing with (58.16), we obtain

$$d^{(j)}_{m'm}(\beta) = (-1)^{m'-m} d^{(j)}_{-m',-m}(\beta). \tag{58.17}$$

[22]The calculation is given in A. R. Edmonds, *Angular Momentum in Quantum Mechanics*, Princeton, 1957 (see also the Russian translation of his article in *Deformation of Atomic Nuclei*, Moscow, 1958). The definition of $D^{(j)}_{m'm}$ in (58.9), (58.10) differs from Edmonds's book by interchanging α and γ (the more natural convention here), and from the article also by reversing the signs of all angles.

[23]For their relation to the hypergeometric series, see § e, formula (e.11).

Equations (58.17), (58.14), and (58.13) imply

$$d^{(j)}_{m'm}(\beta) = (-1)^{m'-m} d^{(j)}_{mm'}(\beta) = (-1)^{m'-m} d^{(j)}_{m'm}(-\beta). \tag{58.18}$$

These relations give corresponding symmetries of the complete functions $D^{(j)}_{m'm}$. In particular,

$$D^{(j)*}_{m'm}(\alpha,\beta,\gamma) = D^{(j)}_{m'm}(-\alpha,\beta,-\gamma) = (-1)^{m'-m} D^{(j)}_{-m',-m}(\alpha,\beta,\gamma). \tag{58.19}$$

Mathematically, the matrices $\hat{D}^{(j)}$ furnish unitary irreducible representations of the rotation group of dimension $2j+1$ (see § 98). Hence the orthogonality and normalization relation is

$$\int D^{(j_1)*}_{m_1' m_1}(\alpha,\beta,\gamma) D^{(j_2)}_{m_2' m_2}(\alpha,\beta,\gamma) \frac{d\omega}{8\pi^2} = \frac{1}{2j_1+1} \delta_{j_1 j_2} \delta_{m_1 m_2} \delta_{m_1' m_2'}, \tag{58.20}$$

where $d\omega = \sin\beta \, d\alpha \, d\beta \, d\gamma$.

Orthogonality in m and m' is supplied by the factor $\exp\{i(m\alpha + m'\gamma)\}$. Orthogonality in j belongs to the functions $d^{(j)}_{m'm}$:

$$\int_0^\pi d^{(j_1)}_{m'm}(\beta) d^{(j_2)}_{m'm}(\beta) \sin\beta \, d\beta = \frac{2}{2j_1+1} \delta_{j_1 j_2}. \tag{58.21}$$

For reference, we finally list the functions $d^{(j)}_{m'm}$ for several special values of their parameters. For $j = 1$,

$d^{(1)}_{m'm}(\beta)$	1	0	-1
1	$\frac{1}{2}(1+\cos\beta)$	$\frac{1}{\sqrt{2}}\sin\beta$	$\frac{1}{2}(1-\cos\beta)$
0	$-\frac{1}{\sqrt{2}}\sin\beta$	$\cos\beta$	$\frac{1}{\sqrt{2}}\sin\beta$
-1	$\frac{1}{2}(1-\cos\beta)$	$-\frac{1}{\sqrt{2}}\sin\beta$	$\frac{1}{2}(1+\cos\beta)$

(58.22)

For integral $j = l$ and $m' = 0$, (58.10), (58.11) give

$$d^{(l)}_{0m}(\beta) = \left[\frac{(l-m)!}{(l+m)!}\right]^{1/2} P_l^m(\cos\beta). \tag{58.23}$$

The origin of this formula is immediate from (58.7). Refer the functions $\psi_{jm'}$ on its right-hand side to the z' axis. For $j = l$,

$$\psi_{lm'}(\mathbf{n}_{z'}) = i^l \sqrt{\frac{2l+1}{4\pi}} \delta_{m'0}. \tag{58.24}$$

The function ψ_{jm} on the left is then the spherical harmonic $Y_{lm}(\beta,\alpha)$ of the spherical angles $\varphi = \alpha$, $\theta = \beta$ of z'. Substitution of (58.24) in (58.7) gives

$$Y_{lm}(\beta,\alpha) = i^l \sqrt{\frac{2l+1}{4\pi}} D^{(l)}_{0m}(\alpha,\beta,\gamma), \tag{58.25}$$

which is equivalent to (58.23).

Finally, when one index m or m' has its largest possible value,

$$d^{(j)}_{jm}(\beta) = (-1)^{j-m} d^{(j)}_{-j,m}(\beta) = \left[\frac{(2j)!}{(j+m)!(j-m)!}\right]^{1/2} \left(\cos\frac{\beta}{2}\right)^{j+m} \left(\sin\frac{\beta}{2}\right)^{j-m}. \tag{58.26}$$

§ 59. Partial polarization of particles

For any given spinor ψ^λ, the wave function of a spin-$1/2$ particle, one component (say ψ^2) can always be made zero by a suitable choice of the z direction. This is already clear because a spatial direction is specified by two quantities (angles), exactly the number of quantities— the real and imaginary parts of the complex ψ^2—that are to vanish.

Physically, if a spin-$1/2$ particle (for definiteness, an electron) is in a state described by a spin wave function, there is a direction along which its spin projection has the definite value $\sigma = 1/2$. In this state the electron is said to be *fully polarized*.

There are also electron states which may be called *partially polarized*. They are described not by wave functions but only by density matrices; they are mixed states with respect to spin (see § 14).

The spin, or *polarization*, density matrix of an electron is the rank-2 spinor $\rho^\lambda{}_\mu$, normalized by

$$\rho^\lambda{}_\lambda = \rho^1{}_1 + \rho^2{}_2 = 1 \tag{59.1}$$

and satisfying the Hermiticity condition

$$(\rho^\lambda{}_\mu)^* = \rho^\mu{}_\lambda. \tag{59.2}$$

For a pure (fully polarized) electron spin state, $\rho^\lambda{}_\mu$ reduces to a product of wave-function components:

$$\rho^\lambda{}_\mu = \psi^\lambda \psi^*_\mu. \tag{59.3}$$

The diagonal components give the probabilities of spin projections $+1/2$ and $-1/2$ on z. Hence

$$\bar{s}_z = \frac{1}{2}(\rho^1{}_1 - \rho^2{}_2),$$

and, using (59.1),

$$\rho^1{}_1 = \frac{1}{2} + \bar{s}_z, \qquad \rho^2{}_2 = \frac{1}{2} - \bar{s}_z. \tag{59.4}$$

In a pure state, the mean values of $s_\pm = s_x \pm is_y$ are

$$\bar{s}_\pm = \psi^*_\lambda \hat{s}_\pm \psi^\lambda.$$

By (55.6) and (55.7),

$$\hat{s}_+ = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad \hat{s}_- = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix},$$

and therefore

$$\bar{s}_+ = \psi^*_1 \psi^2, \qquad \bar{s}_- = \psi^*_2 \psi^1.$$

Correspondingly, in a mixed state,

$$\rho^1{}_2 = \bar{s}_-, \qquad \rho^2{}_1 = \bar{s}_+. \tag{59.5}$$

Equations (59.4) and (59.5) combine, using the Pauli matrices, into

$$\rho^\lambda{}_\mu = \frac{1}{2}(\delta^\lambda{}_\mu + 2\hat{\boldsymbol{\sigma}}^\lambda{}_\mu \bar{\mathbf{s}}). \tag{59.6}$$

Thus every component of the electron polarization density matrix is expressed through the mean components of its spin vector. The real vector $\bar{\mathbf{s}}$ completely specifies the polarization of a spin-1/2 particle. At full polarization, with suitable axes, one component is 1/2 and the other two vanish. In an unpolarized state all three vanish. For an arbitrary partially polarized state in arbitrary coordinates, $0 \leqslant p \leqslant 1$, where

$$p = 2(\bar{s}_x^2 + \bar{s}_y^2 + \bar{s}_z^2)^{1/2}$$

is the electron's *degree of polarization*.

For a particle of arbitrary spin s, the density matrix is the rank-$4s$ spinor $\rho^{\lambda\mu\ldots}{}_{\rho\sigma\ldots}$, symmetric separately in the first $2s$ and last $2s$ indices, and satisfying

$$\rho^{\lambda\mu\ldots}{}_{\lambda\mu\ldots} = 1, \tag{59.7}$$

$$(\rho^{\lambda\mu\ldots}{}_{\rho\sigma\ldots})^* = \rho^{\rho\sigma\ldots}{}_{\lambda\mu\ldots}. \tag{59.8}$$

To count its independent components, note that only $2s + 1$ essentially different sets of values occur among either group of indices. The one relation (59.7) then leaves $(2s+1)^2 - 1 = 4s(s+1)$ different components. Although these components are complex, (59.8) leaves the total number of independent quantities describing partial polarization equal to $4s(s + 1)$[24]. By comparison, full polarization requires only $4s$ quantities: the $2s + 1$ complex components of $\psi^{\lambda\mu\cdots}$, subject to one normalization condition and one overall phase irrelevant to the state.

Like every rank-$4s$ spinor, $\rho^{\lambda\mu\cdots}{}_{\rho\sigma\ldots}$ is equivalent to a set of irreducible tensors of ranks $4s, 4s-2, \ldots, 0$. Here there is just one tensor of each rank, because the symmetry permits each contraction in only one way: between any one index from the first group and any one from the second. The scalar (rank-0 tensor) is absent as an independent quantity, since (59.7) reduces it to unity.

§ 60. Time reversal and Kramers' theorem

In quantum mechanics, symmetry of motion under reversal of the sign of time means that if ψ is the wave function of a stationary state, the time-reversed wave function, denoted by ψ^{rev}, also describes a possible state of the same energy. At the end of § 18 it was stated that ψ^{rev} is the complex conjugate ψ^*. In this simple form the statement applies when particle spin is disregarded; spin requires a qualification.

Represent the wave function of a spin-s particle as a contravariant spinor $\psi^{\lambda\mu\cdots}$ of rank $2s$. Its complex conjugates $\psi^{\lambda\mu\ldots *}$ transform as the components of a covariant spinor. Time reversal therefore takes $\psi^{\lambda\mu\cdots}$ into a new wave function whose covariant components are

$$\psi^{\text{rev}}_{\lambda\mu\ldots} = \psi^*_{\lambda\mu\ldots}. \tag{60.1}$$

For a fixed set of index values, covariant and contravariant components correspond to opposite angular-momentum projections. Thus, in terms of $\psi_{s\sigma}$, time reversal

[24] Specifying these quantities is equivalent to specifying the mean values of the components of $\mathbf{s}$ and all their powers and products of degrees $2, 3, \ldots, 2s$ which cannot be reduced to lower powers (see Problem 3 of § 55).

interchanges $\psi_{s\sigma}$ and $\psi_{s,-\sigma}$, as expected because reversing time reverses angular momentum. Equation (60.1) fixes the precise relation:

$$\psi^{\text{rev}}_{s,-\sigma} = \psi^*_{s\sigma}(-1)^{s-\sigma}. \tag{60.2}$$

In other words, the replacement $\psi_{s\sigma} \longrightarrow \psi^*_{s\sigma}$ required by time reversal amounts to[25]

$$\psi_{s\sigma} \longrightarrow \psi_{s,-\sigma}(-1)^{s-\sigma}. \tag{60.3}$$

Applying the operation twice gives

$$\psi_{s\sigma} \longrightarrow \psi_{s,-\sigma}(-1)^{s-\sigma} \longrightarrow \psi_{s\sigma}(-1)^{s-\sigma}(-1)^{s+\sigma} = \psi_{s\sigma}(-1)^{2s}.$$

Thus two time reversals restore the wave function for integral spin, but change its sign for half-integral spin.

Consider an arbitrary system of interacting particles. When relativistic interactions are included, its orbital and spin angular momenta are generally not conserved separately; only the total J is conserved. With no external field, every energy level is $(2j + 1)$-fold degenerate. An external field generally removes this degeneracy. Can it remove it completely, leaving only simple levels? The answer is closely tied to time-reversal symmetry.

The equations of classical electrodynamics are invariant under reversal of time if the electric field is left unchanged and the magnetic field changes sign[26]. This fundamental property must persist in quantum mechanics. Time-reversal symmetry therefore holds not only for a closed system but also in any external electric field, provided no magnetic field is present.

The system's wave functions are spinors $\psi^{\lambda\mu\cdots}$ of rank $n = 2\sum s_a$, twice the sum of the spins of all particles; this sum need not equal the total spin S of the system. In an arbitrary electric field, a wave function and its time reverse must describe states of equal energy. A nondegenerate level requires these states to be identical, so their wave functions must agree up to a constant factor, both being expressed in the same covariant or contravariant form.

Write $\psi^{\text{rev}}_{\lambda\mu\ldots} = C\psi_{\lambda\mu\ldots}$ or, by (60.1),

$$\psi^*_{\lambda\mu\ldots} = C\psi_{\lambda\mu\ldots}, \tag{60.4}$$

where C is constant. Complex conjugation gives

$$\psi^{\lambda\mu\cdots} = C^*\psi^{\lambda\mu\ldots *}.$$

Lower the indices on the left and correspondingly raise them on the right. That is, multiply both sides by $g_{\alpha\lambda}g_{\beta\mu}\ldots$ and sum over $\lambda, \mu, \ldots$; on the right use

$$g_{\alpha\lambda}g_{\beta\mu}\ldots = (-1)^n g_{\lambda\alpha}g_{\mu\beta}\ldots.$$

[25] Notice the agreement between the complex-conjugation rule for a spherical function, (28.9), and the general rule (60.3).

[26] See II, § 17; also the remark at the end of § 111.

The result is

$$\psi_{\lambda\mu\ldots} = C^*(-1)^n\psi^*_{\lambda\mu\ldots}.$$

Substitution from (60.4) gives

$$\psi_{\lambda\mu\ldots} = (-1)^n CC^*\psi_{\lambda\mu\ldots}.$$

This identity requires $(-1)^n CC^* = 1$. Since $|C|^2$ is positive, this is possible only for even n, that is, integral $\sum s_a$. For odd n, or half-integral $\sum s_a$[27], condition (60.4) cannot hold.

An electric field can therefore remove degeneracy completely only for a system whose particle spins have an integral sum. For a half-integral sum, every level in an arbitrary electric field must be at least doubly degenerate; the two distinct states of equal energy correspond to complex-conjugate spinors[28] (H. A. Kramers, 1930).

One further mathematical remark is useful. A relation of the form (60.4) with real C is the condition that the spinor components can be associated with a set of real quantities; call it a "reality" condition for the spinor[29]. The impossibility of (60.4) for odd n says that no real quantity can correspond to an odd-rank spinor. For even n, by contrast, (60.4) can hold with real C. In particular, a symmetric rank-2 spinor corresponds to a real vector when (60.4) holds with $C = 1$:

$$\psi^{\lambda\mu *} = \psi_{\lambda\mu}$$

(as follows directly from (57.8), (57.9)). More generally, (60.4) with $C = 1$ is the reality condition for a symmetric spinor of any even rank.

[27]When $\sum s_a$ is integral (half-integral), all possible values of the total spin S are likewise integral (half-integral).

[28]If the electric field has high (cubic) symmetry, fourfold degeneracy may also occur; see § 99 and its problem.

[29]It is meaningless to call a spinor literally real, since complex-conjugate spinors have different transformation laws.

CHAPTER IX

Identity of Particles

§ 61. The indistinguishability principle for identical particles

In classical mechanics, identical particles (electrons, for example) retain their "individuality" despite having identical physical properties. One may imagine assigning numbers, at some instant, to the particles comprising a given physical system and thereafter following the motion of each along its own trajectory; the particles could then be identified at any later instant.

Quantum mechanics presents an entirely different situation. We have already noted repeatedly that the uncertainty principle deprives the electron trajectory of all meaning. If an electron's position is known exactly at the present instant, its coordinates have no definite values at all an instant later. Hence localizing and numbering the electrons at one instant does nothing toward identifying them at subsequent instants. If, at another instant, an electron is localized at some point in space, it is impossible to say which particular electron has arrived there. Thus quantum mechanics rules out in principle any means of following identical particles separately and thereby telling them apart. In this sense, identical particles lose their "individuality" completely in quantum mechanics. Their equality in physical properties acquires a far deeper significance here: it entails their complete indistinguishability.

This *indistinguishability principle* for identical particles, as it is called, is fundamental to the quantum theory of systems composed of such particles. Begin with a system containing only two particles. Since the particles are identical, two states related merely by interchanging them must be physically equivalent in every respect. Consequently, such an interchange can alter the system's wave function only by

an immaterial phase factor. Let $\psi(\xi_1, \xi_2)$ be the system's wave function, with ξ_1 and ξ_2 denoting, conventionally, the three coordinates together with the spin projection of each particle. We must then have

$$\psi(\xi_1, \xi_2) = e^{i\alpha}\psi(\xi_2, \xi_1),$$

where α is a real constant. Repeating the interchange restores the initial state, while multiplying ψ by $e^{2i\alpha}$. It follows that $e^{2i\alpha} = 1$, or $e^{i\alpha} = \pm 1$. Thus $\psi(\xi_1, \xi_2) = \pm\psi(\xi_2, \xi_1)$.

There are therefore only two alternatives: the wave function is symmetric (it is wholly unchanged by interchanging the particles), or it is antisymmetric (its sign is reversed by the interchange). Plainly, the wave functions of all states of one and the same system must possess the same symmetry. Otherwise, a state formed by superposing states of different symmetries would have a wave function that was neither symmetric nor antisymmetric.

The result extends immediately to systems containing any number of identical particles. Indeed, particle identity makes it clear that, if one pair is described, say, by symmetric wave functions, every other pair of the same kind must share that property.

The wave function of identical particles must accordingly either remain unchanged when any pair is interchanged (and hence under any permutation whatever), or reverse sign whenever a pair is interchanged. The former is called a *symmetric* wave function and the latter an *antisymmetric* wave function. Which of these two kinds of wave function describes a particle depends on its species. Particles described by antisymmetric functions are said to obey *Fermi–Dirac statistics*, or to be *fermions*; particles described by symmetric functions are said to obey *Bose–Einstein statistics*, or to be *bosons*[1].

Relativistic quantum mechanics permits one to show (see IV, § 25) that a particle's statistics is uniquely tied to its spin: particles of half-integral spin are fermions, whereas particles of integral spin are bosons.

For composite particles the statistics is fixed by the parity of the number of elementary fermions they contain. Indeed, interchanging two identical composite particles amounts to interchanging several pairs of identical elementary particles simultaneously. Interchanging bosons leaves the wave function unchanged, while interchanging fermions reverses its sign. Composite particles containing an odd number of elementary fermions therefore obey Fermi statistics, and those containing an even number obey Bose statistics. This conclusion naturally agrees with the general rule just stated: a composite particle has integral or half-integral spin according as the number of its half-integral-spin constituents is even or odd.

Thus atomic nuclei of odd atomic weight (that is, containing an odd total number of protons and neutrons) obey Fermi statistics, while nuclei of even weight obey Bose statistics. For atoms, which contain electrons in addition to the nucleus, the statistics is evidently determined by whether the sum of the atomic weight and atomic number is even or odd.

Consider a system of N identical particles whose mutual interaction may be neglected. Let $\psi_1, \psi_2, \ldots$ be wave functions for the various stationary states available to each particle separately. The state of the system as a whole can be specified by listing the state numbers occupied by its individual particles. We must determine how the wave function ψ of the entire system is to be constructed from the functions $\psi_1, \psi_2, \ldots$.

Let $p_1, p_2, \ldots, p_N$ denote the state numbers of the individual particles; repetitions among these numbers are allowed. For a boson system, the wave function $\psi(\xi_1, \xi_2, \ldots, \xi_N)$ is a sum of products of the form

$$\psi_{p_1}(\xi_1)\psi_{p_2}(\xi_2)\cdots\psi_{p_N}(\xi_N) \tag{61.1}$$

over all distinct permutations of the indices $p_1, p_2, \ldots$. This sum plainly has the requisite symmetry. For example, for two particles occupying different states ($p_1 \neq p_2$),

$$\psi(\xi_1, \xi_2) = \frac{1}{\sqrt{2}}\left[\psi_{p_1}(\xi_1)\psi_{p_2}(\xi_2) + \psi_{p_1}(\xi_2)\psi_{p_2}(\xi_1)\right]. \tag{61.2}$$

The factor $1/\sqrt{2}$ provides normalization (all the functions $\psi_1, \psi_2, \ldots$ are mutually

[1]The terminology comes from the names of the statistics describing ideal gases made up, respectively, of particles with antisymmetric or symmetric wave functions. In fact, the distinction here is not merely between two statistics, but essentially between two mechanics. Fermi statistics was proposed by E. Fermi for electrons in 1926, and its relation to quantum mechanics was established by Dirac (1926). Bose statistics was proposed by S. Bose for light quanta and generalized by Einstein (1924).

orthogonal and are assumed normalized). More generally, for a system of any number N of particles, the normalized wave function is

$$\psi_{N_1 N_2 \dots} = \left(\frac{N_1! N_2! \cdots}{N!}\right)^{1/2} \sum \psi_{p_1}(\xi_1)\psi_{p_2}(\xi_2)\cdots\psi_{p_N}(\xi_N), \tag{61.3}$$

where the sum runs through all distinct permutations of $p_1, p_2, \dots, p_N$, and N_i gives the number of these indices having the value i (so that $\sum N_i = N$). On integrating $|\psi|^2$ over $d\xi_1 d\xi_2 \cdots d\xi_N$[2], every term vanishes except the squared modulus of each term in the sum. Since the number of terms in the sum (61.3) is evidently $N!/(N_1!N_2!\cdots)$, the normalization coefficient displayed there follows.

For fermions, ψ is the antisymmetric combination of the products (61.1). In a two-particle system, for example,

$$\psi(\xi_1, \xi_2) = \frac{1}{\sqrt{2}}\left[\psi_{p_1}(\xi_1)\psi_{p_2}(\xi_2) - \psi_{p_1}(\xi_2)\psi_{p_2}(\xi_1)\right]. \tag{61.4}$$

For N particles in general, the system's wave function takes the determinant form

$$\psi_{N_1 N_2 \dots} = \frac{1}{\sqrt{N!}} \begin{vmatrix} \psi_{p_1}(\xi_1) & \psi_{p_1}(\xi_2) & \cdots & \psi_{p_1}(\xi_N) \\ \psi_{p_2}(\xi_1) & \psi_{p_2}(\xi_2) & \cdots & \psi_{p_2}(\xi_N) \\ \cdots & \cdots & \cdots & \cdots \\ \psi_{p_N}(\xi_1) & \psi_{p_N}(\xi_2) & \cdots & \psi_{p_N}(\xi_N) \end{vmatrix}. \tag{61.5}$$

Interchanging two particles interchanges two columns of this determinant and therefore reverses its sign.

Formula (61.5) has an important consequence. If two of the numbers $p_1, p_2, \dots$ coincide, two rows of the determinant coincide and the determinant vanishes identically. It can be nonzero only when all the numbers $p_1, p_2, \dots$ are different. Thus no two (or more) particles in a system of identical fermions can occupy the same state simultaneously. This is the so-called *Pauli principle* (W. Pauli, 1925).

§ 62. Exchange interaction

The omission of particle spin from the Schrödinger equation in no way diminishes either that equation or any results obtained from it. The reason is that the electric interaction between particles is independent of their spins[3].

In mathematical terms, the Hamiltonian of a system of electrically interacting particles (with no magnetic field present) contains no spin operators and therefore acts on a wave function without affecting its spin variables. In fact, each component of the wave function consequently satisfies the Schrödinger equation. Equivalently, the wave function of a particle system may be written as the product

$$\psi(\xi_1, \xi_2) = \chi(\sigma_1, \sigma_2, \dots)\varphi(\mathbf{r}_1, \mathbf{r}_2, \dots),$$

[2]Here and in §§ 64, 65, integration over $d\xi$ conventionally means integration over the coordinates together with summation over σ.

[3]This holds only within the nonrelativistic approximation. Once relativistic effects are included, the interaction of charged particles does depend on spin.

where φ depends only on the particle coordinates, and χ only on their spins. We shall refer to the former as the *coordinate* or *orbital* wave function and to the latter as the *spin* wave function.

The Schrödinger equation essentially determines only the coordinate function φ, leaving χ arbitrary. Whenever the particle spins themselves are of no interest, one may therefore apply the Schrödinger equation with the coordinate function alone as the wave function, as was done in the preceding chapters.

Nevertheless, although the electric interaction is independent of spin in the manner just described, the system's energy has a peculiar dependence on its total spin, arising ultimately from the indistinguishability principle for identical particles.

Consider a system of just two identical particles. Solving the Schrödinger equation yields a sequence of energy levels, each associated with a definite symmetric or antisymmetric coordinate wave function $\varphi(\mathbf{r}_1, \mathbf{r}_2)$.

Indeed, particle identity makes the Hamiltonian of the system, and therefore its Schrödinger equation, invariant under interchange of the particles. If an energy level is nondegenerate, exchanging the coordinates $\mathbf{r}_1$ and $\mathbf{r}_2$ can change $\varphi(\mathbf{r}_1, \mathbf{r}_2)$ only

by a constant factor; a second exchange shows that this factor can only be ± 1[4].

Suppose first that the particles have zero spin. There is then no spin factor at all, and the wave function reduces to the coordinate function $\varphi(\mathbf{r}_1, \mathbf{r}_2)$ alone. This function must be symmetric, since zero-spin particles obey Bose statistics. Hence not every energy level produced by a formal solution of the Schrödinger equation can actually occur: levels associated with antisymmetric functions φ are impossible for the system under consideration.

Interchanging two identical particles is equivalent to inverting the coordinate system, whose origin is chosen at the midpoint of the line joining them. Under inversion, on the other hand, φ must acquire the factor $(-1)^l$, where l is the orbital angular momentum of the relative motion of the two particles (see § 30). Combining this fact with the preceding result, we conclude that a pair of identical zero-spin particles can have only even orbital angular momentum.

Next let the system consist of two spin-1/2 particles, electrons for example. The full wave function—the product of $\varphi(\mathbf{r}_1, \mathbf{r}_2)$ and the spin function $\chi(\sigma_1, \sigma_2)$—must necessarily be antisymmetric under interchange of the two particles. A symmetric coordinate function must therefore be accompanied by an antisymmetric spin function, and conversely. Write the spin function in spinor form as a second-rank spinor $\chi^{\lambda\mu}$, with each index corresponding to the spin of one electron. A function symmetric in the spins is represented by a symmetric spinor ($\chi^{\lambda\mu} = \chi^{\mu\lambda}$), and an antisymmetric function by an antisymmetric spinor ($\chi^{\lambda\mu} = -\chi^{\mu\lambda}$). But a symmetric second-rank spinor, as we know, describes a system of total spin one, whereas an antisymmetric spinor reduces to a scalar and thus corresponds to spin zero.

We arrive at the following result. Energy levels whose Schrödinger-equation solutions $\varphi(\mathbf{r}_1, \mathbf{r}_2)$ are symmetric can actually occur

when the total spin of the system is zero, that is, when the two electron spins are

[4]In a degenerate level, linear combinations of its functions can always be chosen so as to satisfy this condition as well.

"antiparallel" and add to zero. Energy values associated with antisymmetric functions $\varphi(\mathbf{r}_1, \mathbf{r}_2)$ require total spin one, so the electron spins must be "parallel."

In other words, the possible energies of an electron system depend on its total spin. One may on this basis speak of a special interaction between the particles that produces this dependence. It is called the *exchange interaction*. This is a purely quantum effect and disappears completely, along with spin itself, in the limiting passage to classical mechanics.

The two-electron system just examined has this characteristic feature: every energy level corresponds to one definite total-spin value, either 0 or 1. As we shall see below (§ 63), this one-to-one correspondence between energy levels and spin values persists in systems containing any number of electrons. It does not, however, hold in systems made up of particles whose spin exceeds $1/2$.

Consider two particles of arbitrary spin s. Their spin wave function is a spinor of rank $4s$,

$$\chi^{\overbrace{\lambda\mu\ldots}^{2s}\overbrace{\rho\sigma\ldots}^{2s}},$$

one half ($2s$) of whose indices correspond to the spin of one particle, and the other half to that of the other. The spinor is symmetric in the indices within each group. Interchanging the particles exchanges every index $\lambda, \mu, \ldots$ in the first group with the indices $\rho, \sigma, \ldots$ in the second. To obtain the spin function for a system state of total spin S, this spinor must be contracted in $2s - S$ pairs of indices, each pair comprising one index from each group, and symmetrized over the remaining indices. The result is a symmetric spinor of rank $2S$.

As we know, contracting a spinor in a pair of indices means forming a combination antisymmetric in those indices. Thus interchange of the particles multiplies the spin wave function by $(-1)^{2s-S}$.

The full wave function of the two-particle system, on the other hand, must acquire the factor $(-1)^{2s}$ upon particle interchange: $+1$ for integral s, and -1 for half-integral s. It follows that the symmetry of the coordinate wave function under particle interchange is determined by the factor $(-1)^S$, which depends only on S.

We have therefore shown that the coordinate wave function for two identical particles is symmetric when their total spin is even and antisymmetric when it is odd.

Recalling the relation established above between particle interchange and inversion of the coordinate system, we also conclude that for even (odd) S the system can possess only even (odd) orbital angular momentum.

Here too, then, the system's possible energies display a dependence on total spin, but the relation is no longer unique. Energy levels associated with symmetric (antisymmetric) coordinate wave functions may occur for every even (odd) value of S.

Let us count all distinct states of a two-particle system having even and odd values of S. The quantity S takes $2s + 1$ values: $2s, 2s - 1, \ldots, 0$. For each fixed S there are $2S + 1$ states, distinguished by the z component of spin, giving $(2s + 1)^2$ states altogether. Let s be integral. Of the $2s + 1$ possible values of S, $s + 1$ are even and s are odd. The total number of states with even S is

$$\sum_{S=0,2,\dots,2s} (2S+1) = (2s+1)(s+1);$$

the remaining $s(2s+1)$ states have odd S. Similarly, for half-integral s there are $s(2s+1)$ states with even S and $(s+1)(2s+1)$ with odd S.

Problem

1. Determine the exchange splitting of the energy levels of a two-electron system, treating the interaction between the electrons as a perturbation.

Solution. Let the particles, with their interaction omitted, occupy states with orbital wave functions $\varphi_1(\mathbf{r}_1)$ and $\varphi_2(\mathbf{r}_2)$. The states of total spin $S = 0$ and $S = 1$ correspond, respectively, to the symmetrized and antisymmetrized products

$$\varphi = \frac{1}{\sqrt{2}}[\varphi_1(\mathbf{r}_1)\varphi_2(\mathbf{r}_2) \pm \varphi_1(\mathbf{r}_2)\varphi_2(\mathbf{r}_1)].$$

In these states the mean values of the particle-interaction operator $U(\mathbf{r}_2 - \mathbf{r}_1)$ are $A \pm J$, where

$$A = \iint U|\varphi_1(\mathbf{r}_1)|^2|\varphi_2(\mathbf{r}_2)|^2, dV_1 dV_2,$$

$$J = \iint U\varphi_1(\mathbf{r}_1)\varphi_1^*(\mathbf{r}_2)\varphi_2(\mathbf{r}_2)\varphi_2^*(\mathbf{r}_1), dV_1 dV_2$$

(the integral J is called the *exchange integral*). Dropping the additive constant A, which is not of exchange character, gives the level shifts $\Delta E_0 = J$ and $\Delta E_1 = -J$; the subscript specifies S. These quantities

may be represented as the eigenvalues of the spin *exchange operator*[5]

$$\widehat{V}_{\text{ex}} = -\frac{1}{2}J(1 + 4\hat{\mathbf{s}}_1\hat{\mathbf{s}}_2) \tag{1}$$

(for the eigenvalues of the product $\mathbf{s}_1\mathbf{s}_2$, see problem 2 of § 55).

If, for example, the two electrons belong to different atoms, the exchange integral decreases exponentially as the interatomic distance R increases. The form of the integrand makes clear that this integral is governed by the "overlap" of the state wave functions $\varphi_1(\mathbf{r}_1)$ and $\varphi_2(\mathbf{r}_2)$. Using the asymptotic decay law for discrete-spectrum wave functions (cf. (21.6)), one finds

$$J \sim \exp[-(\kappa_1 + \kappa_2)R], \qquad \kappa_1 = \frac{1}{\hbar}\sqrt{2m|E_1|}, \qquad \kappa_2 = \frac{1}{\hbar}\sqrt{2m|E_2|},$$

where E_1, E_2 are the electron energy levels in the two atoms.

2. Do the same for a system of three electrons.

[5]This operator was introduced by Dirac.

SOLUTION. From formula (1) of problem 1, the operator for the pairwise exchange interaction of three electrons is

$$\widehat{V}_{ex} = -\sum J_{ab}(1/2 + 2\hat{\mathbf{s}}_a\hat{\mathbf{s}}_b), \tag{1}$$

where the sum is over the particle pairs 12, 13, and 23. The matrix elements of $\hat{\mathbf{s}}_a\hat{\mathbf{s}}_b$ between states with different pairs of values σ_a, σ_b follow from (55.6) and are

$$\langle 1/2, 1/2|\mathbf{s}_a\mathbf{s}_b|1/2, 1/2\rangle = 1/4, \quad \langle 1/2, -1/2|\mathbf{s}_a\mathbf{s}_b|1/2, -1/2\rangle = -1/4,$$

$$\langle 1/2, -1/2|\mathbf{s}_a\mathbf{s}_b| - 1/2, 1/2\rangle = 1/2.$$

First determine the energy corresponding to the largest possible projection of the total spin, $M_S = \sigma_1 + \sigma_2 + \sigma_3$, namely $M_S = 3/2$. This also determines the energy of the state with total spin $S = 3/2$. Evaluation of the corresponding diagonal matrix element of (1) gives

$$\Delta E_{3/2} = -(J_{12} + J_{13} + J_{23}).$$

Now turn to the states with $M_S = 1/2$. This value can be realized in three ways, according to which of $\sigma_1, \sigma_2, \sigma_3$ equals $-1/2$, the other two being $1/2$. These states would therefore lead to a cubic secular equation. The calculation can be shortened at once by observing that one root must correspond to the energy already found for the $S = 3/2$ state. The secular equation must consequently be divisible by $\Delta E - \Delta E_{3/2}$, allowing us in this instance to avoid calculating the constant term of the cubic[6].

Specifically, calculation of the leading terms gives

$$(\Delta E)^3 + (J_{12}+J_{13}+J_{23})(\Delta E)^2 + [J_{12}J_{13}+J_{12}J_{23}+J_{13}J_{23} - (J_{12}^2+J_{13}^2+J_{23}^2)]\Delta E + \ldots = 0,$$

and division by $\Delta E + J_{12} + J_{13} + J_{23}$ yields the two energy levels belonging to states of spin $S = 1/2$:

$$\Delta E_{1/2} = \pm[J_{12}^2 + J_{13}^2 + J_{23}^2 - J_{12}J_{13} - J_{12}J_{23} - J_{13}J_{23}]^{1/2}.$$

There are thus three energy levels in all, in agreement with the count made in problem 1 of § 63.

3. From which states can the ^{8}Be nucleus decay into two α particles?

SOLUTION. Since an α particle has no spin, a system of two α particles can have only even orbital angular momentum (equal to its total angular momentum), and its states have even parity. The stated decay can therefore occur only from even-parity states of ^{8}Be having even total angular momentum.

§ 63. Permutation symmetry

For a system consisting of only two particles, we could assert that the coordinate wave functions of stationary states, $\varphi(\mathbf{r}_1, \mathbf{r}_2)$, must be either symmetric or antisymmetric. In the general case of a system containing an arbitrary number of particles, however,

[6] This device is particularly useful in analogous calculations for systems with more particles.

the solutions of the Schrödinger equation (the coordinate wave functions) need by no means be symmetric or antisymmetric under interchange of every pair of particles, as the complete wave functions (including the spin factor) are. The reason is that interchanging only the coordinates of two particles does not yet constitute their physical interchange. The physical identity of the particles implies here only that the Hamiltonian of the system is invariant under particle interchanges; hence, if a certain function is a solution of the Schrödinger equation, then the functions obtained from it by various permutations of the variables are solutions as well.

We first make a few remarks about permutations in general. For a system of N particles there are $N!$ different permutations in all. If all the particles are imagined to be numbered, each permutation can be represented by a definite sequence of the numbers $1, 2, 3, \ldots$ Every such sequence can be obtained from the natural sequence $1, 2, 3, \ldots$ by successive interchanges of pairs of particles. A permutation is called *even* or *odd* according as it is effected by an even or an odd number of pair interchanges. Let $\widehat{P}$ denote the permutation operators for N particles, and introduce a quantity δ_P equal to $+1$ when $\widehat{P}$ is an even permutation and to -1 when it is odd. If φ is symmetric in all the particles, then

$$\widehat{P}\varphi = \varphi,$$

whereas if the function is antisymmetric in all the particles, then

$$\widehat{P}\varphi = \delta_P \varphi.$$

From an arbitrary function $\varphi(\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_N)$ a symmetric function can be formed by the operation of *symmetrization*, which may be written

$$\varphi_{\text{sym}} = \text{const} \cdot \sum_P \widehat{P}\varphi, \tag{63.1}$$

where the summation extends over all possible permutations. The formation of an antisymmetric function (an operation sometimes called *alternation*) can be written

$$\varphi_{\text{antisym}} = \text{const} \cdot \sum_P \delta_P \widehat{P}\varphi. \tag{63.2}$$

Let us return to the behavior of the wave functions φ of a system of identical particles under permutations[7]. The symmetry of the system Hamiltonian $\widehat{H}$ in all the particles means mathematically that it commutes with every permutation operator $\widehat{P}$. These operators, however, do not commute with one another and therefore cannot be reduced simultaneously to diagonal form. Thus the wave functions φ cannot be chosen so that every one of them is symmetric or antisymmetric under all the separate pair interchanges[8].

[7]Mathematically, the problem is that of finding the irreducible representations of the permutation group. For detailed accounts of the mathematical theory of permutation groups, see: H. Weyl, *The Theory of Groups and Quantum Mechanics*; M. Hamermesh, *Group Theory and Its Application to Physical Problems*; I. G. Kaplan, *Symmetry of Many-Electron Systems*.

[8]Only for a system of two particles is there just one interchange operator, which can be reduced to diagonal form simultaneously with the Hamiltonian.

We shall pose the problem of determining the possible symmetry types, under permutations of the variables, of functions $\varphi(\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_N)$ of N variables (or of sets of several such functions).

The symmetry must be incapable of further increase: every additional operation of symmetrization or alternation applied to these functions must turn them either into linear combinations of the same functions or identically into zero.

We already know two operations that produce functions of maximal symmetry: symmetrization in all the variables and alternation in all the variables. These operations may be generalized as follows.

Divide the set of all N variables $\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_N$ (or, equivalently, the indices $1, 2, 3, \ldots, N$) into several rows containing $N_1, N_2, \ldots$ elements (variables), where $N_1 + N_2 + \ldots = N$. Such a partition can be represented pictorially by a diagram (the so-called *Young diagram*) in which each of the numbers $N_1, N_2, \ldots$ is represented by a row of cells (thus Fig. 21 shows the diagrams for the partitions $6 + 4 + 4 + 3 + 3 + 1 + 1$ and $7 + 5 + 5 + 3 + 1 + 1$ when $N = 22$); one of the numbers $1, 2, 3, \ldots$ is to be placed in every cell. If the rows are arranged in order of decreasing length (as in Fig. 21), the diagram contains not only successive horizontal rows but also vertical columns.

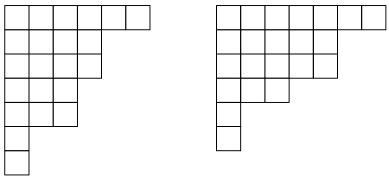


Fig. 21

Symmetrize an arbitrary function $\varphi(\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_N)$ in the variables belonging to each row. After this, alternation can be carried out only with respect to variables belonging to different rows; alternation in a pair of variables lying in the same row would plainly give zero identically.

Select one variable from each row; without loss of generality, these may be regarded as occupying the first cells of the rows (after symmetrization, the order of the variables among the cells of each row is immaterial), and perform alternation in these variables. Next remove the first column and perform alternation in the variables selected one from each row of the diagram thus shortened; these variables may again be regarded as occupying the first cells of the shortened rows. Continuing this procedure, we obtain a function first symmetrized in the variables of each row and then alternated in the variables of each column (after alternation the function, of course, is in general no longer symmetric in the variables of each row; symmetry is retained only with respect to the variables in the cells of the first row that project beyond the other rows).

By distributing the N variables among the rows of a Young diagram in different ways (their distribution among the cells of each row

being immaterial), we obtain in this way a set of functions that transform into one another under arbitrary permutations of the variables[9]. It must be stressed, however, that

[9]Symmetrization and alternation could instead be performed in the reverse order—first alternating in the

not all these functions are linearly independent; in general the number of independent functions is smaller than the number of possible distributions of the variables among the rows of the diagram. We shall not discuss this point further here[10].

Thus every Young diagram determines a certain symmetry type of functions under permutations. By constructing all the Young diagrams possible for a given N, we find all possible symmetry types. This amounts to partitioning the number N in every possible way into a sum of smaller terms, with N itself also counted among the possible partitions (for $N = 4$, for example, the partitions are 4, 3 + 1, 2 + 2, 2 + 1 + 1, and 1 + 1 + 1 + 1).

To each energy level of the system there can be assigned a Young diagram determining the permutation symmetry of the corresponding solutions of the Schrödinger equation; in general, several different functions correspond to each energy value and transform into one another under permutations. This "permutation degeneracy" is associated with the noncommutativity, already mentioned, of the operators $\widehat{P}$, each of which commutes with the Hamiltonian (cf. § 10, p. 48). It should be stressed, however, that this does not signify any additional physical degeneracy of the energy levels. All these different coordinate wave functions, multiplied by spin functions, enter into a single definite combination—the complete wave function—which satisfies, according to the spin of the particles, the condition of symmetry or antisymmetry.

Among the different symmetry types there are always two (for a given N) to which only one function apiece corresponds. One is represented by a function symmetric in all the variables, and the other by an antisymmetric function (in the former case

the Young diagram consists of a single row of N cells, and in the latter of a single column).

We turn now to the spin wave functions $\chi(\sigma_1, \sigma_2, \ldots, \sigma_N)$. Their symmetry types under particle permutations are determined by the same Young diagrams, with the particle spin projections playing the role of the variables. The question arises which diagram should correspond to the spin function for a specified diagram of the coordinate function. Suppose first that the particles have integral spin. The complete wave function ψ must then be symmetric in all particles. For this to hold, the symmetries of the spin and coordinate functions must be described by the same Young diagram, and the complete wave function ψ is expressed as certain bilinear combinations of the two; we shall not consider here how this combination is constructed.

Suppose next that the particles have half-integral spin. The complete wave function must then be antisymmetric in all the particles. It can be shown that the Young diagrams of the coordinate and spin functions must consequently be dual: one is obtained from the other by interchanging rows and columns (the two diagrams shown in Fig. 21 are an example).

We examine in more detail the important case of particles of spin $1/2$ (electrons, for example). Each spin variable $\sigma_1, \sigma_2, \ldots$ then takes only the two values $\pm 1/2$. Since a function antisymmetric in any two variables vanishes when those variables have equal

variables of each column and then symmetrizing in those of the rows. This would yield nothing new, however, since the functions obtained by the two procedures are linear combinations of one another.

[10]The independent functions that transform into one another form a basis of an irreducible representation of the permutation group. Their number is the dimension of the representation. For particles of spin $1/2$, it is found in Problem 1 of this section.

values, it is clear that χ can be alternated only in pairs of variables; if alternation is performed in three variables, two of them must have equal values, and the result is identically zero.

Thus, for a system of electrons, the Young diagrams of the spin functions can have columns only one or two cells long (that is, only one or two rows in all); in the Young diagrams of the coordinate functions, the same restriction applies to the row lengths. The number of possible permutation-symmetry types for a system of N electrons is consequently the number of ways of partitioning N into a sum of ones and twos. For even N this number is $N/2 + 1$ (the partitions with $0, 1, \ldots, N/2$ twos), and for odd N it is $(N + 1)/2$ (the partitions with $0, 1, \ldots, (N - 1)/2$ twos). Figure 22 shows, for example, the possible Young diagrams (coordinate and spin) for $N = 4$.

It is readily seen that each of these symmetry types (that is, each Young diagram) corresponds to a definite total spin S of the electron system. We shall write the spin functions in spinor form, namely as a spinor $\chi^{\lambda\mu\cdots}$ of rank N, whose indices (each corresponding to the spin of a separate particle) are the variables placed in the cells of the Young diagrams.

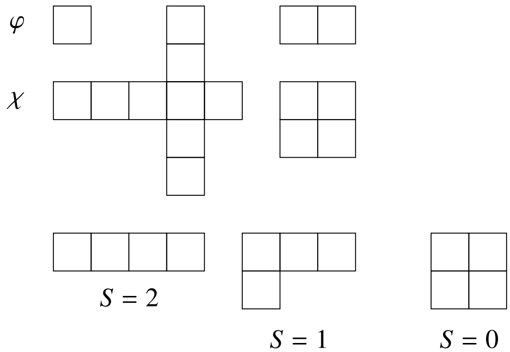


Fig. 22

Consider a spin Young diagram consisting of two rows with N_1 and N_2 cells ($N_1 + N_2 = N$, $N_1 \geq N_2$). Each of the first N_2 columns contains two cells, and the spinor must be antisymmetric in the corresponding pairs of indices. It must be symmetric, on the other hand, in the indices occupying the last $n = N_1 - N_2$ cells of the first row. As we know, however, such a spinor of rank N reduces to a symmetric spinor of rank n, corresponding to the total spin $S = n/2$. Returning to the Young diagrams of the coordinate functions, we can say that the diagram with n rows containing one cell each corresponds to a state of total spin $S = n/2$. For even N the total spin can take the integral values from 0 to $N/2$, and for odd N the half-integral values from $1/2$ to $N/2$, as it should.

It should be emphasized that this one-to-one correspondence between Young diagrams and total spin occurs only for systems of particles with spin $1/2$; for a system of just two particles we already saw it in the preceding section. For a system of N particles of spin s, the spin wave function is constructed from a product of N symmetric spinors of rank $2s$, and is therefore a spinor of rank $2Ns$. If this spinor is symmetrized according to a given Young diagram of N cells, its independent components can usually be combined into several sets of linear combinations, each corresponding to a different value of the total spin S of the system.

Just as the Young diagram of the spin functions for particles of spin $1/2$ cannot contain columns of more than two cells, for particles of arbitrary spin s the column length cannot exceed $2s + 1$ cells.

If the number N of particles in the system is an integral multiple of $2s + 1$, the possible Young diagrams include a rectangular one in which every column contains $2s + 1$ cells. This diagram corresponds to one definite value of the total spin, $S = 0$. It follows that any two (spin) Young diagrams that can be fitted together to form a rectangle of height $2s + 1$ correspond to the same values of S[11]. This conclusion is simply a consequence of the fact that the resultant of two angular momenta can be zero only when the angular momenta being added have equal magnitudes.

To conclude this section, we return to the fact noted earlier (see the note on p. 86) that, for systems of several identical particles, it cannot be asserted that the stationary-state wave function of lowest energy has no nodes. We can now make this statement more precise and explain its origin.

A wave function (here the coordinate function is meant) having no nodes must be symmetric in all the particles; if it were antisymmetric under interchange of any pair of particles 1 and 2, it would vanish when $\mathbf{r}_1 = \mathbf{r}_2$. But if the system consists of three or more electrons, a completely symmetric coordinate wave function is not allowed at all (the Young diagram of the coordinate function cannot have rows of more than two cells). Thus, although the solution of the Schrödinger equation corresponding to the lowest eigenvalue has no nodes (by the theorem of the calculus of variations), this solution may be physically inadmissible. The normal state of the system then corresponds not to the lowest eigenvalue of the Schrödinger equation, and the wave function of this state will in general have nodes. More generally, for particles of half-integral spin s, this situation occurs in systems of more than $2s + 1$ particles. For systems consisting of bosons, however, a completely symmetric coordinate wave function is always possible.

Problem

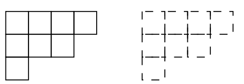
1. Determine the number of energy levels having different values of the total spin S for a system of N particles of spin $1/2$ (F. Bloch, 1929).

Solution. A specified value of the projection of the system's total spin, $M_S = \sum \sigma$, can be realized in

$$f(M_S) = \frac{N!}{(N/2 + M_S)!(N/2 - M_S)!}$$

ways ($N/2 + M_S$ particles are assigned $\sigma = 1/2$, and the rest $\sigma = -1/2$). Each energy level with a specified value of S corresponds to $2S + 1$ states with the values

[11] For example, the following pairs of diagrams have this property (for $s = 1$):



Mutually complementary diagrams are shown by solid and dashed lines.

$M_S = S, S-1, \ldots, -S$. It follows at once that the number of distinct levels with a specified value of S is

$$n(S) = f(S) - f(S+1) = \frac{N!(2S+1)}{(N/2+S+1)!(N/2-S)!}.$$

The total number $n = \sum_S n(S)$ of distinct energy levels is

$$n = f(0) = \frac{N!}{[(N/2)!]^2}$$

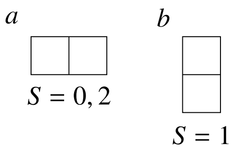
for even N, or

$$n = f(1/2) = \frac{N!}{[(N+1)/2]![(N-1)/2]!}$$

for odd N.

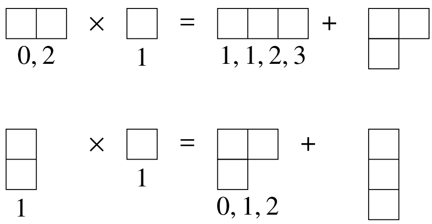
2. Find the values of the total spin S that occur for the different symmetry types of the spin functions of a system of two, three, or four particles of spin 1.

Solution. For two particles, the correspondence is established by requiring that the factor by which the spin function is multiplied under interchange of the particles equal $(-1)^{2s-S}$ (see the end of § 62). For particles of spin $s = 1$, this gives the correspondence



(1)

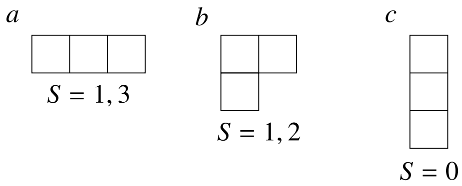
The Young diagrams for a system of three particles are obtained by adding one cell, in every possible way, to the diagrams (1). This may be written as the symbolic equalities



The values of S are indicated beneath the diagrams; the total-spin values for the three-particle system (the diagrams on the right) are obtained from the spins of the two- and one-particle systems (the diagrams on the left) by the rule for addition of angular momenta[12]. The allocation

of the resulting values of S among the individual diagrams on the right can be found by noting that diagram c (a column of three cells) corresponds to $S = 0$. Diagram b therefore corresponds to the remaining values 1 and 2 in the second equality, while diagram a corresponds to the values 1 and 3 left after b in the first equality:

[12]The value 1 is repeated twice beneath the diagrams on the right because this angular-momentum value arises once from adding angular momenta 0 and 1, and once from adding angular momenta 2 and 1.



The Young diagrams for a system of four particles are obtained by adding one cell to the diagrams (2) (subject to the condition that no column contain more than three cells):

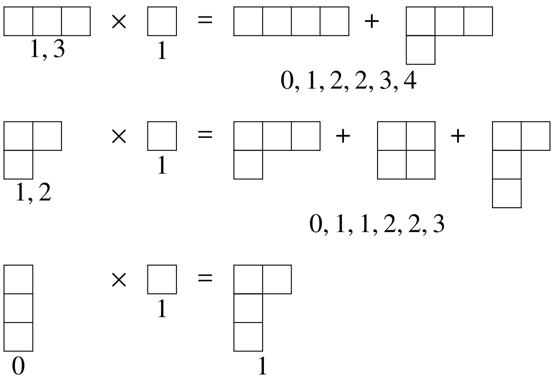
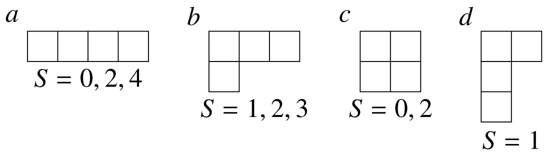


Diagram c combines with diagram 1a to form a rectangle whose columns have three cells; it therefore corresponds to the same values $S = 0, 2$ as 1a. The values of S for diagram b are determined by what remains in the second equality, and those for diagram a then by what remains in the first; the spin value for diagram d is determined uniquely by the third equality:



§ 64. Second quantization: Bose statistics

In the theory of systems containing large numbers of identical particles, a special method known as *second quantization* is widely used. It is particularly indispensable in relativistic theory, where

one must deal with systems in which the number of particles itself is variable[13].

Let $\psi_1(\xi), \psi_2(\xi), \ldots$ be some complete orthonormal set of one-particle stationary-state wave functions[14]. These may be particle states in an arbitrarily chosen external

[13]The second-quantization method was developed by Dirac for photons in the theory of radiation (1927), and was subsequently extended to fermions by Wigner and Jordan (E. Wigner, P. Jordan, 1928).

[14]As in § 61, ξ denotes the collection comprising the particle coordinates and its spin projection σ; integration over $d\xi$ is understood to mean integration over the coordinates together with summation over σ.

field, but usually one simply takes plane waves—the wave functions of a free particle with specified values of momentum (and spin projection). To make the spectrum of states discrete, the motion of the particles is then considered in a large but bounded region of space. For motion in a bounded volume, the eigenvalues of the momentum components form discrete sequences; the intervals between adjacent values are inversely proportional to the linear dimensions of the region and tend to zero as those dimensions increase.

In a system of free particles, the momentum of each particle is separately conserved. Consequently the *occupation numbers* are conserved as well: the numbers $N_1, N_2, \ldots$ specifying how many particles occupy each of the states $\psi_1, \psi_2, \ldots$. In a system of interacting particles, the individual particle momenta are no longer conserved, and neither are the occupation numbers. For such a system one can speak only of the probability distribution of the various occupation-number values. Our aim is to construct a mathematical formalism in which the occupation numbers themselves, rather than particle coordinates and spin projections, serve as the independent variables.

Dirac notation (see the end of § 11) is convenient in this formalism, with $N_1, N_2, \ldots$ chosen as the quantum numbers specifying a state. The states corresponding to the wave functions (61.3) and (61.5) will be denoted by $|N_1, N_2, \ldots\rangle$. Coordinate and spin variables no longer appear explicitly.

With this choice of independent variables, the operators of the various physical quantities, including the Hamiltonian of the system, must likewise be formulated through their action on functions of the occupation numbers. Such a formulation can be reached by starting from the ordinary matrix representation of operators. For this purpose we must consider

operator matrix elements between stationary-state wave functions of a system of noninteracting particles. Since these states can be described by prescribed occupation-number values, the way in which the operators act on these variables will thereby become clear.

We first consider systems of particles obeying Bose statistics.

Let $\hat{f}_a^{(1)}$ be the operator of some quantity associated with one particle (particle a), so that it acts only on functions of the variables ξ_a. Introduce the operator, symmetric in all the particles,

$$\hat{F}^{(1)} = \sum_a \hat{f}_a^{(1)} \tag{64.1}$$

(the sum extends over all particles), and determine its matrix elements between the wave functions (61.3). It is readily seen first of all that the matrix elements can be nonzero only for transitions in which the numbers $N_1, N_2, \ldots$ remain unchanged (the diagonal elements), and for transitions in which one of these numbers increases by unity while another decreases by unity. Indeed, every operator $\hat{f}_a^{(1)}$ acts on only one function in the product $\psi_{p_1}(\xi_1)\psi_{p_2}(\xi_2)\cdots\psi_{p_N}(\xi_N)$, so its matrix elements can be nonzero only for transitions in which the state of one particle changes. This means that the number of particles in one state decreases by unity and the number in another increases by unity. The actual calculation of these matrix elements is quite elementary; it is easier to carry it out than to follow a written account of it. We therefore give only the result. The off-diagonal elements are

$$\langle N_i, N_k - 1|\hat{F}^{(1)}|N_i - 1, N_k\rangle = f_{ik}^{(1)}\sqrt{N_i N_k}. \tag{64.2}$$

Only the indices in which the matrix element is not diagonal have been displayed; the others are suppressed for brevity. Here $f_{ik}^{(1)}$ is the matrix element

$$f_{ik}^{(1)} = \int \psi_i^*(\xi)\hat{f}^{(1)}\psi_k(\xi)\,d\xi; \tag{64.3}$$

since the operators $\hat{f}_a^{(1)}$ differ only in the labels of the variables on which they act, the integrals (64.3) do not depend on the index a, and that index has been omitted. The diagonal matrix elements of $\hat{F}^{(1)}$ are the mean values

of $F^{(1)}$ in the states $\Psi_{N_1 N_2 \ldots}$. Their calculation gives

$$\overline{F^{(1)}} = \sum_i f_{ii}^{(1)} N_i. \tag{64.4}$$

We now introduce the operators fundamental to second quantization, $\hat{a}_i$, which act not on functions of coordinates but on functions of the occupation numbers. By definition, when $\hat{a}_i$ acts on the state $|N_1, N_2, \ldots\rangle$, it lowers the variable N_i by unity and at the same time multiplies the function by $\sqrt{N_i}$[15]:

$$\hat{a}_i|N_1, N_2, \ldots, N_i, \ldots\rangle = \sqrt{N_i}|N_1, N_2, \ldots, N_i - 1, \ldots\rangle. \tag{64.5}$$

Thus $\hat{a}_i$ may be said to reduce by unity the number of particles in state i; it is therefore called the particle *annihilation operator*. It can be represented by a matrix whose sole nonzero element is

$$\langle N_i - 1|\hat{a}_i|N_i\rangle = \sqrt{N_i}. \tag{64.6}$$

The operator $\hat{a}_i^+$ adjoint to $\hat{a}_i$ is represented, by definition (see (11.9)), by a matrix with the single element

$$\langle N_i|\hat{a}_i^+|N_i - 1\rangle = \langle N_i - 1|\hat{a}_i|N_i\rangle^* = \sqrt{N_i}. \tag{64.7}$$

It follows that, when this operator acts on $|N_1, N_2, \ldots\rangle$, it increases N_i by 1:

$$\hat{a}_i^+|N_1, N_2, \ldots, N_i, \ldots\rangle = \sqrt{N_i + 1}|N_1, N_2, \ldots, N_i + 1, \ldots\rangle. \tag{64.8}$$

In other words, $\hat{a}_i^+$ increases by 1 the number of particles in state i; it is called the particle *creation operator*.

When the product $\hat{a}_i^+\hat{a}_i$ acts on a wave function, it can only multiply it by a constant, leaving all the variables $N_1, N_2, \ldots$ unchanged: $\hat{a}_i$ first lowers N_i by 1, after which $\hat{a}_i^+$ restores its original value. Direct multiplication of the matrices (64.6) and (64.7) indeed shows that $\hat{a}_i^+\hat{a}_i$ is represented by a diagonal matrix whose diagonal elements are N_i:

$$\hat{a}_i^+\hat{a}_i = N_i. \tag{64.9}$$

[15]The natural notation $\hat{a}|n\rangle$ has been used for the result of applying the operator $\hat{a}$ to the wave function of the state $|n\rangle$.

Similarly,

$$\hat{a}_i \hat{a}_i^+ = N_i + 1. \tag{64.10}$$

Subtracting these expressions gives the commutation rule for $\hat{a}_i$ and $\hat{a}_i^+$:

$$\hat{a}_i \hat{a}_i^+ - \hat{a}_i^+ \hat{a}_i = 1. \tag{64.11}$$

Operators with different indices i and k, which act on different variables (N_i and N_k), commute:

$$\hat{a}_i \hat{a}_k - \hat{a}_k \hat{a}_i = 0, \qquad \hat{a}_i \hat{a}_k^+ - \hat{a}_k^+ \hat{a}_i = 0, \qquad i \neq k. \tag{64.12}$$

From the properties of $\hat{a}_i$, $\hat{a}_i^+$ just described, it is easy to see that the operator

$$\hat{F}^{(1)} = \sum_{i,k} f_{ik}^{(1)} \hat{a}_i^+ \hat{a}_k \tag{64.13}$$

is identical to the operator (64.1). Indeed, all the matrix elements calculated using (64.6) and (64.7) coincide with (64.2) and (64.4). This is a very important result. In (64.13) the quantities $f_{ik}^{(1)}$ are simply numbers. We have therefore succeeded in expressing an ordinary operator acting on functions of coordinates as an operator acting on functions of the new variables, the occupation numbers N_i.

The result is easily generalized to operators of other forms. Let

$$\hat{F}^{(2)} = \sum_{a>b} \hat{f}_{ab}^{(2)}, \tag{64.14}$$

where $\hat{f}_{ab}^{(2)}$ is the operator of a physical quantity associated with a pair of particles at once, and therefore acts on functions of ξ_a and ξ_b. An analogous calculation shows that such an operator can be expressed in terms of $\hat{a}_i$, $\hat{a}_i^+$ as

$$\hat{F}^{(2)} = \frac{1}{2} \sum_{i,k,l,m} \langle ik|f^{(2)}|lm\rangle \hat{a}_i^+ \hat{a}_k^+ \hat{a}_m \hat{a}_l, \tag{64.15}$$

$$\langle ik|f^{(2)}|lm\rangle = \iint \psi_i^*(\xi_1)\psi_k^*(\xi_2)\hat{f}^{(2)}\psi_l(\xi_1)\psi_m(\xi_2)\, d\xi_1 d\xi_2.$$

The extension of these formulas to operators of any other form symmetric in all particles ($\hat{F}^{(3)} = \sum \hat{f}_{abc}^{(3)}$, and so on) is evident.

The same formulas can be used to express in terms of $\hat{a}_i$, $\hat{a}_i^+$ the Hamiltonian of the physical system under consideration, consisting of N interacting identical particles. The Hamiltonian of such a system is, of course, symmetric in all the particles. In the nonrelativistic approximation[16] it is independent of the particle spins

and can be written in the general form

$$\hat{H} = \sum_a \hat{H}_a^{(1)} + \sum_{a>b} U^{(2)}(\mathbf{r}_a, \mathbf{r}_b) + \sum_{a>b>c} U^{(3)}(\mathbf{r}_a, \mathbf{r}_b, \mathbf{r}_c) + \cdots \tag{64.16}$$

[16] In the absence of a magnetic field.

Here $\hat{H}_a^{(1)}$ is the part of the Hamiltonian that depends on the coordinates of only one particle, particle a:

$$\hat{H}_a^{(1)} = -\frac{\hbar^2}{2m}\Delta_a + U^{(1)}(\mathbf{r}_a), \tag{64.17}$$

where $U^{(1)}(\mathbf{r}_a)$ is the potential energy of one particle in the external field. The remaining terms in (64.16) represent the interaction energy of the particles with one another, with terms depending respectively on the coordinates of two, three, and so forth particles separated out.

Writing the Hamiltonian in this form permits direct application of (64.13), (64.15), and their analogues. Thus,

$$\hat{H} = \sum_{i,k} H_{ik}^{(1)} \hat{a}_i^+ \hat{a}_k + \frac{1}{2}\sum_{i,k,l,m} \langle ik|U^{(2)}|lm\rangle \hat{a}_i^+ \hat{a}_k^+ \hat{a}_m \hat{a}_l + \cdots \tag{64.18}$$

This is the required expression of the Hamiltonian as an operator acting on functions of the occupation numbers.

For a system of noninteracting particles, only the first term remains in (64.18):

$$\hat{H} = \sum_{i,k} H_{ik}^{(1)} \hat{a}_i^+ \hat{a}_k. \tag{64.19}$$

If the functions ψ_i are chosen as eigenfunctions of the one-particle Hamiltonian $\hat{H}^{(1)}$, the matrix $H_{ik}^{(1)}$ is diagonal and its diagonal elements are the particle energy eigenvalues ε_i. Hence

$$\hat{H} = \sum_i \varepsilon_i \hat{a}_i^+ \hat{a}_i;$$

replacing the operator $\hat{a}_i^+ \hat{a}_i$ by its eigenvalues (64.9), we obtain the energy levels of the system in the form

$$E = \sum_i \varepsilon_i N_i,$$

the trivial result that was to be expected.

The formalism developed here may be put in a more compact form by introducing the so-called ψ-operators[17]

$$\hat{\psi}(\xi) = \sum_i \psi_i(\xi)\hat{a}_i, \qquad \hat{\psi}^+(\xi) = \sum_i \psi_i^*(\xi)\hat{a}_i^+, \tag{64.20}$$

where the variables ξ are treated as parameters. From the properties of $\hat{a}_i$, $\hat{a}_i^+$ stated above, it is clear that $\hat{\psi}$ lowers the total number of particles in the system by unity, while $\hat{\psi}^+$ raises it by unity.

It is easy to see that $\hat{\psi}^+(\xi_0)$ creates a particle located at the point ξ_0. Indeed, the action of $\hat{a}_i^+$ creates a particle in the state with wave function $\psi_i(\xi)$. Consequently, the action of $\psi_i^*(\xi_0)\hat{a}_i^+$ creates a particle in a state whose wave function is

$$\sum_i \psi_i^*(\xi)\psi_i(\xi_0) = \delta(\xi - \xi_0)$$

[17]Notice the analogy between (64.20) and the expansion $\psi = \sum a_i\psi_i$ of an arbitrary wave function in a complete set of functions. Here the expansion is, as it were, quantized once again, which is the origin of the name of the method: second quantization.

(formula (5.12) has been used), which corresponds to a particle with definite coordinate and spin values[18]).

The commutation rules for the ψ-operators follow directly from those for $\hat{a}_i$, $\hat{a}_i^+$:

$$\hat{\psi}(\xi)\hat{\psi}(\xi') - \hat{\psi}(\xi')\hat{\psi}(\xi) = 0, \tag{64.21}$$

$$\hat{\psi}(\xi)\hat{\psi}^+(\xi') - \hat{\psi}^+(\xi')\hat{\psi}(\xi) = \sum_i \psi_i(\xi)\psi_i^*(\xi') = \delta(\xi - \xi'). \tag{64.22}$$

In terms of the ψ-operators, the second-quantized operator $\hat{F}^{(1)}$ is

$$\hat{F}^{(1)} = \int \hat{\psi}^+(\xi)\hat{f}^{(1)}\hat{\psi}(\xi)\, d\xi \tag{64.23}$$

(the operator $\hat{f}^{(1)}$ is understood here to act in $\hat{\psi}(\xi)$ on functions of the parameters ξ). Indeed, inserting $\hat{\psi}$ and $\hat{\psi}^+$ from (64.20) and using definition (64.3), we recover formula (64.13). Similarly, in place of (64.15) we have

$$\hat{F}^{(2)} = \frac{1}{2}\iint \hat{\psi}^+(\xi)\hat{\psi}^+(\xi')\hat{f}^{(2)}\hat{\psi}(\xi')\hat{\psi}(\xi)\, d\xi\, d\xi'. \tag{64.24}$$

In particular, the Hamiltonian of the system expressed through the ψ-operators takes the form

$$\begin{aligned} \hat{H} = &\int \left\{-\frac{\hbar^2}{2m}\hat{\psi}^+(\xi)\Delta\hat{\psi}(\xi) + \hat{\psi}^+(\xi)U^{(1)}(\xi)\hat{\psi}(\xi)\right\} d\xi \\ &+ \frac{1}{2}\iint \hat{\psi}^+(\xi)\hat{\psi}^+(\xi')U^{(2)}(\xi,\xi')\hat{\psi}(\xi')\hat{\psi}(\xi)\, d\xi\, d\xi' + \cdots \end{aligned} \tag{64.25}$$

The operator $\hat{\psi}^+(\xi)\hat{\psi}(\xi)$, constructed from the ψ-operators in analogy with the product $\psi^*\psi$ that gives the probability density of a particle in a state with wave function ψ, is called the particle *density operator*. The integral

$$\hat{N} = \int \hat{\psi}^+\hat{\psi}\, d\xi \tag{64.26}$$

plays the role in second quantization of the operator for the total number of particles in the system. Indeed, substituting the ψ-operators from (64.20) and using the normalization and mutual orthogonality of the wave functions gives $\hat{N} = \sum_i \hat{a}_i^+\hat{a}_i$. Every term in this sum is the particle-number operator for state i: according to (64.9), its eigenvalues are the occupation numbers N_i, and their sum is the total number of particles in the system[19].

Finally, if the system contains bosons of different kinds, second quantization requires a separate set of operators $\hat{a}$, $\hat{a}^+$ for every kind of particle. Operators belonging to different kinds of particles plainly commute with one another.

[18] $\delta(\xi - \xi_0)$ is a shorthand for the product

$$\delta(x - x_0)\,\delta(y - y_0)\,\delta(z - z_0)\,\delta_{\sigma\sigma_0}.$$

[19] For systems with a prescribed number of particles, these statements (like the properties of the free-particle Hamiltonian (64.19)) appear trivial. Their extension in relativistic theory, however, leads to new and far from trivial results (cf. Vol. IV, § 11).

§ 65. Second quantization: Fermi statistics

The essential basis of the method of second quantization remains unchanged for systems of identical fermions. The specific formulas for matrix elements of quantities and for the operators $\hat{a}_i$, however, naturally change.

The wave function $\psi_{N_1 N_2 \ldots}$ now has the form (61.5). Its antisymmetry first raises the question of choosing its sign. No such question arose with Bose statistics: because the wave function was symmetric, a sign once chosen was retained under every permutation of the particles.

To fix the sign of (61.5), adopt the following convention. Number all the states ψ_i once and for all in sequence. The rows of the determinant (61.5) are then always filled so that

$$p_1 < p_2 < p_3 < \cdots < p_N, \tag{65.1}$$

while its columns contain functions of the different variables in the order $\xi_1, \xi_2, \ldots, \xi_N$. No two of $p_1, p_2, \ldots$ can be equal, since the determinant would otherwise vanish. Equivalently, every occupation number N_i can have only the value 0 or 1.

Consider again an operator of the form (64.1), $\hat{F}^{(1)} = \sum_a \hat{f}_a^{(1)}$. For the same reasons as in § 64, its matrix elements can be nonzero only for transitions in which all occupation numbers remain unchanged, or in which one of them, N_i, decreases by one (from unity to zero) while another, N_k, increases by one (from zero to unity). For $i < k$ one readily finds

$$\langle 1_i, 0_k | \hat{F}^{(1)} | 0_i, 1_k \rangle = f_{ik}^{(1)} (-1)^{\Sigma(i+1,k-1)}. \tag{65.2}$$

Here $0_i, 1_i$ denote $N_i = 0, N_i = 1$, and $\sum(k,l)$ denotes the sum of the occupation numbers of all states from k through l.[20]

$$\sum(k,l) = \sum_{n=k}^{l} N_n.$$

The diagonal elements are still given by (64.4):

$$\overline{F^{(1)}} = \sum_i f_{ii}^{(1)} N_i. \tag{65.3}$$

For $\hat{F}^{(1)}$ to be representable in the form (64.13), the annihilation operators $\hat{a}_i$ must be defined as matrices

with elements

$$\langle 0_i | \hat{a}_i | 1_i \rangle = \langle 1_i | \hat{a}_i^+ | 0_i \rangle = (-1)^{\Sigma(1,i-1)}. \tag{65.4}$$

Multiplication of these matrices gives, for $k > i$,

$$\begin{aligned}\langle 1_i, 0_k | \hat{a}_i^+ \hat{a}_k | 0_i, 1_k \rangle &= \langle 1_i, 0_k | \hat{a}_i^+ | 0_i, 0_k \rangle \langle 0_i, 0_k | \hat{a}_k | 0_i, 1_k \rangle \\ &= (-1)^{\Sigma(1,i-1)} (-1)^{\Sigma(1,i-1)+\Sigma(i+1,k-1)},\end{aligned}$$

or

$$\langle 1_i, 0_k | \hat{a}_i^+ \hat{a}_k | 0_i, 1_k \rangle = (-1)^{\Sigma(i+1,k-1)}. \tag{65.5}$$

[20] For $i > k$, the exponent in (65.2) must be $\sum(k+1, i-1)$. For $i = k \pm 1$, these sums are to be replaced by zero.

For $i = k$, the matrix $\hat{a}_i^+ \hat{a}_i$ is diagonal, with element unity when $N_i = 1$ and zero when $N_i = 0$. Thus

$$\hat{a}_i^+ \hat{a}_i = N_i. \tag{65.6}$$

Substitution of these expressions in (64.13) indeed gives (65.2), (65.3).

Multiplying $\hat{a}_i^+$ and $\hat{a}_k$ in the opposite order gives

$$\begin{aligned}\langle 1_i, 0_k | \hat{a}_k \hat{a}_i^+ | 0_i, 1_k \rangle &= \langle 1_i, 0_k | \hat{a}_k | 1_i, 1_k \rangle \langle 1_i, 1_k | \hat{a}_i^+ | 0_i, 1_k \rangle \\ &= (-1)^{\sum(1,i-1)+\sum(i+1,k-1)+\sum(1,i-1)+1},\end{aligned}$$

or

$$\langle 1_i, 0_k | \hat{a}_k \hat{a}_i^+ | 0_i, 1_k \rangle = -(-1)^{\sum(i+1,k-1)}. \tag{65.7}$$

Comparison of (65.7) with (65.5) shows that these quantities have opposite signs, and hence

$$\hat{a}_i^+ \hat{a}_k + \hat{a}_k \hat{a}_i^+ = 0, \qquad i \neq k.$$

For the diagonal matrix $\hat{a}_i \hat{a}_i^+$ we find

$$\hat{a}_i \hat{a}_i^+ = 1 - N_i. \tag{65.8}$$

Adding (65.6) gives

$$\hat{a}_i \hat{a}_i^+ + \hat{a}_i^+ \hat{a}_i = 1.$$

Both relations can be written together as

$$\hat{a}_i \hat{a}_k^+ + \hat{a}_k^+ \hat{a}_i = \delta_{ik}. \tag{65.9}$$

An analogous calculation for products of $\hat{a}_i$ and $\hat{a}_k$ yields

$$\hat{a}_i \hat{a}_k + \hat{a}_k \hat{a}_i = 0 \tag{65.10}$$

(in particular, $\hat{a}_i \hat{a}_i = 0$).

Thus operators $\hat{a}_i$ and $\hat{a}_k$ (or $\hat{a}_k^+$) with $i \neq k$ anticommute, whereas under Bose statistics they commuted. This distinction is natural. In the Bose case, $\hat{a}_i$ and $\hat{a}_k$ were completely independent: each $\hat{a}_i$ acted only on the single variable N_i, and its action did not depend on the other occupation numbers. Under Fermi statistics, by contrast, the action of $\hat{a}_i$ depends both on N_i itself and, as (65.4) shows, on the occupation numbers of all preceding states. The actions of different $\hat{a}_i, \hat{a}_k$ therefore cannot be regarded as independent.

Once the properties of $\hat{a}_i, \hat{a}_i^+$ have been defined, all the other formulas (64.13)–(64.18) remain fully valid. So do (64.23)–(64.25), which express operators of physical quantities through the field operators defined by (64.20). The commutation rules (64.21), (64.22), however, are replaced by the anticommutation relations

$$\hat{\psi}^+(\xi')\hat{\psi}(\xi) + \hat{\psi}(\xi)\hat{\psi}^+(\xi') = \delta(\xi - \xi'), \tag{65.11}$$

$$\hat{\psi}(\xi')\hat{\psi}(\xi) + \hat{\psi}(\xi)\hat{\psi}(\xi') = 0. \tag{65.12}$$

If a system contains different kinds of particles, separate second-quantized operators must be introduced for each kind, as mentioned at the end of the preceding section. Operators belonging to bosons commute with those belonging to fermions. Within nonrelativistic theory, operators belonging to different fermions may formally be taken either to commute or to anticommute; the method of second quantization gives the same results under either convention.

With a view to later use in relativistic theory, which permits transformations between different particles, the creation and annihilation operators of different fermions must nevertheless be taken to anticommute. This becomes clear if two different internal states of the same composite particle are regarded as different particles.

CHAPTER X

The Atom

§ 66. Atomic energy levels

In the nonrelativistic approximation, the stationary states of an atom are determined by the Schrödinger equation for a system of electrons moving in the Coulomb field of the nucleus and interacting electrically with one another; electron spin operators do not enter this equation at all. For a system of particles in a centrally symmetric external field, the total orbital angular momentum L and the parity of the state are conserved. Each stationary state of an atom is therefore characterized by a definite value of L and by its parity. Moreover, the coordinate wave functions of stationary states of a system of identical particles have a definite permutation symmetry. We saw in § 63 that, for a system of electrons, every definite type of permutation symmetry (that is, every Young diagram) corresponds to a definite value of the total spin of the system. Thus every stationary state of an atom is also characterized by the total electron spin S.

An energy level with specified S and L is degenerate with respect to the possible directions of the vectors $\mathbf{S}$ and $\mathbf{L}$ in space. The degeneracies associated with the directions of L and S are respectively $2L + 1$ and $2S + 1$. Hence the total degeneracy of a level with specified L and S is $(2L + 1)(2S + 1)$.

In reality, however, the electromagnetic interaction of electrons includes relativistic effects that depend on their spins. As a result, the energy of an atom depends not only on the magnitudes of $\mathbf{L}$ and $\mathbf{S}$, but also on their relative orientation. Strictly speaking, once relativistic interactions are included, neither the orbital angular momentum L nor the spin S of the atom is separately conserved. Only the total angular momentum $\mathbf{J} = \mathbf{L} + \mathbf{S}$ remains conserved. This is a universal exact law following from the isotropy of space for a closed system. The exact energy levels must therefore be characterized by values of the total angular momentum J.

If the relativistic effects are relatively small, as is often the case, they may nevertheless be treated as a perturbation. Under this perturbation, the degenerate level with specified L and S "splits" into a series of distinct, closely spaced levels having different values of J.

These levels are determined, in first approximation, by the corresponding secular equation (§ 39), while their zeroth-order wave functions are particular linear combinations of the wave functions belonging to the original degenerate level with the given L and S.

In this approximation the magnitudes of the orbital angular momentum and spin, though not their directions, may consequently still be regarded as conserved, and the levels may also be characterized by L and S.

Thus, through relativistic effects, a level with given L and S splits into levels having different J. This splitting is called the *fine structure*, or *multiplet splitting*, of the level. The possible values of J run from $L + S$ to $|L - S|$; a level with given L and S therefore splits into $2S + 1$ distinct levels if $L > S$, or $2L + 1$ if $L < S$. Each of these remains degenerate with respect to the direction of $\mathbf{J}$, with degeneracy $2J + 1$. It is readily verified that the sum of $2J + 1$ over all possible J is, as required, $(2L + 1)(2S + 1)$.

Atomic energy levels, also called atomic *spectroscopic terms*, are conventionally denoted by symbols analogous to those used for states of individual particles with definite angular momentum (§ 32). States with different total orbital angular momenta L are denoted by capital Latin letters according to

L	0	1	2	3	4	5	6	7	8	9	10
	S	P	D	F	G	H	I	K	L	M	N

The number $2S + 1$, called the *multiplicity* of the term, is placed at the upper left of the symbol (though it should be remembered that this number equals the number of fine-structure components only when $L \geqslant S$).[1] The value

of the total angular momentum J is placed at the lower right. Thus $^2P_{1/2}$ and $^2P_{3/2}$ denote levels with $L = 1$, $S = 1/2$, and $J = 1/2, 3/2$.

§ 67. Electron states in atoms

An atom with more than one electron is a complicated system of mutually interacting electrons moving in the field of the nucleus. Strictly speaking, only states of the system as a whole can be considered. Nevertheless, the state of each electron separately can be defined to good accuracy as a stationary state of electron motion in an effective centrally symmetric field produced by the nucleus together with all the other electrons. These fields are generally different for different electrons in the atom. They must all be determined simultaneously, since each depends on the states of all the others. Such a field is called *self-consistent*.

Because the self-consistent field is centrally symmetric, each electron state is characterized by a definite orbital angular momentum l. The states of an individual electron for fixed l are numbered, in order of increasing energy, by the *principal quantum number* n, whose values are $n = l + 1, l + 2, \ldots$; this numbering follows the convention for hydrogen. In many-electron atoms, however, levels with different l do not generally occur in the same energy order as in hydrogen. In hydrogen the energy is independent of l, so states with larger n always have higher energy. In a many-electron atom, by contrast, the level $n = 5$, $l = 0$, for example, lies below the level $n = 4$, $l = 2$ (see § 73 for further discussion).

States of individual electrons with different n and l are conventionally denoted by a symbol consisting of a numeral giving the principal quantum number and a letter giving l.[2] Thus $4d$ denotes the state $n = 4$, $l = 2$. A complete specification of an atomic state requires both the total L, S, J and a listing of the states of all its electrons. Thus $1s\,2p\,^3P_0$ denotes a helium state with $L = 1$, $S = 1$, $J = 0$, in which the two electrons occupy

the $1s$ and $2p$ states. If several electrons have the same l and n, this is indicated compactly by an exponent; thus $3p^2$ denotes two electrons in $3p$ states. The distribution of the electrons of an atom among states with different l, n is called its *electron configuration*.

[1] For $2S + 1 = 1, 2, 3, \ldots$, the levels are called singlet, doublet, triplet, and so on.

[2] Another common terminology describes electrons with principal quantum numbers $n = 1, 2, 3, \ldots$ as electrons of the $K, L, M, \ldots$ shells, respectively (see § 74).

For given n and l, an electron may have different projections of orbital angular momentum, m, and spin, σ, on the z axis. For fixed l, m takes $2l + 1$ values, while σ is restricted to the two values $\pm 1/2$. There are consequently $2(2l + 1)$ different states with the same n, l; these states are called *equivalent*. By the Pauli principle, each can contain one electron. Thus no more than $2(2l + 1)$ electrons in an atom can simultaneously have the same n, l. A set of electrons occupying all the states with specified n, l is called a *closed shell* of that type.

The differences between the energies of atomic levels having different L, S but the same electron configuration[3] arise from the electrostatic interaction of the electrons. Usually these energy differences are relatively small, several times smaller than the separations of levels belonging to different configurations. The ordering of levels with the same configuration but different L, S is described by the following empirically established *Hund rule* (F. Hund, 1925):

The term of lowest energy is the one with the greatest value of S permitted by the electron configuration and the greatest value of L permitted for that S.[4]

We now show how to find the atomic terms allowed for a given electron configuration. If the electrons are inequivalent, the possible L, S follow directly from the rule for addition of angular momenta. Thus for the configuration $np, n'p$, with $n \neq n'$, the total L can be 2, 1, or 0 and the total spin S can be 0 or 1; combination of these values gives the terms ${}^{1/3}S$, ${}^{1/3}P$, and ${}^{1/3}D$.

For equivalent electrons, however, the Pauli principle imposes restrictions. Consider a configuration of three equivalent p electrons. For $l = 1$ (a p state), the orbital projection m may be $1, 0, -1$, giving six states with the following pairs m, σ:

$$
\begin{array}{llll}
\text{a)} & 1,, 1/2, & \text{b)}\ 0,, 1/2 & \text{c)}\ -1,, 1/2, \\
\text{a}') & 1,, -1/2, & \text{b}')\ 0,, -1/2 & \text{c}')\ -1,, -1/2.
\end{array}
$$

The three electrons can be placed one apiece in any three of these states. The resulting atomic states have the following projections $M_L = \sum m$ and $M_S = \sum \sigma$ of the total orbital angular momentum and spin:

$$
\begin{array}{lll}
\text{a} + \text{a}' + \text{b})\quad 2,, 1/2, & \text{a} + \text{a}' + \text{c})\ 1,, 1/2, & \text{a} + \text{b} + \text{c})\ 0,, 3/2, \\
 & \text{a} + \text{b} + \text{b}')\ 1,, 1/2, & \text{a} + \text{b} + \text{c}')\ 0,, 1/2, \\
 & \text{a} + \text{b}' + \text{c})\ 0,, 1/2, & \\
 & \text{a}' + \text{b} + \text{c})\ 0,, 1/2. &
\end{array}
$$

(States with negative M_L, M_S need not be written, since they add nothing new.) The existence of a state with $M_L = 2$, $M_S = 1/2$ shows that a ${}^2D$ term must occur; one state of each of $(1, 1/2)$ and $(0, 1/2)$ must also belong to this term. One state $(1, 1/2)$ then remains, so there must be a ${}^2P$ term; one of the $(0, 1/2)$ states also belongs to it. Finally,

[3] The fine structure of each multiplet level is disregarded here.

[4] The requirement that S be maximal may be justified as follows. Consider, for example, a system of two electrons. Here S may be 0 or 1, and spin 1 corresponds to an antisymmetric coordinate wave function $\varphi(\mathbf{r}_1, \mathbf{r}_2)$. This function vanishes when $\mathbf{r}_1 = \mathbf{r}_2$; in other words, in the $S = 1$ state the probability of finding the two electrons near one another is small. Their electrostatic repulsion, and hence their energy, is therefore relatively smaller. Similarly, for a system of several electrons, the greatest spin corresponds to the "most antisymmetric" coordinate wave function.

the remaining states $(0, 3/2)$ and $(0, 1/2)$ belong to a ^4S term. Thus a configuration of three equivalent p electrons permits exactly one term of each type ^2D, ^2P, and ^4S.

Table 1 lists the possible terms for various configurations of equivalent p and d electrons. A number beneath a term symbol gives the number of terms of that type when it exceeds one. A configuration containing the greatest possible number of equivalent electrons ($s^2, p^6, d^{10}, \ldots$) always has the single term ^1S. Notice that the sets of terms are identical for two configurations when the number of electrons in one equals the number lacking from the other to complete the shell. This follows directly because the absence of an electron from a shell can be treated as a *hole*, whose state is specified by the same quantum numbers as the state of the missing electron.

In applying Hund's rule to determine the normal term of an atom from its known electron configuration, only the unfilled shell need be considered, since the angular momenta of electrons in filled shells cancel. Suppose, for example, that an atom has four d electrons outside its closed shells. The magnetic quantum number of a d electron can take the five values $0, \pm1, \pm2$. All four electrons can therefore have the same spin projection $\sigma = 1/2$, making the greatest possible total spin $S = 2$. The electrons must then be assigned distinct values of m that give the greatest $M_L = \sum m$, namely $2, 1, 0, -1$, so that $M_L = 2$. Hence the greatest value of L possible with $S = 2$ is also 2 (the term ^5D).

Table 1. All possible terms for configurations of equivalent electrons

p,	p^5	^2P		
p^2,	p^4	^1SD	^3P	
p^3,		^2PD	^4S	
d,	d^9	^2D		
d^2,	d^8	^1SDG	^3PF	
d^3,	d^7	$^2P\underset{2}{D}FGH$	^4PF	
d^4,	d^6	$^1\underset{2}{S}\underset{2}{D}F\underset{2}{G}I$	$^3\underset{2}{P}D\underset{2}{F}GH$	^5D
d^5,		$^2SP\underset{3}{D}\underset{2}{F}\underset{2}{G}HI$	$^4P\underset{2}{D}\underset{2}{F}G$	^6S

Problem

Find the orbital wave functions of the possible states of a system of three equivalent p electrons.

Solution. In the ^4S state the spin projections σ of all electrons are the same, and hence their m values are distinct. The wave function is the determinant of the form (61.5) constructed from ψ_0, ψ_1, and ψ_{-1} (the subscript specifies m).

For the ^2D term, consider the state with the greatest possible $M_L = 2$. Two of the m projections must then equal 1 and the other 0. Let electrons 2 and 3 have $\sigma = +1/2$ and electron 1 have $\sigma = -1/2$, in accordance with the total spin $S = 1/2$. The corresponding orbital wave function with the required symmetry is

$$\psi = \frac{1}{\sqrt{2}}\psi_1(1)\left[\psi_0(2)\psi_1(3) - \psi_0(3)\psi_1(2)\right]$$

(the numeral in the argument specifies the electron to which the function refers).

For the ^2P term, consider the state with $M_L = 1$ and the same electron spin projections as above. This state can be formed with

two different sets of m values, so its orbital wave function is the linear combination

$$\psi = a\psi_{-111} + b\psi_{100},$$
$$\psi_{-111} = \psi_1(1)[\psi_{-1}(2)\psi_1(3) - \psi_{-1}(3)\psi_1(2)],$$
$$\psi_{100} = \psi_0(1)[\psi_1(2)\psi_0(3) - \psi_1(3)\psi_0(2)].$$

To determine the coefficients, use the relation

$$\hat{L}_+\psi = (\hat{l}_+^{(1)} + \hat{l}_+^{(2)} + \hat{l}_+^{(3)})\psi = 0,$$

which a wave function with $M_L = L$ must satisfy (see (27.8)). From the matrix elements (27.12),

$$\hat{l}_+\psi_1 = 0, \qquad \hat{l}_+\psi_{-1} = \sqrt{2}\psi_0, \qquad \hat{l}_+\psi_0 = \sqrt{2}\psi_1,$$

and hence

$$\hat{L}_+\psi = \sqrt{2}(a - b)\psi_{011} = 0.$$

Thus $a - b = 0$; including normalization gives $a = b = 1/2$.

The wave functions of states with $M_L < L$ are obtained from those found above by applying the operator $\hat{L}_-$.

§ 68. Hydrogen-like energy levels

The only atom for which the Schrödinger equation can be solved exactly is the simplest of all atoms, hydrogen. The energy levels of hydrogen and of the ions He^+, Li^{++}, ..., each containing just one electron, are given by the Bohr formula (36.10):

$$E = -\frac{mZ^2e^4}{2\hbar^2(1 + m/M)}\frac{1}{n^2}. \tag{68.1}$$

Here Ze is the nuclear charge, M the nuclear mass, and m the electron mass. The dependence on the nuclear mass is very weak.

Formula (68.1) includes no relativistic effects. In this approximation hydrogen has the additional, *accidental*, degeneracy discussed in § 36: for fixed principal quantum number n, the energy is independent of the orbital angular momentum l.

Other atoms possess states whose properties resemble those of hydrogen. These are highly excited states in which one electron has a large principal quantum number and therefore spends most of its time far from the nucleus. To some approximation, the motion of this electron can be regarded as taking place in the Coulomb field of the *ionic core*, whose effective charge is unity. The energy levels obtained in this way are, however, too inaccurate; they require a correction for the departure of the field at small distances from a pure Coulomb field. The form of this correction follows readily from the following argument.

Since states with large quantum numbers are semiclassical, their energy levels may be determined from the Bohr–Sommerfeld quantization rules (48.6). The departure from a Coulomb field at distances small compared with the "orbital radius" may be represented formally as a change in the boundary condition imposed on the wave function at $r = 0$. This changes the constant γ in the radial quantization condition. Since the rest of that condition is unchanged, the energy-level expression must differ from the hydrogenic one through the replacement of the radial quantum number, or equivalently the principal quantum number n, by $n + \Delta_l$, where Δ_l is a constant known as the *Rydberg correction*:

$$E = -\frac{me^4}{2\hbar^2}\frac{1}{(n+\Delta_l)^2}. \tag{68.2}$$

By its definition the Rydberg correction is independent of n, but it is of course a function of the azimuthal quantum number l of the excited electron (written as a subscript on Δ), and also of the angular momenta L and S of the atom as a whole. For fixed L and S, Δ_l decreases rapidly as l increases. The larger l is, the less time the electron spends near the nucleus, and the closer the energy levels must therefore approach the hydrogenic levels.[5]

Problem

Find the asymptotic form, at large distances from the ionic core, of the wave function of a hydrogen-like electron in an s state.

Solution. At large distances, where $U = -1/r$ in atomic units, the required function $\psi(r)$ obeys the Schrödinger equation

$$\psi'' + \frac{2}{r}\psi' - \kappa^2\psi + \frac{2}{r}\psi = 0,$$

where $\kappa = \sqrt{2|E|}$. Seek a solution in the form $\psi = \text{const}\cdot r^\nu e^{-\kappa r}$. Neglecting terms in the equation that decrease faster than ψ/r, we find

$$\psi = \text{const}\cdot r^{1/\kappa-1}e^{-\kappa r}.$$

§ 69. The self-consistent field

The Schrödinger equation for atoms containing more than one electron cannot be solved analytically. Approximate methods for calculating the energies and wave functions of stationary atomic states are therefore important. Foremost among them is the method

[5]As an illustration, consider the empirical Rydberg corrections for highly excited states of helium. The total spin of helium may be $S = 0$ or 1, while in the states under consideration the total orbital angular momentum equals the angular momentum l of the excited electron (the other electron is in the $1s$ state). The Rydberg corrections are, for $S = 0$,

$$\Delta_0 = -0{,}140, \qquad \Delta_1 = +0{,}012, \qquad \Delta_2 = -0{,}0022,$$

and, for $S = 1$,

$$\Delta_0 = -0{,}296, \qquad \Delta_1 = -0{,}068, \qquad \Delta_2 = -0{,}0029.$$

known as the self-consistent-field method. Its idea is to regard each electron in an atom as moving in a *self-consistent field* produced jointly by the nucleus and all the other electrons.

As an example, consider the helium atom, restricting attention to terms in which both electrons are in s states (with equal or different n); the states of the whole atom are then S states. Let $\psi_1(r_1)$ and $\psi_2(r_2)$ be the electron wave functions; in s states they depend only on the respective distances r_1, r_2 from the nucleus. The wave function $\psi(r_1, r_2)$ of the atom as a whole is the symmetrized product

$$\psi = \psi_1(r_1)\psi_2(r_2) + \psi_1(r_2)\psi_2(r_1) \tag{69.1}$$

or the antisymmetrized product

$$\psi = \psi_1(r_1)\psi_2(r_2) - \psi_1(r_2)\psi_2(r_1), \tag{69.2}$$

according as the state has total spin $S = 0$ or $S = 1$[6]. We shall consider the latter case; the functions ψ_1 and ψ_2 may then be taken to be mutually orthogonal[7].

Our aim is to determine the function of the form (69.2) that gives the best approximation to the true atomic wave function. It is natural to use

the variational principle for this purpose, admitting only functions of the form (69.2) as trial functions (the method described here was proposed by V. A. Fock, 1930).

As we know, the Schrödinger equation can be obtained from the variational principle

$$\iint \psi^* \hat{H} \psi \, dV_1 dV_2 = \min$$

with the supplementary condition

$$\iint |\psi|^2 dV_1 dV_2 = 1$$

(the integration is over the coordinates of both electrons in the helium atom). Variation gives

$$\iint \delta\psi^* (\hat{H} - E)\psi \, dV_1 dV_2 = 0, \tag{69.3}$$

from which the ordinary Schrödinger equation follows when the variation of ψ is arbitrary. In the self-consistent-field method, however, (69.2) is substituted for ψ in (69.3), and ψ_1 and ψ_2 are varied separately.

In other words, the extremum of the integral is sought among functions ψ of the form (69.2). The resulting energy eigenvalue and wave function are of course inexact, but the wave function is the best of all those representable in this form.

[6]A helium-atom state with $S = 0$ is conventionally called a *parahelium* state, and one with $S = 1$ an *orthohelium* state.

[7]The wave functions $\psi_1, \psi_2, \ldots$ for different electron states obtained by the self-consistent-field method are not in general mutually orthogonal, since they solve different equations rather than one and the same equation. In (69.2), however, ψ_2 may be replaced by $\psi_2' + \text{const} \cdot \psi_1$ without changing the wave function ψ of the atom; the constant can always be chosen so that ψ_1 and ψ_2' are mutually orthogonal.

The Hamiltonian of the helium atom is[8]

$$\hat{H} = \hat{H}_1 + \hat{H}_2 + \frac{1}{r_{12}}, \qquad \hat{H}_1 = -\frac{1}{2}\Delta_1 - \frac{2}{r_1} \tag{69.4}$$

(r_{12} is the distance between the electrons). Substituting (69.2) in (69.3), carrying out the variation, and setting equal to zero the coefficients of $\delta\psi_1$ and $\delta\psi_2$ in the integrand, we obtain

$$\begin{aligned}
&[\tfrac{1}{2}\Delta + \tfrac{2}{r} + E - H_{22} - G_{22}(r)]\psi_1(r) + [H_{12} + G_{12}(r)]\psi_2(r) = 0,\\
&[\tfrac{1}{2}\Delta + \tfrac{2}{r} + E - H_{11} - G_{11}(r)]\psi_2(r) + [H_{12} + G_{12}(r)]\psi_1(r) = 0,
\end{aligned} \tag{69.5}$$

where

$$G_{ab}(r_1) = \int \frac{\psi_a(r_2)\psi_b(r_2)}{r_{12}}\, dV_2, \qquad H_{ab} = \int \psi_a \left(-\frac{1}{2}\Delta - \frac{2}{r}\right)\psi_b\, dV, \quad a, b = 1, 2. \tag{69.6}$$

These are the final equations furnished by the self-consistent-field method; naturally, they can be solved only numerically[9].

The equations for more complicated cases must be derived in an analogous way. The atomic wave function to be inserted in the variational integral is formed as a linear combination of products of the wave functions of individual electrons. This combination must be chosen so that, first, its permutation symmetry corresponds to the total spin S of the atomic state under consideration and, second, it corresponds to the specified total orbital angular momentum L of the atom[10].

By using in the variational principle a wave function with the proper permutation symmetry, we thereby allow for the exchange interaction of the electrons. Simpler equations (though they give less accurate results) are obtained if the exchange interaction and the dependence of the atomic energy on L for a given electron configuration are both neglected (D. R. Hartree, 1928). Again taking the helium atom as an example, we may then write the equations for the electron wave functions directly as ordinary Schrödinger equations

$$[\tfrac{1}{2}\Delta_a + E_a - V_a(r_a)]\psi_a(r_a) = 0, \qquad a = 1, 2, \tag{69.7}$$

in which V_a is the potential energy of one electron moving in the field of the nucleus and in that of the distributed charge of the other electron:

$$V_1(r_1) = -\frac{2}{r_1} + \int \frac{1}{r_{12}}\psi_2^2(r_2)\, dV_2 \tag{69.8}$$

(and similarly for V_2). To find the total energy E of the atom, note that the sum $E_1 + E_2$ counts the electrostatic interaction of the two electrons twice, since it enters the

[8] Atomic units are used in this section and in its problems.

[9] Comparison of energy levels calculated by the self-consistent-field method for light atoms with spectroscopic data indicates an accuracy of roughly 5% (and in some cases even better). For complex atoms, however, the error may be comparable with the intervals between neighboring levels and may therefore give an incorrect level sequence.

[10] General methods for constructing wave functions of a system of electrons in a central field are described in the book by I. G. Kaplan cited on p. 291.

potential energy of the first electron, $V_1(r_1)$, and also that of the second, $V_2(r_2)$. Thus E is obtained from $E_1 + E_2$ by subtracting the mean value of this interaction once:

$$E = E_1 + E_2 - \iint \frac{1}{r_{12}} \psi_1^2(r_1) \psi_2^2(r_2)\, dV_1 dV_2. \tag{69.9}$$

To refine the results of this simplified method, the exchange interaction and the dependence of the energy on L can subsequently be included as perturbations.

Problems

Problem

1. Estimate the energy of the ground level of the helium atom and of helium-like ions (a nucleus of charge Z and two electrons), treating the interaction between the electrons as a perturbation.

Solution. In the ground state of the ion, both electrons are in S states. The unperturbed energy is twice the ground-level energy of a hydrogen-like ion:

$$E^{(0)} = 2(-Z^2/2) = -Z^2.$$

The first-order correction is the mean electron–electron interaction energy evaluated in the state with wave function

$$\psi = \psi_1(r_1)\psi_2(r_2) = \frac{Z^2}{\pi} e^{-Z(r_1+r_2)} \tag{1}$$

(the product of two hydrogenic functions with $l = 0$). The integral

$$E^{(1)} = \iint \psi^2 \frac{1}{r_{12}}\, dV_1 dV_2$$

is most readily calculated as

$$E^{(1)} = 2\int_0^\infty dV_2\, \rho_2 \frac{1}{r_2} \int_0^{r_2} \rho_1\, dV_1, \qquad dV_1 = 4\pi r_1^2 dr_1, \quad dV_2 = 4\pi r_2^2 dr_2$$

(the energy of the charge distribution $\rho_2 = |\psi_2|^2$ in the field of the spherically symmetric distribution $\rho_1 = |\psi_1|^2$; the integrand of the dV_2 integral is the energy of the charge $\rho_2(r_2)$ in the field of the sphere $r_1 < r_2$; the factor 2 before the integral accounts for configurations with $r_1 > r_2$). We thus obtain $E^{(1)} = 5Z/8$, and finally

$$E = E^{(0)} + E^{(1)} = -Z^2 + \frac{5}{8}Z.$$

For helium this gives $-E = 11/4 = 2.75$ (the actual ground-state energy is $-E = 2.90$ atomic units = 78.9 eV).

PROBLEM

2. Do the same by the variational principle, approximating the wave function by the product of two hydrogenic functions with an effective nuclear charge.

SOLUTION. Evaluate

$$\iint \psi \hat{H} \psi \, dV_1 dV_2, \qquad \hat{H} = -\frac{1}{2}(\Delta_1 + \Delta_2) - \frac{Z}{r_1} - \frac{Z}{r_2} + \frac{1}{r_{12}}$$

with the function ψ in equation (1) of the preceding problem, replacing Z there by Z_{eff}. The integral of ψ^2/r_{12} is evaluated as in Problem 1; the integral of $\psi\Delta_1\psi$ can be reduced to that of ψ^2/r_1 by noting that the Schrödinger equation gives

$$\left(-\frac{1}{2}\Delta_1 - \frac{Z_{\text{eff}}}{r_1}\right)\psi_1 = -\frac{1}{2}Z_{\text{eff}}^2\psi_1.$$

The result is

$$\iint \psi \hat{H} \psi \, dV_1 dV_2 = Z_{\text{eff}}^2 - 2ZZ_{\text{eff}} + \frac{5}{8}Z_{\text{eff}}.$$

As a function of Z_{eff}, this expression has its minimum at $Z_{\text{eff}} = Z - 5/16$. The corresponding energy is

$$E = -\left(Z - \frac{5}{16}\right)^2.$$

For the helium atom this gives $-E = 2.85$.

Notice that equation (1), with the value of Z_{eff} just found, is actually the best not only among functions of the form (1), but among all functions depending only on the sum $r_1 + r_2$.

§ 70. The Thomas–Fermi equation

Numerical calculations of the charge and field distributions in an atom by the self-consistent-field method are extremely laborious, especially for complex atoms. For complex atoms in particular, however, there is another approximate method whose merit lies in its simplicity, although it gives results considerably less accurate than those of the self-consistent-field method.

This method (E. Fermi, L. Thomas, 1927) rests on the fact that, in complex atoms containing many electrons, most of the electrons have comparatively large principal quantum numbers. The semiclassical approximation is applicable under these conditions. We may therefore use the notion of "cells in phase space" (§ 48) for the states of the individual atomic electrons.

The phase-space volume corresponding to electrons that have momenta less than p and lie in a physical-space volume element dV is $\frac{4}{3}\pi p^3 dV$.

This volume contains $4\pi p^3 dV/[3(2\pi)^3]$ cells[11], that is, possible states which can contain at most

$$2\frac{4\pi p^3}{3(2\pi)^3}dV = \frac{p^3}{3\pi^2}dV$$

[11] Atomic units are used in this section.

electrons simultaneously (two electrons with mutually opposite spins in each cell). In the normal state of the atom, the electrons present in every volume element dV must occupy, in phase space, the cells corresponding to momenta from zero up to some maximum value p_0. The kinetic energy of the electrons at every point then has the least possible value. If the number of electrons in dV is written as $n\,dV$, where n is the electron number density, the maximum momentum p_0 at each point is related to n by

$$\frac{p_0^3}{3\pi^2} = n.$$

Consequently, the maximum kinetic energy of an electron at a point where the electron density is n is

$$\frac{p_0^2}{2} = \frac{1}{2}(3\pi^2 n)^{2/3}. \tag{70.1}$$

Let $\varphi(r)$ be the electrostatic potential, taken to vanish at infinity. The total energy of an electron is $p^2/2 - \varphi$. Clearly, every electron must have negative total energy; otherwise it would escape to infinity. Denote the maximum total electron energy at every point by $-\varphi_0$, where φ_0 is a positive constant (if it were not constant, electrons would pass from points having smaller φ_0 to points having larger φ_0).

We may thus write

$$\frac{p_0^2}{2} = \varphi - \varphi_0. \tag{70.2}$$

Equating (70.1) and (70.2), we find

$$n = \frac{1}{3\pi^2}[2(\varphi - \varphi_0)]^{3/2}, \tag{70.3}$$

which relates the electron density to the potential at every point of the atom.

When $\varphi = \varphi_0$, the density n vanishes. Evidently, n must also be set equal to zero throughout the region in which $\varphi < \varphi_0$, where (70.2) would give a negative maximum kinetic energy.

The equation $\varphi = \varphi_0$ therefore determines the boundary of the atom. Outside a centrally symmetric charge distribution of zero total charge, however, there is no field. Hence $\varphi = 0$ at the boundary of a neutral atom, and for such an atom the constant φ_0 must be set equal to zero. For an ion, in contrast, φ_0 is nonzero.

We consider a neutral atom below and accordingly put $\varphi_0 = 0$. The electrostatic Poisson equation gives $\Delta\varphi = 4\pi n$; substitution of (70.3) yields the fundamental *Thomas–Fermi equation*

$$\Delta\varphi = \frac{8\sqrt{2}}{3\pi}\varphi^{3/2}. \tag{70.4}$$

The field distribution in the normal state is determined by the centrally symmetric solution of this equation satisfying the following boundary conditions: as $r \to 0$, the field must approach the Coulomb field of the nucleus, so that $\varphi r \to Z$; as $r \to \infty$, $\varphi r \to 0$. Introduce a new variable x in place of r by

$$r = xbZ^{-1/3}, \qquad b = \frac{1}{2}\left(\frac{3\pi}{4}\right)^{2/3} = 0.885, \tag{70.5}$$

and a new unknown function χ in place of φ:

$$\varphi(r) = \frac{Z}{r}\chi\left(\frac{rZ^{1/3}}{b}\right) = \frac{Z^{4/3}}{b}\frac{\chi(x)}{x}\text{[12]}. \tag{70.6}$$

We then obtain

$$x^{1/2}\frac{d^2\chi}{dx^2} = \chi^{3/2} \tag{70.7}$$

with boundary conditions $\chi = 1$ at $x = 0$ and $\chi = 0$ at $x = \infty$. This equation contains no parameters, and hence defines a universal function $\chi(x)$. Table 2 gives this function as obtained by numerical integration of (70.7).

The function $\chi(x)$ decreases monotonically and vanishes only at infinity[13]. In other words, the Thomas–Fermi atom has no boundary and formally extends to infinity. The derivative at $x = 0$ has the value $\chi'(0) = -1.59$. Thus, as $x \to 0$, $\chi(x) \approx 1 - 1.59x$, and the corresponding potential is

$$\varphi(r) \approx Z/r - 1.80\,Z^{4/3}. \tag{70.8}$$

Table 2. Values of the function $\chi(x)$

x	$\chi(x)$	x	$\chi(x)$	x	$\chi(x)$
0.00	1.000	1.4	0.333	6	0.0594
0.02	0.972	1.6	0.298	7	0.0461
0.04	0.947	1.8	0.268	8	0.0366
0.06	0.924	2.0	0.243	9	0.0296
0.08	0.902	2.2	0.221	10	0.0243
0.10	0.882	2.4	0.202	11	0.0202
0.2	0.793	2.6	0.185	12	0.0171
0.3	0.721	2.8	0.170	13	0.0145
0.4	0.660	3.0	0.157	14	0.0125
0.5	0.607	3.2	0.145	15	0.0108
0.6	0.561	3.4	0.134	20	0.0058
0.7	0.521	3.6	0.125	25	0.0035
0.8	0.485	3.8	0.116	30	0.0023
0.9	0.453	4.0	0.108	40	0.0011
1.0	0.424	4.5	0.0919	50	0.00063
1.2	0.374	5.0	0.0788	60	0.00039

The first term is the potential of the nuclear field, and the second is the potential produced by the electrons at the origin. Substitution of (70.6) into (70.3) gives the

[12] In ordinary units: $\varphi(r) = \dfrac{Ze}{r}\chi\left(\dfrac{rZ^{1/3}me^2}{0.885\hbar^2}\right)$.

[13] Equation (70.7) has the exact solution $\chi(x) = 144x^{-3}$, which vanishes at infinity but does not satisfy the boundary condition at $x = 0$. It could be used as an asymptotic expression for $\chi(x)$ at large x. It gives reasonably accurate values, however, only at very large x, whereas the Thomas–Fermi equation itself ceases to be applicable at large distances (see below).

electron density in the form

$$n = Z^2 f\left(\frac{rZ^{1/3}}{b}\right), \qquad f(x) = \frac{32}{9\pi^3}\left(\frac{\chi}{x}\right)^{3/2}. \tag{70.9}$$

We see that in the Thomas–Fermi model the charge-density distributions in different atoms are similar, with

$Z^{-1/3}$ serving as the characteristic length parameter (in ordinary units, $\hbar^2/(me^2Z^{1/3})$, the Bohr radius divided by $Z^{1/3}$). If distances are measured in atomic units, then, in particular, the distances at which the electron density is greatest are the same for all Z. We may therefore say that most of the electrons in an atom of atomic number Z lie at distances of order $Z^{-1/3}$ from the nucleus. Calculation shows that half the total electron charge is contained within a sphere of radius $1.33Z^{-1/3}$.

Similar reasoning shows that the mean electron speed in the atom (regarded in order of magnitude as the square root of the energy) is of order $Z^{2/3}$.

The Thomas–Fermi equation is inapplicable both at excessively small and at excessively large distances from the nucleus. At small r, its range of applicability is bounded by inequality (49.12); at still smaller distances in the Coulomb field of the nucleus, the semiclassical approximation fails. Putting $\alpha = Z$ in (49.12), we find $1/Z$ as the lower limiting distance. The semiclassical approximation also fails in a complex atom at large r. Indeed, it is readily seen that at $r \sim 1$ the de Broglie wavelength of the electron becomes comparable with this distance itself, so the semiclassical condition is completely violated. This can be verified by estimating the terms in (70.2) and (70.4); the result is also evident beforehand, without calculation, since (70.4) contains no Z.

Thus the Thomas–Fermi equation is applicable only at distances large compared with $1/Z$ and small compared with 1. In a complex atom, however, most of the electrons are situated in this region.

The last circumstance means that the "outer boundary" of the atom in the Thomas–Fermi model is at $r \sim 1$, so atomic dimensions are independent of Z. The energy of the outer electrons, and hence the ionization potential of the atom, is likewise independent of Z[14].

The Thomas–Fermi method can be used to calculate the total ionization energy E, that is, the energy required to remove every electron from a neutral atom. For this purpose one must

calculate the electrostatic energy of the Thomas–Fermi charge distribution in the atom. The required total energy equals half this electrostatic energy, since in a system of particles interacting by Coulomb's law, the mean kinetic energy is, by the virial theorem (see I, § 10), minus one half the mean potential energy.

The dependence of E on Z can be found in advance from a simple argument. The electrostatic energy of Z electrons in the field of a nucleus of charge Z, at a mean distance $Z^{-1/3}$ from it, is proportional to $Z \cdot Z/Z^{-1/3} = Z^{7/3}$. Numerical calculation gives $E = 20.8Z^{7/3}$ eV. The dependence on Z agrees well with experimental data; the empirical coefficient, however, is nearer 16.

[14]This model does not, of course, represent the periodic dependence on Z of atomic dimensions and ionization potentials which appears in the periodic system of the elements. Empirical data also show a slight systematic increase of atomic dimensions and decrease of ionization potentials as Z increases.

We have already noted that nonzero positive values of φ_0 correspond to ionized atoms. If χ is defined by $\varphi - \varphi_0 = Z\chi/r$, it obeys the same equation (70.7). We must now consider, however, solutions which vanish not at infinity, as for a neutral atom, but at finite values $x = x_0$; such solutions exist for every x_0. At $x = x_0$ the charge density vanishes together with χ, while the potential remains finite.

The value of x_0 is related to the degree of ionization as follows. By Gauss's theorem, the total charge inside a sphere of radius r is

$$-r^2 \frac{\partial \varphi}{\partial r} = Z[\chi(x) - x\chi'(x)].$$

The total ionic charge z follows on setting $x = x_0$; since $\chi(x_0) = 0$,

$$z = -Zx_0\chi'(x_0). \tag{70.10}$$

In Fig. 23 the heavy curve shows $\chi = \chi(x)$ for a neutral atom, and the two curves below it correspond to ions with different degrees of ionization. Graphically, z/Z is the length intercepted on the ordinate axis by the tangent to a curve at $x = x_0$.

Equation (70.7) also has solutions which nowhere vanish and which diverge at infinity. They may be regarded as corresponding to negative values of φ_0. Two such curves $\chi = \chi(x)$ are also shown in Fig. 23; they lie above the neutral-atom curve. At the point $x = x_1$ where

$$\chi(x_1) - x_1\chi'(x_1) = 0, \tag{70.11}$$

the total charge within the sphere $x < x_1$ vanishes (graphically, this is evidently the point at which the tangent to the curve passes through the origin). If the curve is terminated at this point, it may be said to specify $\chi(x)$ for a neutral atom whose charge density remains nonzero at its boundary. Physically, this corresponds to a "compressed" atom confined within a prescribed finite volume[15].

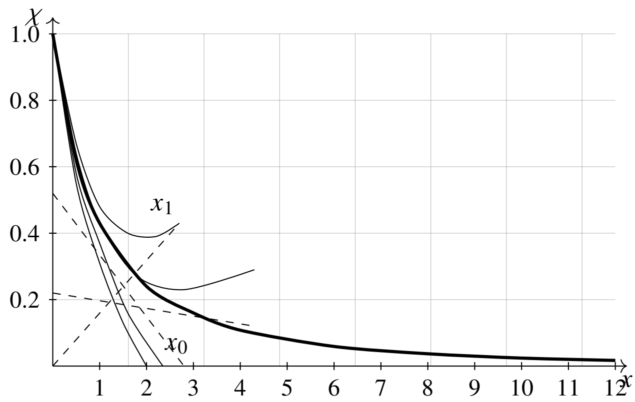


Fig. 23

The Thomas–Fermi equation does not include exchange interaction between the electrons. The effects associated with it are of the next order in $Z^{-2/3}$. Consequently,

[15]This treatment can be useful in studying the equation of state of matter at high compressions.

including exchange interaction in the Thomas–Fermi method requires all effects of this order to be included at the same time[16].

Problem

Find the relation between the electrostatic interaction energy of the electrons with one another and the energy of their interaction with the nucleus in a neutral atom in the Thomas–Fermi model.

Solution. The potential φ_e of the field produced by the electrons is obtained by subtracting the nuclear-field potential Z/r from the total potential φ. Hence the electron–electron interaction energy is

$$U_{ee} = -\frac{1}{2}\int \varphi_e n\, dV = \frac{Z}{2}\int \frac{n}{r}\, dV - \frac{1}{2}\int \varphi n\, dV = \frac{Z}{2}\int \frac{n}{r}\, dV - \frac{(3\pi^2)^{2/3}}{4}\int n^{5/3} dV$$

(where φ has been expressed in terms of n by (70.3)). On the other hand, the interaction energy U_{en} of the electrons with the nucleus and their kinetic energy T are

$$U_{en} = -Z\int \frac{n}{r}\, dV, \qquad T = 2\int\int_0^{p_0} \frac{p^2}{2}\frac{4\pi p^2 dp\, dV}{(2\pi)^3} = \frac{3(3\pi^2)^{2/3}}{10}\int n^{5/3} dV.$$

Comparison of these expressions with the preceding equality gives

$$U_{ee} = -\frac{1}{2}U_{en} - \frac{5}{6}T.$$

At the same time, by the virial theorem (see I, § 10), a system of particles interacting by Coulomb's law obeys $2T = -U = -U_{en} - U_{ee}$. It follows that

$$U_{ee} = -\frac{1}{7}U_{en}.$$

§ 71. Wave functions of outer electrons near the nucleus

We have seen (from the Thomas–Fermi model) that the outer electrons of complex atoms (large Z) are found mainly at distances $r \sim 1$ from the nucleus[17]. A number of atomic properties, however, depend essentially on the electron density close to the nucleus (we shall encounter such properties in §§ 72 and 120). To determine the order of magnitude of this density, let us trace how the electron wave function $\psi(r)$ in an atom changes as r decreases from large distances ($r \sim 1$) to small ones.

In the region $r \sim 1$, the nuclear field is screened by the other electrons, so that the potential energy $U(r) \sim 1/r \sim 1$. The energy E of an electron level in this field is also of order unity. At distances of the order of the Bohr radius in a field of charge Z ($r \sim 1/Z$),

[16] This was done by A. S. Kompaneets and E. S. Pavlovskii (ZhETF. 1956. Vol. 31. P. 427), and by D. A. Kirzhnits (ZhETF. 1957. Vol. 32. P. 115).

[17] Atomic units are used in this section.

on the other hand, the nuclear field may be regarded as unscreened: $U = -Z/r$. In the intermediate region, $1/Z \ll r \ll 1$, the potential energy $|U|$ is already large compared with the electron energy E, and the condition

$$\frac{d}{dr}\frac{1}{p} \sim \frac{d}{dr}\frac{1}{\sqrt{|U|}} \ll 1$$

is satisfied (p is the momentum), so the motion is semiclassical. The spherically symmetric semiclassical wave function is

$$|\psi(r)| \sim \frac{1}{r\sqrt{p}} \sim \frac{1}{r|U|^{1/4}}, \qquad \frac{1}{Z} \ll r \ll 1, \tag{71.1}$$

and the order of magnitude of its coefficient (~ 1) is fixed by matching to the wave function, $\psi \sim 1$, at $r \sim 1$.

Using (71.1) as an order-of-magnitude expression at $r \sim 1/Z$ (with $U = -Z/r$ substituted into it), we obtain the required value of the wave function near the nucleus[18]:

$$\psi\left(\frac{1}{Z}\right) \sim \sqrt{Z}. \tag{71.2}$$

In accordance with the general properties of wave functions in a central field (§ 32), as the distance is reduced further, $\psi(r)$ either remains constant in order of magnitude (for an s electron) or begins to decrease (when $l \neq 0$).

The probability of finding the electron in the region $r \lesssim 1/Z$ is

$$w \sim |\psi|^2 r^3 \sim \frac{1}{Z^2}. \tag{71.3}$$

The formulas (71.2) and (71.3), of course, specify only the systematic trend of these quantities as Z increases, without allowing for nonsystematic changes from one element to the next.

§ 72. Fine structure of atomic energy levels

A systematic derivation of the formulas for relativistic effects in electron interactions belongs to another volume of this course (see IV, §§ 33, 83). Here we give only a general account of these effects as applied to atomic terms.

The relativistic terms in the atomic Hamiltonian fall into two classes: some are linear in the electron spin operators, while others are quadratic in them. The former correspond, as it were, to an interaction between the electrons' orbital motion and their spins; this is called the *spin–orbit interaction*. The latter describe the interaction between the electron spins (the *spin–spin interaction*). Both interactions have the same, second, order in v/c, the ratio of the electron velocity to the velocity of light. In practice, however, in heavy atoms the spin–orbit interaction is considerably greater than the spin–spin interaction.

[18] To determine the coefficient in this formula (given the wave function in the region $r \sim 1$), expression (36.25) would have to be used in the region $r \lesssim 1/Z$.

The reason is that the spin–orbit interaction grows rapidly with atomic number, whereas the spin–spin interaction is, in the main, independent of Z (see below).

The spin–orbit interaction operator has the form

$$\hat{V}_{sl} = \sum_a \hat{\mathbf{s}}_a \hat{\mathbf{A}}_a \tag{72.1}$$

(the sum extends over all electrons in the atom), where $\hat{\mathbf{s}}_a$ are the electron spin operators and $\hat{\mathbf{A}}_a$ are certain "orbital" operators, i.e. operators acting on coordinate functions. In the self-consistent-field approximation the operators $\hat{\mathbf{A}}_a$ are proportional to the electron orbital-angular-momentum operators $\hat{\mathbf{l}}_a$, and $\hat{V}_{sl}$ can then be written

$$\hat{V}_{sl} = \sum_a \alpha_a \hat{\mathbf{s}}_a \hat{\mathbf{l}}_a. \tag{72.2}$$

The coefficients in this sum are expressed in terms of the potential energy $U(r)$ of an electron in the self-consistent field:

$$\alpha_a = \frac{\hbar^2}{2m^2c^2r_a}\frac{dU(r_a)}{dr_a}. \tag{72.3}$$

Since $|U(r)|$ decreases with distance from the nucleus, all the α_a are positive.

Treating (72.2) as a perturbation, we must average it over the unperturbed state to calculate the energy. The principal contribution to this energy comes from the region near the nucleus, at distances of the order of the Bohr radius ($\sim \hbar^2/Zme^2$) for a nucleus of charge Ze. In this region the nuclear field is practically unscreened, and

$$U \sim -\frac{Ze^2}{r} \sim -\frac{Z^2me^4}{\hbar^2},$$

so that

$$\alpha \sim \frac{\hbar^2}{m^2c^2}\frac{U}{r^2} \sim Z^4\left(\frac{e^2}{\hbar c}\right)^2\frac{me^4}{\hbar^2}.$$

The mean value of α follows by multiplying this expression by the probability w of finding the electron near the nucleus. According to (71.3), $w \sim Z^{-2}$, and hence the spin–orbit interaction energy of an electron is ultimately

$$\alpha \sim \left(\frac{Ze^2}{\hbar c}\right)^2\frac{me^4}{\hbar^2},$$

i.e. it differs from the basic energy of an outer electron in the atom ($\sim me^4/\hbar^2$) only by the factor $(Ze^2/\hbar c)^2$. This factor grows rapidly with atomic number and becomes of order unity in heavy atoms.

The actual averaging of the perturbation operator (72.2) over the unperturbed states of an electron shell is carried out in two stages. First we average over the electronic state of an atom having specified magnitudes L and S of its total orbital angular momentum and spin, but not over their directions. After this averaging, $\hat{V}_{sl}$ is still an operator, but one that must now be expressible solely in terms of operators for quantities characterizing the atom as a whole, rather than its individual electrons. These are $\hat{\mathbf{L}}$ and $\hat{\mathbf{S}}$.

Denote the spin–orbit interaction operator averaged in this way by $\hat{V}_{SL}$. Since it is linear in $\hat{\mathbf{S}}$, it has the form

$$\hat{V}_{SL} = A\hat{\mathbf{S}}\hat{\mathbf{L}}, \tag{72.4}$$

where A is a constant characteristic of the given unsplit term: it depends on S and L, but not on the atom's total angular momentum J[19].

To calculate the splitting of the degenerate level we must now solve the secular equation formed from the matrix elements of (72.4). In this case, however, we already know the proper zeroth-order functions in which the matrix of $\hat{V}_{SL}$ is diagonal. They are the wave functions of states with specified total angular momentum J. Averaging over such a state means replacing $\hat{\mathbf{L}}\hat{\mathbf{S}}$ by its eigenvalue, which according to (31.3) is

$$LS = \frac{1}{2}[J(J+1) - L(L+1) - S(S+1)].$$

Since all components of a multiplet have the same L and S, and only their relative positions are of interest, the splitting energy may be written

$$\frac{1}{2}AJ(J+1). \tag{72.5}$$

The intervals between adjacent components, characterized by J and $J-1$, are consequently

$$\Delta E_{J,J-1} = AJ. \tag{72.6}$$

This is the so-called *Landé interval rule* (A. Landé, 1923).

The constant A may be either positive or negative. For $A > 0$, the lowest component of the multiplet level has the smallest possible J, namely $J = |L - S|$; such multiplets are called *normal*. If $A < 0$, the lowest level has $J = L + S$ (an *inverted multiplet*).

The sign of A for normal atomic states is readily found when the electron configuration contains only one incompletely filled shell. If this shell is no more than half filled, Hund's rule (§ 67) says that all its n electrons have parallel spins, so that the total spin has its greatest possible value, $S = n/2$. Substituting $\mathbf{s}_a = \mathbf{S}/n$ in (72.2) and taking α_a, the same for all electrons in one shell, outside the sum gives

$$\hat{V}_{SL} = \frac{\alpha}{2S}\hat{\mathbf{S}}\hat{\mathbf{L}},$$

and hence $A = \alpha/2S > 0$. If the shell is more than half filled, first add and subtract in (72.2) the analogous sum over the vacant places—the holes in the incomplete shell. Since a completely filled shell would have $\hat{V}_{sl} = 0$, the operator becomes $\hat{V}_{sl} = -\sum_a \alpha_a \hat{\mathbf{l}}_a \hat{\mathbf{s}}_a$, summed only over the holes, while the atom's total spin and orbital angular momentum are $\mathbf{S} = -\sum_a \mathbf{s}_a$, $\mathbf{L} = -\sum_a \mathbf{l}_a$. The same argument then gives $A = -\alpha/2S < 0$.

[19] To clarify the meaning of this operation, recall that averaging in quantum mechanics generally means taking the corresponding diagonal matrix element. Partial averaging consists in forming the set of matrix elements diagonal in only some of the quantum numbers specifying the system's state. Thus, here averaging (72.2) means forming the matrix with elements $(nM'_L M'_S|\hat{V}_{sl}|nM_L M_S)$ for all possible M_L, M'_L, M_S, M'_S, diagonal in all the remaining quantum numbers (collectively denoted by n). Correspondingly, $\hat{\mathbf{S}}$ and $\hat{\mathbf{L}}$ are to be understood as the matrices $(M'_S|\mathbf{S}|M_S)$ and $(M'_L|\mathbf{L}|M_L)$, whose elements are given by (27.13). We shall repeatedly use this procedure of averaging in stages below.

It follows that a simple rule determines J in the normal state of an atom with one incompletely filled shell. If that shell contains no more than half the maximum possible number of electrons, then $J = |L - S|$. If it is more than half filled, then $J = L + S$.

As already mentioned, unlike the spin–orbit interaction, the spin–spin interaction is in the main independent of Z. This is clear from its nature as a direct interaction of electrons with one another, unrelated to the nuclear field.

By analogy with (72.4), the averaged spin–spin interaction operator must be quadratic in $\hat{\mathbf{S}}$. The quadratic expressions are S^2 and $(\hat{\mathbf{L}}\hat{\mathbf{S}})^2$. The eigenvalues of the first are independent of J, so it does not split the term. It may therefore be omitted, and we write

$$\hat{V}_{ss} = B(\hat{\mathbf{S}}\hat{\mathbf{L}})^2, \tag{72.7}$$

where B is a constant. The eigenvalues of this operator contain terms independent of J, terms proportional to $J(J+1)$, and finally a term proportional to $J^2(J+1)^2$. The first do not cause splitting and are immaterial; the second may be absorbed into (72.5), which merely changes the constant A. The last give the following contribution to the term energy:

$$\frac{1}{4}BJ^2(J+1)^2. \tag{72.8}$$

The construction of atomic levels described in §§ 66, 67 rests on the assumption that the electron orbital angular momenta combine to form the atom's total orbital angular momentum L, while their spins combine to form its total spin S. As noted above, this description is possible only when relativistic effects are small; more precisely, the fine-structure intervals must be small compared with the separations between levels having different L and S. This approximation is called the *Russell–Saunders case* (H. Russell, F. Saunders, 1925), or LS coupling.

Its actual range of applicability is limited. The levels of light atoms follow LS coupling; as atomic number increases, relativistic interactions in the atom become stronger and the Russell–Saunders approximation ceases to apply[20]. In particular, the approximation also fails for highly excited levels in which one electron has a large n and is therefore mainly at large distances from the nucleus (§ 68). The electrostatic interaction of this electron with the motion of the others is relatively weak, whereas the relativistic interaction in the ionic core does not decrease.

At the opposite limiting case the relativistic interaction is large compared with the electrostatic interaction—more precisely, with the part responsible for the dependence of the energy on L and S. Orbital angular momentum and spin can then no longer be considered separately, since neither is conserved. Individual electrons are characterized by their total angular momenta j, which combine to form the atom's total angular momentum J. This scheme is called jj coupling. In fact, pure jj coupling is not encountered; the levels of very heavy atoms exhibit various forms intermediate between LS and jj coupling[21].

[20]Although the quantitative formulas describing this coupling cease to apply, classification of the levels by this scheme may still be meaningful for heavier atoms, especially for the lowest states, including the normal state.

[21]For further details on coupling types and the quantitative theory, see E. Condon and G. Shortley, *The Theory of Atomic Spectra*.

A distinctive coupling occurs in some highly excited states. The ionic core may then be in a Russell–Saunders state, characterized by L and S, while its coupling to the highly excited electron is of the jj type (again because the electrostatic interaction is weak for this electron).

The fine structure of hydrogen-atom energy levels has certain special features; it is calculated in another volume of this course (see IV, § 34). Here we note only that, for a given principal quantum number n, the energy depends solely on the electron's total angular momentum j. Thus the degeneracy is not completely removed: a level with given n and j corresponds to two states with orbital angular momenta $l = j \pm 1/2$ (unless j has its greatest possible value for the given n, $j = n - 1/2$). Thus the $n = 3$ level splits into three levels: one corresponds to the states $s_{1/2}$ and $p_{1/2}$, another to $p_{3/2}$ and $d_{3/2}$, and the third to $d_{5/2}$.

§ 73. Mendeleev's periodic system of the elements

To explain the origin of the periodic variation of properties established by D. I. Mendeleev (1869) in the sequence of elements arranged by increasing atomic number, we must consider the features of the successive filling of atomic electron shells (N. Bohr, 1922).

In passing from one atom to the next, the nuclear charge increases by one and one electron is added to the shell. At first sight, one might expect the binding energies of the successively added electrons to vary monotonically as the atomic number increases. In fact they do not.

The normal state of the hydrogen atom contains a single electron in the $1s$ state. In the atom of the next element, helium, another electron is added in the same $1s$ state. The binding energy of each $1s$ electron in helium is, however, considerably greater than that of the electron in hydrogen. This follows naturally from the difference between the field experienced by the electron in H and that entered by an electron added to He^+: at large distances the fields are approximately the same, but near the nucleus the field of the He^+ ion with $Z = 2$ is stronger than that of the hydrogen nucleus with $Z = 1$.

In lithium ($Z = 3$), the third electron enters the $2s$ state, since no more than two electrons can occupy $1s$ states simultaneously. At a given Z the $2s$ level lies above $1s$; as the nuclear charge increases, both levels descend. In passing from $Z = 2$ to $Z = 3$, however, the first effect greatly outweighs the second, and the binding energy of the third electron in Li is much smaller than that of the electrons in helium. From Be ($Z = 4$) through Ne ($Z = 10$), another $2s$ electron and then six $2p$ electrons are added in succession. Owing to the increasing nuclear charge, the binding energies of the electrons added along this sequence generally increase. The next electron, added in passing to Na ($Z = 11$), enters a $3s$ state; the effect of passing to a higher shell outweighs the increased nuclear charge, and the binding energy again falls sharply.

This pattern of shell filling is characteristic of the whole sequence of elements. All electron states may be divided into groups filled successively: while a group is filled along the sequence, the binding energy generally increases, but it drops sharply when filling of the next group begins.

Figure 24 shows the ionization potentials of the elements known from spectroscopic data; they determine the binding energies of the electrons added in passing from each element to the next.

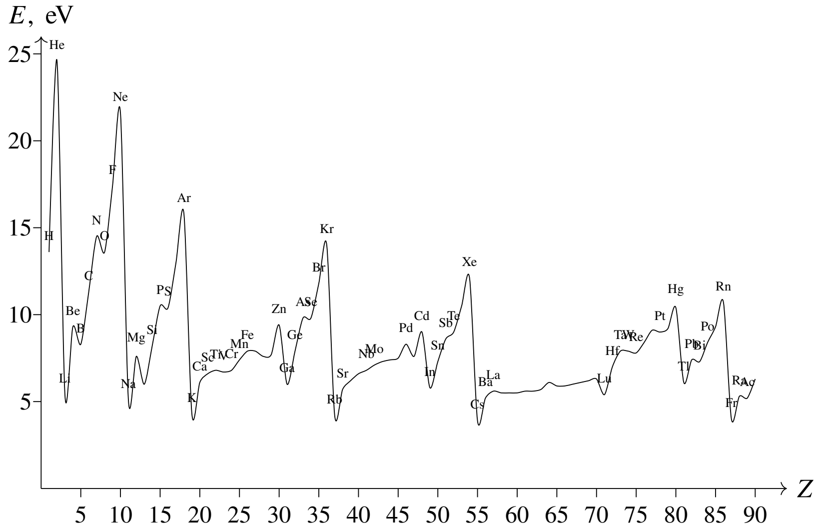


Figure 24

The states are distributed among successively filled groups as follows:

$$
\begin{array}{r}
1s \text{ 2 electrons,} \\
2s,\,,2p \text{ 8 electrons,} \\
3s,\,,3p \text{ 8 electrons,} \\
4s,\,,3d,\,,4p \text{ 18 electrons,} \\
5s,\,,4d,\,,5p \text{ 18 electrons,} \\
6s,\,,4f,\,,5d,\,,6p \text{ 32 electrons,} \\
7s,\,,6d,\,,5f \ldots
\end{array}
\tag{73.1}
$$

The first group is filled in H and He; filling of the second and third corresponds to the first two short periods of the periodic system, each containing eight elements. Next come two long periods of 18 elements and a long period that includes the rare-earth elements and contains 32 elements in all. The last group is not filled completely in naturally occurring elements or in artificial transuranic elements.

To understand the variation of properties as the states of each group are filled, the following feature of d and f states, which distinguishes them from s and p states, is important. After a rapid, almost vertical fall near the origin, the effective-potential-energy curves of the central field (the electrostatic plus centrifugal field) for an electron in a heavy atom have a deep minimum and then rise toward zero asymptotically. For s and p states these curves lie very close together on their rising portions. Thus the electron lies at approximately the same distances from the nucleus in these states. The curves for d and especially f states lie much farther to the left; the classically accessible region bounded by them ends much nearer the nucleus than for s and p states at the same total electron energy. In other words, in d and f states the electron is mainly much closer to the nucleus than in s and p states.

A number of atomic properties, including the chemical properties of the elements (see § 81), depend mainly on the outer regions of the electron shells. The feature of d and f states just described is therefore very important. When $4f$ states are filled in the rare-earth elements (see below), the added electrons lie much closer to the nucleus than electrons in the states already filled. Consequently,

these electrons have almost no effect on chemical properties, and all rare-earth elements are chemically very similar.

Elements with filled d and f shells, or with no such shells, are called *main-group elements*; those in which these states are being filled are called *transition-group elements*. It is convenient to consider the two groups separately.

We begin with the main-group elements. Hydrogen and helium have the normal states

$$_1\mathrm{H}\ 1s\ ^2S_{1/2}; \qquad _2\mathrm{He}\ 1s^2\ ^1S_0$$

(the subscript to the left of a chemical symbol always denotes the atomic number). The electron configurations of the remaining main-group elements are shown in Table 3.

Table 3. Electron configurations of the main-group elements

	s	s^2	s^2p	s^2p^2	s^2p^3	s^2p^4	s^2p^5	s^2p^6	
$n = 2$	$_3$Li	$_4$Be	$_5$B	$_6$C	$_7$N	$_8$O	$_9$F	$_{10}$Ne	$1s^2$
3	$_{11}$Na	$_{12}$Mg	$_{13}$Al	$_{14}$Si	$_{15}$P	$_{16}$S	$_{17}$Cl	$_{18}$Ar	$2s^22p^6$
4	$_{19}$K	$_{20}$Ca							$3s^23p^6$
4	$_{29}$Cu	$_{30}$Zn	$_{31}$Ga	$_{32}$Ge	$_{33}$As	$_{34}$Se	$_{35}$Br	$_{36}$Kr	$3d^{10}$
5	$_{37}$Rb	$_{38}$Sr							$4s^24p^6$
5	$_{47}$Ag	$_{48}$Cd	$_{49}$In	$_{50}$Sn	$_{51}$Sb	$_{52}$Te	$_{53}$I	$_{54}$Xe	$4d^{10}$
6	$_{55}$Cs	$_{56}$Ba							$5s^25p^6$
6	$_{79}$Au	$_{80}$Hg	$_{81}$Tl	$_{82}$Pb	$_{83}$Bi	$_{84}$Po	$_{85}$At	$_{86}$Rn	$4f^{14}5d^{10}$
7	$_{87}$Fr	$_{88}$Ra							$6s^26p^6$
	$^2S_{1/2}$	$^1S_0$	$^2P_{1/2}$	$^3P_0$	$^4S_{3/2}$	$^3P_2$	$^2P_{3/2}$	$^1S_0$	

In each atom the shells shown in the right-hand column on the same line, and those on every line above it, are completely filled. The configuration in the shells being filled is shown at the top; the principal quantum number for these states is the number in the left-hand column on the same line. The normal states of the whole atom are shown at the bottom. Thus Al has the configuration $1s^22s^22p^63s^23p\ ^2P_{1/2}$.

The values of L and S in the normal atomic state can be found from the electron configuration by

Hund's rule (§ 67), and J by the rule given in § 72.

The noble-gas atoms (He, Ne, Ar, Kr, Xe, Rn) occupy a special position: each completes the filling of one of the groups in (73.1). Their electron configurations are particularly stable (their ionization potentials are the largest in the corresponding sequences), which accounts for their chemical inertness.

The filling of states among the main-group elements is very regular: first the s and then the p states for each principal quantum number n are filled. The electron configurations of their ions are equally regular, so long as ionization does not affect electrons in d and f shells: each ion has the configuration of the preceding atom. Thus Mg^+ has the Na configuration, and Mg^{++} that of Ne.

We next consider transition-group elements. The $3d$, $4d$, and $5d$ shells are filled in the groups called, respectively, the *iron*, *palladium*, and *platinum* groups.

Table 4 gives the experimentally determined electron configurations and terms of the atoms in these groups. Filling of the d shells is plainly much less regular than filling of s and p shells in main-group atoms. A characteristic feature is the competition between s and d states. Instead of the regular sequence $d^p s^2$ with increasing p, configurations $d^{p+1}s$ or d^{p+2} are often more favorable. In the iron group, for example, Cr is $3d^5 4s$, not $3d^4 4s^2$; Ni, with eight d electrons, is followed directly by Cu with a completely filled d shell (and thus assigned above to the main groups). The terms of the ions show a similar irregularity: ionic configurations usually do not coincide with those of the preceding atoms. Thus V^+ is $3d^4$, not $3d^2 4s^2$ as in Ti, and Fe^+ is $3d^6 4s$, rather than the $3d^5 4s^2$ configuration of Mn. All ions occurring naturally in crystals and solutions have only d electrons, and no s or p electrons, in their unfilled shells. Iron, for example, occurs in crystals and solutions only as Fe^{++} and Fe^{+++}, with configurations $3d^6$ and $3d^5$ respectively.

Table 4. Electron configurations of atoms in the iron, palladium, and platinum groups

Iron group							
$_{21}$Sc	$_{22}$Ti	$_{23}$V	$_{24}$Cr	$_{25}$Mn	$_{26}$Fe	$_{27}$Co	$_{28}$Ni
$3d4s^2$	$3d^2 4s^2$	$3d^3 4s^2$	$3d^5 4s$	$3d^5 4s^2$	$3d^6 4s^2$	$3d^7 4s^2$	$3d^8 4s^2$
$^2D_{3/2}$	$^3F_2$	$^4F_{3/2}$	$^7S_3$	$^6S_{5/2}$	$^5D_4$	$^4F_{9/2}$	$^3F_4$
Palladium group							
$_{39}$Y	$_{40}$Zr	$_{41}$Nb	$_{42}$Mo	$_{43}$Tc	$_{44}$Ru	$_{45}$Rh	$_{46}$Pd
$4d5s^2$	$4d^2 5s^2$	$4d^4 5s$	$4d^5 5s$	$4d^5 5s^2$	$4d^7 5s$	$4d^8 5s$	$4d^{10}$
$^2D_{3/2}$	$^3F_2$	$^6D_{1/2}$	$^7S_3$	$^6S_{5/2}$	$^5F_5$	$^4F_{9/2}$	$^1S_0$
Platinum group							
$_{57}$La							
$5d6s^2$							
$^2D_{3/2}$							
$_{71}$Lu	$_{72}$Hf	$_{73}$Ta	$_{74}$W	$_{75}$Re	$_{76}$Os	$_{77}$Ir	$_{78}$Pt
$5d6s^2$	$5d^2 6s^2$	$5d^3 6s^2$	$5d^4 6s^2$	$5d^5 6s^2$	$5d^6 6s^2$	$5d^7 6s^2$	$5d^9 6s$
$^2D_{3/2}$	$^3F_2$	$^4F_{3/2}$	$^5D_0$	$^6S_{5/2}$	$^5D_4$	$^4F_{9/2}$	$^3D_3$

The same situation occurs while the $4f$ shell is filled along the series called the *rare-earth elements* (Table 5)[22]. Filling of the $4f$ shell is likewise not entirely regular, being characterized by competition among the $4f$, $5d$, and $6s$ states.

The last group of transition elements begins with actinium. Its $6d$ and $5f$ shells are filled in a manner analogous to the rare-earth series (Table 6).

We conclude this section with an interesting application of the Thomas–Fermi method. We have seen that p-shell electrons first occur in the fifth element (B), d electrons at $Z = 21$ (Sc), and f electrons at $Z = 58$ (Ce). These values of Z can be predicted by the Thomas–Fermi method as follows.

An electron of orbital angular momentum l in a complex atom moves with an

[22] Chemistry courses usually include Lu among the rare-earth elements as well. This is incorrect, however, since its $4f$ shell is already filled; Lu belongs to the platinum group, as in Table 4.

Table 5. Electron configurations of rare-earth atoms

$_{58}$Ce	$_{59}$Pr	$_{60}$Nd	$_{61}$Pm	$_{62}$Sm	$_{63}$Eu	
$4f5d6s^2$	$4f^36s^2$	$4f^46s^2$	$4f^56s^2$	$4f^66s^2$	$4f^76s^2$	
$^1G_4$	$^4I_{9/2}$	$^5I_4$	$^6H_{5/2}$	$^7F_0$	$^8S_{7/2}$	
$_{64}$Gd	$_{65}$Tb	$_{66}$Dy	$_{67}$Ho	$_{68}$Er	$_{69}$Tm	$_{70}$Yb
$4f^75d6s^2$	$4f^96s^2$	$4f^{10}6s^2$	$4f^{11}6s^2$	$4f^{12}6s^2$	$4f^{13}6s^2$	$4f^{14}6s^2$
$^9D_2$	$^6H_{15/2}$	$^5I_8$	$^4I_{15/2}$	$^3H_6$	$^2F_{7/2}$	$^1S_0$

Table 6. Electron configurations of atoms in the actinide group

$_{89}$Ac	$_{90}$Th	$_{91}$Pa	$_{92}$U	$_{93}$Np	$_{94}$Pu	$_{95}$Am	$_{96}$Cm
$6d7s^2$	$6d^27s^2$	$5f^26d7s^2$	$5f^36d7s^2$	$5f^46d7s^2$	$5f^67s^2$	$5f^77s^2$	$5f^76d7s^2$
$^2D_{3/2}$	$^3F_2$	$^4K_{11/2}$	$^5L_6$	$^6L_{11/2}$	$^7F_0$	$^8S_{7/2}$	$^9D_2$

"effective potential energy"[23]

$$U_l(r) = -\varphi(r) + \frac{(l+1/2)^2}{2r^2}.$$

The first term is the potential energy in the electric field described by the Thomas–Fermi potential $\varphi(r)$. The second is the centrifugal energy, in which $(l+1/2)^2$ is used instead of

$l(l+1)$ because the motion is quasiclassical. Since the electron's total energy in the atom is negative, if $U_l(r) > 0$ for all r at given Z and l, the atom cannot contain electrons with this l. If l is fixed and Z varied, then for sufficiently small Z one indeed has $U_l(r) > 0$ everywhere. As Z increases, a point is reached at which the curve $U_l = U_l(r)$ touches the abscissa axis; at larger Z there is a region in which $U_l(r) < 0$. Thus electrons of a given l first appear when the curve $U_l(r)$ is tangent to the axis, i.e. when

$$U_l(r) = -\varphi(r) + \frac{(l+1/2)^2}{2r^2} = 0, \qquad U_l'(r) = -\varphi'(r) - \frac{(l+1/2)^2}{r^3} = 0.$$

Substitution of (70.6) for the potential gives

$$\begin{aligned} Z^{2/3}\frac{\chi(x)}{x} &= \left(\frac{2}{3\pi}\right)^{2/3} \frac{(l+1/2)^2}{2x^2}, \\ Z^{2/3}\frac{x\chi'(x)-\chi(x)}{x^2} &= -2\left(\frac{2}{3\pi}\right)^{2/3} \frac{(l+1/2)^2}{2x^3}. \end{aligned} \tag{73.2}$$

Dividing the second equation term by term by the first gives

$$\frac{\chi'(x)}{\chi(x)} = -\frac{1}{x},$$

after which the first of (73.2) determines Z. Numerical calculation gives

$$Z = 0.155(2l+1)^3.$$

[23] Atomic units are used, as in § 70.

This formula determines the values of Z at which electrons with a given l first appear, with an error of about 10%. The exact integer values result if 0.155 is replaced by 0.17:

$$Z = 0.17(2l + 1)^3. \tag{73.3}$$

For $l = 1, 2, 3$, rounding to the nearest integers gives precisely 5, 21, and 58. For $l = 4$, (73.3) gives $Z = 124$; thus g electrons would first appear only in element 124.

§ 74. X-ray terms

The binding energy of the inner electrons in an atom is so large that, if such an electron passes into an unfilled outer shell (or is removed from the atom altogether), the excited atom (or ion) is mechanically unstable against ionization accompanied by rearrangement of the electron shell and formation of a stable ion. Nevertheless, because electron interactions in the atom are relatively weak, the probability of this transition is rather small, and the lifetime τ of the excited state is long. The level "width" $\hbar/\tau$ (see § 44) is consequently small enough for the energies of an atom with an excited inner electron to be regarded as discrete energy levels of quasistationary atomic states. These levels are called *X-ray terms*[24].

X-ray terms are classified first by specifying the shell from which the electron has been removed, or in which, as one says, a hole has been formed. The electron's destination has almost no effect on the atomic energy and is therefore immaterial.

The total angular momentum of the electrons filling a shell is zero. Removal of one electron leaves the shell with an angular momentum j. For a shell (n, l), j can take the values $l \pm 1/2$. The resulting levels could thus be denoted $1s_{1/2}$, $2s_{1/2}$, $2p_{1/2}$, $2p_{3/2}$, ..., with j written as a subscript on the symbol specifying the hole's location. Special symbols are, however, conventionally used, with the correspondence

$$\begin{array}{ccccccccc} 1s_{1/2} & 2s_{1/2} & 2p_{1/2} & 2p_{3/2} & 3s_{1/2} & 3p_{1/2} & 3p_{3/2} & 3d_{3/2} & 3d_{5/2} \\ K & L_{\mathrm{I}} & L_{\mathrm{II}} & L_{\mathrm{III}} & M_{\mathrm{I}} & M_{\mathrm{II}} & M_{\mathrm{III}} & M_{\mathrm{IV}} & M_{\mathrm{V}} \end{array}$$

Levels with $n = 4, 5, 6$ are similarly denoted by the letters N, O, P.

Levels with the same n, denoted by the same capital letter, lie close together and far from levels having other values of n. This is because the relative proximity of the inner electrons to the nucleus makes their field nearly the unscreened nuclear field. Their states are therefore hydrogen-like, and

to first approximation their energy is $-Z^2/2n^2$ (in atomic units), so it depends only on n.

Allowance for relativistic effects separates terms having different j (compare the discussion of the fine structure of hydrogen levels in § 72), for example L_{I} and L_{II} from L_{III}, and M_{I} and M_{II} from M_{III} and M_{IV}. Such pairs are called *relativistic doublets*.

The separation of terms with different l but the same j (for example, L_{I} from L_{II}, or M_{I} from M_{II}) is due to the departure of the inner-electron field from the Coulomb field of the nucleus, i.e. to the electron's interaction with the other electrons. These doublets are called *screening doublets*. The leading correction to the electron's hydrogen-like

[24] The name reflects the fact that transitions between these levels cause the atom to emit X-rays.

energy comes from the potential produced by the other electrons near the nucleus; it is proportional to $Z^{4/3}$ (see (70.8)). Since this correction depends on neither n nor l, however, it does not affect the intervals between levels. The leading corrections to level differences are therefore associated with the interaction of one electron with those nearest to it. The distances between inner electrons are $r \sim 1/Z$ (the Bohr radius in a field of charge Z), so this interaction energy is $\sim 1/r \sim Z$. To this accuracy an X-ray term energy may thus be written $-(Z-\delta)^2/2n^2$, where $\delta = \delta(n,l)$ is small compared with Z and may be regarded as a measure of screening of the nuclear charge.

Besides X-ray terms with one hole in the electron shells, terms with two and three holes may also exist. Since the spin–orbit interaction is strong for inner electrons, the holes are coupled to one another according to jj coupling.

The width of an X-ray term is determined by the total probability of all possible electron-shell rearrangements that fill the given hole. In heavy atoms the principal processes are transitions of the hole from the given shell to a higher one (that is, reverse electron transitions), accompanied by emission of an X-ray quantum. The probabilities of these radiative transitions, and hence the corresponding part of the level width, increase very rapidly, as Z^4, with atomic number, but at fixed Z decrease in passing from deeper to less deep levels.

For lighter atoms, and for higher levels, an important or even predominant part is played by

nonradiative transitions, in which the energy released when the hole is filled by an electron from a higher level is used to eject another inner electron from the atom (the *Auger effect*); the atom is then left with two holes. To first approximation in $1/Z$, the probabilities of these processes and their contribution to the level width do not depend on atomic number (see the problem)[25].

Problem

Find the limiting dependence of the Auger width of X-ray terms on atomic number when the latter is sufficiently large.

Solution. The probability of an Auger transition is proportional to the square of a matrix element of the form

$$M = \iint \psi_1'^* \psi_2'^* V \psi_1 \psi_2 \, dV_1 dV_2,$$

where ψ_1, ψ_2 and ψ_1', ψ_2' are the initial and final wave functions of the two electrons taking part in the transition, and $V = e^2/r_{12}$ is their interaction energy. For sufficiently large Z, the wave functions of the inner electrons may be treated as hydrogen-like and the screening of the nuclear field by the other electrons neglected (the wave function of the ionized electron is likewise hydrogen-like in the region deep inside the atom that is important for the integral M). If the calculation is made with all quantities expressed in Coulomb units (see § 36), the only Z-dependent quantity in M is $V = e^2/r_{12} = 1/Zr_{12}$,

[25] For example, the Auger width of the K level is about 1 eV, while for higher levels it reaches values of about 10 eV.

and hence $M \sim 1/Z$. The transition probability, and with it the Auger width ΔE, is proportional to Z^{-2}. Returning to ordinary units (the Coulomb unit of energy is $Z^2 me^4/\hbar^2$), we find that ΔE is independent of Z.

§ 75. Multipole moments

In classical theory the electrical properties of a system are characterized by its multipole moments of various orders, expressed in terms of the charges and coordinates of the particles. In quantum theory these quantities retain the same definitions, but must be regarded as operators.

The first multipole moment is the *dipole moment*, the vector

$$\mathbf{d} = \sum e\mathbf{r}$$

(the sum is over all particles in the system; the particle index is omitted for brevity). The matrix of

this operator, like that of any polar vector (see § 30), has nonzero elements only for transitions between states of different parity. All its diagonal elements therefore vanish. In other words, the mean dipole moment of any system of particles, for example an atom, is zero in every stationary state.[26]

The same clearly applies to all 2^l-pole moments with odd l. Their components are odd polynomials of degree l in the coordinates and, like a polar-vector component, change sign under coordinate inversion; the same parity selection rule therefore applies.

The *quadrupole moment* of a system is defined as the symmetric tensor

$$\widehat{Q}_{ik} = \sum e(3x_i x_k - \delta_{ik}\mathbf{r}^2), \tag{75.1}$$

whose diagonal sum is zero. To determine these quantities in a particular state of a system, say an atom, the operator (75.1) must be averaged with the corresponding wave function. It is useful to perform this averaging in two stages (cf. § 72).

Let $\widehat{Q}_{ik}$ denote the quadrupole-moment operator averaged over electron states having a specified total angular momentum J, but not a specified projection M_J. The resulting operator can depend only on operators describing the state of the atom as a whole. The only such vector is $\widehat{\mathbf{J}}$. Hence

$$\widehat{Q}_{ik} = \frac{3Q}{2J(2J-1)}\left(\widehat{J}_i\widehat{J}_k + \widehat{J}_k\widehat{J}_i - \frac{2}{3}\widehat{\mathbf{J}}^2\delta_{ik}\right), \tag{75.2}$$

where the expression in parentheses is symmetric in i, k and contracts to zero over those indices (the meaning of Q is given below). The opera-

tors $\widehat{J}_i$ are the familiar matrices (§ 27, 54) between states of different M_J; $\widehat{\mathbf{J}}^2$ may of course be replaced by its eigenvalue $J(J+1)$.

[26]To avoid misunderstanding, we emphasize that this statement concerns a closed system of particles, or particles in a centrally symmetric external electric field. Thus, if the nuclei are regarded as "fixed," the general assertion applies to the electron system of an atom, but not of a molecule.

It is also assumed that the energy level has no additional, or "accidental," degeneracy besides degeneracy with respect to the direction of the total angular momentum. Otherwise stationary-state wave functions without definite parity can be formed, and the corresponding diagonal elements of the dipole moment need not vanish.

Since the three components of $\mathbf{J}$ cannot have definite values simultaneously, neither can the components of Q_{ik}. For Q_{zz},

$$\widehat{Q}_{zz} = \frac{3Q}{J(2J-1)}\left(\widehat{J_z^2} - \frac{1}{3}\widehat{\mathbf{J}^2}\right).$$

In a state with $\mathbf{J}^2 = J(J+1)$ and $J_z = M_J$, Q_{zz} also has a definite value:

$$Q_{zz} = \frac{3Q}{J(2J-1)}\left[M_J^2 - \frac{1}{3}J(J+1)\right]. \tag{75.3}$$

For $M_J = J$, when the angular momentum is directed "entirely" along z, $Q_{zz} = Q$; this quantity is ordinarily called simply the quadrupole moment.

For $J = 0$ all angular-momentum matrix elements vanish, and so do the operators (75.2). They also vanish identically for $J = 1/2$, as follows directly by multiplying the Pauli matrices (55.7), which represent the components of any angular momentum equal to $1/2$.

This is a special case of the general rule that a 2^l-pole tensor with even l is nonzero only for states whose total angular momentum satisfies

$$J \geqslant l/2. \tag{75.4}$$

Such a multipole tensor is an irreducible tensor of rank l (see II, § 41), and (75.4) follows from the general angular-momentum selection rules for its matrix elements: it is the condition under which diagonal elements may be nonzero (§ 107). As noted above, parity also requires l to be even.

Electric multipole moments are purely "orbital" quantities: their operators contain no spin operators. Thus, if spin–orbit interaction can be neglected and L, S are separately conserved, their matrix elements obey selection rules in L as well as J.

Problem

1. Find the relation between the atomic quadrupole-moment operators in states belonging to different fine-structure components of a level, that is, different J for fixed L, S.

Solution. For fixed L, S, the quadrupole operator, being purely orbital, depends only on $\widehat{\mathbf{L}}$ and has the form (75.2) with $\widehat{\mathbf{J}}$ replaced by $\widehat{\mathbf{L}}$ and with another constant Q. Equation (75.2) results by further averaging over a state of specified J:

$$\begin{aligned}\widehat{Q}_{ik} &= \frac{3Q_J}{2J(J-1)}[\widehat{J}_i\widehat{J}_k + \widehat{J}_k\widehat{J}_i - \tfrac{2}{3}J(J+1)\delta_{ik}] \\ &= \frac{3Q_L}{2L(L-1)}\overline{[\widehat{L}_i\widehat{L}_k + \widehat{L}_k\widehat{L}_i - \tfrac{2}{3}L(L+1)\delta_{ik}]}.\end{aligned} \tag{1}$$

The displayed denominators are retained exactly as printed in the source. To relate Q_J, Q_L, multiply (1) on the left by $\widehat{J}_i$ and on the right by $\widehat{J}_k$, summing over i, k, and pass to eigenvalues of the diagonal operators. Then

$$\widehat{J}_i\widehat{L}_i\widehat{L}_k\widehat{J}_k = (\mathbf{JL})^2, \qquad 2\mathbf{JL} = J(J+1) + L(L+1) - S(S+1).$$

Using

$$\{\hat{L}_i, \hat{L}_k\} = ie_{ikl}\hat{L}_l, \qquad \{\hat{J}_i, \hat{L}_l\} = ie_{ilm}\hat{L}_m,$$

as in the problem to § 29 gives

$$\hat{J}_i\hat{L}_k\hat{L}_i\hat{J}_k = (\mathbf{JL})^2 - (\mathbf{JL}),$$

and similarly

$$\hat{J}_i\hat{J}_i\hat{J}_k\hat{J}_k = (\mathbf{J}^2)^2, \qquad \hat{J}_i\hat{J}_k\hat{J}_i\hat{J}_k = \mathbf{J}^2(\mathbf{J}^2 - 1).$$

Thus

$$Q_J = Q_L \frac{3(\mathbf{JL})(2\mathbf{JL} - 1) - 2J(J+1)L(L+1)}{(J+1)(2J+3)L(2L-1)}. \tag{2}$$

For $S = 1/2$,

$$\begin{aligned} Q_J &= Q_L, & J &= L + \tfrac{1}{2}, \\ Q_J &= Q_L \frac{(L-1)(2L+3)}{L(2L+1)}, & J &= L - \tfrac{1}{2}. \end{aligned} \tag{3}$$

Problem

2. Express the quadrupole moment of an electron of charge $-|e|$ and orbital angular momentum l in terms of its mean-square distance from the centre.

Solution. Average

$$Q_{zz} = -|e|\,\overline{r^2}(3\cos^2\theta - 1) = -|e|\,\overline{r^2}(3n_z^2 - 1)$$

in the state with angular momentum l and projection $m = l$. The angular factor follows directly from the formula obtained in the problem to § 29, with $\hat{l}_z$ replaced by l:

$$Q_l = |e|\,\overline{r^2}\frac{2l}{2l+3}. \tag{1}$$

Its sign is opposite to the electron charge, as it should be: a particle whose angular momentum points along z is found mainly near the plane $z = 0$, and hence $\cos^2\theta < 1/3$. For specified $j = l \pm 1/2$, (3) gives

$$Q_j = |e|\,\overline{r^2}\frac{2j-1}{2j+2}. \tag{2}$$

Problem

3. Determine the quadrupole moment, in its ground state, of an atom in which all ν electrons outside closed shells occupy equivalent states of orbital angular momentum l.

Solution. Since closed shells have zero total quadrupole moment,

$$\hat{Q}_{ik} = \frac{3|e|\,\overline{r^2}}{(2l-1)(2l+3)}\sum[\hat{l}_i\hat{l}_k + \hat{l}_k\hat{l}_i - \tfrac{2}{3}l(l+1)\delta_{ik}],$$

summed over the ν outer electrons (formula (4) was used here).

First suppose $\nu \leqslant 2l + 1$, so at most half the shell is occupied. By Hund's rule (§ 67), all ν spins are parallel, $S = \nu/2$. The spin wave function is symmetric, and the coordinate wave function is therefore antisymmetric in these electrons. Their m values must all differ, giving

$$L = (M_L)_{\max} = \sum_{m=l-\nu+1}^{l} m = \tfrac{1}{2}\nu(2l - \nu + 1).$$

The required Q_L is the Q_{zz} eigenvalue at $M_L = L$:

$$Q_L = \frac{6|e|\overline{r^2}}{(2l-1)(2l+3)} \sum_{m=l-\nu+1}^{l} [m^2 - l(l+1)/3],$$

whence

$$Q_L = \frac{2l(2l - 2\nu + 1)}{(2l-1)(2l+3)}|e|\overline{r^2}. \tag{1}$$

The final passage from Q_L to Q_J uses (2).

For a shell more than half full, use holes instead of electrons. The answer is the same formula (6) with the overall sign reversed, since a hole has charge $+|e|$, and ν now denotes the number of vacancies rather than electrons.

§ 76. Atom in an electric field

When an atom is placed in an external electric field, its energy levels change; this is the *Stark effect*.

In a uniform external electric field the electrons move in an axially symmetric field, formed by the nucleus and the external field. Strictly, the total angular momentum is no longer conserved; only its projection M_J on the field direction remains conserved. States with different M_J have different energies, so the field removes the directional degeneracy. The removal is incomplete: states differing only in the sign of M_J still have equal energies. Indeed, the atom is symmetric under reflection in any plane through the symmetry axis, chosen below as z. States related by such a reflection must have equal energies, while reflection in a plane through an axis reverses the angular momentum about that axis.

Assume the field is weak enough that its additional energy is small compared with separations of neighbouring atomic levels, including fine-structure intervals. The shifts may then be calculated by perturbation theory (§ 38, 39). The perturbation is the energy of the electron system in the uniform field $\boldsymbol{\mathcal{E}}$:

$$\widehat{V} = -\mathbf{d}\boldsymbol{\mathcal{E}} = -\mathcal{E} d_z, \tag{76.1}$$

where $\mathbf{d}$ is the dipole moment. Zeroth-order levels are directionally degenerate, but this degeneracy is immaterial here, and perturbation theory may be applied as for nondegenerate levels. Only matrix elements of d_z for transitions with unchanged M_J can be nonzero (§ 29), so states with different M_J behave independently.

The first-order shift is given by diagonal elements of the perturbation, but diagonal dipole-moment elements vanish (§ 75). Hence the electric-field splitting is second order in the field.[27] Being quadratic in the field, the shift of level E_n has the form

$$\Delta E_n = -\frac{1}{2}\alpha_{ik}^{(n)}\mathcal{E}_i\mathcal{E}_k, \tag{76.2}$$

where $\alpha_{ik}^{(n)}$ is symmetric. With z along the field,

$$\Delta E_n = -\frac{1}{2}\alpha_{zz}^{(n)}\mathcal{E}^2. \tag{76.3}$$

The tensor $\alpha_{ik}^{(n)}$ is also the *polarizability* of the atom. In (11.16), take the parameters λ as the components $\mathcal{E}_i$ and put $\widehat{H} = \widehat{H}_0 - \mathcal{E}_i\widehat{d}_i$. Then the induced mean dipole moment is

$$\overline{d}_i^{(n)} = -\frac{\partial\Delta E_n}{\partial\mathcal{E}_i},$$

so that

$$\overline{d}_i^{(n)} = \alpha_{ik}^{(n)}\mathcal{E}_k. \tag{76.4}$$

By the general second-order formula (38.10),

$$\alpha_{ik}^{(n)} = -2\sum_m{}' \frac{(d_i)_{nm}(d_k)_{mn}}{E_n - E_m}. \tag{76.5}$$

The polarizability depends on the unperturbed state, including M_J. Its dependence on M_J follows generally by regarding the values $\alpha_{ik}^{(n)}$ as eigenvalues of

$$\widehat{\alpha}_{ik}^{(n)} = \alpha_n\delta_{ik} + \beta_n(\widehat{J}_i\widehat{J}_k + \widehat{J}_k\widehat{J}_i - \tfrac{2}{3}\delta_{ik}\widehat{\mathbf{J}}^2). \tag{76.6}$$

This is the general symmetric rank-two tensor depending on $\widehat{\mathbf{J}}$ (cf. § 75). Thus

$$\Delta E_n = -\frac{\mathcal{E}^2}{2}\{\alpha_n + 2\beta_n[M_J^2 - \tfrac{1}{3}J(J+1)]\}. \tag{76.7}$$

Summing over all M_J makes the second term vanish; the first therefore shifts the centre of gravity of the split level. For $J = 1/2$ the level remains unsplit, consistently with Kramers' theorem (§ 60).

In a nonuniform external field that varies little across the atom, a splitting linear in the field may also arise from the atomic quadrupole moment. The quadrupole interaction operator has the classical form (see II, § 42)

$$\widehat{V} = \frac{1}{6}\frac{\partial^2\varphi}{\partial x_i\partial x_k}\widehat{Q}_{ik}, \tag{76.8}$$

where φ is the electric potential and the derivatives are evaluated at the atom.

[27] Hydrogen is an exception: its Stark effect is linear in the field (see the next section). Other atoms in highly excited, and therefore hydrogen-like, states (§ 68) behave similarly in sufficiently strong fields.

Problem

1. Determine how the Stark splitting of the components of a multiplet level depends on J.

Solution. Reverse the order in which the perturbations are introduced: first consider Stark splitting without fine structure, then introduce spin–orbit interaction. Since spin does not interact with the external electric field, a level of orbital angular momentum L splits according to (76.2), with

$$\widehat{\alpha}_{ik} = a\delta_{ik} + b(\widehat{L}_i\widehat{L}_k + \widehat{L}_k\widehat{L}_i - \tfrac{2}{3}\delta_{ik}\widehat{\mathbf{L}}^2).$$

After spin–orbit interaction is introduced, average this operator over states of specified J but unspecified M_J, exactly as in problem 1 of § 75. Equations (76.6), (76.7) result, with

$$\alpha = a, \qquad \beta = b\frac{3(\mathbf{JL})[2(\mathbf{JL})-1] - 2J(J+1)L(L+1)}{J(J+1)(2J-1)(2J+3)}.$$

This determines the J dependence, though not the L, S dependence of a, b.

Problem

2. Determine the splitting of a doublet level ($S = 1/2$) in an arbitrary, not necessarily weak, electric field.

Solution. If the splitting is not small relative to the doublet interval, the electric-field perturbation and spin–orbit interaction must be included together:

$$\widehat{V} = A\widehat{\mathbf{S}}\widehat{\mathbf{L}} - \tfrac{1}{2}\mathcal{E}^2\{a + 2b[\widehat{L}_z^2 - \tfrac{1}{3}L(L+1)]\}.$$

Dropping terms common to all components,

$$\widehat{V} = \frac{A}{2}[\widehat{S}_+\widehat{L}_- + \widehat{S}_-\widehat{L}_+ + 2\widehat{S}_z\widehat{L}_z] - b\mathcal{E}^2\widehat{L}_z^2.$$

For fixed $M = M_J$, form the secular equation in the states $|M_L M_S\rangle = |M \mp 1/2, \pm 1/2\rangle$. Its elements are

$$\langle M-1/2, 1/2|V|M-1/2, 1/2\rangle = \frac{A}{2}(M-\tfrac{1}{2}) - b\mathcal{E}^2(M-\tfrac{1}{2})^2,$$
$$\langle M+1/2, -1/2|V|M+1/2, -1/2\rangle = -\frac{A}{2}(M+\tfrac{1}{2}) - b\mathcal{E}^2(M+\tfrac{1}{2})^2,$$
$$\langle M-1/2, 1/2|V|M+1/2, -1/2\rangle = \frac{A}{2}[(L+M+\tfrac{1}{2})(L-M+\tfrac{1}{2})]^{1/2}.$$

Hence (problem 1, § 39)

$$\Delta E = -b\mathcal{E}^2M^2 \pm \sqrt{\frac{A^2}{4}(L+\tfrac{1}{2})^2 + b\mathcal{E}^2(b\mathcal{E}^2 + A)M^2}. \tag{1}$$

Terms common to the doublet have been omitted. Both signs apply for $|M| \leqslant L - 1/2$. For $|M| = L + 1/2$ only one state exists, and with the same additive constant

$$\Delta E = (\tfrac{A}{2} + b\mathcal{E}^2)(L+\tfrac{1}{2}) - b\mathcal{E}^2(L+\tfrac{1}{2})^2. \tag{2}$$

Problem

3. Determine the quadrupole splitting in an axially symmetric electric field.[28]

Solution. Here

$$\varphi_{xx} = \varphi_{yy} = a, \qquad \varphi_{zz} = -2a,$$

and the other second derivatives vanish. Equation (76.8) becomes

$$\frac{a}{6}(\widehat{Q}_{xx} + \widehat{Q}_{yy} - 2\widehat{Q}_{zz}) = \frac{Qa}{2J(2J-1)}(\widehat{\mathbf{J}}^2 - 3\widehat{J}_z^2),$$

and therefore

$$\Delta E = a\frac{Q}{2J(2J-1)}[J(J+1) - 3M_J^2].$$

Problem

4. Calculate the polarizability of hydrogen in its ground state.

Solution. Spherical symmetry makes $\alpha_{ik} = \alpha\delta_{ik}$, with

$$\alpha = -2e^2 \sum_k{}' \frac{|z_{0k}|^2}{E_0 - E_k}.$$

Introduce $\widehat{b}$ by $z = (m/\hbar)d\widehat{b}/dt$. Then $z_{0k} = (im/\hbar^2)(E_0 - E_k)b_{0k}$ and

$$\alpha = \frac{2ime^2}{\hbar^2}\sum_k z_{0k}b_{k0} = \frac{2ime^2}{\hbar^2}(zb)_{00}. \tag{1}$$

It suffices to know $\widehat{b}\psi_0$. From (9.2), writing $\widehat{b}\psi_0 = b(\mathbf{r})\psi_0$ and using $\widehat{H} = -\hbar^2\Delta/2m + U(r)$,

$$\tfrac{1}{2}\psi_0\Delta b + \nabla b\nabla\psi_0 = iz\psi_0.$$

With $b = \cos\theta f(r)$ this becomes

$$\frac{f''}{2} + \frac{f'}{r} - \frac{f}{r^2} + \frac{\psi_0'}{\psi_0}f' = ir. \tag{2}$$

The solution must keep $f\psi_0$ finite at zero and infinity. For hydrogen's ground state $\psi_0 = e^{-r/a_B}/\sqrt{\pi}$, $a_B = \hbar^2/me^2$, the solution is $f = -ira_B(a_B + r/2)$. Therefore[29]

$$\alpha = \frac{2i}{a_B}(rf\cos^2\theta)_{00} = \frac{2i}{3a_B}(rf)_{00} = \frac{9}{2}a_B^3.$$

[28] For an arbitrary field see problem 6, § 103.

[29] The next section obtains this result by another method.

PROBLEM

5. Calculate the polarizability of an electron in a bound s state in a potential well of range a, where $a\kappa \ll 1$, $\kappa = \sqrt{2m|E_0|}/\hbar$, and $|E_0|$ is the binding energy.

SOLUTION. The region inside the well may be neglected in $(z\widehat{b})_{00}$, and throughout space one may use

$$\psi_0 = \sqrt{\frac{\kappa}{2\pi}}\frac{e^{-\kappa r}}{r}.$$

Equation (2) becomes

$$\frac{f''}{2} - \kappa f' - \frac{f}{r^2} = ir,$$

whose boundary-condition solution is $f = -ir^2/2\kappa$. Equation (1) then gives

$$\alpha = \frac{me^2}{4\hbar^2\kappa^4}.$$

§ 77. Hydrogen atom in an electric field

Unlike the levels of other atoms, hydrogen levels in a uniform electric field undergo a splitting proportional to the first power of the field, the *linear Stark effect*. This follows from the accidental degeneracy of hydrogenic terms: states of different l have equal energies for fixed n. Dipole-moment matrix elements between these states need not vanish, and the secular equation therefore gives a nonzero first-order shift.[30]

Choose the unperturbed wave functions so that the perturbation matrix is diagonal within each degenerate group. This is accomplished by quantizing hydrogen in parabolic coordinates; the functions $\psi_{n_1 n_2 m}$ are given by (37.15), (37.16).

The perturbation is $\mathcal{E}z = \mathcal{E}(\xi - \eta)/2$ (the field points along positive z, while the force on the electron points along negative z).[31] Among transitions $n_1 n_2 m \to n_1' n_2' m'$ with unchanged energy, only diagonal elements are nonzero:

$$\begin{aligned}\int |\psi_{n_1 n_2 m}|^2 \mathcal{E}z\, dV &= \frac{\mathcal{E}}{8}\iiint (\xi^2 - \eta^2)|\psi|^2\, d\varphi\, d\xi\, d\eta \\ &= \frac{\mathcal{E}n}{4}\iint f_{n_1 m}^2(\rho_1) f_{n_2 m}^2(\rho_2)(\rho_1^2 - \rho_2^2)\, d\rho_1 d\rho_2.\end{aligned} \tag{77.1}$$

Here $\xi = n\rho_1$, $\eta = n\rho_2$. Diagonality in m is evident, and that in n_1, n_2 follows from orthogonality of the $f_{n_i m}$. Using integral (f.6) gives[32]

$$E^{(1)} = \frac{3}{2}\mathcal{E}n(n_1 - n_2), \tag{77.2}$$

[30] The following calculation neglects the fine structure of hydrogen levels. The field must be weak enough for perturbation theory, but strong enough that the Stark splitting exceeds the fine structure. For the opposite case see problem 1 in vol. IV, § 52.

[31] Atomic units are used in this section.

[32] Schwarzschild and Epstein (K. Schwarzschild, P. Epstein, 1916) obtained this result from the old quantum theory, and Pauli and Schrödinger (1926) from quantum mechanics.

or in ordinary units

$$E^{(1)} = \frac{3}{2}n(n_1 - n_2)|e|\mathcal{E}\frac{\hbar^2}{me^2}.$$

The extreme components have $(n_1, n_2) = (n-1, 0)$ and $(0, n-1)$; their separation is $3\mathcal{E}n(n-1)$. Thus the total splitting is approximately proportional to n^2, naturally increasing as the electron lies farther from the nucleus and the atomic dipole moment grows.

The linear effect means that the unperturbed atom has mean dipole moment

$$\overline{d}_z = -\frac{3}{2}n(n_1 - n_2). \tag{77.3}$$

This agrees with the asymmetry of a parabolic-coordinate state about $z = 0$ (§ 37). For $n_1 > n_2$ the electron lies mainly at positive z, so the atomic dipole moment opposes the field.

A uniform field cannot remove the degeneracy between states with projections $\pm m$. Equation (77.2) shows that the linear effect leaves still more degeneracy, since the shift is independent of m. Second order removes it further and is especially relevant when $n_1 = n_2$, for which the linear effect vanishes.

Rather than evaluating infinite complicated sums, use a modified method. The Schrödinger equation

$$\frac{1}{2}\Delta\psi + (E + 1/r - \mathcal{E}z)\psi = 0$$

separates in parabolic coordinates. The substitution (37.7) gives

$$\begin{aligned}
&\frac{d}{d\xi}(\xi f_1') + \left(\tfrac{1}{2}E\xi - \tfrac{m^2}{4\xi} - \tfrac{1}{4}\mathcal{E}\xi^2\right)f_1 = -\beta_1 f_1,\\
&\frac{d}{d\eta}(\eta f_2') + \left(\tfrac{1}{2}E\eta - \tfrac{m^2}{4\eta} + \tfrac{1}{4}\mathcal{E}\eta^2\right)f_2 = -\beta_2 f_2, \qquad \beta_1 + \beta_2 = 1.
\end{aligned} \tag{77.4}$$

Treat E as fixed and β_1, β_2 as eigenvalues of the corresponding self-adjoint operators. Their dependence on $E, \mathcal{E}$, together with $\beta_1 + \beta_2 = 1$, determines the energy.

For $\mathcal{E} = 0$,

$$f_1 = f_{n_1 m}(s\xi), \qquad f_2 = f_{n_2 m}(s\eta), \tag{77.5}$$

where

$$s = \sqrt{-2E}, \tag{77.6}$$

and

$$\beta_1^{(0)} = \left(n_1 + \tfrac{|m|+1}{2}\right)s, \qquad \beta_2^{(0)} = \left(n_2 + \tfrac{|m|+1}{2}\right)s. \tag{77.7}$$

The f_i are normalized to unity and orthogonal for different n_i. First order gives

$$\beta_1^{(1)} = \frac{\mathcal{E}}{4s^2}(6n_1^2 + 6n_1|m| + m^2 + 6n_1 + 3|m| + 2),$$

and $\beta_2^{(1)}$ follows by $n_1 \to n_2$ and reversal of sign. In second order,

$$\beta_1^{(2)} = -\frac{\mathcal{E}^2}{16}\sum_{n_1' \neq n_1}\frac{|(\xi^2)_{n_1 n_1'}|^2}{\beta_1^{(0)}(n_1) - \beta_1^{(0)}(n_1')}.$$

Only

$$(\xi^2)_{n_1,n_1-1} = -\frac{4}{s^2}(2n_1 + |m|)\sqrt{n_1(n_1 + |m|)},$$
$$(\xi^2)_{n_1,n_1-2} = \frac{2}{s^2}\sqrt{n_1(n_1 - 1)(n_1 + |m|)(n_1 + |m| - 1)}$$

and their transposes are nonzero, while the denominators equal $s(n_1 - n_1')$. Hence

$$\beta_1^{(2)} = -\frac{\mathcal{E}^2}{16s^5}(|m| + 2n_1 + 1)[4m^2 + 17(2|m|n_1 + 2n_1^2 + |m| + 2n_1) + 18],$$

with $n_1 \to n_2$ for $\beta_2^{(2)}$. Substitution in $\beta_1 + \beta_2 = 1$ yields

$$sn - \frac{\mathcal{E}^2 n}{16s^5}[17n^2 + 5(n_1 - n_2)^2 - 9m^2 + 19] + \frac{3\mathcal{E}n}{2s^2}(n_1 - n_2) = 1,$$

and successive solution gives

$$E = -\frac{1}{2n^2} + \frac{3}{2}\mathcal{E}n(n_1 - n_2) - \frac{\mathcal{E}^2 n^4}{16}[17n^2 - 3(n_1 - n_2)^2 - 9m^2 + 19]. \tag{77.8}$$

The third term is the quadratic Stark effect and is always negative. Differentiation gives, for $n_1 = n_2$,

$$\overline{d}_z = \frac{n^4}{8}(17n^2 - 9m^2 + 19)\mathcal{E}. \tag{77.9}$$

For $n = 1, m = 0$ the polarizability is $9/2$.

At large n the binding energy decreases rapidly while Stark splitting grows. Fields for which the splitting is comparable with the level energy require a semiclassical treatment.[33]

Put

$$f_1 = \chi_1/\sqrt{\xi}, \qquad f_2 = \chi_2/\sqrt{\eta}. \tag{77.10}$$

Then

$$\chi_1'' + (\tfrac{E}{2} + \tfrac{\beta_1}{\xi} - \tfrac{m^2-1}{4\xi^2} - \tfrac{\mathcal{E}}{4}\xi)\chi_1 = 0,$$
$$\chi_2'' + (\tfrac{E}{2} + \tfrac{\beta_2}{\eta} - \tfrac{m^2-1}{4\eta^2} + \tfrac{\mathcal{E}}{4}\eta)\chi_2 = 0. \tag{77.11}$$

These are one-dimensional Schrödinger equations of energy $E/4$ and potentials

$$U_1(\xi) = -\frac{\beta_1}{2\xi} + \frac{m^2 - 1}{8\xi^2} + \frac{\mathcal{E}}{8}\xi,$$
$$U_2(\eta) = -\frac{\beta_2}{2\eta} + \frac{m^2 - 1}{8\eta^2} - \frac{\mathcal{E}}{8}\eta. \tag{77.12}$$

[33] For high levels, perturbation theory requires the perturbation to be small relative to the level's binding energy, not the spacing between levels. In the semiclassical case this is equivalent to requiring the perturbing force to be small relative to the unperturbed forces.

Figures 25 and 26 show their approximate forms for $m > 1$. Bohr–Sommerfeld quantization gives

$$\int_{\xi_1}^{\xi_2} \sqrt{2[E/4 - U_1]}\, d\xi = (n_1 + 1/2)\pi,$$
$$\int_{\eta_1}^{\eta_2} \sqrt{2[E/4 - U_2]}\, d\eta = (n_2 + 1/2)\pi. \tag{77.13}$$

[34] Together with $\beta_1 + \beta_2 = 1$, these equations determine the shifted levels numerically.

Strong-field Stark splitting is accompanied by field ionization (C. Lanczos, 1931). Since $\mathcal{E}z \to -\infty$ as $z \to -\infty$, the allowed region includes both the atomic interior and large distances toward the anode.

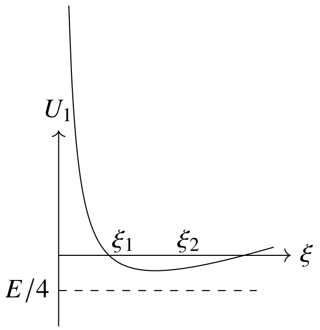


Figure 25

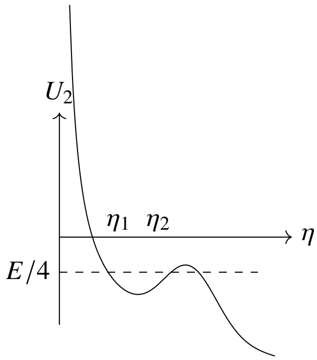


Figure 26

The regions are separated by a barrier whose width decreases with field. Tunnelling through it is ionization. The probability is negligible in weak fields but grows exponentially.[35]

Problem

1. Determine the ionization probability per unit time for ground-state hydrogen when $\mathcal{E} \ll 1$.

Solution. [36] The barrier lies along η. Without the field,

$$\psi = \pi^{-1/2} \exp[-(\xi + \eta)/2]. \tag{1}$$

The ξ dependence may be retained, while $\chi = \sqrt{\eta}\psi$ obeys

$$\chi'' + (-\tfrac{1}{4} + \tfrac{1}{2\eta} + \tfrac{1}{4\eta^2} + \tfrac{1}{4}\mathcal{E}\eta)\chi = 0. \tag{2}$$

[34] A more detailed analysis shows that replacing $m^2 - 1$ by m^2 in U_1, U_2 is more accurate; n_1, n_2 then coincide with the parabolic quantum numbers.

[35] This illustrates how a small perturbation can alter the character of the energy spectrum. Any nonzero field creates a barrier and makes the remote region accessible, rendering the motion strictly infinite and the spectrum continuous. Perturbation theory nevertheless gives physically meaningful energies of almost stationary states, but its series is asymptotic rather than convergent.

[36] Atomic units are used in this problem.

Matching semiclassically at $1 \ll \eta_0 \ll 1/\mathcal{E}$ gives, beyond the barrier,

$$\chi = (\eta_0 |p_0|/\pi p)^{1/2} \exp\left[-(\xi+\eta_0)/2 + i\int_{\eta_0}^{\eta} p\, d\eta + i\pi/4\right],$$

where $p = [-1/4 + 1/(2\eta) + 1/(4\eta^2) + \mathcal{E}\eta/4]^{1/2}$. If $p(\eta_1) = 0$,

$$|\chi|^2 = \frac{\eta_0|p_0|}{\pi p} \exp\left[-\xi - 2\int_{\eta_0}^{\eta_1} |p| d\eta - \eta_0\right]. \tag{3}$$

Keeping the next exponential term yields

$$|\chi|^2 = \frac{4}{\pi\mathcal{E}\sqrt{\mathcal{E}\eta - 1}} \exp(-\xi - 2/3\mathcal{E}). \tag{4}$$

The outward probability flux gives

$$w = \frac{4}{\mathcal{E}} \exp(-2/3\mathcal{E}), \tag{5}$$

or

$$w = \frac{4m^3|e|^9}{\mathcal{E}\hbar^7} \exp\left(-\frac{2m^2|e|^5}{3\mathcal{E}\hbar^4}\right).$$

Problem

2. Determine the probability of removing an electron by an electric field from a short-range potential well in which it occupies a bound s state, assuming $|e|\mathcal{E} \ll \hbar^2\kappa^3/m$, with $\kappa = \sqrt{2m|E|}/\hbar$ (Yu. N. Demkov, G. F. Drukarev, 1964).

Solution. At large distances

$$\psi = A\sqrt{\kappa}\, e^{-\kappa r}/r,$$

where A depends on the well.[37] For $\eta \gg \xi$,

$$\psi \approx \frac{2A\sqrt{\kappa}}{\eta} \exp[-\kappa(\xi + \eta)/2]. \tag{6}$$

Measure mass, length, and time in units m, $1/\kappa$, and $m/\hbar\kappa^2$. Separation gives the second of (77.11) with $E = -1/2$, $m = 0$, $\beta_1 = 1/2$, $\beta_2 = -1/2$. Repeating problem 1 gives

$$w = \pi A^2 \mathcal{E} \exp(-2/3\mathcal{E}),$$

or

$$w = \frac{\pi|e|\mathcal{E}A^2}{\hbar\kappa} \exp\left(-\frac{2\hbar^2\kappa^3}{3m|e|\mathcal{E}}\right).$$

[37] If $a\kappa \ll 1$, then $A = (2\pi)^{-1/2}$; see § 133.

PROBLEM

3. Estimate to exponential accuracy the probability of removing an electron from a potential well by the uniform alternating field $\mathcal{E} = \mathcal{E}_0 \cos \omega t$, assuming $\hbar\omega \ll |E|$ and $|e|\mathcal{E}_0 \ll \hbar^2\kappa^3/m$ (L. V. Keldysh, 1964).[38]

SOLUTION. To exponential accuracy the motion may be treated as one-dimensional along z. Use $A_z = -(c\mathcal{E}_0/\omega)\sin\omega t$ and

$$\widehat{H} = \frac{1}{2m}\left(-i\hbar\partial_z + \frac{|e|\mathcal{E}_0}{\omega}\sin\omega t\right)^2.$$

With

$$\tau = \frac{\hbar\kappa^2}{2m}t, \quad \eta = 2\kappa z, \quad \Omega = \frac{2m\omega}{\hbar\kappa^2}, \quad F = \frac{|e|m\mathcal{E}_0}{\hbar^2\kappa^3},$$

the equation is

$$\frac{i}{4}\Psi_\tau = -\left(\partial_\eta + \frac{iF}{\Omega}\sin\Omega\tau\right)^2\Psi,$$

with

$$\Psi \to e^{i\tau} \quad (\eta \to 0). \tag{7}$$

Seek $\Psi = e^{iS}$, where

$$S = -\int_{\tau_0}^{\tau} H(p,\tau')d\tau' + \eta p + A, \qquad H = 4(p + F\sin\Omega\tau/\Omega)^2, \tag{8}$$

and

$$\eta = \int_{\tau_0}^{\tau} \frac{\partial H}{\partial p}d\tau'. \tag{9}$$

Taking $A(\tau_0) = \tau_0$ and imposing

$$\partial S/\partial\tau_0 = 0 \tag{10}$$

gives

$$H(p,\tau_0) + 1 = 0. \tag{11}$$

At emergence into the classically allowed region $p = 0$, so

$$\Omega\tau_0 = i\,\mathrm{arsinh}\,\gamma, \qquad \gamma = \frac{\Omega}{2F} = \frac{\sqrt{2m|E|}}{|e|\mathcal{E}_0}\omega.$$

Evaluation of the imaginary action gives

$$w \sim \exp\left[-\frac{2|E|}{\hbar\omega}f(\gamma)\right], \qquad f(\gamma) = \left(1 + \frac{1}{2\gamma^2}\right)\mathrm{arsinh}\,\gamma - \frac{\sqrt{1+\gamma^2}}{2\gamma}. \tag{12}$$

The limiting forms are

$$f(\gamma) \approx 2\gamma/3 \quad (\gamma \ll 1), \qquad f(\gamma) \approx \ln 2\gamma - 1/2 \quad (\gamma \gg 1).$$

The first corresponds to a constant field. Formula (12) applies when its exponent is large; in particular, $\hbar\omega \ll |E|$ is required.

[38] For example, this may describe ionization of a singly charged negative ion by a strong light wave; the well is produced by interaction with the neutral ionic core. The first condition permits classical treatment of the wave field.

CHAPTER XI

The Diatomic Molecule

§ 78. Electronic terms of a diatomic molecule

The great disparity between atomic nuclear masses and the electron mass is fundamental to molecular theory. Owing to this mass difference, nuclei in a molecule move slowly in comparison with the electrons. We may therefore study the electronic motion with the nuclei held fixed at prescribed separations. The energy levels U_n found for this system are called the molecule's *electronic terms*. For atoms, the energy levels were definite numbers; here, by contrast, the electronic terms are functions of parameters—the separations of the nuclei in the molecule. The quantity U_n also includes the electrostatic interaction energy of the nuclei. Thus U_n is, in substance, the total energy of the molecule for a specified configuration of fixed nuclei.

We begin the study of molecules with the simplest type, the *diatomic* molecule, for which the fullest theoretical investigation is possible. Its electronic terms depend on just one parameter: the internuclear distance r.

One of the principal ways of classifying atomic terms used the values of the total orbital angular momentum L. For molecules, however, the total orbital angular momentum of the electrons is not conserved in general, since the electric field of several nuclei has no central symmetry.

In a diatomic molecule the field does, nevertheless, possess axial symmetry about the line through both nuclei. The projection of the orbital angular momentum on this axis is consequently conserved, and molecular electronic terms can be classified by its values. The magnitude of this projection on the molecular axis is conventionally denoted by Λ; its possible values are $0, 1, 2, \ldots$ Terms with different Λ are designated by the capital Greek letters corresponding to the Latin symbols for atomic

terms having different L. Thus the cases $\Lambda = 0, 1, 2$ are called, respectively, Σ-, Π-, and Δ-terms; larger values of Λ ordinarily need not be considered.

Every electronic state of a molecule is further characterized by the total spin S of all its electrons. When S is nonzero, the different orientations of the total spin give a degeneracy of multiplicity $2S + 1$[1]. As for atoms, the number $2S + 1$ is called the term's *multiplicity* and is placed to the upper left of the term symbol. For example, $^3\Pi$ denotes a term with $\Lambda = 1$, $S = 1$.

Besides rotations through an arbitrary angle about the axis, the symmetry of the molecule permits reflection in any plane containing that axis. Such a reflection leaves the molecular energy unchanged, although the resulting state need not be wholly identical to the initial one. Under reflection in a plane through the molecular axis, the sign of the angular momentum about that axis changes (angular momentum is an axial vector). It follows that every electronic term with nonzero Λ is doubly degenerate: each

[1]We disregard here the fine structure arising from relativistic interactions (see below, § 83, 84).

energy belongs to two states that differ in the direction of the orbital-angular-momentum projection on the molecular axis. When $\Lambda = 0$, on the other hand, reflection does not alter the molecular state, so Σ-terms are nondegenerate. Reflection can only multiply the wave function of a Σ-term by a constant. Since two reflections in the same plane amount to the identity transformation, this constant is ± 1. We must therefore distinguish the Σ-terms whose wave function remains entirely unchanged under reflection from those whose wave function reverses sign. The former are denoted by Σ^+ and the latter by Σ^-.

For a molecule composed of two identical atoms, a further symmetry arises and supplies an additional characteristic of its electronic terms. A diatomic molecule with identical nuclei also has a center of symmetry at the midpoint of the line joining them (we choose this point as the coordinate origin)[2]. The Hamiltonian is therefore invariant under simultaneous reversal of the coordinates of all the molecular electrons, with the nuclear co-

ordinates left unchanged. The operator of this transformation[3] also commutes with the orbital-angular-momentum operator. Hence terms of definite Λ can be classified additionally by parity: the wave function of an *even* (g) state is unchanged when the electron coordinates reverse sign, whereas that of an *odd* (u) state changes sign. The parity indices u, g are conventionally written as subscripts on the term symbol: Π_u, Π_g, and so forth.

Empirically, the normal electronic state of the overwhelming majority of chemically stable diatomic molecules is known to be totally symmetric: the electronic wave function is invariant under every symmetry transformation of the molecule. In the great majority of cases, the total spin S also vanishes in the normal state. In other words, the molecular ground term is ${}^1\Sigma^+$, or ${}^1\Sigma_g^+$ when the molecule consists of identical atoms. Familiar exceptions to these rules are the molecules O_2 (normal term ${}^3\Sigma_g^-$) and NO (normal term ${}^2\Pi$).

Problem

Separate the variables in the Schrödinger equation for the electronic terms of the H_2^+ ion, using elliptic coordinates.

Solution. The Schrödinger equation for an electron in the field of two fixed protons is

$$\frac{\Delta\psi}{2} + \left(E + \frac{1}{r_1} + \frac{1}{r_2}\right)\psi = 0$$

(atomic units are used). Elliptic coordinates are defined by

$$\xi = \frac{r_1 + r_2}{R}, \qquad \eta = \frac{r_2 - r_1}{R}, \qquad 1 \leqslant \xi < \infty, \quad -1 \leqslant \eta \leqslant 1,$$

and the third coordinate is the angle φ of rotation about the axis through the two nuclei,

[2] It also has a symmetry plane perpendicular to the molecular axis and bisecting it. There is no need to treat this symmetry element separately, since the existence of such a plane follows automatically from the presence of a center and an axis of symmetry.

[3] This must not be confused with inversion of the coordinates of every particle in the system (compare § 86)!

whose separation is R (see I, § 48). In these coordinates the Laplacian is

$$\Delta = \frac{4}{R^2(\xi^2-\eta^2)}\left[\frac{\partial}{\partial\xi}\left[(\xi^2-1)\frac{\partial}{\partial\xi}\right]+\frac{\partial}{\partial\eta}\left[(1-\eta^2)\frac{\partial}{\partial\eta}\right]\right]+\frac{1}{R^2(\xi^2-1)(1-\eta^2)}\frac{\partial^2}{\partial\varphi^2}.$$

Setting

$$\psi = X(\xi)Y(\eta)e^{i\Lambda\varphi},$$

we obtain the following equations for X and Y:

$$\frac{d}{d\xi}\left[(\xi^2-1)\frac{dX}{d\xi}\right]+\left(\frac{ER^2\xi^2}{2}+2R\xi+A-\frac{\Lambda^2}{\xi^2-1}\right)X=0,$$

$$\frac{d}{d\eta}\left[(1-\eta^2)\frac{dY}{d\eta}\right]+\left(-\frac{ER^2\eta^2}{2}-A-\frac{\Lambda^2}{1-\eta^2}\right)Y=0,$$

where A is the separation parameter. Every electronic term $E(R)$ is characterized by three quantum numbers: Λ and two "elliptic quantum numbers" n_ξ, n_η, which specify the numbers of zeros of the functions $X(\xi)$ and $Y(\eta)$.

§ 79. Crossing of electronic terms

The electronic terms of a diatomic molecule may be represented graphically as functions of the distance r between the nuclei, with the energy plotted against r. The question whether the curves representing different terms can cross is of considerable interest.

Let $U_1(r)$ and $U_2(r)$ be two different electronic terms. If they cross at some point, then near that point the functions U_1 and U_2 must have close values. It is convenient to formulate the question of whether such a crossing is possible in the following way.

Consider a point r_0 at which the values of $U_1(r)$ and $U_2(r)$ are very close but not equal; denote them by E_1 and E_2. We ask whether U_1 and U_2 can be made equal by shifting the point through a small amount δr. The energies E_1 and E_2 are eigenvalues of the Hamiltonian $\widehat{H}_0$ for the electrons in the field of nuclei separated by the distance r_0. If r is increased by δr, the Hamiltonian becomes $\widehat{H}_0+\widehat{V}$, where $\widehat{V}=\delta r\,\partial\widehat{H}_0/\partial r$ is a small correction; the values of U_1 and U_2 at $r_0+\delta r$ may be regarded as eigenvalues of the new Hamiltonian. This formulation makes it possible to find the values of the terms $U_1(r)$ and $U_2(r)$ at $r_0+\delta r$ by perturbation theory, with $\widehat{V}$ treated as a perturbation of $\widehat{H}_0$.

The ordinary method of perturbation theory is not applicable here, however, because the energy eigenvalues E_1 and E_2 of the unperturbed problem lie very close to each other, and their difference need not be large compared with the perturbation (condition (38.9) is not fulfilled). As the limit $E_2-E_1=0$ gives the case of degenerate eigenvalues, it is natural to treat close eigenvalues by a method analogous to that developed in § 39.

Let ψ_1 and ψ_2 be eigenfunctions of the unperturbed operator $\widehat{H}_0$, belonging to the energies E_1 and E_2. As the initial zeroth approximation, instead of ψ_1 and ψ_2 themselves, take their linear combinations

$$\psi = c_1\psi_1 + c_2\psi_2. \tag{79.1}$$

Substitution of this expression into the perturbed equation

$$(\widehat{H}_0+\widehat{V})\psi = E\psi, \tag{79.2}$$

gives

$$c_1(E_1 + \widehat{V} - E)\psi_1 + c_2(E_2 + \widehat{V} - E)\psi_2 = 0.$$

Multiplying this equation on the left in turn by ψ_1^* and ψ_2^* and integrating, we obtain the two algebraic equations

$$\left.\begin{aligned} c_1(E_1 + V_{11} - E) + c_2 V_{12} &= 0, \\ c_1 V_{21} + c_2(E_2 + V_{22} - E) &= 0. \end{aligned}\right\} \tag{79.3}$$

Since $\widehat{V}$ is Hermitian, the matrix elements V_{11} and V_{22} are real, while $V_{12} = V_{21}^*$. The compatibility condition for these equations is

$$\begin{vmatrix} E_1 + V_{11} - E & V_{12} \\ V_{21} & E_2 + V_{22} - E \end{vmatrix} = 0,$$

and hence

$$E = \frac{1}{2}(E_1 + E_2 + V_{11} + V_{22}) \pm \sqrt{\frac{1}{4}(E_1 - E_2 + V_{11} - V_{22})^2 + |V_{12}|^2}. \tag{79.4}$$

This formula determines the required energy eigenvalues in the first approximation.

If the energies of the two terms become equal at $r_0 + \delta r$, so that the terms cross, the two values of E given by (79.4) must coincide. The expression under the square root in (79.4) must therefore vanish. Since it is a sum of two squares, the conditions for a crossing point are

$$E_1 - E_2 + V_{11} - V_{22} = 0, \qquad V_{12} = 0. \tag{79.5}$$

But only one arbitrary parameter is available to specify the perturbation $\widehat{V}$, namely the displacement δr. Thus, in general, the two equations (79.5) cannot be fulfilled simultaneously. Here we assume that ψ_1 and ψ_2 have been chosen real, so that V_{12} is real as well.

It may happen, however, that the matrix element V_{12} vanishes identically. Only one equation (79.5) then remains, and it can be fulfilled by a suitable choice of δr. This occurs whenever the two terms under consideration have different symmetries. Here symmetry includes every possible kind: symmetry with respect to rotations about the axis, reflections in planes, inversion, and permutations of electrons. For a diatomic molecule, therefore, the terms may differ in Λ, parity, or multiplicity; for Σ terms they may also be Σ^+ and Σ^-.

This statement follows from the fact that the perturbation operator, like the Hamiltonian itself, commutes with all the symmetry operators of the molecule: the operator of angular momentum about the axis, the reflection and inversion operators, and the electron-permutation operators. It was shown in §§ 29 and 30 that, for a scalar quantity whose operator commutes with the angular-momentum and inversion operators, only matrix elements for transitions between states of equal angular momentum and parity can be nonzero. Essentially the same proof applies to an arbitrary symmetry operator. We shall not repeat it here, especially since another general proof, based on group theory, will be given in § 97.

We therefore reach the result that in a diatomic molecule only terms of different symmetry can cross; terms of the same symmetry cannot cross (E. Wigner, J. von Neumann, 1929). If some approximate calculation were to give two crossing terms of the same symmetry, the next approximation would separate them, as the solid curves in Fig. 27 show.

It should be stressed that this result is not confined to a diatomic molecule. It is in fact a general theorem of quantum mechanics, valid whenever the Hamiltonian contains a parameter and its eigenvalues are consequently functions of that parameter.

In the language of group theory (see § 96), the general requirement for terms to be capable of crossing is that they belong to different irreducible representations of the symmetry group of the system's Hamiltonian[4].

In a polyatomic molecule the electronic terms depend not on one parameter but on several: the distances between the various nuclei.

Let s be the number of independent internuclear distances. In an N-atom molecule ($N > 2$), with an arbitrary arrangement of the nuclei, this number is $s = 3N - 6$. Geometrically, each term $U_n(r_1, \ldots, r_s)$ is a surface in a space of $s + 1$ dimensions. One may then speak of intersections of these surfaces along manifolds of various dimensions, from 0 (intersection at a point) to $s - 1$.

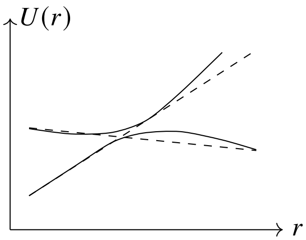


Fig. 27

The foregoing derivation remains fully valid, except that the perturbation $\widehat{V}$ is now determined not by one parameter but by s parameters, the displacements $\delta r_1, \ldots, \delta r_s$. With as few as two parameters, however, the two equations (79.5) can in general be fulfilled. We thus arrive at the result that any two terms in a polyatomic molecule can cross. For terms of the same symmetry, the crossing is specified by the two conditions (79.5), so the manifold along which the crossing occurs has dimension $s - 2$. For terms of different symmetry only one condition remains, and the crossing occurs along a manifold of dimension $s - 1$.

Thus, for $s = 2$, the terms are represented by surfaces in a three-dimensional

[4]The electronic terms of the H_2^+ ion appear to be an exception to this rule. These terms are characterized by the angular-momentum projection Λ and by two elliptic quantum numbers n_ξ, n_η (see the problem in § 78). Since all these numbers are associated with functions of different variables, there is in general no reason to prevent terms $E(R)$ with different pairs n_ξ, n_η but the same Λ from crossing, although such terms have the same symmetry with respect to rotations and reflections. In reality, however, separability of the variables in the Schrödinger equation for this system means that its Hamiltonian has a higher symmetry than its geometrical properties alone suggest; relative to this full symmetry group, states with different values of n_ξ, n_η belong to different types.

coordinate system. If the terms have different symmetries, these surfaces intersect along lines ($s-1=1$); if they have the same symmetry, they intersect at points ($s-2=0$). The shape of the surfaces near a crossing point in the latter case is readily found. The energy values near such points are given by (79.4). In this expression the matrix elements V_{11}, V_{22}, and V_{12} are linear functions of the displacements δr_1, δr_2, and hence linear

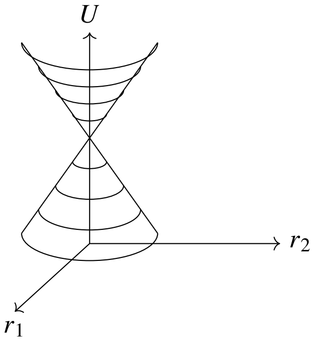


Fig. 28

functions of the distances r_1, r_2 themselves. As is known from analytic geometry, an equation of this form describes an elliptic cone. Thus, near their crossing points the terms are represented by a two-napped elliptic cone of arbitrary orientation (Fig. 28).

§ 80. Relation between molecular and atomic terms

As the distance between the nuclei of a diatomic molecule is increased, the limiting system consists of two isolated atoms (or ions). This raises the question of how an electronic term of the molecule corresponds to the atomic states obtained when the atoms are separated (E. Wigner, E. Witmer, 1928). The correspondence is not unique: bringing together two atoms in specified states may produce a molecule in several different electronic states.

First suppose that the molecule consists of two different atoms. Let the isolated atoms have orbital angular momenta L_1, L_2 and spins S_1, S_2, with $L_1 \geqslant L_2$. The projections of the angular momenta on the line joining the nuclei take the values $M_1 = -L_1, -L_1+1, \ldots, L_1$ and $M_2 = -L_2, -L_2+1, \ldots, L_2$. When the atoms approach, the resulting Λ is the magnitude of $M_1 + M_2$. Combining every possible pair of M_1 and M_2, we find the following numbers of occurrences of the values $\Lambda = |M_1 + M_2|$:

$$
\begin{array}{cl}
\Lambda = L_1 + L_2 & 2 \text{ times}, \\
L_1 + L_2 - 1 & 4 \text{ times}, \\
\dots\dots\dots\dots & \dots\dots\dots\dots\dots\dots \\
L_1 - L_2 & 2(2L_2+1) \text{ times}, \\
L_1 - L_2 - 1 & 2(2L_2+1) \text{ times}, \\
\dots\dots\dots\dots & \dots\dots\dots\dots\dots\dots \\
1 & 2(2L_2+1) \text{ times}, \\
0 & 2L_2 + 1 \text{ times}.
\end{array}
$$

Remembering that every term with $\Lambda \neq 0$ is doubly degenerate, whereas a term with $\Lambda = 0$ is nondegenerate, we find that the possible terms are

$$\begin{array}{rll} 1 \text{ term} & \text{with} & \Lambda = L_1 + L_2, \\ 2 \text{ terms} & \text{with} & \Lambda = L_1 + L_2 - 1, \\ \ldots\ldots\ldots\ldots & & \ldots\ldots\ldots\ldots\ldots \\ 2L_2 + 1 \text{ terms} & \text{with} & \Lambda = L_1 - L_2, \\ 2L_2 + 1 \text{ terms} & \text{with} & \Lambda = L_1 - L_2 - 1, \\ \ldots\ldots\ldots\ldots & & \ldots\ldots\ldots\ldots\ldots \\ 2L_2 + 1 \text{ terms} & \text{with} & \Lambda = 0; \end{array} \tag{80.1}$$

altogether there are $(2L_2 + 1)(L_1 + 1)$ terms with Λ ranging from 0 to $L_1 + L_2$.

The two atomic spins S_1, S_2 combine into the total molecular spin according to the usual angular-momentum addition rule, giving the following

possible values of S:

$$S = S_1 + S_2, \quad S_1 + S_2 - 1, \ldots, |S_1 - S_2|. \tag{80.2}$$

Combining each of these values with all the values of Λ in (80.1) gives the complete list of possible terms of the resulting molecule.

For Σ terms their sign must also be determined. This is easily done by observing that, as $r \to \infty$, the molecular wave functions can be written as products, or sums of products, of the wave functions of the two atoms. The value $\Lambda = 0$ can arise either from nonzero projections with $M_1 = -M_2$ or from $M_1 = M_2 = 0$. Denote the wave functions of the first and second atoms by $\psi^{(1)}_{M_1}$, $\psi^{(2)}_{M_2}$. For $M = |M_1| = |M_2| \neq 0$, form the symmetric and antisymmetric products

$$\psi^+ = \psi^{(1)}_M \psi^{(2)}_{-M} + \psi^{(1)}_{-M} \psi^{(2)}_M, \qquad \psi^- = \psi^{(1)}_M \psi^{(2)}_{-M} - \psi^{(1)}_{-M} \psi^{(2)}_M.$$

Reflection in a vertical plane containing the molecular axis reverses the angular-momentum projection on the axis, so that $\psi^{(1)}_M$, $\psi^{(2)}_M$ become $\psi^{(1)}_{-M}$, $\psi^{(2)}_{-M}$, respectively, and conversely. The function ψ^+ is then unchanged, while ψ^- changes sign. The former therefore belongs to a Σ^+ term and the latter to a Σ^- term. Thus each value of M yields one term of each kind. Since M has L_2 possible values ($M = 1, \ldots, L_2$), there are in all L_2 terms of each sign.

If $M_1 = M_2 = 0$, the molecular wave function has the form $\psi = \psi^{(1)}_0 \psi^{(2)}_0$. To determine the behavior of $\psi^{(1)}_0$ under reflection in a vertical plane, choose coordinates centered on the first atom with the z axis along the molecular axis. Reflection in the vertical xz plane is equivalent to an inversion about the origin followed by a rotation through 180° about the y axis. Under inversion, $\psi^{(1)}_0$ acquires the factor P_1, where $P_1 = \pm 1$ is the parity of the state of the first atom. The effect of an infinitesimal, and hence of any finite, rotation on the wave function is completely fixed by the atom's total orbital angular momentum. It is therefore enough to consider the special case of a one-electron atom with orbital angular momentum l and z component $m = 0$; replacing l by L in the result gives the answer for an arbitrary atom. Apart from a constant factor, the angular part of the electron wave function with $m = 0$ is $P_l(\cos\theta)$ (see (28.8)). Rotation through 180° about the y axis gives $x \to -x$, $y \to y$, $z \to -z$, or, in spherical coordinates, $r \to r$, $\theta \to \pi - \theta$, $\varphi \to \pi - \varphi$. Consequently $\cos\theta \to -\cos\theta$, and $P_l(\cos\theta)$ acquires the factor $(-1)^l$.

Thus reflection in a vertical plane multiplies $\psi_0^{(1)}$ by $(-1)^{L_1}P_1$. Similarly, $\psi_0^{(2)}$ is multiplied by $(-1)^{L_2}P_2$, and the product $\psi = \psi_0^{(1)}\psi_0^{(2)}$ acquires the factor $(-1)^{L_1+L_2}P_1P_2$. The term is Σ^+ or Σ^- according as this factor is $+1$ or -1.

Collecting these results, we find that among the $(2L_2+1)$ Σ terms of each possible multiplicity, L_2+1 are Σ^+ terms and L_2 are Σ^- terms if $(-1)^{L_1+L_2}P_1P_2 = +1$; the numbers are reversed if $(-1)^{L_1+L_2}P_1P_2 = -1$.

We now turn to a molecule composed of identical atoms. The rules for adding the atomic spins and orbital angular momenta to form the molecular S and Λ remain the same as for unlike atoms. The new feature is that the terms may be even or odd. Two cases must be distinguished: the atoms may be in either different or identical states.

If the atoms are in different states[5], the total number of possible terms is doubled relative to the corresponding molecule of unlike atoms. Reflection through the origin, located at the midpoint of the molecular axis, interchanges the states of the two atoms. By symmetrizing or antisymmetrizing the molecular wave function under this interchange, we obtain two terms with the same Λ and S, one even and the other odd. Hence equal numbers of even and odd terms are obtained.

If both atoms are in the same state, the total number of states is the same as for a molecule of unlike atoms. Their parities are given by the following results of an analysis that we omit because of its length[6].

Let N_g and N_u be the numbers of even and odd terms having specified Λ and S. Then

$$\begin{array}{ll} \text{if } \Lambda \text{ is odd,} & N_g = N_u, \\ \text{if } \Lambda \text{ is even and } S \text{ is even } (S = 0, 2, 4, \ldots), & N_g = N_u + 1, \\ \text{if } \Lambda \text{ is even and } S \text{ is odd } (S = 1, 3, \ldots), & N_u = N_g + 1. \end{array}$$

Among the Σ terms, Σ^+ and Σ^- must finally be distinguished. The rule is

$$\begin{array}{ll} \text{if } S \text{ is even,} & N_g^+ = N_u^- + 1 = L + 1, \\ \text{if } S \text{ is odd,} & N_u^+ = N_g^- + 1 = L + 1 \end{array}$$

(where $L_1 = L_2 \equiv L$). Every Σ^+ term has parity $(-1)^S$, and every Σ^- term parity $(-1)^{S+1}$.

Besides the relation considered above between molecular terms and the atomic terms obtained as $r \to \infty$, one may also ask how the molecular terms are related to those of the "united atom" that would result as $r \to 0$, when the two nuclei are brought to the same point; an example is the relation between terms of the H_2 molecule and the He atom. The following rules are readily obtained. From a united-atom term with spin S, orbital angular momentum L, and parity P, separating the constituent atoms gives molecular terms with spin S and axial angular momentum $\Lambda = 0, 1, \ldots, L$, one term for each value of Λ. The parity of the molecular term is the same as P for the atomic term (g for $P = +1$

[5] In particular, this may be the combination of a neutral atom with an ionized one.

[6] See E. Wigner, E. Witmer // Zs. Physik. 1928. V. 51 S. 850.

and u for $P = -1$). The molecular term with $\Lambda = 0$ is a Σ^+ term if $(-1)^L P = +1$, and a Σ^- term if $(-1)^L P = -1$.

Problem

1. Determine the possible terms of the H_2, N_2, O_2, and Cl_2 molecules that can arise when atoms in their normal states combine.

Solution. The rules given above yield the following possible terms:

$$\begin{array}{lll}
\text{molecule } H_2 & (\text{atoms in } {}^2S \text{ states}): & {}^1\Sigma_g^+,\ {}^3\Sigma_u^+; \\
N_2 & (\text{atoms in } {}^4S \text{ states}): & {}^1\Sigma_g^+,\ {}^3\Sigma_u^+,\ {}^5\Sigma_g^+,\ {}^7\Sigma_u^+; \\
Cl_2 & (\text{atoms in } {}^2P \text{ states}): & 2{}^1\Sigma_g^+,\ {}^1\Sigma_u^-,\ {}^1\Pi_g,\ {}^1\Pi_u,\ {}^1\Delta_g, \\
 & & 2{}^3\Sigma_u^+,\ {}^3\Sigma_g^-,\ {}^3\Pi_g,\ {}^3\Pi_u,\ {}^3\Delta_u; \\
O_2 & (\text{atoms in } {}^3P \text{ states}): & 2{}^1\Sigma_g^+,\ {}^1\Sigma_u^-,\ {}^1\Pi_g,\ {}^1\Pi_u,\ {}^1\Delta_g, \\
 & & 2{}^3\Sigma_u^+,\ {}^3\Sigma_g^-,\ {}^3\Pi_u,\ {}^3\Pi_g,\ {}^3\Delta_u, \\
 & & 2{}^5\Sigma_g^+,\ {}^5\Sigma_u^-,\ {}^5\Pi_g,\ {}^5\Pi_u,\ {}^5\Delta_g.
\end{array}$$

(A numeral placed before a term symbol gives the number of terms of that type when it exceeds one.)

Problem

2. Do the same for HCl and CO.

Solution. When unlike atoms combine, the parity of their states is also relevant. Formula (31.6) shows that the normal states of H, O, and C atoms are even, while that of Cl is odd (for the electron configurations of the atoms, see Table 3). The preceding rules give

$$\begin{array}{lll}
\text{molecule HCl} & (\text{atoms in } {}^2S_g, {}^2P_u \text{ states}): & {}^{1,3}\Sigma^+,\ {}^{1,3}\Pi; \\
\text{molecule CO} & (\text{atoms in } {}^3P_g, {}^3P_g \text{ states}): & 2{}^{1,3,5}\Sigma^+,\ {}^{1,3,5}\Sigma^-, \\
 & & 2{}^{1,3,5}\Pi,\ {}^{1,3,5}\Delta.
\end{array}$$

§ 81. Valence

The ability of atoms to combine into molecules is described by the concept of *valence*. Each atom is assigned a definite valence, and when atoms combine their valences must saturate one another: every valence bond of one atom must be matched by a valence bond of another. In methane, CH_4, for example, the four valence bonds of the tetravalent carbon atom are saturated by bonds of four monovalent hydrogen atoms. We begin the physical interpretation of valence with the simplest example, the combination of two hydrogen atoms into an H_2 molecule.

Consider two hydrogen atoms in their ground states (${}^2S$). On approaching, they may form a system in either the ${}^1\Sigma_g^+$ or the ${}^3\Sigma_u^+$ molecular state. The singlet term has an antisymmetric spin wave function, and the triplet term a symmetric one. Conversely, the coordinate wave function is symmetric for the ${}^1\Sigma$ term and antisymmetric for the ${}^3\Sigma$

term. Plainly, only the $^1\Sigma$ term can be the ground term of H_2. Indeed, an antisymmetric wave function $\psi(\mathbf{r}_1, \mathbf{r}_2)$, where $\mathbf{r}_1$ and $\mathbf{r}_2$ are the radius vectors of the two electrons, necessarily has nodes: it vanishes when $\mathbf{r}_1 = \mathbf{r}_2$. It therefore cannot describe the lowest state of the system.

Numerical calculation shows that the $^1\Sigma$ electronic term does have a deep minimum corresponding to a stable H_2 molecule. In the $^3\Sigma$ state, by contrast, the energy $U(r)$ decreases monotonically as the internuclear distance increases, corresponding to mutual repulsion of the two H atoms[7] (Fig. 29).

Thus the total spin of the hydrogen molecule in its ground state is zero, $S = 0$. The molecules of practically all chemically stable compounds of the main-group elements have this property. Among inorganic molecules, the exceptions are the diatomic molecules O_2 (ground state $^3\Sigma$) and NO (ground state $^2\Pi$), and the triatomic molecules NO_2 and ClO_2 (total spin $S = 1/2$). The transition-group elements have special properties, to be discussed below after the valence properties of the main-group elements.

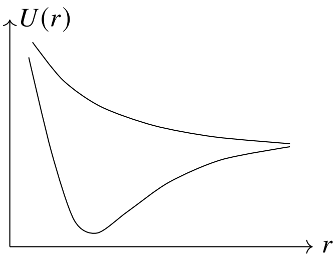


Fig. 29

The ability of atoms to combine is therefore connected with their spin (W. Heitler, H. London, 1927). Combination occurs in such a way that the atomic spins compensate one another. A convenient quantitative measure of an atom's ability to combine is the integer equal to twice its spin. This number coincides with the atom's chemical valence. It must be remembered that the same atom may have different valences according to its state.

Consider from this standpoint the main groups of the periodic table. In their normal state the elements of the first group, the first column of Table 3 and the alkali-metal group, have $S = 1/2$ and hence valence one. A state of higher spin can be obtained only by exciting an electron from a closed shell. Such states lie so high that the excited atom cannot form a stable molecule[8].

In their normal state, atoms of the second-group elements, the second column of Table 3 and the alkaline-earth group, have $S = 0$ and cannot enter chemical compounds in that state. Comparatively close to the ground state, however, there is

an excited state whose unfilled shell has the configuration sp instead of s^2 and whose total spin is $S = 1$. The valence in this state is 2, which is the principal valence of the

[7] We disregard here the van der Waals attraction between the atoms (see § 89). These forces imply that the $U(r)$ curve of the $^3\Sigma$ term also has a minimum, located at large distances. This minimum is, however, very shallow compared with that of the $^1\Sigma$ curve and would not be visible on the scale of Fig. 29.

[8] For Cu, Ag, and Au, see the end of this section.

second-group elements.

The normal configuration of the third-group elements is s^2p, with $S = 1/2$. Excitation of an electron from the filled s shell produces a nearby excited state of configuration sp^2 and spin $S = 3/2$. These elements accordingly behave as both monovalent and trivalent. The first members of the group, B and Al, are only trivalent. The tendency to show valence 1 grows with atomic number, and Tl is about equally monovalent and trivalent, as in TlCl and $TlCl_3$. In the first elements of the group, the energetic advantage of the greater binding energy in compounds of the trivalent element, compared with its monovalent compounds, outweighs the excitation energy of the atom.

For the fourth-group elements the ground state has configuration s^2p^2 and spin 1, while a nearby excited state has configuration sp^3 and spin 2. These states correspond to valences 2 and 4. As in the third group, the first elements C and Si mainly exhibit the higher valence, with CO as an example of an exception, while the tendency toward the lower valence increases with atomic number.

For fifth-group atoms the ground-state configuration is s^2p^3, with $S = 3/2$, so the corresponding valence is three. A state of higher spin can arise only by transferring an electron to the shell with the next principal quantum number. The nearest such state has configuration sp^3s' and $S = 5/2$; here s' denotes an s state whose principal quantum number is one greater than that of s. Although its excitation energy is comparatively large, the excited atom can still enter a stable compound. Fifth-group elements are accordingly tri- and pentavalent: nitrogen is trivalent in NH_3 and pentavalent in HNO_3.

For the sixth group, the ground state, of configuration s^2p^4, has spin 1 and the atom is divalent. Exciting one p electron gives the state s^2p^3s' with spin 2; exciting one more s electron gives the

state $sp^3s'p'$ with spin 3. In both excited states the atom can enter stable molecules, exhibiting valences 4 and 6 respectively. The first member, oxygen, shows only valence 2, whereas later members also show the higher valences: sulfur is respectively di-, tetra-, and hexavalent in H_2S, SO_2, and SO_3.

In the seventh group, the halogens, atoms in the ground state are monovalent, having configuration s^2p^5 and spin $S = 1/2$. They can also enter stable compounds in excited states with configurations s^2p^4s', $s^2p^3s'p'$, and $sp^3s'p'^2$, whose spins are 3/2, 5/2, and 7/2 and whose valences are 3, 5, and 7. The first element F is always monovalent; later elements also exhibit higher valences. Thus chlorine is respectively mono-, tri-, penta-, and heptavalent in HCl, $HClO_2$, $HClO_3$, and $HClO_4$.

Finally, in their ground states the noble-gas atoms have completely filled shells and $S = 0$, while their excitation energies are large. Their valence is accordingly zero and they are chemically inert[9].

One general qualification applies to all these arguments. To say that an atom enters a

[9]Some nevertheless form stable compounds with fluorine or oxygen. These valences may be connected with transitions of electrons from the outer filled shell to energetically nearby unfilled f or d states.

We also mention a distinctive attraction that arises between a noble-gas atom and an excited atom of the same element. It is connected with the doubling of the number of possible states when two identical atoms in different states are brought together (see § 80). Transfer of the excitation between the atoms here replaces the exchange interaction that produces ordinary valence. The He_2 molecule is an example. The same type of bond occurs in molecular ions made of two identical atoms, such as H_2^+.

molecule with the valence belonging to an excited state does not mean that separating the atoms to large distances must produce an excited atom. It means only that the molecular electron-density distribution around that nucleus is close to the distribution in an isolated excited atom. The limiting distribution as the internuclear distance grows may instead correspond to unexcited atoms.

When atoms form a molecule, their filled electron shells change little, but the electron density in unfilled shells may change substantially. In the most pronounced instances of the so-called *heteropolar bond*, all valence electrons pass from some atoms to others, so the molecule may be said to consist of ions whose charges, in units of e, equal their valences. The first-group elements are electropositive: in heteropolar compounds they give up electrons and form positive ions. Through the later groups electropositivity gradually decreases and becomes electronegativity, strongest in the seventh group. The same qualification applies to heteropolarity as to excited atoms in a molecule. A heteropolar molecule need not yield two ions when separated. CsF would indeed give Cs^+ and F^-, whereas NaF tends to neutral Na and F atoms, because fluorine's electron affinity exceeds the ionization potential of cesium but is smaller than that of sodium.

In the opposite limiting case, the so-called *homopolar bond*, the atoms remain neutral on average. Unlike heteropolar molecules, homopolar molecules have no appreciable dipole moment. The distinction between the two bond types is purely quantitative, and every intermediate case can occur.

We now turn to the transition-group elements. In their valence properties, the palladium and platinum groups differ little from the main groups. Since their d electrons lie comparatively deep in the atom, however, these electrons interact more weakly with other atoms in a molecule. Consequently, "unsaturated" compounds whose molecules have nonzero spin, in fact no greater than $1/2$, are relatively common. Each element may display several valences, differing here by one as well as by two. In the main groups, by contrast, a change of valence results from exciting an electron whose spin was compensated, thereby releasing the spins of a pair of electrons at once.

The rare-earth group is characterized by an unfilled f shell. The f electrons lie much deeper than the d electrons and take no part in valence. The valence of a rare-earth element is therefore determined only by the s and p electrons in its unfilled shells[10]. On excitation, however, f electrons may pass into s and p states, increasing the valence by one. Rare-earth elements therefore also show valences differing by one; in practice they are all tri- and tetravalent.

The actinium group occupies a distinctive position. Ac and Th contain no f electrons at all, and their d electrons participate in valence; chemically they therefore resemble the palladium and platinum groups rather than the rare earths. Although a normal U atom contains f electrons, uranium compounds likewise have none. The atoms Np, Pu, Am, and Cm retain f electrons in compounds, but their valence electrons are still s and d electrons; in this sense they are homologous to uranium. The largest possible numbers of unpaired s and d electrons are 1 and 5, respectively, so the maximum valence of the

[10]The d electrons present in unfilled shells of some rare-earth atoms are immaterial, since these atoms in practice always enter compounds in excited states with no d electrons.

actinium group is six. The corresponding maximum for rare-earth elements, whose s and p electrons participate, is $1 + 3 = 4$.

In valence properties the iron group lies between the rare earths and the palladium and platinum groups. Its d electrons are comparatively deep and do not participate in valence bonds in many compounds. In these compounds the iron-group elements behave like rare-earth elements. This includes ionic compounds such as $FeCl_2$ and $FeCl_3$, in which the metal atom occurs as a simple cation. Like the rare earths, iron-group elements may exhibit a wide range of valences in such compounds.

Another class is formed by the so-called complex compounds. The transition-element atom then enters the molecule not as a simple ion but as part of a compound, complex ion, for example MnO_4^- in $KMnO_4$ or $Fe(CN)_6^{4-}$ in $K_4Fe(CN)_6$. In these complex ions the atoms

are closer together than in simple ionic compounds, and d electrons participate in the valence bond. The iron-group elements accordingly behave in complex compounds like the palladium and platinum groups.

Finally, Cu, Ag, and Au, assigned to the main groups in § 73, behave as transition elements in a number of compounds. Their valence can exceed one through promotion of electrons from the d shell into the nearby p shell, for example from $3d$ to $4p$ in Cu. In such compounds the atoms have an unfilled d shell and behave as transition elements: Cu like the iron group, and Ag and Au like the palladium and platinum groups.

Problem

Determine the electronic terms of the molecular ion H_2^+ formed by combining a ground-state hydrogen atom with an H^+ ion, for internuclear distances R large compared with the Bohr radius (L. D. Landau, 1961; C. Herring, 1961)[11].

Solution. This problem is analogous in formulation to problem 3 of § 50: instead of two one-dimensional potential wells, there are two three-dimensional wells around the nuclei, with a common axial symmetry about the line joining them. The level $E_0 = -1/2$, the ground level of hydrogen[12], splits into two levels $U_g(R)$ and $U_u(R)$, the $^2\Sigma_g^+$ and $^2\Sigma_u^+$ terms, with electronic wave functions

$$\psi_{g,u}(x, y, z) = \frac{1}{\sqrt{2}}[\psi_0(x, y, z) \pm \psi_0(-x, y, z)],$$

symmetric and antisymmetric with respect to the plane $x = 0$ midway between the nuclei, which lie at $x = \pm R/2$ on the x axis. Here $\psi_0(x, y, z)$ is the electron wave function in one of the wells. Exactly as in problem 3 of § 50,

$$U_{g,u}(R) - E_0 = \mp \int \psi_0 \frac{\partial \psi_0}{\partial x}\, dy\, dz, \tag{1}$$

[11] For the solution of the analogous problem for H_2, see L. P. Gor'kov, L. P. Pitaevskii // Dokl. Akad. Nauk SSSR. 1963. V. 151. P. 822; C. Herring, M. Flicker // Phys. Rev. 1964. V. 134A. P. 362. The second paper corrects a computational error in the first.

[12] Atomic units are used in this problem.

where the integration is over the plane $x = 0$[13].

Seek the function ψ_0 corresponding, say, to motion about nucleus 1 at $x = R/2$ in the form

$$\psi_0 = \frac{1}{\sqrt{\pi}} e^{-r_1} a, \tag{2}$$

where a varies slowly; for an isolated hydrogen atom $a = 1$. The function ψ_0 obeys

$$\frac{1}{2}\Delta\psi_0 + \left(-\frac{1}{2} - \frac{1}{R} + \frac{1}{r_1} + \frac{1}{r_2}\right)\psi_0 = 0, \tag{3}$$

where r_1, r_2 are the distances from the electron to nuclei 1 and 2. The total electronic energy in this equation is $E_0 - 1/R$, since E_0 itself also includes the nuclear Coulomb-repulsion energy $1/R$.

Because ψ_0 decreases rapidly away from the x axis, only $y, z \ll R$ matter in (1). For these values, substitution of (2) in (3) gives

$$\frac{\partial a}{\partial x} + \frac{R/2 + x}{R} a = 0,$$

where second derivatives of the slowly varying a have been neglected and $r_2 = R/2 + x$ has been used. The solution tending to one as $x \to R/2$, near nucleus 1, is

$$a = \frac{2R}{R + 2x} \exp\left(\frac{x}{R} - \frac{1}{2}\right).$$

Formula (1) now gives

$$U_{g,u} - E_0 = \mp\frac{4}{e}\int_{R/2}^{\infty} e^{-2r_1} 2\pi r_1\, dr_1 = \mp 2Re^{-R-1}.$$

The splitting is[14]

$$U_g - U_u = -4Re^{-R-1}. \tag{4}$$

At sufficiently large distances this exponentially decreasing expression becomes smaller than the second-order dipole interaction between H and H^+. The polarizability of ground-state hydrogen is $9/2$ (see (77.9)), while the field of H^+ is $\mathcal{E} = 1/R^2$; the corresponding interaction energy is $-9/(4R^4)$. Including it gives

$$U_{g,u}(R) - E_0 = \mp 2Re^{-R-1} - \frac{9}{4R^4}. \tag{5}$$

The second term becomes comparable with the first only at $R = 10{,}8$. The term $U_u(R)$ also has a minimum at $R = 12{,}6$, of depth $-5{,}8 \cdot 10^{-5}$ atomic units ($-1{,}6 \cdot 10^{-3}$ eV)[15].

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[13] Thus the effect sought is determined by the region of distances where the electron interacts equally with both nuclei.

[14] The analogous result for H_2 (see the papers cited above) is $U_g - U_u = -1{,}64R^{5/2}e^{-2R}$.

[15] This van der Waals minimum is very shallow compared with the minimum of $U_g(R)$ corresponding to the normal state of stable H_2^+. The main minimum occurs at $R = 2{,}0$ and has depth $-0{,}60$ atomic units ($-16{,}3$ eV).

§ 82. Vibrational and rotational structure of singlet terms in a diatomic molecule

As noted at the beginning of this chapter, the large difference between nuclear and electronic masses permits the problem of molecular energy levels to be divided into two parts. First, the energy levels of the electron system with fixed nuclei are found as functions of the internuclear distance; these are the electronic terms. One then considers the motion of the nuclei for a specified electronic state. This amounts to treating the nuclei as particles interacting according to $U_n(r)$, where U_n is the corresponding electronic term. Molecular motion consists of translation of the molecule as a whole and motion of the nuclei relative to their center of mass. The translational motion is of no interest here, and we may take the center of mass to be at rest.

For clarity, first consider electronic terms for which the total molecular spin is zero (singlet terms). The relative motion of two particles, the nuclei, interacting through $U(r)$ reduces to the motion of one particle of mass M, the reduced mass of the nuclei, in the central field $U(r)$. Here $U(r)$ denotes the energy of the electronic term being considered. Motion in the central field, in turn, reduces to a one-dimensional problem with an effective energy equal to $U(r)$ plus the centrifugal energy.

Let $\mathbf{K}$ be the total angular momentum of the molecule, the sum of the electronic orbital angular momentum $\mathbf{L}$ and the rotational angular momentum of the nuclei. The nuclear centrifugal-energy operator is

$$B(r)(\mathbf{K}-\mathbf{L})^2,$$

where the notation customary in diatomic-molecule theory is

$$B(r) = \frac{\hbar^2}{2Mr^2}. \tag{82.1}$$

Averaging this expression over the electronic state at fixed r gives the centrifugal energy as a function of r, to be included in the effective potential energy $U_K(r)$. Thus

$$U_K(r) = U(r) + B(r)\overline{(\mathbf{K}-\mathbf{L})^2}, \tag{82.2}$$

where the bar denotes this averaging.

Perform the average for a state in which the molecule has a definite squared total angular momentum $\mathbf{K}^2 = K(K+1)$, with integral K, and a definite projection of the electronic angular momentum on the molecular z axis, $L_z = \Lambda$. Expanding (82.2), we have

$$U_K(r) = U(r) + B(r)K(K+1) - 2B(r)\overline{\mathbf{LK}} + B(r)\overline{\mathbf{L}^2}. \tag{82.3}$$

The last term depends only on the electronic state and contains no K; it may simply be included in $U(r)$. We show that the same is true of the preceding term.

Recall that when the projection of an angular momentum on an axis has a definite value, the mean value of the entire angular-momentum vector is directed along the same axis (see the remark at the end of § 27). Let $\mathbf{n}$ be a unit vector along z; then $\overline{\mathbf{L}} = \Lambda\mathbf{n}$. In classical mechanics, the rotational angular momentum of two particles is

$\mathbf{r} \times \mathbf{p}$, where $\mathbf{r} = r\mathbf{n}$ is their relative radius vector and $\mathbf{p}$ their relative momentum. It is perpendicular to $\mathbf{n}$. The same relation holds in quantum mechanics for the nuclear rotational-angular-momentum operator: $(\mathbf{K} - \mathbf{L})\mathbf{n} = 0$, or $\mathbf{Kn} = \mathbf{Ln}$. Equality of the operators entails equality of their eigenvalues; since $\mathbf{nL} = L_z = \Lambda$,

$$\mathbf{Kn} = \Lambda. \tag{82.4}$$

Thus, in the penultimate term of (82.3), $\overline{\mathbf{LK}} = \mathbf{nK}\Lambda = \Lambda^2$ and is independent of K. Redefining $U(r)$, the effective potential energy can finally be written

$$U_K(r) = U(r) + B(r)K(K+1). \tag{82.5}$$

It also follows from $K_z = \Lambda$ that, for a specified Λ, the allowed values of K satisfy

$$K \geqslant \Lambda. \tag{82.6}$$

Solving the one-dimensional Schrödinger equation with potential energy (82.5) gives a sequence of energy levels. For each K, number them in increasing order by $v = 0, 1, 2, \ldots$, with $v = 0$ the lowest. Nuclear motion therefore splits every electronic term into levels characterized by the two quantum numbers K and v.

The number of these levels for a specified electronic term may be finite or infinite. If the electronic state is such that the molecule becomes two isolated neutral atoms as $r \to \infty$, the potential $U(r)$, and hence $U_K(r)$, approaches the constant limiting value $U(\infty)$, the sum of the two isolated-atom energies, faster than $1/r^2$ (see § 89). The number of levels in such a field is finite (see § 18), although in actual molecules it is very large. For each K there is then a definite number of levels distinguished by v. This number decreases as K increases, until a value of K is reached for which no levels remain.

If instead the molecule dissociates into two ions as $r \to \infty$, then at large distances $U(r) - U(\infty)$ becomes the Coulomb attraction energy of the ions, proportional to $1/r$. Such a field has infinitely many levels, accumulating at the limiting value $U(\infty)$. Most molecules in their normal state belong to the first case; only comparatively few yield ions when the nuclei are separated.

The dependence of the energy levels on the quantum numbers cannot be calculated completely in general. Such a calculation is possible only for fairly weakly excited levels not too far above the lowest level[16]. These have small K and v. They are the levels usually encountered in molecular spectra and are therefore of particular interest.

Nuclear motion in weakly excited states may be described as small oscillations about equilibrium. Expand $U(r)$ in powers of $\xi = r - r_e$, where r_e is the position of its minimum. Since $U'(r_e) = 0$, through second order $U(r) = U_e + M\omega_e^2\xi^2/2$, where $U_e = U(r_e)$ and ω_e is the oscillation frequency. In the second term of (82.5), the centrifugal energy, it is enough to put $r = r_e$, since that term already contains the small quantity $K(K+1)$. Hence

$$U_K(r) = U_e + B_e K(K+1) + \frac{M\omega_e^2\xi^2}{2}, \tag{82.7}$$

[16] Throughout, the levels are those arising from one specified electronic term.

where $B_e = \hbar^2/(2Mr_e^2) = \hbar^2/2I$ is the *rotational constant*, and $I = Mr_e^2$ is the molecular moment of inertia.

The first two terms in (82.7) are constants, and the third is the potential of a one-dimensional harmonic oscillator. The required energy levels may therefore be written at once:

$$E = U_e + B_e K(K+1) + \hbar\omega_e \left(v + \frac{1}{2}\right). \tag{82.8}$$

Thus, in this approximation the energy level is the sum of three independent parts:

$$E = E^{el} + E^r + E^v. \tag{82.9}$$

The first, $E^{el} = U_e$, is the electronic energy, including the Coulomb interaction of the nuclei at $r = r_e$. The second,

$$E^r = B_e K(K+1), \tag{82.10}$$

is the *rotational* energy associated with molecular rotation[17]. Finally,

$$E^v = \hbar\omega_e \left(v + \frac{1}{2}\right) \tag{82.11}$$

is the energy of nuclear vibration within the molecule. By definition, v numbers the levels of fixed K in increasing order and is called the *vibrational quantum number*.

For a given shape of $U(r)$, the frequency ω_e is inversely proportional to $\sqrt{M}$. Thus the spacings ΔE^v between vibrational levels are proportional to $1/\sqrt{M}$. The spacings ΔE^r between rotational levels contain the moment of inertia I in the denominator and are proportional to $1/M$. The spacings ΔE^{el} between electronic levels,

like the levels themselves, contain no M. Since m/M, where m is the electron mass, is the small parameter of diatomic-molecule theory,

$$\Delta E^{el} \gg \Delta E^v \gg \Delta E^r. \tag{82.12}$$

These inequalities reflect the distinctive distribution of molecular energy levels. Nuclear vibration splits an electronic term into comparatively closely spaced levels, and molecular rotation in turn produces a still finer splitting of each of these[18].

In subsequent approximations the energy can no longer be separated into independent vibrational and rotational parts. Rotation–vibration terms involving both K and v appear.

[17]The wave function describing rotation of a spinless diatomic molecule essentially coincides with that of a symmetric top (§ 103). Unlike a top, molecular rotation is described by only two angles, $\alpha \equiv \varphi$ and $\beta \equiv \theta$, which determine the direction of the molecular axis. The rotational wave function differs from (103.8) by the absence of the factor $e^{ik\gamma}/\sqrt{2\pi}$ and by the notation for the quantum numbers. Since (82.4) makes Λ equal to the projection of the total angular momentum $\mathbf{K}$ on the molecular axis (the ζ axis of § 103), the replacements are $J, M, k \to K, M, \Lambda$, where now $M = K_z$. Thus $\psi_{\text{rot}}(\varphi, \theta) = i^K \sqrt{(2K+1)/(4\pi)} D^{(K)}_{\Lambda M}(\varphi, \theta, 0)$.

[18]For illustration, the values of U_e, $\hbar\omega_e$, and B_e, in electron volts, for several molecules are

	H_2	N_2	O_2
$-U_e$	4,7	7,5	5,2
$\hbar\omega_e$	0,54	0,29	0,20
$10^3 B_e$	7,6	0,25	0,18

Successive approximations would give E as a series in powers of the quantum numbers K and v.

We calculate the approximation following (82.8). Expand $U(r)$ through fourth order in ξ (cf. the anharmonic-oscillator problem in § 38), and expand the centrifugal energy through terms in ξ^2. This gives

$$\begin{aligned} U_K(r) = U_e + \frac{M\omega_e^2}{2}\xi^2 + \frac{\hbar^2}{2Mr_e^2}K(K+1) - a\xi^3 + b\xi^4 \\ - \frac{\hbar^2}{Mr_e^3}K(K+1)\xi + \frac{3\hbar^2}{2Mr_e^4}K(K+1)\xi^2. \end{aligned} \tag{82.13}$$

Now calculate the correction to the eigenvalues (82.8), treating the last four terms of (82.13) as the perturbation. First-order perturbation theory suffices for the terms in ξ^2 and ξ^4, whereas the terms in ξ and ξ^3 require second order, since their diagonal matrix elements vanish identically. All the matrix elements needed were calculated

in § 23 and problem 3 of § 38. The result is conventionally written

$$\begin{aligned} E = E^{el} + \hbar\omega_e\left(v + \frac{1}{2}\right) - x_e\hbar\omega_e\left(v + \frac{1}{2}\right)^2 \\ + B_v K(K+1) - D_e K^2(K+1)^2, \end{aligned} \tag{82.14}$$

where

$$B_v = B_e - \alpha_e\left(v + \frac{1}{2}\right) \equiv B_0 - \alpha_e v. \tag{82.15}$$

The constants x_e, B_e, α_e, and D_e are related to those entering (82.14) by

$$\begin{aligned} B_e = \frac{\hbar^2}{2I}, \qquad D_e = \frac{4B_e^3}{\hbar^2\omega_e^2}, \\ \alpha_e = \frac{6B_e^2}{\hbar\omega_e}\left(\frac{a\hbar}{M\omega_e^2}\sqrt{\frac{2}{MB_e}} - 1\right), \\ x_e = \frac{3}{2\hbar\omega_e}\left(\frac{\hbar}{M\omega_e}\right)^2\left(\frac{5}{2}\frac{a^2}{M\omega_e^2} - b\right). \end{aligned} \tag{82.16}$$

Terms independent of v and K have been included in E^{el}.

Problem

Estimate the accuracy of the approximation that separates electronic and nuclear motion in a diatomic molecule.

Solution. Write the complete molecular Hamiltonian as $\widehat{H} = \widehat{T}_r + \widehat{H}_{el}$, where $\widehat{T}_r = \widehat{\mathbf{p}}^2/2M$ is the kinetic-energy operator for the relative nuclear motion, $\widehat{\mathbf{p}} = -i\hbar\partial/\partial\mathbf{r}$, $\mathbf{r}$ is the internuclear vector, and M the reduced mass. The Hamiltonian $\widehat{H}_{el}$ includes the electronic kinetic-energy operators, the potential energy of their Coulomb interactions with one another and with the nuclei, and the Coulomb interaction energy of the nuclei[19]. We seek a solution of

$$\widehat{H}\psi = (\widehat{T}_r + \widehat{H}_{el})\psi = E\psi \tag{1}$$

in the form

$$\psi = \sum_m \chi_m(\mathbf{r})\varphi_m(q, \mathbf{r}), \tag{2}$$

where $\varphi_m(q, \mathbf{r})$ are orthonormal solutions of

$$\widehat{H}_{el}\varphi_m(q, \mathbf{r}) = U_m(r)\varphi_m(q, \mathbf{r}), \tag{3}$$

and q denotes all electronic coordinates. The $U_m(r)$ are eigenvalues of $\widehat{H}_{el}$ with r as a parameter. Substituting (2) in (1), multiplying on the left by $\varphi_n^*(q, \mathbf{r})$, and integrating over dq, we obtain

$$\left[\frac{\widehat{\mathbf{p}}^2}{2M} + V''_{nn} + U_n(r) - E\right]\chi_n(\mathbf{r}) = -\sum_m{}' (\widehat{V}'_{nm} + V''_{nm})\chi_m(\mathbf{r}), \tag{4}$$

where

$$\widehat{V}'_{nm} = \frac{1}{M}\mathbf{p}_{nm}\widehat{\mathbf{p}}, \qquad V''_{nm} = \frac{1}{2M}(\mathbf{p}^2)_{nm},$$

and $\mathbf{p}_{nm} = \int \varphi_n^*\widehat{\mathbf{p}}\varphi_m \, dq$ and $(\mathbf{p}^2)_{nm}$ are matrix elements with respect to the electronic wave functions. The diagonal element $\mathbf{p}_{nn}$ vanishes by symmetry.

The electronic functions φ_n vary appreciably only over atomic distances; differentiating them with respect to $\mathbf{r}$ therefore introduces no large parameter M/m, where m is the electron mass. Thus V''_{nn} is smaller than $U_n(r)$ by m/M and may be omitted. If the terms on the right of (4) are regarded as a small perturbation, the zeroth-order functions $\chi_n(\mathbf{R})$ are solutions of

$$\left[\frac{\widehat{\mathbf{p}}^2}{2M} + U_n(r)\right]\chi_{nv} = E_{nv}\chi_{nv}, \tag{5}$$

describing nuclear motion in the field $U_n(r)$; v denotes the quantum numbers of this motion. Perturbation theory is applicable if

$$|\langle nv'|\widehat{V}'_{nm} + V''_{nm}|mv\rangle| \ll |E_{nv'} - E_{mv}|.$$

The right side contains energy differences belonging to different electronic terms and is of zeroth order in the small parameter m/M. The left side consists of matrix elements between nuclear wave functions. The term containing $\widehat{V}''_{nm}$ includes m/M and is certainly small. In the matrix element of $\widehat{V}'_{nm}$, the operator $\widehat{\mathbf{p}}$ acting on χ_{mv} multiplies it by a quantity of the order of the nuclear momentum. For small nuclear oscillations this momentum is $\sim \sqrt{M\hbar\omega_e}$. Since at the same time $\omega_e \propto 1/\sqrt{M}$, the matrix element $\langle nv'|\widehat{V}'_{nm}|mv\rangle$ is of order $(m/M)^{3/4}$.

[19]The Hamiltonian $\widehat{H}$ refers to the frame in which the center of mass of the entire molecule is at rest: $\mathbf{P}_n + \mathbf{P}_e = 0$, where $\mathbf{P}_n$ is the total momentum of the two nuclei and $\mathbf{P}_e$ the total electronic momentum. The term representing the kinetic energy of the nuclear center of mass, $\widehat{\mathbf{P}}_n^2/2(M_1 + M_2) = \mathbf{P}_e^2/2(M_1 + M_2)$, has already been omitted. It is certainly smaller than the electronic kinetic energy by the ratio m/M.

§ 83. Multiplet terms. Case *a*

We now turn to the classification of molecular levels with nonzero spin S. In the zeroth approximation, when relativistic effects are wholly neglected, the energy of a molecule, like that of any system of particles, is independent of the spin direction (the spin is "free"). Every energy level is therefore degenerate with respect to this direction. Relativistic effects remove the degeneracy and make the energy depend on the projection of the spin on the molecular axis. We shall refer to relativistic interactions in molecules as the *spin–axis interaction*. As in atoms, its chief contribution is the interaction of the spins with the orbital motion of the electrons[20].

The nature and classification of molecular levels depend essentially on the relative importance of the spin–orbital-motion interaction and molecular rotation. The importance of the latter is measured by the separations between neighboring rotational levels. Two limiting cases must accordingly be considered. In one, the spin–axis interaction energy is large compared with the rotational-level differences; in the other it is small. The first is conventionally called case (or coupling case) a, and the second case b (F. Hund, 1933).

Case a occurs most often. The main exception is provided by Σ terms, for which case b usually applies because their spin–axis interaction is weak (see below)[21]. For other terms, case b is sometimes encountered in the lightest molecules: there the spin–axis interaction is comparatively weak and the rotational-level spacings are large because the moment of inertia is small.

Intermediate cases between a and b are of course also possible. Moreover, as the rotational quantum number changes, one and the same electronic state may pass continuously from case a to case b. The spacings between neighboring rotational levels grow with the rotational quantum number. At large values they may therefore exceed the spin–axis coupling energy (case b), even though the lower rotational levels belonged to case a.

In case a, the classification differs little in principle from that of terms with zero spin. We first consider the electronic terms with the nuclei fixed, completely neglecting rotation. In addition to the projection Λ

of the electronic orbital angular momentum, we must now consider the projection of the total spin on the molecular axis. This projection is denoted by Σ[22] and has the values $S, S-1, \dots, -S$. We take Σ to be positive when the direction of the spin projection agrees with that of the orbital angular momentum about the axis (recall that Λ denotes the magnitude of the latter). The quantities Λ and Σ add to give the total electronic angular momentum about the molecular axis:

$$\Omega = \Lambda + \Sigma. \tag{83.1}$$

It takes the values $\Lambda + S, \Lambda + S - 1, \dots, \Lambda - S$. Thus an electronic term with orbital angular momentum Λ splits into $2S + 1$ terms distinguished by their values of Ω. As

[20] Besides spin–orbit and spin–spin interactions, there is an interaction of the spin and electronic orbital motion with the rotation of the molecule. This part of the interaction is very small and is of possible interest only for terms with $S = 1/2$ (see § 81).

[21] The normal electronic term $^3\Sigma$ of the O_2 molecule is a special case. Its coupling is intermediate between cases a and b (see problem 3 in § 84).

[22] It must not be confused with the symbol for terms having $\Lambda = 0$.

for atomic terms, this splitting is called the fine structure, or multiplet splitting, of the electronic levels. The value of Ω is written as a subscript on the term symbol; for example, $\Lambda = 1$, $S = 1/2$ gives the terms ${}^2\Pi_{1/2}$ and ${}^2\Pi_{3/2}$.

Allowing for nuclear motion produces vibrational and rotational structure for each of these terms. The rotational levels are characterized by the quantum number J, the total angular momentum of the molecule, including the orbital and spin angular momenta of the electrons and the rotational angular momentum of the nuclei[23]. The number J takes integral values beginning with $|\Omega|$:

$$J \geqslant |\Omega|. \tag{83.2}$$

This is the evident generalization of (82.6).

We derive quantitative formulas for the molecular levels in case a. First consider the fine structure of an electronic term. In treating the fine structure of atomic terms in § 72, we used (72.4), according to which the mean spin–orbit interaction is proportional to the projection of the atom's total spin on the orbital-angular-momentum vector. Quite analogously, the spin–axis interaction in a diatomic molecule, averaged over the electronic state at a specified internuclear distance r, is proportional to the projection Σ of the total molecular spin on its axis. The split electronic term may therefore be written

$$U(r) + A(r)\Sigma,$$

where $U(r)$ is the energy of the original unsplit term and $A(r)$ is a function of r. This function depends on the original term, in particular on Λ, but not on Σ. Since the quantum number Ω, rather than Σ, is normally used, it is more convenient to write $A\Omega$ in place of $A\Sigma$. The two expressions differ by $A\Lambda$, which may be included in $U(r)$. Thus the electronic term is

$$U(r) + A(r)\Omega. \tag{83.3}$$

The components of the split term are equally spaced: neighboring components, whose Ω values differ by one, are separated by $A(r)$ independently of Ω.

General considerations readily show that $A = 0$ for Σ terms. Apply time reversal. The energy must remain unchanged, while the directions of both the orbital and spin angular momenta about the axis are reversed. In the energy $A(r)\Sigma$, Σ changes sign, so $A(r)$ must also change sign if the energy is to remain invariant. For $\Lambda \neq 0$ this says nothing about the value of $A(r)$, since A depends on the orbital angular momentum, which itself reverses. But when $\Lambda = 0$, $A(r)$ must in any event remain unchanged and must therefore vanish identically. Hence, for Σ terms, the spin–orbit interaction causes no first-order splitting. A splitting proportional to Σ^2 would appear only from the spin–orbit interaction in second order or the spin–spin interaction in first order, and is consequently small. This explains the fact noted above that Σ terms generally belong to case b.

Once the multiplet splitting has been determined, molecular rotation can be included as a perturbation, in exact analogy with the derivation at the beginning of the preceding section. The rotational angular momentum of the nuclei is obtained by subtracting the

[23]As usual, K is reserved for the total molecular angular momentum excluding spin. In case a there is no quantum number K, since $\mathbf{K}$ is not even approximately conserved.

electronic orbital angular momentum and spin from the total angular momentum. The centrifugal-energy operator is therefore

$$B(r)(\widehat{\mathbf{J}} - \widehat{\mathbf{L}} - \widehat{\mathbf{S}})^2.$$

Averaging this quantity over the electronic state and adding it to (83.3), we obtain the required effective potential
energy $U_J(r)$:

$$\begin{aligned} U_J(r) &= U(r) + A(r)\Omega + B(r)\overline{(\mathbf{J} - \mathbf{L} - \mathbf{S})^2} \\ &= U(r) + A(r)\Omega + B(r)[\mathbf{J}^2 - 2\mathbf{J}(\overline{\mathbf{L} + \mathbf{S}}) + \overline{\mathbf{L}^2} + 2\overline{\mathbf{LS}} + \overline{\mathbf{S}^2}]. \end{aligned}$$

The eigenvalue of $\mathbf{J}^2$ is $J(J+1)$. By the same reasoning as in § 82,

$$\overline{\mathbf{L}} = \mathbf{n}\Lambda, \qquad \overline{\mathbf{S}} = \mathbf{n}\Sigma, \tag{83.4}$$

and $(\widehat{\mathbf{J}} - \widehat{\mathbf{L}} - \widehat{\mathbf{S}})\mathbf{n} = 0$, so that for the eigenvalues

$$\mathbf{Jn} = (\mathbf{L} + \mathbf{S})\mathbf{n} = \Lambda + \Sigma = \Omega. \tag{83.5}$$

Substitution gives

$$U_J(r) = U(r) + A(r)\Omega + B(r)[J(J+1) - 2\Omega^2 + \overline{\mathbf{L}^2} + 2\overline{\mathbf{LS}} + \overline{\mathbf{S}^2}].$$

The average over the electronic state is taken with the zeroth-approximation wave functions[24]. In this approximation spin is conserved, so $\mathbf{S}^2 = S(S+1)$. The wave function is a product of a spin function and a coordinate function, and the angular momenta $\mathbf{L}$ and $\mathbf{S}$ are consequently averaged independently:

$$\overline{\mathbf{LS}} = \Lambda\mathbf{n}\overline{\mathbf{S}} = \Lambda\Sigma.$$

Finally, the mean square orbital angular momentum $\overline{\mathbf{L}^2}$ is independent of spin and is a function of r characteristic of the given unsplit electronic term. All terms that depend on r but not on J and Σ may be included in $U(r)$, while the term proportional to Σ, or equivalently to Ω, may be absorbed into $A(r)\Omega$. The effective potential energy is therefore

$$U_J(r) = U(r) + A(r)\Omega + B(r)[J(J+1) - 2\Omega^2]. \tag{83.6}$$

The molecular energy levels can now be obtained in the same way as in § 82 from (82.5). Expanding $U(r)$ and $A(r)$ in powers of ξ, retaining terms through second order in $U(r)$ and only zeroth-order terms in the second and third terms, we find

$$E = U_e + A_e\Omega + \hbar\omega_e\left(v + \frac{1}{2}\right) + B_e[J(J+1) - 2\Omega^2], \tag{83.7}$$

where $A_e = A(r_e)$ and B_e are constants characteristic of the given unsplit electronic term. Continuing the expansion would produce further terms of higher degree in the quantum numbers; we shall not write them here.

[24] Zeroth order both in molecular rotation and in the spin–axis interaction.

§ 84. Multiplet terms. Case b

We now turn to case b. Here the effect of molecular rotation predominates over the multiplet splitting. We must therefore consider the rotational effect first, neglecting the spin–axis interaction, and then include the latter as a perturbation.

For a molecule with "free" spin, not only the total angular momentum $\mathbf{J}$ but also the sum $\mathbf{K}$ of the electronic orbital angular momentum and the angular momentum of nuclear rotation is conserved; it is related to $\mathbf{J}$ by

$$\mathbf{J} = \mathbf{K} + \mathbf{S}. \tag{84.1}$$

The quantum number K distinguishes the different states of a rotating molecule with free spin that arise from a given electronic term. The effective potential energy $U_K(r)$ in a state with a specified K is plainly given by the same formula (82.5) as for terms with $S = 0$:

$$U_K(r) = U(r) + B(r)K(K+1), \tag{84.2}$$

where K takes the values $\Lambda, \Lambda + 1, \ldots$

Including the spin–axis interaction generally splits each term into $2S + 1$ terms (or $2K + 1$ if $K < S$), distinguished by their values of the total angular momentum J[25]. By the general rule for addition of angular momenta, for a given K the number J takes the values from $K + S$ to $|K - S|$:

$$|K - S| \leqslant J \leqslant K + S. \tag{84.3}$$

To calculate the splitting energy in first-order perturbation theory, the mean value of the spin–axis interaction-energy operator must be found in the zero-order state with respect to this interaction. In the present case this means averaging both over the electronic state and over molecular rotation (at a fixed r). The first averaging gives an operator of the form $A(r)\mathbf{n}\widehat{\mathbf{S}}$, proportional to the projection of the spin operator on the molecular axis.

We next average this operator over molecular rotation, regarding the direction of the spin vector as arbitrary. Then

$$\overline{\mathbf{nS}} = \overline{\mathbf{n}}\mathbf{S}.$$

The mean value $\overline{\mathbf{n}}$ is a vector which, by symmetry, must have the same direction as the "vector" $\mathbf{K}$, the only vector characterizing the molecular rotation. We may therefore write

$$\overline{\mathbf{n}} = \text{const} \cdot \mathbf{K}.$$

The proportionality coefficient is readily determined by multiplying both sides by $\mathbf{K}$ and noting that the eigenvalues are $\mathbf{nK} = \Lambda$ (see (82.4)) and $\mathbf{K}^2 = K(K+1)$. Thus

$$\overline{\mathbf{nS}} = \frac{\Lambda}{K(K+1)}\mathbf{KS}.$$

[25] In case b, the projection $\mathbf{nS}$ of the spin on the molecular axis has no definite value, so the quantum numbers Σ (and Ω) do not exist.

Finally, by the general formula (31.3), the eigenvalue of the product **KS** is

$$\mathbf{KS} = \frac{1}{2}[J(J+1) - K(K+1) - S(S+1)]. \tag{84.4}$$

We consequently obtain the following expression for the required mean spin–axis interaction energy:

$$\frac{A(r)\Lambda}{2K(K+1)}[J(J+1)-S(S+1)-K(K+1)] = \frac{A(r)\Lambda}{2K(K+1)}(J-S)(J+S+1)-\frac{1}{2}A(r)\Lambda.$$

This expression must be added to the energy (84.2). The term $\frac{1}{2}A(r)\Lambda$, being independent of K and J, can be included in $U(r)$; the final expression for the effective potential energy is therefore

$$U_K(r) = U(r) + B(r)K(K+1) + A(r)\Lambda\frac{(J-S)(J+S+1)}{2K(K+1)}. \tag{84.5}$$

Expansion in powers of $\xi = r - r_e$ gives, in the usual way, the following expression for the molecular energy levels in case b:

$$E = U_e + \hbar\omega_e(v+1/2) + B_eK(K+1) + A_e\Lambda\frac{(J-S)(J+S+1)}{2K(K+1)}. \tag{84.6}$$

As already stated in the preceding section, for Σ-terms the spin–orbit interaction produces no multiplet splitting in first order. To determine the fine structure, the spin–spin interaction must be included; its operator

is quadratic in the electron spins. What interests us here is not this operator itself, but the result of averaging it over the electronic state of the molecule, just as was done for the spin–orbit interaction operator. Symmetry makes it clear that the averaged operator must be proportional to the square of the projection of the total molecular spin on the axis, and can therefore be written

$$a(r)(\mathbf{Sn})^2, \tag{84.7}$$

where $a(r)$ is again a function of the distance r characteristic of the electronic term in question (symmetry also permits a term proportional to $\mathbf{S}^2$, but it is of no interest because the magnitude of the spin is simply a constant). We shall not give here the cumbersome general formula for the splitting produced by the operator (84.7); Problem 1 below derives the formula for triplet Σ-terms.

Doublet Σ-terms are a special case. By Kramers' theorem (§ 60), in a system of particles with total spin $S = 1/2$ a twofold degeneracy necessarily remains even when all internal relativistic interactions in the system are fully included. Hence $^2\Sigma$-terms remain unsplit when both spin–orbit and spin–spin interactions are included to any order.

A splitting would occur here only if the relativistic interaction of the spin with molecular rotation were included; this effect is very small. The averaged operator of this interaction must evidently have the form $\gamma\mathbf{KS}$, and its eigenvalues follow from (84.4) on setting $S = 1/2$, $J = K \pm 1/2$. Thus, for $^2\Sigma$-terms,

$$E = U_e + \hbar\omega_e(v+1/2) + B_eK(K+1) \pm (\gamma/2)(K+1/2) \tag{84.8}$$

(the constant $-\gamma/4$ has been included in U_e).

Problem

Determine the multiplet splitting of a $^3\Sigma$ term in case b (H. Kramers, 1929).

Solution. The required splitting is determined by the operator (84.7), which must be averaged over molecular rotation. Write it as $a_e n_i n_k S_i S_k$, where $a_e = a(r_0)$. Since $\mathbf{S}$ is a conserved vector, only the product $n_i n_k$ is to be averaged. By the analogous formula obtained in the problem to § 29,

$$\overline{n_i n_k} = -\frac{\widehat{K}_i\widehat{K}_k + \widehat{K}_k\widehat{K}_i}{(2K-1)(2K+3)} + \cdots ;$$

terms proportional to δ_{ik} have not been displayed, since they would give an energy contribution independent of J and hence would not produce the splitting of interest. Thus the splitting is determined by the operator

$$-a_e \frac{S_i S_k(\widehat{K}_i\widehat{K}_k + \widehat{K}_k\widehat{K}_i)}{(2K-1)(2K+3)}.$$

Since $\mathbf{S}$ commutes with $\mathbf{K}$,

$$S_i S_k \widehat{K}_i\widehat{K}_k = S_i\widehat{K}_i S_k\widehat{K}_k = (\mathbf{SK})^2,$$

where the eigenvalue of $\mathbf{SK}$ is given by (84.4). Further,

$$\begin{aligned} S_i S_k \widehat{K}_k\widehat{K}_i &= S_i S_k \widehat{K}_i\widehat{K}_k + iS_i S_k e_{kil}\widehat{K}_l \\ &= (\mathbf{SK})^2 - \frac{1}{2}(S_i S_k - S_k S_i)ie_{ikl}\widehat{K}_l \\ &= (\mathbf{SK})^2 + \frac{1}{2}e_{ikl}e_{ikm}S_m K_l = (\mathbf{SK})^2 + \mathbf{SK}. \end{aligned}$$

The three components E_K of the $^3\Sigma$ triplet ($S = 1$) correspond to $J = K, K \pm 1$. The intervals between these components are

$$E_{K+1} - E_K = -a_e\frac{K+1}{2K+3}, \qquad E_{K-1} - E_K = -a_e\frac{K}{2K-1}.$$

Problem

Determine the energy of a doublet term (with $\Lambda \neq 0$) for cases intermediate between a and b (E. Hill, J. van Vleck, 1928).

Solution. Since the rotational energy and the spin–axis interaction energy are assumed to be of the same order, they must be treated simultaneously in perturbation theory, so that the perturbation operator is[26]

$$\widehat{V} = B_e\widehat{\mathbf{K}}^2 + A_e\mathbf{n}\widehat{\mathbf{S}}.$$

[26]The averaging over vibrations must be carried out before the averaging over rotation. We have therefore replaced the functions $B(r)$ and $A(r)$ by the values B_e, A_e, retaining only the first terms of the expansion in ξ. The unperturbed energy levels are $E^{(0)} = U_e + \hbar\omega_e(v + 1/2)$.

For the zero-order wave functions it is convenient to use the wave functions of states in which K and J have definite values (that is, the case-b functions). Since a doublet term has $S = 1/2$, at a given J the quantum number K can have the values $K = J \pm 1/2$. To form the secular equation we must calculate the matrix elements $(nSKJ|V|nSK'J)$ (n denotes the set of quantum numbers specifying the electronic term), where K, K' take these values. The matrix of $\mathbf{K}^2$ is diagonal, with diagonal elements $K(K+1)$. The matrix elements of $\mathbf{nS}$ are calculated from the general formula (109.5), in which S, K, J play the parts of j_1, j_2, J; the reduced matrix elements of $\mathbf{n}$ are given by (87.4). The resulting secular equation is

$$\begin{vmatrix} B_e(J+1/2)(J+3/2) - \dfrac{A_e\Lambda}{2J+1} - E^{(1)} & \dfrac{A_e}{2J+1}\sqrt{(J+1/2)^2-\Lambda^2} \\ \dfrac{A_e}{2J+1}\sqrt{(J+1/2)^2-\Lambda^2} & B_e(J+1/2)(J-1/2) + \dfrac{A_e\Lambda}{2J+1} - E^{(1)} \end{vmatrix} = 0.$$

Solving this equation and adding $E^{(1)}$ to the unperturbed energy, we obtain

$$E = U_e + \hbar\omega_e(v+1/2) + B_eJ(J+1) \pm \sqrt{B_e^2(J+1/2)^2 - A_eB_e\Lambda + A_e^2/4}$$

(the constant $B_e/4$ has been included in U_e). Case a corresponds to $A_e \gg B_eJ$, and case b to the reverse inequality.

Problem

Determine the intervals between the components of a triplet $^3\Sigma$ level in a case intermediate between a and b.

Solution. As in Problem 2, the rotational energy and the spin–spin interaction energy are treated simultaneously in perturbation theory. The perturbation operator is

$$\hat{V} = B_e\hat{\mathbf{K}}^2 + a_e(\mathbf{n}\hat{\mathbf{S}})^2.$$

We use the case-b functions as the zero-order wave functions. The matrix elements $(K|\mathbf{nS}|K')$ (all indices with respect to which the matrix is diagonal are omitted) are again calculated from (109.5) and (87.4), now with $\Lambda = 0$, $S = 1$. The nonzero elements are

$$(J|\mathbf{nS}|J-1) = \sqrt{\frac{J+1}{2J+1}}, \qquad (J|\mathbf{nS}|J+1) = \sqrt{\frac{J}{2J+1}}.$$

For a given J, K can have the values $K = J, J \pm 1$. The matrix elements $(K|V|K')$ are

$$\begin{aligned}
(J|V|J) &= B_eJ(J+1) + a_e, \\
(J-1|V|J-1) &= B_e(J-1)J + a_e\frac{J+1}{2J+1}, \\
(J+1|V|J+1) &= B_e(J+1)(J+2) + a_e\frac{J}{2J+1}, \\
(J-1|V|J+1) = (J+1|V|J-1) &= a_e\frac{J(J+1)}{2J+1}.
\end{aligned}$$

There are no transitions between the state with $K = J$ and the states with $K = J \pm 1$. One of the levels is therefore simply $E_1 = (J|V|J)$. The other two, E_2, E_3, are obtained by solving the quadratic secular equation formed from the matrix elements for transitions between the states $J \pm 1$. Since only the relative positions of the triplet components are of interest, subtract the constant a_e from all three energies $E_{1,2,3}$. We then obtain

$$E_1 = B_e J(J+1), \qquad E_{2,3} = B_e(J^2 + J + 1) - \frac{a_e}{2} \pm \sqrt{B_e^2(2J+1)^2 - a_e B_e + \frac{a_e^2}{4}}.$$

In case b (a_e small), consideration of the three levels with equal K and different J ($J = K, K \pm 1$) again gives the formulas found in Problem 1.

§ 85. Multiplet terms. Cases c and d

Besides coupling cases a and b, and the cases intermediate between them, other coupling types also exist. Their origin is as follows. The appearance of the quantum

number Λ is ultimately due to the electric interaction of the two atoms in the molecule, which gives axial symmetry to the problem of determining the electronic terms (this interaction in a molecule is described as the coupling of the orbital angular momentum to the axis). The magnitude of this interaction is measured by the separations between terms having different values of Λ. Throughout the preceding discussion, this interaction was tacitly assumed to be so strong that these separations are large in comparison both with the multiplet-splitting intervals and with the intervals in the rotational structure of the terms. The reverse situation also occurs, however, when the interaction of the orbital angular momentum with the axis is comparable with, or even small in comparison with, the other effects. In such cases the projection of the orbital angular momentum on the axis cannot be regarded as conserved in any approximation, and the number Λ therefore loses its meaning.

When the coupling of the orbital angular momentum to the axis is weak compared with the spin–orbit coupling, the system is said to be in *case c*. This case occurs in molecules containing a rare-earth atom. Such atoms have f-electrons with uncompensated angular momenta; their interaction with the molecular axis is weakened because the f-electrons lie deep within the atom. Coupling types intermediate between a and c occur in molecules composed of heavy atoms.

When the coupling of the orbital angular momentum to the axis is weak compared with the intervals of the rotational structure, the system is said to be in *case d*. This case is encountered for high rotational levels (with large J) of certain electronic terms of the lightest molecules (H_2, He_2). These terms contain a strongly excited electron whose interaction with the remaining electrons (or, as it is called, with the molecular "core") is so weak that its orbital angular momentum is not quantized along the molecular axis (whereas the core has a definite angular momentum Λ_{core} about that axis).

As the distance r between the nuclei increases, the interaction of the atoms becomes weaker and eventually becomes small in comparison with the spin–orbit interaction within the atoms. Consequently, electronic terms at sufficiently large r belong to case c. This circumstance must be kept in mind when establishing the correspondence between molecular electronic terms and the atomic states obtained as $r \to \infty$.

In § 80 we considered this correspondence while neglecting the spin–orbit interaction. When the fine structure of the terms is included, there is the additional question of the correspondence between the values J_1 and J_2 of the total angular momenta of the isolated atoms

and the values of the molecular quantum number Ω. We give the results here without repeating the reasoning, which is entirely analogous to that used in § 80.

If the molecule consists of unlike atoms, the possible values of Ω[27] formed by joining atoms with angular momenta J_1 and J_2 ($J_1 \geqslant J_2$) are determined by the same scheme (80.1), with J_1, J_2 in place of L_1, L_2 and $|\Omega|$ in place of Λ. The sole difference is that, when $J_1 + J_2$ is half-integral, the smallest value of $|\Omega|$ is $1/2$ rather than the zero shown in the scheme. When $J_1 + J_2$ is integral, there are $2J_2 + 1$ terms with $\Omega = 0$, for which (as with Σ-terms when fine structure is neglected) their sign must be determined. If J_1 and J_2 are both half-integral, the number $2J_2 + 1$ is even, and there are equal numbers of terms conventionally denoted by 0^+ and 0^-. If J_1 and J_2 are both integral, there are $J_2 + 1$ terms 0^+ and J_2 terms 0^- when $(-1)^{J_1+J_2} P_1 P_2 = 1$, and the reverse when $(-1)^{J_1+J_2} P_1 P_2 = -1$.

If the molecule consists of identical atoms in different states, the resulting molecular states are the same as for unlike atoms, except that the total number of terms is doubled: each term occurs once as even and once as odd.

Finally, if the molecule consists of identical atoms in identical states (with angular momenta $J_1 = J_2 = J$), the total number of states remains the same as for unlike atoms, while their parity distribution is

$$\begin{array}{lll} \text{if } J \text{ is integral,} & \Omega \text{ even:} & N_g = N_u + 1, \\ & \Omega \text{ odd:} & N_g = N_u, \\ \text{if } J \text{ is half-integral,} & \Omega \text{ even:} & N_u = N_g, \\ & \Omega \text{ odd:} & N_u = N_g + 1. \end{array}$$

All the 0^+ terms are then even and all the 0^- terms are odd.

As the nuclei approach one another, coupling of type c usually passes into coupling of type a[28]. The following noteworthy situation can then arise.

As stated above, a term with $\Lambda = 0$ belongs to case b; from the standpoint of the case-a classification this means that the multiplet levels having different values of Ω (and the same $\Lambda = 0$) have the same energy. Such levels, however,

may arise as atoms in different fine-structure states approach each other.

Thus, different pairs of atomic fine-structure states may correspond to the same molecular term. An analogous situation may occur for terms with $\Omega = 0$ that, as the nuclei approach, pass into a molecular term with $\Lambda \neq 0$ (and hence $\Sigma = -\Lambda$). Such levels are doubly degenerate because, in case a, the terms 0^+ and 0^- (which may arise from different pairs of atomic states) have the same energy[29].

[27] When the two total atomic angular momenta $\mathbf{J}_1$ and $\mathbf{J}_2$ are combined into the resultant angular momentum $\mathbf{\Omega}$, the sign of Ω is clearly immaterial.

[28] The correspondence between the case-a and case-c classifications of terms cannot be established in general. It requires a specific examination of the potential-energy curves with allowance for the non-crossing rule for levels of the same symmetry (§ 79).

[29] We neglect here the so-called Λ-doubling (see § 88).

§ 86. Symmetry of molecular terms

In § 78 we have already considered some symmetry properties of the terms of a diatomic molecule. Those properties described the behavior of the wave functions under transformations that leave the nuclear coordinates unchanged. Thus, symmetry of the molecule under reflection in a plane through its axis gives the distinction between Σ^+ and Σ^- terms; symmetry under reversal of the signs of all electronic coordinates (for a molecule composed of identical atoms)[30] divides the terms into even and odd ones.

These symmetry properties characterize an electronic term and are the same for all rotational levels belonging to that term.

The states of a molecule (like those of any system of particles; see § 30) are also characterized by their behavior under inversion, that is, simultaneous reversal of the signs of the coordinates of all electrons and nuclei.

Accordingly, all molecular terms are divided into *positive* terms, whose wave functions are unchanged when the signs of the electron and nuclear coordinates are reversed, and *negative* terms, whose wave functions change sign under inversion[31].

For $\Lambda \neq 0$, every term is doubly degenerate, corresponding to the two possible directions of the angular momentum relative to the axis

of the molecule. Under inversion the angular momentum itself does not change sign, but the molecular axis reverses direction (the atoms interchange places), and hence the direction of Λ relative to the molecule also reverses. The two wave functions belonging to a given energy level are therefore transformed into one another, and from them one can always form one linear combination invariant under inversion and another that changes sign under it. Each term thus gives two states, one positive and the other negative. In fact, every term with $\Lambda \neq 0$ does undergo a splitting (see § 88), so these two states correspond to different energies.

The signs of Σ-terms require separate consideration. First, it is clear that spin has no bearing on the sign of a term: inversion acts only on particle coordinates and leaves the spin part of the wave function unchanged. Consequently, all components of the multiplet structure of any one term have the same sign. In other words, the sign of the term depends only on K, not on J[32].

The molecular wave function is a product of electronic and nuclear wave functions. It was shown in § 82 that, in a Σ state, the motion of the nuclei is equivalent to the motion of one particle with orbital angular momentum K in a central field $U(r)$. The nuclear wave function is therefore multiplied by $(-1)^K$ when the signs of the nuclear coordinates are reversed (see (30.7)).

The electronic wave function characterizes the electronic term. To determine its behavior under inversion, consider it in a coordinate system rigidly attached to the nuclei and rotating with them. Let xyz be a space-fixed coordinate system and $\xi\eta\zeta$ a rotating system in which the molecule as a whole is at rest. Choose the directions of the $\xi\eta\zeta$ axes

[30]The origin of coordinates is assumed to lie on the molecular axis, midway between the two nuclei.

[31]We retain the accepted terminology. It is unfortunate, since for an atom the behavior of terms under inversion is described by their parity, rather than by their sign.

For Σ-terms, this sign must not be confused with the + and − signs written as superscripts!

[32]Recall that Σ-terms ordinarily belong to case b, and hence the quantum numbers K and J must be used.

so that the ζ axis coincides with the molecular axis and points, say, from nucleus 1 to nucleus 2, while the relative orientation of the positive $\xi\eta\zeta$ directions is the same as in the xyz system (that is, if xyz is right-handed, $\xi\eta\zeta$ must also be right-handed). Inversion reverses the directions of the xyz axes, turning a right-handed system into a left-handed one. The $\xi\eta\zeta$ system must likewise become left-handed. The ζ axis, however, is rigidly attached to the nuclei and retains its former direction; hence the direction

of either the ξ or the η axis must be reversed. Thus inversion in the fixed coordinate system is equivalent, in the moving system, to reflection in a plane through the molecular axis. Under such a reflection the electronic wave function of a Σ^+ term is unchanged, while that of a Σ^- term changes sign.

It follows that the sign of the rotational components of a Σ^+ term is determined by the factor $(-1)^K$: all levels with even K are positive and those with odd K are negative. For a Σ^- term the sign of the rotational levels is determined by $(-1)^{K+1}$, and all levels with even K are negative while those with odd K are positive.

If the molecule consists of identical atoms[33], its Hamiltonian is also invariant under interchange of the coordinates of the two nuclei. A term is called symmetric with respect to the nuclei if its wave function is unchanged when the nuclei are interchanged, and antisymmetric if the wave function changes sign. Symmetry with respect to the nuclei is closely related to the parity and sign of a term. Interchanging the nuclear coordinates is equivalent to reversing the signs of the coordinates of all particles (electrons and nuclei), followed by reversing only the electronic coordinates. Hence, if a term is even (odd) and at the same time positive (negative), it is symmetric with respect to the nuclei. If a term is even (odd) and at the same time negative (positive), it is antisymmetric with respect to the nuclei.

At the end of § 62 a general theorem was established: the coordinate wave function of a system of two identical particles is symmetric for even total spin and antisymmetric for odd total spin. Applying this result to the two nuclei of a molecule composed of identical atoms, we find that the symmetry of the term is related to the parity of the resultant spin I obtained by combining the spins i of the two nuclei. The term is symmetric for even I and antisymmetric for odd I[34]. In particular, if the nuclei have no spin ($i = 0$), then I is also zero; the molecule will therefore have no antisymmetric terms at all. Thus nuclear spin has an important indirect effect on molecular terms, although its

direct effect (the hyperfine structure of the terms) is altogether negligible.

Allowance for nuclear spin gives an additional degeneracy of the levels. In the same § 62 the numbers of states with even and odd values of I obtained by combining two spins i were counted. Thus, for half-integral i, the number of states with even I is $i(2i+1)$ and that with odd I is $(i+1)(2i+1)$. It follows from the preceding discussion that the ratio of the degeneracies g_s, g_a[35] of the symmetric and antisymmetric terms for half-integral i is

$$\frac{g_s}{g_a} = \frac{i}{i+1}. \tag{86.1}$$

[33] The two atoms must belong not only to the same element but also to the same isotope.

[34] In view of the relation among parity, sign, and symmetry of terms, it follows that for even resultant nuclear spin I, positive levels are even and negative levels are odd; for odd I the reverse holds.

[35] In this connection, the degeneracy of a level is often called its *statistical weight*. Formulas (86.1) and (86.2) determine the ratios of the nuclear statistical weights of symmetric and antisymmetric levels.

For integral i, the corresponding ratio is

$$\frac{g_s}{g_a} = \frac{i+1}{i}. \tag{86.2}$$

We have seen that the sign of the rotational components of a Σ^+ term is determined by $(-1)^K$. Thus, for example, the rotational components of a ${}^1\Sigma_g^+$ term are positive and hence symmetric for even K, while for odd K they are negative and hence antisymmetric. In view of the results just obtained, the nuclear statistical weights of the rotational components of a ${}^1\Sigma_g^+$ level with successive values of K alternate in the ratio (86.1) or (86.2). The same applies to the levels ${}^1\Sigma_u^-$, and also to ${}^1\Sigma_g^-$ and ${}^1\Sigma_u^+$. In particular, for $i = 0$ the statistical weights vanish for levels with even K in the terms ${}^1\Sigma_u^-$, ${}^1\Sigma_g^-$, and for levels with odd K in the terms ${}^1\Sigma_g^+$, ${}^1\Sigma_u^+$. In other words, in the electronic states ${}^1\Sigma_u^-$, ${}^1\Sigma_g^-$ there are no rotational states with even K, while in the states ${}^1\Sigma_g^+$, ${}^1\Sigma_u^+$ there are no rotational states with odd K.

Because the interaction of the nuclear spins with the electrons is extremely weak, the probability of a change in I is very small even in molecular collisions. Molecules differing in the parity of I, and consequently possessing only symmetric or only antisymmetric terms, therefore behave in practice as distinct modifications of the substance. Examples are the so-called orthohydrogen and parahydrogen: in a molecule of the former, the spins $i = 1/2$ of the two nuclei are parallel ($I = 1$), while in the latter they are antiparallel ($I = 0$).

§ 87. Matrix elements for a diatomic molecule

This section gives several general formulas for matrix elements of physical quantities of a diatomic molecule. We begin with matrix elements for transitions between states of zero spin.

Let $\mathbf{A}$ be some vector physical quantity characterizing the molecule with its nuclei fixed (for example, its electric or magnetic dipole moment). First consider this quantity in the coordinate system $\xi\eta\zeta$ rotating with the molecule, with the ζ axis along the molecular axis. The angular momentum of the molecule relative to this system (that is, the electronic angular momentum $\mathbf{L}$) is not conserved in full, but its ζ component is conserved. The selection rules for the quantum number $L_\zeta = \Lambda$ therefore remain valid (and coincide with the selection rules for M in § 29). Thus the nonzero vector matrix elements are

$$\langle n'\Lambda|A_\zeta|n\Lambda\rangle, \quad \langle n'\Lambda|A_\xi + iA_\eta|n, \Lambda - 1\rangle, \quad \langle n', \Lambda - 1|A_\xi - iA_\eta|n\Lambda\rangle \tag{87.1}$$

(n enumerates the electronic terms for a given Λ).

If both terms are Σ terms, one must also take account of the selection rule associated with reflection symmetry in a plane containing the molecular axis. Under such a reflection the ζ component of an ordinary (polar) vector does not change, whereas that of an axial vector changes sign. It follows that for a polar vector A_ζ has nonzero matrix elements only for the transitions $\Sigma^+ \to \Sigma^+$ and $\Sigma^- \to \Sigma^-$, and for an axial vector only for $\Sigma^+ \to \Sigma^-$. We do not discuss the components A_ξ, A_η, since transitions without a change in Λ are impossible for them in any case.

If the molecule consists of identical atoms, there is also a parity selection rule. The components of a polar vector change sign under inversion. Their matrix elements are therefore nonzero only for transitions between states of opposite parity (and, for an axial vector, conversely). In particular, all diagonal matrix elements of the components of a polar vector vanish identically.

The connection between the matrix elements (87.1) and those of the same vector in the fixed coordinate system xyz is furnished by the general formulas derived below, in § 110, for any axially symmetric physical system.

After separating out the dependence on the quantum number M_K common to every vector (the z projection of the molecule's total angular momentum

K), there remain the reduced matrix elements $\langle n'K'\Lambda'\|A\|nK\Lambda\rangle$. Their relation to (87.1) is specified by (110.7), with $k = k' = 1$ (corresponding to a vector) and the appropriate change of notation for the quantum numbers (recall that, by (82.4), Λ equals the ζ component of the total angular momentum **K**). Using the relation (107.1) between the components of a spherical tensor of rank one and the Cartesian vector components, and taking the $3j$ symbols from Table 9 (p. 530), we obtain for the elements diagonal in Λ:

$$\begin{aligned}\langle n'K\Lambda\|A\|nK\Lambda\rangle &= \Lambda\sqrt{\frac{2K+1}{K(K+1)}}\langle n'\Lambda|A_\zeta|n\Lambda\rangle,\\ \langle n',K-1,\Lambda\|A\|nK\Lambda\rangle &= i\sqrt{\frac{K^2-\Lambda^2}{K}}\langle n'\Lambda|A_\zeta|n\Lambda\rangle\end{aligned} \tag{87.2}$$

and for the elements nondiagonal in Λ:

$$\begin{aligned}\langle n'K\Lambda\|A\|nK,\Lambda-1\rangle &= \left[\frac{(2K+1)(K+\Lambda)(K-\Lambda+1)}{4K(K+1)}\right]^{1/2}\\ &\quad\times\langle n'\Lambda|A_\xi+iA_\eta|n,\Lambda-1\rangle,\\ \langle n'K\Lambda\|A\|n,K-1,\Lambda-1\rangle &= i\left[\frac{(K+\Lambda)(K+\Lambda-1)}{4K}\right]^{1/2}\\ &\quad\times\langle n'\Lambda|A_\xi+iA_\eta|n,\Lambda-1\rangle,\\ \langle n',K-1,\Lambda\|A\|nK,\Lambda-1\rangle &= i\left[\frac{(K-\Lambda)(K-\Lambda+1)}{4K}\right]^{1/2}\\ &\quad\times\langle n'\Lambda|A_\xi+iA_\eta|n,\Lambda-1\rangle.\end{aligned} \tag{87.3}$$

The remaining nonzero elements follow from these by using the Hermiticity relations for the reduced matrix elements,

$$\langle nK\Lambda\|A\|n'K'\Lambda'\rangle = \langle n'K'\Lambda'\|A\|nK\Lambda\rangle^*,$$

and for the matrix elements in the $\xi\eta\zeta$ system,

$$\langle n\Lambda|A_\xi - iA_\eta|n'\Lambda'\rangle = \langle n'\Lambda'|A_\xi + iA_\eta|n\Lambda\rangle^*, \qquad \langle n\Lambda|A_\zeta|n'\Lambda'\rangle = \langle n'\Lambda'|A_\zeta|n\Lambda\rangle^*.$$

We record separately the formulas for the matrix elements of the vector $\mathbf{A} = \mathbf{n}$, the unit vector along the molecular axis. Here simply $A_\xi = A_\eta = 0$, $A_\zeta = 1$, so in the $\xi\eta\zeta$

system only the diagonal elements are nonzero: $\langle n\Lambda|A_\zeta|n\Lambda\rangle = 1$. The reduced matrix elements are diagonal in every index except K; displaying only that index, we have

$$\langle K\|n\|K\rangle = \Lambda\sqrt{\frac{2K+1}{K(K+1)}}, \qquad \langle K-1\|n\|K\rangle = i\sqrt{\frac{K^2-\Lambda^2}{K}}. \tag{87.4}$$

(H. Hönl, F. London, 1925). For $\Lambda = 0$ these formulas give

$$\langle K\|n\|K\rangle = 0, \qquad \langle K-1\|n\|K\rangle = i\sqrt{K},$$

which, as expected, are precisely the matrix elements of a unit vector for motion in a centrally symmetric field (see (29.14)).

We now indicate how the formulas must be modified for transitions between states of nonzero spin. It is important here whether the states belong to case a or case b.

If both states belong to case a, the formulas change essentially only in notation. The quantum numbers K, M_K do not exist; in their place are the total angular momentum J and its z projection M_J. In addition there are the numbers S and $\Omega = \Lambda + \Sigma$, so the reduced matrix elements are written

$$\langle n'J'S'\Omega'\Lambda'\|A\|nJS\Omega\Lambda\rangle.$$

Let $\mathbf{A}$ be an orbital vector, that is, one independent of spin. Its operator commutes with the spin operator $\widehat{\mathbf{S}}$, so its matrix is diagonal in S and $S_\zeta = \Sigma$; consequently $\Omega = \Lambda + \Sigma$ changes together with Λ (that is, $\Omega' - \Omega = \Lambda' - \Lambda$). Formulas (87.2)–(87.4) change only by the addition of indices to the matrix elements and the replacement of K, Λ by J, Ω in the remaining factors. For example, the first formula in (87.2) is replaced by

$$\langle n'JS\Omega\Lambda\|A\|nJS\Omega\Lambda\rangle = \Omega\sqrt{\frac{2J+1}{J(J+1)}}\langle n'\Omega\Lambda|A_\zeta|n\Omega\Lambda\rangle$$

(the diagonal index S is omitted).

Now let $\mathbf{A} = \mathbf{S}$. Since the spin operator commutes with the orbital angular momentum and with the Hamiltonian, its matrix is diagonal in n, Λ. It is not, however, diagonal in S and Σ (or Ω). The matrix elements of A_ξ, A_η, A_ζ for

transitions $S, \Sigma \to S', \Sigma'$ are given by (27.13), with S, Σ in place of L, M. The change to the xyz coordinate system is then made by (87.2), (87.3), replacing K, Λ by J, Ω. In this way, for example, we obtain

$$\begin{aligned}\langle J\Omega\|S\|J,\Omega-1\rangle &= \left[\frac{(2J+1)(J+\Omega)(J-\Omega+1)}{4J(J+1)}\right]^{1/2}\langle\Omega|S_\xi + iS_\eta|\Omega-1\rangle \\ &= \left[\frac{(2J+1)(J+\Omega)(J-\Omega+1)(S+\Sigma)(S-\Sigma+1)}{4J(J+1)}\right]^{1/2}.\end{aligned} \tag{87.5}$$

(the diagonal indices n, S, Λ are omitted).

Next let both states belong to case b, and let $\mathbf{A}$ be an orbital vector. The matrix elements are calculated in two stages. First consider the rotating molecule before coupling

S and **K**; the matrix elements are diagonal in S and are given by the same formulas (87.2), (87.3). At the second stage **K** is coupled with **S** to form the total angular momentum **J**, and the passage to the new matrix elements is made by the general formulas (109.3) (with K, S, J playing the roles of j_1, j_2, J in those formulas). Thus, for elements diagonal in J, K, Λ, we first obtain

$$\langle n'JK\Lambda\|A\|nJK\Lambda\rangle = (-1)^{K+J+S+1}(2J+1)\begin{Bmatrix} K & J & S \\ J & K & 1 \end{Bmatrix}\langle n'K\Lambda\|A\|nK\Lambda\rangle,$$

and then, taking the value of the $6j$ symbol from Table 10 (p. 542) and the reduced matrix element from (87.2), finally

$$\langle n'JK\Lambda\|A\|nJK\Lambda\rangle = \Lambda\left[\frac{2J+1}{J(J+1)}\right]^{1/2}\frac{J(J+1)+K(K+1)-S(S+1)}{2K(K+1)}\langle n'\Lambda|A_\zeta|n\Lambda\rangle.$$

Matrix elements for transitions between two states, one belonging to case a and the other to case b, are calculated similarly; we shall not pursue that calculation here.

Problem

Determine the Stark splitting of the terms of a diatomic molecule possessing a permanent dipole moment; the term belongs to case a.

Solution. The energy of a dipole **d** in an electric field $\mathcal{E}$ is $-\mathbf{d}\mathcal{E}$. Symmetry makes it clear that the dipole moment of a diatomic molecule is directed along its axis: $\mathbf{d} = d\mathbf{n}$ (d is constant). Choosing the field direction as the z axis, the perturbation operator is $-dn_z\mathcal{E}$.

Using the formulas derived above to find the diagonal matrix elements of n_z, we obtain for case a the following level splitting:[36]

$$\Delta E = -\mathcal{E}dM_J\frac{\Omega}{J(J+1)}.$$

Problem

The same, for a term belonging to case b (with $\Lambda \neq 0$).

Solution. In the same way we find

$$\Delta E = -\mathcal{E}dM_J\Lambda\frac{J(J+1)-S(S+1)+K(K+1)}{2K(K+1)J(J+1)}.$$

[36]This may appear to contradict the general statement (see § 76) that the linear Stark effect is absent. There is, of course, no contradiction: here the linear effect is associated with the twofold degeneracy of levels having $\Omega \neq 0$. The formula therefore applies provided that the Stark-splitting energy is large compared with the energy of the so-called Λ-doubling (see § 88).

PROBLEM

The same for a $^1\Sigma$ term.

SOLUTION. For $\Lambda = 0$ the linear effect is absent, and second-order perturbation theory is required. In the sum in the general formula (38.10), it is sufficient to retain only terms corresponding to transitions between rotational components of the same electronic term (the energy differences in the denominators of the other terms are large). Thus

$$\Delta E = d^2\mathcal{E}^2\left\{\frac{|\langle KM_K|n_z|K-1, M_K\rangle|^2}{E_K - E_{K-1}} + \frac{|\langle KM_K|n_z|K+1, M_K\rangle|^2}{E_K - E_{K+1}}\right\},$$

where $E_K = BK(K+1)$. A direct calculation gives

$$\Delta E = \frac{d^2\mathcal{E}^2}{B}\frac{K(K+1) - 3M_K^2}{2K(K+1)(2K-1)(2K+3)}.$$

§ 88. Λ-doubling

The twofold degeneracy of terms with $\Lambda \neq 0$ (see § 78) is actually approximate. It holds only insofar as the influence of molecular rotation on the electronic state (and higher-order spin–orbit effects) is neglected, as throughout the preceding theory. Allowing for the interaction of the electronic state with rotation splits a term with $\Lambda \neq 0$ into two closely spaced levels. This phenomenon is called Λ-doubling (E. Hill, J. van Vleck, R. Kronig, 1928).

We again begin the quantitative treatment with singlet terms ($S = 0$). In § 82 the rotational-level energies were calculated to first order in perturbation theory by finding diagonal matrix elements (the mean value) of the operator

$$B(r)(\widehat{\mathbf{K}} - \widehat{\mathbf{L}})^2.$$

For the subsequent orders we must consider elements of this operator nondiagonal in Λ. The operators $\widehat{\mathbf{K}}^2$ and $\widehat{\mathbf{L}}^2$ are diagonal in Λ, so only $2B\widehat{\mathbf{K}}\widehat{\mathbf{L}}$ need be considered.

The matrix elements of $\widehat{\mathbf{K}}\widehat{\mathbf{L}}$ are conveniently calculated with (29.12), setting $\mathbf{A} = \mathbf{K}$, $\mathbf{B} = \mathbf{L}$; K, M_K play the roles of L, M, while n, Λ replaces n, where n denotes the set of quantum numbers (excluding Λ) that specifies the electronic term. Since the matrix of the conserved vector $\mathbf{K}$ is diagonal in n, Λ, while the matrix of $\mathbf{L}$ has nondiagonal elements only for transitions changing Λ by one (compare the discussion of an arbitrary vector $\mathbf{A}$ in § 87), formulas (87.3) give

$$\langle n'\Lambda KM_K|\mathbf{KL}|n, \Lambda-1, KM_K\rangle = \frac{1}{2}\langle n'\Lambda|L_\xi + iL_\eta|n, \Lambda-1\rangle\sqrt{(K+\Lambda)(K+1-\Lambda)}. \tag{88.1}$$

There are no matrix elements corresponding to a larger change in Λ.

The perturbing action of the elements for $\Lambda \to \Lambda - 1$ can produce an energy difference between the states with $\pm\Lambda$ only in order 2Λ of perturbation theory. The effect is consequently proportional to $B^{2\Lambda}$, that is, to $(m/M)^{2\Lambda}$ (M is the nuclear mass and m the electron mass). For $\Lambda > 1$ this quantity is too small to be of interest. Thus Λ-doubling is significant only for Π terms ($\Lambda = 1$), which are considered below.

For $\Lambda = 1$ we must go to second order. The corrections to the energy eigenvalues may be found from the general formula (38.10). The denominators of its summands contain energy differences of the form $E_{n,\Lambda,K} - E_{n',\Lambda-1,K}$. The terms involving K cancel in these differences, because at a fixed internuclear distance r the rotational energy has the same value $B(r)K(K+1)$ for every term.

The dependence of the required splitting ΔE on K is therefore determined entirely by the squares of the matrix elements in the numerators. These include squares of the elements for transitions changing Λ from 1 to 0 and from 0 to -1; by (88.1) both have the same dependence on K. The splitting of a $^1\Pi$ term is consequently

$$\Delta E = \text{const}\ K(K+1), \tag{88.2}$$

where, in order of magnitude, const $\sim B^2/\varepsilon$, and ε is of the order of the separations between neighboring electronic terms.

We turn to terms with nonzero spin ($^2\Pi$ and $^3\Pi$ terms; higher values of S scarcely occur). If the term belongs to case b, the multiplet splitting has no effect at all on the Λ-doubling of the rotational levels, which is still given by (88.2).

In case a, by contrast, spin has an essential influence. Each electronic term is characterized not only by Λ but also by Ω. Merely replacing Λ by $-\Lambda$ changes $\Omega = \Lambda + \Sigma$, and therefore gives an entirely different term. The mutually degenerate states are those with Λ, Ω and $-\Lambda, -\Omega$. This degeneracy can be removed not only by the interaction between orbital angular momentum and molecular rotation considered above, but also by the spin–orbit interaction. Conservation of the projection Ω of the total angular momentum on the molecular axis is an exact conservation law (for fixed nuclei), and hence cannot be violated by the spin–orbit interaction. The latter can, however, change Λ and Σ simultaneously, having matrix elements for the corresponding transitions, in such a way that Ω remains unchanged. Acting alone, or together with the orbit–rotation interaction (which changes Λ without changing Σ), this effect can produce Λ-doubling.

First consider $^2\Pi$ terms. For a $^2\Pi_{1/2}$ term ($\Lambda = 1, \Sigma = -1/2, \Omega = 1/2$), the splitting arises when both the spin–orbit and orbit–rotation interactions are included, each to first order. The former produces the transition $\Lambda = 1, \Sigma = -1/2 \to \Lambda = 0, \Sigma = 1/2$; the latter then takes the state $\Lambda = 0, \Sigma = 1/2$ into $\Lambda = -1, \Sigma = 1/2$, which differs from the initial state by reversal of the signs of Λ and Ω. The spin–orbit matrix elements do not depend on the rotational quantum number J, while the dependence of the orbit–rotation matrix elements is given by (88.1), with K, Λ under the radical replaced by J, Ω. Thus the Λ-doubling of the $^2\Pi_{1/2}$ term is

$$\Delta E_{1/2} = \text{const}\,(J + 1/2), \tag{88.3}$$

where const $\sim AB/\varepsilon$. For a $^2\Pi_{3/2}$ term, however, the splitting can arise only in higher orders, so in practice $\Delta E_{3/2} = 0$.

Finally consider $^3\Pi$ terms. For a $^3\Pi_0$ term ($\Lambda = 1$, $\Sigma = -1$), the splitting is produced by the spin–orbit interaction in second order (through the transitions $\Lambda = 1$, $\Sigma = -1 \to \Lambda = 0$, $\Sigma = 0 \to \Lambda = -1$, $\Sigma = 1$). Hence in this case the Λ-doubling is entirely independent of J:

$$\Delta E_0 = \text{const}, \tag{88.4}$$

where const $\sim A^2/\varepsilon$. For a $^3\Pi_1$ term, $\Sigma = 0$, so spin does not affect the splitting and we recover a formula of the form (88.2), with K replaced by J:

$$\Delta E_1 = \text{const}\ J(J+1). \tag{88.5}$$

For a $^3\Pi_2$ term still higher orders are required, and one may take $\Delta E_2 = 0$.

Of the two levels produced by Λ-doubling, one is always positive and the other negative; this was already discussed in § 6. An analysis of the molecular wave functions determines the pattern in which positive and negative levels alternate. We give only the result of that analysis here.[37] If for some J the positive level lies below the negative one, their order is reversed in the doublet with $J+1$: the positive level lies above the negative one, and so on. The ordering alternates with successive values of the total angular momentum (for case a terms; in case b the same statement holds for successive values of K).

Problem

Determine the Λ-splitting of a $^1\Delta$ term.

Solution. Here the effect first appears in fourth-order perturbation theory. Its dependence on K is determined by products of four matrix elements (88.1), for transitions changing Λ as $2 \to 1$, $1 \to 0$, $0 \to -1$, $-1 \to -2$. This gives

$$\Delta E = \text{const}\,(K-1)K(K+1)(K+2),$$

where const $\sim B^4/\varepsilon^3$.

§ 89. Interaction of atoms at large separations

Consider two atoms separated by a distance large compared with their dimensions, and determine their interaction energy. In other words, we seek the form of the electronic terms at large internuclear distances.

We apply perturbation theory, regarding the two isolated atoms as the unperturbed system and the potential energy of their electrical interaction as the perturbation operator. As is known (see Vol. II, §§ 41, 42), the electrical interaction of two systems of charges separated by a large distance r can be expanded in powers of $1/r$. Successive terms describe the interaction of the total charges, dipole moments, quadrupole moments, and so forth of the two systems. For neutral atoms the total charges vanish. The expansion then begins with the dipole–dipole interaction ($\sim 1/r^3$), followed by dipole–quadrupole terms ($\sim 1/r^4$), quadrupole–quadrupole (and dipole–octupole) terms ($\sim 1/r^5$), and so on.

Suppose first that both atoms are in S states. It is readily seen that their interaction has no effect in first-order perturbation theory. Indeed, to first order the interaction

[37] See the article by Wigner and Witmer cited on p. 376.

energy is the diagonal matrix element of the perturbation operator evaluated with the unperturbed wave functions of the system (which are themselves products of the wave functions of the two atoms).[38] But in S states the diagonal matrix elements, that is, the mean dipole, quadrupole, and higher moments of the atoms, vanish directly as a consequence of the spherical symmetry of the atomic charge distributions. Hence every term in the expansion of the perturbation operator in powers of $1/r$ gives zero in first-order perturbation theory.[39]

In second order it is enough to retain the dipole interaction in the perturbation operator, since it decreases most slowly as r increases, namely

$$V = \frac{\mathbf{d}_1\mathbf{d}_2 - 3(\mathbf{d}_1\mathbf{n})(\mathbf{d}_2\mathbf{n})}{r^3} \tag{89.1}$$

($\mathbf{n}$ is the unit vector directed from atom 1 to atom 2). Since the nondiagonal matrix elements of the dipole moment are in general nonzero, second-order perturbation theory gives a nonzero result which, being quadratic in V, is proportional to $1/r^6$. The second-order correction to the lowest eigenvalue is always negative (see § 38). Thus for atoms in their normal states the interaction energy has the form

$$U(r) = -\frac{\text{const}}{r^6}, \tag{89.2}$$

where const is positive[40] (F. London, 1928).

Consequently two atoms in normal S states, at a large separation, attract with a force $(-dU/dr)$ inversely proportional to the seventh power of the distance. Attractive forces between atoms at large distances are usually called *van der Waals forces*. These forces also produce wells in the potential-energy curves of electronic terms for atoms that do not form a stable molecule. Such wells are very shallow, however (their depths are only tenths or even hundredths of an electron volt), and occur at distances several times larger than the interatomic distances in stable molecules.

If only one atom is in an S state, the same result (89.2) is obtained for their interaction energy, since the vanishing of the dipole and higher moments of just one atom is enough to make the first-order result zero. The constant in the numerator of (89.2) then depends not only on the states of both atoms but also on their relative orientation, that is, on the magnitude Ω of the projection of angular momentum on the line joining them.

If both atoms have nonzero orbital and total angular momenta, the situation changes. The mean dipole moment is zero in every atomic state (see § 75). The mean quadrupole moments (in states with $L \neq 0$, $J \neq 0, 1/2$), however, are nonzero.

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[38] Exchange effects that decrease exponentially with distance (compare Problem 1 of § 62 and the problem in § 81) are discarded here.

[39] This does not, of course, mean that the mean interaction energy of the atoms is exactly zero. It decreases exponentially with distance, faster than every finite power of $1/r$, which is why every term in the expansion vanishes. The multipole expansion of the interaction operator assumes that the charges of the two atoms are separated by the large distance r. Yet the quantum-mechanical electron-density distribution has finite, though exponentially small, values even at large distances.

[40] For example, the values of this constant (in atomic units) for pairs of atoms are: hydrogen, 6.5; helium, 1.5; argon, 68; krypton, 130.

The quadrupole–quadrupole term in the perturbation operator therefore contributes already in first order, and the interaction energy decreases as the fifth rather than the sixth power of the distance:

$$U(r) = \frac{\text{const}}{r^5}. \tag{89.3}$$

The constant may have either sign, so either attraction or repulsion may occur. As before, it depends not only on the atomic states but also on the state of the combined system formed by the two atoms.

A special case is the interaction between two identical atoms in different states. The unperturbed system (the two isolated atoms) then has an additional degeneracy associated with interchanging the states between the atoms. Accordingly, the first-order correction is determined by a secular equation containing both diagonal and nondiagonal matrix elements of the perturbation. If the two atomic states have opposite parities and angular momenta L differing by ± 1 or 0, but not both zero (and likewise for J), the nondiagonal dipole-moment matrix elements for transitions between them are in general nonzero. A first-order effect then arises already from the dipole term in the perturbation operator. The atomic interaction energy is consequently proportional to $1/r^3$:

$$U(r) = \frac{\text{const}}{r^3}; \tag{89.4}$$

the constant may have either sign.

Usually, however, the interaction averaged over all possible orientations of the atomic angular momenta is of interest (as, for example, for atoms interacting in a gas). This averaging makes the mean values of all multipole moments vanish. With them vanish all first-order effects linear in these moments in the atomic interaction. Hence in every case the averaged long-range forces between atoms obey (89.2).[41]

We conclude with the related question of the interaction between a neutral atom and an ion.

In first-order perturbation theory this interaction is the mean value of the operator (76.8), the energy of a quadrupole in the Coulomb field of the ion. Since the ion's potential $\varphi \sim 1/r$, the atom–ion interaction energy is proportional to $1/r^3$. This effect exists only if the atom has a nonzero mean quadrupole moment. Even then, it disappears on averaging over all directions of the atomic angular momentum $\mathbf{J}$.

The next term in powers of $1/r$, which is always nonzero, is the second-order interaction from the dipole operator (76.1). Since the ion's field strength is $\sim 1/r^2$, this interaction energy is proportional to $1/r^4$. In terms of the polarizability α of an atom in an S state, it is

$$U = -\alpha e^2/2r^4. \tag{89.5}$$

If the atom is in its normal state, this energy (like every correction to the ground-state energy) is negative, so an attractive force acts between atom and ion.[42]

[41] This law, obtained from nonrelativistic theory, applies only while retardation of the electromagnetic interactions is unimportant. For this purpose the separation r must be small compared with c/ω_{0n}, where ω_{0n} are the transition frequencies between the ground and excited atomic states. For the interaction including retardation, see Vol. IV, § 85.

[42] The same attraction occurs at large distances between an atom and an electron. It is the reason atoms can

Problem

For two identical atoms in S states, derive a formula expressing the van der Waals forces in terms of matrix elements of their dipole moments.

Solution. The result follows from the general perturbation-theory formula (38.10), applied to the operator (89.1). Because atoms in S states are isotropic, it is clear in advance that, on summing over all intermediate states, the squared matrix elements of the three components of each vector $\mathbf{d}_1$ and $\mathbf{d}_2$ make equal contributions, whereas terms containing products of different components vanish.

The result is

$$U(r) = -\frac{6}{r^6}\sum_{n,n'}\frac{|\langle n|d_z|0\rangle|^2|\langle n'|d_z|0\rangle|^2}{E_n + E_{n'} - E_0},$$

where E_0, E_n are the unperturbed energies of the ground and excited atomic states. Since, by assumption, $L = 0$ in the ground state, the matrix elements $(d_z)_{0n}$ are nonzero only for transitions to P states ($L = 1$). With formulas (29.7), we rewrite $U(r)$ finally as

$$U(r) = -\frac{2}{3r^6}\sum_{n,n'}\frac{\langle n1\|d\|00\rangle^2\langle n'1\|d\|00\rangle^2}{E_{n1} + E_{n'1} - 2E_{00}},$$

where in the indices nL of the energy levels and reduced matrix elements, the second symbol gives L, while the first denotes the set of all remaining quantum numbers specifying the energy level.

§ 90. Predissociation

The theory of diatomic molecules developed in this chapter rests chiefly on the assumption that the molecular wave function separates into a product of an electronic wave function (depending parametrically on the internuclear distance) and a wave function for the nuclear motion. This assumption amounts to neglecting certain small terms in the exact molecular Hamiltonian that describe the coupling of nuclear motion to the electrons.

When these terms are included as perturbations, transitions arise between different electronic states. Of particular physical importance are transitions for which at least one state belongs to the continuous spectrum.

Figure 30 shows the potential-energy curves of two electronic terms (more precisely, the effective potential energy U_J in specified rotational states of the molecule). The energy E' is that of a particular vibrational level of a stable molecule in electronic state 2. In state 1 this energy lies in the continuous spectrum. Thus a transition from state 2 to state 1 causes spontaneous breakup of the molecule; this phenomenon is called *predissociation*.[43] As a result of predissociation, the discrete-spectrum state represented by curve 2 actually has a finite lifetime. This means that the discrete

energy level is broadened: it acquires a certain width (see the end of § 44).

form negative ions by attaching an electron (with binding energies from fractions to several electron volts). Not every atom has this property, however. In a field decreasing at large distances as $1/r^4$ (or $1/r^3$), the number of levels corresponding to bound electron states is finite in every case and may, in particular cases, be zero.

[43] Curve 1 need not have a minimum at all if it corresponds to purely repulsive forces between the atoms.

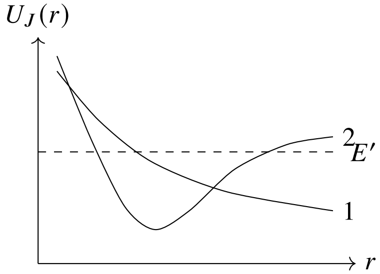


Fig. 30

If instead the total energy E lies above the dissociation limit in both states, a transition from one state to the other corresponds to a so-called *collision of the second kind*. A transition $1 \to 2$, for example, means that two colliding atoms pass into excited states and separate with reduced kinetic energy (as $r \to \infty$, curve 1 lies below curve 2; the difference $U_2(\infty) - U_1(\infty)$ is the atomic excitation energy).

Because of the large nuclear mass, the nuclear motion is semiclassical. The problem of finding the probability of these transitions therefore belongs to the class discussed in § 52. The general considerations given there show that the transition probability is governed by the point at which the transition could occur classically.[44] Since the total energy of the two-atom system (the molecule) is conserved in the transition, its "classical possibility" requires equality of the effective potential energies: $U_{J1}(r) = U_{J2}(r)$. The total molecular angular momentum is also conserved, so the centrifugal energies are the same in the two states. The condition therefore reduces to equality of the potential energies,

$$U_1(r) = U_2(r), \tag{90.1}$$

and contains no angular momentum at all.

If (90.1) has no real root in the classically accessible region (where $E > U_{J1}, U_{J2}$), then by § 52 the transition probability is exponentially small.[45] Transitions have appreciable probability only if the potential-energy curves cross within the classically accessible region (as in Fig. 30). In that event the exponent in (52.1) vanishes (so that formula is, of course, inapplicable), and the transition probability is determined by a nonexponential expression, derived below. Condition (90.1) may

also be interpreted directly as follows. Equal potential (and total) energies imply equal momenta. Thus (90.1) may be written

$$r_1 = r_2, \qquad p_1 = p_2, \tag{90.2}$$

where p is the momentum of the relative radial motion of the nuclei, and the indices 1 and 2 refer to the two electronic states. Hence at the instant of transition the internuclear

[44] Or the point $r = 0$, at which the potential energy becomes infinite.

[45] A peculiar situation occurs for a transition involving a molecular term that may be formed from two different pairs of atomic states (see the end of § 85), so that toward increasing distances the potential-energy curve seems to divide into two branches. The transition probability is then appreciably enhanced; for an example see A. I. Voronin and E. E. Nikitin, *Optics and Spectroscopy* **25**, 803 (1968).

distance and nuclear momentum may be said to remain unchanged (the so-called *Franck–Condon principle*). Physically, this is because electronic velocities are large compared with nuclear velocities, so "during the electronic transition" the nuclei have no time to change their positions or velocities appreciably.

The selection rules for these transitions are easily established. First, there are two evident exact rules: the total angular momentum J and the sign (positive or negative; see § 86) of the term must not change. This follows directly because conservation of total angular momentum and invariance of the wave-function character under inversion of the coordinate system are exact laws for every closed system of particles.

There is also, to high accuracy, a rule forbidding transitions between states of different parity in a molecule of identical atoms. Indeed, the parity of the state is uniquely fixed by the nuclear spin and the sign of the term. The sign is conserved exactly, while the nuclear spin is conserved to high accuracy because of its weak interaction with the electrons.

The requirement that the potential-energy curves have a crossing point means that the terms must have different symmetries (see § 79). Consider transitions that arise already in first-order perturbation theory. We first note that the Hamiltonian terms producing these transitions are precisely those responsible for the Λ-doubling of the levels. They include, in the first place, terms describing the spin–orbit interaction. These are products of two axial vectors, one spin-like and the other coordinate-like; it should be stressed, however, that these vectors are by no means simply $\widehat{\mathbf{S}}$ and $\widehat{\mathbf{L}}$. They therefore have nonzero matrix elements for transitions in which S and Λ change by $0, \pm 1$.

The case $\Delta S = \Delta\Lambda = 0$ with $\Lambda \neq 0$ must be excluded, since the symmetry of the term would then not change at all in the transition. A transition between two Σ terms is possible if one is a Σ^+ term and the other a Σ^- term (an axial vector has matrix elements only for transitions between Σ^+ and Σ^-; see § 87).

The Hamiltonian term describing the coupling between molecular rotation and orbital angular momentum is proportional to $\widehat{\mathbf{J}}\widehat{\mathbf{L}}$. Its matrix elements are nonzero for transitions with $\Delta\Lambda = \pm 1$ and no change in spin (matrix elements with $\Delta\Lambda = 0$ are possessed only by the ζ component of the vector, namely L_ζ, but L_ζ is diagonal in the electronic states).

Besides these terms there is a perturbation due to the nuclear kinetic-energy operator acting not only on the nuclear wave function but also on the electronic function, which depends parametrically on r. The corresponding Hamiltonian terms have the same symmetry as the unperturbed Hamiltonian. They can therefore produce transitions only between electronic terms of the same symmetry, whose probability is negligible because such terms do not cross.

We now calculate the transition probability explicitly. For definiteness, consider a collision of the second kind. By the general formula (43.1), the required probability is

$$w = \frac{4\pi^2}{\hbar^2}\left|\int \chi_{\mathrm{nuc},2} V(r) \chi_{\mathrm{nuc},1}\, dr\right|^2, \tag{90.3}$$

where $\chi_{\mathrm{nuc}} = r\psi_{\mathrm{nuc}}$, and $V(r)$ is the perturbing energy. The final wave function $\chi_{\mathrm{nuc},2}$ must be normalized to an energy delta function. With this normalization, the semiclassical function is

$$\chi_{\mathrm{nuc},2} = \sqrt{\frac{2}{\pi\hbar v_2}}\cos\left(\frac{1}{\hbar}\int_{a_2}^{r} p_2\, dr - \frac{\pi}{4}\right). \tag{90.4}$$

The initial-state wave function is written

$$\chi_{\mathrm{nuc},1} = \frac{2}{\sqrt{v_1}} \cos\left(\frac{1}{\hbar}\int_{a_1}^{r} p_1\, dr - \frac{\pi}{4}\right). \tag{90.5}$$

It is normalized so that the current density in each of the two traveling waves is unity. Substitution of these functions in (90.3) gives the dimensionless transition probability w, which may be regarded as the probability for two passages of the nuclei through $r = r_0$.

The matrix element contains a product of cosines, which we expand into cosines of the sum and difference of their arguments. Near $r = r_0$ only the latter cosine is relevant, so that

$$w = \frac{4}{\hbar^2}\left|\int \frac{V(r)}{\sqrt{v_1 v_2}} \cos\left(\frac{1}{\hbar}\int_{a_1}^{r} p_1 dr - \frac{1}{\hbar}\int_{a_2}^{r} p_2 dr\right) dr\right|^2.$$

The integral converges rapidly away from the crossing point. Expanding the argument in powers of $\xi = r - r_0$ and recalling that $p_1 = p_2$ at the crossing, we find

$$\int^{r} p_1 dr - \int^{r} p_2 dr \approx S_0 + \frac{1}{2}\left(\frac{dp_1}{dr} - \frac{dp_2}{dr}\right)_0 \xi^2.$$

Differentiating $p_1^2/2\mu + U_1 = p_2^2/2\mu + U_2$ gives

$$v_1\frac{dp_1}{dr} - v_2\frac{dp_2}{dr} = F_1 - F_2,$$

and hence

$$\int^{r} p_1 dr - \int^{r} p_2 dr \approx S_0 + \frac{F_1 - F_2}{2v}\xi^2.$$

Using the familiar integral, we finally obtain

$$w = \frac{8\pi V^2}{\hbar v|F_2 - F_1|}\cos^2\left(\frac{S_0}{\hbar} + \frac{\pi}{4}\right). \tag{90.6}$$

The quantity $S_0/\hbar$ is large and changes rapidly with the energy. On averaging, the squared cosine is replaced by its mean value, and

$$w = \frac{4\pi V^2}{\hbar v|F_2 - F_1|}. \tag{90.7}$$

(L. D. Landau, 1932). All quantities on the right are evaluated at the crossing point.

For predissociation, the decay probability per unit time is obtained by multiplying w by $\omega/2\pi$:

$$\frac{1}{\tau} = \frac{2\omega V^2}{\hbar v|F_2 - F_1|}. \tag{90.8}$$

In speaking of crossing terms, we meant eigenvalues of the unperturbed Hamiltonian $\widehat{H}_0$, without the terms $\widehat{V}$ that produce transitions. When these terms are included, the crossing disappears and the curves separate as in Fig. 31.

Let $U_{J1}(r)$ and $U_{J2}(r)$ be two eigenvalues of $\widehat{H}_0$. The method of § 79 gives

$$U_{b,a}(r) = \frac{1}{2}(U_{J1} + U_{J2}) \pm \frac{1}{2}\sqrt{(U_{J1} - U_{J2})^2 + 4V^2}. \tag{90.9}$$

The level separation is

$$\Delta U = \sqrt{(U_{J1} - U_{J2})^2 + 4V^2}, \tag{90.10}$$

and its minimum at $r = r_0$ is

$$(\Delta U)_{\min} = 2|V(r_0)|. \tag{90.11}$$

Near this point,

$$\Delta U = \sqrt{(F_2 - F_1)^2(r - r_0)^2 + 4V^2}. \tag{90.12}$$

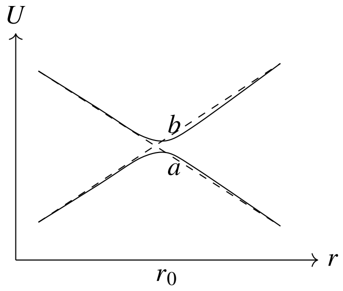


Fig. 31

These formulas require $(\Delta U)_{\min}$ to be small compared with the distance to the other terms. If condition (90.19) is not satisfied, ordinary perturbation theory for the transition probability is inapplicable and a more general treatment is required.

In the neighborhood of the crossing, treating the nuclear motion semiclassically, we replace the velocity operator by the constant v and put $\xi = r - r_0 = vt$. The problem reduces to the equation

$$i\hbar\frac{\partial \Psi}{\partial t} = [\widehat{H}_0(t) + \widehat{V}(t)]\Psi. \tag{90.13}$$

Let ψ_a, ψ_b be the electronic states of curves a, b. We seek the solution as

$$\Psi = a(t)\psi_a + b(t)\psi_b. \tag{90.14}$$

With the boundary condition $a = 1, b = 0$ as $t \to -\infty$, the quantity $|b(\infty)|^2$ is the probability of a transition from curve a to curve b in one passage through the point. For two passages the transition proceeds by the paths $a \to b \to b$ or $a \to a \to b$, so

$$w = 2|b(\infty)|^2[1 - |b(\infty)|^2]. \tag{90.15}$$

The value of $b(\infty)$ can be found by the method of § 53.[46]

The curves $U_a(t)$ and $U_b(t)$ cross at the imaginary points

$$t_0^{(\pm)} = \pm i\frac{2V}{|F_2 - F_1|v} = \pm i\tau_0. \tag{90.16}$$

[46]In § 53 the process was assumed adiabatic throughout. Here adiabaticity may fail in the immediate vicinity of r_0, but only adiabaticity at large $|t|$ and the possibility of restricting the treatment to two levels are essential.

Passing into the complex t plane and going around the point $t_0^{(+)}$, with

$$\Delta U = \sqrt{(F_2 - F_1)^2 v^2 t^2 + 4V^2}, \tag{90.17}$$

we finally obtain

$$w = 2\exp\left(-\frac{2\pi V^2}{\hbar v|F_2 - F_1|}\right)\left[1 - \exp\left(-\frac{2\pi V^2}{\hbar v|F_2 - F_1|}\right)\right]. \tag{90.18}$$

(C. Zener, 1932). The probability is small in both limiting cases. For $V^2 \gg \hbar v|F_2 - F_1|$ it is

exponentially small (the adiabatic case), while for

$$V^2 \ll \hbar v|F_2 - F_1| \tag{90.19}$$

formula (90.18) reduces to (90.7). Equation (90.17) shows that $\tau \sim |V|/(|F_2 - F_1|v)$ is the "time taken by the nuclei to pass" the crossing point.

We conclude with a phenomenon related to predissociation, the so-called perturbations in the spectrum of diatomic molecules. If two discrete molecular levels E_1, E_2 are close, the possibility of a transition shifts them. By (79.4),

$$E = \frac{E_1 + E_2}{2} \pm \sqrt{\left(\frac{E_1 - E_2}{2}\right)^2 + |V_{12}^{\text{nuc}}|^2}. \tag{90.20}$$

The two levels move apart, and their separation increases as $|E_1 - E_2|$ decreases. The matrix element V_{12}^{nuc} is calculated as above, except that the discrete-spectrum nuclear functions are normalized to unity. Comparison gives

$$|V_{12}^{\text{nuc}}|^2 = w\frac{\hbar\omega_1}{2\pi}\frac{\hbar\omega_2}{2\pi}. \tag{90.21}$$

Problem

Determine the total cross section for collisions of the second kind as a function of the kinetic energy E of the colliding atoms, for transitions due to the spin–orbit interaction (L. D. Landau, 1932).

Solution. Since the nuclear motion is semiclassical, we may introduce the impact parameter ρ and write $d\sigma = 2\pi\rho\, d\rho\, w(\rho)$. For the spin–orbit interaction the matrix element $V(r)$ is independent of angular momentum. The velocity at the crossing point is

$$v = \sqrt{\frac{2}{\mu}\left(E - U - \frac{\rho^2 E}{r_0^2}\right)}.$$

Integration over ρ gives

$$\sigma = \frac{4\sqrt{2\mu}\pi^2 V^2 r_0^2}{\hbar|F_2 - F_1|}\frac{\sqrt{E - U}}{E}.$$

Problem

The same for transitions due to the coupling of molecular rotation to orbital angular momentum (L. D. Landau, 1932).

Solution. The matrix element has the form $V(r) = MD/\mu r^2$, where $D(r)$ is a matrix element of the electronic orbital angular momentum. In the same way we obtain

$$\sigma = \frac{16\sqrt{2}\pi^2 D^2 (E-U)^{3/2}}{3\hbar\mu r_0^2 |F_2 - F_1| E}.$$

Problem

Determine the transition probability for energies E close to the value U of the potential energy at the crossing point.

Solution. For small $E-U$, formula (90.7) is inapplicable. Near the crossing we replace the curves by the straight lines $U_1 = U - F_1\xi$, $U_2 = U - F_2\xi$, $\xi = r - r_0$. The wave functions coincide with those for one-dimensional motion in a uniform field. Calculation in the momentum representation gives

$$w = \frac{4\pi V^2 (2\mu)^{2/3}}{\hbar^{4/3}(F_1F_2)^{1/2}(F_2-F_1)^{2/3}}\Phi^2\left[-\frac{E-U}{\hbar^{2/3}}\left(\frac{2\mu(F_2-F_1)^2}{F_1^2F_2^2}\right)^{1/3}\right], \tag{90.22}$$

where Φ is the Airy function (see § b of the Mathematical Appendices). For large $E-U$, this formula reduces to (90.7).

Problem

Determine the charge-transfer probability in a distant slow ($v \ll 1$) collision between a hydrogen atom and a proton (O. V. Firsov, 1951).[47]

Solution. Regard the system $H + H^+$ as a hydrogen molecular ion (see the problem in § 81). Charge transfer consists in the electron passing from the state ψ_1, localized at the first nucleus, to the state ψ_2 near the second. The stationary states are

$$\psi_{g,u} = \frac{1}{\sqrt{2}}(\psi_1 \pm \psi_2).$$

Their energies are $U_{g,u}(R)$. For a prescribed slow classical motion of the nuclei, the time dependence is specified by the factors $\exp\left[-i\int U_{g,u}(t)dt\right]$. As $t \to \infty$, the charge-transfer probability is

$$w = \sin^2\eta, \qquad \eta = \frac{1}{2}\int_{-\infty}^{\infty}(U_u - U_g)dt.$$

[47] Atomic units are used in this problem.

For large impact parameters ρ, put $R = \sqrt{\rho^2 + v^2t^2}$ and use formula (4) from the problem in § 81:

$$\eta = \frac{4}{v}\int_{\rho}^{\infty} \frac{R^2 e^{-R-1}}{\sqrt{R^2-\rho^2}}\,dR.$$

For $\rho \gg 1$, putting $R = \rho(1+x)$, we obtain

$$\eta = \frac{2\sqrt{2\pi}}{ev}\rho^{3/2}e^{-\rho}.$$

CHAPTER XII

Theory of Symmetry

§ 91. Symmetry transformations

As in the diatomic case, the classification of the terms of a polyatomic molecule is closely connected with its symmetry. We therefore begin by studying the kinds of symmetry a molecule may possess.

The symmetry of a body is specified by the collection of motions that bring the body into coincidence with itself; these motions are called *symmetry transformations*. Every possible symmetry transformation can be represented as a combination of one or more of three basic types. These three essentially distinct types are: a *rotation* of the body through a definite angle about an axis, a *reflection* in a plane, and a *translation* of the body through some distance. Of these, the last type can clearly occur only for an unbounded medium (a crystal lattice). A body of finite dimensions (in particular, a molecule) can be symmetric only under rotations and reflections.

If a body coincides with itself after a rotation through $2\pi/n$ about an axis, that axis is called an *n-fold symmetry axis*. The number n may have any integral value $n = 2, 3, \ldots$; $n = 1$ corresponds to a rotation through 2π, or equivalently through zero, and hence to the identity transformation. We denote the operation of rotation through $2\pi/n$ about the given axis by C_n. Repeating this operation two, three, ... times gives rotations through $2(2\pi/n), 3(2\pi/n), \ldots$, which likewise bring the body into coincidence with itself; these rotations may be denoted by $C_n^2, C_n^3, \ldots$ Clearly, if n is divisible by p, then

$$C_n^p = C_{n/p}. \tag{91.1}$$

In particular, after performing the rotation n times we return to the initial position, i.e. carry out the *identity transformation*; the latter is conventionally denoted by E, so that

$$C_n^n = E. \tag{91.2}$$

If a body coincides with itself after reflection in a plane, that plane is called a *plane of symmetry*. We denote the operation of reflection in a plane by σ. Evidently, two successive reflections in the same plane give the identity transformation,

$$\sigma^2 = E. \tag{91.3}$$

The combined application of a rotation and a reflection leads to the so-called *rotation–reflection axes*. A body has an n-fold rotation–reflection axis if it coincides with itself after rotation through $2\pi/n$ about this axis followed by reflection in a plane perpendicular to the axis (Fig. 32). It is readily seen that this is a new kind of symmetry only when n is even. Indeed, if n is odd, repeating the rotation–reflection transformation n times is equivalent simply to reflection in the plane perpendicular to the axis (the total angle of rotation is 2π, while an odd number of reflections in the same plane is a single reflection). Repeating the transformation a further n times shows that the rotation–reflection axis reduces to the simultaneous presence of an independent n-fold symmetry axis and a plane of symmetry perpendicular to it. If n is even, however, n repetitions of the rotation–reflection transformation return the body to its initial position.

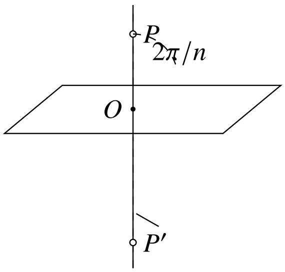


Figure 32

We denote a rotation–reflection transformation by S_n. Denoting reflection in the plane perpendicular to the given axis by σ_h, we may write, by definition,

$$S_n = C_n\sigma_h = \sigma_h C_n \tag{91.4}$$

(the order in which C_n and σ_h are performed clearly does not affect the result).

An important special case is a twofold rotation–reflection axis. A rotation through π followed by reflection in a plane perpendicular to the rotation axis is readily seen to constitute the transformation of *inversion*: a point P of the body is carried into a point P' on the continuation of the line joining P to the point O where the axis meets the plane, with $OP = OP'$.

A body symmetric under this transformation is said to possess a center of symmetry (we denote inversion by I):

$$I \equiv S_2 = C_2\sigma_h. \tag{91.5}$$

It is also evident that $I\sigma_h = C_2$ and $IC_2 = \sigma_h$; in other words, a twofold axis, a plane of symmetry perpendicular to it, and a center of symmetry at their intersection are mutually dependent: the presence of any two automatically entails the third.

We now state several purely geometrical properties of rotations and reflections that are useful in studying the symmetry of bodies.

The product of two rotations about axes intersecting at a point is a rotation about a third axis through the same point. The product of two reflections in intersecting planes is equivalent to a rotation; its axis clearly coincides with the line in which the planes intersect, and a simple geometrical construction shows that its angle is twice the angle between them. Denote rotation through an angle φ about the axis by $C(\varphi)$ and reflections in two planes containing the axis by σ_v and σ_v'[1]. The statement can then be written

$$\sigma_v\sigma_v' = C(2\varphi), \tag{91.6}$$

[1]The subscript v conventionally distinguishes reflection in a plane containing the given axis (a "vertical" plane), while h denotes reflection in a plane perpendicular to the axis (a "horizontal" plane).

where φ is the angle between the two planes. The order of the two reflections is important: $\sigma_v\sigma_v'$ gives a rotation directed from the plane σ_v' toward σ_v, whereas interchanging the factors gives rotation in the opposite direction. Multiplying (91.6) from the left by σ_v, we obtain

$$\sigma_v' = \sigma_v C(2\varphi); \tag{91.7}$$

thus, the product of a rotation and reflection in a plane containing its axis is equivalent to reflection in another plane meeting the first at an angle equal to half the angle of rotation. It follows in particular that a twofold symmetry axis and two mutually perpendicular planes of symmetry containing it are mutually dependent: the presence of two requires the third.

We shall show that the product of rotations through π about two axes meeting at an angle φ (Oa and Ob in Fig. 33) is a rotation through 2φ about an axis perpendicular to both (PP' in Fig. 33). It is clear in advance that the resulting transformation is also a rotation. After the first rotation (about Oa), the point P goes to P', and after the second (about Ob) it returns to its initial position. The line PP' therefore remains fixed and is consequently the rotation axis. To determine the angle of rotation, it suffices to note that the axis Oa remains fixed during the first rotation and is carried by the second into the position Oa', which makes an angle 2φ with Oa. The same argument shows that reversing the order of the two transformations gives a rotation in the opposite direction.

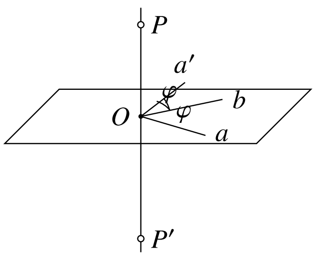


Figure 33

Although the result of two successive transformations generally depends on their order, in a number of cases the order is immaterial: the transformations commute. This occurs for the following transformations:

1) two rotations about the same axis;

2) two reflections in mutually perpendicular planes (they are equivalent to a rotation through π about the line in which the planes intersect);

3) two rotations through π about mutually perpendicular axes (they are equivalent to a rotation through the same angle about the third perpendicular axis);

4) a rotation and reflection in a plane perpendicular to the rotation axis;

5) any rotation (or reflection) and inversion in a point lying on the rotation axis (or in the reflection plane); this follows from 1) and 4).

§ 92. Transformation groups

The collection of all symmetry transformations of a given body is called its group of symmetry transformations, or simply its *symmetry group*. Above we treated these transformations as geometrical motions of the body. In quantum-mechanical applications, however, it is more convenient to regard symmetry transformations as coordinate transformations that leave the Hamiltonian of the system invariant. Clearly, if a system coincides with itself after a rotation or reflection, the corresponding coordinate transformation leaves its Schrödinger equation unchanged. We shall therefore speak of the group of transformations under which the given Schrödinger equation is invariant[2].

Symmetry groups can conveniently be studied with the general mathematical apparatus known as *group theory*, whose foundations are set out below. We begin with groups containing a finite number of distinct transformations (the so-called *finite groups*). Each transformation belonging to a group is called an *element of the group*.

Symmetry groups have the following evident properties. Every group contains the identity transformation E (called the *identity element of the group*). Elements of a group can be *multiplied*; the product of two (or more) transformations means the result of applying them successively. The product of any two elements of a group is clearly an element of that same group. Multiplication of elements obeys the associative law $(AB)C = A(BC)$, where A, B, C are elements of the group. The commutative law, however, does not hold in general; generally $AB \neq BA$. For every group element A, the same group contains an *inverse element* A^{-1} (the inverse transformation) such that $AA^{-1} = E$. In some

cases an element may coincide with its inverse; in particular, $E^{-1} = E$. Mutually inverse elements A and A^{-1} evidently commute.

The element inverse to a product AB is

$$(AB)^{-1} = B^{-1}A^{-1},$$

and similarly for a product of more elements; this is readily verified by multiplication and use of associativity.

If all elements of a group commute, the group is called *Abelian*. A special class of Abelian groups consists of the so-called *cyclic groups*. A cyclic group is one whose elements can all be obtained as successive powers of a single element, i.e. a group consisting of

$$A, A^2, A^3, \ldots, A^n = E,$$

[2]This viewpoint makes it possible to include not only the groups of rotations and reflections considered here, but also other types of transformations that leave the Schrödinger equation unchanged. Among these are permutations of the coordinates of identical particles belonging to the system (a molecule or atom). The collection of all possible permutations of identical particles in a given system is called its *permutation group* (we encountered these already in § 63). The general properties of groups developed below apply to permutation groups as well; we shall not pursue a more detailed study of this kind of group.

The notation used in this chapter calls for the following remark. Symmetry transformations are essentially operators of the same kind as those used throughout this book and ought therefore to be marked with hats. We omit them in deference to standard notation and because no ambiguity can result in the present chapter. For the same reason we use the conventional symbol E, rather than 1 as in the remaining chapters, for the identity transformation. Finally, in this chapter the inversion operator is denoted by I rather than by the symbol P used in § 30 and customary in modern quantum-mechanical literature.

where n is an integer.

Let $\boldsymbol{G}$ be a group[3]. If a collection of its elements $\boldsymbol{H}$ can be selected which itself forms a group, then $\boldsymbol{H}$ is called a *subgroup* of $\boldsymbol{G}$. The same group element may belong to several of its subgroups.

Starting with any element A and taking successive powers, we eventually obtain the identity element (since the group has only finitely many elements). If n is the smallest number for which $A^n = E$, then n is called the *order of the element* A, and the collection $A, A^2, \dots, A^n = E$ is called the *period* of A. The period is denoted by $\{A\}$; it forms a group in its own right, and hence is a cyclic subgroup of the original group.

To check whether a given collection of group elements is a subgroup, it is enough to verify that the product of any two of its elements is again contained in the collection. Indeed, all powers of every element A will then be present, including A^{n-1} (n being the order of the element), which serves as its inverse since $A^{n-1}A = A^n = E$; the identity element is plainly present as well.

The total number of elements in a group is called its *order*. It is easy to see that the order of a subgroup divides the order of the whole

group. Consider a subgroup $\boldsymbol{H}$ of $\boldsymbol{G}$, and let G_1 be an element of $\boldsymbol{G}$ not belonging to $\boldsymbol{H}$. Multiplying all elements of $\boldsymbol{H}$ by G_1 (on the right, for example), we obtain a set (or, as it is called, a complex) of elements denoted by $\boldsymbol{H}G_1$. All elements of this complex clearly belong to $\boldsymbol{G}$, but none belongs to $\boldsymbol{H}$. For if $H_a G_1 = H_b$ for some two H_a, H_b in $\boldsymbol{H}$, then $G_1 = H_a^{-1} H_b$ would follow, so that G_1 too would belong to $\boldsymbol{H}$, contrary to the assumption. Similarly, if G_2 belongs neither to $\boldsymbol{H}$ nor to $\boldsymbol{H}G_1$, all elements of $\boldsymbol{H}G_2$ belong to neither $\boldsymbol{H}$ nor $\boldsymbol{H}G_1$. Continuing in this way, we eventually exhaust all elements of the finite group $\boldsymbol{G}$. Thus they are partitioned into sets (called *cosets* of $\boldsymbol{H}$ in $\boldsymbol{G}$)

$$\boldsymbol{H},\ \boldsymbol{H}G_1,\ \boldsymbol{H}G_2, \dots, \boldsymbol{H}G_m,$$

each containing h elements, where h is the order of $\boldsymbol{H}$. It follows that the order of $\boldsymbol{G}$ is $g = hm$, proving the assertion. The integer $m = g/h$ is called the *index* of $\boldsymbol{H}$ in $\boldsymbol{G}$.

If the order of a group is prime, the result just proved immediately shows that the group has no subgroups apart from E and itself. The converse is also true: every group without subgroups must have prime order and must moreover be cyclic (otherwise it would contain elements whose periods formed subgroups).

We introduce the important notion of *conjugate elements*. Two elements A and B are called conjugate if

$$A = CBC^{-1},$$

where C is also an element of the group (multiplying this equality on the right by C and on the left by C^{-1} gives the inverse relation $B = C^{-1}AC$). An essential property of conjugacy is that if A is conjugate to B, and B to C, then A is conjugate to C. Indeed, from $B = P^{-1}AP$ and $C = Q^{-1}BQ$ (where P, Q are group elements), it follows that $C = (PQ)^{-1}A(PQ)$. We may therefore speak of sets of mutually conjugate group elements. Such sets are called *classes of conjugate elements*, or simply classes of the group. Each class is completely determined by any one of its elements A: once A is given, the entire

[3] Groups will be denoted by italic bold letters.

class is obtained by forming the products GAG^{-1} as G runs through all elements of the group (each element of the class may, of course, occur more than once). The whole group can thus be partitioned into classes; every group element plainly belongs to only one class. The identity element forms a class by itself, since $GEG^{-1} = E$ for every group element. In an Abelian group the same is true of every element: because all elements commute by definition, each element is conjugate only to itself and therefore constitutes a class on its own. We emphasize that a class of a group (unless it is E) is by no means a subgroup; this is already evident because it does not contain the identity element.

All elements of a given class have the same order. Indeed, if n is the order of A (so that $A^n = E$), then for an element $B = CAC^{-1}$ conjugate to it we likewise have $(CAC^{-1})^n = CA^nC^{-1} = E$.

Let $\boldsymbol{H}$ be a subgroup of $\boldsymbol{G}$, and let G_1 be an element of $\boldsymbol{G}$ not in $\boldsymbol{H}$. It is readily verified that the collection $G_1\boldsymbol{H}G_1^{-1}$ satisfies all the requirements for a group and is therefore another subgroup of $\boldsymbol{G}$. The subgroups $\boldsymbol{H}$ and $G_1\boldsymbol{H}G_1^{-1}$ are called conjugate; every element of one is conjugate to an element of the other. Assigning different values to G_1 gives a series of conjugate subgroups, some of which may coincide. It may happen that all subgroups conjugate to $\boldsymbol{H}$ coincide with $\boldsymbol{H}$. In that event $\boldsymbol{H}$ is called a *normal divisor* (or *invariant subgroup*) of $\boldsymbol{G}$. Every subgroup of an Abelian group, for example, is evidently a normal divisor.

Consider a group $\boldsymbol{A}$ with n elements $A, A', A'', \dots$ and a group $\boldsymbol{B}$ with m elements $B, B', B'', \dots$, and suppose that all elements of $\boldsymbol{A}$ (apart from the identity E) differ from and commute with the elements of $\boldsymbol{B}$. Multiplying each element of $\boldsymbol{A}$ by each element of $\boldsymbol{B}$ gives a collection of nm elements which also forms a group. Indeed, for any two elements of this collection, $AB \cdot A'B' = AA' \cdot BB' = A''B''$, again an element of the same collection. The resulting group of order nm is denoted by $\boldsymbol{A} \times \boldsymbol{B}$ and is called the *direct product* of $\boldsymbol{A}$ and $\boldsymbol{B}$.

Finally, we introduce the notion of *isomorphism* of groups. Two groups $\boldsymbol{A}$ and $\boldsymbol{B}$ of the same order are called isomorphic if a one-to-one correspondence can be set up between their elements such that, if A corresponds to B

and A' to B', then $A'' = AA'$ corresponds to $B'' = BB'$. Considered abstractly, two such groups plainly have identical properties, although the concrete meaning of their elements differs.

§ 93. Point groups

The transformations belonging to the symmetry group of a body of finite dimensions (in particular, a molecule) must leave at least one point of the body fixed under every transformation. In other words, all symmetry axes and planes of a molecule must have at least one common point of intersection. Indeed, successive rotations about two nonintersecting axes, or reflections in nonintersecting planes, produce a translation of the body, which clearly cannot bring it into coincidence with itself. Symmetry groups having this property are called *point groups*.

Before constructing the possible kinds of point groups, we describe a simple geometrical method for distributing the group elements among classes. Let Oa be an axis, and let the group element A be rotation through a definite angle about it. Further, let G be a transformation in the same group (a rotation or reflection) which carries the axis Oa itself into the position Ob. We shall show that $B = GAG^{-1}$ is then rotation about Ob through the same angle as A rotates about Oa. Consider the action of GAG^{-1} on the axis Ob. The inverse transformation G^{-1} carries Ob into Oa, so the subsequent rotation A leaves it there; finally, G returns it to its initial position. Thus Ob remains fixed, and B is a rotation about this axis. Since A and B belong to the same class, they have the same order, and hence rotate through the same angle.

Thus two rotations through the same angle belong to one class if the group contains a transformation that brings one rotation axis into coincidence with the other. In exactly the same way, two reflections in different planes belong to one class if some group transformation carries one plane into the other. Symmetry axes or planes whose directions can be brought into coincidence are called

equivalent.

Some further remarks are needed when both rotations are about the same axis. The inverse of C_n^k $(k = 1, 2, \ldots, n-1)$ is $C_n^{-k} = C_n^{n-k}$, i.e. rotation through $(n-k)(2\pi/n)$ in the same sense, or equivalently through $2k\pi/n$ in the opposite sense. If the group contains rotation through π about a perpendicular axis (which reverses the direction of the axis under consideration), then the general rule just proved shows that C_n^k and C_n^{-k} belong to one class. Reflection σ_h in a plane perpendicular to the axis also reverses its direction, but reflection reverses the sense of rotation as well. Thus the presence of σ_h does not make C_n^k and C_n^{-k} conjugate. Reflection σ_v in a plane containing the axis does not reverse the axis, but does reverse the sense of rotation; hence $C_n^{-k} = \sigma_v C_n^k \sigma_v$, and in the presence of such a plane C_n^k and C_n^{-k} belong to one class. If rotations about an axis through the same angle in opposite senses are conjugate, we shall call the axis *two-sided*.

The classes of a point group can often be found more easily by the following rule. Let $\boldsymbol{G}$ be a group not containing inversion I, and let $\boldsymbol{C}_i$ be the two-element group consisting of I and E. Then $\boldsymbol{G} \times \boldsymbol{C}_i$ has twice as many elements as $\boldsymbol{G}$; half coincide with the elements of $\boldsymbol{G}$, while the others are obtained by multiplying them by I. Since I commutes with every point-group transformation, $\boldsymbol{G} \times \boldsymbol{C}_i$ plainly has twice as many classes as $\boldsymbol{G}$: each class A of $\boldsymbol{G}$ corresponds to two classes, A and AI. In particular, inversion I always constitutes a class by itself.

We now enumerate all possible point groups, constructing them from the simplest by adjoining new symmetry elements. Point groups will be denoted by bold Latin letters with the appropriate subscripts.

I. T h e g r o u p $\boldsymbol{C}_n$

The simplest type of symmetry contains only one n-fold axis. The group $\boldsymbol{C}_n$ consists of rotations about this axis and is evidently cyclic. Each of its n elements is a class by itself. The group $\boldsymbol{C}_1$

contains only the identity transformation E and corresponds to the absence of any symmetry.

II. T h e g r o u p $\boldsymbol{S}_{2n}$

This is the group of transformations about an even-order rotation–reflection axis $2n$. It contains $2n$ elements and is evidently cyclic. In particular, $\boldsymbol{S}_2$ contains only

E and I and is also denoted by $\boldsymbol{C}_i$. If the group order has the form $2n = 4p + 2$, its elements include inversion, since $(S_{4p+2})^{2p+1} = C_2\sigma_h = I$. Such a group can be written $\boldsymbol{S}_{4p+2} = \boldsymbol{C}_{2p+1} \times \boldsymbol{C}_i$ and is also denoted by $\boldsymbol{C}_{2p+1,i}$.

III. The group $\boldsymbol{C}_{nh}$

This group is obtained by adjoining to an n-fold axis a plane of symmetry perpendicular to it. It contains $2n$ elements: the n rotations of $\boldsymbol{C}_n$ and the n rotation–reflection transformations $C_n^k\sigma_h$ $(k = 1, 2, 3, \ldots, n)$, including $C_n^n\sigma_h = \sigma_h$. All its elements commute, so it is Abelian and has as many classes as elements. If $n = 2p$ is even, the group contains a center of symmetry, since $C_{2p}^p\sigma_h = C_2\sigma_h = I$. The simplest group $\boldsymbol{C}_{1h}$ contains only E and σ_h and is also denoted by $\boldsymbol{C}_s$.

IV. The group $\boldsymbol{C}_{nv}$

Adjoining to an n-fold axis a plane of symmetry through it automatically produces another $n - 1$ planes intersecting along the axis at angles π/n (this follows directly from (91.7)[4]). The resulting group $\boldsymbol{C}_{nv}$ therefore has $2n$ elements: n rotations about the n-fold axis and n reflections σ_v in vertical planes. Figure 34 shows, as examples, the axes and planes of $\boldsymbol{C}_{3v}$ and $\boldsymbol{C}_{4v}$.

The planes through the symmetry axis make it two-sided. The actual distribution into classes differs for even and odd n.

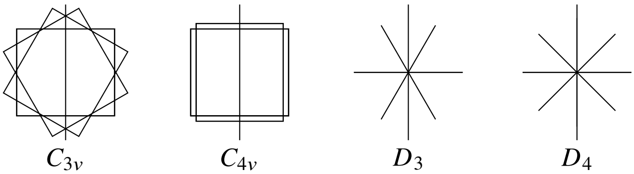


Figure 34

If $n = 2p + 1$ is odd, successive rotations C_{2p+1} carry each plane into all the other $2p$ planes, so all symmetry planes are equivalent and their reflections form one class. The $2p$ nonidentity rotations are conjugate in pairs, forming p two-element classes (C_{2p+1}^k and C_{2p+1}^{-k}, $k = 1, 2, \ldots, p$); E is one further class. There are therefore $p + 2$ classes.

If $n = 2p$ is even, successive rotations C_{2p} can bring only alternate planes into coincidence; adjacent planes cannot be so matched. Thus there are two sets of p equivalent planes and correspondingly two classes of p reflections each. Among the rotations, $C_{2p}^{2p} = E$ and $C_{2p}^p = C_2$ each form a class, while the other $2p - 2$ rotations are conjugate in pairs and give $p - 1$ further two-element classes. Hence $\boldsymbol{C}_{2p,v}$ has $p + 3$ classes.

V. The group $\boldsymbol{D}_n$

Adjoining to an n-fold axis a perpendicular twofold axis produces another $n - 1$ such axes, giving n horizontal twofold axes meeting at angles π/n. The group $\boldsymbol{D}_n$ has $2n$

[4] A finite group cannot contain two planes of symmetry meeting at an angle that is not a rational part of 2π. The presence of two such planes would imply infinitely many others, all meeting along the same line and obtained by reflecting one plane in the other without limit. Thus two such planes immediately imply complete axial symmetry.

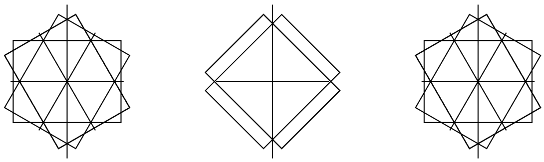
elements: n rotations about the n-fold axis and n rotations through π about the horizontal axes (we denote the latter by U_2, reserving C_2 for rotation through π about the vertical axis). Figure 34 shows the axes of $\boldsymbol{D}_3$ and $\boldsymbol{D}_4$.

As before, the n-fold axis is two-sided; the horizontal twofold axes are all equivalent when n is odd and form two inequivalent sets when n is even. Thus $\boldsymbol{D}_{2p}$ has $p+3$ classes: E, two classes of p rotations U_2, the rotation C_2, and $p-1$ classes of two rotations about the vertical axis. The group $\boldsymbol{D}_{2p+1}$ has $p+2$ classes: E, a class of $2p+1$ rotations U_2, and p classes of two rotations about the vertical axis.

An important special case is $\boldsymbol{D}_2$, whose axes are three mutually perpendicular twofold axes. This group is also denoted by $\boldsymbol{V}$.

VI. The group $\boldsymbol{D}_{nh}$

If a horizontal symmetry plane containing the n twofold axes is added to $\boldsymbol{D}_n$, then n vertical planes automatically appear, each containing the vertical axis and one horizontal axis. The resulting group $\boldsymbol{D}_{nh}$ contains $4n$ elements: in addition to the $2n$ elements of $\boldsymbol{D}_n$, it has n reflections σ_v and n rotation–reflection transformations $C_n^k\sigma_h$. Figure 35 shows the axes and planes of $\boldsymbol{D}_{3h}$.



D_{3h} D_{2d} D_{3d}

Figure 35

Reflection σ_h commutes with all other group elements; hence $\boldsymbol{D}_{nh} = \boldsymbol{D}_n \times \boldsymbol{C}_s$, where $\boldsymbol{C}_s$ consists of E and σ_h. For even n the group contains inversion, and we may also write $\boldsymbol{D}_{2p,h} = \boldsymbol{D}_{2p} \times \boldsymbol{C}_i$.

Consequently, $\boldsymbol{D}_{nh}$ has twice as many classes as $\boldsymbol{D}_n$. Half coincide with the classes of $\boldsymbol{D}_n$ (rotations about axes), while the rest are obtained by multiplication by σ_h. The vertical reflections σ_v form one class for odd n and two classes for even n. The rotation–reflection transformations $\sigma_h C_n^k$ and $\sigma_h C_n^{-k}$ are conjugate in pairs.

VII. The group $\boldsymbol{D}_{nd}$

Planes of symmetry can be adjoined to the axes of $\boldsymbol{D}_n$ in another way: vertically through the n-fold axis, midway between each pair of adjacent horizontal twofold axes. Again, adjoining one plane entails another $n-1$. The resulting system defines $\boldsymbol{D}_{nd}$ (Fig. 35 shows $\boldsymbol{D}_{2d}$ and $\boldsymbol{D}_{3d}$).

The group contains $4n$ elements. To the $2n$ elements of $\boldsymbol{D}_n$ are added n reflections in vertical planes (denoted σ_d, “diagonal” planes) and n transformations $G = U_2\sigma_d$. To determine the latter, write $U_2 = \sigma_h\sigma_v$ by (91.6), where σ_v is reflection in a vertical plane through the given twofold axis. Then $G = \sigma_h\sigma_v\sigma_d$ (neither σ_v nor σ_h separately belongs to the group). The planes σ_v and σ_d meet along the n-fold axis at an angle $(\pi/2n)(2k+1)$, $k = 1, \dots, n-1$, since adjacent planes here are separated by $\pi/2n$.

By (91.6), $\sigma_v \sigma_d = C_{2n}^{2k+1}$. Hence $G = \sigma_h C_{2n}^{2k+1} = S_{2n}^{2k+1}$: these elements are rotation–reflection transformations about the vertical axis, which is therefore not merely an n-fold symmetry axis but a $2n$-fold rotation–reflection axis.

Diagonal planes reflect adjacent horizontal twofold axes into one another, so all twofold axes are equivalent for both even and odd n; likewise all diagonal planes are equivalent. The transformations S_{2n}^{2k+1} and S_{2n}^{-2k-1} are conjugate in pairs[5].

Applying these results to $\boldsymbol{D}_{2p,d}$, we find $2p + 3$ classes: E; the rotation C_2 about the n-fold axis; $p - 1$ classes of two conjugate rotations about that axis; a class of $2p$ rotations U_2; a class of $2p$ reflections σ_d; and p classes of two rotation–reflection transformations.

For odd $n = 2p + 1$, the elements include inversion (one horizontal axis is then perpendicular to a vertical plane). Thus $\boldsymbol{D}_{2p+1,d} = \boldsymbol{D}_{2p+1} \times \boldsymbol{C}_i$, and its $2p + 4$ classes follow directly from the $p + 2$ classes of $\boldsymbol{D}_{2p+1}$.

VIII. T h e g r o u p $\boldsymbol{T}$ (t e t r a h e d r a l g r o u p)

Its axes are the symmetry axes of a tetrahedron. They may be obtained by adding to the axes of $\boldsymbol{V}$ four inclined threefold axes, rotations about which permute the three twofold axes. It is convenient to picture the three twofold axes as passing through centers of opposite faces of a cube, and the threefold axes as its body diagonals. Figure 36 shows one axis of each kind in the cube and tetrahedron.

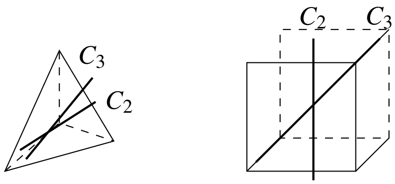


Figure 36

The three twofold axes are equivalent. The threefold axes are likewise equivalent, being carried into one another by C_2 rotations, but are not two-sided. Thus the 12 elements of $\boldsymbol{T}$ form four classes: E, three rotations C_2, four rotations C_3, and four rotations C_3^2.

IX. T h e g r o u p $\boldsymbol{T}_d$

This group contains every symmetry transformation of a tetrahedron. Its axes and planes are obtained by adding to $\boldsymbol{T}$ symmetry planes each containing one twofold and two threefold axes. The twofold axes thereby become fourfold rotation–reflection axes (as in $\boldsymbol{D}_{2d}$). It is convenient to picture the three rotation–reflection axes through centers of opposite cube faces, the four threefold axes as its body diagonals, and the six symmetry planes

through pairs of opposite edges (Fig. 37 shows one of each kind).

[5]Indeed,

$$\sigma_d S_{2n}^{2k+1} \sigma_d = \sigma_d \sigma_h C_{2n}^{2k+1} \sigma_d = \sigma_h \sigma_d C_{2n}^{2k+1} \sigma_d = \sigma_h C_{2n}^{-2k-1} = S_{2n}^{-2k-1}.$$

Because the symmetry planes are vertical relative to the threefold axes, those axes are two-sided. All axes and planes of each kind are equivalent. The 24 elements therefore form five classes: E; eight rotations C_3 and C_3^2; six plane reflections; six rotation–reflection transformations S_4 and S_4^3; and three rotations $C_2 = S_4^2$.

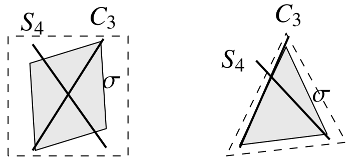


Figure 37

X. T h e g r o u p $\boldsymbol{T_h}$

This group is obtained from $\boldsymbol{T}$ by adding a center of symmetry: $\boldsymbol{T}_h = \boldsymbol{T} \times \boldsymbol{C}_i$. Three mutually perpendicular symmetry planes appear, each containing two twofold axes, and the threefold axes become sixfold rotation–reflection axes (Fig. 38 shows one such axis and plane).

The group has 24 elements in eight classes obtained directly from those of $\boldsymbol{T}$.

XI. T h e g r o u p $\boldsymbol{O}$ (o c t a h e d r a l g r o u p)

Its axes are the symmetry axes of a cube: three fourfold axes through centers of opposite faces, four threefold axes through opposite vertices, and six twofold axes through midpoints of opposite edges (Fig. 39).

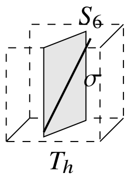


Figure 38

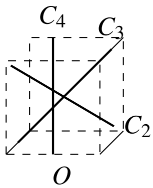


Figure 39

All axes of a given order are equivalent and two-sided. Thus the 24 elements form five classes: E; eight rotations C_3 and C_3^2; six rotations C_4 and C_4^3; three rotations C_4^2; and six rotations C_2.

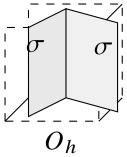


Figure 40

XII. T h e g r o u p $\boldsymbol{O}_h$

This is the group of all symmetry transformations of a cube[6]. It is obtained by adding a center of symmetry to $\boldsymbol{O}$: $\boldsymbol{O}_h = \boldsymbol{O} \times \boldsymbol{C}_i$. The threefold axes become sixfold rotation–reflection axes (the body diagonals of the cube); there also appear six symmetry planes through pairs of opposite edges and three planes parallel to the cube faces (Fig. 40). The group has 48 elements in ten classes obtained directly from those of $\boldsymbol{O}$. Five coincide with the classes of $\boldsymbol{O}$; the others are: I; eight rotation–reflection transformations S_6 and S_6^5; six transformations $C_4\sigma_h$, $C_4^3\sigma_h$ about the fourfold axes; three reflections σ_h in planes horizontal relative to those axes; and six reflections σ_v in planes vertical relative to them.

XIII, XIV. T h e g r o u p s $\boldsymbol{Y}$, $\boldsymbol{Y}_h$ (i c o s a h e d r a l g r o u p s)

Only exceptionally do these groups occur in nature as molecular symmetry groups. We therefore merely note that $\boldsymbol{Y}$ is the group of 60 rotations about the symmetry axes of an icosahedron (a regular 20-faced polyhedron with triangular faces) or a pentagonal dodecahedron (a regular 12-faced polyhedron with pentagonal faces): there are six fivefold, ten threefold, and fifteen twofold axes. Adding a center of symmetry gives $\boldsymbol{Y}_h = \boldsymbol{Y} \times \boldsymbol{C}_i$, the complete symmetry group of these polyhedra.

This exhausts all possible types of point groups with finitely many elements. They must be supplemented by the so-called continuous point groups, containing infinitely many elements; these will be considered in § 98.

§ 94. Group representations

Consider some symmetry group, and let ψ_1 be a single-valued function of the coordinates in the configuration space of the physical system. Under a transformation of the coordinate system corresponding to a group element G, this function is transformed into another function.

Applying in turn all g transformations of the group, where g is its order, will in general produce g different functions from ψ_1. For particular choices of ψ_1, however, some of these functions may be linearly dependent. The result is a set of f ($f \leqslant g$) linearly independent functions $\psi_1, \psi_2, \ldots, \psi_f$ which transform linearly into one another under the symmetry transformations in the group. In other words, a transformation G

[6]The groups $\boldsymbol{T}$, $\boldsymbol{T}_d$, $\boldsymbol{T}_h$, $\boldsymbol{O}$, and $\boldsymbol{O}_h$ are called *cubic*.

carries each ψ_i $(i = 1, 2, \ldots, f)$ into a linear combination

$$\sum_{k=1}^{f} G_{ki}\psi_k,$$

where the constants G_{ki} depend on G. The set of these constants is called the *transformation matrix*[7].

It is convenient here to regard the group elements G as operators acting on the functions ψ_i, so that

$$\widehat{G}\psi_i = \sum_k G_{ki}\psi_k. \tag{94.1}$$

The functions ψ_i can always be chosen mutually orthogonal and normalized. The transformation matrix then coincides with the matrix of an operator as defined in § 11:

$$G_{ik} = \int \psi_i^* \widehat{G}\psi_k \, dq. \tag{94.2}$$

The product of two group elements G and H corresponds to the matrix obtained from their matrices by the ordinary rule of matrix multiplication (11.12):

$$(GH)_{ik} = \sum_l G_{il}H_{lk}. \tag{94.3}$$

The collection of matrices of all group elements is called a *representation of the group*. The functions $\psi_1, \ldots, \psi_f$ used to define these matrices are called a *basis of the representation*. Their number f is the *dimension of the representation*.

Consider the integrals $\int \psi_i^*\psi_k \, dq$. Since integration is over the entire space, their values plainly remain unchanged under any rotation or reflection of the coordinate system. Thus symmetry transformations preserve the orthonormality of the basis functions, which means (see § 12) that the operators $\widehat{G}$ are unitary[8].

The matrices representing the group elements in a representation with an orthonormal basis are correspondingly unitary.

Apply a linear unitary transformation to the functions $\psi_1, \ldots, \psi_f$,

$$\psi_i' = \sum_k S_{ki}\psi_k. \tag{94.4}$$

This gives a new set $\psi_1', \ldots, \psi_f'$ which is also orthonormal (see § 12)[9]. Taking the ψ_i' as basis gives a new representation of the same dimension. Representations obtainable from one another by a linear transformation of the basis functions are called *equivalent*; plainly they are not essentially different.

---

[7] Since the functions ψ_i are assumed single-valued, every group element corresponds to one definite matrix.

[8] This argument essentially uses the fact that the integrals either vanish (for $i \neq k$) or are certainly nonzero (for $i = k$), because the integrand $|\psi_i|^2$ is positive.

[9] Recall (see (12.12)) that, because the transformations are unitary, the sum of the squared moduli of the basis functions is invariant.

The matrices of equivalent representations are related simply. According to (12.7), the matrix of $\widehat{G}$ in the new representation equals the matrix of the operator

$$\widehat{S}^{-1}\widehat{G}\widehat{S} \tag{94.5}$$

in the old representation.

The sum of the diagonal elements, or trace, of a matrix representing the group element G is called its *character*; characters will be denoted by $\chi(G)$. It is crucial that matrices in equivalent representations have equal characters (see (12.11)). This makes the specification of a group representation by its characters especially important: it immediately distinguishes essentially different representations from equivalent ones. Below, only inequivalent representations will be called different.

If S in (94.5) is taken to be the group element relating conjugate elements G and G', we find

that, in any given representation, matrices representing elements of the same class have equal characters.

The identity element E corresponds to the identity transformation. Its matrix in every representation is therefore diagonal, with all diagonal entries equal to unity. Consequently its character is simply the representation dimension:

$$\chi(E) = f. \tag{94.6}$$

Consider a representation of dimension f. It may happen that after a suitable linear transformation (94.4) the basis functions divide into sets of $f_1, f_2, \ldots$ functions ($f_1 + f_2 + \cdots = f$), such that all group elements transform the functions within each set only among themselves, without involving functions from the other sets. The representation is then said to be *reducible*.

If no linear transformation can reduce the number of basis functions transformed into one another, their representation is called *irreducible*. Every reducible representation can be decomposed into irreducible representations. That is, a suitable linear transformation divides the basis functions into sets, each of which transforms under the group elements according to some irreducible representation. Several distinct sets may transform according to the same irreducible representation; one then says that the reducible representation contains that irreducible representation the corresponding number of times.

Irreducible representations are an essential characteristic of a group and play the central part in every quantum-mechanical application of group theory. We shall state their principal properties[10].

It can be shown that the number of different irreducible representations of a group equals the number r of its classes. We distinguish the characters of different irreducible representations by superscripts: the characters of the matrix of G in the different representations are $\chi^{(1)}(G), \chi^{(2)}(G), \ldots, \chi^{(r)}(G)$.

The matrix elements of irreducible representations obey a number of orthogonality relations. First, for two distinct irreducible representations,

[10]Proofs of these properties may be found in any specialized course on group theory.

$$\sum_G G_{ik}^{(\alpha)} G_{lm}^{(\beta)*} = 0, \tag{94.7}$$

where $\alpha \neq \beta$ distinguish the representations and the sum runs over all elements of the group. For each irreducible representation,

$$\sum_G G_{ik}^{(\alpha)} G_{lm}^{(\alpha)*} = \frac{g}{f_\alpha} \delta_{il} \delta_{km}. \tag{94.8}$$

Thus only the sums of squared moduli of matrix elements are nonzero:

$$\sum_G |G_{ik}^{(\alpha)}|^2 = \frac{g}{f_\alpha}.$$

Relations (94.7) and (94.8) can be combined as

$$\sum_G G_{ik}^{(\alpha)} G_{lm}^{(\beta)*} = \frac{g}{f_\alpha} \delta_{\alpha\beta} \delta_{il} \delta_{km}. \tag{94.9}$$

In particular, an important orthogonality relation for the characters follows. Summing both sides of (94.9) over the index pairs i, k and l, m gives

$$\sum_G \chi^{(\alpha)}(G) \chi^{(\beta)}(G)^* = g \delta_{\alpha\beta}. \tag{94.10}$$

For $\alpha = \beta$,

$$\sum_G |\chi^{(\alpha)}(G)|^2 = g;$$

the sum of the squared moduli of the characters of an irreducible representation equals the group order. This relation may be used as a criterion of irreducibility: for a reducible representation the sum is necessarily greater than g (it equals ng if the representation contains n mutually distinct irreducible parts).

It also follows from (94.10) that equality of the characters of two irreducible representations is both a necessary and a sufficient condition for their equivalence.

Since characters belonging to elements of the same class are equal, the sum in (94.10) actually contains only r independent terms and can be written

$$\sum_C g_C \chi^{(\alpha)}(C) \chi^{(\beta)}(C)^* = g \delta_{\alpha\beta}, \tag{94.11}$$

where the sum is over the r classes of the group, denoted schematically by C, and g_C is the number of elements in class C.

Since the number of irreducible representations equals the number of classes, the quantities $f_{\alpha C} = \sqrt{g_C/g}\, \chi^{(\alpha)}(C)$ form a square matrix of r^2 quantities.

The orthogonality relations in the first index, $\sum_C f_{\alpha C} f_{\beta C}^* = \delta_{\alpha\beta}$, then automatically imply orthogonality in the second index: $\sum_\alpha f_{\alpha C} f_{\alpha C'}^* = \delta_{CC'}$. Thus, alongside (94.11),

$$\sum_\alpha \chi^{(\alpha)}(C) \chi^{(\alpha)}(C')^* = \frac{g}{g_C} \delta_{CC'}. \tag{94.12}$$

Among the irreducible representations of every group there is always one trivial representation, realized by a single basis function invariant under all group transformations. This one-dimensional representation is called the *identity representation*; all its characters equal unity. If one representation in (94.10) or (94.11) is the identity representation, the other satisfies

$$\sum_G \chi^{(\alpha)}(G) = \sum_C g_C \chi^{(\alpha)}(C) = 0. \tag{94.13}$$

Thus the sum of the characters of all group elements vanishes for every nonidentity representation.

Relation (94.10) makes it particularly simple to decompose any reducible representation into irreducible ones when the characters of both are known.

Let $\chi(G)$ be the characters of a reducible representation of dimension f, and let $a^{(1)}, a^{(2)}, \ldots, a^{(r)}$ give the numbers of times the corresponding irreducible representations occur in it, so that

$$\sum_{\beta=1}^{r} a^{(\beta)} f_\beta = f \tag{94.14}$$

(f_β are the dimensions of the irreducible representations). The characters $\chi(G)$ can then be written

$$\chi(G) = \sum_{\beta=1}^{r} a^{(\beta)} \chi^{(\beta)}(G). \tag{94.15}$$

Multiplying this equality by $\chi^{(\alpha)}(G)^*$ and summing over all G, relation (94.10) gives

$$a^{(\alpha)} = \frac{1}{g} \sum_G \chi(G) \chi^{(\alpha)}(G)^*. \tag{94.16}$$

Consider the representation of dimension $f = g$ realized by the g functions $\widehat{G}\phi$, where ϕ is some general function of the coordinates, so that all g functions obtained from it are linearly independent. This representation is called *regular*. Clearly, none of its matrices has diagonal elements except the matrix corresponding to the identity element. Thus $\chi(G) = 0$ for $G \neq E$, while $\chi(E) = g$. On decomposing the representation into irreducible ones, (94.16) gives $a^{(\alpha)} = (1/g) g f_\alpha = f_\alpha$. Hence every irreducible representation occurs in the regular representation a number of times equal to its dimension. Substitution in (94.14) gives

$$f_1^2 + f_2^2 + \cdots + f_r^2 = g; \tag{94.17}$$

the sum of the squares of the dimensions of a group's irreducible representations equals its order[11].

It follows in particular that for Abelian groups, where $r = g$, all irreducible representations are one-dimensional ($f_1 = f_2 = \cdots = f_r = 1$).

We also state without proof that the dimensions of a group's irreducible representations divide its order.

[11] For point groups, it may be noted that for given r and g, equation (94.17) can in fact be satisfied by a set of integers $f_1, \ldots, f_r$ in only one way.

The actual decomposition of the regular representation into irreducible parts is effected by the formula

$$\psi_i^{(\alpha)} = \frac{f_\alpha}{g} \sum_G G_{ik}^{(\alpha)*} \widehat{G} \phi. \tag{94.18}$$

It is readily verified that, for fixed k, the functions $\psi_i^{(\alpha)}$ $(i = 1, 2, \ldots, f_\alpha)$ defined by this formula transform among themselves according to

$$\widehat{G} \psi_i^{(\alpha)} = \sum_l G_{li}^{(\alpha)} \psi_l^{(\alpha)},$$

and hence form a basis of the αth irreducible representation. Giving k its different values produces f_α distinct sets of basis functions $\psi_i^{(\alpha)}$ for the same irreducible representation, in agreement with the fact that each irreducible representation occurs f_α times in the regular representation.

An arbitrary function ψ can be expressed as a sum of functions transforming according to irreducible representations of the group. The solution is

$$\psi = \sum_\alpha \sum_i \psi_i^{(\alpha)}, \qquad \psi_i^{(\alpha)} = \frac{f_\alpha}{g} \sum_G G_{ii}^{(\alpha)*} \widehat{G} \psi. \tag{94.19}$$

To prove this, substitute the second formula into the first and sum over i, obtaining

$$\psi = \frac{1}{g} \sum_\alpha f_\alpha \chi^{(\alpha)*}(G) \widehat{G} \psi. \tag{94.20}$$

Noting that f_α equals the character $\chi^{(\alpha)}(E)$ of the identity element and using (94.12), we find that $\sum_\alpha f_\alpha \chi^{(\alpha)*}(G)$ is nonzero, and equals g, only when G is the identity element. The right-hand side of (94.20) therefore coincides identically with ψ.

Consider two different systems of functions $\psi_1^{(\alpha)}, \ldots, \psi_{f_\alpha}^{(\alpha)}$ and $\psi_1^{(\beta)}, \ldots, \psi_{f_\beta}^{(\beta)}$ realizing two irreducible representations of the group. The products $\psi_i^{(\alpha)} \psi_k^{(\beta)}$ form a system of $f_\alpha f_\beta$ new functions which can serve as a basis for a new representation of dimension $f_\alpha f_\beta$. This is called the *direct (or Kronecker) product* of the first two representations. It is irreducible only if at least one of f_α or f_β equals unity. The characters of the direct product are plainly the products of the characters of its two factors. Indeed, if

$$\widehat{G} \psi_i^{(\alpha)} = \sum_l G_{li}^{(\alpha)} \psi_l^{(\alpha)}, \qquad \widehat{G} \psi_k^{(\beta)} = \sum_m G_{mk}^{(\beta)} \psi_m^{(\beta)},$$

then

$$\widehat{G} \psi_i^{(\alpha)} \psi_k^{(\beta)} = \sum_{l,m} G_{li}^{(\alpha)} G_{mk}^{(\beta)} \psi_l^{(\alpha)} \psi_m^{(\beta)};$$

hence, denoting its character by $(\chi^{(\alpha)} \times \chi^{(\beta)})(G)$, we have

$$(\chi^{(\alpha)} \times \chi^{(\beta)})(G) = \sum_{i,k} G_{ii}^{(\alpha)} G_{kk}^{(\beta)} = \sum_i G_{ii}^{(\alpha)} \sum_k G_{kk}^{(\beta)},$$

and therefore

$$(\chi^{(\alpha)} \times \chi^{(\beta)})(G) = \chi^{(\alpha)}(G) \chi^{(\beta)}(G). \tag{94.21}$$

The two irreducible representations being multiplied may in particular coincide. In that case there are two distinct sets of functions $\psi_1, \ldots, \psi_f$ and $\phi_1, \ldots, \phi_f$ realizing the same representation; its direct product with itself is realized by the f^2 functions $\psi_i \phi_k$ and has characters

$$(\chi \times \chi)(G) = [\chi(G)]^2.$$

This reducible representation can immediately be divided into two representations of smaller dimension, though in general they are still reducible. One is realized by the $f(f+1)/2$ functions $\psi_i\phi_k + \psi_k\phi_i$, and the other by the $f(f-1)/2$ functions $\psi_i\phi_k - \psi_k\phi_i$ $(i \neq k)$; evidently the functions in each set transform only among themselves. The first is called the *symmetric product* of the representation with itself, and its characters are denoted by $[\chi^2](G)$. The second is the *antisymmetric product*, with characters denoted by $\{\chi^2\}(G)$.

To determine the characters of the symmetric product, write

$$\begin{aligned}\widehat{G}(\psi_i\phi_k + \psi_k\phi_i) &= \sum_{l,m} G_{li}G_{mk}(\psi_l\phi_m + \psi_m\phi_l) \\ &= \frac{1}{2}\sum_{l,m}(G_{li}G_{mk} + G_{mi}G_{lk})(\psi_l\phi_m + \psi_m\phi_l).\end{aligned}$$

Its character is therefore

$$[\chi^2](G) = \frac{1}{2}\sum_{i,k}(G_{ii}G_{kk} + G_{ik}G_{ki}).$$

But

$$\sum_i G_{ii} = \chi(G), \qquad \sum_{i,k} G_{ik}G_{ki} = \chi(G^2);$$

thus finally

$$[\chi^2](G) = \frac{1}{2}\{[\chi(G)]^2 + \chi(G^2)\}, \tag{94.22}$$

which gives the characters of the symmetric square from those of the original representation. In precisely the same way, the characters of the antisymmetric product are[12]

$$\{\chi^2\}(G) = \frac{1}{2}\{[\chi(G)]^2 - \chi(G^2)\}. \tag{94.23}$$

If the functions ψ_i and ϕ_i coincide, only the symmetric product can plainly be defined from them; it is realized by the squares ψ_i^2 and products $\psi_i\psi_k$ $(i \neq k)$. Symmetric products of higher degree also occur in applications, and their characters may be obtained in the same way.

We note a property of direct products important below. The decomposition of the direct product of two distinct irreducible representations into irreducible parts contains the identity representation, exactly once, only if the two factors are complex conjugates. For real representations the identity representation occurs only in the direct product of

[12]For representations of dimension 2, it is useful to note that the characters $\{\chi^2\}(G)$ coincide with the determinants of the linear transformations G, as direct calculation readily verifies.

an irreducible representation with itself, and plainly in its symmetric part. To determine whether (94.21) contains the identity representation, one need only sum its characters over G and divide by the group order g, according to (94.16). The assertion then follows directly from the orthogonality relations (94.10).

Finally, consider the irreducible representations of a group that is itself the direct product of two other groups; this must not be confused with the direct product of two representations of the same group. If $\psi_i^{(\alpha)}$ realize an irreducible representation of group A, and $\psi_k^{(\beta)}$ do the same for group B, then the products $\psi_k^{(\beta)}\psi_i^{(\alpha)}$ form a basis for an $f_\alpha f_\beta$-dimensional representation of $A \times B$, and this representation is irreducible. Its characters are products of the corresponding characters of the original representations (compare the derivation of (94.21)); the element $C = AB$ of $A \times B$ has character

$$\chi(C) = \chi^{(\alpha)}(A)\chi^{(\beta)}(B). \tag{94.24}$$

Multiplying in this way every irreducible representation of A by every irreducible representation of B gives all irreducible representations of $A \times B$.

§ 95. Irreducible representations of point groups

We now turn to the explicit determination of the irreducible representations of point groups. The overwhelming majority of molecules possess only symmetry axes of orders two, three, four, and six. We shall therefore omit the icosahedral groups Y, Y_h; consider C_n, C_{nh}, C_{nv}, D_n, D_{nh} only for $n = 1, 2, 3, 4, 6$; and consider S_{2n}, D_{nd} only for $n = 1, 2, 3$.

The characters of the representations of these groups are given in Table 7. Isomorphic groups have the same representations and are placed together in one array. Numbers preceding the group-element symbols in the first rows give the numbers of elements in the corresponding classes (see § 93). The first columns contain the conventional labels for the representations. One-dimensional representations are denoted by A, B, two-dimensional ones by E, and three-dimensional ones by F (the symbol E for a two-dimensional irreducible representation must not be confused with E for the identity element!)[13]. Basis functions of the A representations are symmetric, and those of the B representations antisymmetric, under rotations about the principal axis of order n. Functions having different symmetry under reflection σ_h are distinguished by the number of primes (one or two), while the subscripts g and u specify symmetry under inversion. Alongside the representation labels, the letters x, y, z indicate the representation according to which the coordinates themselves transform; the z axis is everywhere chosen along the principal symmetry axis. The letters ε and ω mean

$$\varepsilon = e^{2\pi i/3}, \qquad \omega = e^{2\pi i/6} = -\omega^4, \qquad \varepsilon + \varepsilon^2 = -1, \qquad \omega^2 - \omega = -1.$$

Clearly, when $\widehat{G}$ acts on a basis function ψ, the latter can only be multiplied by a gth

[13]The reason why two complex-conjugate one-dimensional representations are denoted as a single two-dimensional representation will become clear in § 96.

Table 7. Characters of irreducible representations of point groups

C_i	E	I
A_g	1	1
$A_u; x, y, z$	1	-1

C_2	E	C_2
$A; z$	1	1
$B; x, y$	1	-1

C_s	E	σ
$A'; x, y$	1	1
$A''; z$	1	-1

C_{2h}	E	C_2	σ_h	I
A_g	1	1	1	1
B_g	1	-1	-1	1
$A_u; z$	1	1	-1	-1
$B_u; x, y$	1	-1	1	-1

C_{2v}	E	C_2	σ_v	σ'_v
$A_1; z$	1	1	1	1
$B_2; y$	1	-1	-1	1
A_2	1	1	-1	-1
$B_1; x$	1	-1	1	-1

D_2	E	C_2	C'_2	C''_2
A	1	1	1	1
$B_3; x$	1	-1	-1	1
$B_1; z$	1	1	-1	-1
$B_2; y$	1	-1	1	-1

C_{3v}	E	$2C_3$	$3\sigma_v$
$A_1; z$	1	1	1
A_2	1	1	-1
$E; x, y$	2	-1	0

D_3	E	$2C_3$	$3U_2$
A_1	1	1	1
$A_2; z$	1	1	-1
$E; x, y$	2	-1	0

Table 7 (continued)

C_6	E	C_6	C_3	C_2	C_6^5	C_6^4
$A; z$	1	1	1	1	1	1
B	1	-1	1	-1	1	-1
E_1	1	ω	ω^2	-1	$-\omega$	$-\omega^2$
	1	$-\omega$	ω^2	-1	ω	$-\omega^2$
$E_2; x \pm iy$	1	ω^2	$-\omega$	1	ω^2	$-\omega$
	1	$-\omega^2$	$-\omega$	1	$-\omega^2$	ω

C_{4v}, D_4, D_{2d}	E	C_2	$2C_4$	$2U_2$	$2U'_2$
A_1	1	1	1	1	1
A_2	1	1	1	-1	-1
B_1	1	1	-1	1	-1
B_2	1	1	-1	-1	1
$E; x, y$	2	-2	0	0	0

Table 7 (continued)

D_6, C_{6v}, D_{3h}	E	C_2	$2C_3$	$2C_6$	$3U_2$	$3U'_2$
A_1	1	1	1	1	1	1
A_2	1	1	1	1	-1	-1
B_1	1	-1	1	-1	1	-1
B_2	1	-1	1	-1	-1	1
E_2	2	2	-1	-1	0	0
$E_1; x, y$	2	-2	-1	1	0	0

T	E	$3C_2$	$4C_3$	$4C_3^2$
A	1	1	1	1
E	1	1	ε	ε^2
	1	1	ε^2	ε
$F; x, y, z$	3	-1	0	0

O, T_d	E	$8C_3$	$3C_2$	$6\sigma_d$	$6S_4$
A_1	1	1	1	1	1
A_2	1	1	1	-1	-1
E	2	-1	2	0	0
$F_2; x, y, z$	3	0	-1	1	-1
F_1	3	0	-1	-1	1

root of unity, i.e.[14]

$$\widehat{G}\psi = e^{2\pi ik/g}\psi, \qquad k = 1, 2, \ldots, g.$$

The group C_{2h} (and the groups C_{2v} and D_2 isomorphic to it) is Abelian, so all its irreducible representations are one-dimensional; their characters can only be ± 1 (since the square of every element is E).

Consider next the group C_{3v}. Relative to C_3, it also contains reflections σ_v in vertical planes (all belonging to one class). A function invariant under rotation about the axis (a basis function of the A representation of C_3) may be symmetric or antisymmetric under the reflections σ_v. Functions multiplied by ε and ε^2 under a C_3 rotation (basis functions of the complex-conjugate E representations) are interchanged by reflection[15]. It follows that C_{3v} (and the isomorphic group D_3) has two one-dimensional irreducible representations and one two-dimensional irreducible representation with the characters displayed in the

table. That these are all the irreducible representations follows from $1^2 + 1^2 + 2^2 = 6$, equal to the order of the group.

The characters of representations of the other groups of the same type (C_{4v}, C_{6v}) are obtained by analogous arguments.

The group T is obtained from $D_2 \equiv V$ by adjoining rotations about four inclined axes of order three. A function invariant under transformations of V (the basis of representation A) may be multiplied by 1, ε, or ε^2 under a C_3 rotation. The basis functions of the three one-dimensional representations B_1, B_2, B_3 of V are interchanged by rotations about the third-order axes (as is evident if the coordinates x, y, z themselves are chosen as these functions). We thus obtain three one-dimensional and one three-dimensional irreducible representations ($1^2 + 1^2 + 1^2 + 3^2 = 12$).

Finally, consider the isomorphic groups O and T_d. The group T_d is obtained from T by adjoining reflections σ_d in planes, each of which passes through two third-order axes. A basis function of the identity representation A of T may be symmetric or antisymmetric under these reflections (all of which form one class), giving two one-dimensional representations of T_d. Functions multiplied by ε or ε^2 under rotation about a third-order axis (the bases of the complex-conjugate E representations of T) are interchanged under reflection in a plane containing that axis, so that a single two-dimensional representation results.

Of the three basis functions of representation F of T, one transforms into itself under reflection (and may either remain unchanged or change sign), while the other two are interchanged. Altogether this gives two one-dimensional, one two-dimensional, and two three-dimensional representations[16].

The representations of the remaining point groups of interest follow directly from those already written down, since these groups are direct products of groups already

[14] For the point group C_n, possible choices are, for example, $\psi = e^{ik\varphi}$ ($k = 1, 2, \ldots, n$), where φ is the angle of rotation about the axis measured from some fixed direction.

[15] They may be chosen, for example, as $\psi_1 = e^{i\varphi}$, $\psi_2 = e^{-i\varphi}$. Reflection in a vertical plane changes the sign of φ.

[16] Irreducible representations of higher dimensions (four and five) occur in the icosahedral groups.

considered with C_i (or C_s). Specifically,

$$C_{3h} = C_3 \times C_s, \quad D_{2h} = D_2 \times C_i, \quad D_{3d} = D_3 \times C_i, \quad O_h = O \times C_i,$$
$$C_{4h} = C_4 \times C_i, \quad D_{4h} = D_4 \times C_i, \quad D_{6h} = D_6 \times C_i,$$
$$C_{6h} = C_6 \times C_i, \quad S_6 = C_3 \times C_i, \quad T_h = T \times C_i.$$

Each of these direct products has twice as many irreducible representations as the original group; half are symmetric (denoted by the subscript g) and half antisymmetric (subscript u) under inversion.

Their characters are obtained from the characters of the original group's representations by multiplication by ± 1 (in accordance with (94.24)). Thus, for D_{3d} we obtain

D_{3d}	E	$2C_3$	$3U_2$	I	$2S_6$	$3\sigma_d$
A_{1g}	1	1	1	1	1	1
A_{2g}	1	1	−1	1	1	−1
E_g	2	−1	0	2	−1	0
A_{1u}	1	1	1	−1	−1	−1
A_{2u}	1	1	−1	−1	−1	1
E_u	2	−1	0	−2	1	0

§ 96. Irreducible representations and the classification of terms

Applications of group theory in quantum mechanics rest on the invariance of the Schrödinger equation for a physical system (atom or molecule) under the symmetry transformations of that system[17]. It follows directly that applying group elements to a function satisfying the Schrödinger equation at a particular energy (eigenvalue) must again give solutions of the same equation with the same energy. In other words, under symmetry transformations, the wave functions of stationary states belonging to one energy level transform into one another and hence realize a representation of the group. Crucially, this representation is irreducible. Indeed, functions that necessarily transform into one another under symmetry operations must in any event belong to the same energy level; equality of energy eigenvalues belonging to several sets of functions (into which the basis of a reducible

representation can be divided) that do not transform into one another would be an extremely unlikely accident[18].

Thus every energy level of a system corresponds to an irreducible representation of its symmetry group. The dimension of this representation determines the degeneracy of the level, i.e. the number of distinct states with the given energy. Specifying the irreducible representation fixes all symmetry properties of the state—its behavior under the various symmetry transformations.

[17] Group-theoretical methods were first introduced into quantum mechanics by Wigner (E. P. Wigner, 1926).

[18] Unless special reasons exist for it. Recall here the "accidental" degeneracy arising because the Hamiltonian of a system may have a symmetry higher than the purely geometrical symmetry discussed in this chapter (cf. the end of § 36).

Irreducible representations of dimension greater than one occur only in groups containing noncommuting elements (Abelian groups have only one-dimensional irreducible representations). It is appropriate to recall that the connection between degeneracy and the presence of operators that do not commute with one another (but do commute with the Hamiltonian) was established earlier without group-theoretical reasoning (see § 10).

All these statements require an important qualification. As noted earlier (see § 18), symmetry under reversal of the sign of time (present in the absence of a magnetic field) implies in quantum mechanics that complex-conjugate wave functions belong to the same energy eigenvalue. Hence, if a set of functions and the set of their complex conjugates realize distinct (inequivalent) irreducible representations, these two complex-conjugate representations must be considered together as a single "physically irreducible" representation of twice the dimension (as will always be understood below). The preceding section supplied examples. The group C_3, for instance, has only one-dimensional representations; two of them, however, are complex conjugates and physically correspond to doubly degenerate energy levels. (In a magnetic field there is no time-reversal symmetry, and complex-conjugate representations therefore correspond to different energy levels[19]).

Suppose that a physical system is acted on by a perturbation (the system is placed in an external field). To what extent can the perturbation split degenerate levels? The external field has a symmetry of its own[20]. If this symmetry is the same as, or higher than, that of the unperturbed system[21], then the symmetry of the perturbed Hamiltonian $\widehat{H} = \widehat{H}_0 + \widehat{V}$ coincides with that of the unperturbed operator $\widehat{H}_0$. Clearly, no splitting of degenerate levels occurs in this case. If the perturbation has lower symmetry than the unperturbed system, the symmetry of $\widehat{H}$ coincides with that of the perturbation $\widehat{V}$. Wave functions that realized an irreducible representation of the symmetry group of $\widehat{H}_0$ also realize a representation of the symmetry group of the perturbed operator $\widehat{H}$, but this representation may be reducible, signifying a splitting of the degenerate level. An example will show how the group-theoretical apparatus answers the specific question of how a given level splits.

Let the unperturbed system have symmetry T_d. Consider a triply degenerate level corresponding to the irreducible representation F_2 of this group; its characters are

	E	$8C_3$	$3C_2$	$6\sigma_d$	$6S_4$
F_2	3	0	−1	1	−1

Suppose the system is acted on by a perturbation of symmetry C_{3v} (with its third-order axis coinciding with one such axis of T_d). The three wave functions of the degenerate level realize a representation of C_{3v} (a sub-

[19] Strictly, real characters (i.e. equivalence of complex-conjugate representations) are not a sufficient condition for choosing real basis functions of a representation. This is nevertheless true for irreducible representations of point groups (but not for "double" point groups—see § 99).

[20] One may have in mind, for example, energy levels of the d and f shells of ions in a crystal lattice, weakly interacting with the surrounding atoms. The perturbation (external field) is then the field exerted on the ion by the other atoms.

[21] If a symmetry group H is a subgroup of G, the symmetry corresponding to H is said to be lower than the higher symmetry of G. Evidently, the symmetry of a sum of two expressions having symmetries G and H is the lower symmetry H.

group of T_d), whose characters are simply those of the same elements in the original representation of T_d, namely

E	$2C_3$	$3\sigma_v$
3	0	1

This representation is reducible. From the characters of the irreducible representations of C_{3v} it is readily decomposed into irreducible parts (by the general rule (94.16)). It is thereby found to split into the A_1 and E representations of C_{3v}. Thus the triply degenerate F_2 level splits into one nondegenerate A_1 level and one doubly degenerate E level. If the same system is subjected to a perturbation with symmetry C_{2v} (also a subgroup of T_d), the wave functions of the same F_2 level give a representation with characters

E	C_2	σ_v	σ_v'
3	-1	1	1

Decomposition into irreducible parts shows that it contains the representations A_1, B_1, and B_2. In this case, therefore, the level splits completely into three nondegenerate levels.

§ 97. Selection rules for matrix elements

Group theory not only permits the classification of the terms of any symmetric physical system; it also provides a simple method for finding the selection rules for matrix elements of the various quantities characterizing the system.

The method rests on the following general theorem. Let $\psi_i^{(\alpha)}$ be one basis function of an irreducible, nonidentity representation of the symmetry group. Its integral over the entire space[22] then vanishes identically:

$$\int \psi_i^{(\alpha)}\, dq = 0. \tag{97.1}$$

The proof uses the evident fact that an integral over the whole space is invariant under every transformation of the coordinate system, including

every symmetry transformation. Hence

$$\int \psi_i^{(\alpha)} dq = \int \widehat{G}\psi_i^{(\alpha)} dq = \int \sum_k G_{ki}^{(\alpha)} \psi_k^{(\alpha)} dq.$$

Summing this equality over all elements of the group merely multiplies the integral on the left by the group order g, and gives

$$g \int \psi_i^{(\alpha)} dq = \sum_k \int \psi_k^{(\alpha)} \sum_G G_{ki}^{(\alpha)} dq.$$

For every nonidentity irreducible representation, however, $\sum_G G_{ki}^{(\alpha)} = 0$ identically (this is the special case of the orthogonality relations (94.7) in which one irreducible representation is the identity representation). The theorem is thereby proved.

[22]The configuration space of the physical system in question is meant.

If ψ belongs to the basis of some reducible representation of the group, the integral $\int \psi \, dq$ can be nonzero only if that representation contains the identity representation. This follows directly from the preceding theorem.

The matrix elements of a physical quantity f are the integrals

$$\langle \beta k | f | \alpha i \rangle = \int \psi_k^{(\beta)} \hat{f} \psi_i^{(\alpha)} dq, \tag{97.2}$$

where α and β distinguish different energy levels of the system, while i and k enumerate wave functions belonging to the same degenerate level[23]. Denote by $D^{(\alpha)}$ and $D^{(\beta)}$ the irreducible representations of the symmetry group realized by $\psi_i^{(\alpha)}$ and $\psi_k^{(\beta)}$. Let D_f denote the representation of the same group corresponding to the symmetry of f; it depends on the tensor character of f. If f is a true scalar, for example, its operator $\hat{f}$ is invariant under all symmetry transformations, and D_f is the identity representation. The same is true of a pseudoscalar if the group contains symmetry axes only; if reflections are also present, D_f is one-dimensional but nonidentity. If f is a vector, D_f is the representation realized by its three mutually transforming

components; in general this representation differs for polar and axial vectors.

The products $\psi_k^{(\beta)} \hat{f} \psi_i^{(\alpha)}$ realize the representation $D^{(\beta)} \times D_f \times D^{(\alpha)}$. The matrix elements are nonzero if this representation contains the identity representation, or equivalently if $D^{(\beta)} \times D^{(\alpha)}$ contains D_f. In practice it is more convenient to decompose $D^{(\alpha)} \times D_f$ into irreducible parts; this immediately gives all types $D^{(\beta)}$ of states for which transitions from a state of type $D^{(\alpha)}$ have nonzero matrix elements.

In the simplest case, that of a scalar quantity, D_f is the identity representation. It follows immediately that matrix elements are nonzero only for transitions between states of the same type. Indeed, the direct product $D^{(\alpha)} \times D^{(\beta)}$ of two distinct irreducible representations does not contain the identity representation, whereas the direct product of an irreducible representation with itself always does. This is the most general statement of a theorem whose special cases have arisen repeatedly.

Matrix elements diagonal in energy require separate consideration: these are elements for transitions between states belonging to the same term, as distinct from two different terms of the same type. Here there is only one, rather than two distinct, sets of functions $\psi_1^{(\alpha)}, \psi_2^{(\alpha)}, \ldots$ The selection rules are found differently according to the behavior of f under time reversal.

Consider a state with wave function $\psi = \sum_i c_i \psi_i^{(\alpha)}$. The mean value of f in this state is

$$\bar{f} = \sum_{i,k} c_k^* c_i \langle \alpha k | f | \alpha i \rangle.$$

For the state with complex-conjugate wave function $\psi^* = \sum_i c_i^* \psi_i^{(\alpha)}$ we have

$$\bar{f} = \sum_{i,k} c_k c_i^* \langle \alpha k | f | \alpha i \rangle = \sum_{i,k} c_i c_k^* \langle \alpha i | f | \alpha k \rangle.$$

If f is invariant under time reversal, the two states not only belong to the same

[23]After passing to "physically irreducible" representations the basis functions can be chosen real, and we therefore make no distinction in (97.2) between wave functions and their complex conjugates.

energy level but must also have the same value $\bar{f}$. Since the coefficients c_i are arbitrary, this means that

$$\langle \alpha k|f|\alpha i\rangle = \langle \alpha i|f|\alpha k\rangle.$$

It is readily shown that the selection rules must then be found not from the full direct product $D^{(\alpha)} \times D^{(\alpha)}$, but only from its symmetric part $[D^{(\alpha)2}]$; nonzero matrix elements exist if $[D^{(\alpha)2}]$ contains D_f[24].

If f changes sign under time reversal, the replacement $\psi \to \psi^*$ must be accompanied by a change of sign of $\bar{f}$. The same argument then gives

$$\langle \alpha k|f|\alpha i\rangle = -\langle \alpha i|f|\alpha k\rangle.$$

In this case the selection rules are determined by decomposing the antisymmetric part of the direct product, $\{D^{(\alpha)2}\}$.

Problems

PROBLEM

1. Find the selection rules for matrix elements of the electric dipole moment $\mathbf{d}$ and magnetic dipole moment $\boldsymbol{\mu}$ in symmetry O.

SOLUTION. The group O contains no reflections, so the polar vector $\mathbf{d}$ and the axial vector $\boldsymbol{\mu}$ transform according to the same irreducible representation, F_1. The decompositions of direct products of F_1 with the other representations of O are

$$\begin{aligned} &F_1 \times A_1 = F_2, && F_1 \times A_2 = F_1, && F_1 \times E = F_1 + F_2, \\ &F_1 \times F_1 = A_1 + E + F_1 + F_2, && F_1 \times F_2 = A_2 + E + F_1 + F_2. && \end{aligned} \tag{1}$$

The nondiagonal (in energy) matrix elements are therefore nonzero for the transitions

$$F_1 \leftrightarrow A_1, E, F_1, F_2; \qquad F_2 \leftrightarrow A_2, E, F_1, F_2.$$

The symmetric and antisymmetric products of irreducible representations of O are

$$\begin{aligned} &[A_1^2] = [A_2^2] = A_1, && [E^2] = A_1 + E, && [F_1^2] = [F_2^2] = A_1 + E + F_2, \\ &\{E^2\} = A_2, && \{F_1^2\} = \{F_2^2\} = F_1. && \end{aligned} \tag{2}$$

The symmetric products do not contain F_1; hence the energy-diagonal matrix elements of $\mathbf{d}$, which is invariant under time reversal, vanish. The magnetic moment, which changes sign under time reversal, has diagonal matrix elements in states F_1 and F_2.

[24]The product $[D^{(\alpha)2}]$ always contains the identity representation, so for a scalar quantity the diagonal elements (as well as nondiagonal elements between states of the same type) are nonzero.

Problem

2. The same for symmetry D_{3d}.

Solution. The transformation laws of **d** and $\boldsymbol{\mu}$ in D_{3d} differ:

$$d_x, d_y \sim E_u, \quad d_z \sim A_{2u}, \qquad \mu_x, \mu_y \sim E_g, \quad \mu_z \sim A_{2g}$$

(here and below in the problems, ~ means "transforms according to the representation"). We have

$$\begin{aligned} E_u \times A_{1g} &= E_u \times A_{2g} = E_u, & E_u \times A_{1u} &= E_u \times A_{2u} = E_g, \\ E_u \times E_u &= A_{1g} + A_{2g} + E_g, & E_u \times E_g &= A_{1u} + A_{2u} + E_u. \end{aligned} \tag{3}$$

Thus the nondiagonal matrix elements of d_x, d_y are nonzero for $E_u \leftrightarrow A_{1g}, A_{2g}, E_g$ and $E_g \leftrightarrow A_{1u}, A_{2u}, E_u$. In the same way we obtain

$$\begin{aligned} \text{for } d_z: &\quad A_{1g} \leftrightarrow A_{2u};\ A_{2g} \leftrightarrow A_{1u};\ E_g \leftrightarrow E_u; \\ \text{for } \mu_x, \mu_y: &\quad E_g \leftrightarrow A_{1g}, A_{2g}, E_g;\ E_u \leftrightarrow A_{1u}, A_{2u}, E_u; \\ \text{for } \mu_z: &\quad A_{1g} \leftrightarrow A_{2g};\ A_{1u} \leftrightarrow A_{2u};\ E_g \leftrightarrow E_g;\ E_u \leftrightarrow E_u. \end{aligned}$$

The symmetric and antisymmetric products of irreducible representations of D_{3d} are

$$\begin{aligned} &[A_{1g}^2] = [A_{1u}^2] = [A_{2g}^2] = [A_{2u}^2] = A_{1g}, \\ &[E_g^2] = [E_u^2] = E_g + A_{1g}, \\ &\{E_g^2\} = \{E_u^2\} = A_{2g}. \end{aligned} \tag{4}$$

It follows that all components of **d** have no energy-diagonal matrix elements. For $\boldsymbol{\mu}$, μ_z has diagonal matrix elements between states belonging to a degenerate level of type E_g or E_u.

Problem

3. Find the selection rules for matrix elements of the electric quadrupole-moment tensor Q_{ik} in symmetry O.

Solution. The components of Q_{ik}, a symmetric tensor with $\sum_i Q_{ii} = 0$, transform under O as

$$\begin{gathered} Q_{xy},\ Q_{xz},\ Q_{yz} \sim F_2, \\ Q_{xx} + \epsilon Q_{yy} + \epsilon^2 Q_{zz}, \quad Q_{xx} + \epsilon^2 Q_{yy} + \epsilon Q_{zz} \sim E \quad (\epsilon = e^{2\pi i/3}). \end{gathered}$$

Decomposing the direct products of F_2 and E with every representation of the group gives the selection rules for nondiagonal matrix elements:

$$\begin{aligned} \text{for } Q_{xy}, Q_{xz}, Q_{yz}: &\quad F_1 \leftrightarrow A_2, E, F_1, F_2;\ F_2 \leftrightarrow A_1, E, F_1, F_2; \\ \text{for } Q_{xx}, Q_{yy}, Q_{zz}: &\quad E \leftrightarrow A_1, A_2, E;\ F_1 \leftrightarrow F_1, F_2;\ F_2 \leftrightarrow F_2. \end{aligned}$$

As follows from (2), diagonal matrix elements occur in the states

$$\begin{aligned} \text{for } Q_{xy}, Q_{xz}, Q_{yz}: &\quad F_1, F_2, \\ \text{for } Q_{xx}, Q_{yy}, Q_{zz}: &\quad E, F_1, F_2. \end{aligned}$$

Problem

4. The same for symmetry D_{3d}.

Solution. The components of Q_{ik} transform under D_{3d} according to

$$Q_{zz} \sim A_{1g}; \quad Q_{xx} - Q_{yy},\ Q_{xy} \sim E_g; \quad Q_{xz},\ Q_{yz} \sim E_g.$$

The component Q_{zz} behaves as a scalar. Decomposing the direct products of E_g with all representations of the group gives the selection rules for nondiagonal matrix elements of the remaining components:

$$E_g \leftrightarrow A_{1g}, A_{2g}, E_g; \qquad E_u \leftrightarrow A_{1u}, A_{2u}, E_u.$$

The diagonal elements are nonzero, as follows from (4), only for states E_g and E_u.

§ 98. Continuous groups

Besides the finite point groups listed in § 93, there are *continuous point groups* having infinitely many elements. These are the groups of *axial* and *spherical* symmetry.

The simplest axial-symmetry group is C_∞, which contains rotations $C(\varphi)$ through an arbitrary angle φ about the symmetry axis (it is called the *two-dimensional rotation group*). It can be regarded as the limiting case of the groups C_n as $n \to \infty$. In the same way, the continuous groups $C_{\infty h}$, $C_{\infty v}$, D_∞, and $D_{\infty h}$ arise as limiting cases of C_{nh}, C_{nv}, D_n, and D_{nh}.

A molecule has axial symmetry only when all its atoms lie on one straight line. If it is not symmetric with respect to its midpoint, its point group is $C_{\infty v}$; in addition to rotations about the axis, this group contains reflections σ_v in every plane through the axis. If the molecule is symmetric with respect to its midpoint, its point group is $D_{\infty h} = C_{\infty v} \times C_i$. The groups C_∞, $C_{\infty h}$, and D_∞, on the other hand, cannot be realized at all as molecular symmetry groups.

The group of full spherical symmetry contains rotations through arbitrary angles about every axis through the center, and reflections in every plane through the same point. This group, denoted by K_h, is the symmetry group of an isolated atom. As a subgroup it contains K, the group of all spatial rotations (called the three-dimensional rotation group, or simply the *rotation group*). The group K_h is obtained by adjoining a center of symmetry to K ($K_h = K \times C_i$).

The elements of a continuous point group can be distinguished by one or more parameters ranging continuously over their possible values. For the rotation group, for example, three Euler angles specifying a rotation of the coordinate system may be used as parameters.

The general properties of finite groups described in § 92, and the associated concepts (subgroups, conjugate elements, classes, and so forth), extend directly to continuous groups. Statements tied directly to the order of a group naturally cease to apply; one example is the assertion that the order of a subgroup divides the order of the group.

In $C_{\infty v}$ all symmetry planes are equivalent, so all reflections σ_v form a single class containing a continuous set of elements. The symmetry axis is two-sided, and there is consequently a continuous set of classes, each containing the two elements $C(\pm\varphi)$. Since $D_{\infty h} = C_{\infty v} \times C_i$, the classes of $D_{\infty h}$ follow directly from those of $C_{\infty v}$.

In the rotation group K all axes are equivalent and two-sided. Its classes therefore consist of rotations through a prescribed absolute angle $|\varphi|$ about any axis. The classes of K_h follow directly from those of K.

The concept of representations, reducible and irreducible, likewise extends directly to continuous groups. Each irreducible representation contains a continuous set of matrices, whereas the number of basis functions transformed into one another (the dimension of the representation) is finite. These functions can always be chosen so that the representation is unitary. A continuous group has infinitely many different irreducible representations, but they form a discrete sequence and can therefore be enumerated consecutively. Orthogonality relations hold for their matrix elements and characters, generalizing the corresponding relations for finite groups. In place of (94.9) we now have

$$\int G_{ik}^{(\alpha)} G_{lm}^{(\beta)*}\, d\tau_G = \frac{1}{f_\alpha}\delta_{\alpha\beta}\delta_{il}\delta_{km}\int d\tau_G, \tag{98.1}$$

and in place of (94.10),

$$\int \chi^{(\alpha)}(G)\chi^{(\beta)}(G)^*\, d\tau_G = \delta_{\alpha\beta}\int d\tau_G. \tag{98.2}$$

The integration in these formulas is the so-called invariant integration over the group. The integration element $d\tau_G$ is expressed in terms of the group parameters and their differentials in such a way that every group transformation carries it back into the same integration element[25]. For the rotation group one may take $d\tau_G = \sin\beta\, d\alpha\, d\beta\, d\gamma$, where α, β, and γ are the Euler angles defining the rotation of the coordinate system (§ 58); then $\int d\tau_G = 8\pi^2$.

We have in substance already found the irreducible representations of the three-dimensional rotation group (without using group-theoretical

terminology), when determining the eigenvalues and eigenfunctions of the total angular momentum. The angular-momentum component operators coincide, apart from a constant factor, with the operators of infinitesimal rotations[26], and the angular-momentum eigenvalues characterize the behavior of the wave functions under spatial rotations. An angular-momentum value j has $2j + 1$ distinct eigenfunctions ψ_{jm}, distinguished by the values m of the angular-momentum projection and belonging to one $(2j + 1)$-fold degenerate energy level. Rotations of the coordinate system transform these functions into one another, so that they realize an irreducible representation of the rotation group. In group-theoretical terms, therefore, the numbers j enumerate the irreducible representations of the rotation group, with one $(2j + 1)$-dimensional representation for every j. Since j takes integral and half-integral values, the representation dimension $2j + 1$ takes all positive integral values $1, 2, 3, \ldots$

[25]The stated properties of irreducible representations of continuous groups require convergence of the integrals (98.1) and (98.2); in particular, the "group volume" $\int d\tau_G$ must be finite. This condition holds for continuous point groups. It does not hold, for example, for the so-called Lorentz group, which will occur in relativistic theory.

[26]In mathematical terminology these operators are called the *generators* of the rotation group.

The basis functions of these representations were already examined in substance in §§ 56 and 57 (and the representation matrices were found in § 58). A basis for the representation with given j is formed by the $2j + 1$ independent components of a symmetric spinor of rank $2j$ (equivalently, by the set of $2j + 1$ functions ψ_{jm}).

The irreducible representations of the rotation group associated with half-integral j have an essential special feature. Under a rotation through 2π, their basis functions (the components of a spinor of odd rank) change sign. A 2π rotation, however, is the identity element of the group. It follows that the representations with half-integral j are, as one says, *double-valued*: in such a representation every group element (a rotation through an angle φ, $0 \leqslant \varphi \leqslant 2\pi$, about some axis) corresponds not to one matrix but to two matrices whose characters have opposite signs[27].

As noted already, an isolated atom has the symmetry $K_h = K \times C_i$. In group-theoretical terms, each atomic term is therefore associated with an irreducible representation of the rotation group K (which specifies the value

of the total atomic angular momentum J) and with an irreducible representation of C_i (which specifies the parity of the state)[28].

When an atom is placed in an external electric field, its energy levels split. The number of distinct resulting levels and the symmetry of their states can be determined by the method of § 96. For this purpose, the reducible $(2J + 1)$-dimensional representation of the symmetry group of the external field, realized by the functions ψ_{JM}, must be decomposed into irreducible representations of that group. This requires the characters of the representation realized by ψ_{JM}. Since the characters of elements in the same class are equal, it is enough to consider rotations about one axis, chosen as the z axis. Under a rotation through φ about z, the wave functions ψ_{JM} are multiplied by $e^{iM\varphi}$, where M is the angular-momentum projection on that axis. Hence the transformation matrix of the functions ψ_{JM} is diagonal, with character

$$\chi^{(J)}(\varphi) = \sum_{M=-J}^{J} e^{iM\varphi} = \frac{e^{i(J+1)\varphi} - e^{-iJ\varphi}}{e^{i\varphi} - 1},$$

or[29]

$$\chi^{(J)}(\varphi) = \frac{\sin(J + 1/2)\varphi}{\sin(\varphi/2)}. \tag{98.3}$$

[27] Strictly, double-valued representations are not representations of a group in the true sense, since they are realized by multivalued basis functions; see also § 99.

[28] The atomic Hamiltonian is also invariant under permutations of the electrons. In the nonrelativistic approximation the coordinate and spin wave functions separate, and one can speak of representations of the permutation group realized by the coordinate functions. Specification of an irreducible representation of the permutation group determines the total atomic spin S (see § 63). When relativistic interactions are included, the wave function cannot be separated into coordinate and spin parts. Symmetry under simultaneous permutations of the particle coordinates and spins supplies no characteristic of a term, because the Pauli principle permits only total wave functions antisymmetric in all electrons. Correspondingly, when relativistic interactions are included, spin is not, strictly speaking, conserved (only the total angular momentum J is conserved).

[29] To avoid misunderstanding, we emphasize that this formula uses a parametrization of the group elements different from parametrization by Euler angles: a transformation is specified by the direction of the rotation axis and the angle φ about it. It can be shown that with this parametrization, integration in (98.2), for example, must use $2(1 - \cos\varphi)\, d\varphi\, do$, where do is the solid-angle element for the direction of the rotation axis.

Under inversion I, however, all functions ψ_{JM} with different M behave alike: they are multiplied by $+1$ or -1 according as the atomic state is even or odd.

Consequently,

$$\chi^{(J)}(I) = \pm(2J+1). \tag{98.4}$$

Finally, the characters for reflection in a plane σ and for a rotation-reflection through an angle φ are calculated by expressing these symmetry operations as

$$\sigma = IC_2, \qquad S(\varphi) = IC(\pi+\varphi).$$

We shall also consider the irreducible representations of the axial-symmetry group $C_{\infty v}$. This question was in substance already answered when we classified the electronic terms of a diatomic molecule, whose symmetry is precisely $C_{\infty v}$ when the two atoms differ. The 0^+ and 0^- terms (terms with $\Omega = 0$) correspond to two one-dimensional representations: the identity representation A_1, and A_2, in which the basis function is invariant under all rotations and changes sign under reflection in the planes σ_v. The doubly degenerate terms with $\Omega = 1, 2, \ldots$ correspond to two-dimensional representations denoted by $E_1, E_2, \ldots$ Under a rotation through φ about the axis their basis functions are multiplied by $e^{\pm i\Omega\varphi}$, while reflection in a plane σ_v interchanges them. The characters of all these representations are

$C_{\infty v}$	E	$2C(\varphi)$	$\infty\sigma_v$
A_1	1	1	1
A_2	1	1	-1
E_k	2	$2\cos k\varphi$	0

(98.5)

The irreducible representations of $D_{\infty h} = C_{\infty v} \times C_i$ follow directly from those of $C_{\infty v}$ (and correspond to the classification of the terms of a diatomic molecule with identical nuclei).

If half-integral values are taken for Ω, the functions $e^{\pm i\Omega\varphi}$ realize double-valued irreducible representations of $C_{\infty v}$, corresponding to molecular terms with half-integral spin[30].

§ 99. Double-valued representations of finite point groups

States of a system with half-integral spin (and therefore half-integral total angular momentum) correspond to double-valued representations of the point symmetry group of the system. This is a general property of spinors and hence applies to continuous and finite point groups alike. It is therefore necessary to determine the *double-valued* irreducible representations of finite point groups.

As already noted, double-valued representations are not, strictly speaking, true representations of a group at all. In particular, the relations discussed in § 94 do not

[30] Unlike the three-dimensional rotation group, here one could obtain not only single- and double-valued representations but also triple-valued and higher representations by a suitable choice of fractional Ω. The physically possible eigenvalues of angular momentum, as the operator of a three-dimensional infinitesimal rotation, are determined by representations of the three-dimensional rotation group itself. Thus triple-valued and higher representations of the two-dimensional rotation group (and of any finite symmetry group), although they can be defined mathematically, have no physical meaning.

apply to them. Thus, whenever those relations referred to all irreducible representations (for example, relation (94.17) for the sum of the squares of their dimensions), only true, single-valued representations were understood to be included.

The following formal device is convenient for finding double-valued representations (H. A. Bethe, 1929). We introduce, in a purely formal manner, a new group element Q: rotation through 2π about an arbitrary axis. It is regarded as distinct from the identity, but as becoming identical with E when applied twice, so that $Q^2 = E$. Accordingly, rotations C_n about n-fold symmetry axes give the identity only after $2n$ applications, rather than n:

$$C_n^n = Q, \qquad C_n^{2n} = E. \tag{99.1}$$

Inversion I, as an element commuting with every rotation, must still give E when applied twice. Two successive reflections in a plane, however, now give Q, not E:

$$\sigma^2 = Q, \qquad \sigma^4 = E \tag{99.2}$$

(this follows because a reflection can be written $\sigma = IC_2$). We thus obtain a collection of elements forming a fictitious point symmetry group whose order is twice that of the original group; such groups will be called *double point groups*. The double-valued representations of the actual point group are plainly single-valued, true representations of the corresponding double group, and can therefore be found by ordinary methods.

The number of classes in a double group is greater than in the original group (though not generally twice as great). The element Q commutes with all other group elements[31] and therefore always constitutes a class by itself. If a symmetry axis is two-sided, this means in the double group that C_n^k and $C_n^{2n-k} = QC_n^{n-k}$ are conjugate. Consequently, when twofold axes are present, the division of the elements into classes also depends on whether these axes are two-sided (this is immaterial in ordinary point groups, since C_2 coincides with the inverse rotation C_2^{-1}).

In the group T, for example, the twofold axes are equivalent and each is two-sided, while the threefold axes are equivalent but not two-sided. The 24 elements of the double group T'[32] are therefore divided among seven classes: E, Q, a class containing the three rotations C_2 and the three C_2Q, and the classes $4C_3$, $4C_3^2$, $4C_3Q$, $4C_3^2Q$.

All irreducible representations of a double point group include, first, representations identical with the single-valued representations of the ordinary group (where Q, like E, is represented by the unit matrix), and, second, the double-valued representations of the ordinary group, where Q is represented by the negative unit matrix. It is the latter that concern us here.

The double groups C_n' $(n = 1, 2, 3, 4, 6)$ and S_4', like their corresponding ordinary groups, are cyclic[33]. All their irreducible representations are one-dimensional and can be found immediately, as explained in § 95.

The irreducible representations of D_n' (or of the groups C_{nv}' isomorphic to them) can be found in the same way as for the corresponding ordinary groups. They are

[31] This is evident for rotations and inversion; for reflection in a plane it follows by representing the reflection as the product of inversion and a rotation.

[32] A prime on the symbol of an ordinary group will distinguish its double group.

[33] The groups $S_2' \equiv C_i'$ and $S_6' \equiv C_{3i}'$, which contain inversion I, are Abelian but not cyclic.

realized by functions $e^{\pm ik\varphi}$, where φ is the angle of rotation about the n-fold axis and k takes half-integral values (integral values correspond to the ordinary single-valued representations). Rotations about the horizontal twofold axes interchange these functions, while C_n multiplies them by $e^{\pm 2\pi ik/n}$.

The representations of the double cubic groups are somewhat harder to find. The 24 elements of T' are divided among seven

classes. There are consequently seven irreducible representations in all, four of which coincide with those of the ordinary group T. The sum of the squares of the dimensions of the remaining three must be 12, showing that all three are two-dimensional. Since C_2 and C_2Q belong to the same class, $\chi(C_2) = \chi(C_2Q) = -\chi(C_2)$, and hence $\chi(C_2) = 0$ in all three representations. Further, at least one of the three representations must be real, since complex representations can occur only in mutually conjugate pairs. Consider this representation and suppose that the matrix of C_3 has been reduced to diagonal form, with diagonal elements a_1, a_2. Since $C_3^3 = Q$, we have $a_1^3 = a_2^3 = -1$. For $\chi(C_3) = a_1 + a_2$ to be real, we must take $a_1 = e^{\pi i/3}$ and $a_2 = e^{-\pi i/3}$. We then find $\chi(C_3) = 1$ and $\chi(C_3^2) = a_1^2 + a_2^2 = -1$. One of the required representations has thus been found. Its direct products with the two mutually complex-conjugate one-dimensional representations of T give the other two.

Similar arguments, which we shall not give here, determine the representations of O'. Table 8 summarizes the characters of the representations of the double groups just listed (only the representations corresponding to double-valued representations of the ordinary groups are included). Double groups isomorphic to these have the same representations.

The remaining point groups are either isomorphic to those considered or are obtained by taking their direct products with C_i, so their representations require no separate calculation.

For the same reasons as with ordinary representations, two mutually complex-conjugate double-valued representations must be treated as one physically irreducible representation of twice the dimension. One-dimensional double-valued representations must be doubled even when their characters are real. The reason is that (see § 60) complex-conjugate wave functions are linearly independent in systems of half-integral spin. Thus, if a double-valued one-dimensional representation has real characters[34] and is realized by a function ψ, then although the complex-conjugate function ψ^* transforms according to an equivalent representation, ψ and ψ^* are nevertheless linearly independent. Since,

[34] Such representations occur for the groups C_n' with odd n; their characters are $\chi(C_n^k) = (-1)^k$.

Table 8. Double-valued representations of point groups

D_2'	E	Q	$C_2^{(x)}$	$C_2^{(y)}$	$C_2^{(z)}Q$				
			$C_2^{(x)}Q$	$C_2^{(y)}Q$	$C_2^{(z)}Q$				
E'	2	-2	0	0	0				
D_3'	E	Q	C_3	C_3^2	$3U_2$	$3U_2Q$			
			C_3^2Q	C_3Q					
E_1'	1	-1	-1	1	i	$-i$			
	1	-1	-1	1	$-i$	i			
E_2'	2	-2	1	-1	0	0			
D_6'	E	Q	C_2	C_3	C_3^2	C_6	C_6^5	$3U_2$	$3U_2'$
			C_2Q	C_3^2Q	C_3Q	C_6^5Q	C_6Q	$3U_2Q$	$3U_2'Q$
E_1'	2	-2	0	1	-1	$\sqrt{3}$	$-\sqrt{3}$	0	0
E_2'	2	-2	0	1	-1	$-\sqrt{3}$	$\sqrt{3}$	0	0
E_3'	2	-2	0	-2	2	0	0	0	0
D_4'	E	Q	C_2	C_4	C_4^3	$2U_2$	$2U_2'$		
			C_2Q	C_4^3Q	C_4Q	$2U_2Q$	$2U_2'Q$		
E_1'	2	-2	0	$\sqrt{2}$	$-\sqrt{2}$	0	0		
E_2'	2	-2	0	$-\sqrt{2}$	$\sqrt{2}$	0	0		
T'	E	Q	$4C_3$	$4C_3^2$	$4C_3Q$	$4C_3^2Q$	$3C_2$		
							$3C_2Q$		
E'	2	-2	1	-1	-1	1	0		
G'	2	-2	ϵ	$-\epsilon^2$	$-\epsilon$	ϵ^2	0		
	2	-2	ϵ^2	$-\epsilon$	$-\epsilon^2$	ϵ	0		
O'	E	Q	$4C_3$	$4C_3^2$	$3C_4^2$	$3C_4$	$3C_4^3$	$6C_2$	
			$4C_3^2Q$	$4C_3Q$	$3C_4^2Q$	$3C_4^3Q$	$3C_4Q$	$6C_2Q$	
E_1'	2	-2	1	-1	0	$\sqrt{2}$	$-\sqrt{2}$	0	
E_2'	2	-2	1	-1	0	$-\sqrt{2}$	$\sqrt{2}$	0	
G'	4	-4	-1	1	0	0	0	0	

on the other hand, complex-conjugate wave functions must belong to the same energy level, we see that such a representation must be doubled in physical applications.

Everything said in § 97 about finding selection rules for matrix elements of physical quantities f remains valid for states of a system with half-integral spin, with a change only for matrix elements diagonal in energy. Repeating the argument at the end of § 97, now taking formulas (60.2) and (60.3) into account, we find that if f is even or odd under time reversal, the selection rules are obtained by considering, respectively, the antisymmetric product $\{D^{(\alpha)2}\}$ or symmetric product $[D^{(\alpha)2}]$ of $D^{(\alpha)}$ with itself. This is the reverse of the rule stated in § 97 for systems with integral spin[35].

Problem

Determine how the levels of an atom (with specified values of the total angular momentum J) split when the atom is placed in a field of cubic symmetry O[36].

Solution. The wave functions of the atomic states with angular momentum J and different values of M_J realize a $(2J+1)$-dimensional reducible representation of O, with characters given by (98.3). Decomposing this representation into irreducible parts (single-valued for integral J and double-valued for half-integral J) determines the required splitting (cf. § 96). The irreducible parts for the first several values of J are

$J=0$	$1/2$	1	$3/2$	2	$5/2$	3
A_1	E_1'	F_1	G'	$E+F_2$	$E_2'+G'$	$A_2+F_1+F_2$

[35]For application of these rules, note that with double-valued representations the identity representation is contained not in the symmetric but in the antisymmetric product of a representation with itself. For a two-dimensional double-valued representation, $\{D^{(\alpha)2}\}$ is simply the identity representation.

[36]One example is an atom in a crystal lattice. Notice also that whether the symmetry group of the external field has a center of symmetry is immaterial here, since the behavior of the wave function under inversion (the evenness or oddness of the level) is unrelated to J.

CHAPTER XIII

Polyatomic Molecules

§ 100. Classification of Molecular Vibrations

For polyatomic molecules, group theory first of all resolves the question of classifying their electronic terms, that is, the energy levels at a specified arrangement of the nuclei. They are classified according to the irreducible representations of the point symmetry group possessed by the nuclear configuration under consideration. It must, however, be stressed that the classification obtained in this way pertains precisely to this particular arrangement of the nuclei, since a displacement of the nuclei generally destroys the symmetry of the configuration. Usually the arrangement in question is the one corresponding to the equilibrium positions of the nuclei. In that case the classification retains a certain meaning also for small vibrations of the nuclei, but naturally loses its meaning when the vibrations cannot be regarded as small.

For a diatomic molecule we did not encounter this question, since its axial symmetry is of course preserved under any displacement of the nuclei. The situation is similar for triatomic molecules. Three nuclei always lie in one plane, which is a symmetry plane of the molecule. Consequently, the electronic terms of a triatomic molecule can always be classified with respect to this plane (by whether the wave functions are symmetric or antisymmetric under reflection in the plane).

There is an empirical rule for the normal electronic terms of polyatomic molecules: in the overwhelming majority of molecules, the wave function of the normal electronic state is totally symmetric (this rule was already mentioned for diatomic molecules in § 78). In other words, it is invariant under every element of the molecular symmetry group, and thus belongs to the identity irreducible representation of the group.

Group-theoretical methods are especially important in the study of molecular vibrations (E. Wigner, 1930). A purely classical treatment of the vibrations

of a molecule as a system of a number of interacting particles (nuclei) must precede the quantum-mechanical study of this question.

As is known from mechanics (see I, § 23, 24), a system of N particles (not lying on one straight line) has $3N-6$ vibrational degrees of freedom; of the total $3N$ degrees of freedom, three correspond to translational and three to rotational motion of the system as a whole[1]. The energy of a system of particles executing small vibrations can be written as

$$E = \frac{1}{2}\sum_{i,k} m_{ik}\dot{u}_i\dot{u}_k + \frac{1}{2}\sum_{i,k} k_{ik}u_iu_k. \qquad (100.1)$$

Here m_{ik} and k_{ik} are constant coefficients, while u_i are components of the vectors giving the particles' displacements from their equilibrium positions (the indices i, k

[1] If all the particles lie on one straight line, the number of vibrational degrees of freedom is $3N-5$ (rotation then has only two corresponding coordinates, so that rotation of a linear molecule about its own axis has no meaning).

enumerate both vector components and particles). By a suitable linear transformation of the quantities u_i, the coordinates corresponding to translation and rotation of the system can be eliminated from (100.1), and the vibrational coordinates can be chosen so that both quadratic forms in (100.1) become sums of squares. Normalizing these coordinates so that all coefficients in the kinetic-energy expression become unity, we obtain the vibrational energy in the form

$$E = \frac{1}{2}\sum_{i,\alpha} \dot{Q}_{\alpha i}^2 + \frac{1}{2}\sum_{\alpha} \omega_\alpha^2 \sum_i Q_{\alpha i}^2. \tag{100.2}$$

The vibrational coordinates $Q_{\alpha i}$ are called *normal coordinates*; ω_α are the frequencies of the corresponding independent vibrations. Several normal coordinates may have one and the same frequency (the frequency is then said to be *multiple*); the index α of a normal coordinate specifies the frequency, while $i = 1, 2, \ldots, f_\alpha$ enumerates the coordinates belonging to that same frequency (f_α is the multiplicity of the frequency).

Expression (100.2) for the molecular energy must be invariant under symmetry transformations. This means that, under every transformation belonging to the molecule's point symmetry group, the normal coordinates $Q_{\alpha i}$, $i = 1, 2, \ldots, f_\alpha$ (for each fixed α), transform linearly

among themselves in such a way that the sum of squares $\sum_i Q_{\alpha i}^2$ remains unchanged.

In other words, the normal coordinates belonging to each given natural frequency of the molecule realize an irreducible representation of its symmetry group; the frequency multiplicity determines the dimension of the representation. Irreducibility follows from the same considerations given in § 96 for solutions of the Schrödinger equation. Equality of frequencies corresponding to two different irreducible representations would be an extraordinarily unlikely coincidence. Here again a qualification is required: since physical normal coordinates are intrinsically real quantities, two complex-conjugate representations correspond to one natural frequency having twice the multiplicity.

These considerations make it possible to classify the natural vibrations of a molecule without solving the difficult problem of explicitly determining its normal coordinates. To do so, we must first find (by the method described below) the representation realized by all the vibrational coordinates together; we shall call it the complete *vibrational representation*. This representation is reducible, and by decomposing it into irreducible parts we thereby determine the multiplicities of the natural frequencies and the symmetry properties of the corresponding vibrations. The same irreducible representation may occur several times in the complete representation; this means that there are several different frequencies of the same multiplicity whose vibrations have the same symmetry.

To find the complete vibrational representation, we use the fact that the characters of a representation are invariant under a linear transformation of the basis functions. Hence, in calculating them, we may use as basis functions not the normal coordinates but simply the components u_i of the vectors giving the displacements of the nuclei from their equilibrium positions.

It is clear first of all that, when calculating the character of an element G of the point group, only those nuclei which remain fixed (more precisely, whose equilibrium positions remain fixed) under the given symmetry transformation need be considered. Indeed, if under the rotation or reflection G a nucleus 1 is moved to a new position previously occupied by another identical nucleus 2, the displacement of nucleus 1 is transformed under G into the displacement of nucleus 2. In other words, the rows of the matrix G_{ik} corresponding to this nucleus (that is, to its displacement u_i) will in any event contain no diagonal elements. The components of the displacement vector of a nucleus

whose equilibrium position is unaffected by the operation G, on the other hand, transform only among themselves and can therefore be considered independently of the displacement vectors of the other nuclei.

Consider first a rotation $C(\varphi)$ through an angle φ about a symmetry axis. Let u_x, u_y, u_z be the components of the displacement vector of a nucleus whose equilibrium position lies on the axis itself and is therefore unaffected by the rotation. Under the rotation these components transform like those of any ordinary (polar) vector, according to the formulas (the z axis coincides with the symmetry axis)

$$u'_x = u_x \cos\varphi + u_y \sin\varphi, \qquad u'_y = -u_x \sin\varphi + u_y \cos\varphi, \qquad u'_z = u_z.$$

The character, that is, the sum of the diagonal elements of the transformation matrix, is $1 + 2\cos\varphi$. If a total of N_C nuclei lie on this axis, the total character is

$$N_C(1 + 2\cos\varphi). \tag{100.3}$$

This character, however, corresponds to the transformation of all $3N$ displacements u_i; the part corresponding to the transformation of the translational displacement and the (small) rotation of the molecule as a whole must therefore be separated off. The translational displacement is specified by the displacement vector $\mathbf{U}$ of the molecular center of mass; its contribution to the character is consequently $1 + 2\cos\varphi$. The rotation of the molecule as a whole is specified by the rotation-angle vector $\delta\boldsymbol{\Omega}$[2].

The vector $\delta\boldsymbol{\Omega}$ is axial; under rotations of the coordinate system, however, an axial vector behaves just like a polar vector. Thus $\delta\boldsymbol{\Omega}$ also has the character $1 + 2\cos\varphi$. In all, therefore, we must subtract $2(1 + 2\cos\varphi)$ from (100.3). We finally obtain the character $\chi(C)$ of the rotation $C(\varphi)$ in the complete vibrational representation:

$$\chi(C) = (N_C - 2)(1 + 2\cos\varphi). \tag{100.4}$$

The character of the identity element E is evidently just the total number of vibrational degrees of freedom: $\chi(E) = 3N - 6$ (this also follows from (100.4) with $N_C = N$, $\varphi = 0$).

In the same way we calculate the character of the rotation–reflection transformation $S(\varphi)$ (a rotation through φ about the z axis followed by reflection in the xy plane). Under this transformation a vector changes according to

$$u'_x = u_x \cos\varphi + u_y \sin\varphi, \qquad u'_y = -u_x \sin\varphi + u_y \cos\varphi, \qquad u'_z = -u_z,$$

which has the character $(-1 + 2\cos\varphi)$. The character of the representation realized by all $3N$ displacements u_i is therefore

$$N_S(-1 + 2\cos\varphi), \tag{100.5}$$

[2]As is known, a small rotation angle can be regarded as a vector $\delta\boldsymbol{\Omega}$, whose magnitude equals the angle of rotation and whose direction is along the rotation axis in the sense fixed by the screw rule. A vector $\delta\boldsymbol{\Omega}$ defined in this way is evidently axial.

where N_S is the number of nuclei unaffected by the operation $S(\varphi)$ (this number can evidently be either zero or one). The displacement vector $\mathbf{L}$ of the center of mass has the character $(-1 + 2\cos\varphi)$. As for $\delta\boldsymbol{\Omega}$, being an axial vector it is unchanged by inversion of the coordinate system; on the other hand, the rotation–reflection transformation $S(\varphi)$ can be written

$$S(\varphi) = C(\varphi)\sigma_h = C(\varphi)C_2 I = C(\pi + \varphi)I,$$

that is, as a rotation through $\pi + \varphi$ followed by inversion. Consequently, the character of $S(\varphi)$ acting on $\delta\boldsymbol{\Omega}$ is equal to the character of $C(\pi + \varphi)$ acting on an ordinary vector, namely $1 + 2\cos(\pi + \varphi) = 1 - 2\cos\varphi$. The sum $(-1 + 2\cos\varphi) + (1 - 2\cos\varphi) = 0$, and we thus reach the result that expression (100.5) itself is the required character $\chi(S)$ of the rotation–reflection transformation $S(\varphi)$ in the complete vibrational representation:

$$\chi(S) = N_S(-1 + 2\cos\varphi). \tag{100.6}$$

In particular, the character of reflection in a plane ($\varphi = 0$) is $\chi(\sigma) = N_\sigma$, while the character of inversion ($\varphi = \pi$) is $\chi(I) = -3N_I$.

Once the characters χ of the complete vibrational representation have been found, it remains only to decompose it into irreducible representations. This is done from formula (94.16), using the character tables given in § 95 (see the problems to this section).

Group theory is not needed to classify the vibrations of a linear molecule. The total number of vibrational degrees of freedom is $3N - 5$. Among the vibrations one must distinguish those in which the atoms remain on one straight line from

those in which they do not[3]. The number of degrees of freedom for the motion of N particles along a line is N; one of them corresponds to translational motion of the molecule as a whole. Hence the number of normal coordinates for vibrations that leave the atoms on a straight line is $N - 1$; in general, these correspond to $N - 1$ different natural frequencies. The remaining $(3N - 5) - (N - 1) = 2N - 4$ normal coordinates belong to vibrations that destroy the linearity of the molecule; they correspond to $N - 2$ different double frequencies (each frequency has two normal coordinates, describing identical vibrations in two mutually perpendicular planes)[4].

Problems

Problem

1. Classify the normal vibrations of the NH_3 molecule (a regular pyramid with the N atom at its apex and the H atoms at the corners of its base; Fig. 41).

Solution. The molecular point symmetry group is C_{3v}. Rotations about the threefold axis leave only one atom (N) fixed, while reflections in the planes leave two atoms fixed (N and one H). From formulas (100.4), (100.6), the characters of the complete vibrational

[3] If the molecule is symmetric about its midpoint, there is one further characteristic of the vibrations; see problem 10 to this section.

[4] In the notation for the irreducible representations of the group $C_{\infty v}$ (§ 98), there are $N - 1$ vibrations of type A_1 and $N - 2$ vibrations of type E_1.

representation are

	E	$2C_3$	$3\sigma_v$
	6	0	2

Decomposing this representation into irreducible parts, we find that it contains the

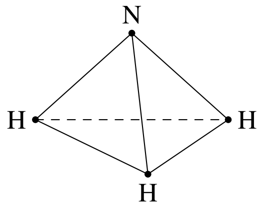


Figure 41

representation A_1 twice and E twice. There are therefore two simple frequencies corresponding to vibrations of type A_1, which preserve the full symmetry of the molecule (the so-called totally symmetric vibrations), and two double frequencies corresponding to normal coordinates that transform into one another according to the representation E.

Problem

2. Do the same for the H_2O molecule (Fig. 42).

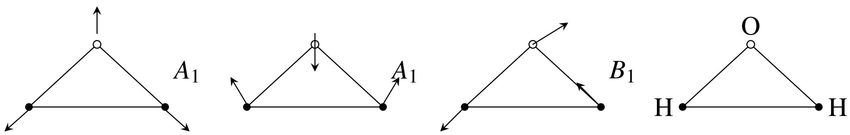


Figure 42

Solution. The symmetry group is C_{2v}. The transformation C_2 leaves the O atom fixed, the transformation σ_v (reflection in the molecular plane) leaves all three atoms fixed, and the reflection σ_v' leaves only the O atom fixed. The characters of the complete vibrational representation are

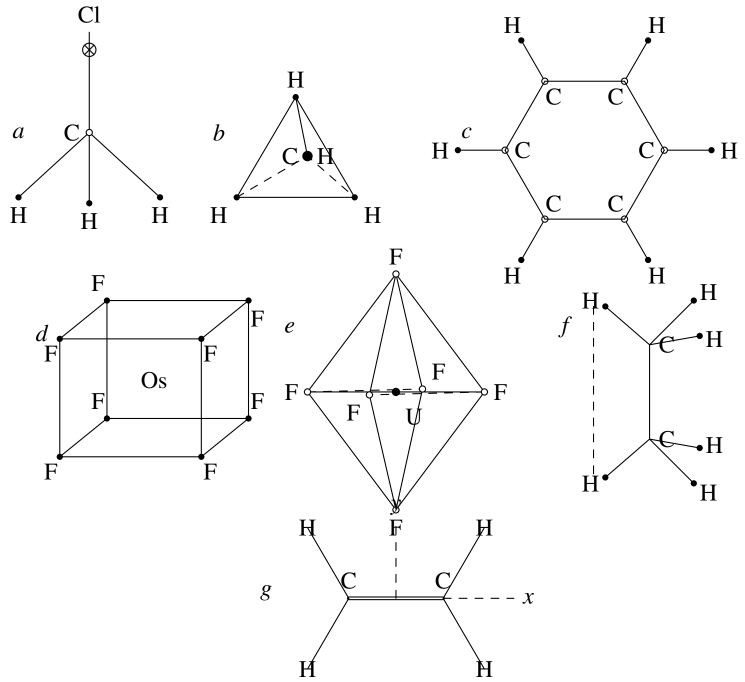
	E	C_2	σ_v	σ_v'
	3	1	3	1

This representation decomposes into the irreducible representations $2A_1$, $1B_1$: there are two totally symmetric vibrations and one whose symmetry is specified by the representation B_1; all the frequencies are simple (Fig. 42 shows the corresponding normal vibrations).

Problem

3. Do the same for the CH_3Cl molecule (Fig. 43*a*).

Solution. The molecular symmetry group is C_{3v}. The same method shows that there are three totally symmetric vibrations A_1 and three double vibrations of type E.


Cl
a
C
H
H
H
b
H
C
H
H
H
c
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H
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C
H
C
C
H
H
d
F
F
F
F
F
Os
F
F
F
F
e
F
F
F
F
F
U
F
f
H
H
C
H
C
H
H
H
g
H
H
C
C
x
H
H

Figure 43

PROBLEM

4. Do the same for the CH_4 molecule (the C atom is at the center and the H atoms are at the vertices of a tetrahedron; Fig. 43*b*).

SOLUTION. The molecular symmetry is T_d. The vibrations are $1A_1$, $1E$, $2F_2$.

PROBLEM

5. Do the same for the C_6H_6 molecule (Fig. 43*c*).

SOLUTION. The molecular symmetry is D_{6h}. The vibrations are

$$2A_{1g}, 1A_{2g}, 1A_{2u}, 1B_{1g}, 1B_{1u}, 1B_{2g}, 3B_{2u}, 1E_{1g}, 3E_{1u}, 4E_{2g}, 2E_{2u}.$$

PROBLEM

6. Do the same for the OsF_8 molecule (the Os atom is at the center and the F atoms are at the vertices of a cube; Fig. 43*d*).

SOLUTION. The molecular symmetry is O_h. The vibrations are

$$1A_{1g}, 1A_{2u}, 1E_g, 1E_u, 2F_{1u}, 2F_{2g}, 2F_{2u}.$$

PROBLEM

7. Do the same for the UF_6 molecule (the U atom is at the center and the F atoms are at the vertices of an octahedron; Fig. 43*e*).

SOLUTION. The molecular symmetry is O_h. The vibrations are

$$1A_{1g}, 1E_g, 2F_{1u}, 1F_{2g}, 1F_{2u}.$$

PROBLEM

8. Do the same for the C_2H_6 molecule (Fig. 43*f*).

SOLUTION. The molecular symmetry is D_{3d}. The vibrations are

$$3A_{1g}, 1A_{1u}, 2A_{2u}, 3E_g, 3E_u.$$

PROBLEM

9. Do the same for the C_2H_4 molecule (Fig. 43*g*; all the atoms lie in one plane).

SOLUTION. The molecular symmetry is D_{2h}. The vibrations are

$$3A_{1g}, 1A_{1u}, 2B_{1g}, 1B_{1u}, 2B_{3u}, 1B_{2g}, 2B_{2u}$$

(the coordinate axes are chosen as shown in the figure).

PROBLEM

10. Do the same for a linear molecule of N atoms symmetric about its midpoint.

SOLUTION. To the classification of the vibrations of a linear molecule considered in the text, one must add classification by their behavior under inversion in the center. The cases of even and odd N must be distinguished.

If N is even ($N = 2p$), there is no atom at the midpoint of the molecule. Give the p atoms in one half of the molecule independent displacements along the line, and the other p atoms equal and opposite displacements. We then find that p of the vibrations which leave the atoms on the line are symmetric about the center, while the remaining $(2p-1)-p = p-1$ vibrations of this type are antisymmetric about the center. Next, the p atoms have $2p$ degrees of freedom for motions in which the atoms are not constrained to the line. Giving symmetrically situated atoms equal and opposite displacements would produce $2p$ symmetric vibrations; from this number, however, two corresponding to rotation of the molecule must be subtracted. There are therefore $p-1$ double frequencies for vibrations that carry the atoms off the line and are symmetric about the center, and the same number ($(2p-2)-(p-1) = p-1$) that are antisymmetric. In the notation for the irreducible representations of the group $D_{\infty h}$ (see the end of § 98), there are p vibrations of type A_{1g} and $(p-1)$ vibrations of each of the types A_{1u}, E_{1g}, E_{1u}.

If N is odd ($N = 2p+1$), analogous reasoning shows that there are p vibrations of each of the types A_{1g}, A_{1u}, E_{1u}, and $(p-1)$ vibrations of type E_{1g}.

§ 101. Vibrational energy levels

In a quantum-mechanical treatment, the vibrational energy of a molecule is determined by the eigenvalues of the Hamiltonian

$$\widehat{H}^{(v)} = \frac{1}{2}\sum_{\alpha}\sum_{i=1}^{f_\alpha}\left(\widehat{P}_{\alpha i}^2 + \omega_\alpha^2 Q_{\alpha i}^2\right), \tag{101.1}$$

where $\widehat{P}_{\alpha i} = -i\hbar\,\partial/\partial Q_{\alpha i}$ are the momentum operators corresponding to the normal coordinates $Q_{\alpha i}$. Since this Hamiltonian separates into a sum of independent terms (the expressions in parentheses), the energy levels are given by the sums

$$E^{(v)} = \hbar\sum_{\alpha}\omega_\alpha\sum_{i}\left(v_{\alpha i}+\frac{1}{2}\right) = \sum_{\alpha}\hbar\omega_\alpha\left(v_\alpha + \frac{f_\alpha}{2}\right), \tag{101.2}$$

where $v_\alpha = \sum_i v_{\alpha i}$, while f_α is the multiplicity of the frequency ω_α. The wave

functions, in turn, are products of the corresponding wave functions of linear harmonic oscillators:

$$\psi = \prod_{\alpha}\psi_\alpha, \tag{101.3}$$

where

$$\psi_\alpha = \mathrm{const}\cdot\exp\left(-\frac{1}{2}c_\alpha^2\sum_i Q_{\alpha i}^2\right)\prod_i H_{v_{\alpha i}}(c_\alpha Q_{\alpha i}); \tag{101.4}$$

H_v denotes the Hermite polynomial of degree v, and $c_\alpha = \sqrt{\omega_\alpha/\hbar}$.

If some of the frequencies ω_α are multiple, the vibrational energy levels are, in general, degenerate. The energy (101.2) depends only on the sum $v_\alpha = \sum_i v_{\alpha i}$. Consequently, the degeneracy of a level equals the number of ways in which the given set of numbers v_α can be formed from the numbers $v_{\alpha i}$. For a single number v_α, this number is[5]

$$\frac{(v_\alpha + f_\alpha - 1)!}{v_\alpha!(f_\alpha - 1)!}.$$

Thus the total degeneracy is

$$\prod_\alpha \frac{(v_\alpha + f_\alpha - 1)!}{v_\alpha!(f_\alpha - 1)!}. \tag{101.5}$$

For doubly occurring frequencies, the factors in this product are $v_\alpha + 1$, and for triply occurring ones they are $(1/2)(v_\alpha + 1)(v_\alpha + 2)$.

It must be kept in mind that this degeneracy occurs only insofar as the vibrations are treated as purely harmonic. When terms of higher degree in the normal coordinates are included in the Hamiltonian (anharmonicity of the vibrations), the degeneracy is generally removed, though not entirely (see § 104 for more detail).

The wave functions (101.3) belonging to one and the same degenerate vibrational term furnish a certain representation (in general, a reducible one) of the molecular symmetry group. Functions belonging to different frequencies, however, transform independently of one another. Hence the representation furnished by all the functions (101.3) is the product of the representations furnished by the functions (101.4), and it is therefore sufficient to consider only the latter.

The exponential factor in (101.4) is invariant under every symmetry transformation. In the Hermite polynomials, terms of each fixed degree transform only among themselves (a symmetry transformation plainly does not change the degree of any term). On the other hand, since each Hermite polynomial is determined completely by its highest-degree term, after writing

$$\prod_{i=1}^{f_\alpha} H_{v_{\alpha i}}(c_\alpha Q_{\alpha i}) = \text{const} \cdot Q_{\alpha 1}^{v_{\alpha 1}} Q_{\alpha 2}^{v_{\alpha 2}} \ldots Q_{\alpha f_\alpha}^{v_{\alpha f_\alpha}} + \text{terms of lower degree},$$

it is enough to consider only the highest-degree term.

The same term contains the functions for which the sum $v_\alpha = \sum_i v_{\alpha i}$ has the same value. Thus we have a representation furnished by products of v_α quantities $Q_{\alpha i}$; this is precisely the symmetric product (see § 94), taken v_α times, of the irreducible representation furnished by the quantities $Q_{\alpha i}$ with itself (L. Tisza, 1933).

For one-dimensional representations, finding the characters of their v-fold symmetric products with themselves is trivial[6]:

$$\chi_v(G) = [\chi(G)]^v.$$

[5]This is the number of ways of distributing v_α balls among f_α boxes.

[6]Here we use the notation $\chi_v(G)$ in place of the cumbersome $[\chi^v](G)$.

For two- and three-dimensional representations, the following mathematical device is convenient[7]. The sum of the squares of the basis functions of an irreducible representation is invariant under all symmetry transformations. We may therefore formally regard them as the components of a two- or three-dimensional vector, and the symmetry transformations as certain rotations (or reflections) acting on these vectors. We emphasize that, in general, these rotations and reflections have nothing in common with the actual symmetry transformations and that, for every given element of the group G, they also depend on the particular representation under consideration.

Let us examine two-dimensional representations in greater detail. Let $\chi(G)$ be the character of some group element in a given two-dimensional representation, with $\chi(G) \neq 0$. Under a rotation through an angle φ in the plane, the sum of the diagonal elements of the transformation matrix for the components x, y of a two-dimensional vector equals $2\cos\varphi$. By setting

$$2\cos\varphi = \chi(G), \tag{101.6}$$

we find the angle of the rotation formally corresponding to the element G in the given irreducible representation. The v-fold symmetric product of the representation with itself is a representation with a basis consisting of the $v + 1$ quantities $x^v, x^{v-1}y, \ldots, y^v$. The characters

of this representation are[8]

$$\chi_v(G) = \frac{\sin(v+1)\varphi}{\sin\varphi}. \tag{101.7}$$

The case $\chi(G) = 0$ requires separate consideration, since a zero character corresponds both to a rotation through $\pi/2$ and to a reflection. If $\chi(G^2) = -2$, the transformation is a rotation through $\pi/2$, and for $\chi_v(G)$ we obtain

$$\chi_v(G) = (-1)^{v/2}\frac{1+(-1)^v}{2}. \tag{101.8}$$

If, however, $\chi(G^2) = 2$, then $\chi(G)$ must be regarded as the character of a reflection (that is, the transformation $x \to x$, $y \to -y$); in this case

$$\chi_v(G) = \frac{1+(-1)^v}{2}. \tag{101.9}$$

Formulas for symmetric products of three-dimensional representations can be obtained in the same way. The rotation (or reflection) that formally corresponds to a group element in a given representation is readily found with the aid of Table 7. It

[7] Applied to this purpose by A. S. Kompaneets (1940).

[8] For the calculation it is convenient to choose the basis functions in the form

$$(x+iy)^v, \quad (x+iy)^{v-1}(x-iy), \ldots, (x-iy)^v;$$

the rotation matrix is then diagonal, and the sum of its diagonal elements has the form

$$e^{iv\varphi} + e^{i(v-2)\varphi} + \ldots + e^{-iv\varphi}.$$

is the transformation corresponding to the given $\chi(G)$ in that one of the isomorphic groups in which the coordinates transform according to this representation. Thus, for the representation F_1 of the groups O and T_d, the transformation must be taken from the group O, and for the representation F_2, from the group T_d. We shall not pursue the derivation of the corresponding formulas for the characters $\chi_v(G)$ here.

§ 102. Stability of symmetric molecular configurations

For a symmetric arrangement of the nuclei, a molecular electronic term can be degenerate if the symmetry group has irreducible representations of dimension greater than one.

We ask whether such a symmetric configuration can be a stable equilibrium configuration

of the molecule. We shall neglect the effect of spin (if there is any), which is generally negligible in polyatomic molecules. The degeneracy of electronic terms discussed below is therefore purely orbital and unrelated to spin.

For the configuration to be stable, the molecular energy, regarded as a function of the internuclear distances, must have a minimum at this arrangement of the nuclei. This means that the change of energy under a small displacement of the nuclei must contain no term linear in the displacements.

Let $\widehat{H}$ be the Hamiltonian of the molecule's electronic state, with the internuclear distances regarded as parameters. Denote by $\widehat{H}_0$ this Hamiltonian at the specified symmetric configuration. The normal vibrational coordinates $Q_{\alpha i}$ may be used as quantities describing small nuclear displacements. The expansion of $\widehat{H}$ in powers of $Q_{\alpha i}$ is

$$\widehat{H} = \widehat{H}_0 + \sum_{\alpha,i} V_{\alpha i} Q_{\alpha i} + \sum_{\alpha,\beta,i,k} W_{\alpha i,\beta k} Q_{\alpha i} Q_{\beta k} + \dots \tag{102.1}$$

The expansion coefficients $V, W, \dots$ are functions only of the electron coordinates. Under a symmetry transformation, the quantities $Q_{\alpha i}$ transform among themselves, and the sums in (102.1) thereby pass into other sums of the same form. We may therefore regard a symmetry operation formally as transforming the coefficients in these sums while leaving the $Q_{\alpha i}$ fixed. In particular, for each fixed α, the coefficients $V_{\alpha i}$ then transform according to the same representation of the symmetry group as the corresponding coordinates $Q_{\alpha i}$. This follows at once from the invariance of the Hamiltonian under every symmetry operation: the collection of terms of each fixed order in its expansion, and in particular the linear terms, must possess the same invariance[9].

Consider an electronic term E_0 that is degenerate at the symmetric configuration. A displacement of the nuclei that breaks the molecular symmetry will generally split the term. To first order in the nuclear displacements, the splitting is determined by the secular equation formed from matrix elements of the linear part of (102.1),

$$V_{\rho\sigma} = \sum_{\alpha,i} Q_{\alpha i} \int \psi_\rho V_{\alpha i} \psi_\sigma \, dq, \tag{102.2}$$

[9] Strictly, the quantities $V_{\alpha i}$ should transform according to the representation complex-conjugate to that of the $Q_{\alpha i}$. As noted earlier, however, if two complex-conjugate representations do not coincide, physically they must in any event be considered together as a single representation of twice the dimension. The qualification is therefore immaterial.

where ψ_ρ, ψ_σ are wave functions of electronic states belonging to this degenerate term (chosen to be real). Stability of the symmetric configuration requires the absence of splitting linear in Q: all roots of the secular equation must vanish identically, which means that the entire matrix $V_{\rho\sigma}$ must vanish. Naturally, only normal vibrations that break the molecular symmetry are to be considered; the totally symmetric vibrations (corresponding to the unit representation of the group) must be omitted.

Since the $Q_{\alpha i}$ are arbitrary, the matrix elements (102.2) vanish only if all the integrals

$$\int \psi_\rho V_{\alpha i} \psi_\sigma \, dq \tag{102.3}$$

vanish.

Let $D^{(el)}$ be the irreducible representation according to which the electronic wave functions ψ_ρ transform, and let D_α denote the corresponding representation for the quantities $V_{\alpha i}$; as already stated, the representations D_α are those according to which the corresponding normal coordinates $Q_{\alpha i}$ transform. From the results of § 97, the integrals (102.3) can be nonzero if the product $[D^{(el)2}] \times D_\alpha$ contains the unit representation, or equivalently if $[D^{(el)2}]$ contains D_α. Otherwise every integral vanishes.

Thus the symmetric configuration is stable if $[D^{(el)2}]$ contains none of the irreducible representations D_α characterizing the molecular vibrations, apart from the unit representation. For a nondegenerate electronic state this condition is always satisfied, since the symmetric square of a one-dimensional representation is the unit representation.

As an example, consider a molecule of the CH_4 type, with one atom (C) at the center and four atoms (H) at the vertices of a tetrahedron. This configuration has symmetry T_d. Its degenerate electronic terms correspond to the representations E, F_1, F_2 of this group. The molecule has one normal vibration of type A_1 (a totally symmetric vibration), one double vibration E,

and two triple vibrations F_2 (see Problem 4 of § 100). The symmetric squares of E, F_1, F_2 are

$$[E^2] = A_1 + E, \qquad [F_1^2] = [F_2^2] = A_1 + E + F_2.$$

Each contains at least one of the representations E, F_2; hence the tetrahedral configuration under consideration is unstable for a degenerate electronic state.

This result is a general rule known as the *Jahn–Teller theorem* (H. A. Jahn, E. Teller, 1937): for a degenerate electronic state, every symmetric arrangement of nuclei is unstable, with the sole exception of an arrangement on one straight line. Owing to this instability, the nuclei move in such a way that the symmetry of their configuration is broken sufficiently to remove the term's degeneracy completely. In particular, the normal electronic term of a symmetric nonlinear molecule can only be nondegenerate[10].

As already mentioned, only linear molecules form an exception. This can easily be seen without group theory. A displacement by which a nucleus leaves the molecular axis is an ordinary vector with ξ- and η-components (the ζ-axis is directed along the molecular

[10]The physical idea that symmetry is destroyed in an electronic state whose degeneracy is due to that very symmetry was put forward by Landau (1934). Jahn and Teller (1937) proved the theorem by enumerating all possible types of symmetric nuclear arrangements in a molecule and examining each by the method given above.

axis). We saw in § 87 that such vectors have matrix elements only for transitions in which the angular momentum Λ about the axis changes by one. A degenerate term of a linear molecule, however, corresponds to states with angular momenta Λ and $-\Lambda$ about the axis, where $\Lambda \geqslant 1$. A transition between them entails a change in angular momentum of at least 2, and the matrix elements must therefore vanish. Thus a linear arrangement of nuclei in a molecule may be stable even for a degenerate electronic state.

A constructive general proof of the theorem rests on the following observation (E. Ruch, 1957).

Degeneracy of electronic states due to the symmetry of the nuclear arrangement can occur only for molecular point groups containing at least one rotation axis (C_n) or rotation–reflection axis (S_n) of order $n > 2$. Among the wave functions

of the mutually degenerate states (that is, among the basis functions of the corresponding representation $D^{(el)}$), there is then at least one whose electron density $\rho = |\psi|^2 = \psi^2$ is not invariant under rotations about this axis; the electric field created by the electrons is likewise not symmetric about the axis. At the same time, a nonlinear molecule contains equivalent nuclei away from the axis—nuclei carried into one another by the rotations C_n (or S_n). Equivalent nuclei thus lie at inequivalent points of the electric field. Yet equilibrium positions of charged particles in a field cannot possess an equivalence not required by the field's symmetry, except through an exceedingly improbable accident.

The full proof gives a concrete mathematical embodiment of this physical situation. We show how such a proof is constructed (E. Ruch, A. Schönhofer, 1965)[11].

In a nonlinear molecule, consider a nucleus (call it a) that lies outside the "center" of the molecule (that is, outside the fixed point of the transformations of its symmetry group) and is not on the principal symmetry axis, if one exists[12]. Let H be the set of symmetry operations of the molecule that leave nucleus a fixed. It is a subgroup of the full molecular symmetry group G and may be one of the point groups C_1, C_s, C_n, C_{nv}. The operations in G that are not in H carry nucleus a into other equivalent nuclei $a', a'', \ldots$; let s be the number of nuclei in this set. Clearly, the order of H is g/s, where g is the order of the full group G (that is, s is the index of H in G)[13].

Certainly $s \geqslant 3$, since the presumed existence of a multidimensional irreducible representation $D^{(el)}$ requires, as noted above, at least one symmetry axis of order higher than 2, while by assumption nucleus a does not lie on it.

When restricted from the group G to the lower-symmetry group H, the representation $D^{(el)}$ is generally reducible. Suppose that its decomposition into irreducible representations of H contains a one-dimensional representation; denote it by $d^{(el)}$. It is realized by an electronic wave function ψ, one of the basis functions of $D^{(el)}$. Because $d^{(el)}$ is one-dimensional, the square $\rho = \psi^2$ is invariant under all operations in H, and hence realizes the unit irreducible representation of this group.

The same unit representation of H can be realized by choosing as a basis one of the displacements Q_a of atom a—the displacement along the radius vector drawn from the

[11] For details see E. Ruch, A. Schönhofer // Theoret. chim. acta (Berl.). 1965. Bd. 3. S. 291.

[12] In noncubic and nonicosahedral symmetry groups, the principal axis means an axis C_n or S_n of order $n > 2$.

[13] All elements of the group G can be partitioned into s cosets $H, G'H, G''H, \ldots$, where G', G'' are elements of the group carrying nucleus a into $a', a'', \ldots$

molecular center to nucleus a.

Applying every operation of G to this displacement gives a basis for a certain representation of G, generally reducible; call it D_Q. Every transformation in G but not in H carries Q_a into a displacement of one of the other $s-1$ equivalent nuclei $a', a'', \ldots$, and displacements of different nuclei are plainly linearly independent. The dimension of D_Q is therefore s. Moreover, the displacements $Q_a, Q_{a'}, \ldots$ forming the basis of D_Q cannot correspond either to a pure translation or to a pure rotation of the molecule as a whole: with three or more equivalent nuclei, no such motion can be composed from their radial displacements.

A representation of G can likewise be obtained by applying all its operations to the function $\rho = \psi^2$; call it D_ρ. The dimension of D_ρ may equal s, but it may also be smaller, since there is no prior reason for all s functions $\rho, G'\rho, G''\rho, \ldots$ to be linearly independent. Nevertheless, one may assert that D_ρ, if it does not coincide with D_Q, is in any case contained wholly in it[14]. Furthermore, D_ρ is not the unit representation, since ψ^2 is certainly not invariant under the whole group G (only the sum of the squares of all basis functions of the multidimensional irreducible representation $D^{(el)}$ is invariant).

The properties of D_Q and D_ρ just established immediately give the desired result. Indeed, D_Q is part of the complete vibrational representation, while D_ρ is a part of $[D^{(el)2}]$ that does not contain the unit representation. Since D_ρ is contained in D_Q, it follows that $[D^{(el)2}]$ contains at least one nonunit vibrational representation D_α, as was to be proved.

The preceding argument still assumed, however, that the decomposition of $D^{(el)}$ into irreducible representations of the subgroup H contains a one-dimensional representation. This assumption holds in the overwhelming majority of cases. It is certainly true for $H = C_1, C_s, C_2, C_{2v}$, since all irreducible representations of these groups are one-dimensional. It is also certainly true for $H = C_n, C_{nv}$ with $n > 2$ when the dimension of $D^{(el)}$ is odd, because the groups C_n, C_{nv} have only one- and two-dimensional irreducible representations. Inspection of the character tables for irreducible representations of the point groups shows that the exceptions are two-dimensional representations of the cubic groups $G = O, T_d, O_h$ relative to the subgroups $H = C_3, C_{3v}$.

For definiteness, take $G = O$ and $H = C_3$; this choice affects only the notation for the representations. The two electronic functions ψ_1, ψ_2 realize the representation $D^{(el)} = E$ of O, and also the representation $d^{(el)} = E$ of C_3. The representation of C_3 realized by the products $\psi_1^2, \psi_2^2, \psi_1\psi_2$ is $[E^2] = A + E$. The three components of the vector of an arbitrary displacement $\mathbf{Q}_a$ of nucleus a realize the same representation of C_3. In this case the representation D_ρ of O is $D_\rho = [D^{(el)2}] = A_1 + E$; it does not contain the representation F_2 corresponding to a vector of translation or rotation of the molecule as a whole, and, besides the unit representation, it contains a nonunit one. Therefore the inclusion of D_ρ in D_Q (for the same reasons as above), which here has dimension $3s$,

[14]The general assertion is as follows. Suppose that the same representation (of dimension f) of a subgroup H is realized by different sets of basis functions, and that applying every transformation of G to one of these sets generates a representation of G of dimension sf, where s is the index of H in G. Then the representation of G generated in the same manner from any other such set either coincides with the first or is contained wholly in it. A rigorous proof is given in the paper cited on the preceding page.

proves the instability of the molecule in this case as well[15].

In keeping with the qualification at the beginning of this section, the degeneracy of the electronic states was understood throughout the preceding discussion to have a purely orbital origin. The Jahn–Teller theorem nevertheless remains valid when spin–orbit and spin–spin interactions are included, with the single difference that, in nonlinear molecules with half-integral spin, twofold Kramers degeneracy does not cause instability, in accordance with the general theorem proved in § 60. This case corresponds to two-dimensional double-valued irreducible representations of the double point groups. The absence of instability here can already be seen by the following formal argument. To determine the selection rules for the matrix elements (102.3) when $D^{(el)}$ is double-valued, one must consider not symmetric but antisymmetric products $\{D^{(el)2}\}$ (see § 99). For every two-dimensional double-valued irreducible representation, however, these products coincide with the unit representation, and therefore cannot contain representations corresponding to any molecular vibrations that are not totally symmetric.

§ 103. Quantization of rigid-rotor rotation

The study of the rotational levels of a polyatomic molecule is often complicated by the need to consider rotation together with vibration. As a preliminary problem, we shall consider the rotation of a molecule as a rigid body, that is, with its atoms "rigidly fixed" (a *rigid rotor*).

Let $\xi\eta\zeta$ be a coordinate system whose axes are directed along the three principal axes of inertia of the rotor and which rotates with it. The corresponding Hamiltonian is obtained by replacing the components J_ξ, J_η, J_ζ of its angular momentum in the classical expression for the energy by the corresponding operators:

$$\hat{H} = \frac{\hbar^2}{2}\left(\frac{\hat{J}_\xi^2}{I_A} + \frac{\hat{J}_\eta^2}{I_B} + \frac{\hat{J}_\zeta^2}{I_C}\right), \tag{103.1}$$

where I_A, I_B, I_C are the principal moments of inertia of the rotor.

The commutation rules for the operators $\hat{J}_\xi, \hat{J}_\eta, \hat{J}_\zeta$, the components of angular momentum in the rotating coordinate system, are not evident, since the usual derivation of the commutation rule concerns the components $\hat{J}_x, \hat{J}_y, \hat{J}_z$ in a fixed coordinate system. They are readily obtained, however, from the formula

$$(\hat{\mathbf{J}}\mathbf{a})(\hat{\mathbf{J}}\mathbf{b}) - (\hat{\mathbf{J}}\mathbf{b})(\hat{\mathbf{J}}\mathbf{a}) = -i\hat{\mathbf{J}}(\mathbf{a}\times\mathbf{b}), \tag{103.2}$$

where $\mathbf{a}, \mathbf{b}$ are any two vectors characterizing the given body (and commuting with one another). This formula is easily

verified by evaluating the left-hand side in the fixed coordinate system xyz, using the general commutation rules of the angular-momentum components with one another and with the components of an arbitrary vector.

[15]Four-dimensional representations of the icosahedral groups constitute one further exceptional case. It is treated analogously and leads to the same result.

Let **a** and **b** be unit vectors along the ξ and η axes. Then $\mathbf{a} \times \mathbf{b}$ is a unit vector along the ζ axis, and (103.2) gives

$$\hat{J}_\xi \hat{J}_\eta - \hat{J}_\eta \hat{J}_\xi = -i\hat{J}_\zeta. \tag{103.3}$$

The other two relations follow similarly.

Thus the commutation rules for the operators of the angular-momentum components in the rotating coordinate system differ from those in the fixed system only by the sign on the right-hand side[16]. It follows that all the results previously derived from the commutation rules for eigenvalues and matrix elements also hold for J_ξ, J_η, J_ζ, with the sole difference that every expression must be replaced by its complex conjugate. In particular, the eigenvalues of J_ζ (which in this section we denote by k, to distinguish them from the eigenvalues $J_z = M$) take the values $k = -J, \ldots, +J$, where J (an integer!) is the magnitude of the rotor angular momentum.

Spherical rotor. The energy eigenvalues of a rotating rotor are found most simply when all three principal moments of inertia are equal: $I_A = I_B = I_C = I$. For a molecule this occurs when it has the symmetry of one of the cubic point groups. The Hamiltonian (103.1) becomes

$$\hat{H} = \frac{\hbar^2}{2I}\hat{\mathbf{J}}^2,$$

and its eigenvalues are

$$E = \frac{\hbar^2}{2I}J(J+1). \tag{103.4}$$

Each of these energy levels is degenerate with respect to the $2J+1$ directions of the angular momentum relative to the rotor itself (that is, with respect to the values $J_\zeta = k$)[17].

Symmetric rotor. The energy levels are also readily calculated when only two moments of inertia of the rotor coincide: $I_A = I_B \neq I_C$. This occurs for molecules having a single symmetry axis of order greater than two. The Hamiltonian (103.1) takes the form

$$\hat{H} = \frac{\hbar^2}{2I_A}(\hat{J}_\xi^2 + \hat{J}_\eta^2) + \frac{\hbar^2}{2I_C}\hat{J}_\zeta^2 = \frac{\hbar^2}{2I_A}\hat{\mathbf{J}}^2 + \frac{\hbar^2}{2}\left(\frac{1}{I_C} - \frac{1}{I_A}\right)\hat{J}_\zeta^2. \tag{103.5}$$

It follows that in a state with definite J and k the energy is

$$E = \frac{\hbar^2}{2I_A}J(J+1) + \frac{\hbar^2}{2}\left(\frac{1}{I_C} - \frac{1}{I_A}\right)k^2, \tag{103.6}$$

which determines the energy levels of a symmetric rotor.

The degeneracy in k present for the spherical rotor is partially removed here. The energies coincide only for values of k differing in sign alone, corresponding to mutually

[16] This circumstance expresses the fact that, as regards the action on the rotor wave function, a rotation of the xyz system is equivalent to the opposite rotation of the $\xi\eta\zeta$ system.

[17] Here and below we disregard the always present, physically inessential $(2J+1)$-fold degeneracy with respect to the directions of the angular momentum relative to the fixed coordinate system. When it is included, the total degeneracy of the energy levels of a spherical rotor is $(2J+1)^2$.

opposite directions of the angular momentum relative to the rotor axis. Thus the levels of a symmetric rotor with $k \neq 0$ are doubly degenerate.

The stationary states of a symmetric rotor are therefore characterized by three quantum numbers: the angular momentum J and its projections on the rotor axis ($J_\zeta = k$) and on the space-fixed z axis ($J_z = M$); the rotor energy is independent of the last number. In this connection, the very fact that the magnitude of the angular momentum and its projections on a space-fixed axis and on an axis rigidly attached to the physical system are simultaneously measurable[18] follows because the operators $\hat{\mathbf{J}}^2$ and $\hat{J}_z$ commute not only with one another but also with $\hat{J}_\zeta = \hat{\mathbf{J}}\mathbf{n}$ ($\mathbf{n}$ is a unit vector along the ζ axis). This is readily checked by direct calculation, but is also evident beforehand. The angular-momentum operator reduces to the operator of an infinitesimal rotation, while the scalar product $\mathbf{Jn}$ of two vectors attached to the rotor is invariant under any rotation of the coordinate system.

The problem of determining the stationary-state wave functions of a symmetric rotor is consequently reduced to finding the common eigenfunctions of $\hat{\mathbf{J}}^2, \hat{J}_z, \hat{J}_\zeta$. This question, in turn, is mathematically closely connected with the transformation law of angular-momentum eigenfunctions under

finite rotations. Changing the notation for the quantum numbers, we write this law (58.7) as

$$\psi_{JM} = \sum_k D^{(J)}_{kM}(\alpha, \beta, \gamma)\psi_{Jk}. \tag{103.7}$$

We understand ψ_{JM} to be the wave function of a rotor state described relative to the fixed coordinate axes xyz, and ψ_{Jk} to be the wave functions of states described relative to the rotor-fixed axes $\xi\eta\zeta$. In coordinates rigidly attached to the physical system (the rotor), however, the quantities ψ_{Jk} have definite values independent of the orientation of the system in space; denote them by $\psi^{(0)}_{Jk}$. Formula (103.7) then gives the angular dependence of the functions ψ_{JM}. Suppose now that the state $|JM\rangle$ also has a definite value k of the angular-momentum projection on the ζ axis. This means that only one of all the quantities $\psi^{(0)}_{Jk}$, the one with the specified k, is nonzero. The sum in (103.7) then reduces to one term:

$$\psi_{JMk} = \psi^{(0)}_{Jk} D^{(J)}_{kM}(\alpha, \beta, \gamma).$$

We have thereby found the dependence of the wave functions of the states $|JMk\rangle$ on the Euler angles specifying the rotation of the rotor axes relative to the fixed axes. Normalizing the wave function by

$$\int |\psi_{JMk}|^2 \sin\beta \, d\alpha \, d\beta \, d\gamma = 1,$$

we obtain

$$\psi_{JMk} = i^J \sqrt{\frac{2J+1}{8\pi^2}} D^{(J)}_{kM}(\alpha, \beta, \gamma); \tag{103.8}$$

the phase factor is chosen so that for $k = 0$ the function (103.8) becomes the eigenfunction of a free integral angular momentum J (not attached in any way to the ζ axis) with

[18]These must not be confused with the projections (which are not simultaneously measurable) on two space-fixed axes!

projection M, that is, the usual spherical function (cf. (58.25)[19]).

Asymmetric rotor. When $I_A \neq I_B \neq I_C$, the energy levels cannot be calculated in general form. The degeneracy with respect to the directions of angular momentum relative to the rotor is completely removed, so that a given J corresponds to $2J + 1$ distinct nondegenerate levels. To calculate these levels (for a specified J), one must start from the Schrödinger equation written in matrix form (O. Klein, 1929). This is done as follows.

The wave functions ψ_{Jk} of rotor states with definite J and definite ζ-component of the angular momentum are the functions (103.8) found above (the index M of the z-component, on which the energy does not depend, will be omitted below for brevity); in these states the energy of the asymmetric rotor has no definite value. Conversely, in stationary states the projection J_ζ has no definite value, so definite values of k cannot be assigned to the energy levels. We seek the wave functions of these states as linear combinations

$$\psi_J = \sum_k c_k \psi_{Jk} \tag{103.9}$$

(all the functions are understood to have some one value of M, the same for every term). Substitution in the Schrödinger equation $\hat{H}\psi_J = E_J\psi_J$ gives the system

$$\sum_{k'} \big(\langle Jk|H|Jk'\rangle - E\delta_{kk'}\big)c_{k'} = 0, \tag{103.10}$$

and the solvability condition for this system gives the secular equation

$$|\langle Jk|H|Jk'\rangle - E\delta_{kk'}| = 0. \tag{103.11}$$

The roots of this equation determine the rotor energy levels; the system (103.10) then gives the linear combinations (103.9) that diagonalize the Hamiltonian, that is, the stationary-state wave functions of the rotor for specified J (and M). The calculation of matrix elements of any physical quantity with these wave functions is thus reduced to matrix elements for a symmetric rotor.

The operators $\hat{J}_\xi, \hat{J}_\eta$ have matrix elements only for transitions in which k changes by one, while $\hat{J}_\zeta$ has diagonal elements only (see formulas (27.13), with J, k in place of L, M). Consequently $\hat{J}_\xi^2, \hat{J}_\eta^2, \hat{J}_\zeta^2$, and hence $\hat{H}$, have

matrix elements only for transitions $k \to k, k \pm 2$. The absence of matrix elements for transitions between states with even and odd k causes the secular equation of degree $2J + 1$ to separate at once into two independent equations of degrees J and $J + 1$. One is formed from matrix elements for transitions among states with even k, and the other from those among states with odd k.

Each of these equations can in turn be reduced to two equations of lower degree. For this purpose one must use matrix elements defined not with the functions ψ_{Jk} but with

[19] For a direct derivation of (103.8), without recourse to the theory of finite rotations, see Problem 1 of this section. For the calculation of matrix elements of various quantities with the wave functions (103.8), see § 110, 87 (the corresponding formulas differ from those for a diatomic molecule without spin only in the notation for the quantum numbers; cf. the note on p. 389).

the functions

$$\begin{aligned} \psi^+_{Jk} &= \frac{1}{\sqrt{2}}(\psi_{Jk} + \psi_{J,-k}), \\ \psi^-_{Jk} &= \frac{1}{\sqrt{2}}(\psi_{Jk} - \psi_{J,-k}) \quad (k \neq 0), \\ \psi^+_{J0} &= \psi_{J0}. \end{aligned} \tag{103.12}$$

Functions distinguished by the indices + and − have different symmetry (with respect to reflection in a plane through the ζ axis, which changes the sign of k), and hence matrix elements for transitions between them vanish. Secular equations may therefore be formed separately for the + and − states.

The Hamiltonian (103.1), together with the commutation rules (103.3), has a specific symmetry: it is invariant under simultaneous reversal of the signs of any two of the operators $\hat{J}_\xi, \hat{J}_\eta, \hat{J}_\zeta$. This symmetry formally corresponds to the group D_2. The levels of an asymmetric rotor may therefore be classified by the irreducible representations of this group. There are thus four types of nondegenerate levels, corresponding to the representations A, B_1, B_2, B_3 (see Table 7, p. 460).

It is easy to establish which asymmetric-rotor states belong to each type. To do so, we must determine the symmetry properties of the functions ψ_{Jk} and of the functions (103.12) formed from them. This could be done directly from (103.8). It is simpler, however, to start from the more familiar spherical functions, noting that, in their symmetry properties, wave functions of states with definite angular-momentum projection on the ζ axis coincide with angular-momentum eigenfunctions

$$\psi_{Jk} \sim Y^*_{Jk}(\theta, \varphi) \sim e^{-ik\varphi}\Theta_{Jk}(\theta), \tag{103.13}$$

where θ, φ are spherical angles in the $\xi\eta\zeta$ axes, and the sign ~ here means "transforms as"; the complex conjugation in (103.13) is connected with the reversed sign on the right-hand sides of the commutation relations (103.3).

A rotation through π about the ζ axis (that is, the symmetry operation $C_2^{(\zeta)}$) multiplies the function (103.13) by $(-1)^k$:

$$C_2^{(\zeta)}: \qquad \psi_{Jk} \to (-1)^k \psi_{Jk}.$$

The operation $C_2^{(\eta)}$ may be regarded as a successive inversion and reflection in the $\xi\zeta$ plane; the first multiplies ψ_{Jk} by $(-1)^J$, while the second (reversal of the sign of φ) is equivalent to reversing the sign of k. Taking account of the definition of $\Theta_{J,-k}$ (28.6), we therefore obtain

$$C_2^{(\eta)}: \qquad \psi_{Jk} \to (-1)^{J+k} \psi_{J,-k}.$$

Finally, under $C_2^{(\xi)} = C_2^{(\eta)} C_2^{(\zeta)}$,

$$C_2^{(\xi)}: \qquad \psi_{Jk} \to (-1)^J \psi_{J,-k}.$$

Using these transformation laws, we find that the states corresponding to the functions

(103.12) have the following symmetry types:

$$\psi^+_{Jk}\begin{cases}\text{even } J, \text{ even } k & -A,\\ \text{even } J, \text{ odd } k & -B_3,\\ \text{odd } J, \text{ even } k & -B_1,\\ \text{odd } J, \text{ odd } k & -B_2,\end{cases}\qquad \psi^-_{Jk}\begin{cases}\text{even } J, \text{ even } k & -B_1,\\ \text{even } J, \text{ odd } k & -B_2,\\ \text{odd } J, \text{ even } k & -A,\\ \text{odd } J, \text{ odd } k & -B_3.\end{cases} \tag{103.14}$$

A simple count gives the number of states of each type for a specified J. Specifically, the numbers of states of type A and of each type B_1, B_2, B_3 are

	A	B_1, B_2, B_3
Even J	$J/2+1$	$J/2$
Odd J	$(J-1)/2$	$(J+1)/2$

(103.15)

For an asymmetric rotor there are selection rules for matrix elements governing transitions among states of types A, B_1, B_2, B_3; these follow in the usual way from symmetry. Thus, for the components of a vector physical quantity $\mathbf{A}$, the selection rules are

$$\begin{aligned}&\text{for } A_\xi: \quad A \leftrightarrow B_3^{(\xi)}, \quad B_1^{(\zeta)} \leftrightarrow B_2^{(\eta)},\\ &\text{for } A_\eta: \quad A \leftrightarrow B_2^{(\eta)}, \quad B_1^{(\zeta)} \leftrightarrow B_3^{(\xi)},\\ &\text{for } A_\zeta: \quad A \leftrightarrow B_1^{(\zeta)}, \quad B_2^{(\eta)} \leftrightarrow B_3^{(\xi)}\end{aligned} \tag{103.16}$$

(for clarity, an index on the representation symbol indicates the axis about which a rotation has character +1 in that representation).

Problems

Problem

1. Find the wave functions of the states $|JMk\rangle$ of a symmetric rotor by direct calculation as common eigenfunctions of the operators $\hat{\mathbf{J}}^2, \hat{J}_z, \hat{J}_\zeta$ (F. Reiche, H. Rademacher, 1926).

Solution. . To obtain ψ_{JMk} as a function of the Euler angles α, β, γ, the operators of the angular-momentum projections on the fixed axes x, y, z must be expressed in terms of these angles. Since the operator of the projection on any axis is $-i\partial/\partial\varphi$, where φ is the angle of rotation about that axis, we may write

$$\hat{J}_x = -i\frac{\partial}{\partial\varphi_x}, \qquad \hat{J}_y = -i\frac{\partial}{\partial\varphi_y}, \qquad \hat{J}_z = -i\frac{\partial}{\partial\varphi_z},$$

where $\varphi_x, \varphi_y, \varphi_z$ are the angles of rotation about the corresponding axes. Derivatives with respect to these angles can be expressed in terms of derivatives with respect to α, β, γ by recalling that infinitesimal rotations add as vectors (directed along the axes of rotation). The directions of the vectors $\delta\alpha, \delta\beta, \delta\gamma$ of the infinitesimal rotations described by the Euler angles are shown in Fig. 20 (p. 270). Projecting them on the fixed axes xyz, we find the rotation angles about these axes in the form

$$\begin{aligned}
\delta\varphi_x &= -\sin\alpha\,\delta\beta + \cos\alpha\sin\beta\,\delta\gamma,\\
\delta\varphi_y &= \cos\alpha\,\delta\beta + \sin\alpha\sin\beta\,\delta\gamma,\\
\delta\varphi_z &= \delta\alpha + \cos\beta\,\delta\gamma.
\end{aligned}$$

Conversely,

$$\begin{aligned}
\delta\alpha &= -\cot\beta\cos\alpha\,\delta\varphi_x - \cot\beta\sin\alpha\,\delta\varphi_y + \delta\varphi_z,\\
\delta\beta &= -\sin\alpha\,\delta\varphi_x + \cos\alpha\,\delta\varphi_y,\\
\delta\gamma &= \frac{\cos\alpha}{\sin\beta}\delta\varphi_x + \frac{\sin\alpha}{\sin\beta}\delta\varphi_y.
\end{aligned}$$

These expressions give

$$\begin{aligned}
\hat{J}_x &= -i\left(-\cos\alpha\cot\beta\frac{\partial}{\partial\alpha} - \sin\alpha\frac{\partial}{\partial\beta} + \frac{\cos\alpha}{\sin\beta}\frac{\partial}{\partial\gamma}\right),\\
\hat{J}_y &= -i\left(-\sin\alpha\cot\beta\frac{\partial}{\partial\alpha} + \cos\alpha\frac{\partial}{\partial\beta} + \frac{\sin\alpha}{\sin\beta}\frac{\partial}{\partial\gamma}\right),\\
\hat{J}_z &= -i\frac{\partial}{\partial\alpha}.
\end{aligned}$$

When acting on ψ_{JMk}, the operators $\hat{J}_z = -i\partial/\partial\alpha$ and $\hat{J}_\zeta = -i\partial/\partial\gamma$ (γ is the angle of rotation about the ζ axis!) may be replaced by M and k (the corresponding dependence of the wave function on the Euler angles α and γ is given by the factor $\exp(i\alpha M + i\gamma k)$). We then have

$$\begin{aligned}
\hat{J}_+ &= \hat{J}_x + i\hat{J}_y = e^{i\alpha}\left(\frac{\partial}{\partial\beta} - M\cot\beta + \frac{k}{\sin\beta}\right),\\
\hat{J}_- &= \hat{J}_x - i\hat{J}_y = e^{-i\alpha}\left(-\frac{\partial}{\partial\beta} - M\cot\beta + \frac{k}{\sin\beta}\right).
\end{aligned}$$

The rest of the derivation is exactly analogous to that given at the end of § 28. We begin with $\hat{J}_+\psi_{JJk} = 0$, which holds for the wave function with $M = J$. Hence

$$\left(\frac{\partial}{\partial\beta} - J\cot\beta + \frac{k}{\sin\beta}\right)\psi_{JJk} = 0.$$

The normalized solution of this equation is

$$\psi_{JJk} = i^J(-1)^{J-k}\left[\frac{(2J+1)!}{2(J+k)!(J-k)!}\right]^{1/2}\left(\cos\frac{\beta}{2}\right)^{J+k}\left(\sin\frac{\beta}{2}\right)^{J-k}\frac{e^{i(J\alpha+k\gamma)}}{2\pi}$$

(the normalization integral reduces to Euler's beta integral). This expression does indeed coincide, apart from a phase factor, with the function

$$\sqrt{\frac{2J+1}{8\pi^2}}D^{(J)}_{kJ}(\alpha,\beta,\gamma)$$

(cf. (58.26)); the phase factor has been chosen in accordance with the definition in (103.7).

The wave functions with $M < J$ are then calculated by repeated application to ψ_{JJk} of the formula

$$\hat{J}_{-}\psi_{J,M+1,k} = \sqrt{(J-M)(J+M+1)}\psi_{JMk}.$$

The final result coincides with (103.8), where the functions $D^{(J)}_{kM}$ are given by formulas (58.9)–(58.11), with the symmetry property (58.18) also taken into account.

Problem

2. Calculate the matrix elements $\langle Jk'|H|Jk\rangle$ for an asymmetric rotor.

Solution. . From formulas (27.13) we find

$$\langle k|J_\xi^2|k\rangle = \langle k|J_\eta^2|k\rangle = \frac{1}{2}[J(J+1)-k^2],$$

$$\langle k|J_\xi^2|k+2\rangle = \langle k+2|J_\xi^2|k\rangle = -\langle k|J_\eta^2|k+2\rangle = -\langle k+2|J_\eta^2|k\rangle$$
$$= \frac{1}{4}\sqrt{(J-k)(J-k-1)(J+k+1)(J+k+2)}$$

(the diagonal indices J, J on the matrix elements are omitted throughout for brevity). Hence the required matrix elements

of the Hamiltonian are[20]

$$\langle k|H|k\rangle = \frac{\hbar^2}{4}(a+b)[J(J+1)-k^2] + \frac{\hbar^2}{2}ck^2,$$
$$\langle k|H|k+2\rangle = \langle k+2|H|k\rangle = \frac{\hbar^2}{8}(a-b)\sqrt{(J-k)(J-k-1)(J+k+1)(J+k+2)}. \tag{1}$$

The matrix elements with respect to the functions (103.12) are expressed in terms of the elements (1) by

$$\begin{aligned}
\langle k\pm|H|k\pm\rangle &= \langle k|H|k\rangle, \quad k \neq 1,\\
\langle 1\pm|H|1\pm\rangle &= \langle 1|H|1\rangle \pm \langle 1|H|-1\rangle,\\
\langle k\pm|H|k+2,\pm\rangle &= \langle k|H|k+2\rangle, \quad k\neq 0,\\
\langle 0+|H|2+\rangle &= \sqrt{2}\langle 0|H|2\rangle.
\end{aligned} \tag{2}$$

[20] In Problems 2–5, to simplify the notation in the formulas, we use $a = 1/I_A$, $b = 1/I_B$, $c = 1/I_C$.

Problem

3. Determine the energy levels of an asymmetric rotor for $J = 1$.

Solution. . The secular equation of third degree separates into three equations of first degree. One of them gives

$$E_1 = \langle 0+|H|0+\rangle = \frac{\hbar^2}{2}(a+b). \tag{3}$$

The other two energy levels can be written down at once, since it is evident beforehand that the three parameters a, b, c enter the problem symmetrically. Thus

$$E_2 = \frac{\hbar^2}{2}(a+c), \qquad E_3 = \frac{\hbar^2}{2}(b+c). \tag{4}$$

The levels E_1, E_2, E_3 belong respectively to the symmetry types B_1, B_2, B_3[21]. The wave functions of these states are

$$\psi_1 = \psi_{10}^+, \qquad \psi_2 = \psi_{11}^-, \qquad \psi_3 = \psi_{11}^+.$$

Problem

4. Do the same for $J = 2$.

Solution. . The secular equation of fifth degree separates into three equations of first degree and one of second degree. One of the first-degree equations gives

$$E_1 = \langle 2-|H|2-\rangle = 2\hbar^2 c + \frac{\hbar^2}{2}(a+b) \tag{5}$$

(a level of type B_1). We infer that there must be two further levels (of types B_2 and B_3):

$$E_2 = 2\hbar^2 b + \frac{\hbar^2}{2}(a+c), \qquad E_3 = 2\hbar^2 a + \frac{\hbar^2}{2}(b+c).$$

The wave functions corresponding to these three levels are

$$\psi_1 = \psi_{22}^-, \qquad \psi_2 = \psi_{21}^-, \qquad \psi_3 = \psi_{21}^+.$$

The equation of second degree is

$$\begin{vmatrix} \langle 0+|H|0+\rangle - E & \langle 2+|H|0+\rangle \\ \langle 2+|H|0+\rangle & \langle 2+|H|2+\rangle - E \end{vmatrix} = 0. \tag{6}$$

Solving it, we obtain

$$E_{4,5} = \hbar^2(a+b+c) \pm \hbar^2[(a+b+c)^2 - 3(ab+bc+ac)]^{1/2}. \tag{7}$$

These levels are of type A. Their wave functions are linear combinations of ψ_{20}^+ and ψ_{22}^+.

[21] This follows directly from symmetry considerations. For example, the energy E_1 is symmetric in the parameters a and b; this must be so for the energy of a state whose symmetry with respect to the ξ and η axes is the same (a state of type B_1).

PROBLEM

5. Do the same for $J = 3$.

SOLUTION. . The secular equation of seventh degree separates into one equation of first degree and three of second degree. The first-degree equation gives

$$E_1 = \langle 2-|H|2-\rangle = 2\hbar^2(a+b+c) \tag{8}$$

(a level of type A). One of the second-degree equations is equation (6) of the preceding problem (with a different value of J). Its roots are

$$E_{2,3} = \frac{5\hbar^2}{2}(a+b) + \hbar^2 c \pm \hbar^2[4(a-b)^2 + c^2 + ab - ac - bc]^{1/2} \tag{9}$$

(levels of type B_1). The remaining levels are obtained from these by permuting the parameters a, b, c.

PROBLEM

6. Determine the splitting of the levels of a system possessing a quadrupole moment in an arbitrary external electric field.

SOLUTION. . Choosing the principal axes of the tensor $\partial^2\varphi/\partial x_i \partial x_k$ as coordinate axes (cf. Problem 3 of § 76), we reduce the quadrupole part of the Hamiltonian of the system to the form

$$\hat{H} = A\hat{J}_x^2 + B\hat{J}_y^2 + C\hat{J}_z^2, \qquad A + B + C = 0.$$

Because this expression is completely analogous in form to the Hamiltonian (103.1), the stated problem is equivalent to finding the energy levels of an asymmetric rotor, except that the sum of the coefficients is now $A + B + C = 0$ and the angular momentum may also take half-integral values. For the latter, the calculations must be carried out anew by the same method; for integral J, the results of Problems 3–5 may be used. We finally obtain the following energy shifts ΔE for the first few values of J:

$$\begin{aligned} J = 1: &\qquad \Delta E = -A, -B, -C, \\ J = 3/2: &\qquad \Delta E = \pm\sqrt{(3/2)(A^2+B^2+C^2)}, \\ J = 2: &\qquad \Delta E = 3A, 3B, 3C, \ \pm\sqrt{6(A^2+B^2+C^2)}. \end{aligned}$$

For $J = 3/2$ the energy levels remain doubly degenerate, in accordance with Kramers' theorem (§ 60).

§ 104. Interaction between molecular vibration and rotation

Up to this point, rotation and vibration have been treated as independent motions of a molecule. In fact, when both occur simultaneously, a distinctive interaction arises between them (E. Teller, L. Tisza, O. Placzek, 1932–1933).

We begin with linear polyatomic molecules. A linear molecule can undergo two kinds of vibration (see the end of § 100): longitudinal vibrations with nondegenerate frequencies, and transverse vibrations with doubly degenerate frequencies. It is the latter that concern us here.

A molecule undergoing transverse vibrations generally has angular momentum. This follows at once from elementary mechanical considerations[22], but it can also be demonstrated by a quantum-mechanical treatment. Such a treatment also determines the possible values of this momentum in a specified vibrational state.

Suppose that one particular doubly degenerate frequency ω_α is excited in the molecule. The energy level with vibrational quantum number v_α has degeneracy $v_\alpha + 1$. It is represented by $v_\alpha + 1$ wave functions

$$\psi_{v_{\alpha1}v_{\alpha2}} = \text{const} \cdot \exp\left[-\frac{1}{2}c_\alpha^2(Q_{\alpha1}^2 + Q_{\alpha2}^2)\right] H_{v_{\alpha1}}(c_\alpha Q_{\alpha1}) H_{v_{\alpha2}}(c_\alpha Q_{\alpha2})$$

(where $v_{\alpha1} + v_{\alpha2} = v_\alpha$), or by any independent linear combinations of them. In every one of these functions, the polynomial multiplying the exponential has the same highest total degree in $Q_{\alpha1}$ and $Q_{\alpha2}$, namely v_α. Linear combinations of the functions $\psi_{v_{\alpha1}v_{\alpha2}}$ can evidently always be chosen as basis functions in the form

$$\begin{aligned} \psi_{v_\alpha l_\alpha} = {} & \text{const} \cdot \exp\left[-\frac{1}{2}c_\alpha^2(Q_{\alpha1}^2 + Q_{\alpha2}^2)\right] \times \\ & \times \left[(Q_{\alpha1} + iQ_{\alpha2})^{(v_\alpha + l_\alpha)/2}(Q_{\alpha1} - iQ_{\alpha2})^{(v_\alpha - l_\alpha)/2} + \ldots\right]. \end{aligned} \tag{104.1}$$

The square brackets contain a definite polynomial of which only the highest-degree term has been displayed; l_α is an integer having the $v_\alpha + 1$ possible values $l_\alpha = v_\alpha$, $v_\alpha - 2$, $v_\alpha - 4, \ldots, -v_\alpha$.

The normal coordinates $Q_{\alpha1}$ and $Q_{\alpha2}$ of a transverse vibration describe two mutually perpendicular displacements from the molecular axis. Under a rotation through an angle φ about this axis, the highest-degree polynomial term, and hence the whole function $\psi_{v_\alpha l_\alpha}$, is multiplied by

$$\exp\left\{i\varphi\left(\frac{v_\alpha + l_\alpha}{2}\right) - i\varphi\left(\frac{v_\alpha - l_\alpha}{2}\right)\right\} = \exp(il_\alpha\varphi).$$

It follows that the function (104.1) describes a state whose angular momentum about the axis is l_α.

We thus obtain the following result. In a state in which a doubly degenerate frequency ω_α is excited with quantum number v_α, the molecule has an angular momentum about its own axis that takes the values

$$l_\alpha = v_\alpha, v_\alpha - 2, v_\alpha - 4, \ldots, -v_\alpha. \tag{104.2}$$

This is called the molecule's *vibrational angular momentum*. If several transverse vibrations are excited at once, the total vibrational angular momentum is $\sum l_\alpha$. Together with the electronic orbital angular momentum, it gives the total angular momentum l of the molecule about its axis.

[22]For example, two mutually perpendicular transverse vibrations whose phases differ by $\pi/2$ may be regarded as a pure rotation of the bent molecule about its longitudinal axis.

As in a diatomic molecule, the total molecular angular momentum J cannot be smaller than the angular momentum about the axis. Thus J assumes the values

$$J = |l|, |l| + 1, \ldots$$

In other words, states with $J = 0, 1, \ldots, |l| - 1$ do not exist.

For harmonic vibrations the energy depends only on the numbers v_α, not on l_α. Anharmonicity removes the degeneracy of the vibrational levels with respect to l_α, though only partially: the levels remain doubly degenerate, states related by simultaneous reversal of the signs of every l_α and of l having equal energy. In the approximation following the harmonic one, the energy acquires a term quadratic in the momenta l_α, of the form

$$\sum_{\alpha,\beta} g_{\alpha\beta} l_\alpha l_\beta$$

(the $g_{\alpha\beta}$ are constants). The remaining twofold degeneracy is removed by an effect analogous to Λ-doubling in diatomic molecules.

Before turning to nonlinear molecules, we must make a purely mechanical observation. For an arbitrary nonlinear system of particles, one must ask how vibrational motion can be separated from rotation at all, or, equivalently, what is meant by a "nonrotating system." At first sight, vanishing angular momentum might appear to provide the criterion for the absence of rotation:

$$\sum m(\mathbf{r} \times \mathbf{v}) = 0 \tag{104.3}$$

(the sum is over the particles of the system). The expression on the left, however, is not the total time derivative of any function of the coordinates. Consequently, this equality cannot be integrated with respect to time and recast as the vanishing of some coordinate function. Yet just such a formulation is needed if the concepts of "pure vibration" and "pure rotation" are to be defined meaningfully.

The condition to be adopted as the definition of no rotation is therefore

$$\sum m(\mathbf{r}_0 \times \mathbf{v}) = 0, \tag{104.4}$$

where $\mathbf{r}_0$ is the radius vector of the equilibrium position of a particle. Writing $\mathbf{r} = \mathbf{r}_0 + \mathbf{u}$, with $\mathbf{u}$ the displacement in a small vibration, gives $\mathbf{v} = \dot{\mathbf{r}} = \dot{\mathbf{u}}$. Integration of (104.4) with respect to time then yields

$$\sum m(\mathbf{r}_0 \times \mathbf{u}) = 0. \tag{104.5}$$

We shall regard molecular motion as the combination of a purely vibrational motion satisfying (104.5) and a rotation of the molecule as a whole[23]. On writing the angular momentum as

$$\sum m(\mathbf{r} \times \mathbf{v}) = \sum m(\mathbf{r}_0 \times \mathbf{v}) + \sum m(\mathbf{u} \times \mathbf{v}),$$

we see from the definition (104.4) of no rotation that the sum $\sum m(\mathbf{u} \times \mathbf{v})$ is to be understood as the vibrational angular momentum. It must be remembered, however, that this momentum is only one part of the total momentum of the system and is not itself

[23]Translational motion is assumed to have been separated from the outset by choosing a coordinate system in which the molecular center of mass is at rest.

conserved at all. Thus only a mean value of the vibrational angular momentum can be assigned to each vibrational state.

Molecules having no symmetry axis of order higher than two are of the asymmetric-top type. All vibrational frequencies of molecules of this kind are nondegenerate, since their symmetry groups have only one-dimensional irreducible representations. Hence every vibrational level is nondegenerate. In any nondegenerate state, however, the mean angular momentum vanishes (see § 26). Asymmetric-top molecules therefore have zero mean vibrational angular momentum in every state.

If the symmetry elements of a molecule include one axis of order higher than two, the molecule is of the symmetric-top type. Such a molecule has both nondegenerate and doubly degenerate vibrational frequencies. The mean vibrational angular momentum again vanishes for the former. A doubly degenerate frequency, on the other hand, corresponds to a nonzero mean projection of angular momentum on the molecular axis.

The energy of rotational motion of a symmetric-top molecule can readily be found with the vibrational angular momentum taken into account. Its operator differs from (103.5) by replacing the top's rotational angular momentum by the difference between the molecule's total conserved angular momentum $\mathbf{J}$ and its vibrational angular momentum $\mathbf{J}^{(v)}$:

$$\widehat{H}_{\text{rot}} = \frac{\hbar^2}{2I_A}(\widehat{\mathbf{J}} - \widehat{\mathbf{J}}^{(v)})^2 + \frac{\hbar^2}{2}\left(\frac{1}{I_C} - \frac{1}{I_A}\right)(\widehat{J}_\zeta - \widehat{J}_\zeta^{(v)})^2. \tag{104.6}$$

The energy sought is the mean value $\overline{H}_{\text{rot}}$. Terms in (104.6) containing squares of components of $\mathbf{J}$ give the purely rotational energy (103.6). Those containing squares of components of $\mathbf{J}^{(v)}$ give constants independent of the rotational quantum numbers and may be omitted. The terms containing products of components of $\mathbf{J}$ and $\mathbf{J}^{(v)}$ represent the effect of the interaction between molecular vibration and rotation that is of interest here. It is called the *Coriolis interaction*, in view of its correspondence to the Coriolis forces of classical mechanics. In averaging these terms, the mean values of the transverse (ξ, η) components of the vibrational angular momentum must be taken to vanish. The resulting mean Coriolis-interaction energy is

$$E_{\text{Cor}} = -\frac{\hbar^2}{I_C} k k_v, \tag{104.7}$$

where the integer k, as in § 103, is the projection of the total angular momentum on the molecular axis, while $k_v = \overline{J_\zeta^{(v)}}$ is the mean projection of the vibrational angular momentum characterizing the given vibrational state. Unlike k, k_v need not be an integer at all.

Finally, consider spherical-top molecules. These are molecules possessing the symmetry of one of the cubic groups. They have nondegenerate, doubly degenerate, and triply degenerate frequencies, corresponding to the occurrence of one-, two-, and three-dimensional irreducible representations among those of the cubic groups. As always, anharmonicity partially removes the degeneracy of the vibrational levels;

after these effects have been allowed for, only nondegenerate, doubly degenerate, and triply degenerate levels remain. We shall discuss precisely these levels after their splitting by anharmonicity.

For a spherical-top molecule, the mean vibrational angular momentum plainly vanishes not only in nondegenerate vibrational states but also in doubly degenerate ones. This follows directly from elementary symmetry considerations. Indeed, the vectors of the mean momenta in the two states belonging to one degenerate energy level would have to transform into each other under every symmetry transformation of the molecule. No cubic symmetry group, however, permits two directions that transform only into one another; only sets containing at least three directions are transformed among themselves.

The same argument shows that the mean vibrational angular momentum is nonzero in states belonging to triply degenerate vibrational levels. After averaging over the vibrational state, this momentum is represented by an operator whose matrix elements correspond to transitions among the three mutually degenerate states. In accordance with their number, this operator must have the form $\zeta\widehat{\mathbf{I}}$, where $\widehat{\mathbf{I}}$ is the operator of an angular momentum equal to unity (for which $2I + 1 = 3$), and ζ is a constant characteristic of the given vibrational level. After this averaging, the Hamiltonian of the rotational motion,

$$\widehat{H}_{\text{rot}} = \frac{\hbar^2}{2I}(\widehat{\mathbf{J}} - \widehat{\mathbf{J}}^{(v)})^2,$$

becomes the operator

$$\widehat{H}_{\text{rot}} = \frac{\hbar^2}{2I}\widehat{\mathbf{J}}^2 + \frac{\hbar^2}{2I}\overline{\widehat{\mathbf{J}}^{(v)2}} - \frac{\hbar^2}{I}\zeta\widehat{\mathbf{J}}\widehat{\mathbf{I}}. \tag{104.8}$$

The eigenvalues of the first term give the usual rotational energy (103.4), while the second term contributes an inessential constant independent of the rotational quantum number. The last term in (104.8) gives the required Coriolis splitting energy of the vibrational level. The eigenvalues of $\mathbf{JI}$ are obtained in the usual way. For a specified J, it can have three distinct values, corresponding to values $J+1$, $J-1$, and J of the vector $\mathbf{I}+\mathbf{J}$:

$$E_{\text{Cor}}^{(J+1)} = -\frac{\hbar^2}{I}\zeta J, \qquad E_{\text{Cor}}^{(J-1)} = \frac{\hbar^2}{I}\zeta(J+1), \qquad E_{\text{Cor}}^{(J)} = \frac{\hbar^2}{I}\zeta. \tag{104.9}$$

§ 105. Classification of molecular terms

The wave function of a molecule is the product of the electronic wave function, the wave function of the vibrational motion of the nuclei, and the rotational wave function. We have already discussed separately the classification and symmetry types of these functions. It remains to consider the classification of molecular terms as a whole, that is, the possible symmetry of the complete wave function.

Clearly, specifying the symmetry of all three factors under particular transformations also determines the symmetry of their product under those transformations. A complete characterization of the symmetry of a state must further specify the behavior of the complete wave function under simultaneous inversion of the coordinates of all particles (electrons and nuclei) in the molecule. A state is called *negative* or *positive* according as its wave function changes sign or remains unchanged under this transformation[24].

[24] As is customary, we use the same infelicitous terminology as for diatomic molecules (see § 86).

It must be borne in mind, however, that characterizing a state with respect to inversion is meaningful only for molecules without stereoisomers. Stereoisomerism means that inversion brings a molecule into a configuration that cannot be superposed on the original one by any rotation in space (the "right" and "left" modifications of a substance)[25]. Thus, in the presence of stereoisomerism, wave functions obtained from one another by inversion belong essentially to different molecules and there is no point in comparing them[26].

We saw in § 86 that in diatomic molecules nuclear spin has an important indirect influence on the scheme of molecular terms: it determines their degeneracies and, in some cases, altogether forbids levels of a particular symmetry. The same occurs in polyatomic molecules. Here, however, the analysis is considerably more complicated and requires group-theoretical methods in each particular case.

The idea of the method is as follows. In addition to the coordinate part

(the only part considered so far), the complete wave function must also contain a spin factor, which is a function of the projections of the spins of all nuclei on some chosen direction in space. The projection σ of a nuclear spin takes $2i + 1$ values (i is the nuclear spin); on assigning all possible values to all $\sigma_1, \sigma_2, \ldots, \sigma_N$ (N is the number of atoms in the molecule), we obtain a total of $(2i_1 + 1)(2i_2 + 1) \ldots (2i_N + 1)$ different values of the spin factor. Under each symmetry transformation, certain nuclei (of the same kind) exchange places. If the spin values are imagined to "remain in place," the transformation is equivalent to permuting the spin values among the nuclei. The different spin factors are correspondingly transformed into one another and thus realize a certain representation (in general reducible) of the molecular symmetry group. By decomposing it into irreducible parts, we find the possible symmetry types of the spin wave function.

A general formula for the characters $\chi_{\text{sp}}(G)$ of the representation realized by the spin factors is readily written down. It is enough to observe that only those spin factors for which the nuclei exchanging places have equal σ_a remain unchanged by the transformation; otherwise one spin factor passes into another and contributes nothing to the character. Since σ_a takes $2i_a + 1$ values, we find

$$\chi_{\text{sp}}(G) = \prod (2i_a + 1), \tag{105.1}$$

where the product is over the groups of atoms that exchange places with one another under the given transformation G (one factor in the product for each group).

What interests us, however, is not so much the symmetry of the spin function as that of the coordinate wave function (symmetry under permutations of the nuclear coordinates while the electron coordinates remain fixed). These symmetries are directly related, since the complete wave function must remain unchanged or change sign when each pair of nuclei obeying Bose or Fermi statistics, respectively, is interchanged (in other words, it must be multiplied by $(-1)^{2i}$, where i is the spin of the nuclei being interchanged). Introducing the corresponding factor into the characters (105.1), we obtain the set

[25] For stereoisomerism to be possible, the molecule must possess no symmetry element involving reflection (an inversion center, a symmetry plane, or a rotation–reflection axis).

[26] Strictly speaking, quantum mechanics always gives a nonzero probability of transition from one modification to the other. This probability, however, which is associated with passage of the nuclei through a barrier, is extremely small.

of characters $\chi(G)$ of a representation containing all the irreducible representations according to which the coordinate wave functions transform:

$$\chi(G) = \prod (2i_a + 1)(-1)^{2i_a(n_a-1)}. \tag{105.2}$$

Here n_a is the number of nuclei in each group that exchange places with one another under the given transformation. Decomposition of this representation into irreducible parts gives the possible symmetry types of the molecular coordinate wave functions, together with the degeneracies of the corresponding energy levels (here and below, degeneracy with respect to the different spin states of the nuclear system is meant)[27].

Each symmetry type of states is associated with definite values of the total spins of groups of equivalent nuclei in the molecule (that is, groups of nuclei interchanged by some symmetry transformations of the molecule). The correspondence is not one-to-one: in general, each symmetry type of states may occur with different values of the spins of groups of equivalent nuclei. This correspondence can likewise be established in each particular case by group-theoretical methods.

As an example, consider an asymmetric-rotor molecule, ethylene $^{12}C_2{}^1H_4$ (Fig. 43g, symmetry group D_{2h}). The superscript on a chemical symbol indicates the isotope to which the nucleus belongs; this specification is necessary because nuclei of different isotopes may have different spins. Here the spin of a ^1H nucleus is one half, while the ^{12}C nucleus has zero spin. We therefore need consider only the hydrogen atoms.

Choose the coordinate system as shown in Fig. 43g (the z axis is perpendicular to the molecular plane and the x axis is directed along its axis). Reflection in the plane $\sigma(xy)$ leaves all atoms in place, while the other reflections and rotations interchange the hydrogen atoms pairwise. Formula (105.2) gives the following characters of the representation:

E	$\sigma(xy)$	$\sigma(xz)$	$\sigma(yz)$	I	$C_2(z)$	$C_2(y)$	$C_2(x)$
16	16	4	4	4	4	4	4

Decomposing this representation into irreducible parts, we find that it contains the following irreducible representations of the group D_{2h}: $7A_g$, $3B_{1g}$, $3B_{2u}$, $3B_{3u}$. The number indicates the multiplicity with which the given irreducible representation occurs in the reducible one; these numbers are precisely the nuclear statistical weights of levels of the corresponding symmetry[28].

The classification obtained for the states of the ethylene molecule concerns the symmetry of the complete (coordinate) wave function, containing electronic, vibrational, and rotational parts. Usually, however, it is useful to approach these results from another direction. Namely, once the possible symmetries of the complete wave function are known, one can directly find which rotational levels are possible (and with which statistical weights) for a specified electronic and vibrational state.

Consider, for example, the rotational structure of the lowest vibrational level (with vibrations unexcited) of the normal electronic term, assuming the electronic wave function

[27] In this connection the degeneracy of a level is often called its nuclear statistical weight (cf. the note on p. 407).

[28] For the relation between the symmetry of states and the values of the total spin of the four H nuclei in the ethylene molecule, see Problem 1.

of the normal state to be totally symmetric (as is true in practice for nearly all polyatomic molecules). The symmetry of the complete wave function under rotations about symmetry axes then coincides with that of the rotational wave function. Comparison with the results above therefore shows that in ethylene the rotational levels of types A and B_1 (see § 103) are positive and have statistical weights 7 and 3, whereas levels of types B_2 and B_3 are negative and have statistical weight 3.

As in diatomic molecules (see the end of § 86), transitions between ethylene states of different nuclear symmetry practically do not occur, owing to the extreme weakness of the interaction of the nuclear spins with the electrons. Molecules in these states therefore behave as distinct modifications of the substance, so that ethylene $^{12}C_2{}^1H_4$ has four modifications with nuclear statistical weights 7, 3, 3, 3. Essential to this conclusion is that states of different symmetry belong to different energy levels (whose separations are large compared with the nuclear-spin interaction energy). It is therefore not valid for molecules having states of different nuclear symmetry that belong to the same degenerate energy level.

Consider the symmetric-rotor ammonia molecule $^{14}N^1H_3$ (Fig. 41, symmetry group C_{3v}). The spin of the ^{14}N nucleus is 1, and that of ^1H is one half. Formula (105.2) gives the characters of the representation of C_{3v} that interests us:

E	$2C_3$	$3\sigma_v$
24	6	-12

It contains the following irreducible representations of the group

C_{3v}: $12A_2, 6E$. Thus two types of levels are possible; their nuclear statistical weights are 12 and 6[29].

The rotational levels of a symmetric rotor are classified (for a given J) by the values of the quantum number k. As in the preceding example, consider the rotational structure of the normal electronic and vibrational states of the NH_3 molecule (that is, assume the electronic and vibrational wave functions to be totally symmetric). In determining the symmetry of the rotational wave function, one must remember that its behavior is meaningful only with respect to rotations about axes. We therefore replace the symmetry planes by second-order symmetry axes perpendicular to them (reflection in a plane is equivalent to rotation about such an axis followed by inversion). Thus, in place of C_{3v}, we must here consider the isomorphic point group D_3.

Under a rotation C_3 about the vertical third-order axis, the rotational wave functions with $k = \pm|k|$ are multiplied by $e^{\pm 2\pi i|k|/3}$; under a rotation U_2 about a horizontal second-order axis, they pass into one another, and hence realize a two-dimensional representation of D_3. When $|k|$ is not a multiple of three, this representation is irreducible: it is the representation E. The representation of C_{3v} corresponding to the complete wave function is obtained by multiplying the character $\chi(U_2)$ by $+1$ or -1 according as the term is positive or negative. But since $\chi(U_2) = 0$ in the representation E, in either case we again obtain the same representation E (now as a representation of C_{3v} rather than

[29]Terms of symmetry A_2 correspond to total spin $3/2$ of the hydrogen nuclei, and terms E to spin $1/2$. The occurrence of the two-dimensional representation E among the irreducible representations does not imply an additional degeneracy of the molecular energy levels. This is an instance of permutation degeneracy, discussed in § 63.

D_3). In view of the results above, we therefore conclude that when $|k|$ is not a multiple of three, both positive and negative levels with nuclear statistical weight 6 are possible (levels for which the complete coordinate wave function has symmetry type E). When $|k|$ is a nonzero multiple of three, the rotational functions realize a representation (of D_3) with characters

$$\begin{array}{c|ccc} & E & 2C_3 & 3U_2 \\ \hline & 2 & 2 & 0 \end{array}.$$

This representation is reducible and decomposes into A_1 and A_2. For the complete wave function to belong to the representation A_2 of C_{3v}, the rotational level A_1 must be negative and A_2 positive. Thus, for nonzero $|k|$ divisible by three, both positive and negative levels with nuclear statistical weight 12 are possible (levels of type A_2).

For the projection $k = 0$ there is just one rotational function, realizing a representation with characters[30]

$$\begin{array}{c|ccc} & E & 2C_3 & 3U_2 \\ \hline & 1 & 1 & (-1)^J \end{array}.$$

For the complete wave function to have symmetry A_2, its behavior under inversion must consequently be determined by the factor $-(-1)^J$. Thus, at $k = 0$, levels with even (odd) J can only be negative (positive); in both cases the statistical weight is 12 (levels of type A_2).

Collecting these results, we obtain the following table of possible states at different values of the quantum number k for the normal electronic and vibrational term of the molecule $^{14}N^{1}H_3$ (+ and – denote positive and negative states):

	(+)	(−)
$\lvert k\rvert$ not divisible by three	$6E$	$6E$
$\lvert k\rvert$ divisible by three	$12A_2$	$12A_2$
$k = 0,\ J$ even	–	$12A_2$
$k = 0,\ J$ odd	$12A_2$	–

For given J and k, the energy levels of the NH_3 molecule are in general degenerate (see also the table for ND_3 in Problem 3). This degeneracy is partially removed by a distinctive effect associated with the flattened shape of the ammonia molecule and the small mass of the hydrogen atoms. A comparatively small vertical displacement of the atoms in this molecule can produce a transition between two configurations related by reflection in a plane parallel to the base of the pyramid (Fig. 44). These transitions split the levels,

separating the positive and negative levels (an effect analogous to the one-dimensional case considered in Problem 3 of § 50).

The magnitude of the splitting is proportional to the probability for the atoms to pass through the "potential barrier" separating the two molecular configurations. Although this probability is comparatively large in ammonia because of the properties just described, the splitting is nevertheless small ($1 \cdot 10^{-4}$ eV). An example of a spherical-rotor molecule is treated in Problem 5 of this section.

[30] Under a rotation through an angle π, an angular-momentum eigenfunction with magnitude J and zero projection is multiplied by $(-1)^J$.

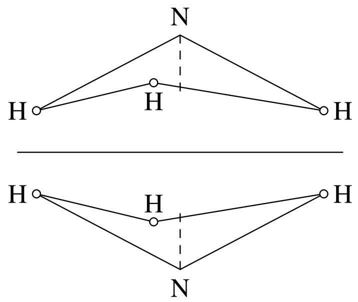

Figure 44

Problems

PROBLEM

1. Establish the relation between the symmetry of states of the molecule $^{12}C_2{}^1H_4$ and the total spin of its hydrogen nuclei.

SOLUTION.[31] The total spin of the four ^{1}H nuclei can have the values $I = 2, 1, 0$, and its projection M_I ranges from 2 to -2. Consider the representations realized by the spin factors belonging to each individual value of M_I, beginning with the maximum value.

For $M_I = 2$ there is only one spin factor, in which all nuclei have spin projection $+1/2$. For $M_I = 1$ there are four different spin factors, distinguished by which of the four nuclei is assigned spin projection $-1/2$. Finally, $M_I = 0$ is realized by six spin factors, according to which pair of nuclei is assigned spin projection $-1/2$. The characters of the corresponding three representations are

	E	$\sigma(xy)$	$\sigma(xz)$	$\sigma(yz)$	I	$C_2(z)$	$C_2(y)$	$C_2(x)$
$M_I = 2$	1	1	1	1	1	1	1	1
$M_I = 1$	4	4	0	0	0	0	0	0
$M_I = 0$	6	6	2	2	2	2	2	2

The first is the identity representation A_g. Since $M_I = 2$ can occur only for $I = 2$, it follows that spin $I = 2$ corresponds to a state of symmetry A_g.

The value $M_I = 1$ can occur for both $I = 1$ and $I = 2$. Accordingly subtracting the first representation from the second and decomposing the result into irreducible parts, we find that spin $I = 1$ corresponds to the states B_{1g}, B_{2u}, B_{3u}.

Finally, $M_I = 0$ can occur in all the cases in which $M_I = 1$ is possible, and also for $I = 0$. Accordingly subtracting the second representation from the third, we find two states A_g corresponding to spin $I = 0$.

[31] For the method of solving such problems based on permutation-group theory, see the book by I. G. Kaplan cited on p. 291, Ch. VI, § 2.

Problem

2. Determine the symmetry types of the complete (coordinate) wave functions and the statistical weights of the corresponding levels for the molecules

$^{12}C_2{}^2H_4$, $^{13}C_2{}^1H_4$, $^{14}N_2{}^{16}O_4$ (all the molecules have the same shape); the spins are $i(^2H) = 1, i(^{13}C) = 1/2, i(^{14}N) = 1$.

SOLUTION. By the same method as was applied in the text to $^{12}C_2{}^1H_4$, we find the following states (the coordinate axes are chosen as in the text):

Molecule	(+)	(−)
$^{12}C_2{}^2H_4$	$27A_g,\ 18B_{1g}$	$18B_{2u},\ 18B_{3u}$
$^{13}C_2{}^1H_4$	$16A_g,\ 12B_{1g}$	$12B_{2u},\ 24B_{2u}$
$^{14}N_2{}^{16}O_4$	$6A_g$	$3B_{3u}$

Problem

3. Do the same for the molecule $^{14}N^2H_3$.

SOLUTION. Just as for $^{14}N^1H_3$ in the text, we find the states $30A_1$, $3A_2$, $24E$. For the normal electronic and vibrational term, the following states are possible at different values of the quantum number k:

	(+)	(−)
$\|k\|$ not divisible by three	$24E$	$24E$
$\|k\|$ divisible by three	$30A_1, 3A_2$	$30A_1, 3A_2$
$k = 0,\ J$ even	$30A_1$	$3A_2$
$k = 0,\ J$ odd	$3A_2$	$30A_1$

Problem

4. Do the same for the molecule $^{12}C_2{}^1H_6$ (see Fig. 43f, symmetry D_{3d}).

SOLUTION. The following types of states are possible: $7A_{1g}$, $1A_{1u}$, $3A_{2g}$, $13A_{2u}$, $9E_g$, $11E_u$.

For the normal electronic and vibrational term, the following states are obtained:

	(+)	(−)
$\|k\|$ not divisible by three	$9E_g$	$11E_u$
$\|k\|$ divisible by three	$7A_{1g}, 3A_{2g}$	$1A_{1u}, 13A_{2u}$
$k = 0,\ J$ even	$7A_{1g}$	$1A_{1u}$
$k = 0,\ J$ odd	$3A_{2g}$	$13A_{2u}$

Problem

5. Do the same for the methane molecule $^{12}C^1H_4$ (the H atoms are at the vertices and the C atom at the center of a tetrahedron).

Solution. The molecule is a spherical rotor and has symmetry T_d. By the same method, we find that states of the following types are possible: $5A_2$, $1E$, $3F_1$ (the corresponding total molecular spins are, respectively, 2, 0, 1). The rotational states of a spherical rotor are classified by the values of the total angular momentum J. The $(2J + 1)$ rotational functions belonging to a given J realize a $(2J + 1)$-dimensional representation of the group O, which is isomorphic to T_d and is obtained from it

by replacing every symmetry plane by a second-order axis perpendicular to it. The characters of this representation are determined by formula (98.3). Thus, for example, for $J = 3$ we obtain a representation with characters

E	$8C_3$	$6C_2$	$6C_4$	$3C_4^2$
7	1	−1	−1	−1

It contains the following irreducible representations of O: A_2, F_1, F_2. Considering again the rotational structure of the normal electronic and vibrational term, it follows that at $J = 3$ states for which the complete wave function has symmetry A_2 can only be positive, whereas levels of the state F_1 may be either positive or negative. For the first few values of J, the following states are obtained in the same way (shown together with their nuclear statistical weights):

	(+)	(−)
$J = 0$	−	$5A_2$
$J = 1$	$3F_1$	−
$J = 2$	$1E$	$1E, 3F_1$
$J = 3$	$5A_2, 3F_1$	$3F_1$
$J = 4$	$1E, 3F_1$	$5A_2, 1E, 3F_1$

CHAPTER XIV

Addition of Angular Momenta

§ 106. $3j$ Symbols

The rule for adding angular momenta obtained in § 31 determines the possible values of the total angular momentum of a system composed of two particles (or more complex parts) having angular momenta j_1 and j_2[1]. This rule is in fact closely connected with the behavior of wave functions under spatial rotations and follows directly from the properties of spinors.

The wave functions of particles with angular momenta j_1 and j_2 are symmetric spinors of ranks $2j_1$ and $2j_2$, while the system's wave function is their product

$$\psi^{(1)}_{\lambda\mu\ldots}\psi^{(2)}_{\rho\sigma\ldots}. \tag{106.1}$$

Symmetrizing this product over all indices gives a symmetric spinor of rank $2(j_1 + j_2)$, corresponding to a state of total angular momentum $j_1 + j_2$. Next contract the product (106.1) over one pair of indices, one belonging to $\psi^{(1)}$ and the other to $\psi^{(2)}$ (otherwise the result is zero). Because each spinor is symmetric, it is immaterial which indices are chosen from $\lambda, \mu, \ldots$ and $\rho, \sigma, \ldots$ After symmetrization we obtain a symmetric spinor of rank $2(j_1 + j_2 - 1)$, corresponding to angular momentum $j_1 + j_2 - 1$[2].

Continuing this procedure, we find in accordance with the rule already known that j assumes, once each, the values from $j_1 + j_2$ down to $|j_1 - j_2|$.

Mathematically, this is the decomposition of the direct product $D^{(j_1)} \times D^{(j_2)}$ of two irreducible representations of the rotation group (of dimensions $2j_1 + 1$ and $2j_2 + 1$) into irreducible parts. In these terms the angular-momentum addition rule is

$$D^{(j_1)} \times D^{(j_2)} = D^{(j_1+j_2)} + D^{(j_1+j_2-1)} + \cdots + D^{(|j_1-j_2|)}.$$

To solve the angular-momentum addition problem completely, we must also determine how to construct the wave function of a system with a specified total angular momentum from the wave functions of its two constituent particles.

We begin with the simplest case: two angular momenta adding to zero. Evidently $j_1 = j_2$ and $m_1 = -m_2$. Let ψ_{jm} be normalized wave functions for the states of one

[1] Strictly, throughout this discussion we tacitly mean a system whose parts interact so weakly that their angular momenta may, in the first approximation, be regarded as conserved. The results below apply, of course, not only to the addition of the total angular momenta of two particles (or systems), but also to the addition of the orbital angular momentum and spin of one and the same system when the spin–orbit coupling is sufficiently weak.

[2] To avoid confusion, the following observation is useful. The wave function of a two-particle system is always a spinor of rank $2(j_1 + j_2)$, which in general differs from $2j$, where j is the system's total angular momentum. Such a spinor may, however, be equivalent to one of lower rank. For example, the wave function of two particles with $j_1 = j_2 = 1/2$ is a second-rank spinor; if the total angular momentum is $j = 0$, this spinor is antisymmetric and therefore reduces to a scalar. In general, j determines the symmetry of the system's spinor wave function: it is symmetric in $2j$ indices and antisymmetric in the remaining indices.

particle with angular momentum j and projection m (in a nonspinor representation). The required system wave function ψ_0 is a sum of products of the two particles' wave functions with opposite values of m:

$$\psi_0 = \frac{1}{\sqrt{2j+1}} \sum_{m=-j}^{j} (-1)^{j-m} \psi^{(1)}_{jm} \psi^{(2)}_{j,-m}. \tag{106.2}$$

Here j is the common value of j_1 and j_2. The factor before the sum follows from normalization. All coefficients in the sum must have the same absolute value, since all projections m of the particle angular momenta are equally probable. Their alternating signs in (106.2) are readily found from the spinor representation. In spinor notation the sum in (106.2) is the scalar (the system's total angular momentum is zero)

$$\psi^{(1)}_{\lambda\mu\ldots} \psi^{(2)\lambda\mu\ldots}, \tag{106.3}$$

formed from two spinors of rank $2j$. The signs in (106.2) then follow directly from (57.3).

Only the relative signs of the terms in (106.2) are in general unambiguous; the overall sign may depend on the "order of addition" of the angular momenta. Indeed, lowering every spinor index of $\psi^{(1)}$ (there are $j+m$ ones and $j-m$ twos) and raising those of $\psi^{(2)}$ multiplies the scalar (106.3) by $(-1)^{2j}$, so its sign changes for half-integral j.

Now consider a system of three particles, with angular momenta j_1, j_2, j_3 and projections m_1, m_2, m_3, whose total angular momentum is zero. This condition implies $m_1 + m_2 + m_3 = 0$ and requires each j_i to be obtainable by vector addition of the other two. Geometrically, j_1, j_2, j_3 must be the sides of a closed triangle; thus each is no smaller than the difference and no greater than the sum of the other two:

$$|j_1 - j_2| \leq j_3 \leq j_1 + j_2 \quad \text{and so on.}$$

The algebraic sum $j_1 + j_2 + j_3$ is therefore an integer.

The wave function of this system has the form

$$\psi_0 = \sum_{m_1 m_2 m_3} \begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} \psi^{(1)}_{j_1 m_1} \psi^{(2)}_{j_2 m_2} \psi^{(3)}_{j_3 m_3}, \tag{106.4}$$

where each m_i is summed from $-j_i$ to j_i. The coefficients in this expression are called Wigner $3j$ symbols. By definition, they are nonzero only when $m_1 + m_2 + m_3 = 0$.

Permuting the indices 1, 2, 3 can change (106.4) only by an inessential phase factor. The $3j$ symbols can in fact be defined as purely real (see below), so the only ambiguity in ψ_0 is its overall sign, as for (106.2). Hence a permutation of the columns of a $3j$ symbol either leaves it unchanged or reverses its sign.

The customary definition uses the following, maximally symmetric prescription. In spinor notation ψ_0 is a scalar formed from the product of three spinors $\psi^{(1)}_{\lambda\mu\ldots}$, $\psi^{(2)}_{\lambda\mu\ldots}$, $\psi^{(3)}_{\lambda\mu\ldots}$ by contracting all index pairs, each pair belonging to two different spinors. For every pair belonging to particles 1 and 2, write the index upstairs on $\psi^{(1)}$ and downstairs on $\psi^{(2)}$; for a pair belonging to particles 2 and 3, upstairs on $\psi^{(2)}$ and downstairs on

$\psi^{(3)}$; and for a pair belonging to particles 3 and 1, upstairs on $\psi^{(3)}$ and downstairs on $\psi^{(1)}$. There are respectively $j_1 + j_2 - j_3$, $j_2 + j_3 - j_1$, and $j_1 + j_3 - j_2$ pairs of these three kinds. This prescription fixes the sign of ψ_0 uniquely.

A cyclic permutation of 1, 2, 3 plainly leaves ψ_0 unchanged, and therefore does not change the $3j$ symbol. Interchanging any two indices requires raising the lower and lowering the upper indices in all $j_1 + j_2 + j_3$ pairs. Thus ψ_0 is multiplied by $(-1)^{j_1+j_2+j_3}$, and

$$\begin{pmatrix} j_2 & j_1 & j_3 \\ m_2 & m_1 & m_3 \end{pmatrix} = (-1)^{j_1+j_2+j_3} \begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} \quad \text{and so on.} \tag{106.5}$$

Thus the sign changes when two columns are interchanged if $j_1 + j_2 + j_3$ is odd.

Finally,

$$\begin{pmatrix} j_1 & j_2 & j_3 \\ -m_1 & -m_2 & -m_3 \end{pmatrix} = (-1)^{j_1+j_2+j_3} \begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix}. \tag{106.6}$$

Indeed, reversing the signs of all z components may be regarded as a rotation through π about the y axis. This transformation is equivalent to raising every lower and lowering every upper spinor index; see (58.5).

Equation (106.4) leads to an important formula for the wave function ψ_{jm} of a two-particle system with prescribed j and m. Regard particles 1 and 2 together as a single system. Since its angular momentum j combines with that of particle 3 to give zero, $j = j_3$ and $m = -m_3$. From (106.2),

$$\psi_0 = \frac{1}{\sqrt{2j+1}} \sum_m (-1)^{j-m} \psi_{jm} \psi^{(3)}_{j,-m}. \tag{106.7}$$

This must be compared with (106.4), with $j, -m$ written for j_3, m_3. First, however, the contraction convention in (106.7), as specified by (106.3), must be reconciled with that in (106.4). This requires interchanging upper and lower indices in the pairs belonging to particles 1 and 3, and introduces an additional factor $(-1)^{j_1-j_2+j_3}$. We obtain[3]

$$\psi_{jm} = (-1)^{j_1-j_2+m} \sqrt{2j+1} \sum_{m_1 m_2} \begin{pmatrix} j_1 & j_2 & j \\ m_1 & m_2 & -m \end{pmatrix} \psi^{(1)}_{j_1 m_1} \psi^{(2)}_{j_2 m_2}, \tag{106.8}$$

where m_1, m_2 are summed subject to $m_1 + m_2 = m$.

Formula (106.8) is the desired expansion of the system wave function in wave functions of the two particles with specified angular momenta j_1, j_2. It may be written

$$\psi_{jm} = \sum_{m_1 m_2} (m_1 m_2 \mid jm) \psi^{(1)}_{j_1 m_1} \psi^{(2)}_{j_2 m_2}, \qquad m_2 = m - m_1. \tag{106.9}$$

The coefficients

$$(m_1 m_2 \mid jm) = (-1)^{j_1-j_2+m} \sqrt{2j+1} \begin{pmatrix} j_1 & j_2 & j \\ m_1 & m_2 & -m \end{pmatrix} \tag{106.10}$$

[3] Under time reversal the wave functions transform, according to (60.2), as $\psi_{jm} \to (-1)^{j-m}\psi_{j,-m}$. It is readily verified that applying this transformation to $\psi_{j_1 m_1}$ and $\psi_{j_2 m_2}$ on the right-hand side of (106.8) transforms ψ_{jm} on the left in the same way.

form the transformation matrix from the complete orthonormal set of $(2j_1+1)(2j_2+1)$ state wave functions $|m_1m_2\rangle$ to the corresponding set $|jm\rangle$, for fixed j_1, j_2. They are called vector-addition coefficients, or Clebsch–Gordan coefficients. The notation $(m_1m_2 \mid jm)$ follows the general notation (11.18) for expansion coefficients between two systems of functions. For brevity the quantum numbers j_1, j_2, common to both systems, have been omitted; when needed they are included as $(j_1m_1j_2m_2 \mid j_1j_2jm)$[4].

The transformation (106.9) is unitary (see § 12). Hence the coefficients of the inverse transformation

$$\psi^{(1)}_{j_1m_1}\psi^{(2)}_{j_2m_2} = \sum_{j=|j_2-j_1|}^{j_1+j_2} (jm \mid m_1m_2)\psi_{jm} \tag{106.11}$$

are the complex conjugates of those in (106.9). As shown below, these coefficients are real; therefore

$$(m_1m_2 \mid jm) = (jm \mid m_1m_2).$$

By the general rules of quantum mechanics, the squares of the expansion coefficients in (106.11) give the probabilities for the system to have the various values j, m, when j_1, m_1 and j_2, m_2 are specified.

Unitarity of (106.9) gives the orthogonality relations

$$\sum_{m_1m_2} (m_1m_2 \mid jm)(m_1m_2 \mid j'm') = \delta_{jj'}\delta_{mm'}, \tag{106.12}$$

$$\sum_{jm} (m_1m_2 \mid jm)(m_1'm_2' \mid jm) = \delta_{m_1m_1'}\delta_{m_2m_2'}. \tag{106.13}$$

The explicit general expression for a $3j$ symbol is rather cumbersome. It can be written[5]

$$\begin{aligned}\begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} = {} & \left[\frac{(j_1+j_2-j_3)!(j_1-j_2+j_3)!(-j_1+j_2+j_3)!}{(j_1+j_2+j_3+1)!}\right]^{1/2} \\ & \times [(j_1+m_1)!(j_1-m_1)!(j_2+m_2)!(j_2-m_2)!(j_3+m_3)!(j_3-m_3)!]^{1/2} \\ & \times \sum_z \frac{(-1)^{z+j_1-j_2-m_3}}{z!(j_1+j_2-j_3-z)!(j_1-m_1-z)!(j_2+m_2-z)!} \\ & \qquad \times \frac{1}{(j_3-j_2+m_1+z)!(j_3-j_1-m_2+z)!}. \end{aligned} \tag{106.14}$$

The sum extends over all integral z; since the factorial of a negative number is infinite, only finitely many terms actually contribute. The factor preceding the sum is manifestly symmetric in 1, 2, 3; symmetry of the sum appears after an appropriate change of summation variable.

[4]The notations $C^{jm}_{j_1m_1j_2m_2}$ and $C^{j_1j_2j}_{m_1m_2m}$ are also used for Clebsch–Gordan coefficients.

[5]The expansion coefficients (106.9) were first calculated by Wigner (1931). Their symmetry properties and the symmetric expression (106.14) were found by Racah (G. Racah, 1942). Probably the most direct calculation passes from the suitably normalized spinor representation of ψ_0 to the sum (106.4), using the correspondence formula (57.6); the reality of its coefficient automatically establishes the reality of the $3j$ symbols. Another derivation is given in Edmonds's book (see the note on p. 272). The table below is also taken from that book.

Besides (106.5) and (106.6), which follow simply from the definition, $3j$ symbols possess further symmetry properties whose derivation is more involved and will not be given here. They are conveniently stated by introducing the square (3×3) array

$$\begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} \longleftrightarrow \begin{bmatrix} j_2 + j_3 - j_1 & j_3 + j_1 - j_2 & j_1 + j_2 - j_3 \\ j_1 - m_1 & j_2 - m_2 & j_3 - m_3 \\ j_1 + m_1 & j_2 + m_2 & j_3 + m_3 \end{bmatrix}. \tag{106.15}$$

The sum of the entries in every row and column is $j_1 + j_2 + j_3$. Then: 1) interchanging any two columns multiplies the $3j$ symbol by $(-1)^{j_1+j_2+j_3}$, reproducing (106.5); 2) the same holds for any two rows (for the last two rows this reproduces (106.6)); 3) the symbol is unchanged when rows and columns are interchanged[6].

Some simpler special cases are

$$\begin{pmatrix} j & j & 0 \\ m & -m & 0 \end{pmatrix} = \frac{(-1)^{j-m}}{\sqrt{2j+1}}, \tag{106.16}$$

which corresponds to (106.2), and

$$\begin{pmatrix} j_1 & j_2 & j_1 + j_2 \\ m_1 & m_2 & -m_1 - m_2 \end{pmatrix} = (-1)^{j_1 - j_2 + m_1 + m_2} \times \left[\frac{(2j_1)!(2j_2)!(j_1 + j_2 + m_1 + m_2)!(j_1 + j_2 - m_1 - m_2)!}{(j_1 + j_2 + 1)!(j_1 + m_1)!(j_1 - m_1)!(j_2 + m_2)!(j_2 - m_2)!} \right]^{1/2}, \tag{106.17}$$

which follows directly from (106.14).

The derivation of

$$\begin{pmatrix} j_1 & j_2 & j_3 \\ 0 & 0 & 0 \end{pmatrix} = (-1)^p \left[\frac{(j_1 + j_2 - j_3)!(j_1 - j_2 + j_3)!(-j_1 + j_2 + j_3)!}{(2p+1)!} \right]^{1/2} \times \frac{p!}{(p - j_1)!(p - j_2)!(p - j_3)!}, \tag{106.18}$$

where $2p = j_1 + j_2 + j_3$ is even, requires additional calculation[7]. For odd $2p$ this $3j$ symbol vanishes by (106.6).

For reference, Table 9 lists $3j$ symbols for $j_3 = 1/2, 1, 3/2, 2$. For every j_3 it gives the smallest set from which all remaining symbols follow using (106.5) and (106.6).

Problem

Determine the angular dependence of the wave functions of a spin-1/2 particle in states with specified orbital angular momentum l, total angular momentum j, and projection m.

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[6] See T. Regge, Nuovo Cimento, 1958, vol. 10, p. 544; 1959, vol. 11, p. 116. For deeper mathematical aspects of (106.15), and of the $6j$-symbol property (108.3) below, see the review by Ya. A. Smorodinsky and L. A. Shelepin, Sov. Phys. Usp., 1972, vol. 15, p. 1.

[7] See the book by Edmonds cited above.

Table 9. Formulas for $3j$ symbols

j_3	j_1	Value of $(-1)^{j-m}\begin{pmatrix} j_1 & j & j_3 \\ m_1 & -m-m_3 & m_3 \end{pmatrix}$
$\frac{1}{2}$	j	$m_3 = \frac{1}{2}:\quad -\left[\dfrac{j-m}{(2j+1)(2j+2)}\right]^{1/2}$
	$j+1$	$\left[\dfrac{j+m+1}{(2j+1)(2j+2)}\right]^{1/2}$
1	j	$m_3 = 0:\quad \dfrac{2m}{[2j(2j+1)(2j+2)]^{1/2}}$
	$j+1$	$\left[\dfrac{2(j+m+1)(j-m+1)}{(2j+1)(2j+2)(2j+3)}\right]^{1/2}$
	j	$m_3 = 1:\quad \left[\dfrac{2(j-m)(j+m+1)}{2j(2j+1)(2j+2)}\right]^{1/2}$
	$j+1$	$\left[\dfrac{(j-m)(j-m+1)}{(2j+1)(2j+2)(2j+3)}\right]^{1/2}$

Table 9 (continued)

j_3	j_1	Value
$\frac{3}{2}$	$j+\frac{1}{2}$	$m_3 = \frac{1}{2}:\quad -(j+3m+\frac{3}{2})\left[\dfrac{j-m+\frac{1}{2}}{2j(2j+1)(2j+2)(2j+3)}\right]^{1/2}$
	$j+\frac{3}{2}$	$\left[\dfrac{3(j-m+\frac{1}{2})(j-m+\frac{3}{2})(j+m+\frac{3}{2})}{(2j+1)(2j+2)(2j+3)(2j+4)}\right]^{1/2}$
	$j+\frac{1}{2}$	$m_3 = \frac{3}{2}:\quad -\left[\dfrac{3(j-m-\frac{1}{2})(j-m+\frac{1}{2})(j+m+\frac{3}{2})}{2j(2j+1)(2j+2)(2j+3)}\right]^{1/2}$
	$j+\frac{3}{2}$	$\left[\dfrac{(j-m-\frac{1}{2})(j-m+\frac{1}{2})(j-m+\frac{3}{2})}{(2j+1)(2j+2)(2j+3)(2j+4)}\right]^{1/2}$
2	j	$m_3 = 0:\quad \dfrac{2[3m^2-j(j+1)]}{[(2j-1)2j(2j+1)(2j+2)(2j+3)]^{1/2}}$
	$j+1$	$-2m\left[\dfrac{6(j+m+1)(j-m+1)}{2j(2j+1)(2j+2)(2j+3)(2j+4)}\right]^{1/2}$
	$j+2$	$\left[\dfrac{6(j+m+2)(j+m+1)(j-m+2)(j-m+1)}{(2j+1)(2j+2)(2j+3)(2j+4)(2j+5)}\right]^{1/2}$
	j	$m_3 = 1:\quad (1+2m)\left[\dfrac{6(j+m+1)(j-m)}{(2j-1)2j(2j+1)(2j+2)(2j+3)}\right]^{1/2}$
	$j+1$	$-2(j+2m+2)\left[\dfrac{(j-m+1)(j-m)}{2j(2j+1)(2j+2)(2j+3)(2j+4)}\right]^{1/2}$
	$j+2$	$2\left[\dfrac{(j+m+2)(j-m+2)(j-m+1)(j-m)}{(2j+1)(2j+2)(2j+3)(2j+4)(2j+5)}\right]^{1/2}$
	j	$m_3 = 2:\quad \left[\dfrac{6(j-m-1)(j-m)(j+m+1)(j+m+2)}{(2j-1)2j(2j+1)(2j+2)(2j+3)}\right]^{1/2}$
	$j+1$	$-2\left[\dfrac{(j-m-1)(j-m)(j-m+1)(j+m+2)}{2j(2j+1)(2j+2)(2j+3)(2j+4)}\right]^{1/2}$
	$j+2$	$\left[\dfrac{(j-m-1)(j-m)(j-m+1)(j-m+2)}{(2j+1)(2j+2)(2j+3)(2j+4)(2j+5)}\right]^{1/2}$

Solution. The general formula (106.8) solves the problem if $\psi^{(1)}$ denotes an orbital-angular-momentum eigenfunction (the spherical function Y_{lm_l}) and $\psi^{(2)}$ the spin wave function $\chi(\sigma)$, where $\sigma = \pm 1/2$:

$$\psi_{jm} = (-1)^{l+m-1/2}\sqrt{2j+1}\sum_{\sigma}\begin{pmatrix} l & \frac{1}{2} & j \\ m-\sigma & \sigma & -m\end{pmatrix} Y_{l,m-\sigma}\chi(\sigma).$$

Substitution of the $3j$ symbols gives

$$\psi_{l+1/2,m} = \sqrt{\frac{j+m}{2j}}\,\chi\left(\frac{1}{2}\right) Y_{l,m-1/2} + \sqrt{\frac{j-m}{2j}}\,\chi\left(-\frac{1}{2}\right) Y_{l,m+1/2},$$
$$\psi_{l-1/2,m} = -\sqrt{\frac{j-m+1}{2j+2}}\,\chi\left(\frac{1}{2}\right) Y_{l,m-1/2} + \sqrt{\frac{j+m+1}{2j+2}}\,\chi\left(-\frac{1}{2}\right) Y_{l,m+1/2}.$$

§ 107. Matrix elements of tensors

In § 29 we obtained formulas giving the dependence of the matrix elements of a vector physical quantity on the value of the angular-momentum projection. In fact, those formulas are a special case of general formulas that solve the same problem for an irreducible tensor (see p. 268) of any rank[8].

The set of $2k+1$ components of an irreducible tensor of rank k (k is an integer) is equivalent, in its transformation properties, to the set of $2k+1$ spherical functions Y_{kq} ($q = -k, \ldots, k$) (see the note on p. 268). This means that suitable linear combinations of the tensor components can be formed so as to give a set of quantities transforming under rotations in the same way as the functions Y_{kq}. We shall call such a set of quantities, denoted here by f_{kq}, a *spherical tensor* of rank k.

Thus a vector corresponds to $k = 1$, and the quantities f_{1q} are related to the vector components by

$$f_{10} = ia_z, \qquad f_{1,\pm 1} = \mp\frac{i}{\sqrt{2}}(a_x \pm ia_y) \tag{107.1}$$

(cf. (57.7)). The analogous formulas for a tensor of second rank are

$$f_{20} = -\sqrt{3/2}\,a_{zz}, \qquad f_{2,\pm 1} = \pm(a_{xz} \pm ia_{yz}),$$
$$f_{2,\pm 2} = -\frac{1}{2}(a_{xx} - a_{yy} \pm 2ia_{xy}), \tag{107.2}$$

where $a_{xx} + a_{yy} + a_{zz} = 0$[9].

Tensor products of two (or more) spherical tensors $f_{k_1q_1}, g_{k_2q_2}$ are formed according to the rules for addition of angular momenta, with k_1, k_2 formally playing the role of the "angular momenta" associated with these tensors. Thus, from two spherical tensors of

[8]The treatment of the questions considered in § 107–109, and most of the results presented there, are due to Racah (1942/1943).

[9]It is understood that the quantities f_{kq} are complex only because of the change to spherical coordinates; that is, the original Cartesian tensor components are real.

ranks k_1 and k_2 one can form spherical tensors of ranks $K = k_1 + k_2, \dots, |k_1 - k_2|$ by

$$\begin{aligned}(f_{k_1} g_{k_2})_{KQ} &= \sum_{q_1 q_2} (q_1 q_2 | KQ) f_{k_1 q_1} g_{k_2 q_2} = \\ &= (-1)^{k_1 - k_2 + Q} \sqrt{2K+1} \sum_{q_1 q_2} \begin{pmatrix} k_1 & k_2 & K \\ q_1 & q_2 & -Q \end{pmatrix} f_{k_1 q_1} g_{k_2 q_2}\end{aligned} \tag{107.3}$$

(cf. (106.9)). The scalar product of two spherical tensors of the same rank k is, however, conventionally defined by

$$(f_k g_k)_{00} = \sum_q (-1)^{k-q} f_{kq} g_{k,-q}, \tag{107.4}$$

which differs by a factor $\sqrt{2k+1}$ from the definition supplied by (107.3) with $K = Q = 0$ (cf. (106.2))[10]. This definition may also be written

$$(f_k g_k)_{00} = \sum_q f_{kq} g^*_{kq},$$

on observing that complex conjugation of a spherical tensor is performed according to

$$f^*_{kq} = (-1)^{k-q} f_{k,-q}$$

(cf. (28.9))[11].

Representing physical quantities as spherical tensors is especially convenient when their matrix elements are calculated, since the results of the theory of angular-momentum addition can then be used.

By the definition of matrix elements,

$$\hat{f}_{kq} \psi_{njm} = \sum_{n'j'm'} \langle n'j'm' | f_{kq} | njm \rangle \psi_{n'j'm'}, \tag{107.5}$$

where ψ_{njm} are the wave functions of stationary states of the system, specified by the magnitude j of its angular momentum, its projection m, and the set n of the remaining quantum numbers. In their transformation properties, the functions on the right- and left-hand sides of (107.5) correspond respectively to the functions on the right- and left-hand sides of (106.11). The selection rules follow at once.

The matrix elements of the components f_{kq} of an irreducible tensor of rank k can be nonzero only for transitions $jm \to j'm'$ satisfying the "angular-momentum addition rule" $\mathbf{j}' = \mathbf{j} + \mathbf{k}$; the numbers j', j, k must satisfy the "triangle rule" (that is, they can be the lengths of the sides of a closed triangle), and $m' = m + q$. In particular, diagonal matrix elements can be nonzero only if $2j \geqslant k$.

[10] If $\mathbf{A}$ and $\mathbf{B}$ are two vectors corresponding, by (107.1), to the spherical tensors f_{1q} and g_{1q}, then

$$(f_1 g_1)_{00} = \mathbf{AB}.$$

[11] We repeat here the observation made above in connection with (106.8): with this rule, complex conjugation of the tensors of ranks k_1 and k_2 on the right-hand side of (107.3) produces the corresponding conjugation of the tensor of rank K on its left-hand side.

The same analogy of transformation properties further shows that the coefficients in the sum (107.5) must be proportional to those in (106.11) (the *Wigner–Eckart theorem*). This determines their dependence on m and m'. Accordingly, we write the matrix elements as

$$\langle n'j'm'|f_{kq}|njm\rangle = i^k(-1)^{j_{\max}-m'}\begin{pmatrix} j' & k & j \\ -m' & q & m \end{pmatrix}\langle n'j'\|f_k\|nj\rangle, \tag{107.6}$$

where $j_{\max}$ is the larger of j and j', while $\langle n'j'\|f_k\|nj\rangle$ are quantities independent of m, m', and q'; they are called *reduced matrix elements*. This formula answers the question of determining the dependence of matrix elements on angular-momentum projections. That dependence is entirely governed by symmetry with respect to the rotation group, whereas dependence on the other quantum numbers is determined by the physical nature of the quantities f_{kq} themselves[12].

The operators $\widehat{f}_{kq}$ are related by

$$\widehat{f}^{+}_{kq} = (-1)^{k-q}\widehat{f}_{k,-q}. \tag{107.7}$$

Their matrix elements therefore satisfy

$$\langle n'j'm'|f_{kq}|njm\rangle^* = (-1)^{k-q}\langle njm|f_{k,-q}|n'j'm'\rangle. \tag{107.8}$$

Substitution of (107.6), together with the properties (106.5), (106.6) of the $3j$ symbols, gives the following "Hermiticity" relation for the reduced matrix elements[13]:

$$\langle n'j'\|f_k\|nj\rangle = \langle nj\|f_k\|n'j'\rangle^*. \tag{107.9}$$

The matrix elements of the scalar (107.4) are diagonal in j and m. The rule for matrix multiplication gives

$$\begin{aligned}\langle n'jm|(f_k g_k)_{00}|njm\rangle = \sum_q (-1)^{k-q} \sum_{n''j''m''} &\langle n'jm|f_{kq}|n''j''m''\rangle \\ &\times \langle n''j''m''|g_{k,-q}|njm\rangle.\end{aligned}$$

Substituting (107.6) here and summing over q and m'' by means of the orthogonality relation (106.12) for the $3j$ symbols, we obtain

$$\langle n'jm|(f_k g_k)_{00}|njm\rangle = \frac{1}{2j+1}\sum_{n''j''}\langle n'j\|f_k\|n''j''\rangle\langle n''j''\|g_k\|nj\rangle. \tag{107.10}$$

In the same way one readily obtains the following formulas for sums of squared matrix elements:

$$\sum_{qm'}|\langle n'j'm'|f_{kq}|njm\rangle|^2 = \frac{1}{2j+1}|\langle n'j'\|f_k\|nj\rangle|^2, \tag{107.11}$$

[12] Among the consequences of these results are the selection rules for vector matrix elements stated in § 29 and the corresponding formulas (29.7)–(29.9).

[13] The phase factor in the definition (107.6) was chosen precisely so that this equality holds.

$$\sum_{mm'} |\langle n'j'm'|f_{kq}|njm\rangle|^2 = \frac{1}{2k+1}|\langle n'j'\|f_k\|nj\rangle|^2. \tag{107.12}$$

In the first, the sum is over q and m' at fixed m; in the second, it is over m and m' at fixed q (with $m' = m + q$ throughout).

For reference, consider the case in which the quantities f_{kq} are the spherical functions Y_{kq} themselves, and give their matrix elements for transitions between states of one particle with integral orbital angular momenta l_1 and l_2, that is, the integrals

$$\langle l_1 m_1|Y_{lm}|l_2 m_2\rangle = \int Y^*_{l_1 m_1} Y_{lm} Y_{l_2 m_2}\, do. \tag{107.13}$$

Besides the selection rule arising from angular-momentum addition ($\mathbf{l} + \mathbf{l}_2 = \mathbf{l}_1$), these matrix elements obey the additional rule that $l + l_1 + l_2$ must be even. This is connected with parity conservation: the product $(-1)^{l_1+l_2}$ of the parities of the two states must equal the parity $(-1)^l$ of the physical quantity under consideration (see § 30).

The matrix elements (107.13) are a special case of a more general integral that will be evaluated in § 110 (see the note on p. 547). They are

$$\langle l_1 m_1|Y_{lm}|l_2 m_2\rangle = (-1)^{m_1} i^{-l_1+l_2+l} \begin{pmatrix} l_1 & l & l_2 \\ -m_1 & m & m_2 \end{pmatrix} \begin{pmatrix} l_1 & l & l_2 \\ 0 & 0 & 0 \end{pmatrix} \times \left[\frac{(2l+1)(2l_1+1)(2l_2+1)}{4\pi}\right]^{1/2}. \tag{107.14}$$

In particular, for $m_1 = m_2 = m = 0$ we find the following value of the integral of the product of three Legendre polynomials:

$$\int_{-1}^{1} P_l(\mu) P_{l_1}(\mu) P_{l_2}(\mu)\, d\mu = 2\begin{pmatrix} l_1 & l & l_2 \\ 0 & 0 & 0 \end{pmatrix}^2. \tag{107.15}$$

§ 108. $6j$ symbols

In § 106 we defined the $3j$ symbols as the coefficients in the sum (106.4), which represents the wave function of a system of three particles having zero total angular momentum. From the standpoint of transformation properties under rotations, this sum is a scalar. It follows that the set of $3j$ symbols for specified values of j_1, j_2, j_3 (and all possible m_1, m_2, m_3) may be regarded as a collection of quantities which, under rotations, transform by a law contragredient to the transformation law of the products $\psi_{j_1 m_1}\psi_{j_2 m_2}\psi_{j_3 m_3}$, thereby ensuring invariance of the whole sum.

This point of view suggests the question of constructing a scalar made solely from $3j$ symbols. Such a scalar must depend only on the numbers j, and not on the numbers m, which change under rotations. In other words, it must be expressible as sums over all the numbers m. Each such summation consists in contracting a product of two $3j$ symbols according to the rule

$$\sum_m (-1)^{j-m} \begin{pmatrix} j & \cdots \\ m & \cdots \end{pmatrix} \begin{pmatrix} j & \cdots \\ -m & \cdots \end{pmatrix} \tag{108.1}$$

(compare the construction of the scalar (106.2)).

Since each contraction involves a pair of numbers m, products of an even number of $3j$ symbols must be considered in order to form a complete scalar. By their orthogonality property, contraction of the product of two $3j$ symbols gives the trivial result

$$\sum_{m_1 m_2 m_3} \begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} \begin{pmatrix} j_1 & j_2 & j_3 \\ -m_1 & -m_2 & -m_3 \end{pmatrix} (-1)^{j_1+j_2+j_3-m_1-m_2-m_3}$$
$$= \sum_{m_1 m_2 m_3} \begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix}^2 = 1$$

(here the equality $m_1 + m_2 + m_3 = 0$ and formulas (106.6) and (106.12) have been used). Hence the smallest number of factors needed to obtain a nontrivial scalar is four. In every $3j$ symbol the three numbers j form a geometrically closed triangle. Since, in a contraction, each number j must occur in two $3j$ symbols, it is clear that a scalar constructed from a product of four $3j$ symbols contains six numbers j. Geometrically, they must be represented by the edge lengths of an irregular tetrahedron (Fig. 45), with each $3j$ symbol corresponding to one face. The definition of the required scalar adopts a definite convention for carrying out the contractions, expressed by the formula

$$\begin{Bmatrix} j_1 & j_2 & j_3 \\ j_4 & j_5 & j_6 \end{Bmatrix} = \sum_{\text{all } m} (-1)^{\sum_i (j_i - m_i)} \begin{pmatrix} j_1 & j_2 & j_3 \\ -m_1 & -m_2 & -m_3 \end{pmatrix} \begin{pmatrix} j_1 & j_5 & j_6 \\ m_1 & -m_5 & m_6 \end{pmatrix}$$
$$\times \begin{pmatrix} j_4 & j_2 & j_6 \\ m_4 & m_2 & -m_6 \end{pmatrix} \begin{pmatrix} j_4 & j_5 & j_3 \\ -m_4 & m_5 & m_3 \end{pmatrix}. \tag{108.2}$$

The summation here extends over all possible values of all the numbers m; since, however, the sum of the three m's in each $3j$ symbol must vanish, only

three of the six m's are actually independent. The quantities defined by (108.2) are called $6j$ symbols, or Racah coefficients[14].

From definition (108.2), with the symmetry properties of the $3j$ symbols taken into account, it is readily verified that a $6j$ symbol is unchanged by any permutation of its three columns, and that the upper and lower entries may be interchanged in any pair of columns. Owing to these symmetry properties, the sequence of numbers $j_1 \ldots j_6$ in a $6j$ symbol can be displayed in 24 equivalent forms[15].

The $6j$ symbols also possess another, less evident, symmetry property, which gives an equality between symbols with different sets of numbers j:

$$\begin{Bmatrix} j_1 & j_2 & j_3 \\ j_4 & j_5 & j_6 \end{Bmatrix} = \begin{Bmatrix} j_1 & \frac{1}{2}(j_2 + j_5 + j_3 - j_6) & \frac{1}{2}(j_3 + j_6 + j_2 - j_5) \\ j_4 & \frac{1}{2}(j_2 + j_5 + j_6 - j_3) & \frac{1}{2}(j_3 + j_6 + j_5 - j_2) \end{Bmatrix}. \tag{108.3}$$

(*T. Regge*, 1959)[16].

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[14] The notation

$$W(j_1 j_2 j_4 j_5; j_3 j_6) = (-1)^{j_1+j_2+j_4+j_5} \begin{Bmatrix} j_1 & j_2 & j_3 \\ j_4 & j_5 & j_6 \end{Bmatrix}$$

is also used in the literature.

[15] If the tetrahedron in Fig. 45 is imagined to be regular, the 24 equivalent permutations of the numbers j can be obtained by applying the 24 symmetry transformations (rotations and reflections) of the tetrahedron.

[16] See the note on p. 529.

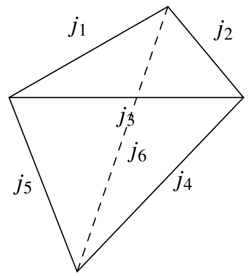

Figure 45

We give a useful relation between the $6j$ and $3j$ symbols which follows from definition (108.2):

$$\sum_{m_4 m_5 m_6} (-1)^{j_4+j_5+j_6-m_4-m_5-m_6} \begin{pmatrix} j_1 & j_5 & j_6 \\ m_1 & -m_5 & m_6 \end{pmatrix} \begin{pmatrix} j_4 & j_2 & j_6 \\ m_4 & m_2 & -m_6 \end{pmatrix} \times \begin{pmatrix} j_4 & j_5 & j_3 \\ -m_4 & m_5 & m_3 \end{pmatrix} = \begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} \begin{Bmatrix} j_1 & j_2 & j_3 \\ j_4 & j_5 & j_6 \end{Bmatrix}. \tag{108.4}$$

The expression summed on the left-hand side differs from the sum in (108.2) by the absence of one factor (a $3j$ symbol). The sum in (108.4) may therefore be represented by a tetrahedron (Fig. 45) with one face removed; this is what distinguishes the sum from a scalar. In other words, in its transfor-

mation properties it corresponds to a single $3j$ symbol—the one on the right-hand side of (108.4), to which it must be proportional. The coefficient of proportionality (the $6j$ symbol on the right-hand side) is readily obtained by multiplying both sides by

$$\begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix}$$

and summing over the remaining numbers m_1, m_2, m_3.

The $6j$ symbols arise naturally in the following question connected with the addition of three angular momenta.

Let three angular momenta j_1, j_2, j_3 combine to give a resultant angular momentum J. Specification of J (and of its projection M) does not, however, determine the state of the system uniquely; the state also depends on the manner in which the angular momenta are added (or, as it is called, on their coupling scheme).

Consider, for example, the following two coupling schemes: 1) first j_1 and j_2 are combined into the total angular momentum j_{12}, after which j_{12} and j_3 are combined into the final angular momentum J; 2) j_2 and j_3 are combined into j_{23}, and then j_{23} and j_1 into J. The first scheme corresponds to states in which (besides j_1, j_2, j_3, J, M) the quantity j_{12} has a definite value; denote their wave functions by $\psi_{j_{12}JM}$, omitting for brevity the repeated indices $j_1 j_2 j_3$. Similarly, denote the wave functions of the second coupling scheme by $\psi_{j_{23}JM}$. In both cases the values of the "intermediate" angular momentum (j_{12} or j_{23}) are in general not unique. Thus, for specified J, M, there are two different sets of states distinguished by the values of j_{12} or j_{23}. By the general rules, the functions of these two sets are related by a certain unitary transformation

$$\psi_{j_{23}JM} = \sum_{j_{12}} \langle j_{12}|j_{23}\rangle \psi_{j_{12}JM}. \tag{108.5}$$

It is evident on physical grounds that the coefficients of this transformation do not depend on M: they cannot depend on the orientation of the entire system in space. Thus they depend only on the values of the six angular momenta $j_1, j_2, j_3, j_{12}, j_{23}, J$, not on their projections; that is, they are scalar quantities in the sense stated above. These coefficients can be evaluated directly as follows.

Applying formula (106.9) twice, we find

$$\begin{aligned}
\psi_{j_{23}JM} &= \sum_{(m)} \langle m_1 m_{23}|JM\rangle \psi_{j_1 m_1} \psi_{j_{23} m_{23}} \\
&= \sum_{(m)} \langle m_1 m_{23}|JM\rangle \langle m_2 m_3|j_{23} m_{23}\rangle \psi_{j_1 m_1} \psi_{j_2 m_2} \psi_{j_3 m_3}, \\
\psi_{j_{12}JM} &= \sum_{(m)} \langle m_3 m_{12}|JM\rangle \langle m_1 m_2|j_{12} m_{12}\rangle \psi_{j_1 m_1} \psi_{j_2 m_2} \psi_{j_3 m_3}
\end{aligned}$$

(the symbol (m) beneath a summation sign means that the summation is over all the numbers $m_1, m_2, \ldots$ occurring in the expression). Using the orthonormality of the functions ψ_{jm}, we now obtain

$$\begin{aligned}
\langle j_{12}|j_{23}\rangle &\equiv \int \psi^*_{j_{12}JM} \psi_{j_{23}JM}\, dq \\
&= \sum_{(m)} \langle m_3 m_{12}|JM\rangle \langle m_1 m_{23}|JM\rangle \langle m_1 m_2|j_{12} m_{12}\rangle \langle m_2 m_3|j_{23} m_{23}\rangle.
\end{aligned}$$

The sum on the right-hand side is taken at a specified value of M, but the result is in fact independent of M, for the reason given above. The summation may therefore be extended over the values of M if a factor $1/(2J+1)$ is placed before the sum. Then, expressing the coefficients $\langle m_1 m_2|jm\rangle$ in terms of $3j$ symbols according to (106.10), we obtain

$$\langle j_{12}|j_{23}\rangle = (-1)^{j_1+j_2+j_3+J} \sqrt{(2j_{12}+1)(2j_{23}+1)} \begin{Bmatrix} j_1 & j_2 & j_{12} \\ j_3 & J & j_{23} \end{Bmatrix}. \tag{108.6}$$

The relation between the $6j$ symbols and the transformation coefficients (108.5) makes it easy to obtain several useful summation formulas for products of $6j$ symbols.

First, by the unitarity of transformation (108.5), and since its coefficients are real, we have

$$\sum_j (2j+1)(2j''+1) \begin{Bmatrix} j_1 & j_2 & j' \\ j_3 & j_4 & j \end{Bmatrix} \begin{Bmatrix} j_3 & j_2 & j \\ j_1 & j_4 & j'' \end{Bmatrix} = \delta_{j'j''}. \tag{108.7}$$

Next consider three coupling schemes for three angular momenta, with intermediate sums j_{12}, j_{23}, and j_{31}, respectively. By the rule of matrix multiplication, the corresponding transformation coefficients (108.6) obey

$$\sum_{j_{23}} \langle j_{12}|j_{23}\rangle\langle j_{23}|j_{31}\rangle = \langle j_{12}|j_{31}\rangle.$$

Substitution of (108.6), followed by a change of index notation, gives

$$\sum_j (-1)^{j+j_3+j_6}(2j+1)\begin{Bmatrix} j_2 & j_4 & j_6 \\ j_1 & j_5 & j \end{Bmatrix}\begin{Bmatrix} j_4 & j_1 & j_6 \\ j_2 & j_5 & j_3 \end{Bmatrix} = \begin{Bmatrix} j_1 & j_2 & j_3 \\ j_4 & j_5 & j_6 \end{Bmatrix}. \qquad (108.8)$$

Finally, consideration of different coupling schemes for four angular momenta yields[17] the following addition formula for products of three $6j$ symbols:

$$\begin{aligned}\sum_j (-1)^{j+\Sigma_1^9 j_i}(2j+1)&\begin{Bmatrix} j_4 & j_2 & j_6 \\ j_9 & j_8 & j \end{Bmatrix}\begin{Bmatrix} j_2 & j_1 & j_3 \\ j_7 & j & j_9 \end{Bmatrix}\begin{Bmatrix} j_4 & j_3 & j_5 \\ j_7 & j_8 & j \end{Bmatrix}\\ &= \begin{Bmatrix} j_1 & j_2 & j_3 \\ j_4 & j_5 & j_6 \end{Bmatrix}\begin{Bmatrix} j_6 & j_1 & j_5 \\ j_7 & j_8 & j_9 \end{Bmatrix}.\end{aligned} \qquad (108.9)$$

(*L. C. Biedenharn*, *J. P. Elliott*, 1953).

For reference, we give some explicit expressions for $6j$ symbols. In the general case a $6j$ symbol can be represented by the following sum:

$$\begin{aligned}\begin{Bmatrix} j_1 & j_2 & j_3 \\ j_4 & j_5 & j_6 \end{Bmatrix} &= \Delta(j_1 j_2 j_3)\Delta(j_1 j_5 j_6)\Delta(j_4 j_2 j_6)\Delta(j_4 j_5 j_3)\\ &\times \sum_z \frac{(-1)^z (z+1)!}{(z-j_1-j_2-j_3)!(z-j_1-j_5-j_6)!(z-j_4-j_2-j_6)!(z-j_4-j_5-j_3)!}\\ &\times (j_1+j_2+j_4+j_5-z)!(j_2+j_3+j_5+j_6-z)!(j_3+j_1+j_6+j_4-z)!,\end{aligned} \qquad (108.10)$$

where

$$\Delta(abc) = \left[\frac{(a+b-c)!(a-b+c)!(-a+b+c)!}{(a+b+c+1)!}\right]^{1/2},$$

and the sum is taken over all positive integral values of z for which none of the factorials in the denominator has a negative argument. Table 10 gives formulas for $6j$ symbols in the cases where one of the parameters is 0, 1/2, or 1.

We conclude with several remarks on scalars of higher order constructed from $3j$ symbols.

The scalar next in complexity after the $3j$ symbol is formed by contracting a product of six $3j$ symbols. These $3j$ symbols contain 18 numbers j equal in pairs,

so that the resulting scalar depends on nine parameters j. It is customarily called a

[17] See the book by Edmonds cited above.

$9j$ symbol and is defined as follows (*E. Wigner*, 1951)[18]:

$$\begin{Bmatrix} j_{11} & j_{12} & j_{13} \\ j_{21} & j_{22} & j_{23} \\ j_{31} & j_{32} & j_{33} \end{Bmatrix} = \sum_{\text{all } m} \begin{pmatrix} j_{11} & j_{12} & j_{13} \\ m_{11} & m_{12} & m_{13} \end{pmatrix} \begin{pmatrix} j_{21} & j_{22} & j_{23} \\ m_{21} & m_{22} & m_{23} \end{pmatrix} \times \begin{pmatrix} j_{31} & j_{32} & j_{33} \\ m_{31} & m_{32} & m_{33} \end{pmatrix} \begin{pmatrix} j_{11} & j_{21} & j_{31} \\ m_{11} & m_{21} & m_{31} \end{pmatrix} \times \begin{pmatrix} j_{12} & j_{22} & j_{32} \\ m_{12} & m_{22} & m_{32} \end{pmatrix} \begin{pmatrix} j_{13} & j_{23} & j_{33} \\ m_{13} & m_{23} & m_{33} \end{pmatrix}. \tag{108.11}$$

This quantity can also be represented as a sum of products of three $6j$ symbols:

$$\begin{Bmatrix} j_{11} & j_{12} & j_{13} \\ j_{21} & j_{22} & j_{23} \\ j_{31} & j_{32} & j_{33} \end{Bmatrix} = \sum_{j} (-1)^{2j}(2j+1) \begin{Bmatrix} j_{11} & j_{21} & j_{31} \\ j_{32} & j_{33} & j \end{Bmatrix} \times \begin{Bmatrix} j_{12} & j_{22} & j_{32} \\ j_{21} & j & j_{23} \end{Bmatrix} \begin{Bmatrix} j_{13} & j_{23} & j_{33} \\ j & j_{11} & j_{12} \end{Bmatrix}. \tag{108.12}$$

The equivalence of (108.11) and (108.12) can be verified by substituting definition (108.2) into (108.12) and using the orthogonality properties of the $3j$ symbols.

The $9j$ symbol has a high degree of symmetry, following directly from definition (108.11) and the symmetry properties of the $3j$ symbols. It is readily verified that interchange of any two of its rows or any two of its columns multiplies the $9j$ symbol by $(-1)^{\Sigma j}$. Moreover, the $9j$ symbol is unchanged by transposition, that is, by interchange of rows and columns.

Scalars of still higher orders depend on still more parameters j. Clearly, their number must always be a multiple of three (the $3nj$ symbols). We shall not discuss the properties of these quantities here. We mention only that for every $n > 3$ there is more than one distinct, mutually irreducible type of $3nj$ symbol. Thus there are two different types of $12j$ symbols[19].

§ 109. Matrix elements in the addition of angular momenta

Consider once more a system made up of two parts (to be called subsystems 1 and 2), and let $f^{(1)}_{kq}$ be a spherical tensor characterizing the first subsystem. According to (107.6), its

[18] According to the general contraction rule (108.1), the arguments m in the last three $3j$ symbols should be written with minus signs and a factor $(-1)^{\Sigma(j-m)}$ should be inserted under the summation sign. Using property (106.6) of the $3j$ symbols, however, and noting that in the present case the sum $\sum m$ of all nine numbers m is zero, as is readily seen, we arrive at definition (108.11).

[19] For a fuller account of the theory of $9j$ symbols and the properties of $3nj$ symbols, see the book by Edmonds cited on p. 272 and the books: A. P. Yutsis, I. B. Levinson, V. V. Vanagas, *The Mathematical Apparatus of the Theory of Angular Momentum*, Vilnius, 1960; D. A. Varshalovich, A. N. Moskalev, V. K. Khersonskii, *Quantum Theory of Angular Momentum*, Moscow: Nauka, 1975.

Table 10. Formulas for $6j$ symbols

$\begin{Bmatrix} a & b & c \\ 0 & c & b \end{Bmatrix} = \frac{(-1)^s}{\sqrt{(2b+1)(2c+1)}}, \qquad s = a+b+c$
$\begin{Bmatrix} a & b & c \\ \frac{1}{2} & c-\frac{1}{2} & b+\frac{1}{2} \end{Bmatrix} = (-1)^s \left[\frac{(s-2b)(s-2c+1)}{(2b+1)(2b+2)2c(2c+1)} \right]^{1/2}$
$\begin{Bmatrix} a & b & c \\ \frac{1}{2} & c-\frac{1}{2} & b-\frac{1}{2} \end{Bmatrix} = (-1)^s \left[\frac{(s+1)(s-2a)}{2b(2b+1)2c(2c+1)} \right]^{1/2}$
$\begin{Bmatrix} a & b & c \\ 1 & c-1 & b-1 \end{Bmatrix} = (-1)^s \left[\frac{s(s+1)(s-2a-1)(s-2a)}{(2b-1)2b(2b+1)(2c-1)2c(2c+1)} \right]^{1/2}$
$\begin{Bmatrix} a & b & c \\ 1 & c-1 & b \end{Bmatrix} = (-1)^s \left[\frac{2(s+1)(s-2a)(s-2b)(s-2c+1)}{2b(2b+1)(2b+2)(2c-1)2c(2c+1)} \right]^{1/2}$
$\begin{Bmatrix} a & b & c \\ 1 & c-1 & b+1 \end{Bmatrix} = (-1)^s \left[\frac{(s-2b-1)(s-2b)(s-2c+1)(s-2c+2)}{(2b+1)(2b+2)(2b+3)(2c-1)2c(2c+1)} \right]^{1/2}$
$\begin{Bmatrix} a & b & c \\ 1 & c & b \end{Bmatrix} = (-1)^{s+1} \frac{2[b(b+1)+c(c+1)-a(a+1)]}{[2b(2b+1)(2b+2)2c(2c+1)(2c+2)]^{1/2}}$

matrix elements with respect to the wave functions of that same subsystem are defined by

$$\langle n_1' j_1' m_1' | f_{kq}^{(1)} | n_1 j_1 m_1 \rangle = i^k (-1)^{j_{1\,\max} - m_1'} \begin{pmatrix} j_1' & k & j_1 \\ -m_1' & q & m_1 \end{pmatrix} \times \langle n_1' j_1' \| f_k^{(1)} \| n_1 j_1 \rangle. \tag{109.1}$$

We now ask how to calculate matrix elements of these same quantities with respect to wave functions of the complete system. We shall show how they can be expressed in terms of the same reduced matrix elements that occur in (109.1).

The states of the complete system are specified by the quantum numbers $j_1 j_2 J M n_1 n_2$ (J and M are the magnitude and projection of the angular momentum of the complete system). Since $f_{kq}^{(1)}$ belongs to subsystem 1, its operator commutes with the angular-momentum operator of subsystem 2. Its matrix is consequently diagonal in j_2, and also in the other quantum numbers n_2 of that subsystem. For brevity we shall omit these indices (j_2, n_2) and write the matrix elements sought as

$$\langle n_1' j_1' J' M' | f_{kq}^{(1)} | n_1 j_1 J M \rangle.$$

By (107.6), their dependence on M is given by

$$\langle n_1' j_1' J' M' | f_{kq}^{(1)} | n_1 j_1 J M \rangle = i^k (-1)^{J_{\max} - M'} \begin{pmatrix} J' & k & J \\ -M' & q & M \end{pmatrix} \times \langle n_1' j_1' J' \| f_k^{(1)} \| n_1 j_1 J \rangle. \tag{109.2}$$

To establish the relation between the reduced matrix elements on the right-hand sides of (109.1) and (109.2), we write, by the definition of matrix elements,

$$\begin{aligned} \langle n_1' j_1' J' M' | f_{kq}^{(1)} | n_1 j_1 J M \rangle &= \int \psi_{J'M'}^* \hat{f}_{kq}^{(1)} \psi_{JM} \, dq = \\ &= \sum_{m_1 m_1'} (-1)^{j_1' - j_2 + M' - M} \sqrt{(2J'+1)(2J+1)} \begin{pmatrix} j_1' & j_2 & J' \\ m_1' & m_2 & -M' \end{pmatrix} \\ &\quad \times \begin{pmatrix} j_1 & j_2 & J \\ m_1 & m_2 & -M \end{pmatrix} \langle n_1' j_1' m_1' | f_{kq}^{(1)} | n_1 j_1 m_1 \rangle. \end{aligned}$$

Substitution of (109.1) and (109.2) into this expression, followed by comparison of the resulting relation with (108.4), shows that the ratio of the reduced matrix elements in (109.1) and (109.2) must be proportional to a particular $6j$ symbol. Careful comparison of the two relations gives the following final formula:

$$\begin{aligned} \langle n_1' j_1' J' \| f_k^{(1)} \| n_1 j_1 J \rangle &= (-1)^{j_{1\,\max} + j_2 + J_{\min} + k} \sqrt{(2J+1)(2J'+1)} \\ &\quad \times \begin{Bmatrix} j_1' & J' & j_2 \\ J & j_1 & k \end{Bmatrix} \langle n_1' j_1' \| f_k^{(1)} \| n_1 j_1 \rangle \end{aligned} \tag{109.3}$$

(here $j_{1\,\max}$ is the larger of j_1, j_1', while $J_{\min}$ is the smaller of J, J'). The corresponding formula for reduced matrix elements of a spherical tensor belonging to the second subsystem is

$$\begin{aligned} \langle n_2' j_2' J' \| f_k^{(2)} \| n_2 j_2 J \rangle &= (-1)^{j_1 + j_{2\,\min} + J_{\max} + k} \sqrt{(2J+1)(2J'+1)} \\ &\quad \times \begin{Bmatrix} j_2' & J' & j_1 \\ J & j_2 & k \end{Bmatrix} \langle n_2' j_2' \| f_k^{(2)} \| n_2 j_2 \rangle. \end{aligned} \tag{109.4}$$

The lack of complete symmetry between (109.3) and (109.4), in the exponent of -1, is connected with the dependence of the wave-function phase on the order in which the angular momenta are added. This difference must be kept in mind when matrix elements have to be evaluated for both subsystems at once.

We next derive a useful formula for matrix elements, with respect to wave functions of the complete system, of the scalar product (see definition (107.4)) of two spherical tensors of the same rank k which belong to different subsystems (and therefore commute with one another). According to (107.10), these matrix elements are expressed through the reduced matrix elements of each tensor (with respect to wave functions of the complete system) as follows:

$$\begin{aligned}\langle n_1'n_2'j_1'j_2'JM|(f_k^{(1)}f_k^{(2)})_{00}|n_1n_2j_1j_2JM\rangle &= \\ &= \frac{1}{2J+1}\sum_{J''}\langle n_1'j_1'J\|f_k^{(1)}\|n_1j_1J''\rangle\langle n_2'j_2'J''\|f_k^{(2)}\|n_2j_2J\rangle\end{aligned}$$

(we have used here the fact that the matrix of a quantity belonging to either subsystem is diagonal in the quantum numbers of the other subsystem). Substituting (109.3) and (109.4) here and using the summation formula (108.8), we obtain the desired formula, which expresses the matrix elements of the scalar product through the reduced matrix elements of each tensor with respect to the wave functions of the corresponding subsystems:

$$\begin{aligned}\langle n_1'n_2'j_1'j_2'JM|(f_k^{(1)}f_k^{(2)})_{00}|n_1n_2j_1j_2JM\rangle &= \\ &= (-1)^{j_{1\,\mathrm{min}}+j_{2\,\mathrm{max}}+J}\begin{Bmatrix}J & j_2' & j_1'\\ k & j_1 & j_2\end{Bmatrix} \\ &\times\langle n_1'j_1'\|f_k^{(1)}\|n_1j_1\rangle\langle n_2'j_2'\|f_k^{(2)}\|n_2j_2\rangle.\end{aligned} \tag{109.5}$$

§ 110. Matrix elements in axially symmetric systems

The integral of a product of three D-functions is the starting point for calculating matrix elements of quantities that characterize systems of the symmetric-top type.

To derive the required expression, consider again expansion (106.11),

$$\psi_{j_1m_1}\psi_{j_2m_2} = \sum_j \langle jm|m_1m_2\rangle\psi_{jm}, \qquad m = m_1+m_2,$$

and subject both sides to a finite rotation of the coordinate system. Each ψ-function transforms by (58.7), and hence

$$\sum_{m_1'm_2'} D^{(j_1)}_{m_1'm_1}D^{(j_2)}_{m_2'm_2}\psi_{j_1m_1'}\psi_{j_2m_2'} = \sum_j\sum_{m'}\langle jm|m_1m_2\rangle D^{(j)}_{m'm}\psi_{jm'}.$$

We now expand the functions $\psi_{jm'}$ on the right by means of (106.9). Comparing the coefficients of corresponding products $\psi_{j_1m_1}$, $\psi_{j_2m_2}$ then gives

$$D^{(j_1)}_{m_1'm_1}(\omega)D^{(j_2)}_{m_2'm_2}(\omega) = \sum_j \langle m_1'm_2'|jm'\rangle D^{(j)}_{m'm}(\omega)\langle m_1m_2|jm\rangle \tag{110.1}$$

with $m = m_1 + m_2$ and $m' = m_1' + m_2'$; here ω collectively denotes the three Euler angles α, β, and γ. In terms of $3j$-symbols, the same result reads

$$D^{(j_1)}_{m_1'm_1}(\omega)D^{(j_2)}_{m_2'm_2}(\omega) = \sum_j (2j+1)\begin{pmatrix} j_1 & j_2 & j \\ m_1' & m_2' & -m' \end{pmatrix}\times \\ \times\begin{pmatrix} j_1 & j_2 & j \\ m_1 & m_2 & -m \end{pmatrix} D^{(j)*}_{-m',-m}(\omega), \tag{110.2}$$

where property (58.19) of the D-functions has also been used.

Multiply the two sides of (110.2) by $D^{(j)}_{-m',-m}(\omega)$ and integrate over $d\omega$, applying the orthogonality relation (58.20). The outcome is

$$\int D^{(j_1)}_{m_1'm_1}(\omega)D^{(j_2)}_{m_2'm_2}(\omega)D^{(j_3)}_{m_3'm_3}(\omega)\frac{d\omega}{8\pi^2} = \\ = \begin{pmatrix} j_1 & j_2 & j_3 \\ m_1' & m_2' & m_3' \end{pmatrix}\begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix}. \tag{110.3}$$

An evident relabelling of the indices has been made here to display the symmetry more fully. Equation (110.3) is the desired formula[20].

Let $f_{kq'}$ be a spherical tensor of rank k describing the top in the body-fixed coordinate axes $x'y'z' \equiv \xi\eta\zeta$, with the ζ-axis directed along the top axis. An electric or magnetic multipole-moment tensor is one possible example. Denote by f_{kq} the components of this same tensor referred to the fixed coordinate axes xyz. The matrix of finite rotations relates the two sets of components according to

$$f_{kq} = \sum_{q'} D^{(k)}_{q'q}(\omega) f_{kq'}. \tag{110.4}$$

The wave functions for rotation of the system as a whole differ from D-functions only through their normalization:

$$\psi_{jm\mu} = i^j \sqrt{\frac{2j+1}{8\pi^2}} D^{(j)}_{\mu m}(\omega). \tag{110.5}$$

Here j is the system's total angular momentum, m is its projection on the fixed z-axis, and μ is its projection on the system axis. The phase factor is chosen so that, when j is integral and $\mu = 0$, (110.5) becomes an eigenfunction of free angular momentum (compare (103.8)). Evaluation, between these functions, of the matrix element of (110.4), with (110.3) and the representation (58.19) for the complex-conjugate D-function, yields

$$\langle j'\mu'm'|f_{kq}|j\mu m\rangle = i^{j-j'}(-1)^{\mu'-m'}\sqrt{(2j+1)(2j'+1)}\times \\ \times\begin{pmatrix} j' & k & j \\ -\mu' & q' & \mu \end{pmatrix}\begin{pmatrix} j' & k & j \\ -m' & q & m \end{pmatrix}\langle \mu'|f_{kq'}|\mu\rangle, \tag{110.6}$$

[20]For integral values $j_1 = l_1$, $j_2 = l_2$, $j_3 = l_3$ and for $m_1' = m_2' = m_3' = 0$, the functions $D^{(l)}_{0m}$ reduce, by (58.25), to spherical functions. Formula (110.3) then supplies the expression for the integral of three spherical functions (107.14).

where $q' = \mu' - \mu$ and $q = m' - m$.

Formula (110.6) answers the question posed. It fixes the dependence of the matrix elements on j, j' and on their projections m, m'. Their dependence on the quantum numbers μ, μ' necessarily remains unspecified: the possible values of these numbers are determined by the system's "internal" states, between which the "internal" matrix element $\langle\mu'|f_{kq'}|\mu\rangle$ is taken.

The m, m' dependence of the matrix elements in (110.6) is the one common to every system of specified total angular momentum. Separating it by introducing reduced matrix elements as in (107.6), we obtain

$$\begin{aligned}\langle j'\mu'\|f_k\|j\mu\rangle = i^{j-j'-k}(-1)^{j_{\max}-\mu'}&\sqrt{(2j+1)(2j'+1)}\times\\ \times&\begin{pmatrix} j' & k & j\\ -\mu' & q' & \mu\end{pmatrix}\langle\mu'|f_{kq'}|\mu\rangle.\end{aligned}\tag{110.7}$$

For fixed m, sum the squared modulus of (110.6) over every value of the final quantum number m' (and over $q = m' - m$). The result is independent of m and, by the general rule (107.11), is

$$\begin{aligned}\sum_{qm'}|\langle j'\mu'm'|f_{kq}|j\mu m\rangle|^2 &= \frac{1}{2j+1}|\langle j'\mu'\|f_k\|j\mu\rangle|^2 =\\ &= (2j'+1)\begin{pmatrix} j' & k & j\\ -\mu' & q' & \mu\end{pmatrix}^2|\langle\mu'|f_{kq'}|\mu\rangle|^2.\end{aligned}\tag{110.8}$$

As they must be, the Hermiticity relations (107.9) for the reduced matrix elements (110.7) in the xyz coordinates agree with relations (107.8),

$$\langle\mu|f_{kq'}|\mu'\rangle = (-1)^{k-q'}\langle\mu'|f_{k,-q'}|\mu\rangle^*,$$

for the matrix elements in the $\xi\eta\zeta$ coordinates.

The rotation of an axially symmetric system such as a diatomic molecule (or an axial nucleus) is specified by only two angles, $\alpha\equiv\varphi$ and $\beta\equiv\theta$, which determine the direction of the system axis. Its rotational wave function differs from (110.5) by the omission of the factor $e^{i\mu\gamma}/\sqrt{2\pi}$ (compare the note on p. 389). This difference nevertheless leaves the matrix elements unchanged. Indeed, since the entire γ dependence of $D^{(j)}_{m'm}(\alpha,\beta,\gamma)$ is contained in the factor $e^{im'\gamma}$, relation (110.3) may be written

in the form

$$\begin{aligned}\delta_{m',0}\int D^{(j_1)}_{m_1'm_1}(\alpha,\beta,0)D^{(j_2)}_{m_2'm_2}(\alpha,\beta,0)D^{(j_3)}_{m_3'm_3}(\alpha,\beta,0)\frac{\sin\alpha\, d\alpha\, d\beta}{4\pi} &=\\ = \begin{pmatrix} j_1 & j_2 & j_3\\ m_1 & m_2 & m_3\end{pmatrix}\begin{pmatrix} j_1 & j_2 & j_3\\ m_1' & m_2' & m_3'\end{pmatrix}&,\end{aligned}$$

where $m' = m_1' + m_2' + m_3'$, and the value of the integral is unaltered. The selection rule for the projection of angular momentum on the system axis likewise retains its former form, $\mu' - \mu = q'$. It now follows from the orthogonality of the electronic wave functions, as a consequence of the molecule's symmetry about the ζ-axis. In (110.6) and (110.7), $\langle\mu'|f_{kq'}|\mu\rangle$ must in this case be understood as a matrix element between electronic states with the nuclei held fixed.

CHAPTER XV

Motion in a Magnetic Field

§ 111. The Schrödinger equation in a magnetic field

A particle with spin also has a definite "intrinsic" magnetic moment $\boldsymbol{\mu}$. The corresponding quantum-mechanical operator is proportional to the spin operator $\hat{\mathbf{s}}$ and can therefore be written as

$$\hat{\boldsymbol{\mu}} = \frac{\mu}{s}\hat{\mathbf{s}}, \tag{111.1}$$

where s is the magnitude of the particle's spin, while μ is a constant characteristic of the particle. The eigenvalues of the projection of the magnetic moment are $\mu_z = \mu\sigma/s$. It follows that the coefficient μ (which is usually referred to simply as the magnitude of the magnetic moment) is the largest possible value of μ_z, attained for the spin projection $\sigma = s$.

The ratio $\mu/(\hbar s)$ is the ratio of the particle's intrinsic magnetic moment to its intrinsic mechanical angular momentum (when both point along the z axis). For ordinary (orbital) angular momentum this ratio is known to be $e/(2mc)$ (see II, § 44). The proportionality coefficient between the intrinsic magnetic moment and the particle's spin, however, is different. For the electron it is $-|e|/mc$, twice the ordinary value (this value follows theoretically from the relativistic Dirac wave equation; see IV, § 33). Consequently, the intrinsic magnetic moment of the electron (spin $1/2$) is $-\mu_B$, where

$$\mu_B = \frac{|e|\hbar}{2mc} = 0.927 \cdot 10^{-20}\,\frac{\text{erg}}{\text{G}}. \tag{111.2}$$

This quantity is called the *Bohr magneton*.

Magnetic moments of heavy particles are customarily measured in *nuclear magnetons*, defined as $e\hbar/(2m_p c)$, where m_p is the proton mass. Experiment gives 2.79 nuclear magnetons for the proton's intrinsic magnetic moment, with the moment directed along the spin. The neutron's magnetic moment is directed opposite to its spin and is equal to 1.91 nuclear magnetons.

Notice that the quantities $\boldsymbol{\mu}$ and $\mathbf{s}$ appearing on the two sides of (111.1) have, as they should, the same vector character: both are axial vectors. An analogous equality for the electric dipole moment $\mathbf{d}$ ($\mathbf{d} = \text{const} \cdot \mathbf{s}$) would conflict with symmetry under coordinate inversion: inversion would change the relative sign of the two sides of the equality[1].

Within nonrelativistic quantum mechanics a magnetic field can be treated only as an external field. The magnetic interaction among particles is a relativistic effect, whose inclusion requires a consistently relativistic theory.

[1] Such an equality (and hence the existence of an electric moment of an elementary particle) would also conflict with symmetry under time reversal. Changing the sign of time leaves $\mathbf{d}$ unchanged but reverses the spin. This is evident, for example, from the definitions of these quantities for orbital motion: only coordinates enter the definition of $\mathbf{d}$, whereas the particle's velocity also enters that of the angular momentum.

In classical theory the Hamiltonian function of a charged particle in an electromagnetic field is

$$H = \frac{1}{2m}\left(\mathbf{p} - \frac{e}{c}\mathbf{A}\right)^2 + e\varphi,$$

where φ and $\mathbf{A}$ are respectively the scalar and vector potentials of the field, and $\mathbf{p}$ is the particle's generalized momentum (see II, § 16). If the particle has no spin, passage to quantum mechanics is made in the usual way: the generalized momentum is replaced by the operator $\hat{\mathbf{p}} = -i\hbar\boldsymbol{\nabla}$, giving the Hamiltonian[2]

$$\hat{H} = \frac{1}{2m}\left(\hat{\mathbf{p}} - \frac{e}{c}\mathbf{A}\right)^2 + e\varphi. \tag{111.3}$$

For a particle with spin, this procedure is insufficient. Its intrinsic magnetic moment interacts directly with the magnetic field. This interaction is altogether absent from the classical Hamiltonian function, since spin itself, being a purely quantum effect, vanishes in the classical limit. The correct Hamiltonian is obtained by adding to (111.3) the term $-\hat{\boldsymbol{\mu}}\mathbf{H}$, corresponding to the energy of a magnetic moment $\boldsymbol{\mu}$ in a field $\mathbf{H}$. Thus,

the Hamiltonian of a particle with spin is[3]

$$\hat{H} = \frac{1}{2m}\left(\hat{\mathbf{p}} - \frac{e}{c}\mathbf{A}\right)^2 - \hat{\boldsymbol{\mu}}\mathbf{H} + e\varphi. \tag{111.4}$$

When expanding the square $(\hat{\mathbf{p}} - (e/c)\mathbf{A})^2$, one must bear in mind that the operator $\hat{\mathbf{p}}$ does not in general commute with the vector $\mathbf{A}$, which is a function of the coordinates. Hence one must write

$$\hat{H} = \frac{1}{2m}\hat{\mathbf{p}}^2 - \frac{e}{2mc}(\hat{\mathbf{p}}\mathbf{A} + \mathbf{A}\hat{\mathbf{p}}) + \frac{e^2}{2mc^2}\mathbf{A}^2 - \frac{\mu}{s}\hat{\mathbf{s}}\mathbf{H} + e\varphi. \tag{111.5}$$

By the commutation rule (16.4) for the momentum operator and an arbitrary function of the coordinates,

$$\hat{\mathbf{p}}\mathbf{A} - \mathbf{A}\hat{\mathbf{p}} = -i\hbar\nabla\cdot\mathbf{A}. \tag{111.6}$$

Thus $\hat{\mathbf{p}}$ and $\mathbf{A}$ commute if $\nabla\cdot\mathbf{A} = 0$. In particular, this is so for a uniform field if its vector potential is chosen as

$$\mathbf{A} = (1/2)(\mathbf{H}\times\mathbf{r}). \tag{111.7}$$

The equation $i\hbar\,\partial\Psi/\partial t = \hat{H}\Psi$, with the Hamiltonian (111.4), generalizes the Schrödinger equation to the presence of a magnetic field. The wave functions on which the Hamiltonian acts in this equation are symmetric spinors of rank $2s$.

The wave functions of a particle in an electromagnetic field have an ambiguity associated with the ambiguity of the field potentials. As is known (see II, § 18), the latter are defined only up to a *gauge transformation*

$$\mathbf{A} \to \mathbf{A} + \boldsymbol{\nabla} f, \qquad \varphi \to \varphi - \frac{1}{c}\frac{\partial f}{\partial t}, \tag{111.8}$$

[2]Here we denote the generalized momentum by the same letter $\mathbf{p}$ as the ordinary momentum (rather than by $\mathbf{P}$ as in II, § 16), in order to emphasize that it corresponds to the same operator.

[3]Using the same letter for the magnetic field and the Hamiltonian cannot cause confusion: the letter denoting the Hamiltonian carries a hat.

where f is an arbitrary function of the coordinates and time. Such a transformation does not affect the field strengths. It is therefore clear that it must not change the solutions of the wave equation in any essential way; in particular, $|\Psi|^2$ must remain unchanged. Indeed, it is readily verified that the original equation is recovered if, together with the substitution (111.8) in the Hamiltonian, the wave function is also replaced according to

$$\Psi \to \Psi \exp\left(\frac{ie}{\hbar c} f\right). \tag{111.9}$$

This ambiguity of the wave function has no effect on any physically meaningful quantity whose definition does not explicitly involve the potentials.

In classical mechanics the generalized momentum of a particle is related to its velocity by $m\mathbf{v} = \mathbf{p} - e\mathbf{A}/c$. To find the operator $\hat{\mathbf{v}}$ in quantum mechanics, one must commute the vector $\mathbf{r}$ with the Hamiltonian. A straightforward calculation gives

$$m\hat{\mathbf{v}} = \hat{\mathbf{p}} - \frac{e}{c}\mathbf{A}, \tag{111.10}$$

exactly analogous to the classical result. The operators for the velocity components obey the commutation rules

$$\left\{\hat{v}_x, \hat{v}_y\right\} = i\frac{e\hbar}{m^2 c}H_z, \qquad \left\{\hat{v}_y, \hat{v}_z\right\} = i\frac{e\hbar}{m^2 c}H_x, \qquad \left\{\hat{v}_z, \hat{v}_x\right\} = i\frac{e\hbar}{m^2 c}H_y, \tag{111.11}$$

as is easily checked by direct calculation. Thus, in a magnetic field, the operators of the three velocity components of a charged particle do not commute. This means that the particle cannot simultaneously have definite velocity values in all three directions.

For motion in a magnetic field, time-reversal symmetry holds only if the sign of the field $\mathbf{H}$ (and of the vector potential $\mathbf{A}$) is also reversed. This means (see § 18 and 60) that the Schrödinger equation $\hat{H}\psi = E\psi$ must retain its form upon passage to the complex-conjugate quantities and reversal of the sign of $\mathbf{H}$. This is directly evident for every term in the Hamiltonian (111.4) except $-\hat{\mathbf{s}}\mathbf{H}$. Under the stated transformation, the term $-\hat{\mathbf{s}}\mathbf{H}\psi$ in the Schrödinger equation becomes $\hat{\mathbf{s}}^*\mathbf{H}\psi^*$ and, at first sight, violates the required invariance because the operator $\hat{\mathbf{s}}^*$ is not identical to $-\hat{\mathbf{s}}$.

It must be taken into account, however, that the wave function is in fact a contravariant spinor $\psi^{\lambda\mu\cdots}$, which under complex conjugation becomes the covariant spinor $\psi^{\lambda\mu\ldots *}$ (see § 60). The spinor $\psi^*_{\lambda\mu\ldots}$, on the other hand, is contravariant. Using definitions (57.4) and (57.5) to find the components of $(\hat{\mathbf{s}}\mathbf{H}\psi)^*$ and expressing them in terms of $\psi^*_{\lambda\mu\ldots}$, one verifies that time reversal leads to a Schrödinger equation for the components $\psi^*_{\lambda\mu\ldots}$ of the same form as the original equation for the components $\psi^{\lambda\mu\cdots}$.

§ 112. Motion in a uniform magnetic field

We shall determine the energy levels of a particle in a constant uniform magnetic field (L. D. Landau, 1930).

For a uniform field it is convenient here to choose the vector potential not in the form (111.7), but as follows:

$$A_x = -Hy, \qquad A_y = A_z = 0. \tag{112.1}$$

(the z axis is chosen along the field). The Hamiltonian then takes the form

$$\hat{H} = \frac{1}{2m}\left(\hat{p}_x + \frac{eH}{c}y\right)^2 + \frac{\hat{p}_y^2}{2m} + \frac{\hat{p}_z^2}{2m} - \frac{\mu}{s}\hat{s}_z H. \tag{112.2}$$

First, we note that the operator $\hat{s}_z$ commutes with the Hamiltonian (since the latter contains no operators for the other spin components). Thus the z projection of the spin is conserved, and $\hat{s}_z$ may consequently be replaced by its eigenvalue $s_z = \sigma$. The spin dependence of the wave function then ceases to be relevant, and ψ in the Schrödinger equation may be understood as an ordinary coordinate function. For this function we have

$$\frac{1}{2m}\left[\left(\hat{p}_x + \frac{eH}{c}y\right)^2 + \hat{p}_y^2 + \hat{p}_z^2\right]\psi - \frac{\mu}{s}\sigma H\psi = E\psi. \tag{112.3}$$

The Hamiltonian in this equation contains neither x nor z explicitly. Hence the operators $\hat{p}_x$ and $\hat{p}_z$ (differentiation with respect to x and z) also commute with it; that is, the x and z components of the generalized momentum are conserved. Accordingly, we seek ψ in the form

$$\psi = \exp\left[\frac{i}{\hbar}(p_x x + p_z z)\right]\chi(y). \tag{112.4}$$

The eigenvalues p_x and p_z range over all values from $-\infty$ to ∞. Since $A_z = 0$, the z component of the generalized momentum is identical with the component mv_z of the ordinary momentum. Thus the particle velocity along the field may have any value; one may say that motion along the field is "not quantized."

Substitution of (112.4) into (112.3) gives the following equation for $\chi(y)$:

$$\chi'' + \frac{2m}{\hbar^2}\left[\left(E + \frac{\mu\sigma}{s}H - \frac{p_z^2}{2m}\right) - \frac{m}{2}\omega_H^2(y - y_0)^2\right]\chi = 0, \tag{112.5}$$

where we have introduced $y_0 = -cp_x/eH$ and

$$\omega_H = \frac{|e|H}{mc}. \tag{112.6}$$

Equation (112.5) has the same form as the Schrödinger equation (23.6) for a linear oscillator of frequency ω_H. We can therefore conclude at once that the expression in parentheses in (112.5), which plays the part of the oscillator energy, can take the values $(n + 1/2)\hbar\omega_H$, where $n = 0, 1, 2, \ldots$

We thus obtain the following expression for the energy levels of a particle in a uniform magnetic field:

$$E = \left(n + \frac{1}{2}\right)\hbar\omega_H + \frac{p_z^2}{2m} - \frac{\mu\sigma}{s}H. \tag{112.7}$$

The first term in this expression gives the discrete energy values associated with motion in the plane perpendicular to the field; they are called *Landau levels*. For an electron, $\mu/s = -|e|\hbar/mc$, and (112.7) becomes

$$E = \left(n + \frac{1}{2} + \sigma\right)\hbar\omega_H + \frac{p_z^2}{2m}. \tag{112.8}$$

The eigenfunctions $\chi_n(y)$ belonging to the energy levels (112.7) are given by (23.12), with the corresponding change of notation ($a_H = \sqrt{\hbar/m\omega_H}$):

$$\chi_n(y) = \frac{1}{\pi^{1/4} a_H^{1/2} \sqrt{2^n n!}} \exp\left(-\frac{(y-y_0)^2}{2a_H^2}\right) H_n\left(\frac{y-y_0}{a_H}\right). \tag{112.9}$$

In classical mechanics, particles move in a circle with a fixed center in the plane perpendicular to the field **H** (the xy plane). The quantity y_0, which is conserved in the quantum case, corresponds to the classical y coordinate of the center of the circle. Along with it, the quantity $x_0 = (cp_y/eH) + x$ is also conserved (it is readily verified that its operator commutes with the Hamiltonian (112.2)). This quantity corresponds to the classical x coordinate of the center of the circle[4].

The operators $\hat{x}_0$ and $\hat{y}_0$, however, do not commute with each other, and hence the coordinates x_0 and y_0 cannot simultaneously have definite values.

Since (112.7) does not contain p_x, which ranges continuously, the energy levels have a continuous degeneracy. The degeneracy becomes finite, however, if motion in the xy plane is restricted to a large but finite area $S = L_x L_y$. The number of distinct (now discrete) values of p_x in an interval Δp_x is $(L_x/2\pi\hbar)\Delta p_x$. All values of p_x for which the orbit center lies within S are allowed (we neglect the orbit radius in comparison with the large length L_y). From the conditions $0 < y_0 < L_y$, we have $\Delta p_x = eHL_y/c$. The number of states (for given n and p_z) is therefore $eHS/2\pi\hbar c$. If the region of motion is also bounded along the z axis (with length L_z), the number of possible values of p_z in an interval Δp_z is $(L_z/2\pi\hbar)\Delta p_z$, and the number of states in that interval is

$$\frac{eHS}{2\pi\hbar c}\frac{L_z}{2\pi\hbar}\Delta p_z = \frac{eHV}{4\pi^2\hbar^2 c}\Delta p_z. \tag{112.10}$$

For an electron there is a further degeneracy: the energy levels (112.8) coincide for states with quantum numbers n, $\sigma = 1/2$ and $n+1$, $\sigma = -1/2$.

Problems

Problem

1. Find the wave functions of an electron in a uniform magnetic field for states in which it has definite values of the momentum and angular momentum along the direction of the field.

Solution. In cylindrical coordinates ρ, φ, z, with the z axis directed along the field, the vector potential in the gauge (111.7) has components $A_\varphi = H\rho/2$, $A_z = A_\rho = 0$, and the

[4] Indeed, for classical motion in a circle of radius cmv_t/eH (v_t is the projection of the velocity onto the xy plane; see II, § 21), we have

$$y_0 = -cp_x/eH = -(cm/eH)v_x + y.$$

It is evident from this expression that y is the coordinate of the center of the circle. Its other coordinate is

$$x_0 = (cm/eH)v_y + x = cp_y/eH + x.$$

Schrödinger equation is[5]

$$-\frac{\hbar^2}{2M}\left[\frac{1}{\rho}\frac{\partial}{\partial\rho}\left(\rho\frac{\partial\psi}{\partial\rho}\right)+\frac{\partial^2\psi}{\partial z^2}+\frac{1}{\rho^2}\frac{\partial^2\psi}{\partial\varphi^2}\right]-\frac{i\hbar\omega_H}{2}\frac{\partial\psi}{\partial\varphi}+\frac{M\omega_H^2}{8}\rho^2\psi=E\psi. \tag{1}$$

We seek a solution in the form

$$\psi=\frac{e^{im\varphi}}{\sqrt{2\pi}}e^{ip_z z/\hbar}R(\rho),$$

and obtain for the radial function the equation

$$\frac{\hbar^2}{2M}\left(R''+\frac{1}{\rho}R'-\frac{m^2}{\rho^2}R\right)+\left[E-\frac{p_z^2}{2M}-\frac{M\omega_H^2}{8}\rho^2-\frac{\hbar\omega_H m}{2}\right]R=0.$$

Introducing the new independent variable $\xi=(M\omega_H/2\hbar)\rho^2$, we rewrite this equation as

$$\xi R''+R'+\left(-\frac{\xi}{4}+\beta-\frac{m^2}{4\xi}\right)R=0,\qquad \beta=\frac{1}{\hbar\omega_H}\left(E-\frac{p_z^2}{2M}\right)-\frac{m}{2}.$$

As $\xi\to\infty$, the required function behaves as $e^{-\xi/2}$, while as $\xi\to 0$ it behaves as $\xi^{|m|/2}$. We therefore seek a solution in the form

$$R(\xi)=e^{-\xi/2}\xi^{|m|/2}w(\xi),$$

and for $w(\xi)$ obtain the confluent hypergeometric function equation:

$$w=F\left\{-\left(\beta-\frac{|m|+1}{2}\right),|m|+1,\xi\right\}.$$

For the wave function to be finite everywhere, $\beta-(|m|+1)/2$ must be a nonnegative integer n_ρ. The energy levels are then

$$E=\hbar\omega_H\left(n_\rho+\frac{|m|+m+1}{2}\right)+\frac{p_z^2}{2M},$$

which is equivalent to (112.7). The radial wave functions belonging to these levels are

$$R_{n_\rho m}(\rho)=\frac{1}{a_H^{1+|m|}}\left[\frac{(|m|+n_\rho)!}{2^{|m|}n_\rho!|m|!^2}\right]^{1/2}\exp\left(-\frac{\rho^2}{4a_H^2}\right)\rho^{|m|}F\left(-n_\rho,|m|+1,\frac{\rho^2}{2a_H^2}\right), \tag{2}$$

where $a_H=\sqrt{\hbar/M\omega_H}$; the functions are normalized by $\int_0^\infty R^2\rho\,d\rho=1$. Here the hypergeometric function is a generalized Laguerre polynomial.

[5]We write the electron charge as $e=-|e|$, and denote its mass here by M in order to distinguish it from the angular momentum m. The particle-spin term, which is immaterial for this problem, is omitted.

PROBLEM

2. Determine the lowest energy level corresponding to a bound state of an electron in a shallow potential well $U(r)$ ($|U| \ll \hbar^2/ma^2$, where a is the range of the forces in the well), in the additional presence of a uniform magnetic field (Yu. A. Bychkov, 1960).

SOLUTION. The condition imposed on the field $U(r)$ ensures that, in the absence of the magnetic field, perturbation theory is applicable to it; there are then no bound states in the well (see § 45). When the magnetic field is also present, $U(r)$ may be treated as a perturbation only for the motion in the plane transverse to $\mathbf{H}$, whose (discrete) energy-spectrum character is not changed by adding U. The character of the motion along $\mathbf{H}$, however, does change: as will become clear, infinite motion becomes finite, and the spectrum correspondingly changes from continuous to discrete. The well field therefore cannot be treated by perturbation theory for this motion.

Accordingly, on separating variables in the Schrödinger equation (equation (1) of the preceding problem, with the term $U\psi$ added on the left), we retain the radial functions $R(\rho)$ in their former form (2); the lowest level corresponds to the quantum numbers $n_\rho = m = 0$. Substituting $\psi = R_{00}(\rho)\chi(z)$ into the Schrödinger equation, then multiplying it by $R_{00}(\rho)$ and integrating over $\rho\, d\rho$, we obtain for $\chi(z)$ the equation

$$-\frac{\hbar^2}{2m}\chi'' + \overline{U}(z)\chi = \varepsilon\chi, \tag{3}$$

where $\varepsilon = E - \hbar\omega_H/2$ and

$$\overline{U}(z) = \int_0^\infty U\left(\sqrt{z^2+\rho^2}\right) R_{00}^2(\rho)\rho\, d\rho$$

(m again denotes the particle mass). This equation has the form of the Schrödinger equation for one-dimensional motion in the potential well $\overline{U}(z)$, with ε as the energy of that motion. We may therefore use directly the result of Problem 1 in § 45, according to which the discrete energy level is

$$\varepsilon = -\frac{m}{2\hbar^2}\left[\int_{-\infty}^\infty \overline{U}(z)\, dz\right]^2 = -\frac{m}{2\hbar^2}\left[\int_{-\infty}^\infty \int_0^\infty U\left(\sqrt{z^2+\rho^2}\right) R_{00}^2(\rho)\rho\, d\rho\, dz\right]^2. \tag{4}$$

The wave function $R_{00}(\rho)$ decays over distances $\rho \sim a_H$. If the magnetic field is so weak that $a_H \gg a$, the integral over $d\rho$ is determined by the region $\rho \lesssim a$, where we may put $R_{00}(\rho) \approx R_{00}(0) = 1/a_H$. Then

$$\varepsilon = -\frac{me^2H^2}{8\pi^2\hbar^4c^2}\left(\int U(r)\, dV\right)^2 \tag{5}$$

($dV = 2\pi\rho\, d\rho\, dz \to 4\pi r^2\, dr$). In the opposite case of a strong magnetic field, when $a_H \ll a$, the integral in (4) is determined by the region $\rho \lesssim a_H$, where we may put $U(\sqrt{\rho^2+z^2}) \approx U(z)$. The integral over $d\rho$ then reduces to the normalization integral for R_{00} and is equal to 1, so that

$$\varepsilon = -\frac{2m}{\hbar^2}\left(\int_0^\infty U(z)\, dz\right)^2. \tag{6}$$

In both cases, estimation of the integrals shows that $\varepsilon \ll \hbar\omega_H$.

Problem

3. Determine the energy levels of the hydrogen atom in a magnetic field so strong that $a_H \ll a_B$, where a_B is the Bohr radius (R. J. Elliott, R. Loudon, 1960).

Solution. Under the stated condition, $\hbar\omega_H \gg me^4/\hbar^2$, the effect of the nuclear Coulomb field on the electron's motion in the plane transverse to **H** may be treated as a small perturbation. We therefore return to the situation considered in Problem 2 and may use equation (3), with

$$\overline{U}(z) = -e^2 \int_0^\infty \frac{R_{00}^2(\rho)\rho}{\sqrt{\rho^2 + z^2}}\, d\rho. \tag{7}$$

Having inserted the radial function R_{00} in this expression, below we restrict ourselves to the energy levels of the longitudinal motion associated with the lowest Landau level ($\hbar\omega_H/2$) of the transverse motion.

The ground-state wave function $\chi_0(z)$ extends over distances $|z| \lesssim a_B$ and varies slowly across them (having no nodes, it does not vanish at $z = 0$). The conditions used in the solution of Problem 1 in § 45 therefore hold for the ground level, and we can use formula (6), which was based on that solution. The logarithmically divergent integral is cut off at the upper end at distances $|z| \sim a_B$, and at the lower end at distances $|z| \sim a_H$ (where $|z| \sim \rho$ and the replacement of $\sqrt{\rho^2 + z^2}$ by $|z|$ in (7) is not permissible). We thereby find

$$\varepsilon_0 = -\frac{2me^4}{\hbar^2} \ln^2 \frac{a_B}{a_H} = -\frac{me^4}{2\hbar^2} \ln^2 \frac{\hbar^3 H}{m^2 c|e|^3}. \tag{8}$$

This formula is said to have logarithmic accuracy: it is assumed that not only the ratio a_B/a_H itself but also its logarithm is large; the numerical factor in the logarithm's argument then remains undetermined.

The excited states of the discrete spectrum are obtained as solutions of the Schrödinger equation (3) with the field $\overline{U}(z) \approx -e^2/z$ (which follows from (7) when $z \sim a_B \gg \rho$). But by the substitution $\chi = z\varphi(z)$ this equation is reduced to

$$-\frac{\hbar^2}{2m}\frac{1}{z^2}\frac{d}{dz}\left(z^2 \frac{d\varphi}{dz}\right) - \frac{e^2}{z}\varphi = \varepsilon\varphi, \tag{9}$$

which has the same form as the equation for the radial wave functions of s states in the three-dimensional Coulomb problem. The required levels are therefore given by (36.10):

$$\varepsilon_n = -\frac{me^4}{2\hbar^2 n^2}, \tag{10}$$

where $n = 1, 2, 3, \ldots$ This expression too has only logarithmic accuracy: the next correction would be small relative to the leading term only by the ratio $1/\ln(a_B/a_H)$.

Equation (9) determines the wave function only for $z > 0$; it may be continued into $z < 0$ either as $\chi(-z) = \chi(z)$ or as $\chi(-z) = -\chi(z)$. Accordingly, in the approximation considered, the levels (10) are twofold degenerate. This degeneracy is, however, lifted in higher orders in a_H/a_B.

§ 113. The atom in a magnetic field

Consider an atom placed in a uniform magnetic field **H**. Its Hamiltonian is

$$\hat{H} = \frac{1}{2m}\sum_a \left[\hat{\mathbf{p}}_a + \frac{|e|}{c}\mathbf{A}(\mathbf{r}_a)\right]^2 + U + \frac{|e|\hbar}{mc}\mathbf{H}\hat{\mathbf{S}}, \tag{113.1}$$

where the sum extends over all electrons (the electron charge is written as $-|e|$); U is the energy of interaction of the electrons with the nucleus and with one another; $\hat{\mathbf{S}} = \sum_a \hat{\mathbf{s}}_a$ is the operator of the atom's total (electronic) spin.

If the vector potential of the field is chosen in the form (111.7), then, as already noted, the operator $\hat{\mathbf{p}}$ commutes with **A**. Using this fact when expanding the square in (113.1), and denoting the Hamiltonian of the atom in the absence of a field by $\hat{H}_0$, we find

$$\hat{H} = \hat{H}_0 + \frac{|e|}{mc}\sum_a \mathbf{A}_a\hat{\mathbf{p}}_a + \frac{e^2}{2mc^2}\sum_a \mathbf{A}_a^2 + \frac{|e|\hbar}{mc}\mathbf{H}\hat{\mathbf{S}}.$$

On substituting **A** from (111.7), this becomes

$$\hat{H} = \hat{H}_0 + \frac{|e|}{2mc}\mathbf{H}\sum_a(\mathbf{r}_a \times \hat{\mathbf{p}}_a) + \frac{e^2}{8mc^2}\sum_a(\mathbf{H}\times\mathbf{r}_a)^2 + \frac{|e|\hbar}{mc}\mathbf{H}\hat{\mathbf{S}}.$$

But the vector product $\mathbf{r}_a \times \hat{\mathbf{p}}_a$ is the operator of the electron's orbital angular momentum, and summation over all the electrons gives the operator $\hbar\hat{\mathbf{L}}$ of the atom's total orbital angular momentum. Hence

$$\hat{H} = \hat{H}_0 + \mu_B(\hat{\mathbf{L}} + 2\hat{\mathbf{S}})\mathbf{H} + \frac{e^2}{8mc^2}\sum_a(\mathbf{H}\times\mathbf{r}_a)^2 \tag{113.2}$$

(μ_B is the Bohr magneton). The operator

$$\hat{\boldsymbol{\mu}}_{\text{at}} = -\mu_B(\hat{\mathbf{L}} + 2\hat{\mathbf{S}}) \tag{113.3}$$

may be regarded as the operator of the atom's "intrinsic" magnetic moment, which it possesses in the absence of a field.

An external magnetic field splits the atomic levels by removing their degeneracy with respect to the directions of the total angular momentum (the *Zeeman effect*). Let us determine the splitting energy for atomic levels characterized by definite values of the quantum numbers J, L, and S (that is, assuming LS coupling for the levels; see § 72).

We shall suppose that the magnetic field is so weak that $\mu_B H$ is small compared with the separations between the atom's energy levels, including the fine-structure intervals. The second and third terms in (113.2) may then be treated as a perturbation, the unperturbed levels being the individual components of the multiplets. In first approximation, the third term, which is quadratic in the field, may be neglected in comparison with the linear second term.

In this approximation the splitting energy ΔE is determined by the mean values of the perturbation in the (unperturbed) states differing in the value of the projection of the total angular momentum along the field. Taking this direction as the z axis, we have

$$\Delta E = \mu_B H(\overline{L}_z + 2\overline{S}_z) = \mu_B H(\overline{J}_z + \overline{S}_z). \tag{113.4}$$

The mean value $\overline{J}_z$ is simply the specified eigenvalue $J_z = M_J$. The mean value $\overline{S}_z$ can be found as follows by means of averaging in stages (cf. § 72).

First average the operator $\hat{\mathbf{S}}$ over a state of the atom with specified S, L, and J, but not M_J. The operator $\overline{\mathbf{S}}$ obtained by this averaging can be "directed" only along $\hat{\mathbf{J}}$, the sole conserved "vector" characterizing the free atom. We may therefore write

$$\overline{\mathbf{S}} = \text{const} \cdot \mathbf{J}.$$

In this form, however, the equality has only a conditional meaning, since the three components of $\mathbf{J}$ cannot simultaneously have definite values. Its z projection does have a literal meaning:

$$\overline{S}_z = \text{const} \cdot J_z = \text{const} \cdot M_J,$$

as does the equality

$$\overline{\mathbf{SJ}} = \text{const} \cdot \mathbf{J}^2 = \text{const} \cdot J(J+1),$$

obtained by multiplying both sides by $\mathbf{J}$. Bringing the conserved vector $\mathbf{J}$ under the averaging sign, we write $\overline{\mathbf{S}}\mathbf{J} = \overline{\mathbf{SJ}}$. The mean value $\overline{\mathbf{SJ}}$ is equal to the eigenvalue

$$\mathbf{SJ} = \frac{1}{2}[J(J+1) - L(L+1) + S(S+1)],$$

which it has in a state with definite values of $\mathbf{L}^2$, $\mathbf{S}^2$, and $\mathbf{J}^2$ (cf. (31.4)). Determining the constant from the second equality and substituting it in the first, we thus obtain

$$\overline{S}_z = M_J \frac{\mathbf{JS}}{\mathbf{J}^2}. \tag{113.5}$$

Combining these results and substituting them in (113.4), we find the following final expression for the splitting energy:

$$\Delta E = \mu_B g M_J H, \tag{113.6}$$

where

$$g = 1 + \frac{J(J+1) - L(L+1) + S(S+1)}{2J(J+1)} \tag{113.7}$$

is called the *Landé factor*, or *gyromagnetic factor*. Notice that $g = 1$ when there is no spin ($S = 0$, and hence $J = L$), while $g = 2$ when $L = 0$ (and hence $J = S$)[6].

Formula (113.6) gives different energies for all $2J+1$ values $M_J = -J, -J+1, \ldots, J$. In other words, the magnetic field removes completely the degeneracy of the levels with respect to angular-momentum directions, unlike an electric field, which left the levels with $M_J = \pm|M_J|$ unsplit (§ 76)[7]. The linear splitting given by (113.6) is nevertheless absent if $g = 0$ (which is possible even for $J \neq 0$, for example in the state ${}^4D_{1/2}$).

[6]The splitting described by the general formulas (113.6), (113.7) is sometimes called the anomalous Zeeman effect. This infelicitous name arose historically because, before the discovery of electron spin, the effect described by (113.6) with $g = 1$ was regarded as normal.

[7]The reasoning used in this connection for the electric case in § 76 does not apply to a magnetic field. Indeed, $\mathbf{H}$ is an axial vector and therefore changes sign under reflection in a plane containing its direction. Consequently, states transformed into one another by this operation actually refer to an atom in different fields.

We saw in § 76 that the displacement of an atomic energy level in an electric field is related to its mean electric dipole moment. A corresponding relation exists in the magnetic case. In classical theory the potential energy of a system of charges is $-\boldsymbol{\mu}\mathbf{H}$, where $\boldsymbol{\mu}$ is the system's magnetic moment. In quantum theory this is replaced by the corresponding operator, so the system's Hamiltonian is

$$\hat{H} = \hat{H}_0 - \hat{\boldsymbol{\mu}}\mathbf{H} = \hat{H}_0 - \hat{\mu}_z H.$$

Applying formula (11.16), with the field H as the parameter λ, we find for the mean magnetic moment

$$\overline{\mu}_z = -\frac{\partial \Delta E}{\partial H}, \tag{113.8}$$

where ΔE is the displacement of the energy level of the atomic state in question. Substitution of (113.6) shows that an atom in a state having a definite value M_J of the angular-momentum projection along some direction z has a mean magnetic moment along that same direction:

$$\overline{\mu}_z = -\mu_B g M_J. \tag{113.9}$$

If the atom has neither spin nor orbital angular momentum ($S = L = 0$), the second term in (113.2) produces no level displacement in first or any higher approximation (since all matrix elements of $\mathbf{L}$ and $\mathbf{S}$ vanish). The whole effect in this case is therefore due to the third term in (113.2), and in first-order perturbation theory the level displacement is the mean value

$$\Delta E = \frac{e^2}{8mc^2}\sum_a \overline{(\mathbf{H}\times\mathbf{r}_a)^2}. \tag{113.10}$$

Writing $(\mathbf{H}\times\mathbf{r}_a)^2 = H^2 r_a^2 \sin^2\theta$, where θ is the angle between $\mathbf{r}_a$ and $\mathbf{H}$, and averaging over the directions of $\mathbf{r}_a$, we obtain $\overline{\sin^2\theta} = 1 - \overline{\cos^2\theta} = 2/3$ (the wave function of a state with $L = S = 0$ is spherically symmetric, so averaging over directions is performed independently of averaging over the distances r_a). Thus

$$\Delta E = \frac{e^2}{12mc^2} H^2 \sum_a \overline{r_a^2}. \tag{113.11}$$

The magnetic moment calculated from (113.8) is now proportional to the magnitude of the field (an atom with $L = S = 0$ naturally has no magnetic moment in the absence of a field). Writing it as χH, we may regard the coefficient χ as the magnetic susceptibility of the atom. For it we obtain the following *Langevin formula* (P. Langevin, 1905):

$$\chi = -\frac{e^2}{6mc^2}\sum_a \overline{r_a^2}. \tag{113.12}$$

This quantity is negative; that is, the atom is diamagnetic[8].

[8]We mention that the Thomas–Fermi model cannot be used to calculate the mean square distance of the electrons from the nucleus. Although the integral $\int nr^2\,dr$ with the Thomas–Fermi density $n(r)$ does converge, it converges too slowly, with the result that the values obtained differ greatly from empirical ones.

If $J = 0$ but $S = L \neq 0$, the level displacement linear in the field is likewise absent, but the quadratic second-order effect from the perturbation $-\hat{\boldsymbol{\mu}}_{\mathrm{at}}\mathbf{H}$ exceeds the effect (113.11)[9]. This is because, according to the general formula (38.10), the second-order correction to an energy eigenvalue is determined by a sum of expressions whose denominators contain differences of unperturbed energy levels—in this case the fine-structure intervals of the level, which are small quantities. It was noted in § 38 that the second-order correction to a normal level is always negative. The magnetic moment in the normal state will therefore be positive; that is, an atom in the normal state with $J = 0$, $L = S \neq 0$ is paramagnetic (J. H. van Vleck, 1928).

In strong magnetic fields, when $\mu_B H$ is comparable to or greater than the fine-structure intervals, the level splitting departs from that predicted by (113.6), (113.7); this phenomenon is called the *Paschen–Back effect*.

The splitting energy is particularly easy to calculate when the Zeeman splitting is large compared with the fine-structure intervals but remains, of course, small compared with the separations between different multiplets (so that the third term in the Hamiltonian (113.2) may still be neglected in comparison with the second). In other words, the energy in the magnetic field greatly exceeds the spin–orbit interaction[10]. This interaction may therefore be neglected in first approximation. Then not only the projection of the total angular momentum but also the projections M_L and M_S of the orbital angular momentum and spin are conserved, and the splitting is given by

$$\Delta E = \mu_B H(M_L + 2M_S). \tag{113.13}$$

The multiplet splitting is superposed on the magnetic-field splitting. It is determined by the mean value of the operator $A\hat{\mathbf{L}}\hat{\mathbf{S}}$ (72.4) in a state with given M_L, M_S (we are considering the multiplet splitting associated with the spin–orbit interaction). For a specified value of one angular-momentum component, the mean values of the other two vanish. Hence $\overline{\mathbf{LS}} = M_L M_S$, and in the next approximation the level energies are given by

$$\Delta E = \mu_B H(M_L + 2M_S) + AM_L M_S. \tag{113.14}$$

The Zeeman splitting cannot be calculated in general for an arbitrary coupling scheme (not LS coupling). One can assert only that the splitting in a weak field is linear in the field and proportional to the projection M_J of the total angular momentum, that is, that it has the form

$$\Delta E = \mu_B g_{nJ} H M_J, \tag{113.15}$$

where g_{nJ} are coefficients characteristic of the term in question (the letter n denotes the set of quantum numbers, apart from J, that characterize the term). Although these coefficients cannot be calculated individually, it is possible to derive a useful formula for their sum over all possible states of the atom with the given electron configuration and total angular momentum.

[9]For $S = L \neq 0$, the off-diagonal matrix elements of L_z, S_z for the transitions $S, L, J \to S, L, J \pm 1$ are, in general, nonzero.

[10]For intermediate cases, in which the influence of the magnetic field is comparable to the spin–orbit interaction, the splitting cannot be calculated in general form (the case $S = 1/2$ is worked out in Problem 1).

By definition,

$$g_{nJ} M_J = \langle nJM_J | L_z + 2S_z | nJM_J \rangle.$$

The quantities $g_{SLJ} M_J$ (where g_{SLJ} is the Landé factor (113.7) corresponding to LS coupling) are the diagonal matrix elements

$$g_{SLJ} M_J = \langle SLJM_J | L_z + 2S_z | SLM_J \rangle,$$

calculated in another complete system of wave functions. The functions of either system are obtained from those of the other by a linear unitary transformation. Such a transformation, however, leaves the sum of the diagonal matrix elements unchanged (§ 12). It follows that

$$\sum_n g_{nJ} M_J = \sum_{SL} g_{SLJ} M_J,$$

or, since g_{nJ} and g_{SLJ} do not depend on M_J,

$$\sum_n g_{nJ} = \sum_{SL} g_{SLJ}. \tag{113.16}$$

The summation is over all states with the given value of J that are possible for the specified electron configuration. This is the required relation.

Problems

PROBLEM

1. Determine the splitting of a term with $S = 1/2$ in the Paschen–Back effect.

SOLUTION. The magnetic field and the spin–orbit interaction must be included simultaneously in perturbation theory; thus the perturbation operator has the form[11]

$$\hat{V} = A\hat{\mathbf{L}}\hat{\mathbf{S}} + \mu_B(\hat{L}_z + 2\hat{S}_z)H.$$

As the initial zeroth-approximation wave functions we choose functions corresponding to states with definite values of L, $S = 1/2$, M_L, and M_S (L is specified, $M_L = -L, \ldots, L$; $M_S = \pm 1/2$). In the perturbed states only the sum $M \equiv M_J = M_L + M_S$ is conserved ($\hat{V}$ commutes with $\hat{J}_z$), so definite values of M can be assigned to the components of the split term.

The values $M = \pm(L + 1/2)$ can be realized in only one way: respectively with $|M_L M_S\rangle = |L, 1/2\rangle$ and $|-L, -1/2\rangle$. The energy corrections for states with these values of M are therefore simply the diagonal matrix elements $\langle M_L M_S | \hat{V} | M_L M_S \rangle$ at the indicated values of $|M_L M_S\rangle$. The remaining values of M can be realized in two ways, $|M - 1/2, 1/2\rangle$ and $|M + 1/2, -1/2\rangle$. Each M then corresponds to two different energy values, found from the secular equation formed from the matrix elements for transitions between these two states.

[11] We do not write in $\hat{V}$ a term proportional to $(\hat{\mathbf{L}}\hat{\mathbf{S}})^2$ (spin–spin interaction). It must be kept in mind, however, that for spin $S = 1/2$ the expression $(\hat{\mathbf{L}}\hat{\mathbf{S}})^2$, by virtue of the special properties of the Pauli matrices (see § 55), reduces to an expression in $\hat{\mathbf{L}}\hat{\mathbf{S}}$ and is therefore included in the formula displayed.

The matrix elements of **LS** are obtained by directly multiplying the matrices $\langle M_L|\mathbf{L}|M_L'\rangle$ and $\langle M_S|\mathbf{S}|M_S'\rangle$, and are

$$\langle M_L M_S|\mathbf{LS}|M_L M_S\rangle = M_L M_S,$$

$$\begin{aligned}\langle M+1/2,-1/2|\mathbf{LS}|M-1/2,1/2\rangle &= \langle M-1/2,1/2|\mathbf{LS}|M+1/2,-1/2\rangle\\ &= \frac{1}{2}\sqrt{(L+M+1/2)(L-M+1/2)}.\end{aligned}$$

In the absence of a magnetic field the term is a doublet whose components are separated by $\epsilon = A(L+1/2)$ (see (72.6)). Choose the lower of these levels as the energy origin. The final formulas for the levels in a magnetic field are then

$$E = \epsilon \pm \mu_B H(L+1) \quad \text{for } M = \pm(L+1/2),$$

$$E^{\pm} = \frac{\epsilon}{2} + \mu_B HM \pm \left[\frac{1}{4}(\epsilon^2 + \mu_B^2 H^2) + \frac{\mu_B HM\epsilon}{2L+1}\right]^{1/2}$$

for $M = L-1/2, \ldots, -(L-1/2)$.

For $\mu_B H/\epsilon \ll 1$, we obtain

$$E^+ = \epsilon + \mu_B HM\frac{2(L+1)}{2L+1}, \qquad E^- = \mu_B HM\frac{2L}{2L+1},$$

in agreement with (113.6), (113.7), in which $S = 1/2$, $J = L \pm 1/2$ are to be substituted. For $\mu_B H/\epsilon \gg 1$,

$$E^{\pm} = \mu_B H(M \pm 1/2) + \frac{\epsilon}{2} \pm \frac{M\epsilon}{2L+1},$$

in agreement with (113.14).

Problem

2. Determine the Zeeman splitting of the terms of a diatomic molecule in case *a*.

Solution. The magnetic moment due to the motion of the nuclei is very small compared with the magnetic moment of the electrons. The magnetic-field perturbation for the molecule must therefore be written as for a system of electrons, that is, still in the form $\hat{V} = \mu_B \mathbf{H}(\hat{\mathbf{L}} + 2\hat{\mathbf{S}})$, where **L**, **S** are the electronic orbital and spin angular momenta.

Averaging the perturbation over the electronic state, we obtain in case *a*

$$\mu_B H n_z(\Lambda + 2\Sigma) = \mu_B H n_z(2\Omega - \Lambda).$$

The mean value of n_z over the molecular rotation is the diagonal matrix element

$$\langle JM|n_z|JM\rangle = \frac{\Omega M}{J(J+1)},$$

where $M \equiv M_J$ (the matrix element is calculated from the reduced matrix element given by (87.4), with $K, \Lambda \to J, \Omega$).

Thus the required splitting is

$$\Delta E = \mu_B HM\frac{\Omega(2\Omega - \Lambda)}{J(J+1)}.$$

PROBLEM

3. The same for case b.

SOLUTION. The diagonal matrix elements $\langle \Lambda KJ|V|\Lambda KJ\rangle$ that determine the required splitting could be calculated by the general rules set out in § 87. It is simpler, however, to carry out the calculation in a more direct form. Averaging the perturbation operator over the orbital and electronic states gives

$$\mu_B H(\Lambda n_z + 2\hat{S}_z)$$

(the spin operator is unaffected by this averaging). Next average over the molecular rotation (the mean value of n_z is found with (87.4)):

$$\mu_B H\left(\frac{\Lambda^2}{K(K+1)}\hat{K}_z + 2\hat{S}_z\right).$$

Finally, we average over the spin wave function. After the complete averaging, the mean vectors can be directed only along the sole conserved vector, the total angular momentum **J**. We therefore obtain (cf. (113.5))

$$\frac{\mu_B H}{J(J+1)}\left[\frac{\Lambda^2}{K(K+1)}(\mathbf{KJ}) + 2(\mathbf{SJ})\right]M$$

($M \equiv M_J$), or, finally,

$$\Delta E = \frac{\mu_B}{J(J+1)}\left\{\frac{\Lambda^2}{2K(K+1)}[J(J+1)+K(K+1)-S(S+1)]\right.$$
$$\left. + [J(J+1)-K(K+1)+S(S+1)]\right\}HM.$$

PROBLEM

4. A diamagnetic atom is in an external magnetic field. Determine the strength of the induced magnetic field at the center of the atom.

SOLUTION. For $S = L = 0$, the Hamiltonian contains no perturbation linear in the field at all, and consequently the atomic wave function has no first-order correction in the magnetic field. The change $\mathbf{j}'$ in the electron current in the atom induced by the external magnetic field is associated (in the same first approximation in H) only with the addition of the term $(|e|/mc)\mathbf{A}$ to the electron velocity operators. Hence[12]

$$\mathbf{j}' = -\rho\frac{e^2}{mc}\mathbf{A} = -\rho\frac{e^2}{2mc}(\mathbf{H}\times\mathbf{r}), \tag{1}$$

where ρ is the electron density in the atom. The strength of the magnetic field produced at the center of the atom by this additional current is

$$\mathbf{H}_{\text{ind}} = \frac{1}{c}\int\frac{\mathbf{r}\times\mathbf{j}'}{r^3}\,dV$$

[12]This expression corresponds to the Larmor precession of the atom's electron shell about the direction of the external magnetic field (see II, § 45).

(cf. (121.8) below). Substituting (1) here and averaging over the directions of **r** under the integral, we obtain

$$\mathbf{H}_{\text{ind}} = -\frac{e^2}{3mc^2}\mathbf{H}\int \frac{\rho}{r}\,dV = \frac{e}{3mc^2}\varphi_e(0)\mathbf{H}, \tag{2}$$

where $\varphi_e(0)$ is the potential of the field produced at its center by the atom's electron shell.

In the Thomas–Fermi model $\varphi_e(0) = -1{,}80Z^{4/3}me^3/\hbar^2$ (see (70.8)), and hence

$$\mathbf{H}_{\text{ind}} = -0{,}60\left(\frac{e^2}{\hbar c}\right)^2 Z^{4/3}\mathbf{H} = -3{,}2\cdot 10^{-5}Z^{4/3}\mathbf{H}.$$

§ 114. Spin in a time-dependent magnetic field

Consider an electrically neutral particle with a magnetic moment, placed in a uniform magnetic field which varies with time. The particle may be elementary (a neutron, for example) or composite (an atom). We suppose the field to be so weak that the particle's magnetic energy in it is small compared with the separations between its energy levels. The motion of the particle as a whole may then be considered for a specified internal state.

Let $\hat{\mathbf{s}}$ denote the operator of the particle's "intrinsic" angular momentum: its spin for an elementary particle, or the total angular momentum **J** for an atom. Write the magnetic-moment operator in the form (111.1). The Hamiltonian governing the motion of the neutral particle as a whole is

$$\widehat{H} = -\frac{\mu}{s}\hat{\mathbf{s}}\mathbf{H} \tag{114.1}$$

(only the spin-dependent part of the Hamiltonian has been displayed).

In a uniform field this operator contains no explicit coordinates[13]. The particle's wave function therefore separates into coordinate and spin factors. The former is merely the wave function for free motion; below we are concerned only with the spin factor. We shall show that the problem for a particle of arbitrary angular momentum s reduces to the simpler problem of the motion of a spin-1/2 particle (E. Majorana). For this purpose we use the device already employed in § 57. In place of one particle with spin s, we formally introduce a system of $2s$ spin-1/2 "particles." The operator $\hat{\mathbf{s}}$ is then represented by the sum $\sum_a \hat{\mathbf{s}}_a$ of their spin operators, while the wave function is represented by a product of $2s$ spinors of rank one. The Hamiltonian (114.1) accordingly separates into a sum of $2s$ independent Hamiltonians:

$$\widehat{H} = \sum_a \widehat{H}_a, \qquad \widehat{H}_a = -\frac{\mu}{s}\mathbf{H}\hat{\mathbf{s}}_a, \tag{114.2}$$

[13]The same reasoning also applies when a particle (including a charged one) moves through a nonuniform magnetic field and its motion can be treated semiclassically. The magnetic field encountered as the particle advances along its trajectory may then simply be regarded as a function of time, and the same equations may be used for the change of the spin wave function.

so that each of the $2s$ "particles" moves independently of all the others. Once this has been done, the products of components of the $2s$ rank-one spinors need only be replaced again by the components of an arbitrary symmetric spinor of rank $2s$.

Problems

Problem

1. Find the spin wave function of a neutral spin-$1/2$ particle in a uniform magnetic field whose direction is fixed but whose magnitude follows an arbitrary function $H(t)$.

SOLUTION. The wave function is a spinor ψ^ν satisfying the wave equation

$$i\hbar\dot{\psi}^\nu = -2\mu\mathbf{H}\hat{\mathbf{s}}\psi^\nu. \tag{1}$$

Choose the field direction as the z axis. In spinor components, this equation then becomes

$$i\hbar\dot{\psi}^1 = -\mu H\psi^1, \qquad i\hbar\dot{\psi}^2 = \mu H\psi^2.$$

It follows that

$$\psi^1 = c_1 \exp\left(\frac{i\mu}{\hbar}\int H\,dt\right), \qquad \psi^2 = c_2 \exp\left(-\frac{i\mu}{\hbar}\int H\,dt\right).$$

The constants c_1, c_2 are fixed by the initial conditions and the normalization condition $|\psi^1|^2 + |\psi^2|^2 = 1$.

Problem

2. Solve the same problem for a uniform magnetic field of constant magnitude whose direction rotates uniformly (with angular velocity ω) about the z axis, maintaining an angle θ with it.

SOLUTION. The components of the magnetic field are

$$H_x = H\sin\theta\cos\omega t, \qquad H_y = H\sin\theta\sin\omega t, \qquad H_z = H\cos\theta,$$

and (1) gives the system

$$\dot{\psi}^1 = i\omega_H(\cos\theta\cdot\psi^1 + \sin\theta\cdot e^{-i\omega t}\psi^2),$$
$$\dot{\psi}^2 = i\omega_H(\sin\theta\cdot e^{i\omega t}\psi^1 - \cos\theta\cdot\psi^2),$$

where $\omega_H = \mu H/\hbar$. The substitution

$$\psi^1 = e^{-i\omega t/2}\varphi^1, \qquad \psi^2 = e^{i\omega t/2}\varphi^2$$

reduces these equations to a system of linear equations with constant coefficients. Solving it gives

$$\psi^1 = e^{-i\omega t/2}\left(c_1 e^{i\Omega t/2} + c_2 e^{-i\Omega t/2}\right),$$

$$\psi^2 = 2\omega_H\sin\theta\, e^{i\omega t/2}\left[\frac{c_1}{\Omega+\omega+2\omega_H\cos\theta}e^{i\Omega t/2} - \frac{c_2}{\Omega-\omega-2\omega_H\cos\theta}e^{-i\Omega t/2}\right],$$

where

$$\Omega = \sqrt{(\omega+2\omega_H\cos\theta)^2 + 4\omega_H^2\sin^2\theta}.$$

§ 115. Current density in a magnetic field

We shall derive the quantum-mechanical expression for the current density of a charged particle moving in a magnetic field. We start from the formula[14]

$$\delta H = -\frac{1}{c}\int \mathbf{j}\,\delta\mathbf{A}\,dV, \tag{115.1}$$

which determines the change in the Hamiltonian function of charges distributed in space when the vector potential is varied.[15] In quantum mechanics this formula must be applied to the mean value of the Hamiltonian of the charged particle:

$$\overline{H} = \int \Psi^* \left[\frac{1}{2m}\left(\hat{\mathbf{p}} - \frac{e}{c}\mathbf{A}\right)^2 - \frac{\mu}{s}\mathbf{H}\hat{\mathbf{s}}\right]\Psi\,dV. \tag{115.2}$$

Performing the variation and recalling that $\delta\mathbf{H} = \nabla\times\delta\mathbf{A}$, we find

$$\begin{aligned}\delta\overline{H} = &\int \Psi^*\left[-\frac{e}{2mc}(\hat{\mathbf{p}}\,\delta\mathbf{A} + \delta\mathbf{A}\hat{\mathbf{p}}) + \frac{e^2}{mc^2}\mathbf{A}\,\delta\mathbf{A}\right]\Psi\,dV\\ &-\frac{\mu}{s}\int \nabla\times\delta\mathbf{A}\cdot\Psi^*\hat{\mathbf{s}}\Psi\,dV.\end{aligned} \tag{115.3}$$

We transform the term containing $\hat{\mathbf{p}}\,\delta\mathbf{A}$ by integration by parts:

$$\int \Psi^*\hat{\mathbf{p}}\,\delta\mathbf{A}\Psi\,dV = -i\hbar\int \Psi^*\nabla(\delta\mathbf{A}\Psi)\,dV = i\hbar\int \delta\mathbf{A}\Psi\nabla\Psi^*\,dV$$

(the integral over the infinitely distant surface vanishes, as usual). We also integrate by parts in the last term of (115.3), using the familiar formula of vector analysis

$$\mathbf{a}\cdot\nabla\times\mathbf{b} = -\nabla\cdot\mathbf{a}\times\mathbf{b} + \mathbf{b}\cdot\nabla\times\mathbf{a}.$$

The integral of the term containing $\nabla\cdot$ vanishes, leaving

$$\int \Psi^*\hat{\mathbf{s}}\Psi\cdot\nabla\times\delta\mathbf{A}\,dV = \int \delta\mathbf{A}\cdot\nabla\times(\Psi^*\hat{\mathbf{s}}\Psi)\,dV.$$

The final result is therefore

$$\begin{aligned}\delta\overline{H} = &-\frac{ie\hbar}{2mc}\int \delta\mathbf{A}(\Psi\nabla\Psi^* - \Psi^*\nabla\Psi)\,dV\\ &+\frac{e^2}{mc^2}\int \mathbf{A}\,\delta\mathbf{A}\Psi\Psi^*\,dV - \frac{\mu}{s}\int \delta\mathbf{A}\cdot\nabla\times(\Psi^*\hat{\mathbf{s}}\Psi)\,dV.\end{aligned}$$

[14] In this section $\mathbf{j}$ denotes the electric-current density: the particle-flux density multiplied by the charge e.

[15] The Lagrangian function of a charge in a magnetic field contains the term $(e/c)\mathbf{vA}$, or, if the charge is represented as distributed in space, $(1/c)\int \mathbf{jA}\,dV$. Consequently, the change in the Lagrangian function under a variation of $\mathbf{A}$ is

$$\delta L = \frac{1}{c}\int \mathbf{j}\,\delta\mathbf{A}\,dV.$$

An infinitesimal change in the Hamiltonian function is the negative of the change in the Lagrangian function (see I, § 40).

Comparing this with (115.1), we obtain the following expression for the current density:

$$\mathbf{j} = \frac{ie\hbar}{2m}[(\nabla\Psi^*)\Psi - \Psi^*\nabla\Psi] - \frac{e^2}{mc}\mathbf{A}\Psi^*\Psi + \frac{\mu}{s}c\nabla\times(\Psi^*\hat{\mathbf{s}}\Psi). \tag{115.4}$$

We emphasize that although the vector potential occurs explicitly in this expression, the expression is nevertheless, as it must be, entirely unambiguous. This is readily verified by direct calculation: along with the gauge transformation of the vector potential, one must, by (111.8), also transform the wave function according to (111.9).

It is also readily checked that the current (115.4), together with the charge density $\rho = e|\Psi|^2$, obeys the continuity equation, as required:

$$\frac{\partial\rho}{\partial t} + \nabla\cdot\mathbf{j} = 0.$$

The last term in (115.4) contributes to the current density through the particle's magnetic moment. It has the form $c\nabla\times\mathbf{m}$, where

$$\mathbf{m} = \frac{\mu}{s}\Psi^*\hat{\mathbf{s}}\Psi = \Psi^*\hat{\boldsymbol{\mu}}\Psi \tag{115.5}$$

is the spatial density of magnetic moment.

Expression (115.4) is the mean value of the current density. It may be regarded as a diagonal matrix element of an operator—the current-density operator $\hat{\mathbf{j}}$. This operator is written most simply in the second-quantized representation: one replaces Ψ and Ψ^* by the operators $\hat{\Psi}$ and $\hat{\Psi}^+$ (and, by the general rule, $\hat{\Psi}^+$ must stand to the left of $\hat{\Psi}$ in every term). The off-diagonal matrix elements of this operator may also be defined:

$$\mathbf{j}_{nm} = \frac{ie\hbar}{2m}[(\nabla\Psi_n^*)\Psi_m - \Psi_n^*\nabla\Psi_m] - \frac{e^2}{mc}\mathbf{A}\Psi_n^*\Psi_m + \frac{\mu}{s}c\nabla\times(\Psi_n^*\hat{\mathbf{s}}\Psi_m). \tag{115.6}$$

CHAPTER XVI

Structure of the Atomic Nucleus

CHAPTER XVII

Elastic Collisions

CHAPTER XVIII

Inelastic Collisions

Mathematical Supplements

§ a. Hermite Polynomials

The equation

$$y'' - 2xy' + 2ny = 0 \tag{a.1}$$

belongs to a class of equations soluble by the *Laplace method*[1].

In general, the method applies to linear equations of the form

$$\sum_{m=0}^{n} (a_m + b_m x) \frac{d^m y}{dx^m} = 0,$$

whose coefficients are at most linear in x. Its procedure is as follows. Form the polynomials

$$P(t) = \sum_{m=0}^{n} a_m t^m, \qquad Q(t) = \sum_{m=0}^{n} b_m t^m$$

and from them construct

$$Z(t) = \frac{1}{Q} \exp \int \frac{P}{Q}\, dt,$$

which is determined up to a constant factor. A solution of the equation can then be written as the complex integral

$$y = \int_C Z(t) e^{xt}\, dt.$$

The path C is to be chosen so that this integral is finite and nonzero, and the function

$$V = e^{xt} QZ$$

must return to its initial value after t has traversed the whole of C; the contour may be either closed or open. For equation (a.1),

$$P = t^2 + 2n; \qquad Q = -2t, \qquad Z = -\frac{1}{2t^{n+1}} e^{-t^2/4}, \qquad V = \frac{1}{t^n} e^{xt - t^2/4},$$

and hence its solution takes the form

$$y = \int \exp\left(xt - \frac{t^2}{4}\right) \frac{dt}{t^{n+1}}. \tag{a.2}$$

For physical applications it is sufficient to consider $n > -1/2$. At these values either C_1 or C_2 in Fig. 52 may serve as the integration path. The required conditions are satisfied because V vanishes at their endpoints, $t = +\infty$ or $t = -\infty$[2].

[1] See, for example, É. Goursat, *Course in Mathematical Analysis*, vol. II, Moscow: Gostekhizdat, 1933; V. I. Smirnov, *Course of Higher Mathematics*, vol. III, part 2, Moscow: Nauka, 1974.

[2] These paths cannot be used for negative integral n, since the integral (a.2) along either path would then vanish identically.

We shall determine the values of n for which (a.1) has a solution that is finite at every finite x and, as $x \longrightarrow \pm\infty$, grows no faster than a finite power of x. Begin with nonintegral n. Integration in (a.2) along C_1 and C_2 gives two independent solutions. Transform the C_1 integral by introducing u through $t = 2(x - u)$. On omitting a constant factor, the result is

$$y = e^{x^2} \int_{C_1'} \frac{e^{-u^2}}{(u - x)^{n+1}}\, du, \tag{a.3}$$

where C_1' is the contour in the complex u plane shown in Fig. 53.

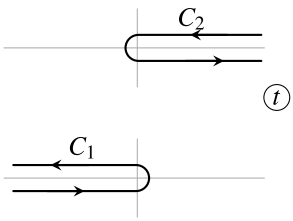

Figure 52

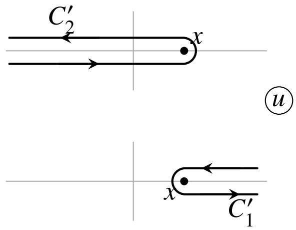

Figure 53

As $x \longrightarrow +\infty$, the whole contour C_1' recedes to infinity, and the integral in (a.3) tends to zero as e^{-x^2}. For $x \longrightarrow -\infty$, however, the integration path extends along the entire real axis and the integral in (a.3) does not decrease exponentially; the leading growth of $y(x)$ is therefore e^{x^2}. In the same way, one readily verifies that the integral (a.2) over C_2 diverges exponentially as $x \longrightarrow \infty$.

When n is a nonnegative integer, the integrals over the straight parts of the path cancel pairwise. Both versions of (a.3), over C_1' and C_2', then reduce to an integral over a closed contour surrounding the point $u = x$. We thus obtain the solution

$$y(x) = e^{x^2} \oint \frac{e^{-u^2}}{(u - x)^{n+1}}\, du,$$

which has the required properties. By Cauchy's familiar formula for the derivatives of an analytic function,

$$f^{(n)}(x) = \frac{n!}{2\pi i} \oint \frac{f(t)}{(t - x)^{n+1}}\, dt,$$

this solution, apart from a constant factor, is the *Hermite polynomial*

$$H_n(x) = (-1)^n e^{x^2} \frac{d^n}{dx^n} e^{-x^2}. \tag{a.4}$$

Expanded and ordered by descending powers of x, H_n is

$$H_n(x) = (2x)^n - \frac{n(n-1)}{1}(2x)^{n-2} + \frac{n(n-1)(n-2)(n-3)}{1 \cdot 2}(2x)^{n-4} - \dots \tag{a.5}$$

It contains only powers of x having the same parity as n. The first few Hermite polynomials are

$$\begin{gathered} H_0 = 1, \quad H_1 = 2x, \quad H_2 = 4x^2 - 2, \quad H_3 = 8x^3 - 12x, \\ H_4 = 16x^4 - 48x^2 + 12. \end{gathered} \tag{a.6}$$

To evaluate the normalization integral, substitute the expression (a.4) for $e^{-x^2}H_n$ and integrate by parts n times:

$$\int_{-\infty}^{+\infty} e^{-x^2} H_n^2(x)\,dx = \int_{-\infty}^{+\infty} (-1)^n H_n(x) \frac{d^n}{dx^n} e^{-x^2}\,dx = \int_{-\infty}^{+\infty} e^{-x^2} \frac{d^n H_n}{dx^n}\,dx.$$

But $d^n H_n/dx^n$ is the constant $2^n n!$, and consequently

$$\int_{-\infty}^{+\infty} e^{-x^2} H_n^2(x)\,dx = 2^n n! \sqrt{\pi}. \tag{a.7}$$

Subject Index